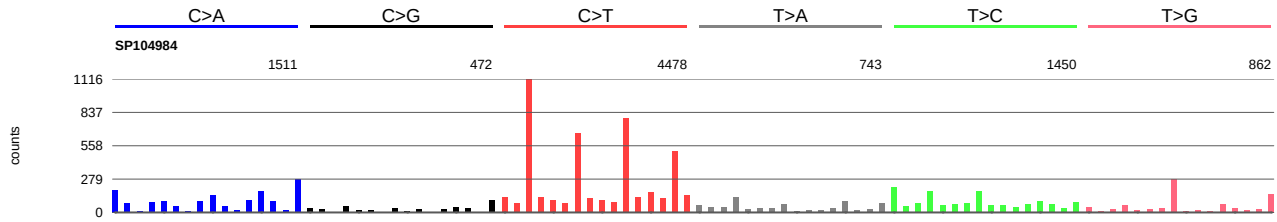
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SP104984
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862
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837
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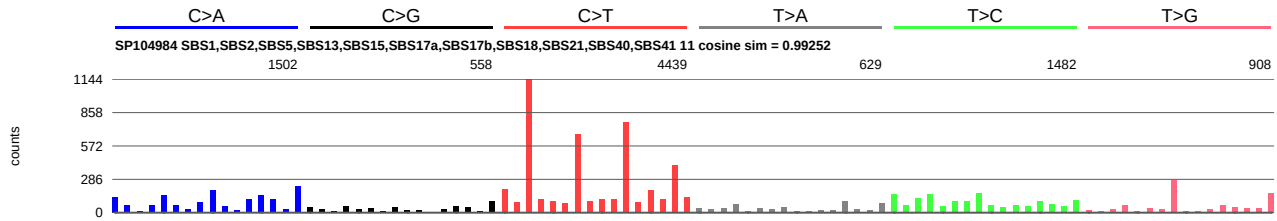
counts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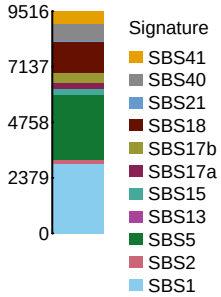
558
4439
629
1482
908
1144
858
572
286
0
counts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7137
4758
2379
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Signature
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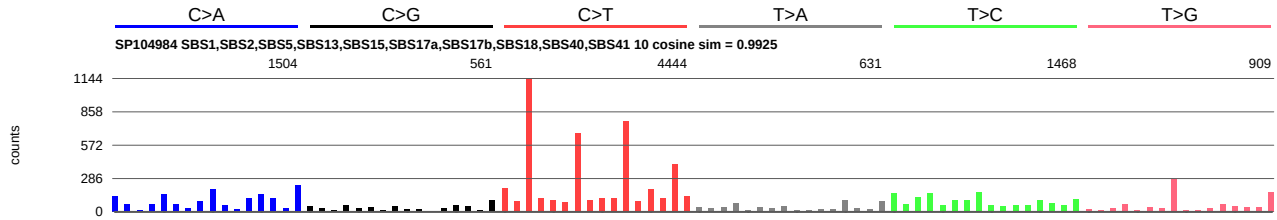
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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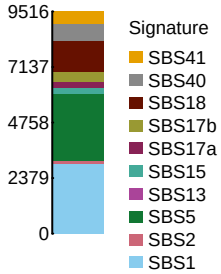
561
4444
631
1468
909
1144
858
572
286
0
counts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7137
4758
2379
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Signature
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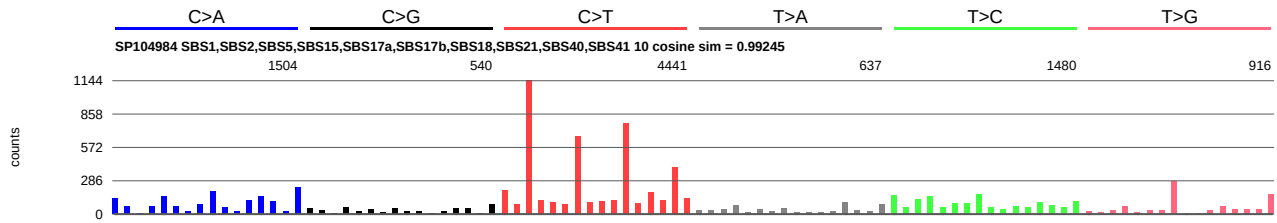
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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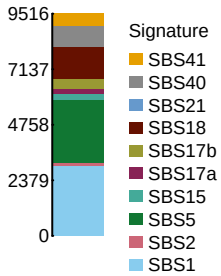
540
4441
637
1480
916
1144
858
572
286
0
counts


9516
7137
4758
2379
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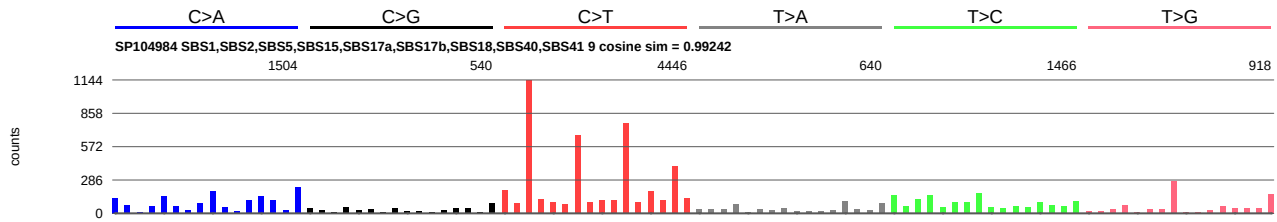
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99242
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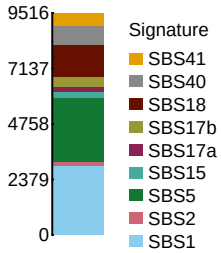
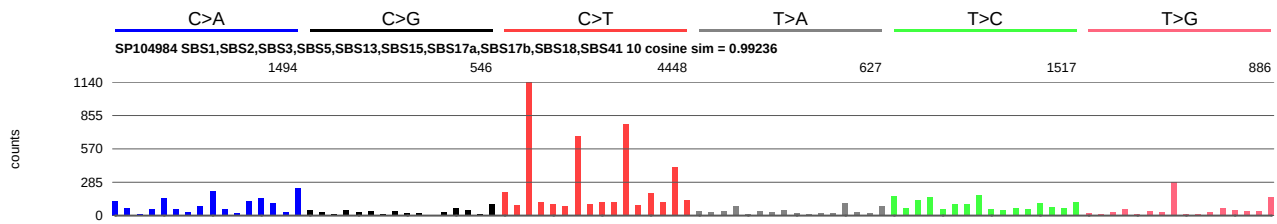
540
4446
640
1466
918
1144
858
572
286
0
counts


9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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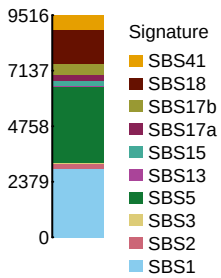
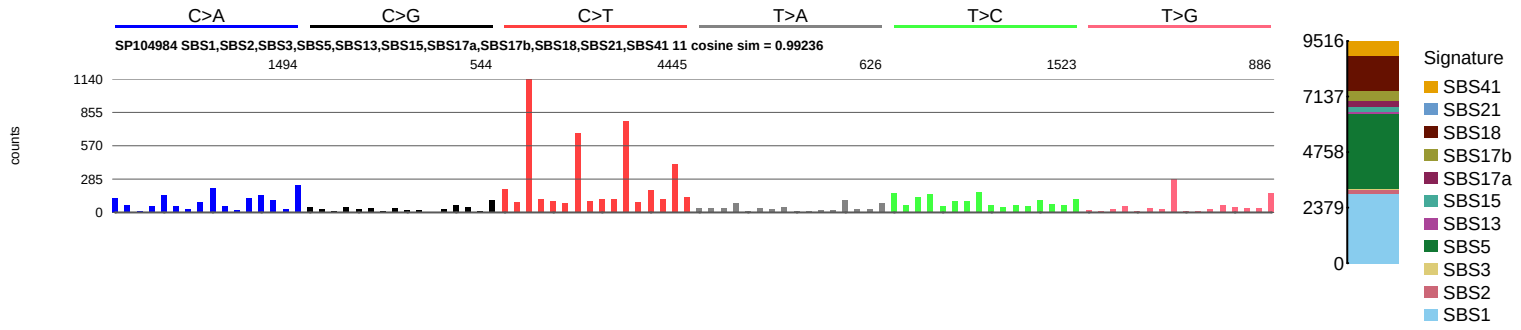
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.99236
1494
546
4448
627
1517
886
1140
855
570
285
0
counts


9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.99236
1494
544
4445
626
1523
886
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
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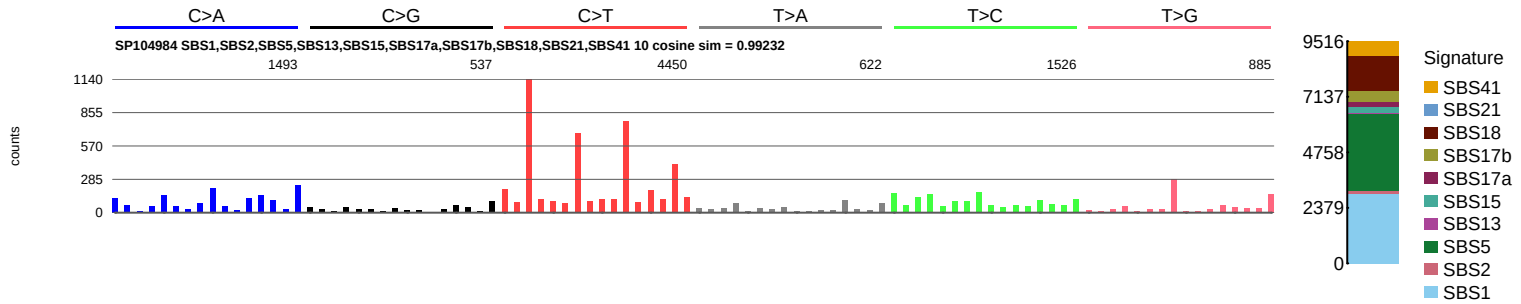
SBS13
SBS5
SBS3
SBS2
SBS1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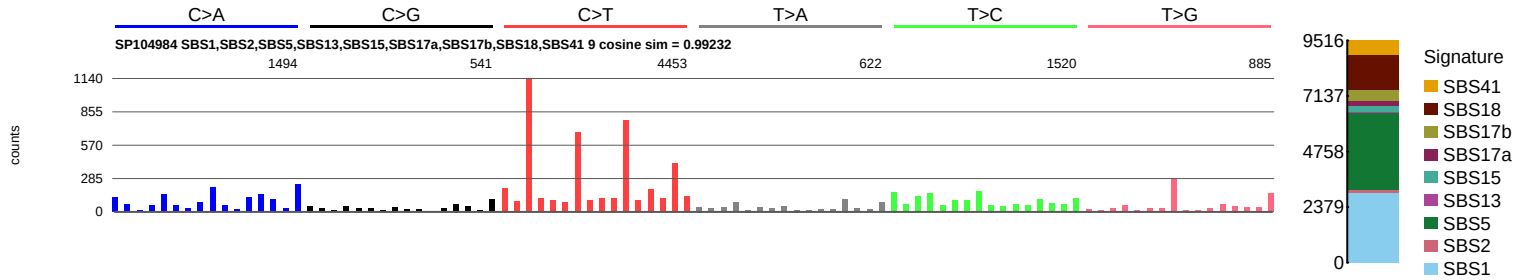
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99232
1493
537
4450
622
1526
885
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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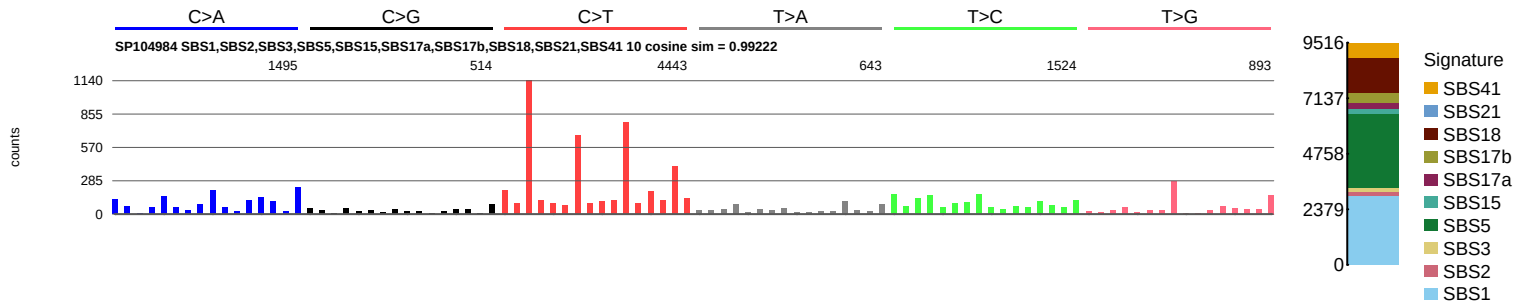
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99232
1494
541
4453
622
1520
885
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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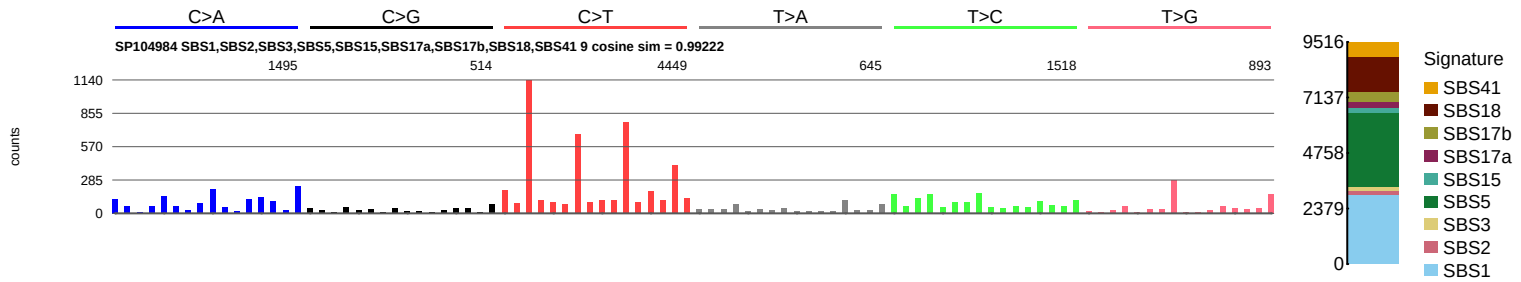
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99222
1495
514
4443
643
1524
893
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99222
1495
514
4449
645
1518
893
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
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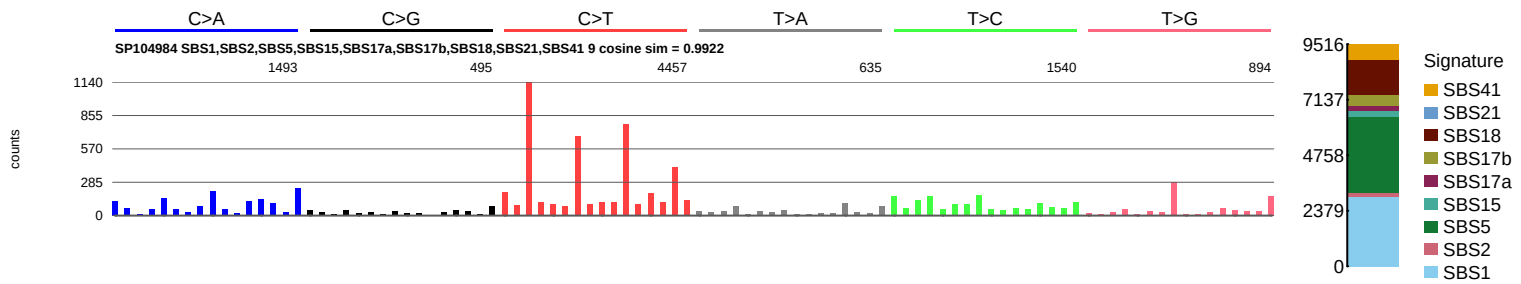
SBS2
SBS1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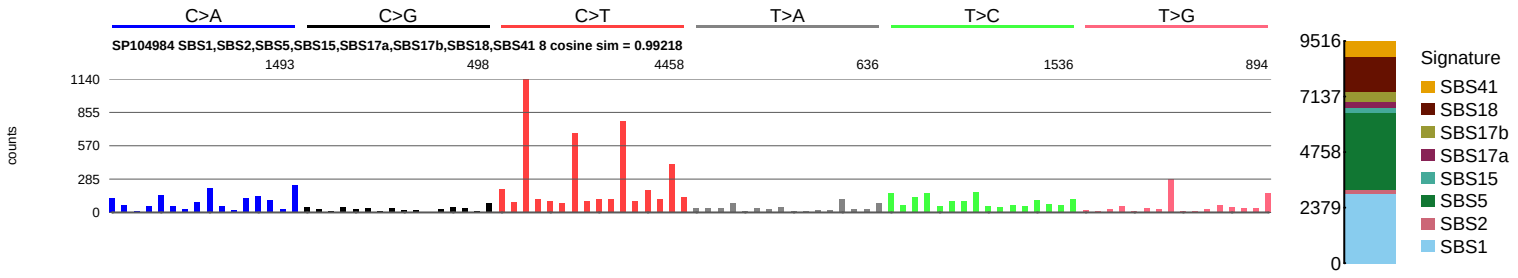
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9922
1493
495
4457
635
1540
894
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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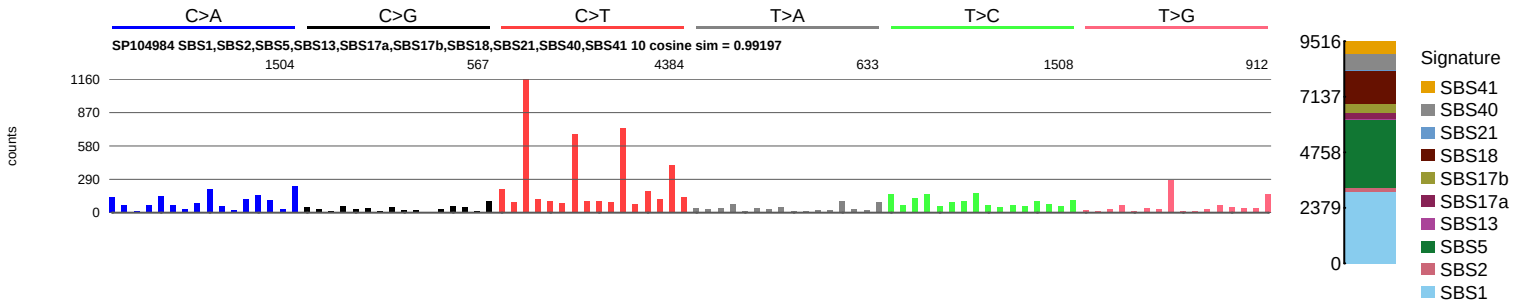
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99218
1493
498
4458
636
1536
894
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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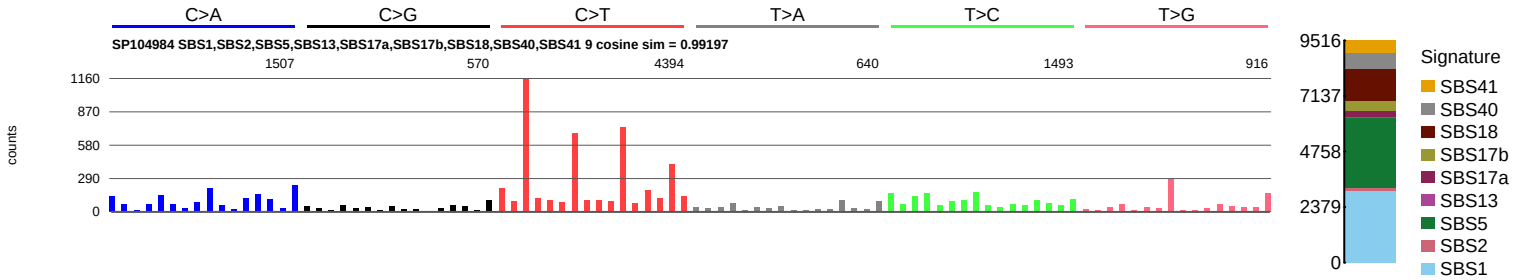
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99197
1504
567
4384
633
1508
912
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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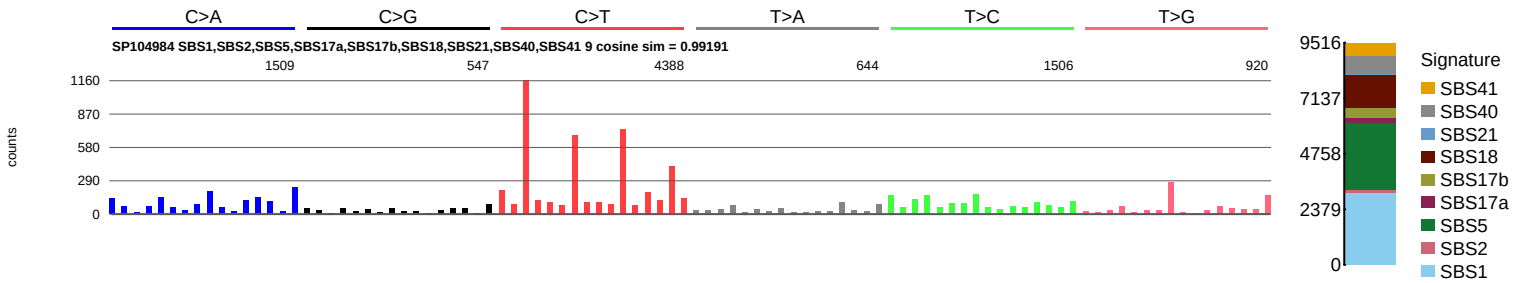
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99197
1507
570
4394
640
1493
916
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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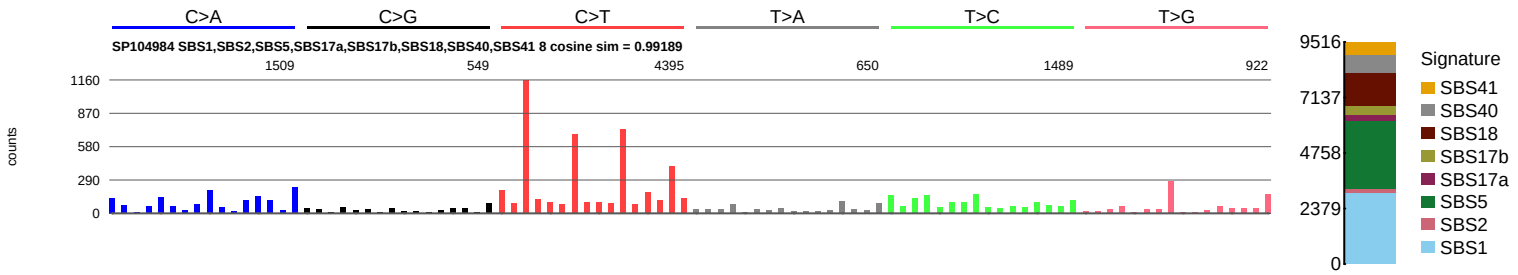
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99191
1509
547
4388
644
1506
920
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99189
1509
549
4395
650
1489
922
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
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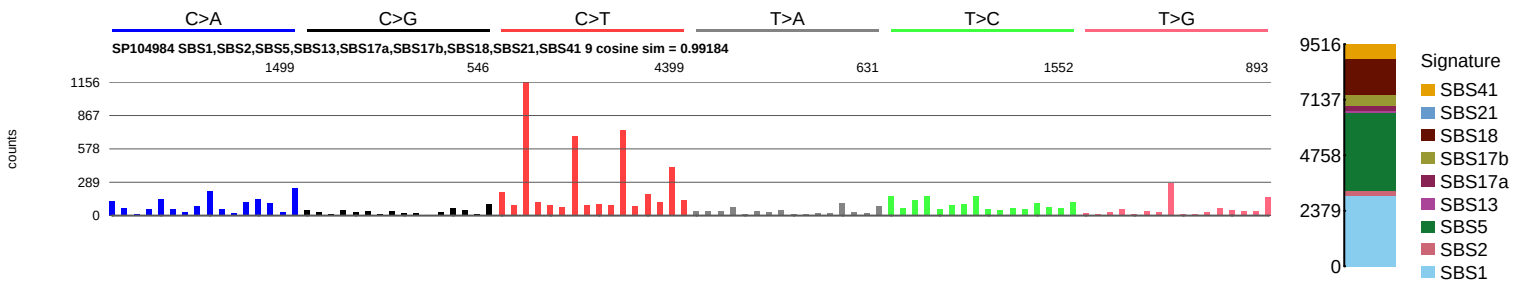
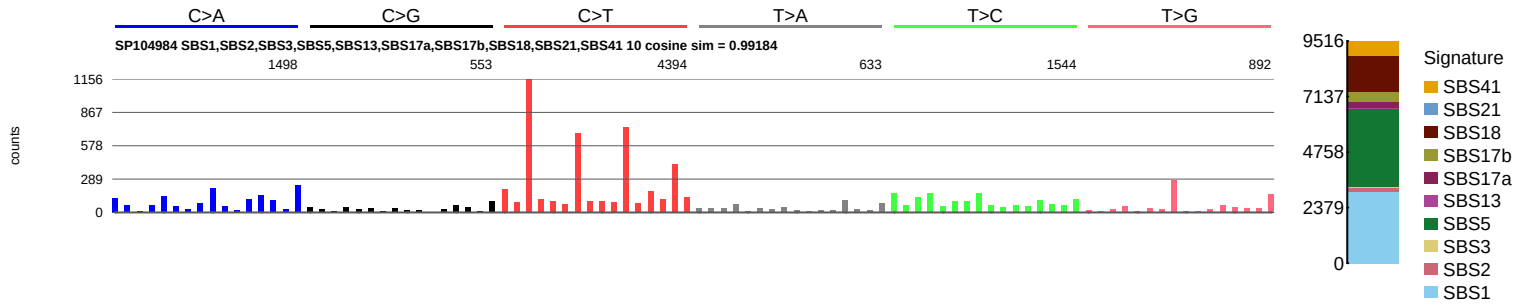
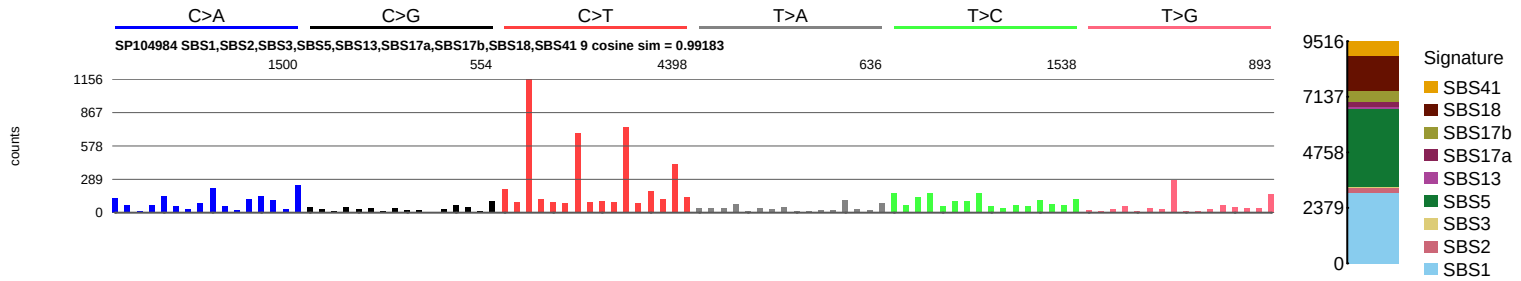
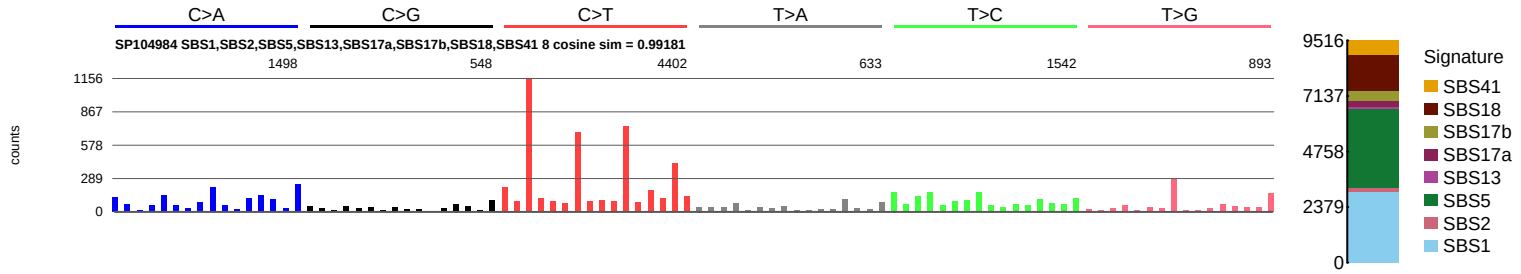
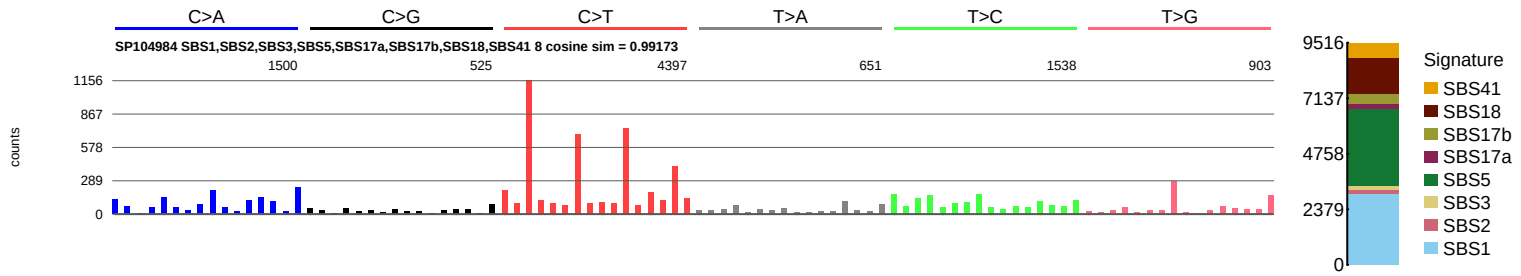
SBS2
SBS1

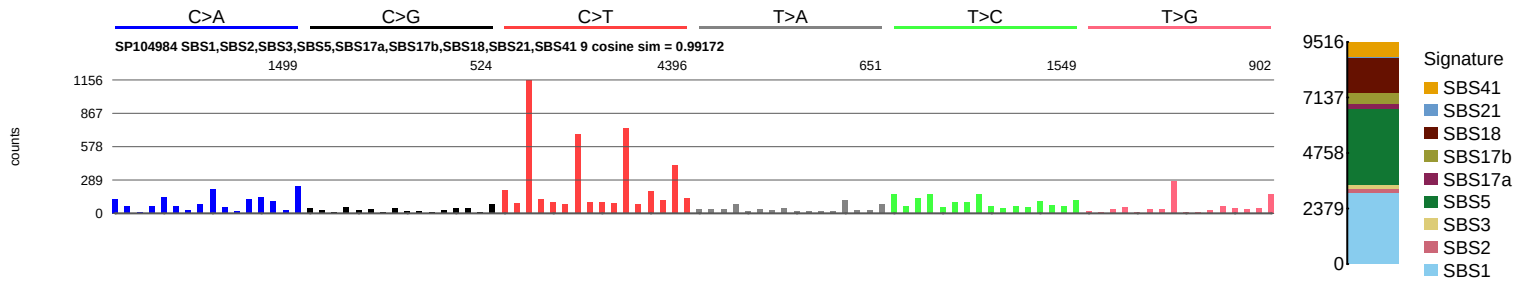

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99184
1499
546
4399
631
1552
893
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99184
1498
553
4394
633
1544
892
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99183
1500
554
4398
636
1538
893
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99181
1498
548
4402
633
1542
893
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99173
1500
525
4397
651
1538
903
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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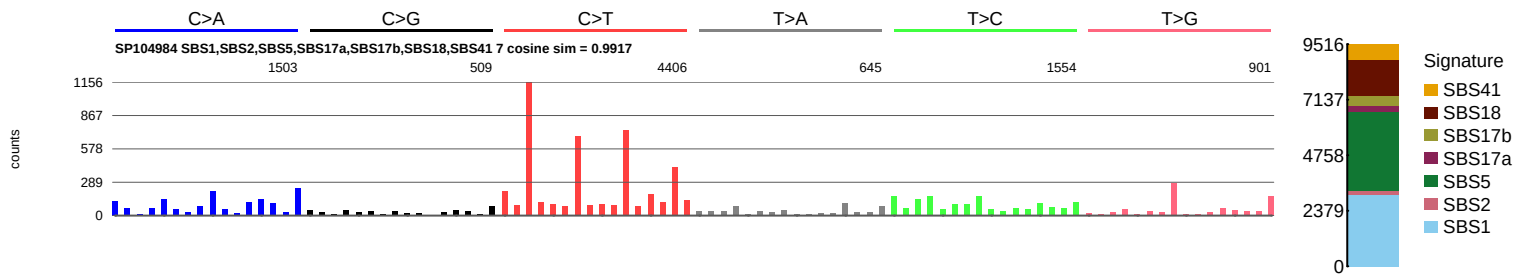
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99172
1499
524
4396
651
1549
902
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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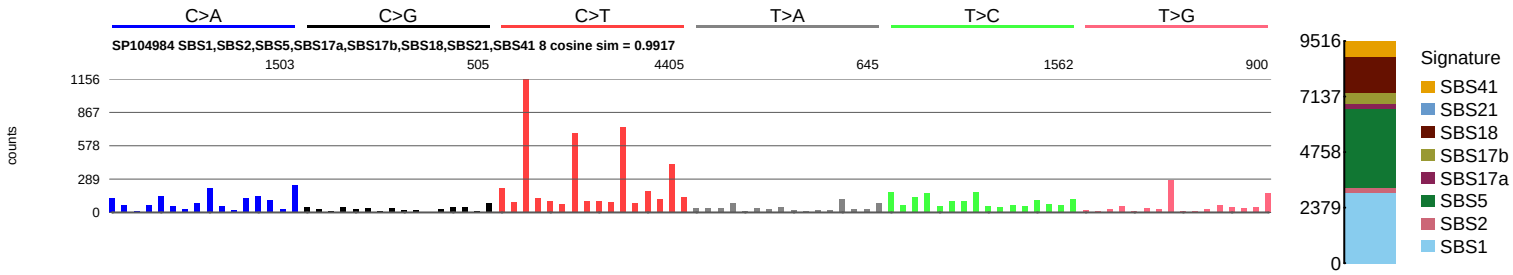
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9917
1503
509
4406
645
1554
901
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>T
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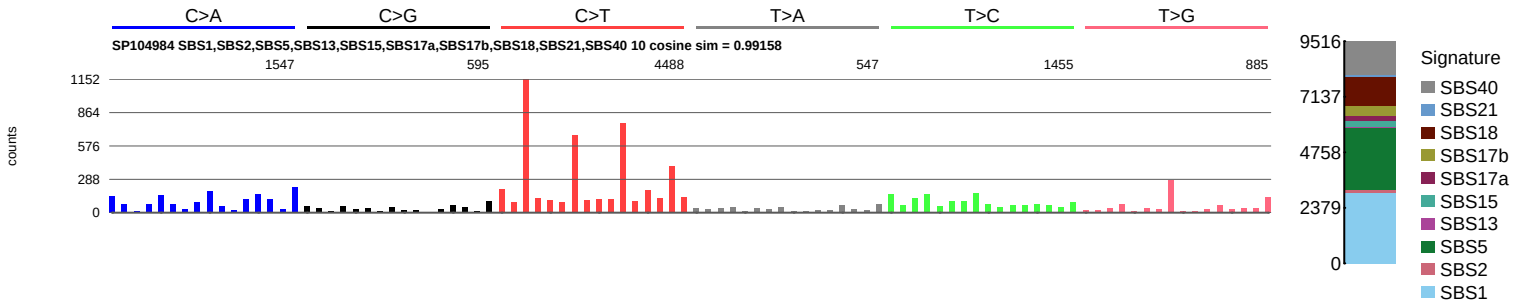
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9917
1503
505
4405
645
1562
900
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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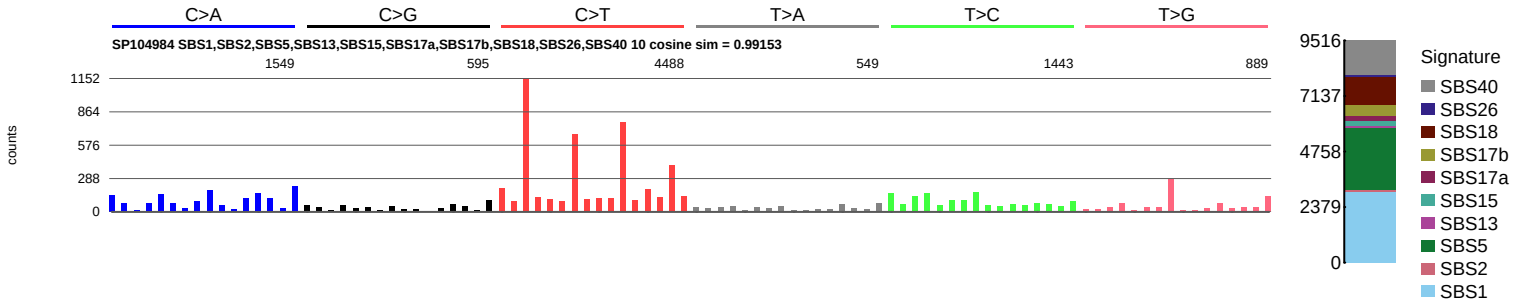
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.99158
1547
595
4488
547
1455
885
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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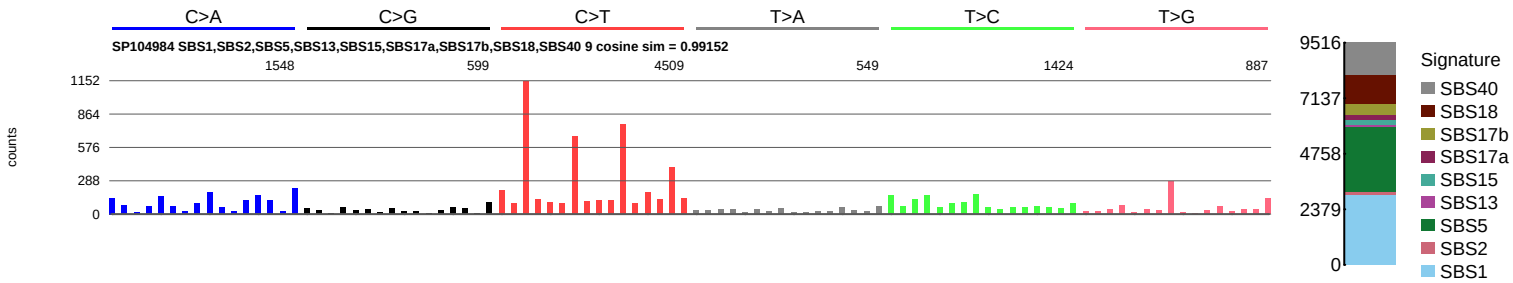
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.99153
1549
595
4488
549
1443
889
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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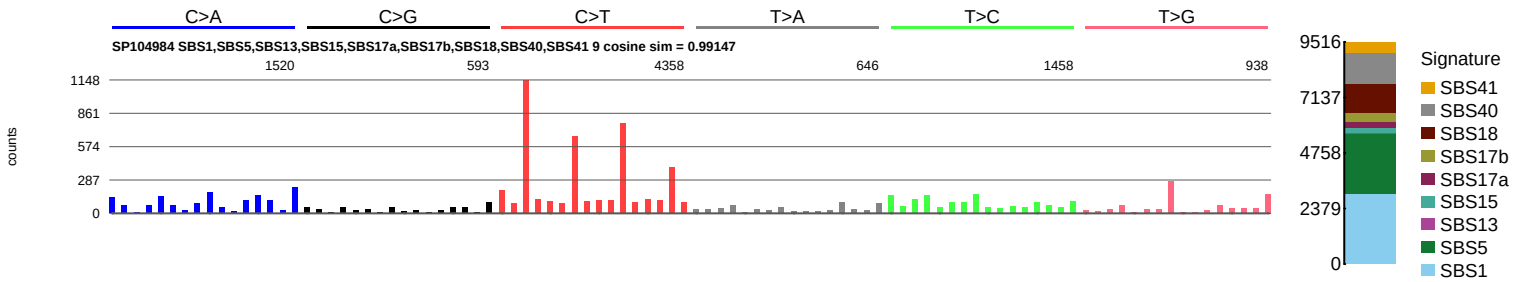
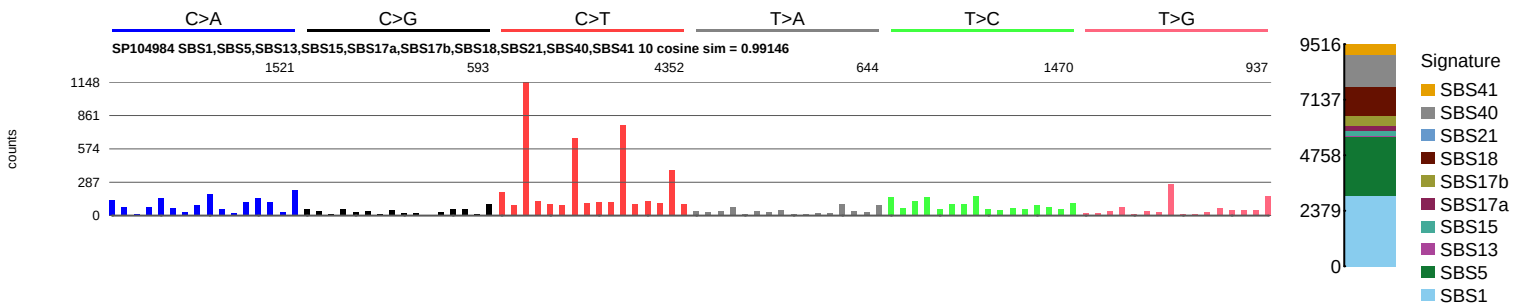
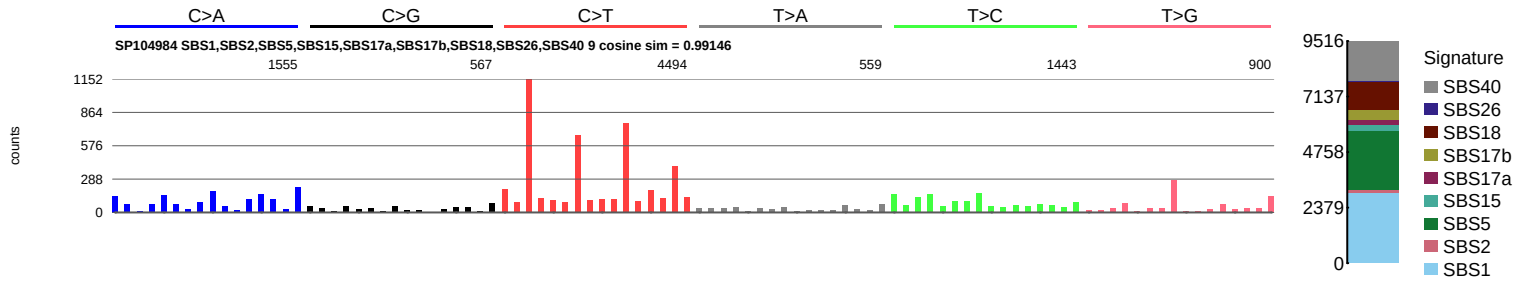
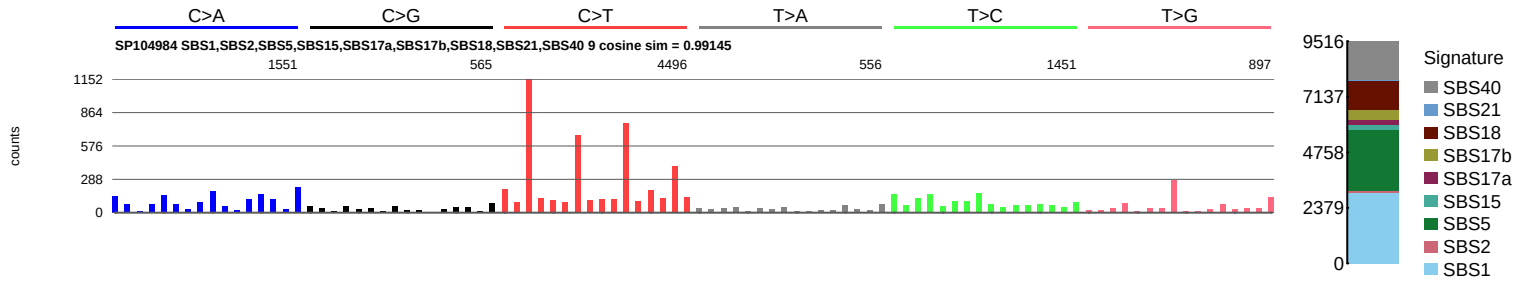
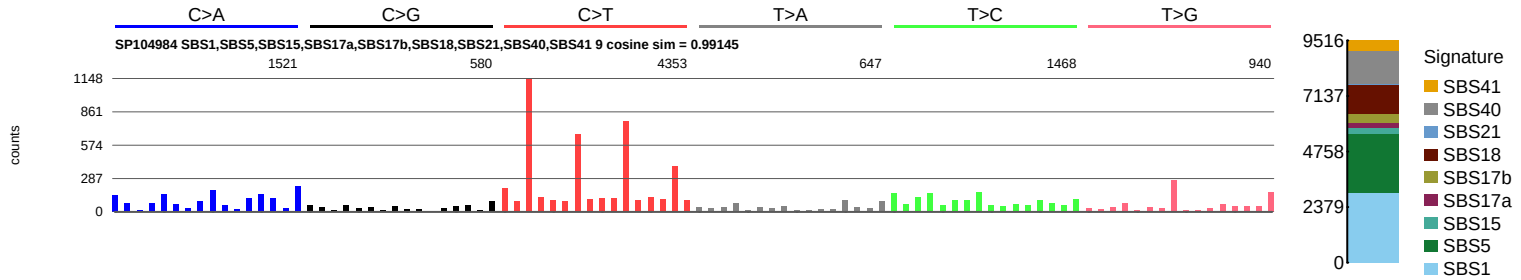
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.99152
1548
599
4509
549
1424
887
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

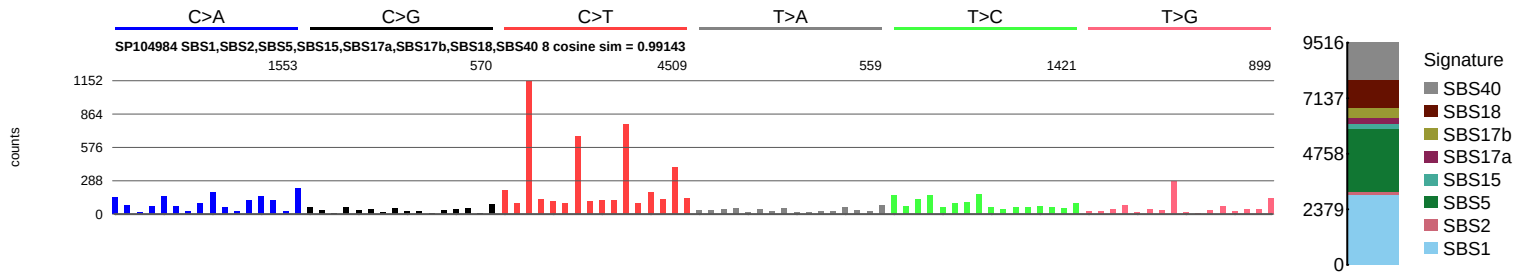

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99147
1520
593
4358
646
1458
938
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99146
1521
593
4352
644
1470
937
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99146
1555
567
4494
559
1443
900
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99145
1551
565
4496
556
1451
897
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99145
1521
580
4353
647
1468
940
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99143
1553
570
4509
559
1421
899
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
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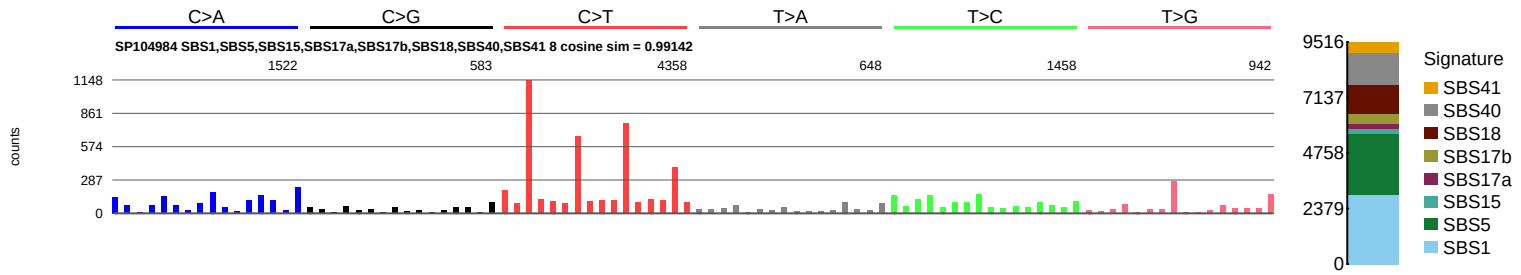
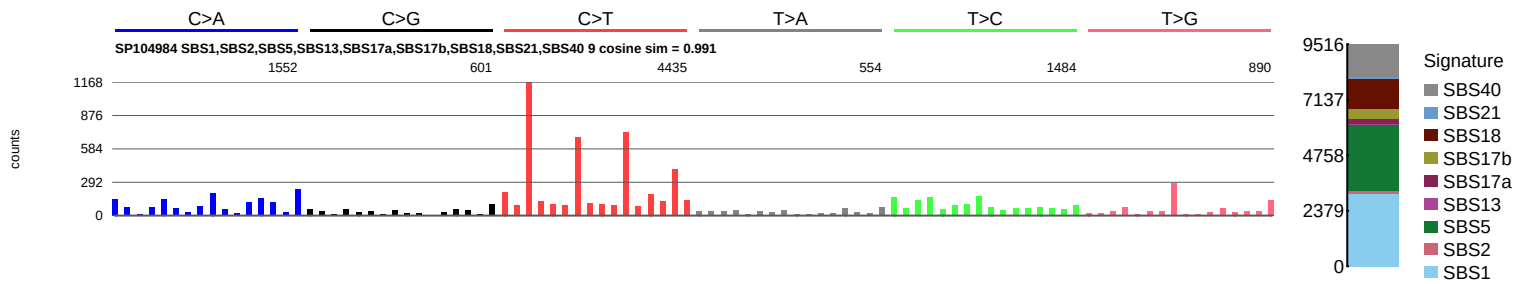
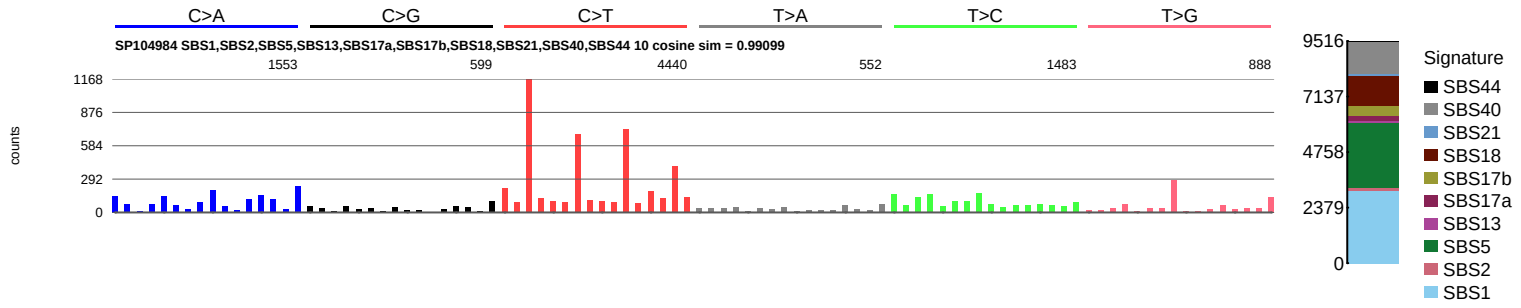
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99142
1522
583
4358
648
1458
942
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.991
1552
601
4435
554
1484
890
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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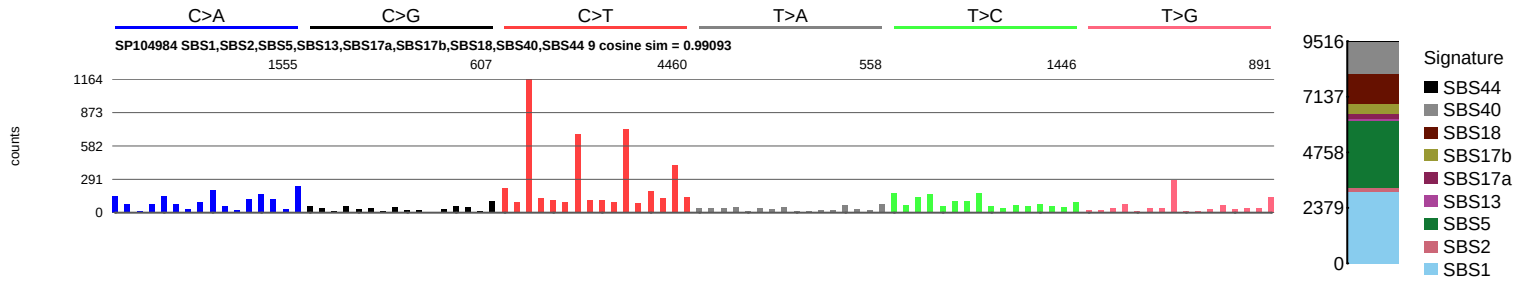
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.99099
1553
599
4440
552
1483
888
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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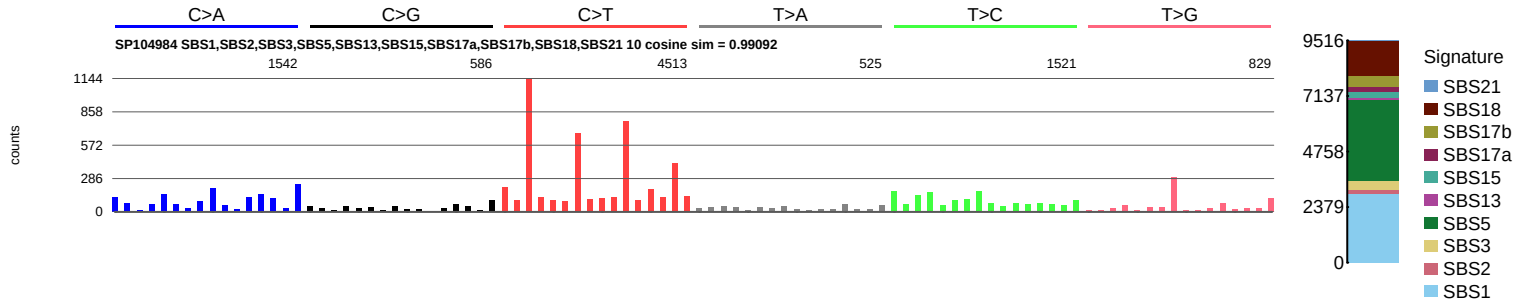
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.99093
1555
607
4460
558
1446
891
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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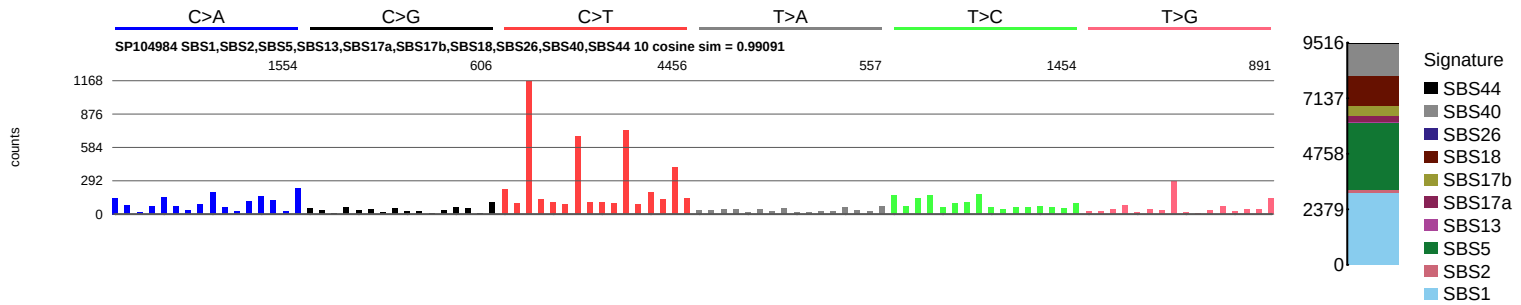
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.99092
1542
586
4513
525
1521
829
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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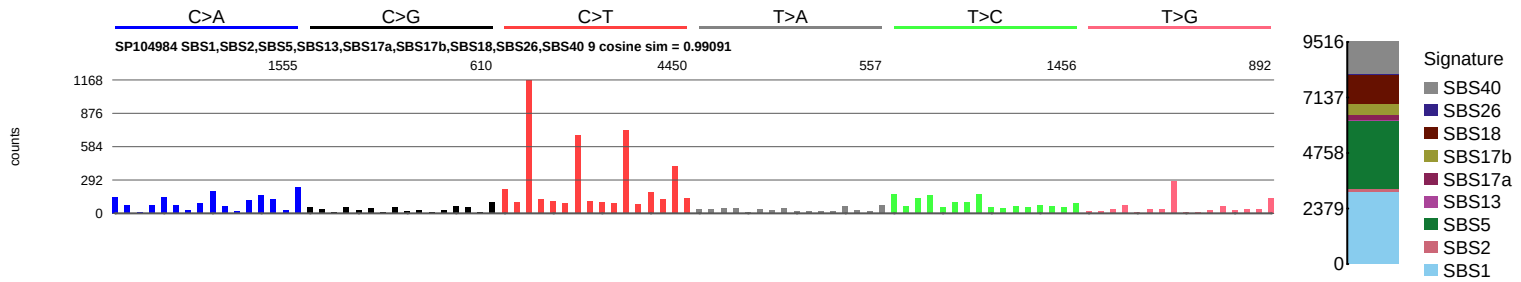
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.99091
1554
606
4456
557
1454
891
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99091
1555
610
4450
557
1456
892
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
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SBS17b
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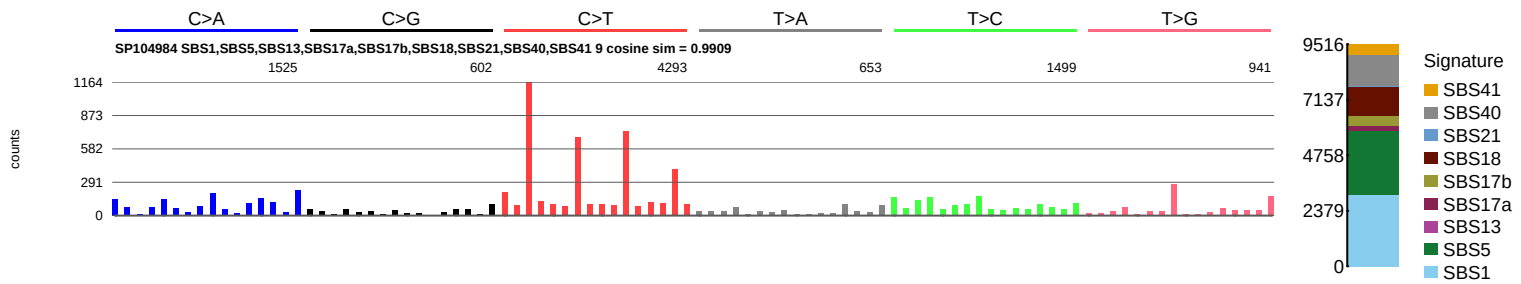
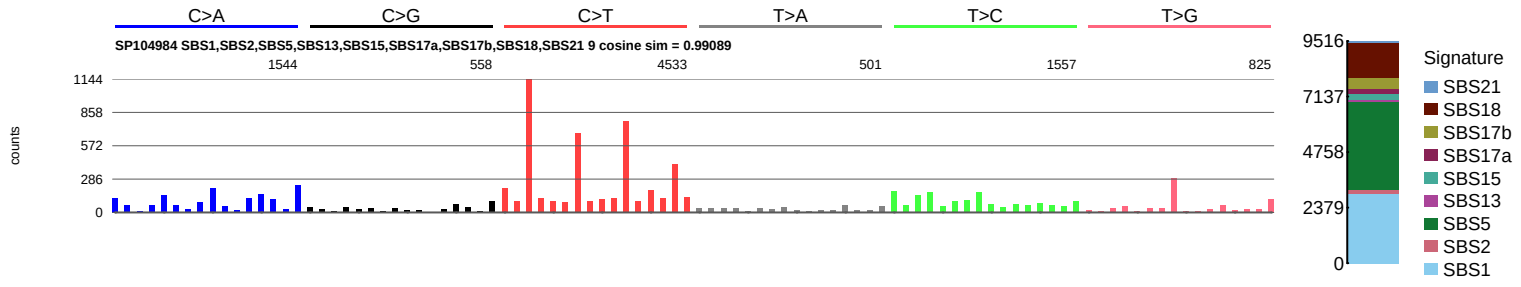
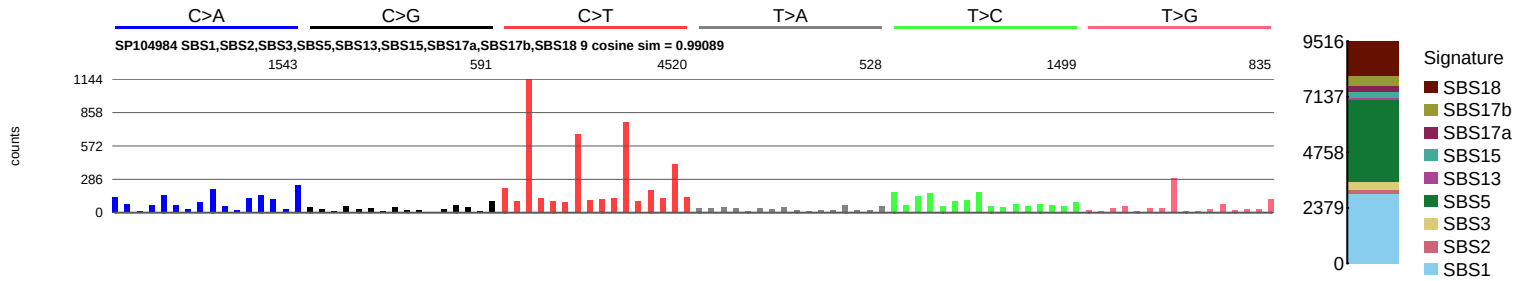
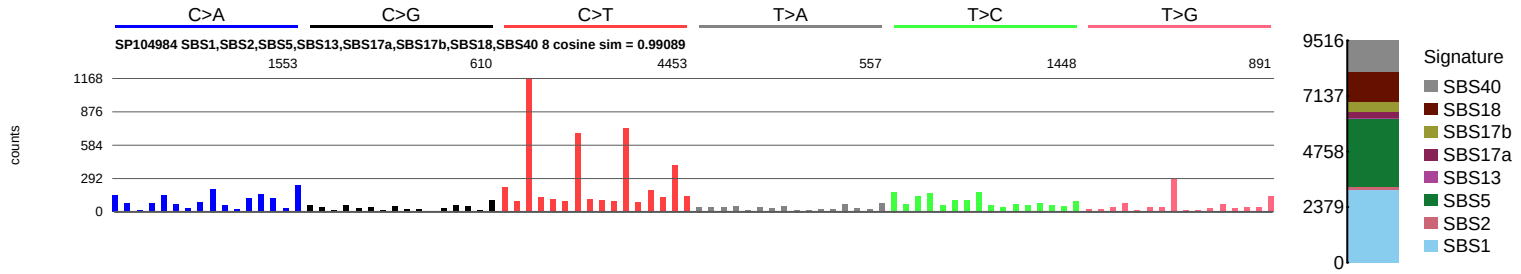
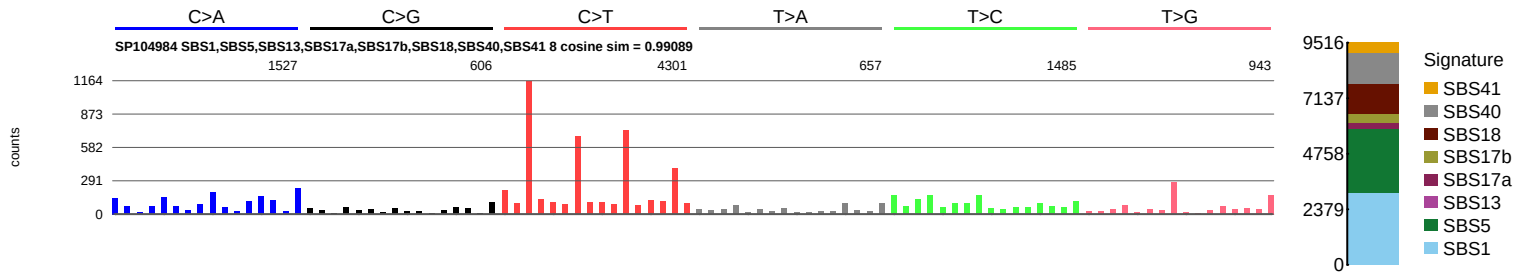
SBS5
SBS2
SBS1

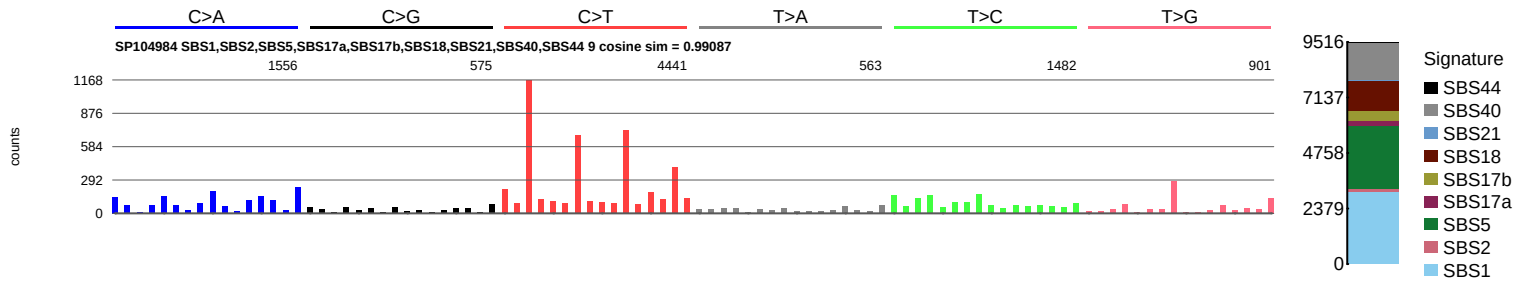

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9909
1525
602
4293
653
1499
941
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99089
1544
558
4533
501
1557
825
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.99089
1543
591
4520
528
1499
835
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99089
1553
610
4453
557
1448
891
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99089
1527
606
4301
657
1485
943
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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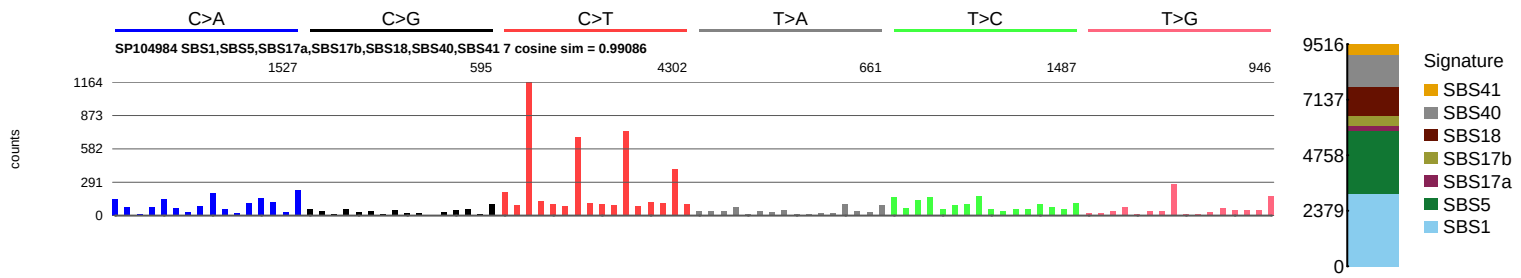
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.99087
1556
575
4441
563
1482
901
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
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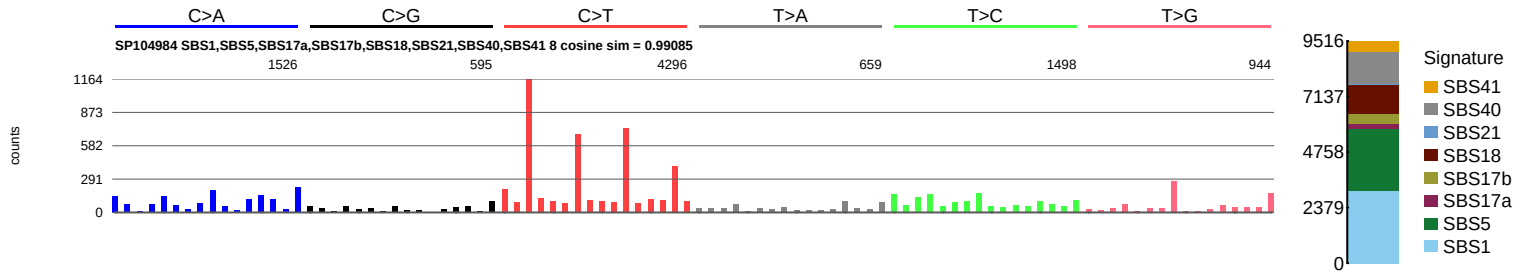
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99086
1527
595
4302
661
1487
946
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>T
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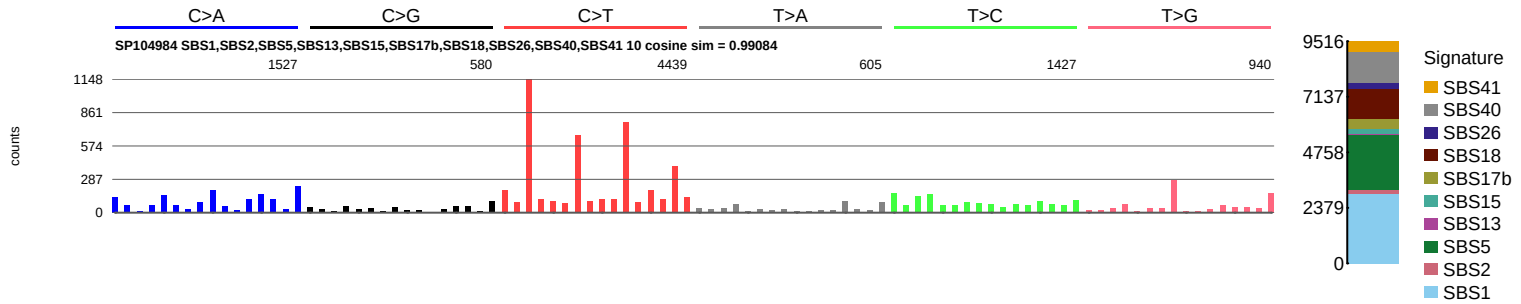
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99085
1526
595
4296
659
1498
944
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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T>C
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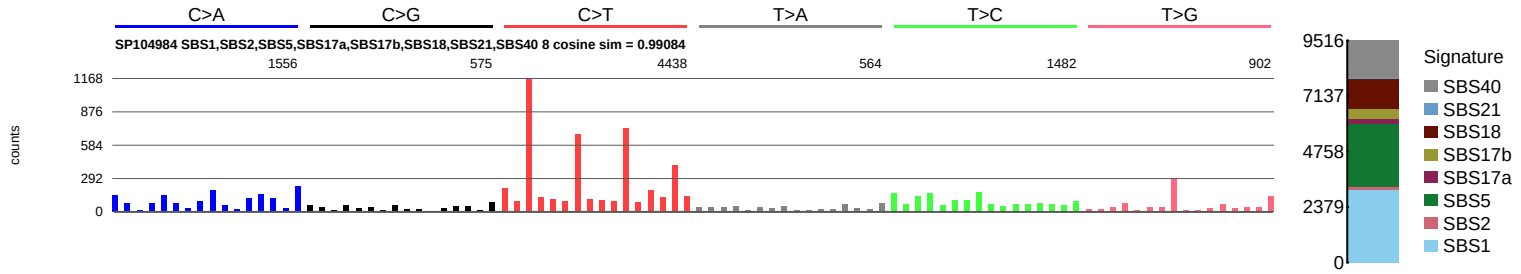
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99084
1527
580
4439
605
1427
940
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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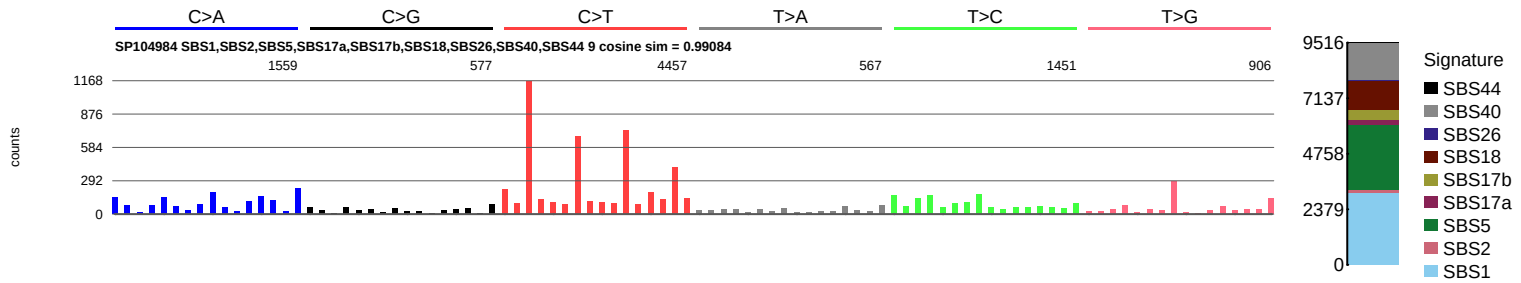
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99084
1556
575
4438
564
1482
902
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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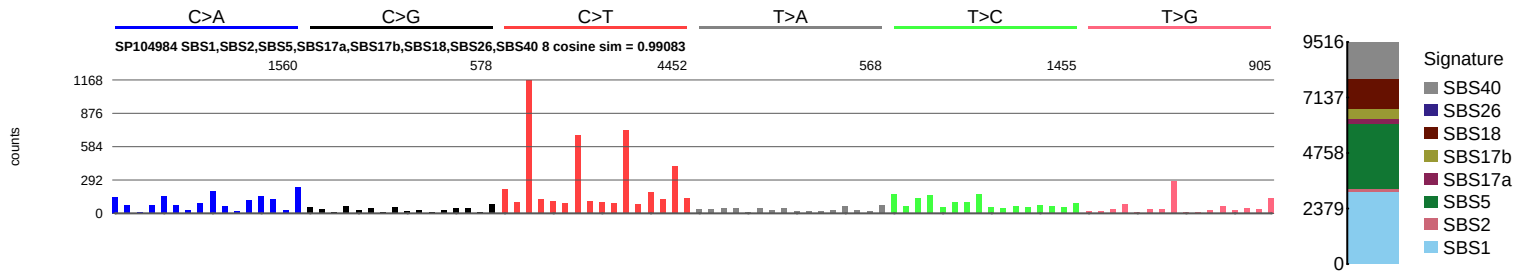
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.99084
1559
577
4457
567
1451
906
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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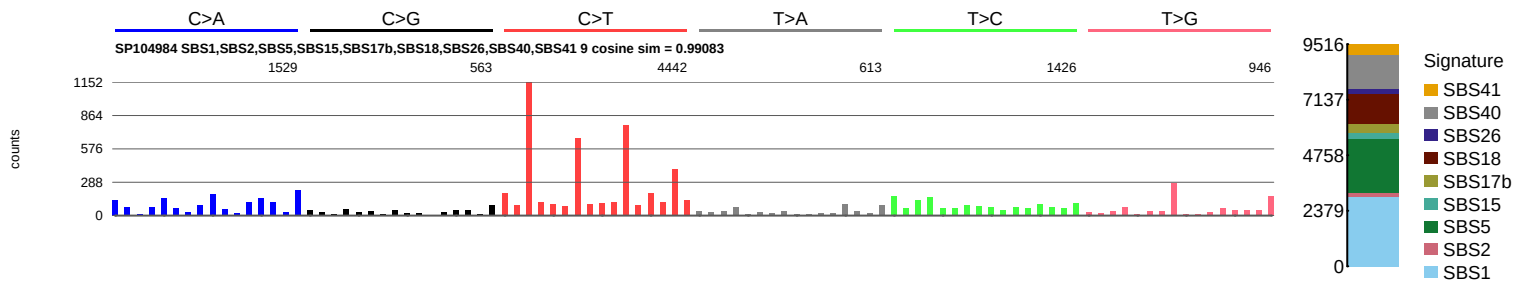
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99083
1560
578
4452
568
1455
905
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99083
1529
563
4442
613
1426
946
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
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SBS26
SBS18
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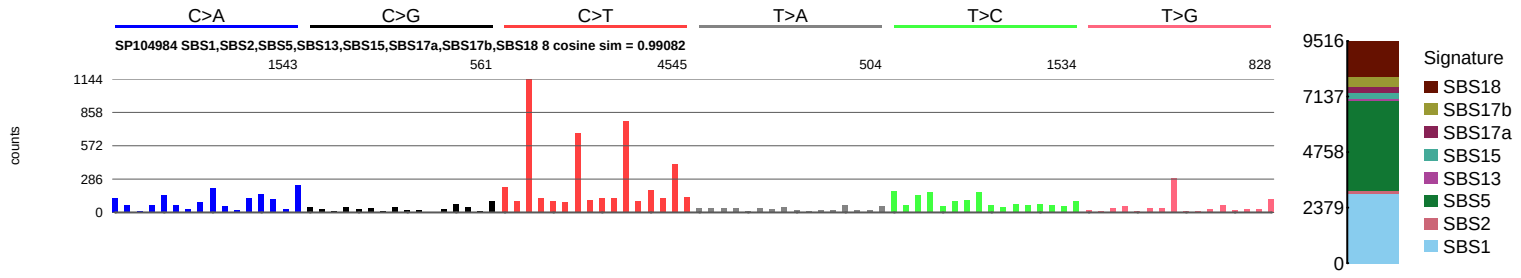
SBS5
SBS2
SBS1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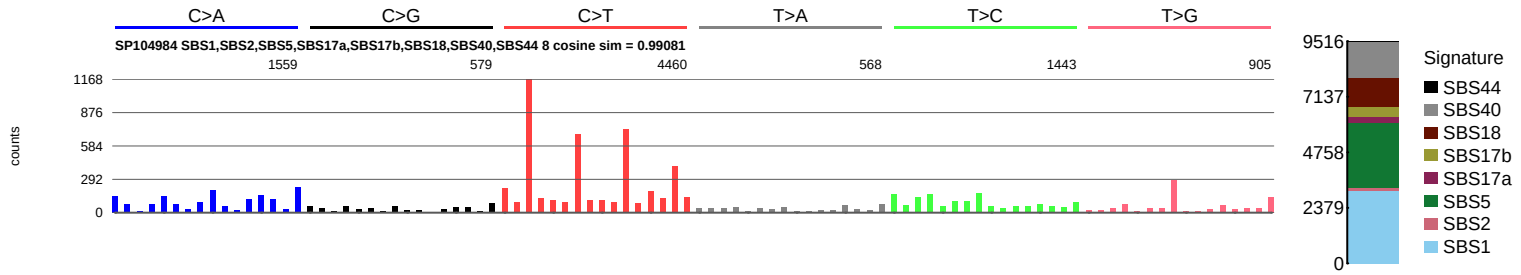
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99082
1543
561
4545
504
1534
828
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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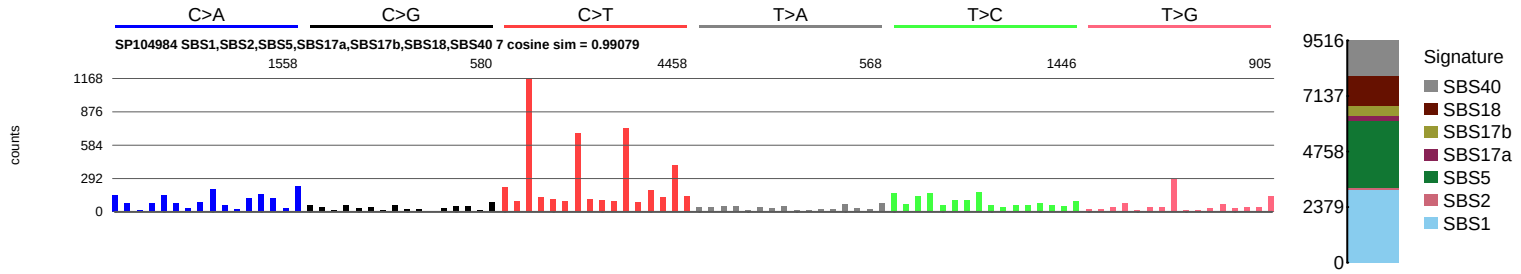
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.99081
1559
579
4460
568
1443
905
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99079
1558
580
4458
568
1446
905
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
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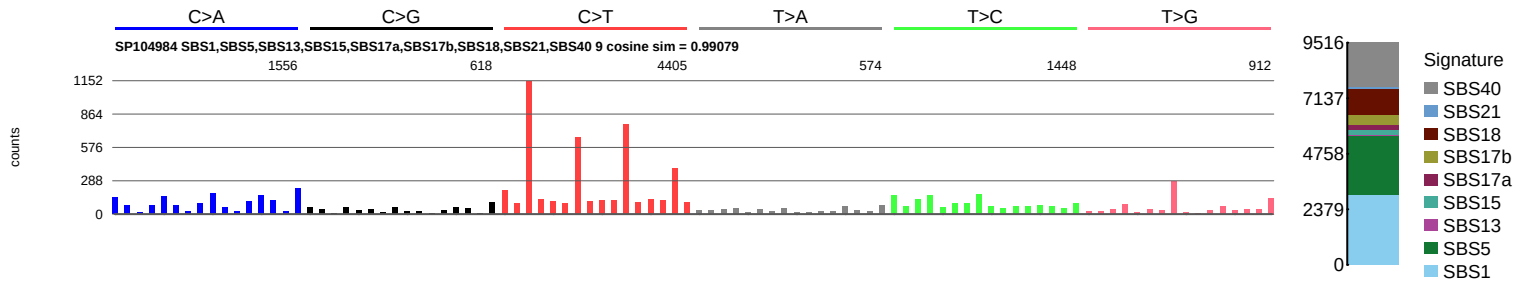
SBS5
SBS2
SBS1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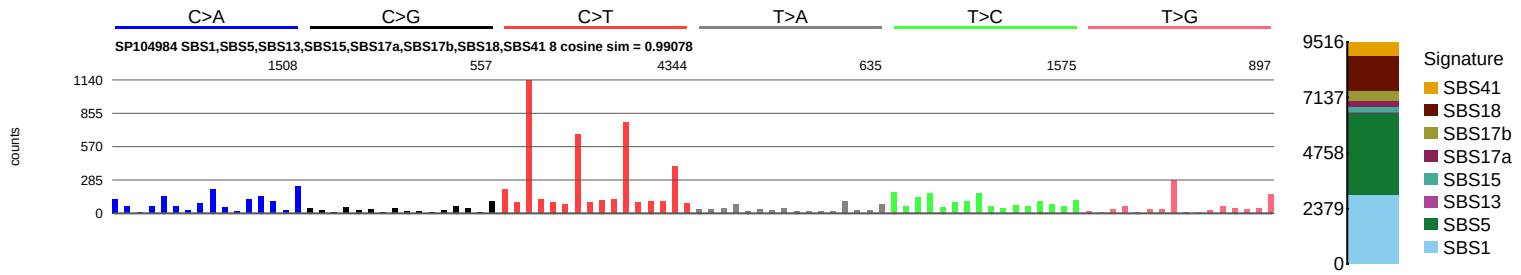
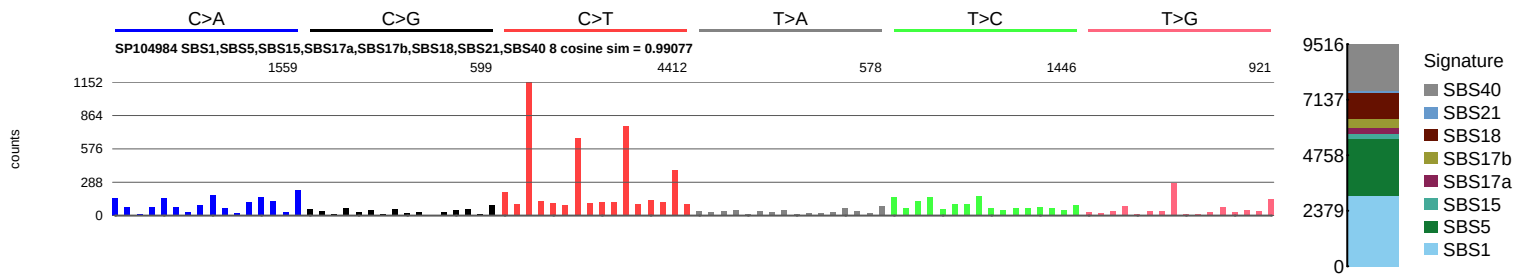
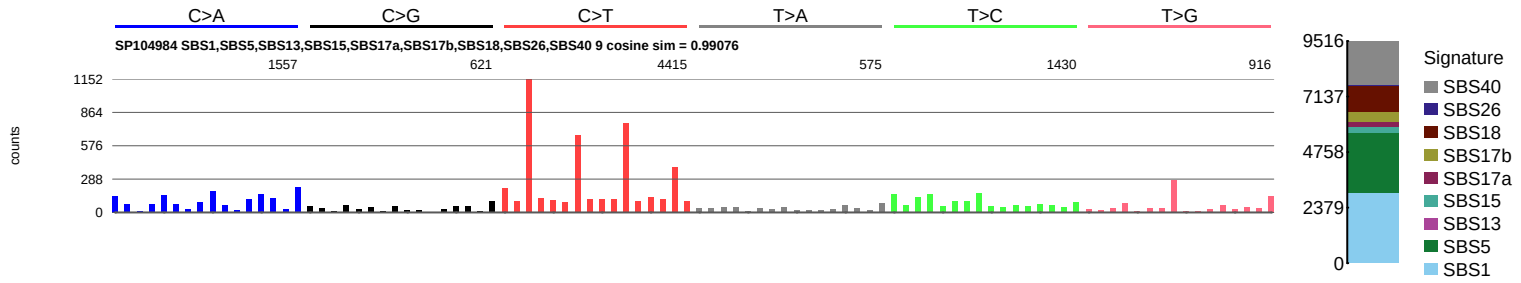
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.99079
1556
618
4405
574
1448
912
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99078
1508
557
4344
635
1575
897
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99077
1559
599
4412
578
1446
921
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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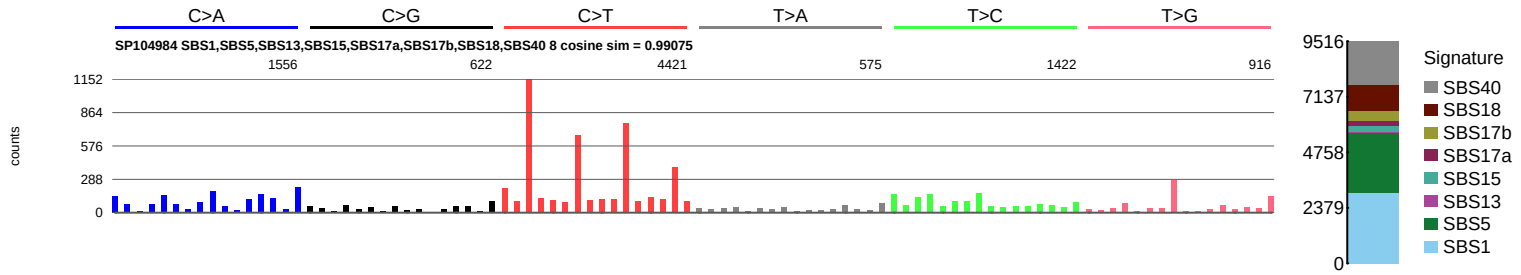
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99076
1557
621
4415
575
1430
916
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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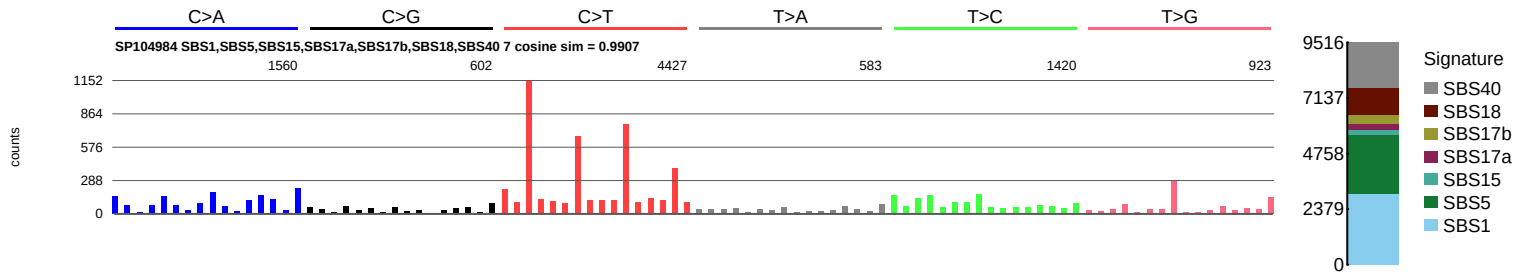
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.99075
1556
622
4421
575
1422
916
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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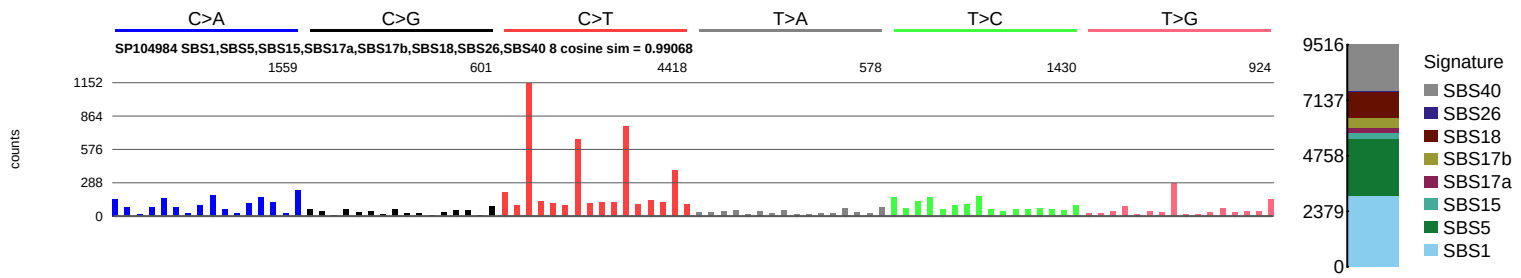
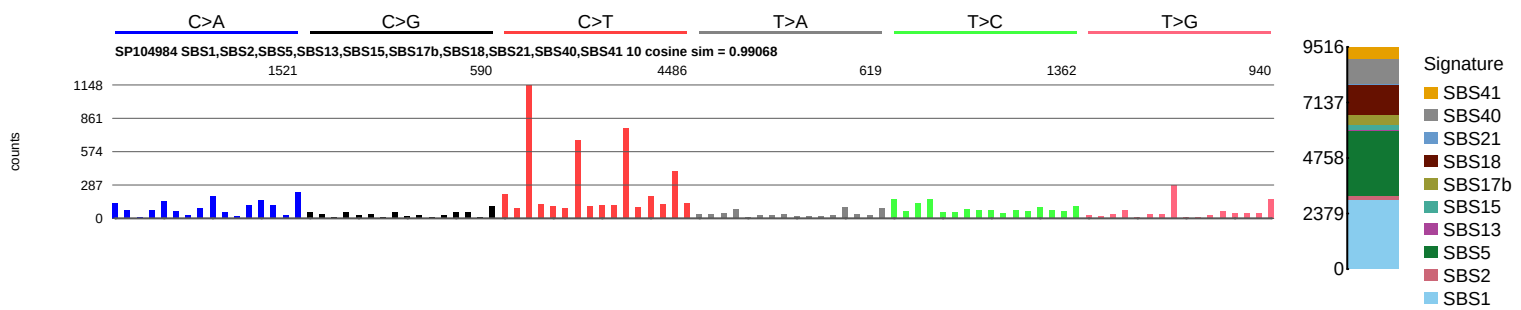
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.9907
1560
602
4427
583
1420
923
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99068
1559
601
4418
578
1430
924
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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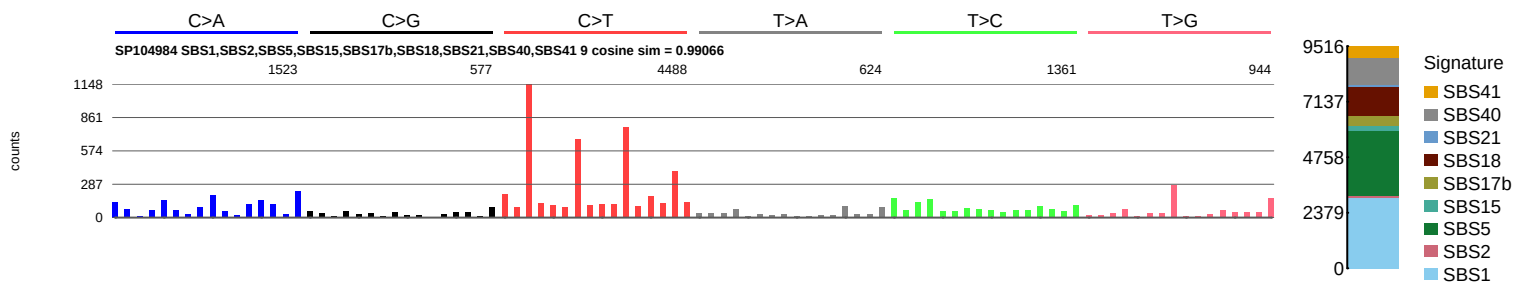
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99068
1521
590
4486
619
1362
940
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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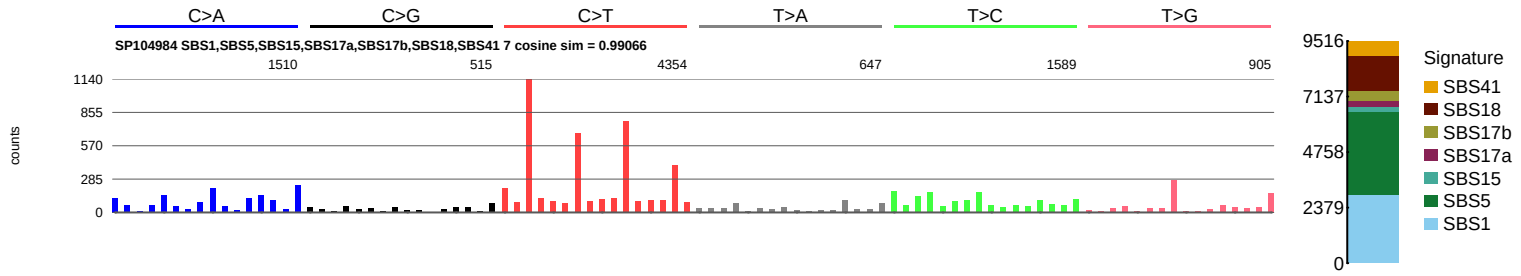
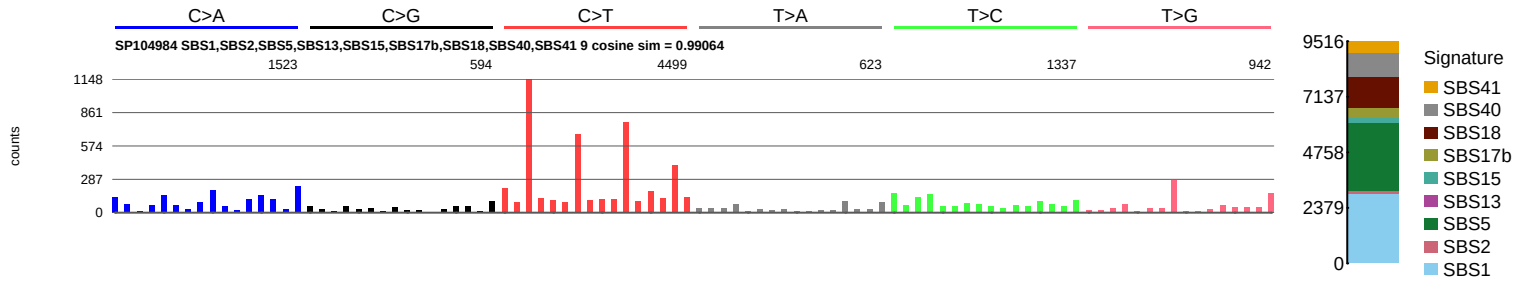
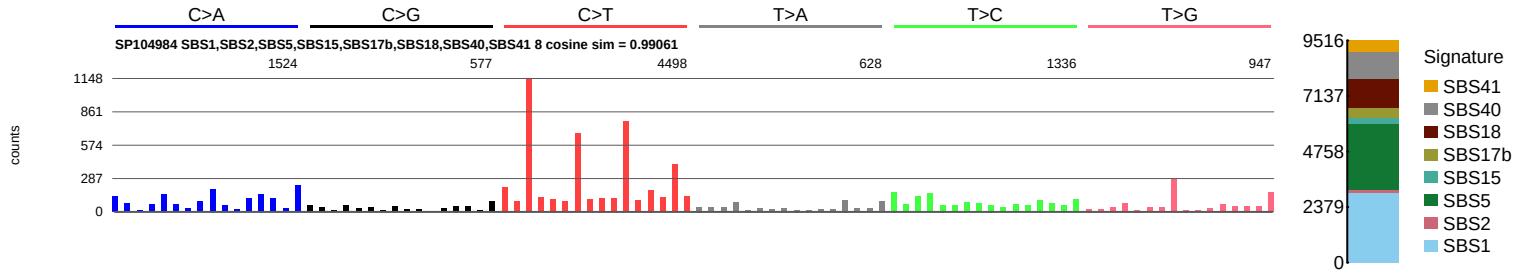
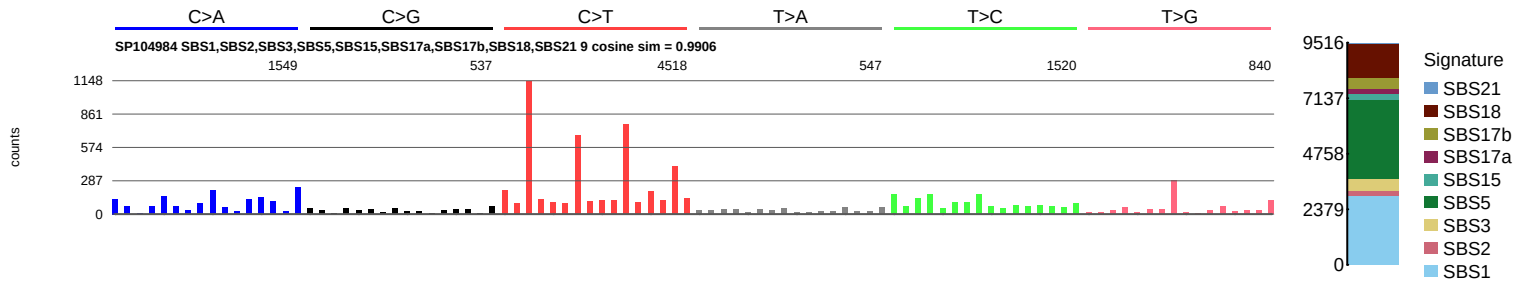
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99066
1523
577
4488
624
1361
944
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99066
1510
515
4354
647
1589
905
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99064
1523
594
4499
623
1337
942
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99061
1524
577
4498
628
1336
947
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.9906
1549
537
4518
547
1520
840
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
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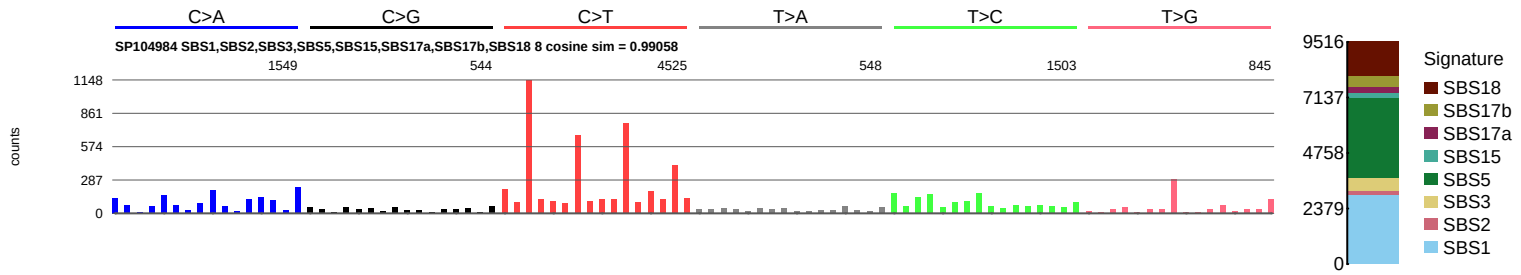
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99058
1549
544
4525
548
1503
845
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
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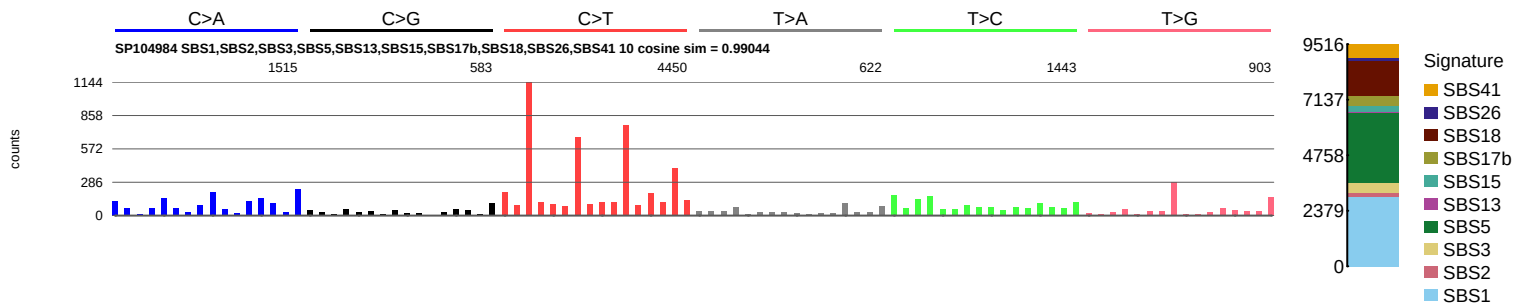
SBS2
SBS1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T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99044
1515
583
4450
622
1443
903
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
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SBS18
SBS17b
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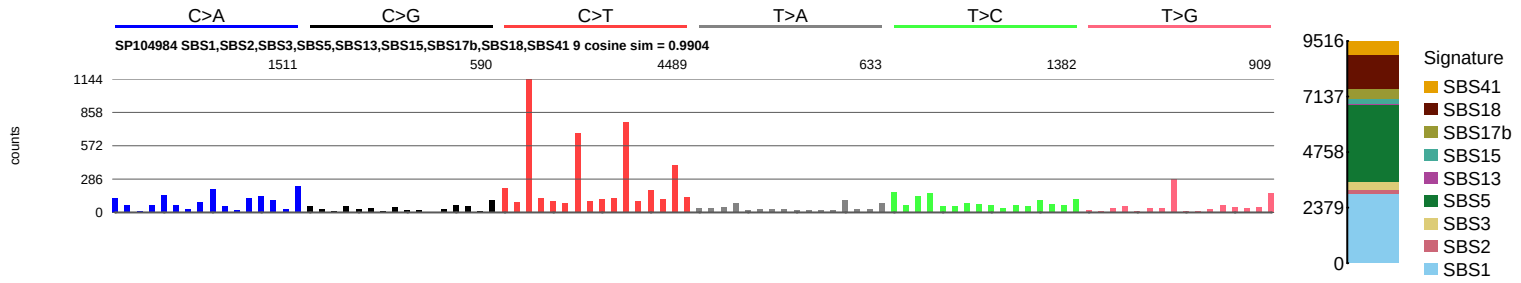
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
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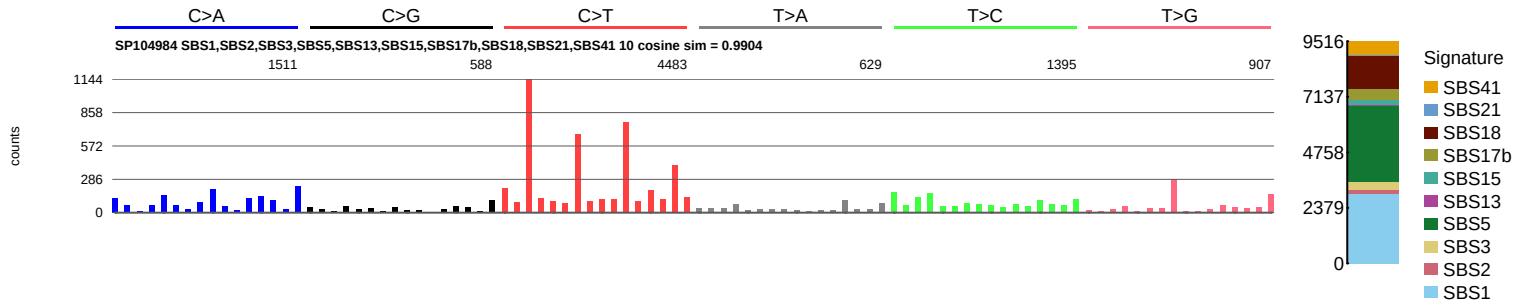
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.9904
1511
590
4489
633
1382
909
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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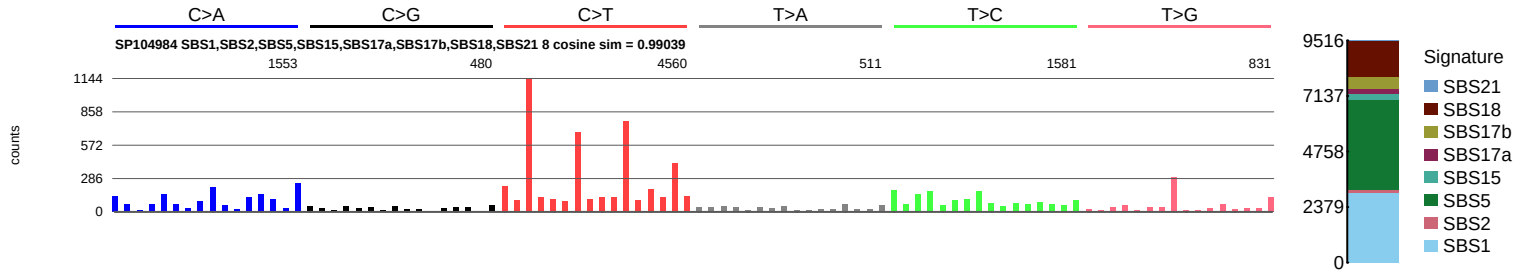
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.9904
1511
588
4483
629
1395
907
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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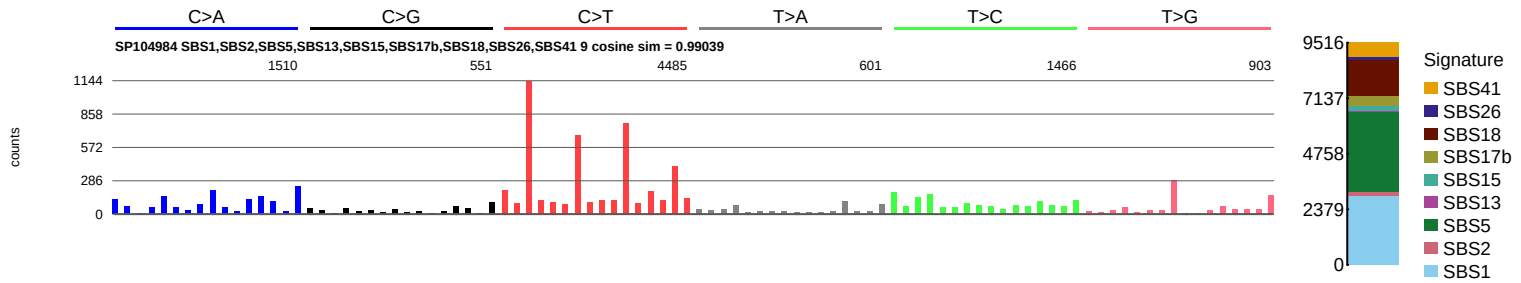
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99039
1553
480
4560
511
1581
831
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99039
1510
551
4485
601
1466
903
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
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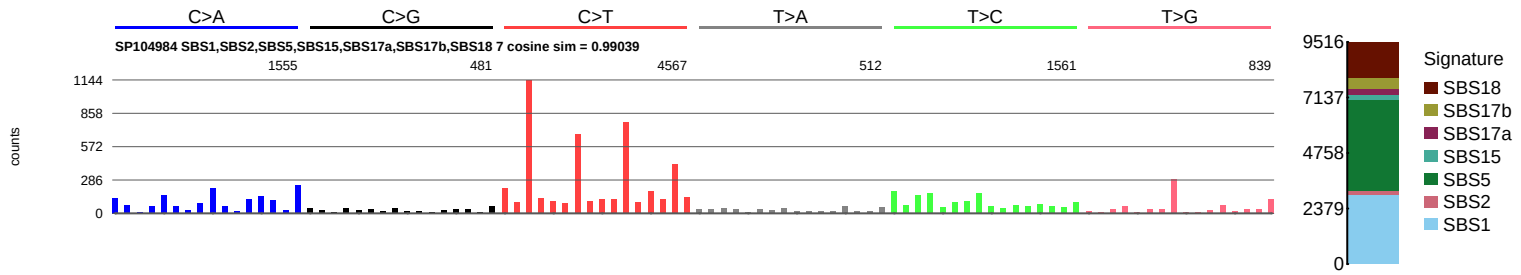
SBS2
SBS1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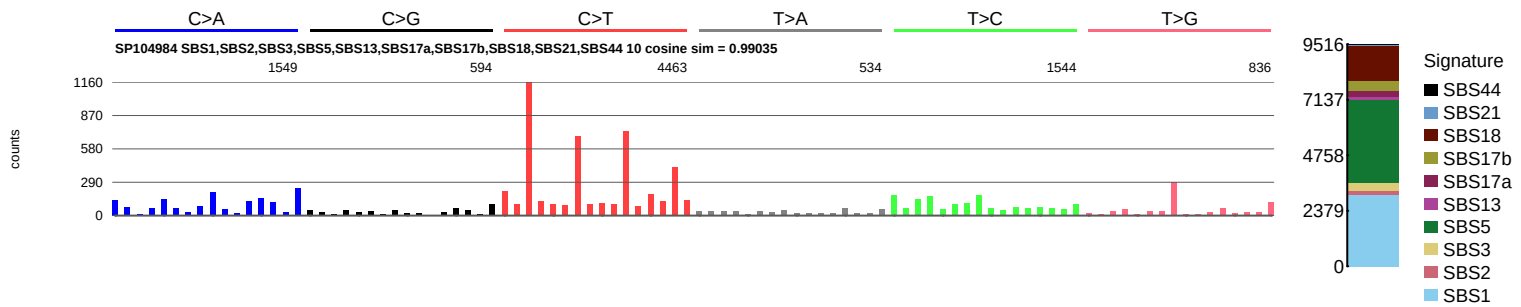
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99039
1555
481
4567
512
1561
839
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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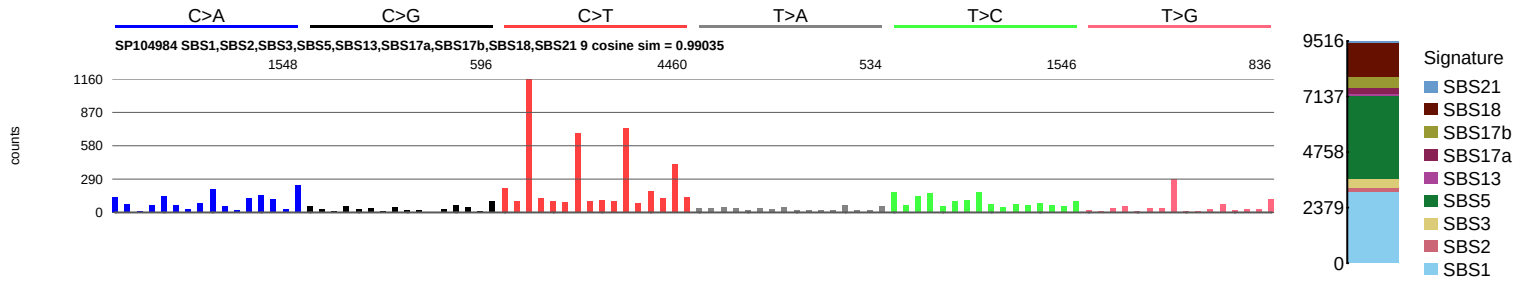
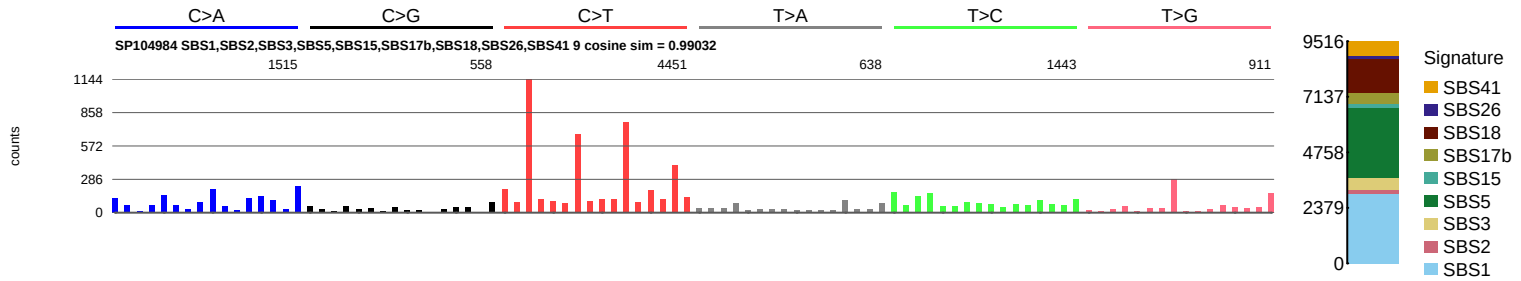
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.99035
1549
594
4463
534
1544
836
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.99035
1548
596
4460
534
1546
836
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99032
1515
558
4451
638
1443
911
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
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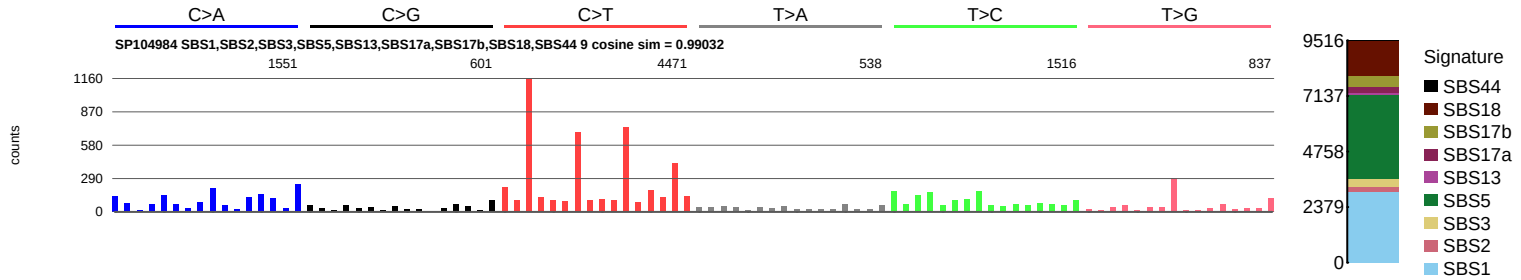
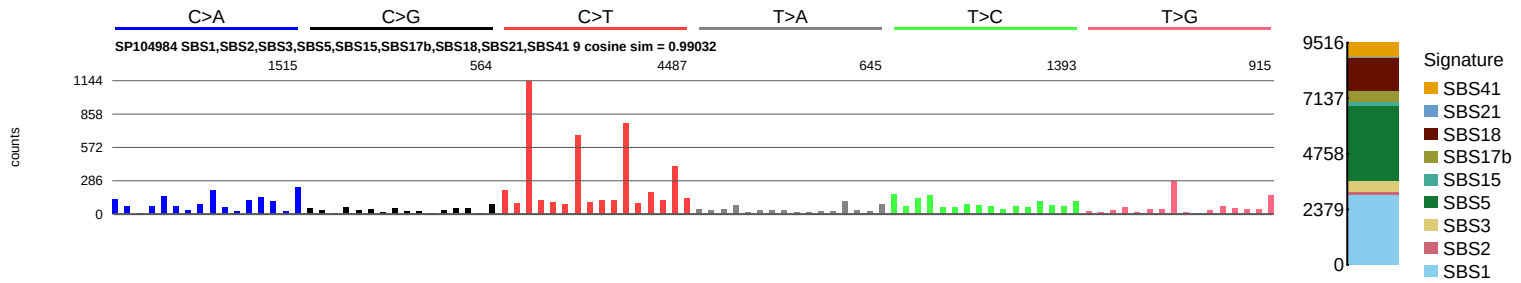
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.99032
1551
601
4471
538
1516
837
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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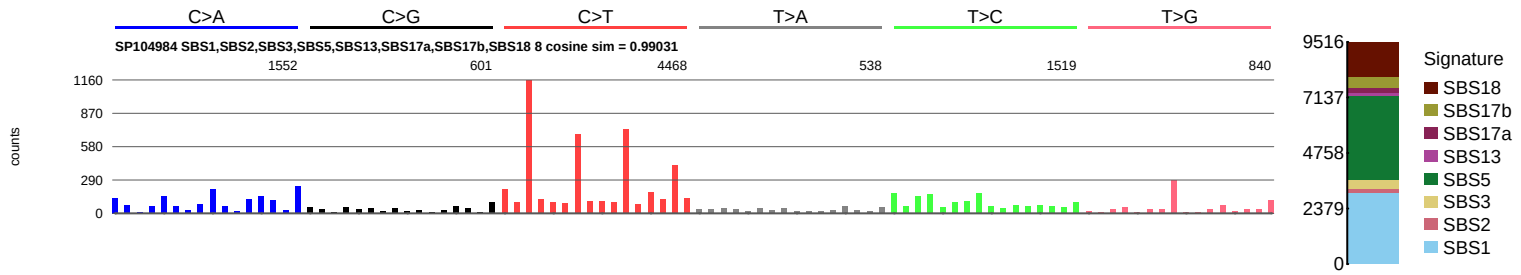
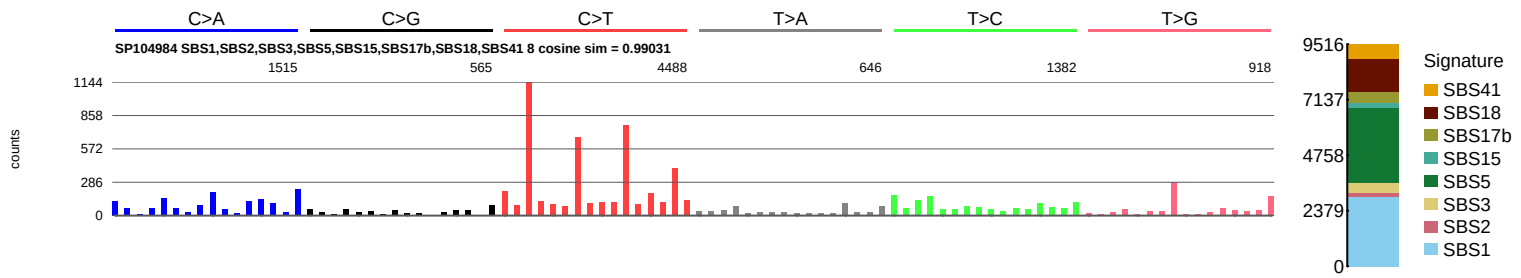
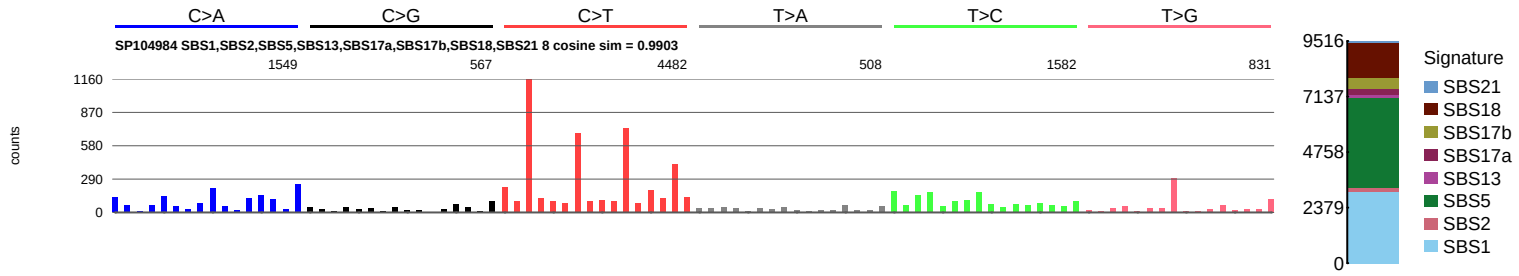
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99032
1515
564
4487
645
1393
915
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.99031
1552
601
4468
538
1519
840
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.99031
1515
565
4488
646
1382
918
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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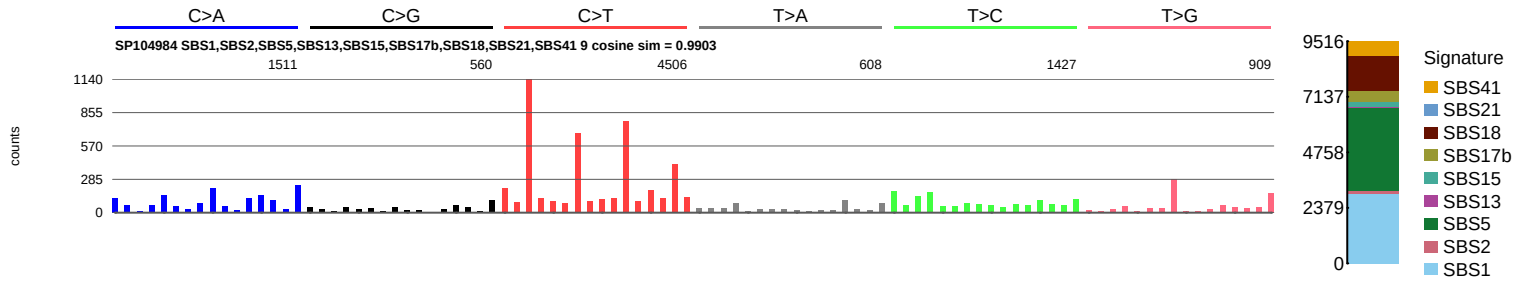
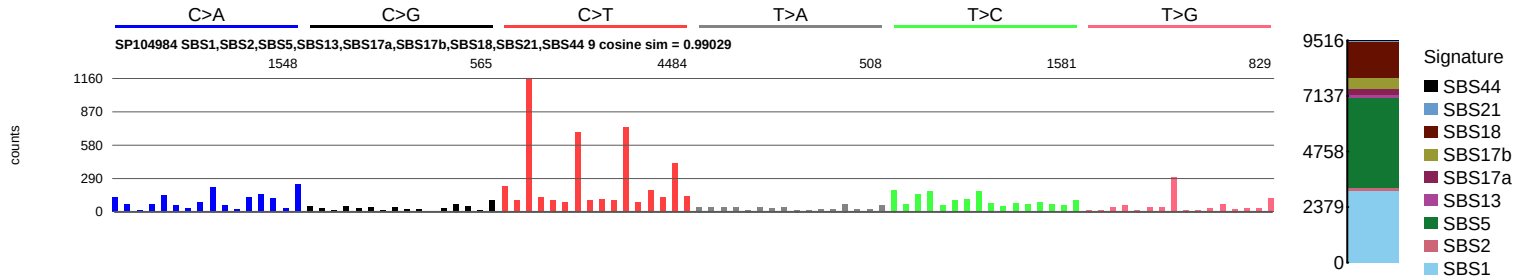
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.9903
1549
567
4482
508
1582
831
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9903
1511
560
4506
608
1427
909
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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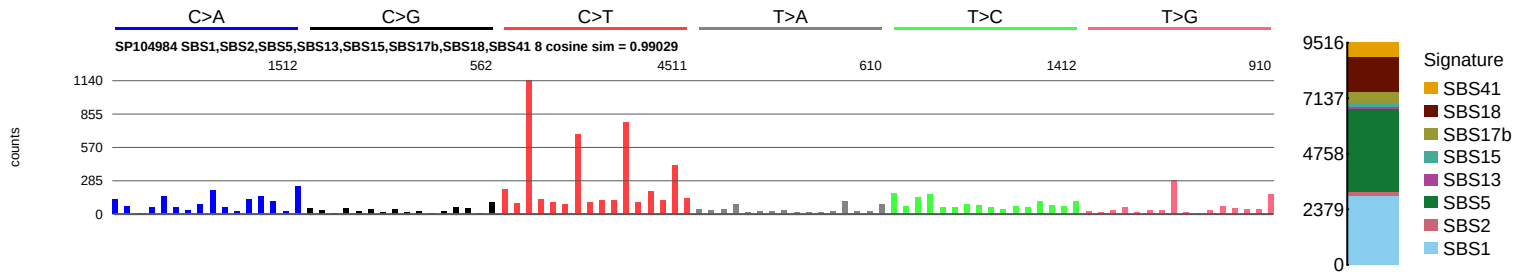
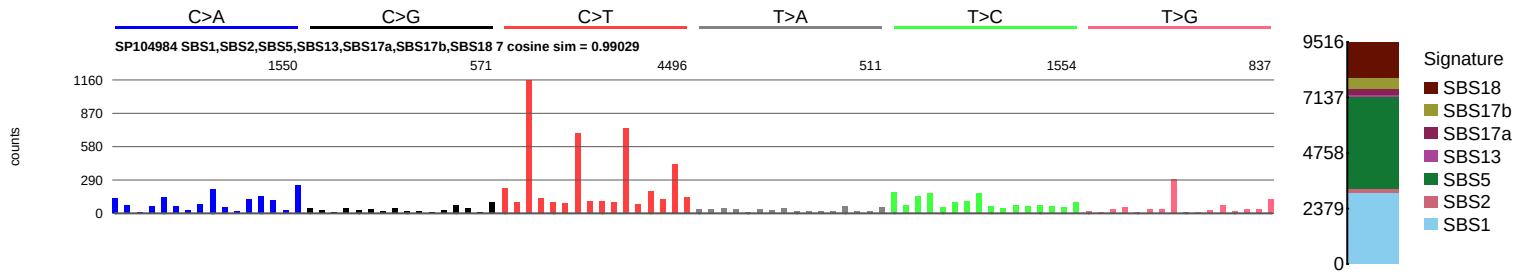
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.99029
1548
565
4484
508
1581
829
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.99029
1512
562
4511
610
1412
910
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99029
1550
571
4496
511
1554
837
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
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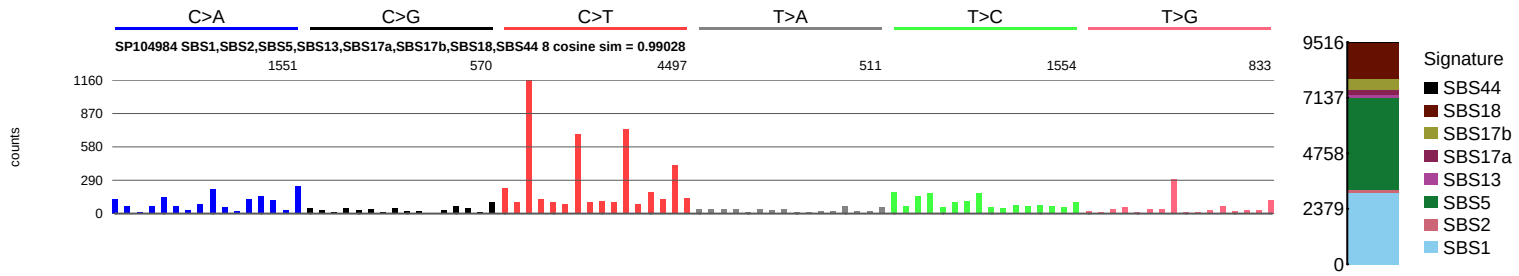
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99028
1551
570
4497
511
1554
833
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS44
SBS18
SBS17b
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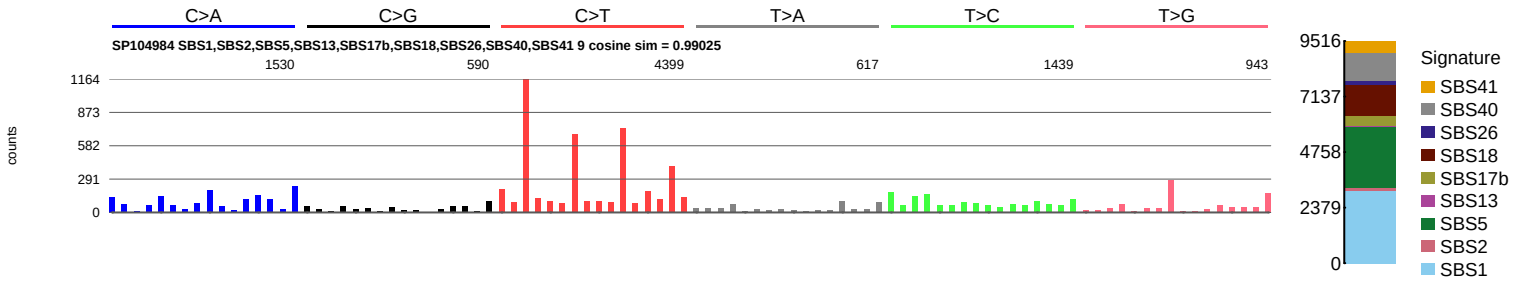
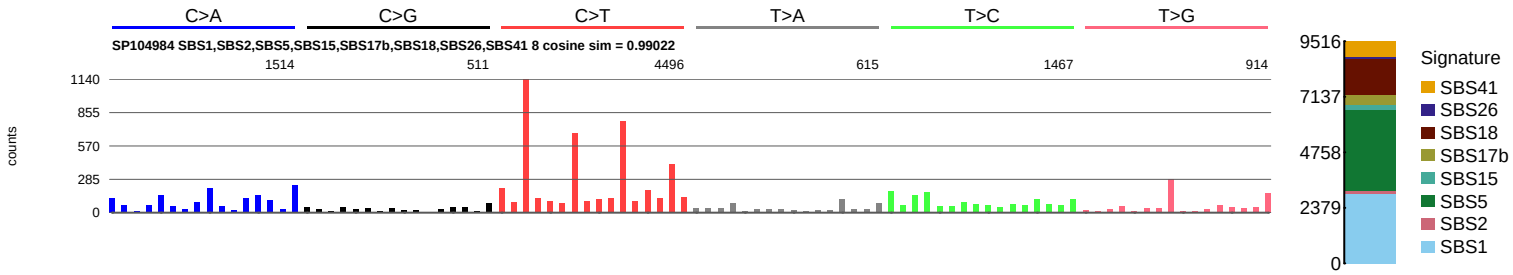
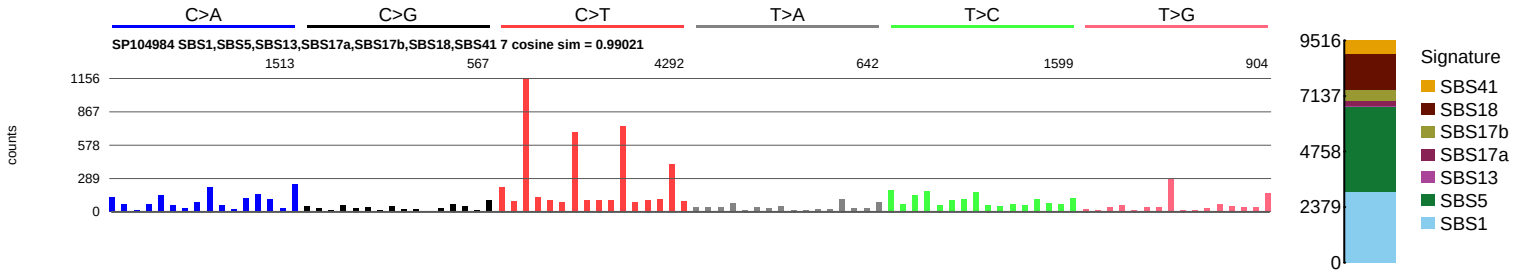
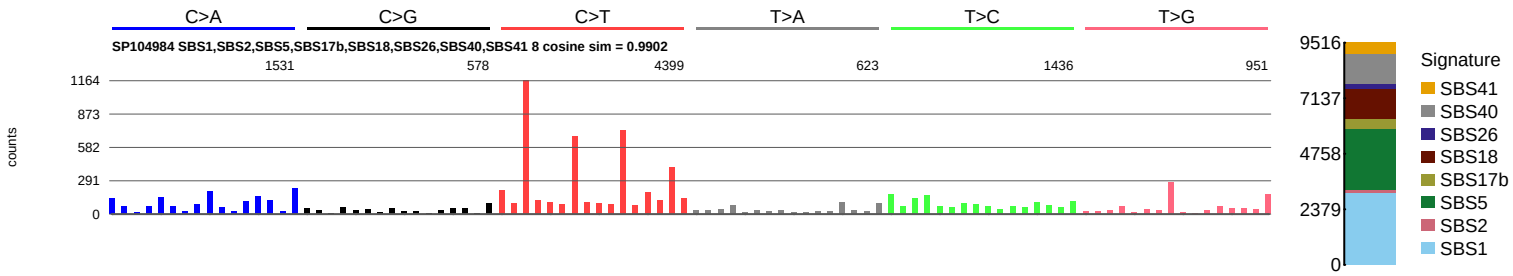
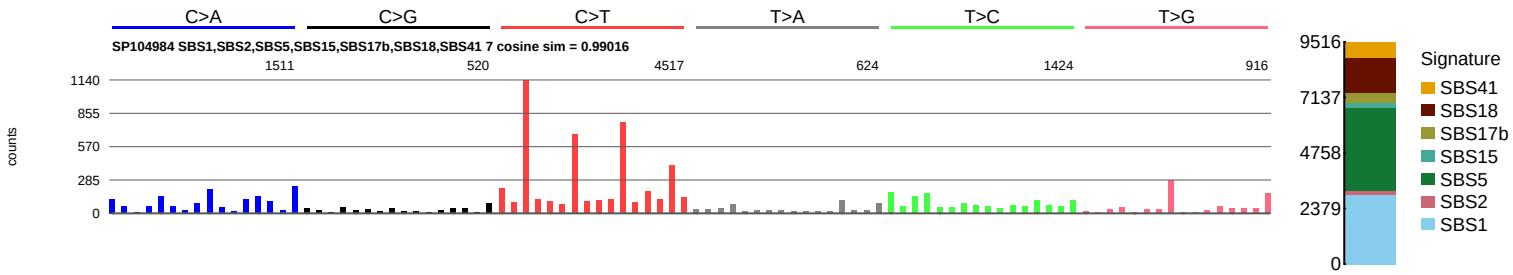
SBS13
SBS5
SBS2
SBS1

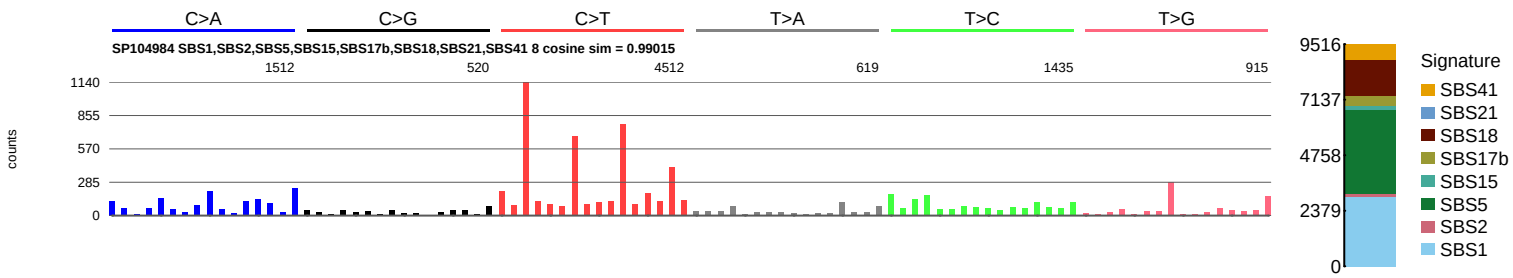

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99025
1530
590
4399
617
1439
943
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

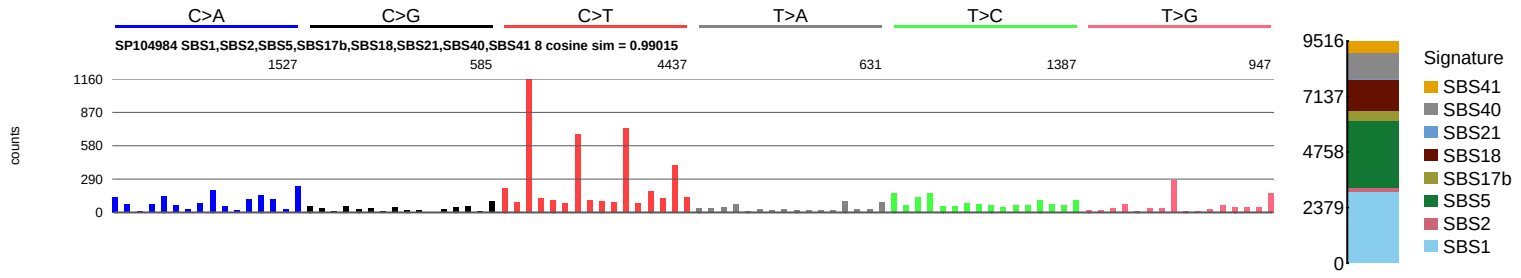

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99022
1514
511
4496
615
1467
914
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99021
1513
567
4292
642
1599
904
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9902
1531
578
4399
623
1436
951
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS41 7 cosine sim = 0.99016
1511
520
4517
624
1424
916
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99015
1512
520
4512
619
1435
915
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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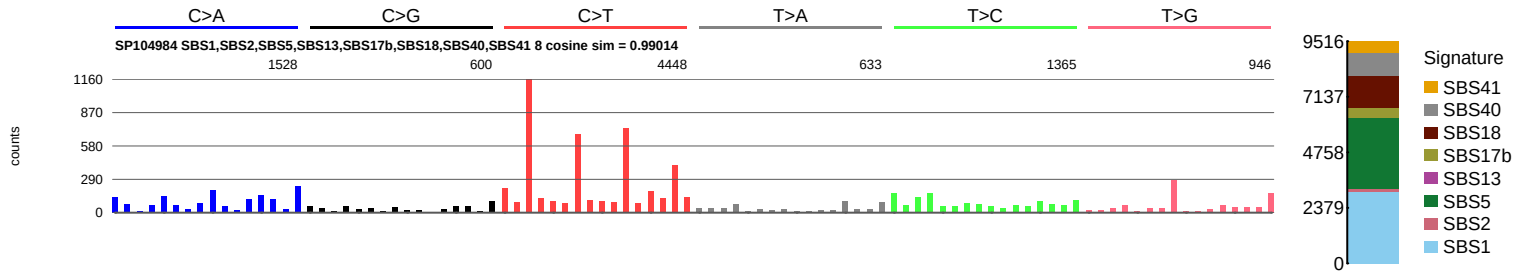
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99015
1527
585
4437
631
1387
947
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1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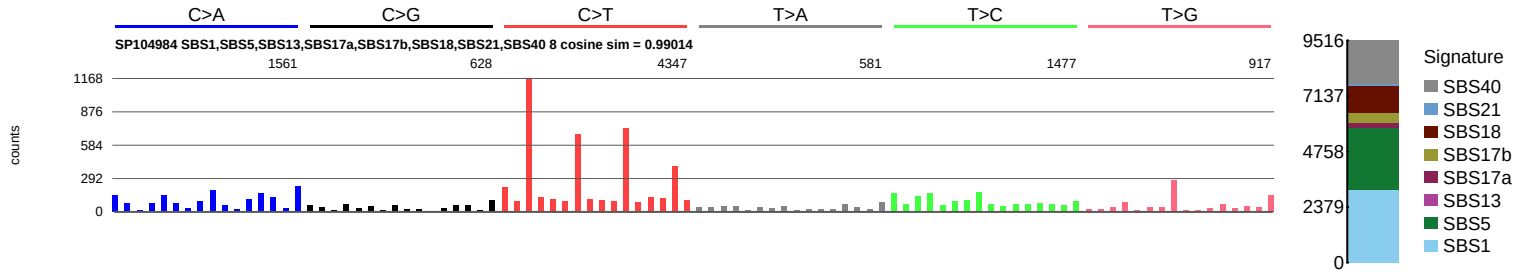
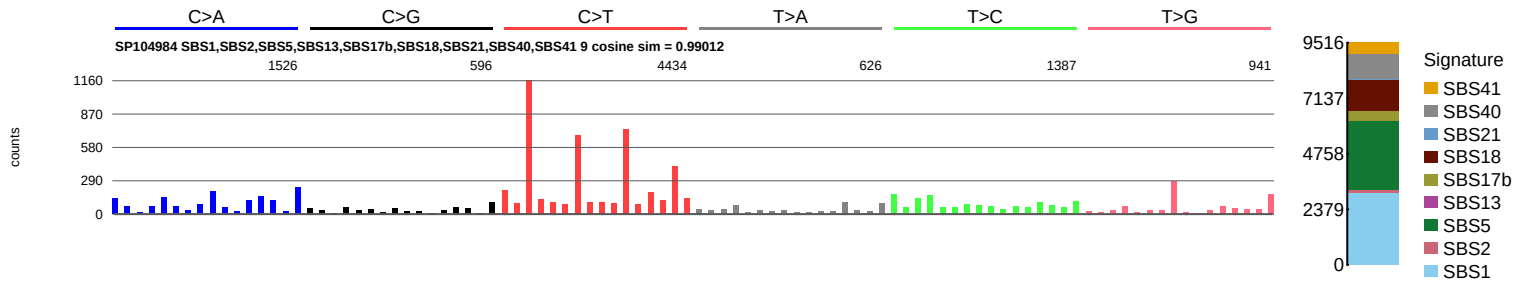
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99014
1528
600
4448
633
1365
946
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.99014
1561
628
4347
581
1477
917
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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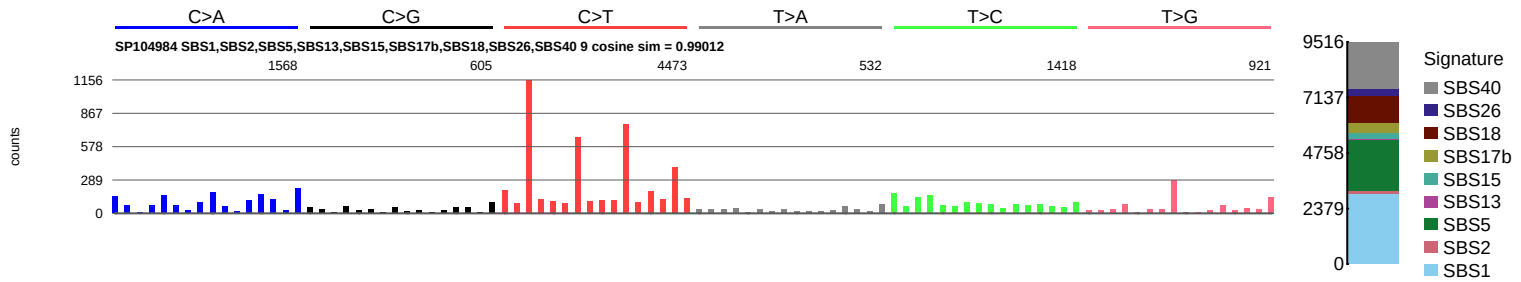
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99012
1526
596
4434
626
1387
941
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.99012
1568
605
4473
532
1418
921
counts
1156
867
578
289
0
9516
7137
4758
2379
0
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SBS17b
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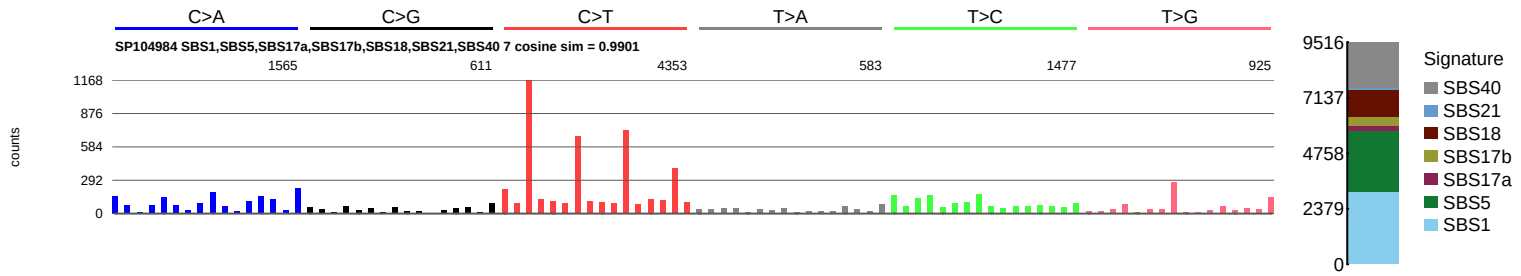
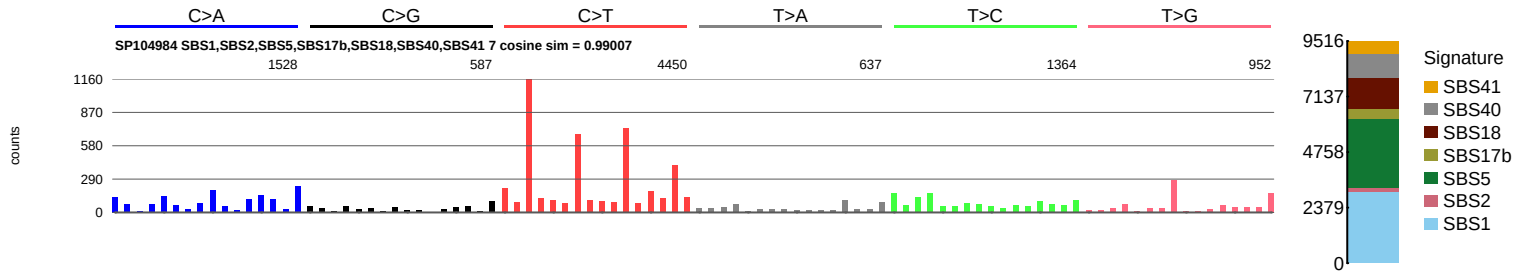
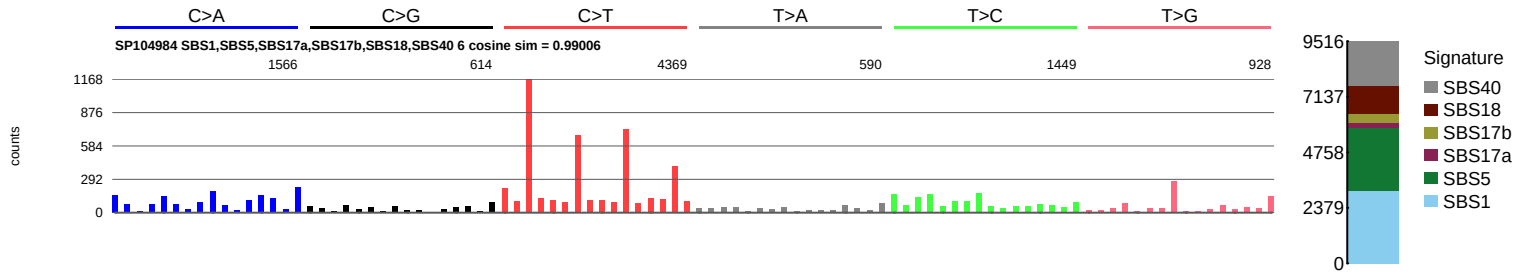
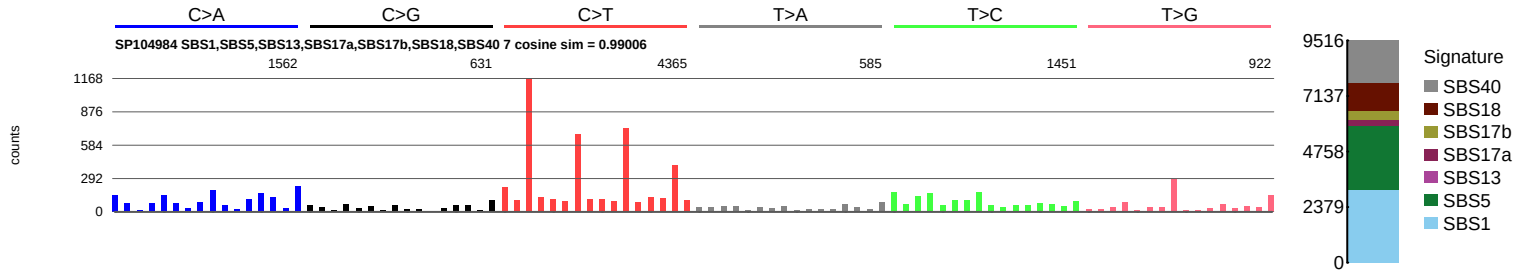
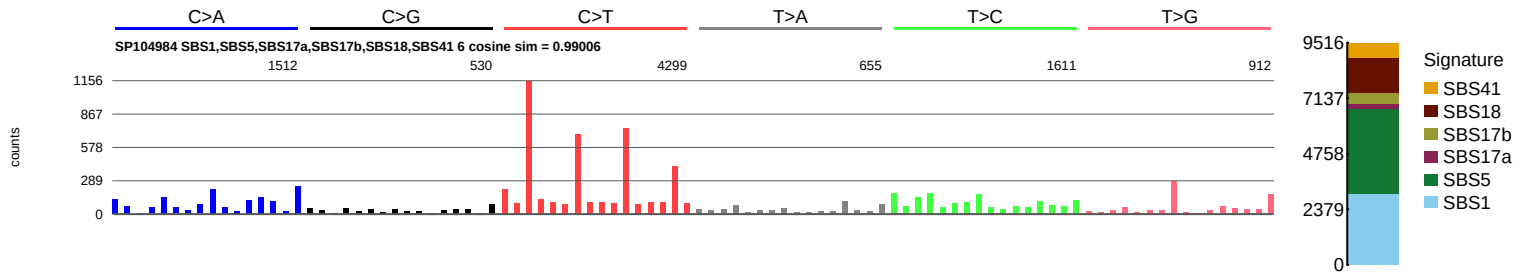
SBS5
SBS2
SBS1

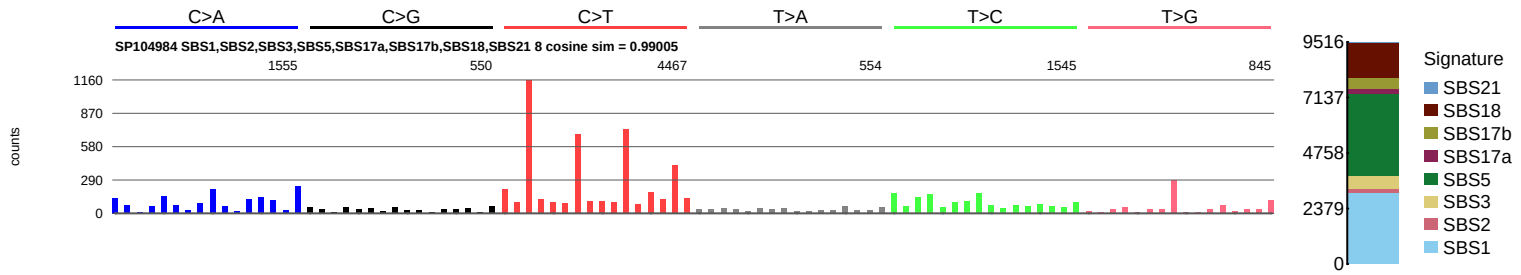

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.9901
1565
611
4353
583
1477
925
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

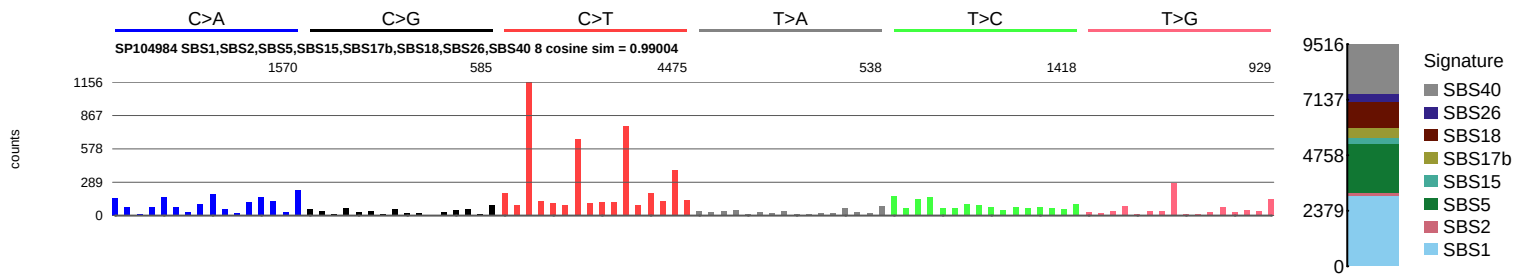

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99007
1528
587
4450
637
1364
952
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1

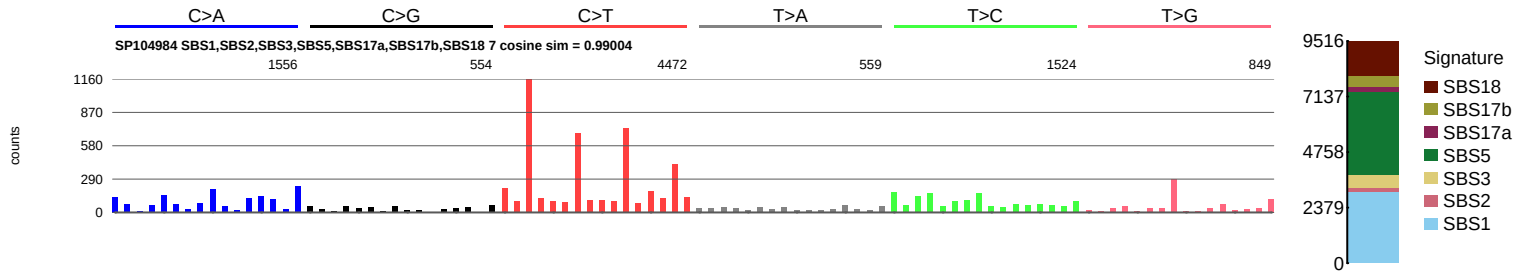

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.99006
1566
614
4369
590
1449
928
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

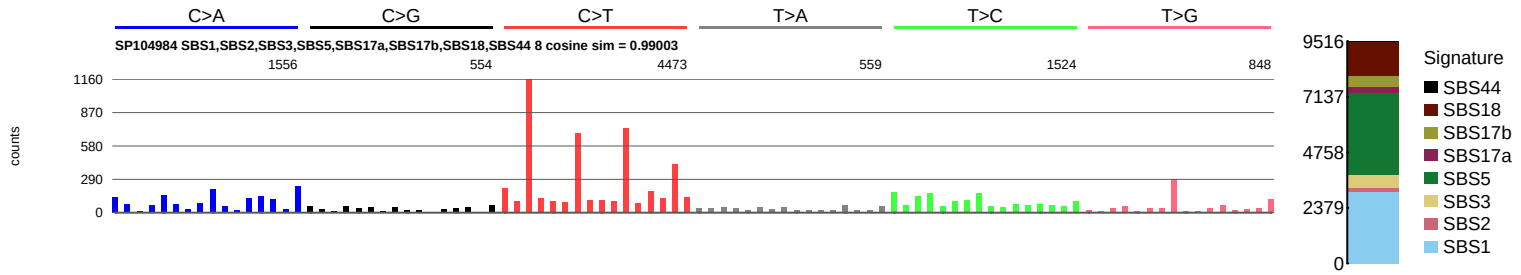

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.99006
1562
631
4365
585
1451
922
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

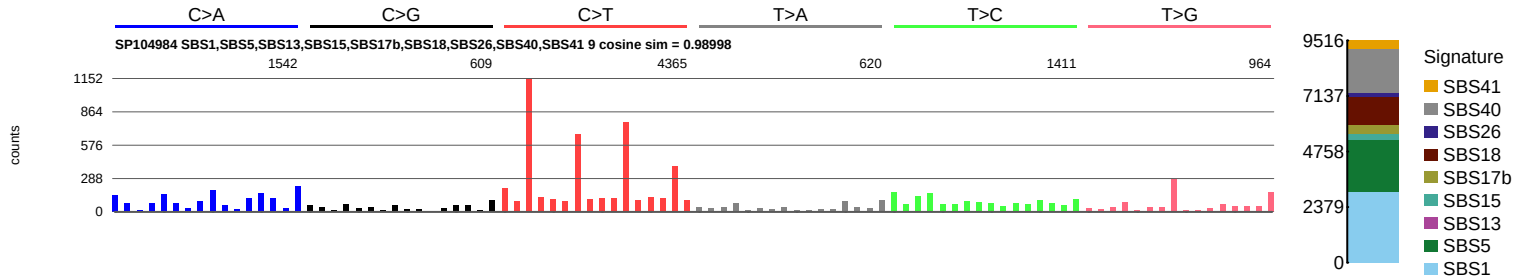

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99006
1512
530
4299
655
1611
912
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.99005
1555
550
4467
554
1545
845
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.99004
1570
585
4475
538
1418
929
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.99004
1556
554
4472
559
1524
849
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.99003
1556
554
4473
559
1524
848
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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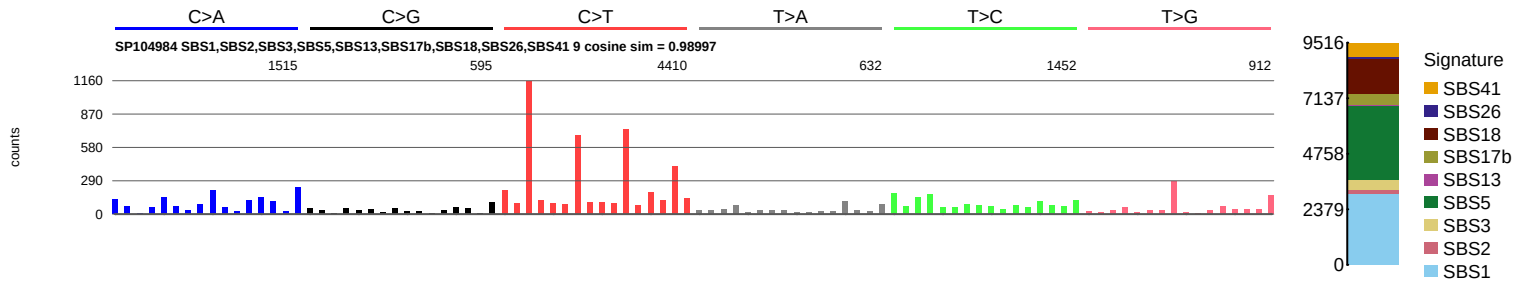
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98998
1542
609
4365
620
1411
964
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1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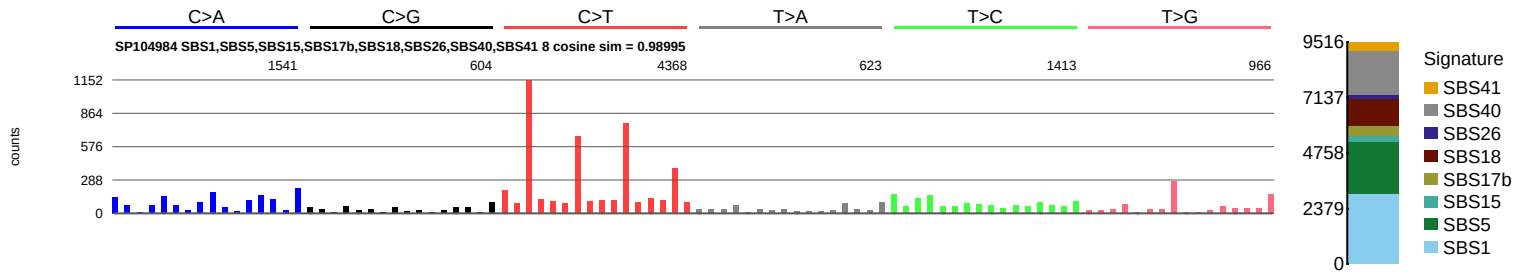
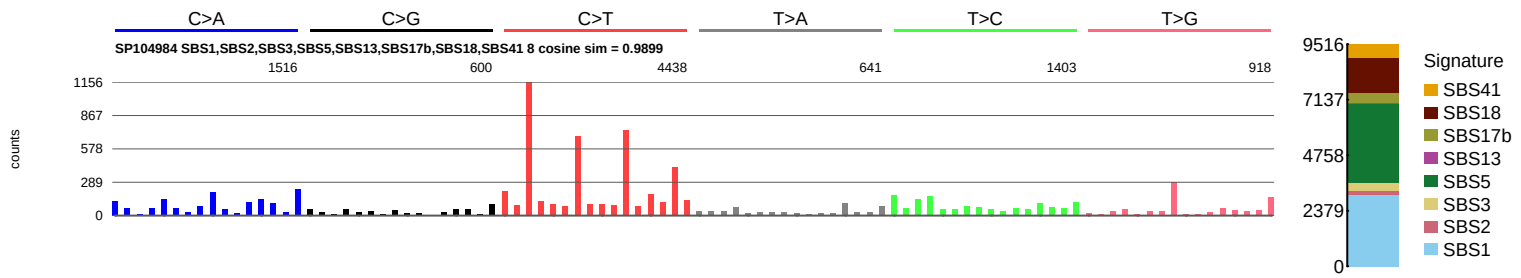
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98997
1515
595
4410
632
1452
912
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98995
1541
604
4368
623
1413
966
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.9899
1516
600
4438
641
1403
918
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
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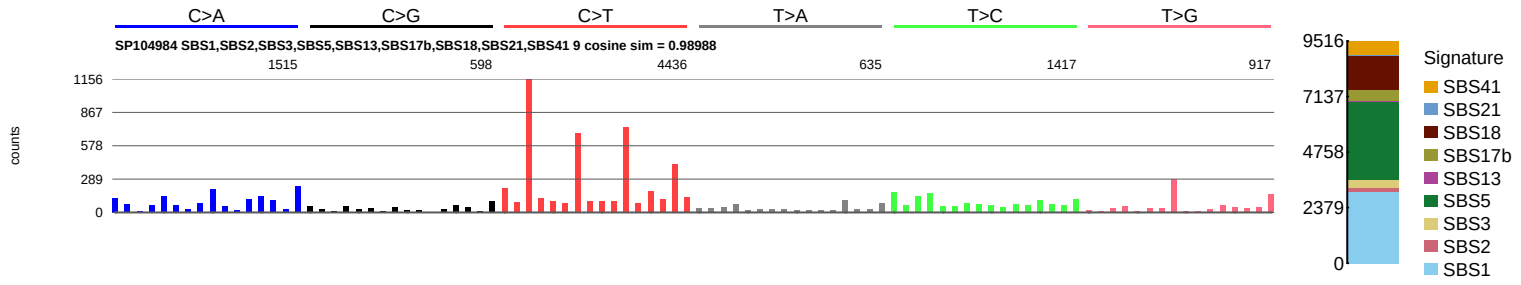
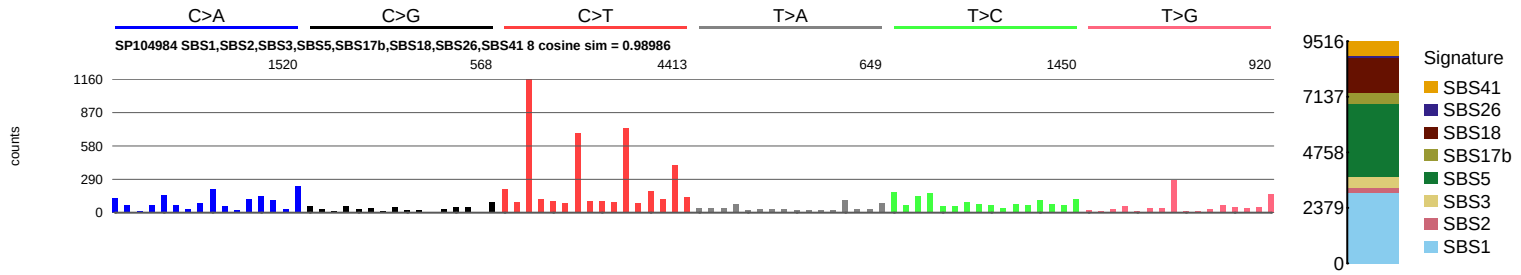
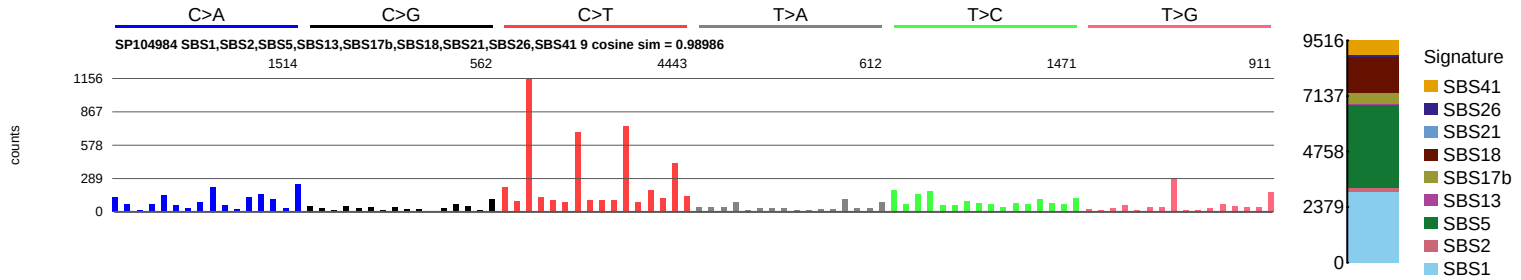
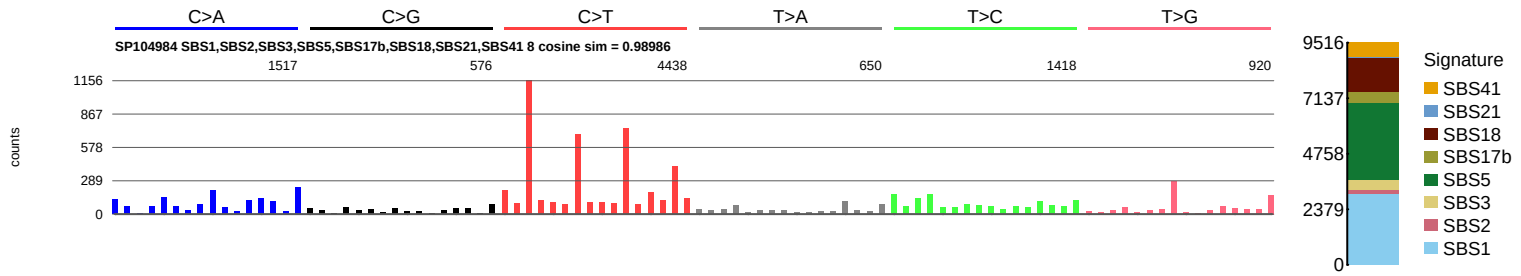
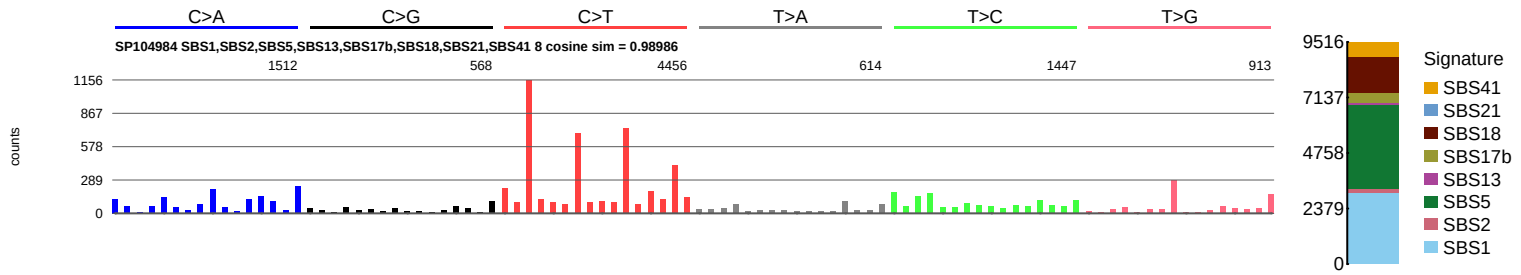
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98988
1515
598
4436
635
1417
917
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98986
1520
568
4413
649
1450
920
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98986
1514
562
4443
612
1471
911
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98986
1517
576
4438
650
1418
920
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98986
1512
568
4456
614
1447
913
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
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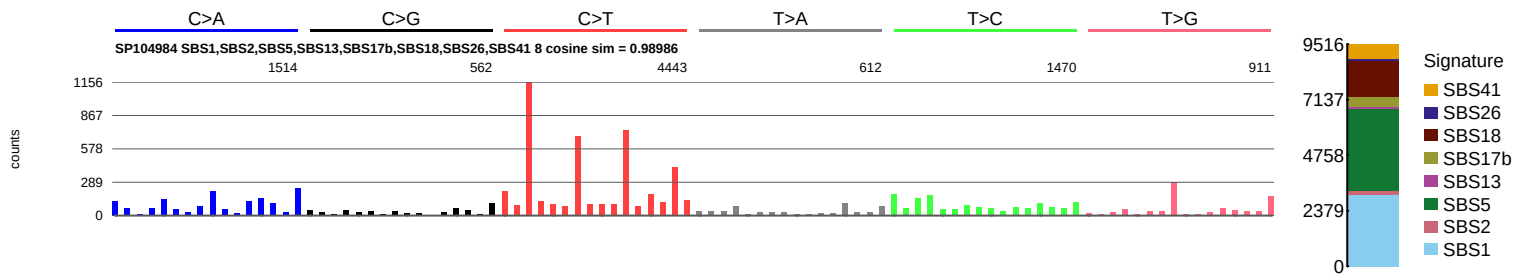
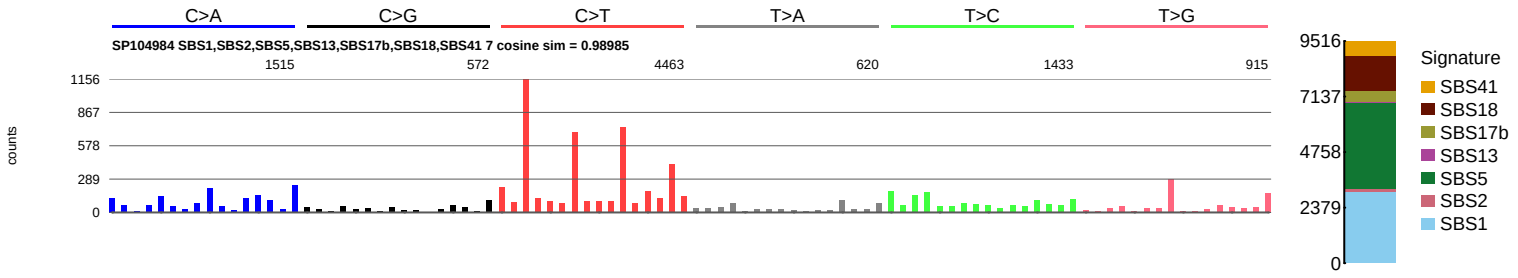
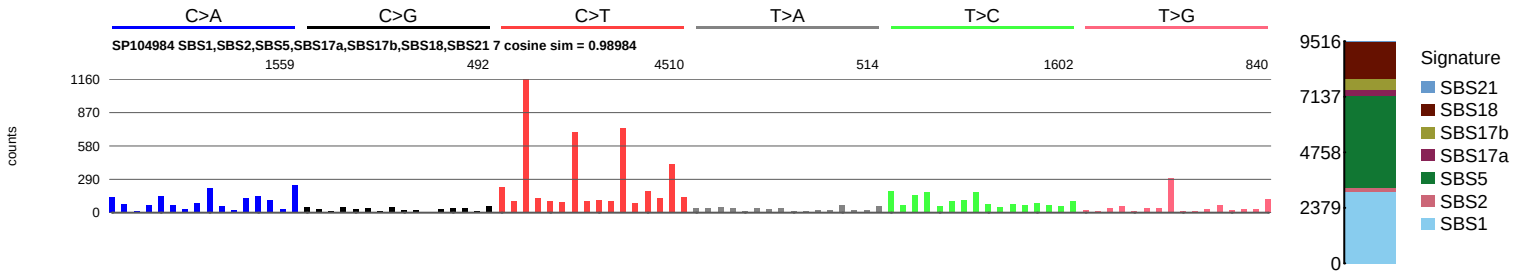
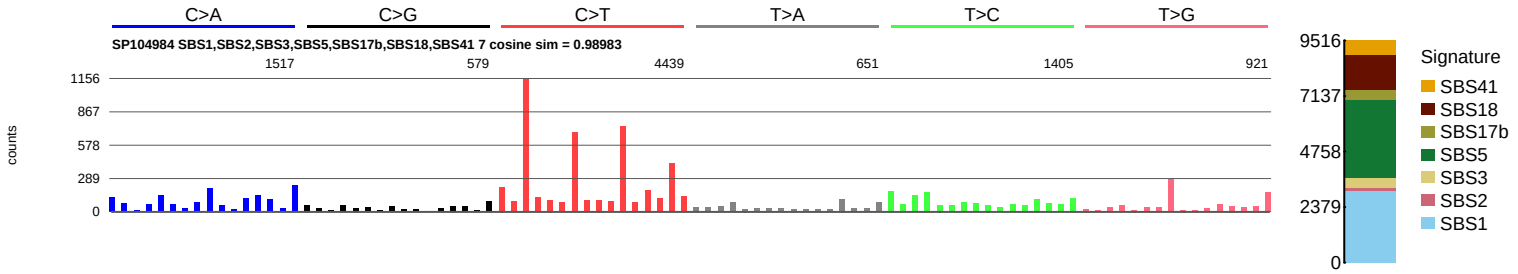
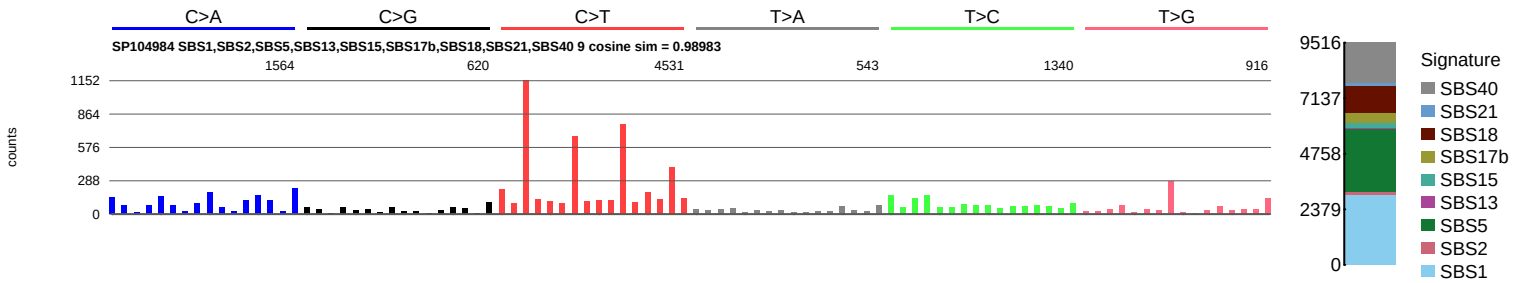
SBS2
SBS1

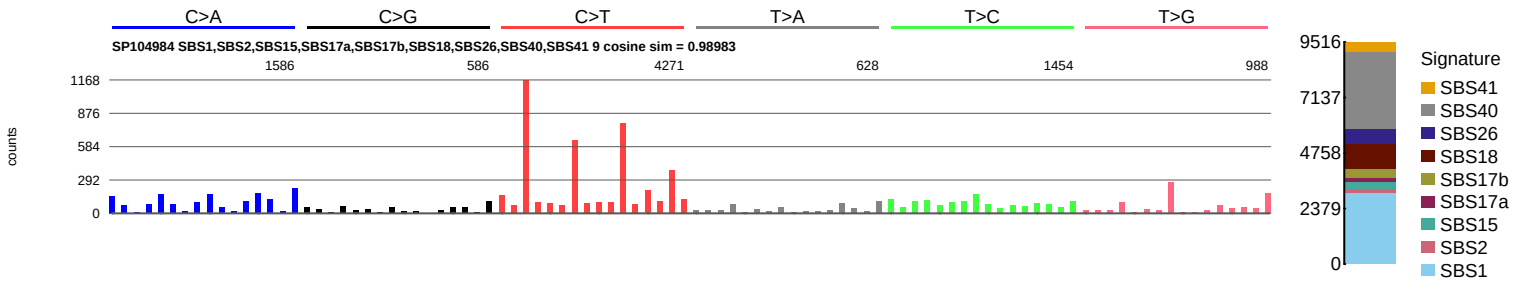

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98986
1514
562
4443
612
1470
911
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

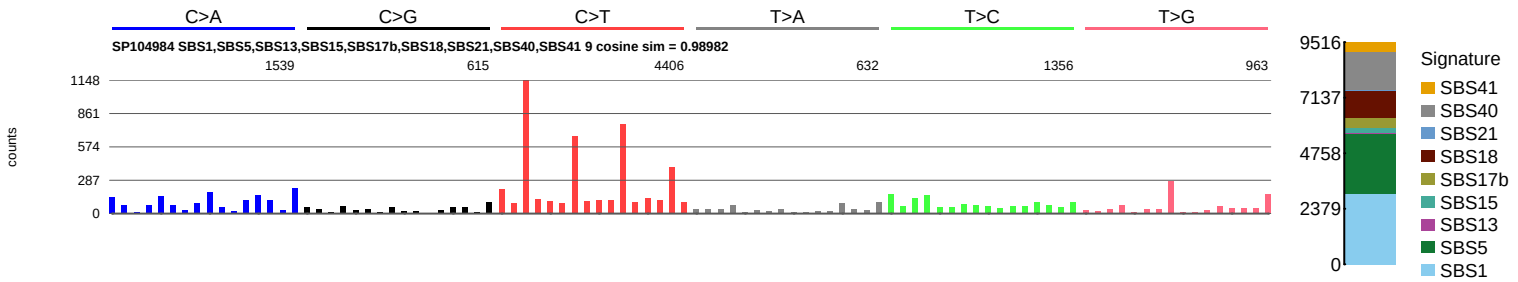

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.98985
1515
572
4463
620
1433
915
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

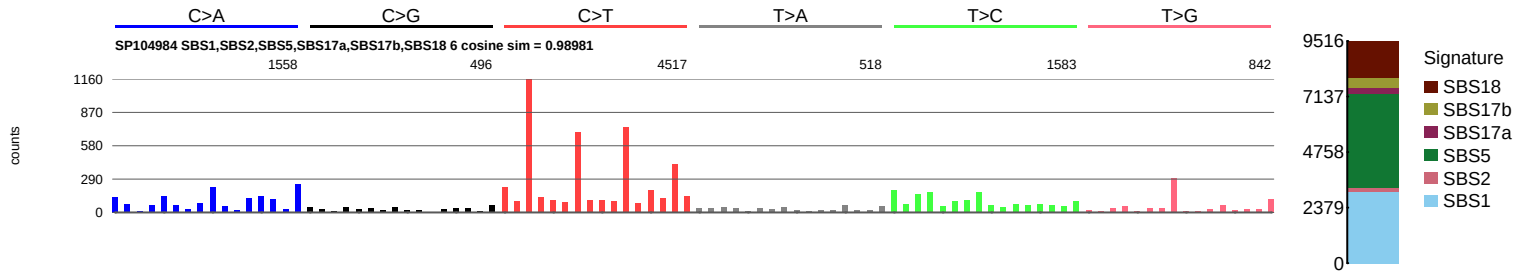

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98984
1559
492
4510
514
1602
840
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

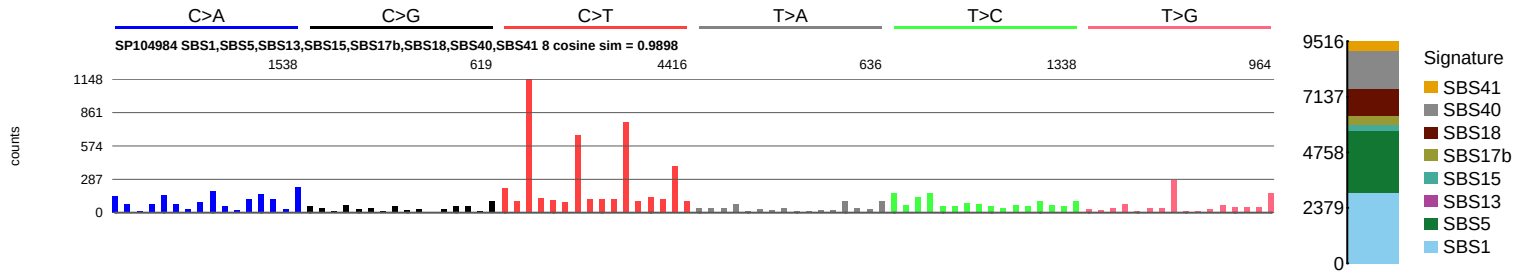

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41 7 cosine sim = 0.98983
1517
579
4439
651
1405
921
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

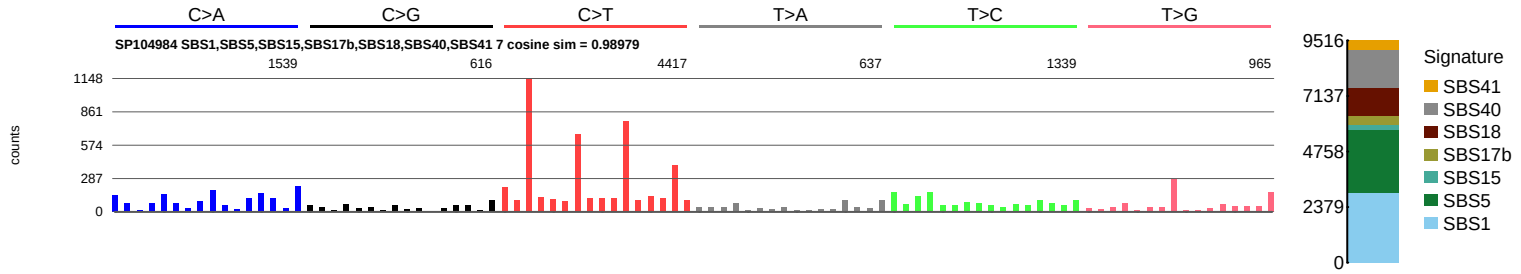

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98983
1564
620
4531
543
1340
916
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

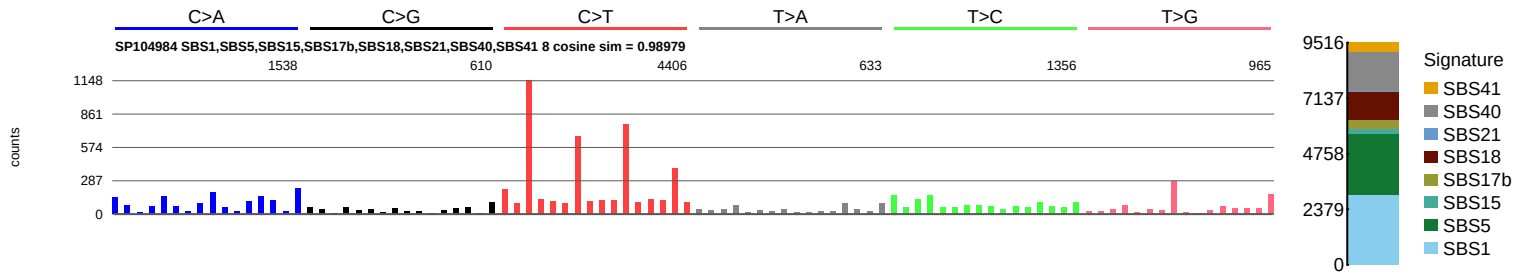

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98983
1586
586
4271
628
1454
988
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

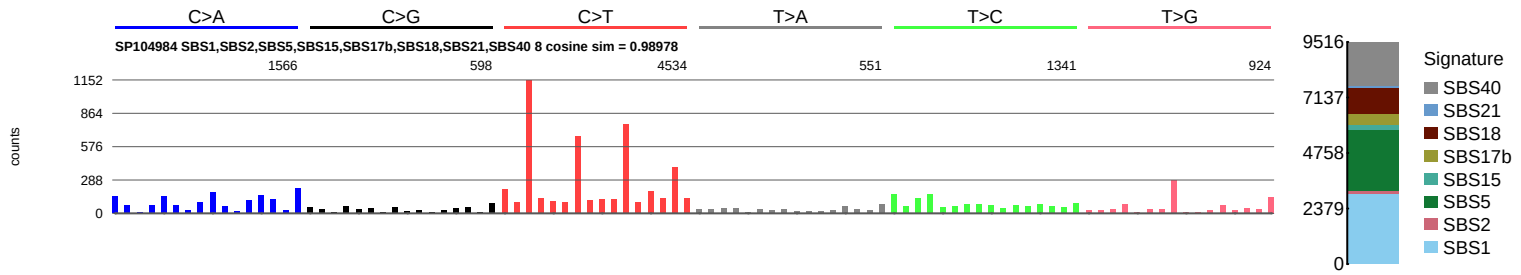

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98982
1539
615
4406
632
1356
963
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1

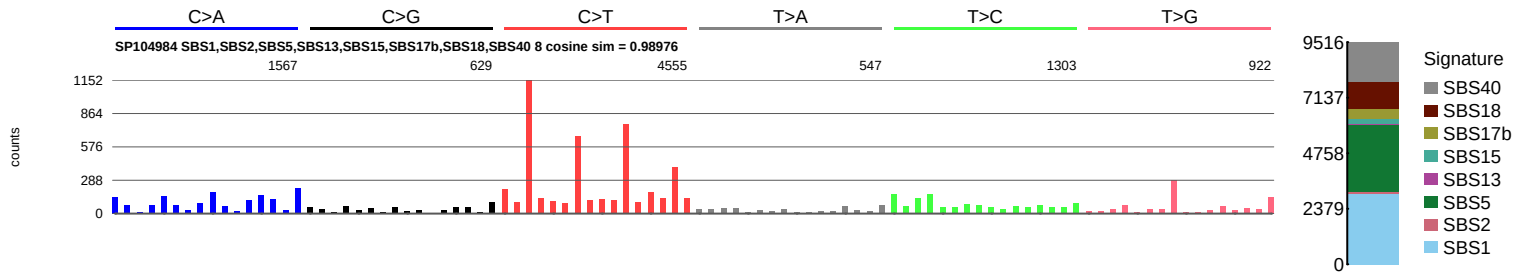

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98981
1558
496
4517
518
1583
842
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9898
1538
619
4416
636
1338
964
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98979
1539
616
4417
637
1339
965
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98979
1538
610
4406
633
1356
965
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98978
1566
598
4534
551
1341
924
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98976
1567
629
4555
547
1303
922
counts
1152
864
576
288
0
9516
7137
4758
2379
0
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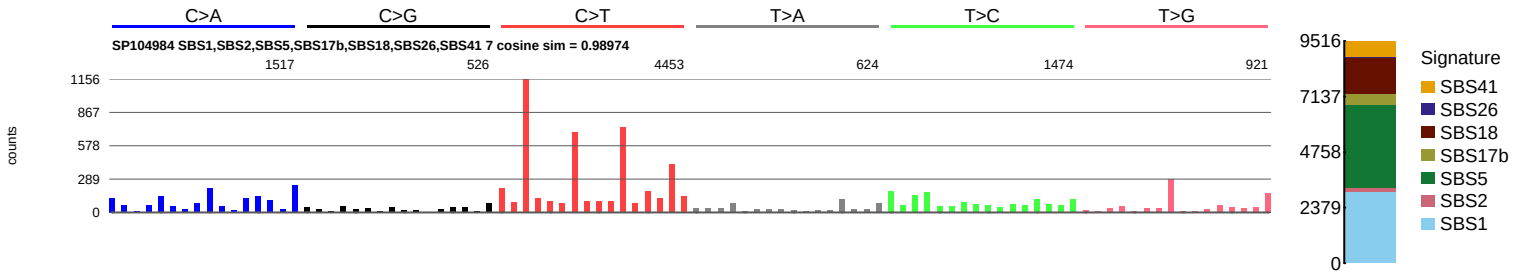
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98974
1517
526
4453
624
1474
921
counts
1156
867
578
289
0
9516
7137
4758
2379
0
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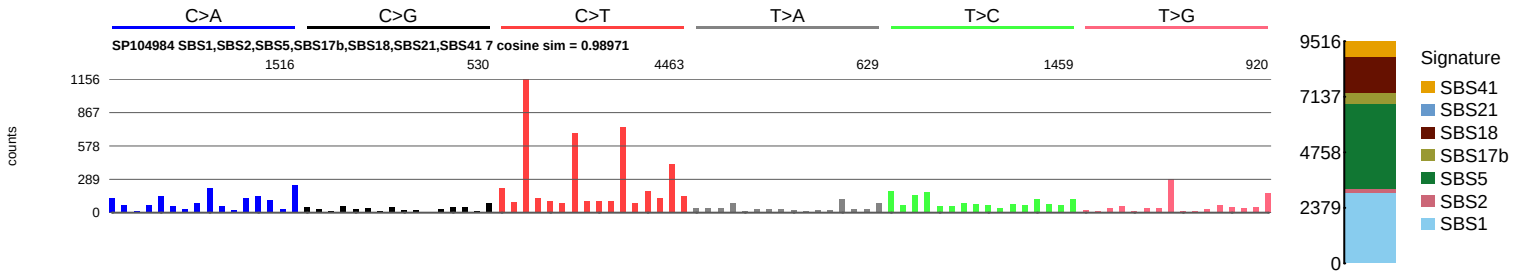
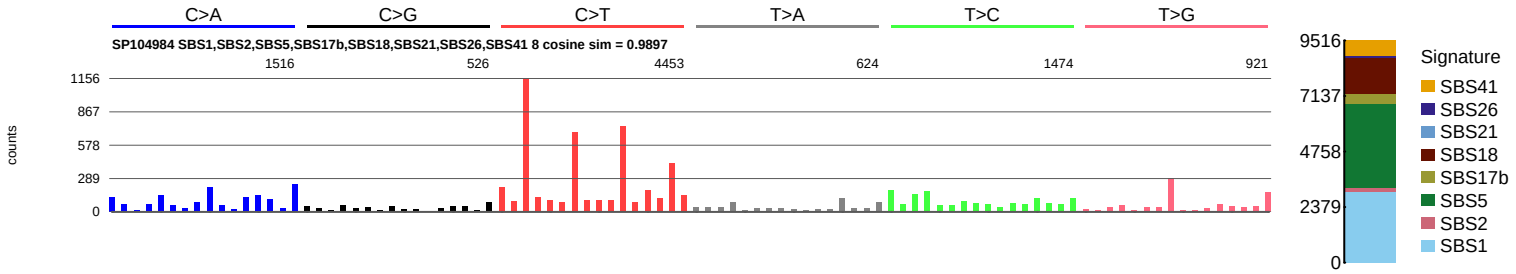
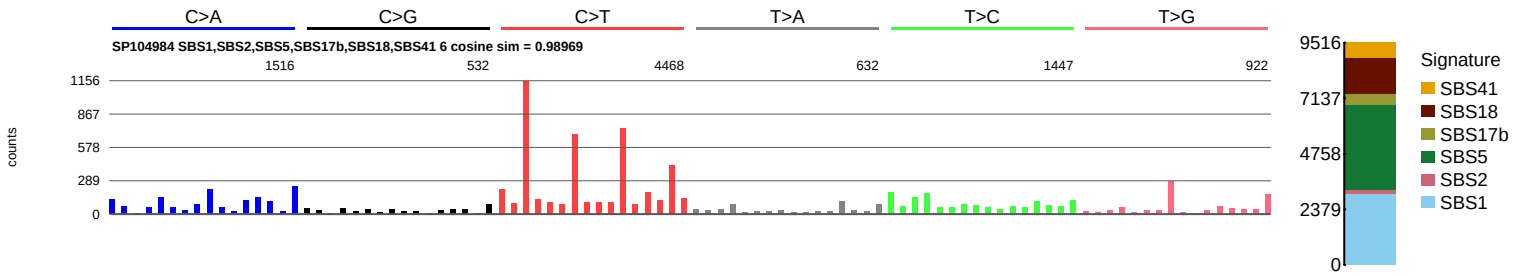
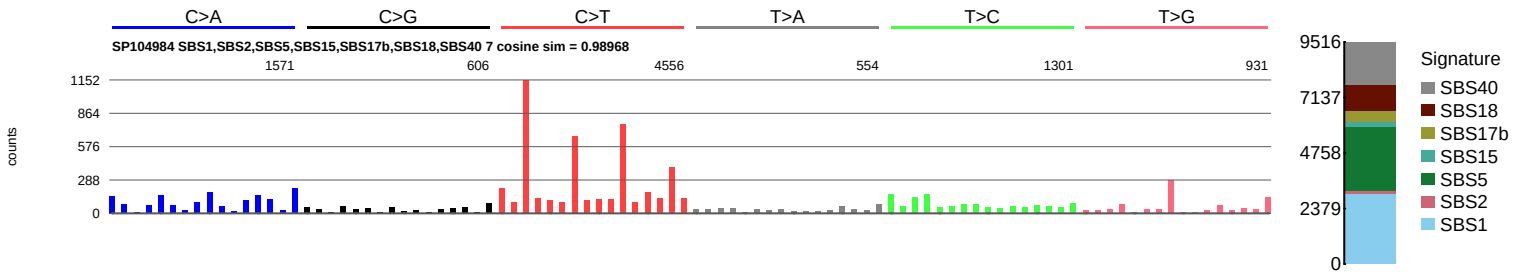
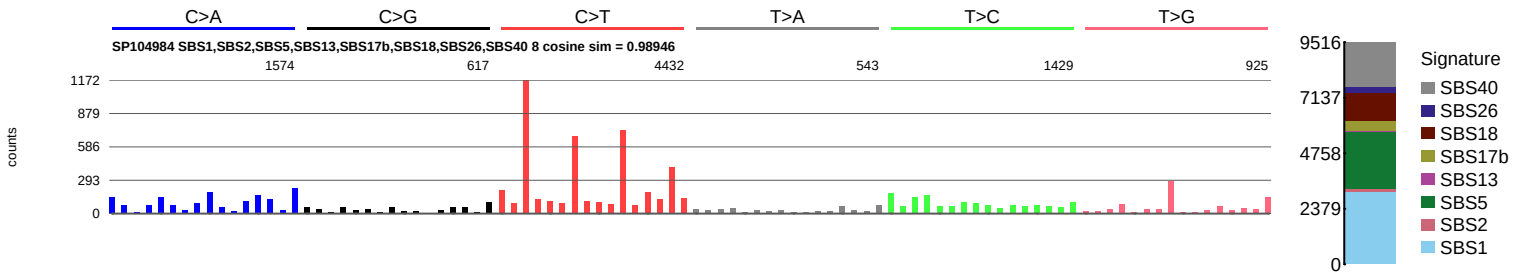
SBS5
SBS2
SBS1

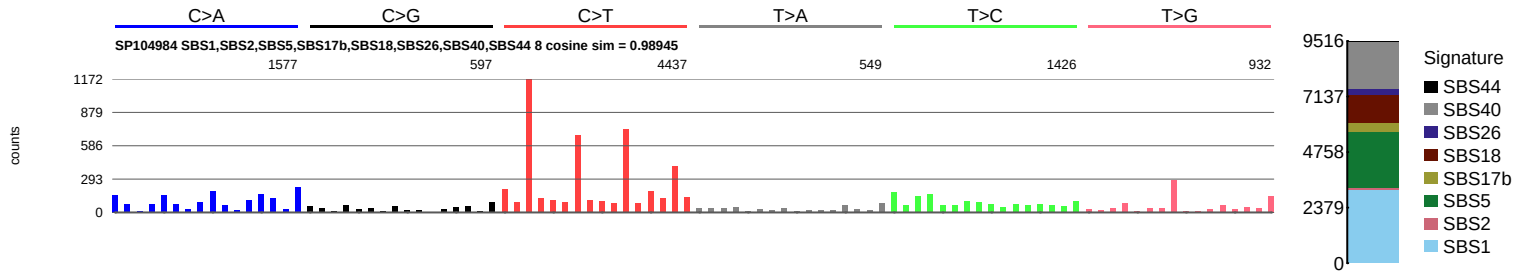

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98971
1516
530
4463
629
1459
920
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

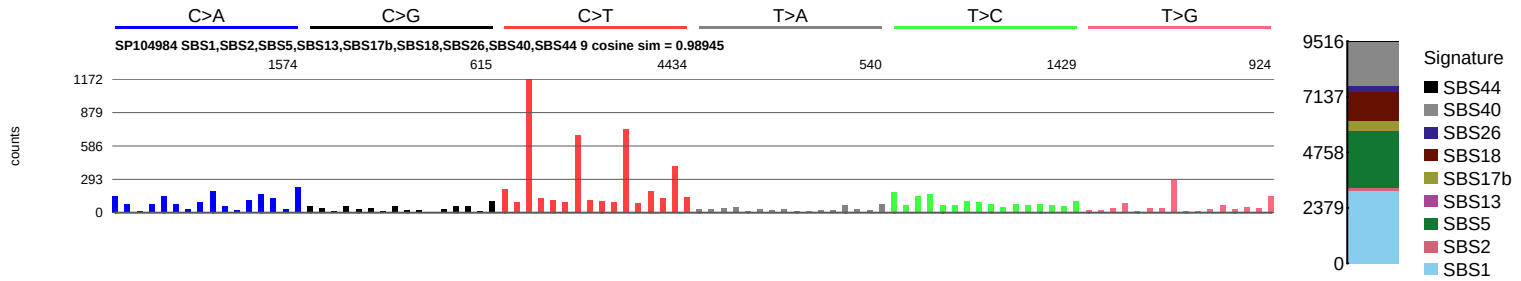

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.9897
1516
526
4453
624
1474
921
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

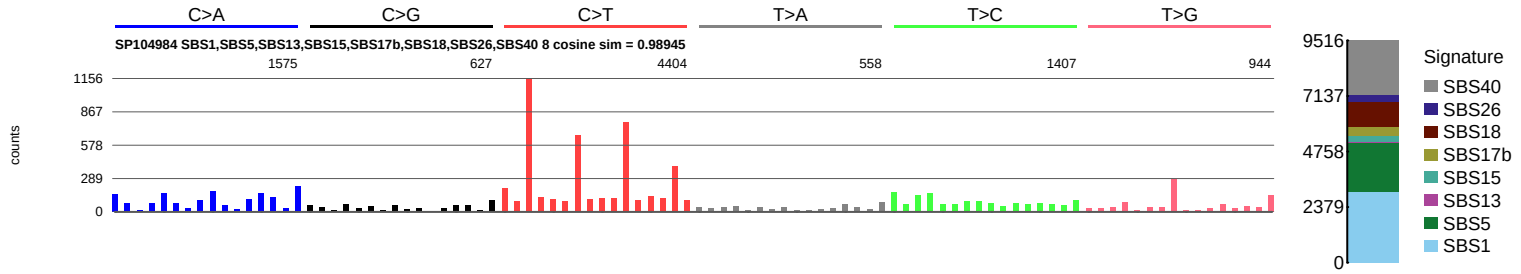

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS41 6 cosine sim = 0.98969
1516
532
4468
632
1447
922
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS2
SBS1

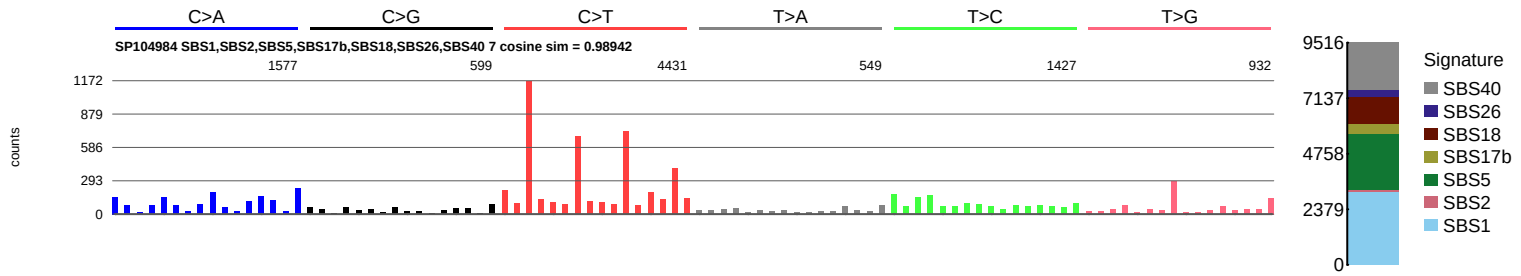

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98968
1571
606
4556
554
1301
931
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

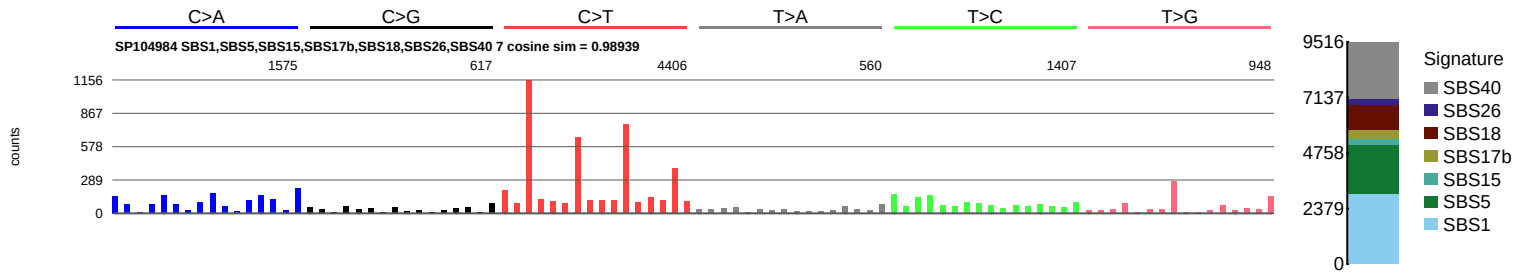

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98946
1574
617
4432
543
1429
925
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

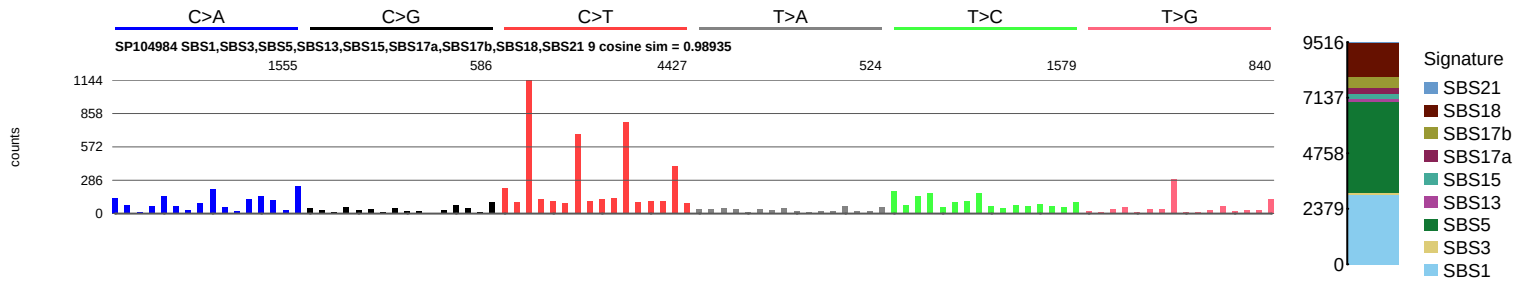

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98945
1577
597
4437
549
1426
932
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1

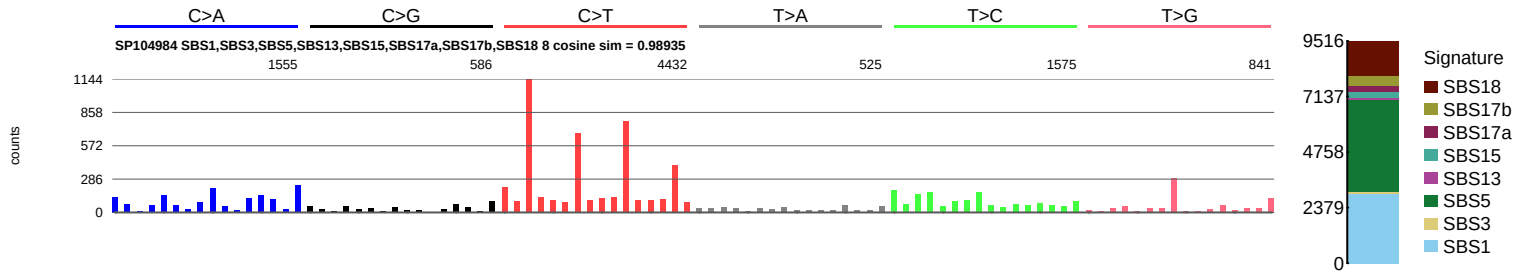

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98945
1574
615
4434
540
1429
924
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

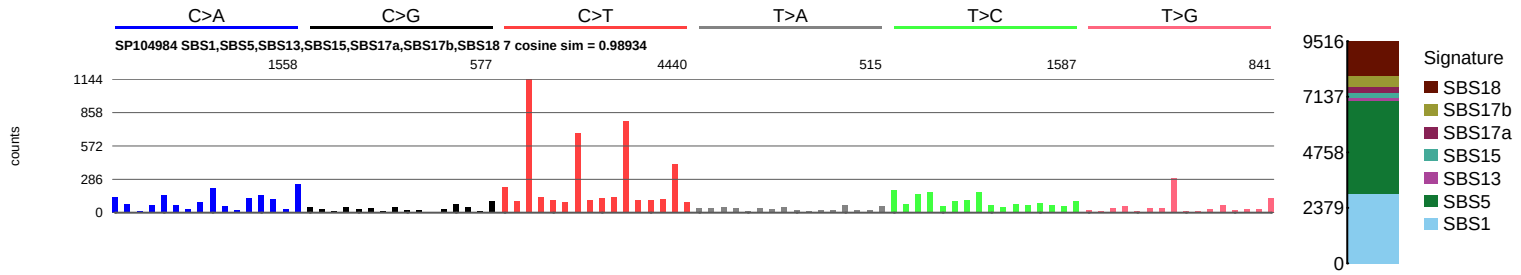

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98945
1575
627
4404
558
1407
944
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98942
1577
599
4431
549
1427
932
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98939
1575
617
4406
560
1407
948
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98935
1555
586
4427
524
1579
840
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98935
1555
586
4432
525
1575
841
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98934
1558
577
4440
515
1587
841
counts
1144
858
572
286
0
9516
7137
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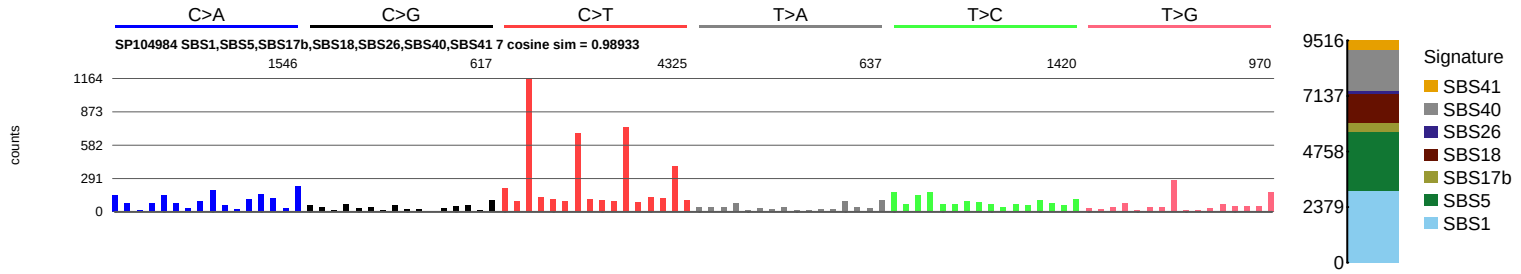
SBS1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SP104984 SBS1,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98933
1546
617
4325
637
1420
970
counts
1164
873
582
291
0
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7137
4758
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0
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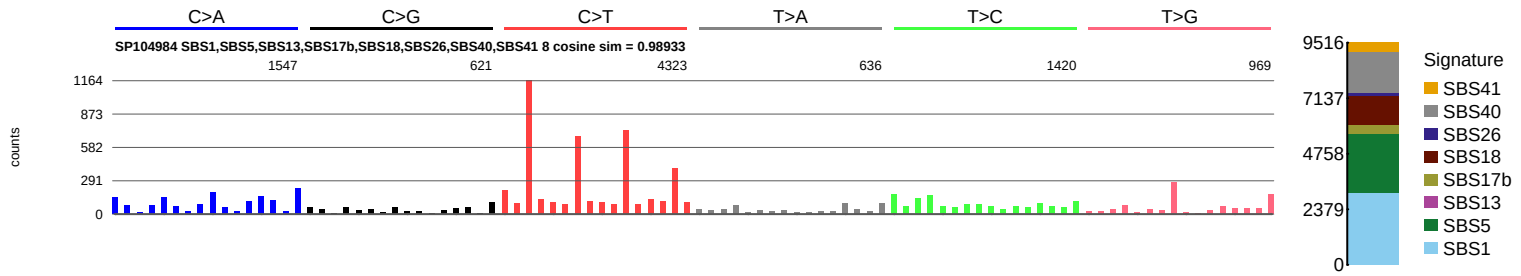
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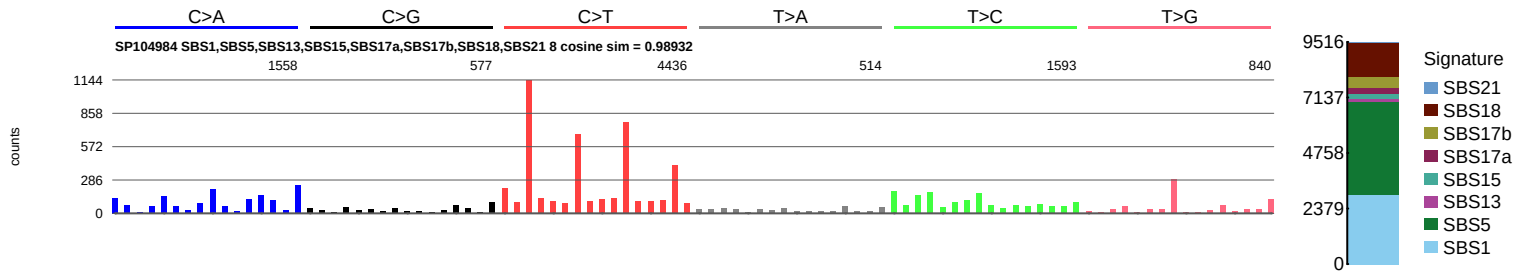
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98933
1547
621
4323
636
1420
969
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS1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SP104984 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98932
1558
577
4436
514
1593
840
counts
1144
858
572
286
0
9516
7137
4758
2379
0
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SBS21
SBS18
SBS17b
SBS17a
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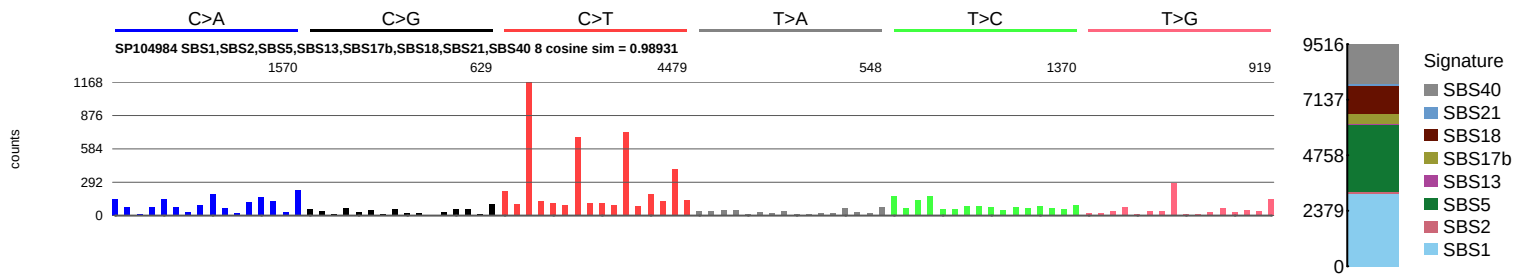
SBS13
SBS5
SBS1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1570
629
4479
548
1370
919
counts
1168
876
584
292
0
9516
7137
4758
2379
0
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SBS40
SBS21
SBS18
SBS17b
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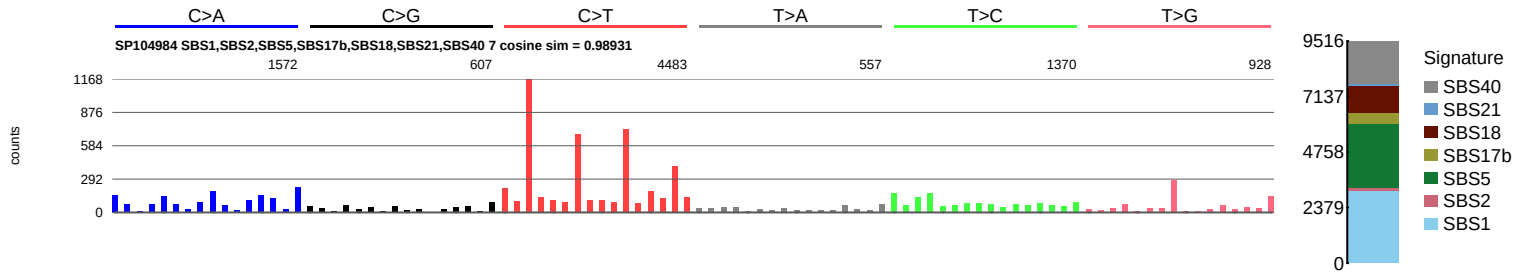
SBS2
SBS1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SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98931
1572
607
4483
557
1370
928
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
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SBS18
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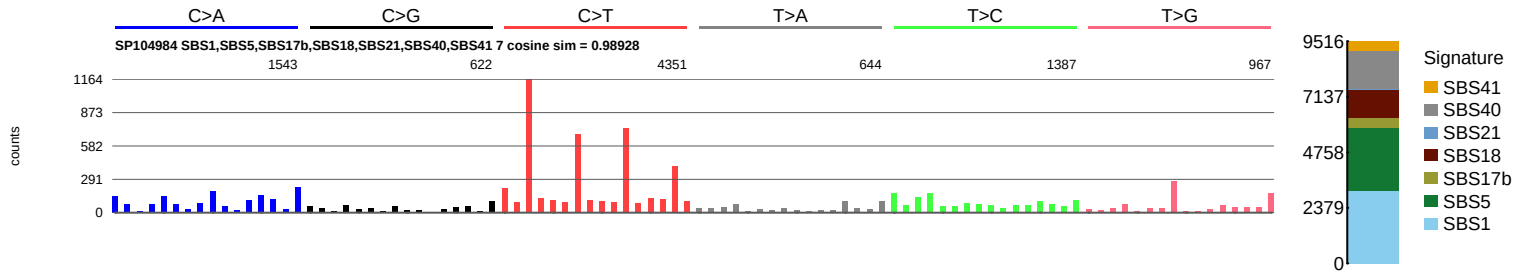
SBS5
SBS2
SBS1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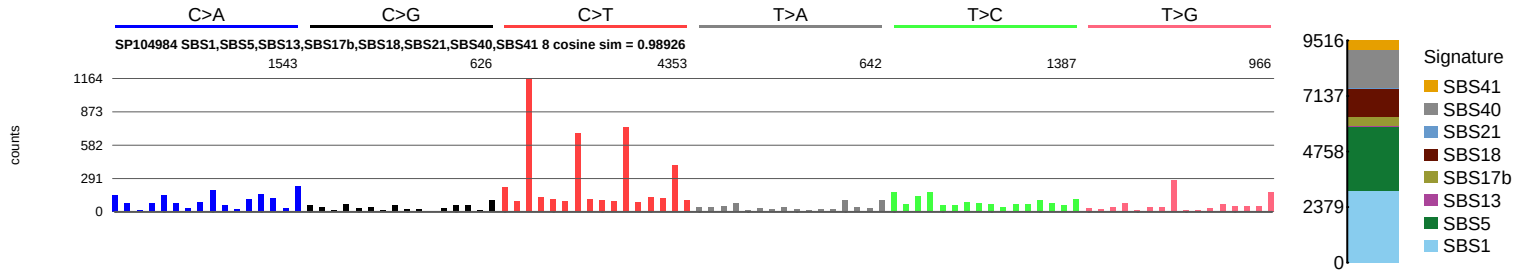
1543
622
4351
644
1387
967
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1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C>T
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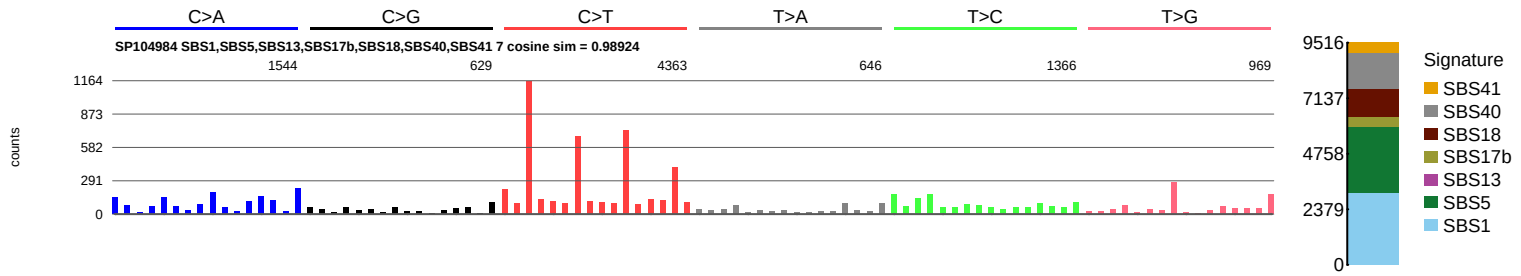
1543
626
4353
642
1387
966
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1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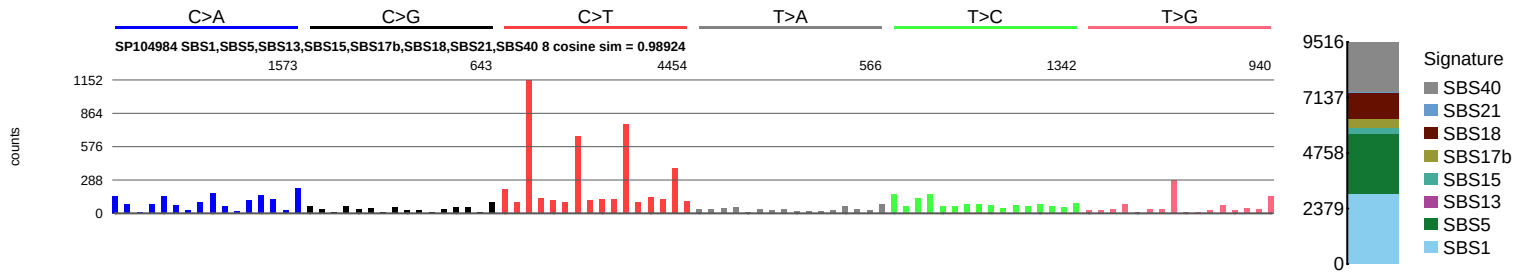
T>G
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98924
1544
629
4363
646
1366
969
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS1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SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98924
1573
643
4454
566
1342
940
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
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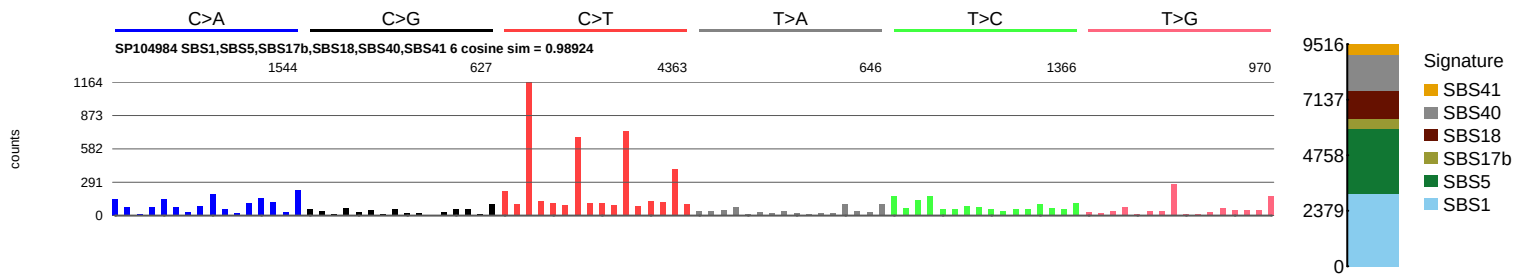
SBS13
SBS5
SBS1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C>G
C>T
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T>C
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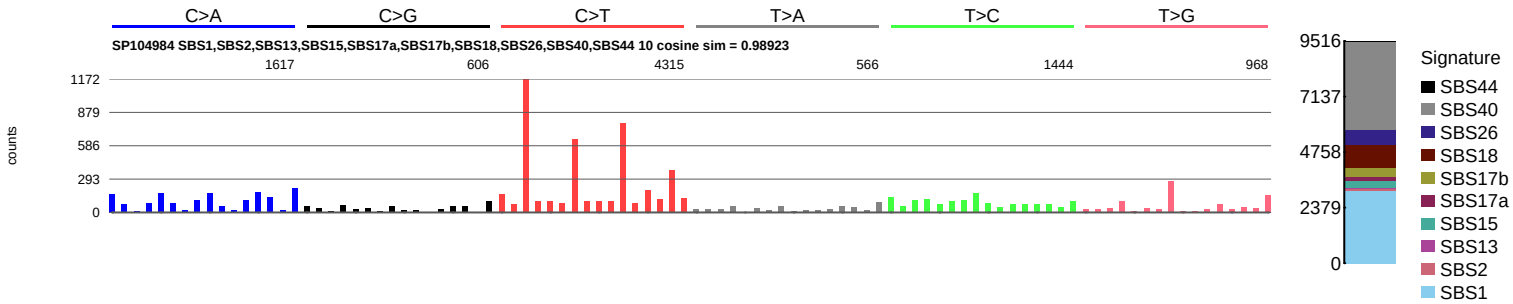
SP104984 SBS1,SBS5,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98924
1544
627
4363
646
1366
970
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS1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T>C
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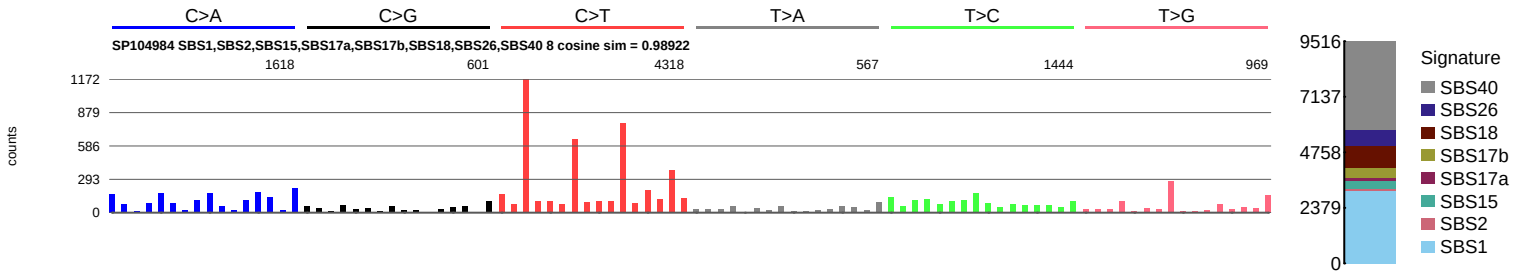
SP104984 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98923
1617
606
4315
566
1444
968
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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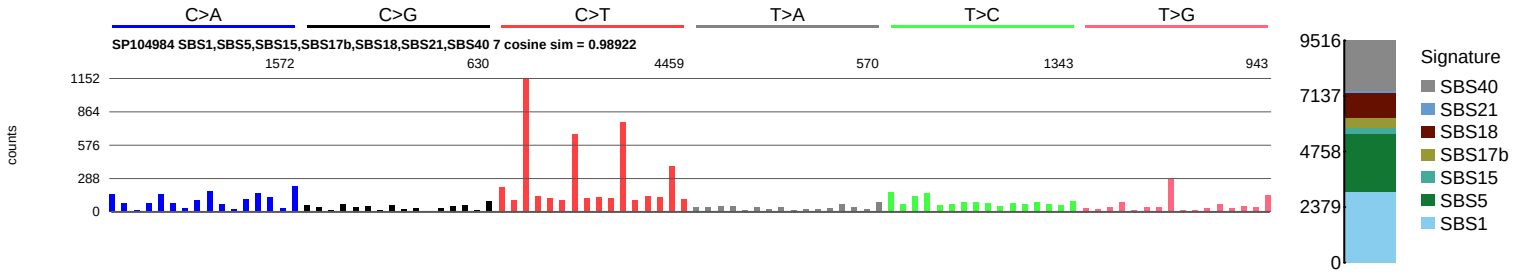
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98922
1618
601
4318
567
1444
969
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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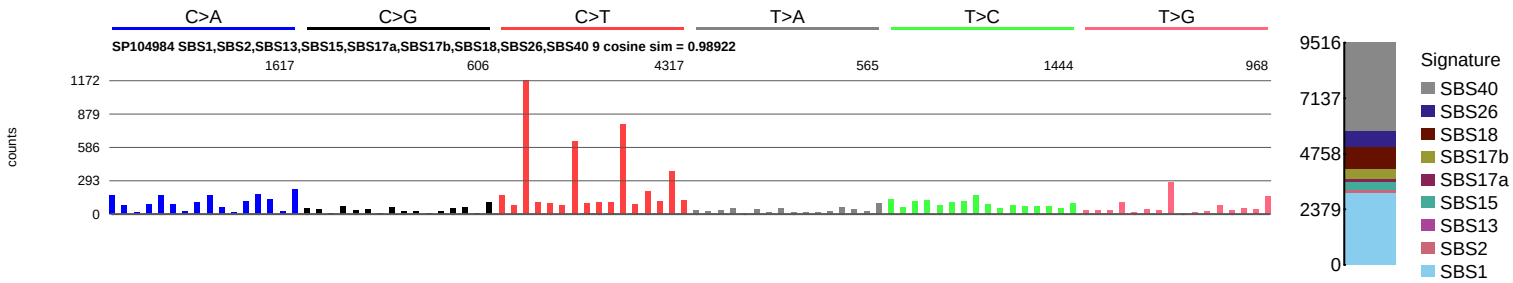
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98922
1572
630
4459
570
1343
943
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1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T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98922
1617
606
4317
565
1444
968
counts
1172
879
586
293
0
9516
7137
4758
2379
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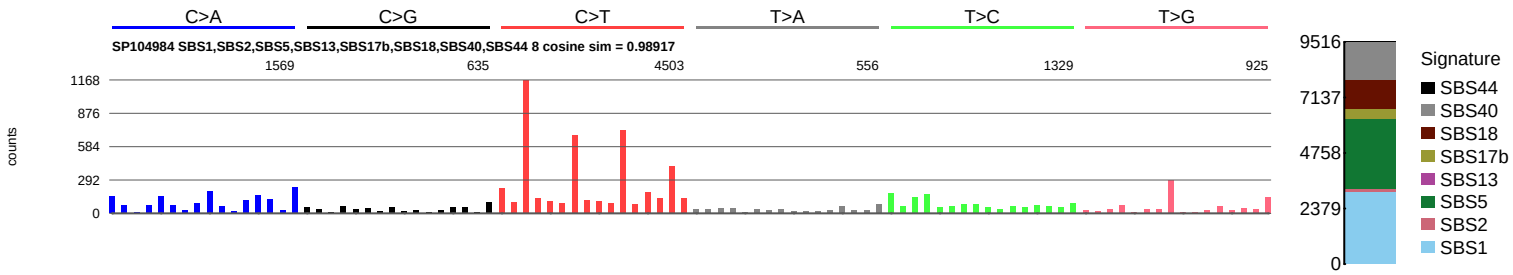
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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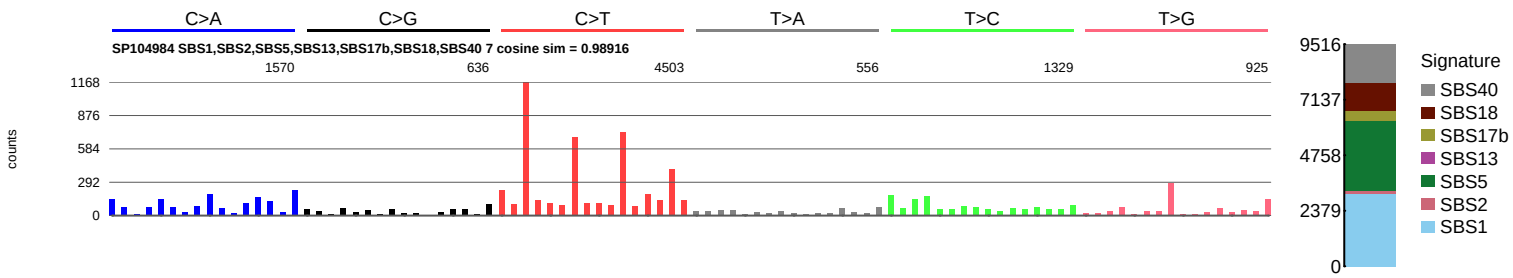
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98917
1569
635
4503
556
1329
925
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98916
1570
636
4503
556
1329
925
counts
1168
876
584
292
0
9516
7137
4758
2379
0
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SBS18
SBS17b
SBS13
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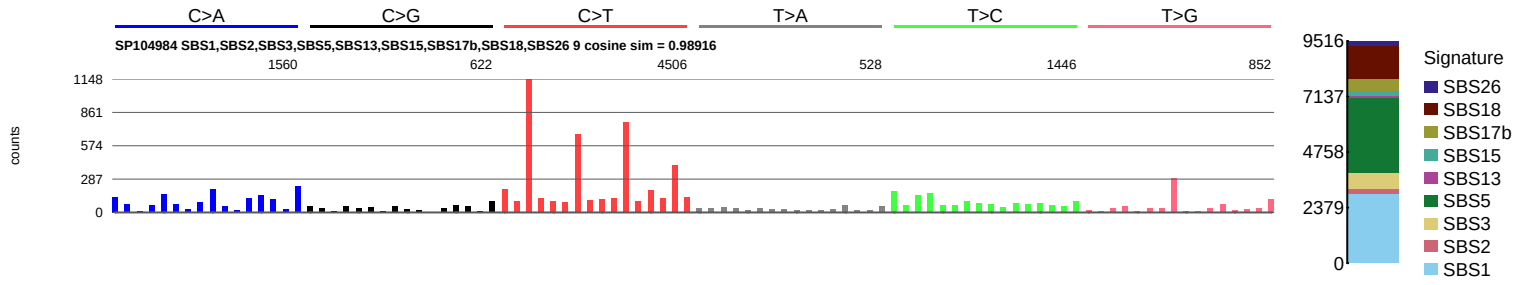
SBS2
SBS1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C>T
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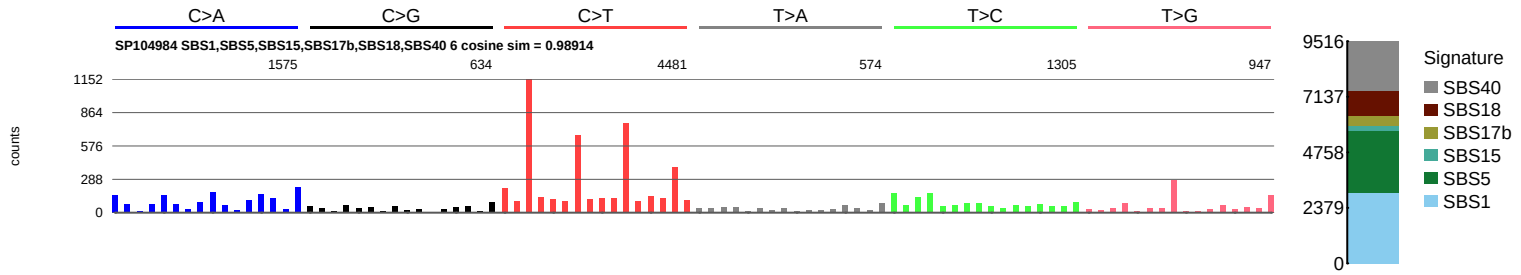
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26 9 cosine sim = 0.98916
1560
622
4506
528
1446
852
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS40 6 cosine sim = 0.98914
1575
634
4481
574
1305
947
counts
1152
864
576
288
0
9516
7137
4758
2379
0
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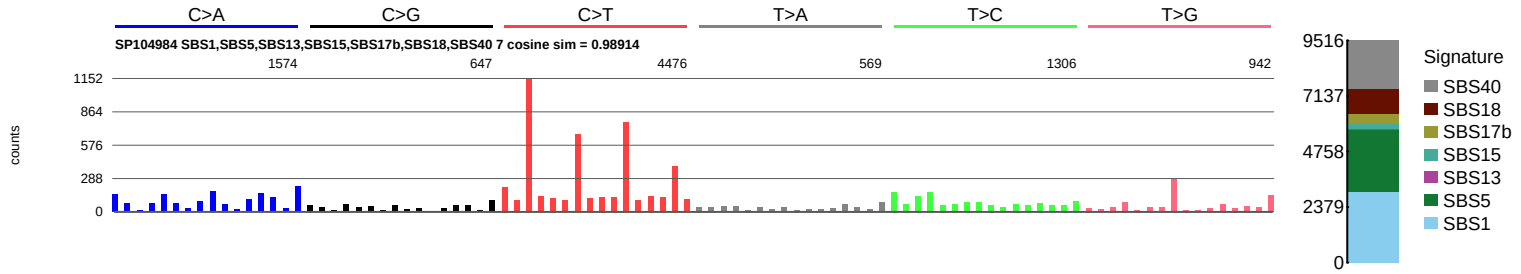
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98914
1574
647
4476
569
1306
942
counts
1152
864
576
288
0
9516
7137
4758
2379
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Signature
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SBS17b
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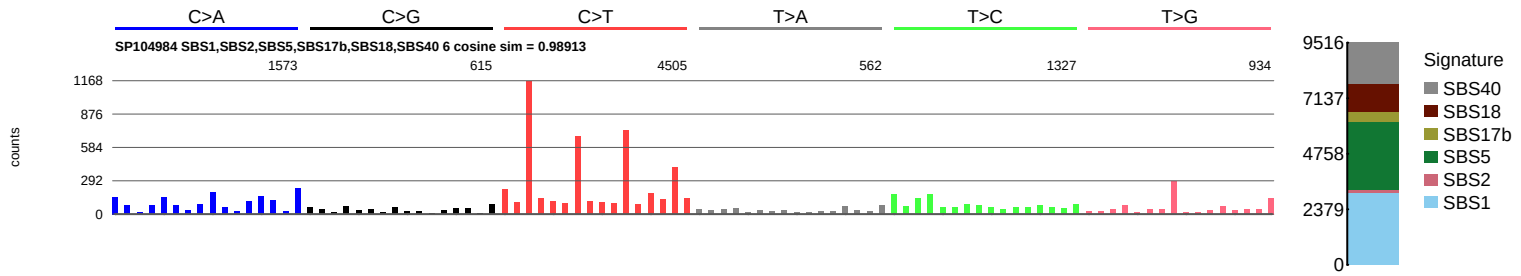
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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40 6 cosine sim = 0.98913
1573
615
4505
562
1327
934
counts
1168
876
584
292
0
9516
7137
4758
2379
0
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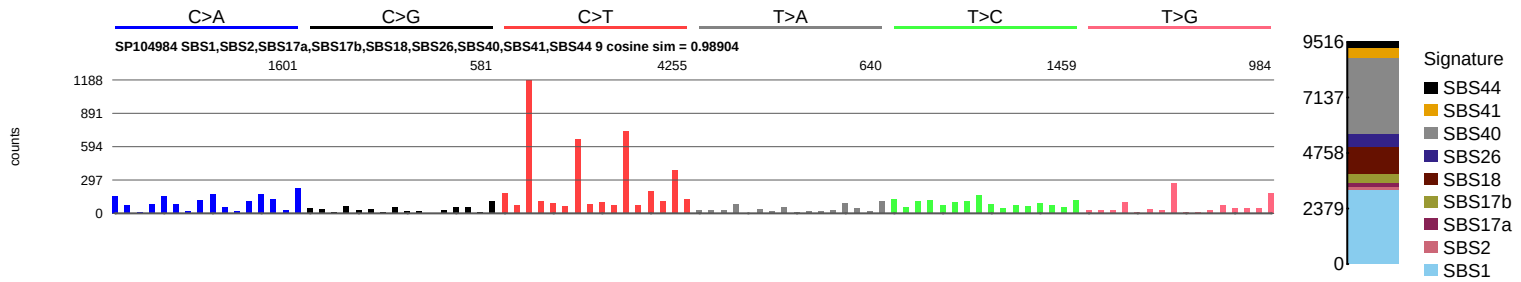
SBS18
SBS17b
SBS5
SBS2
SBS1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T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98904
1601
581
4255
640
1459
984
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
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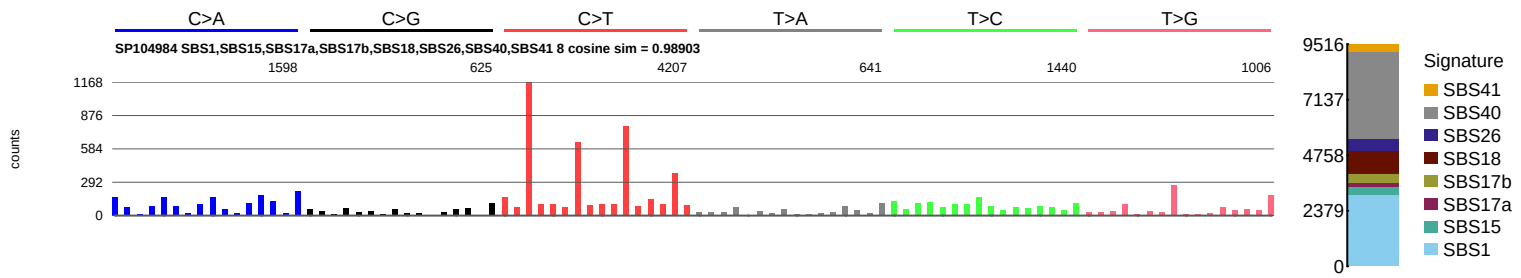
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98903
1598
625
4207
641
1440
1006
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
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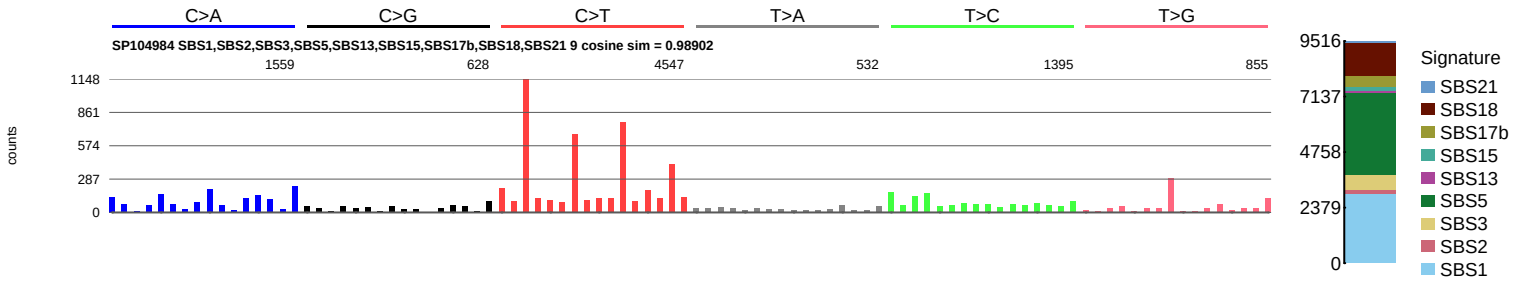
SBS15
SBS1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C>G
C>T
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T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98902
1559
628
4547
532
1395
855
counts
1148
861
574
287
0
9516
7137
4758
2379
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Signature
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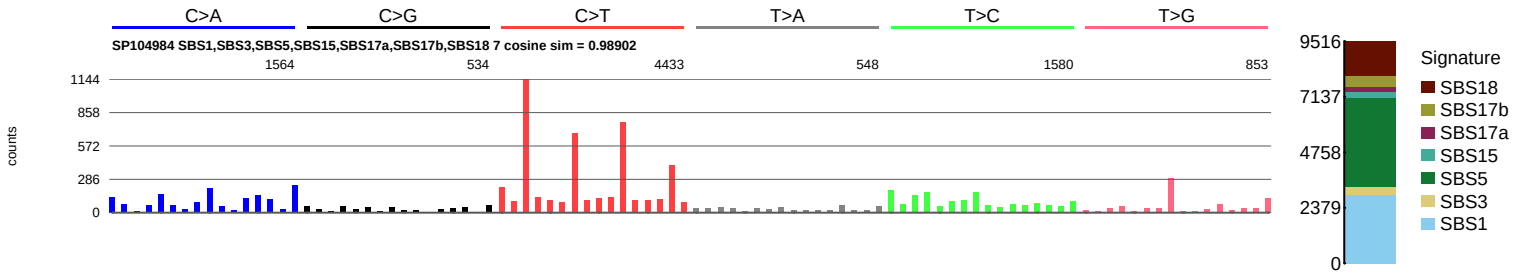
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98902
1564
534
4433
548
1580
853
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
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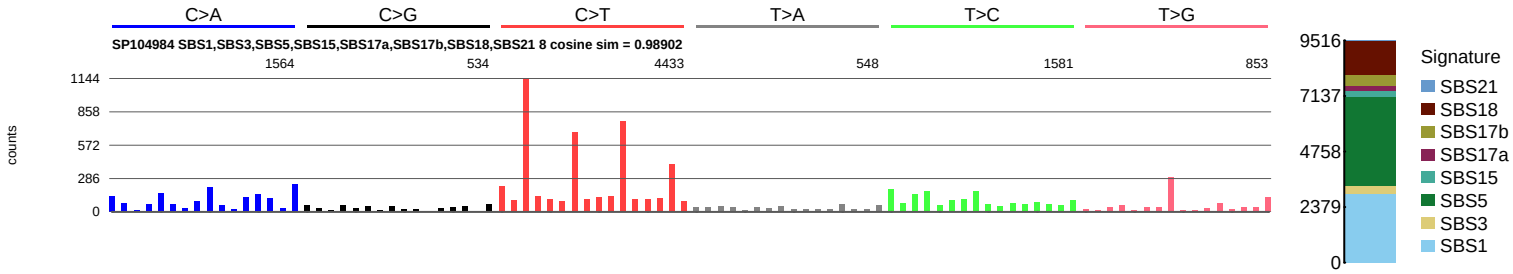
SBS3
SBS1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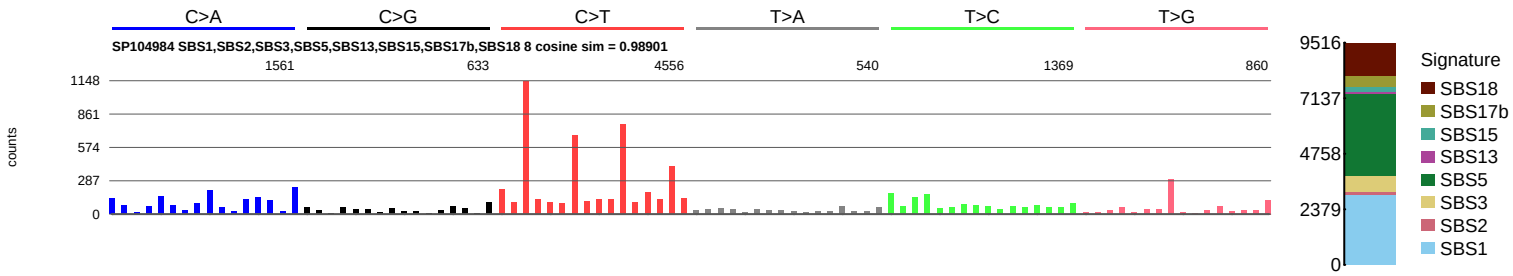
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98902
1564
534
4433
548
1581
853
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98901
1561
633
4556
540
1369
860
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS15
SBS13
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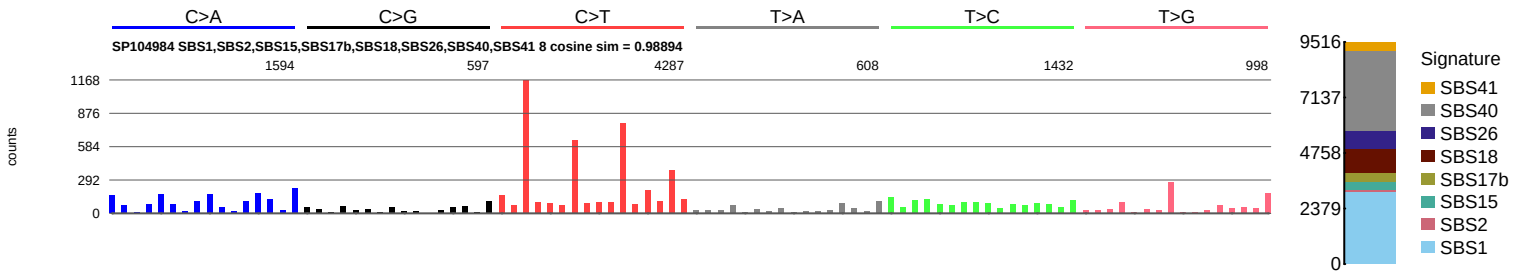
SBS3
SBS2
SBS1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SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98894
1594
597
4287
608
1432
998
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
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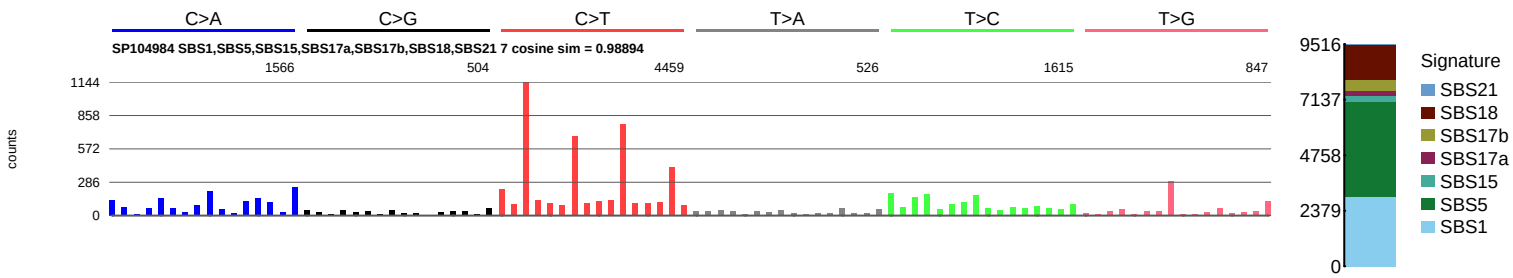
SBS18
SBS17b
SBS15
SBS2
SBS1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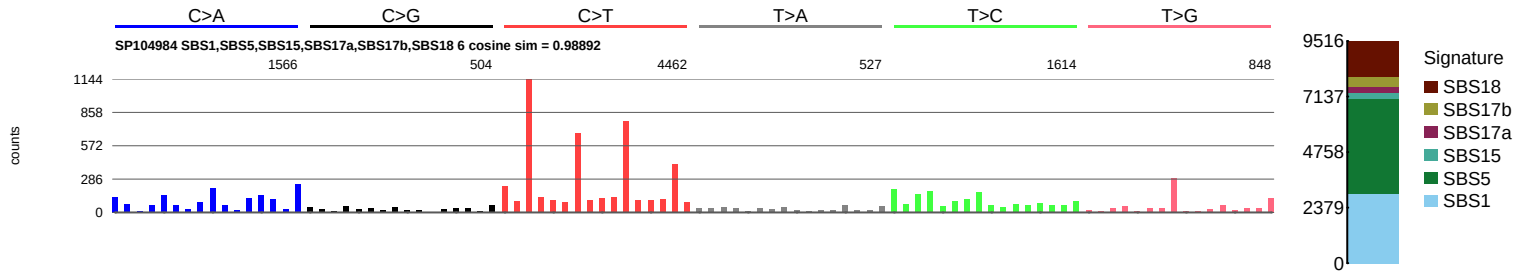
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98894
1566
504
4459
526
1615
847
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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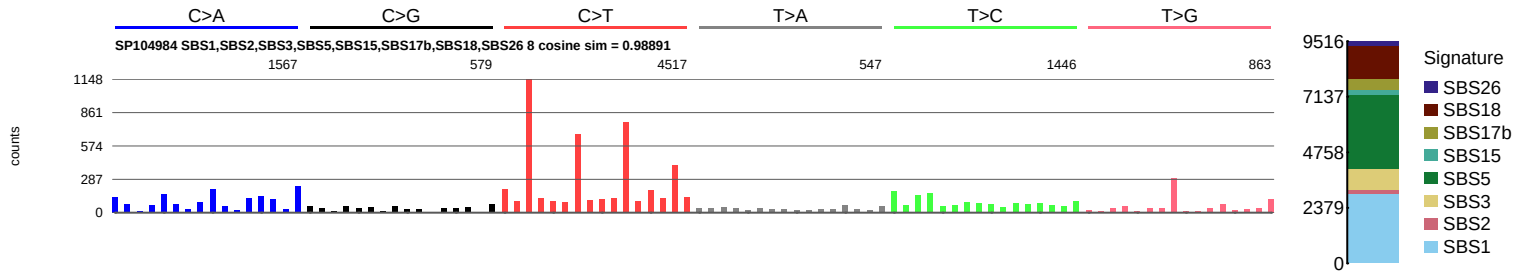
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98892
1566
504
4462
527
1614
848
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>T
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T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26 8 cosine sim = 0.98891
1567
579
4517
547
1446
863
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
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SBS17b
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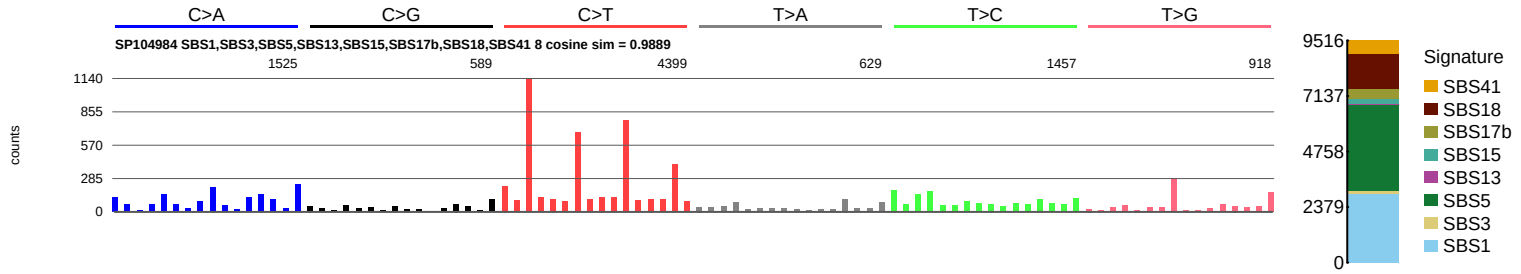
SBS3
SBS2
SBS1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C>T
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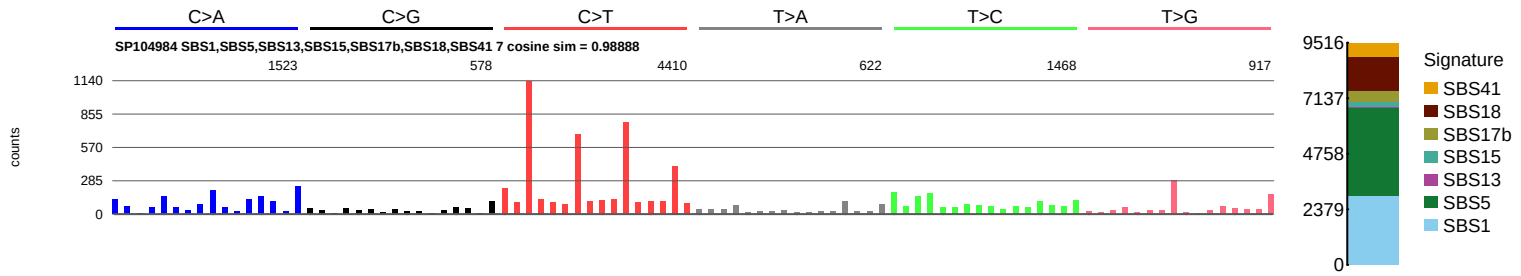
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.9889
1525
589
4399
629
1457
918
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
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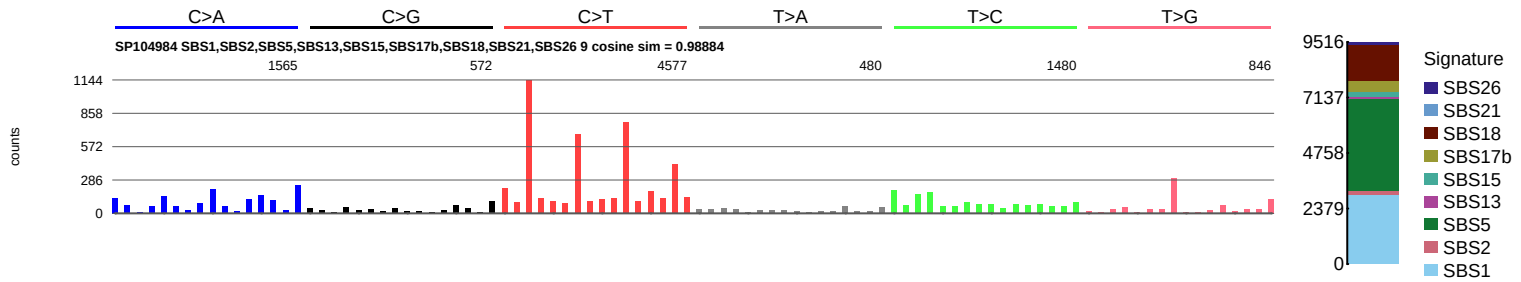
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 7 cosine sim = 0.98888
1523
578
4410
622
1468
917
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1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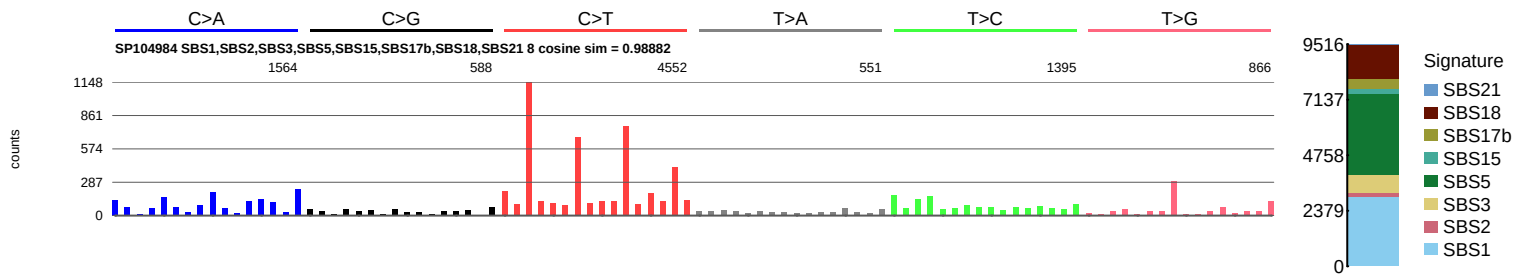
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98884
1565
572
4577
480
1480
846
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98882
1564
588
4552
551
1395
866
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
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SBS17b
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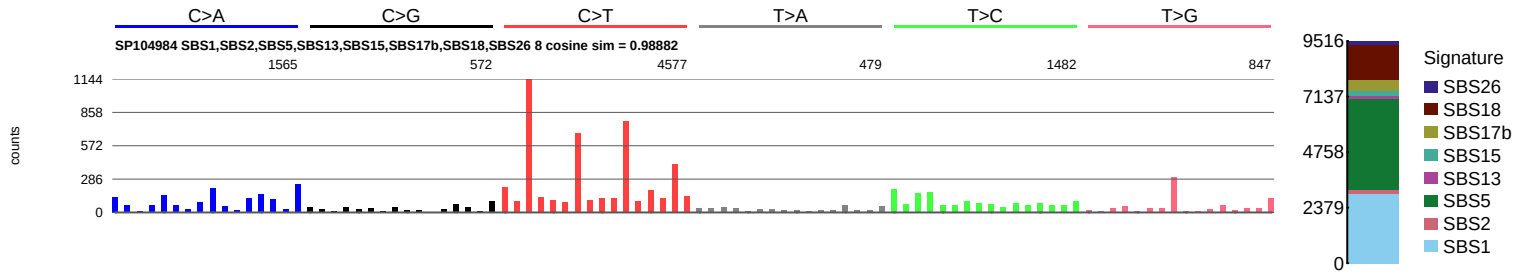
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26 8 cosine sim = 0.98882
1565
572
4577
479
1482
847
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
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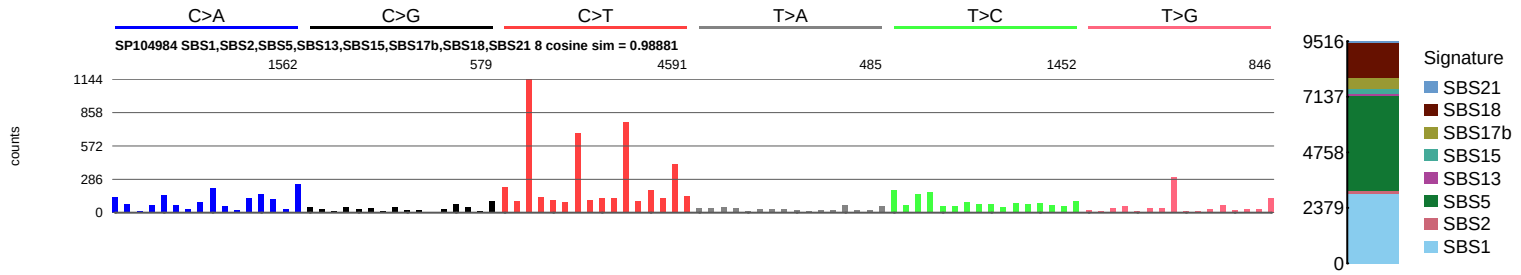
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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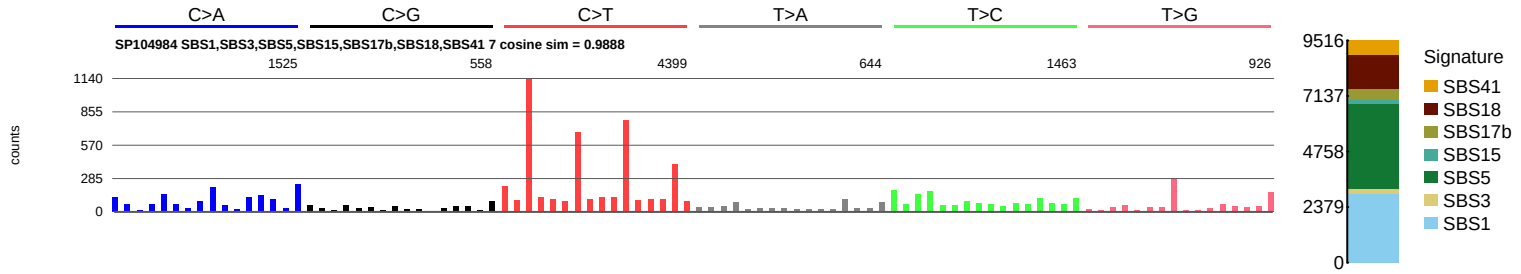
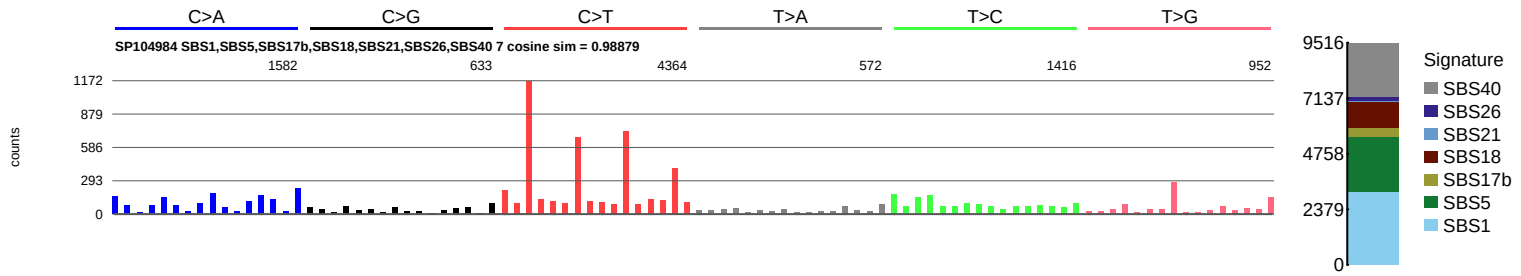
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98881
1562
579
4591
485
1452
846
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 7 cosine sim = 0.9888
1525
558
4399
644
1463
926
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
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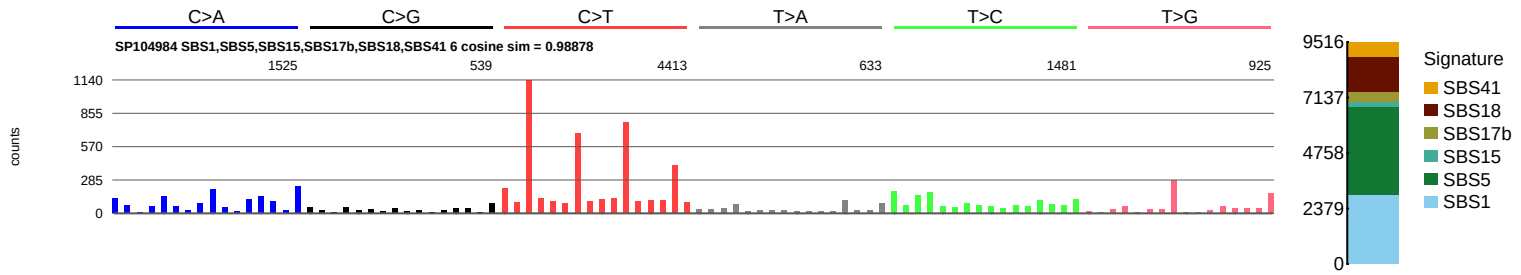
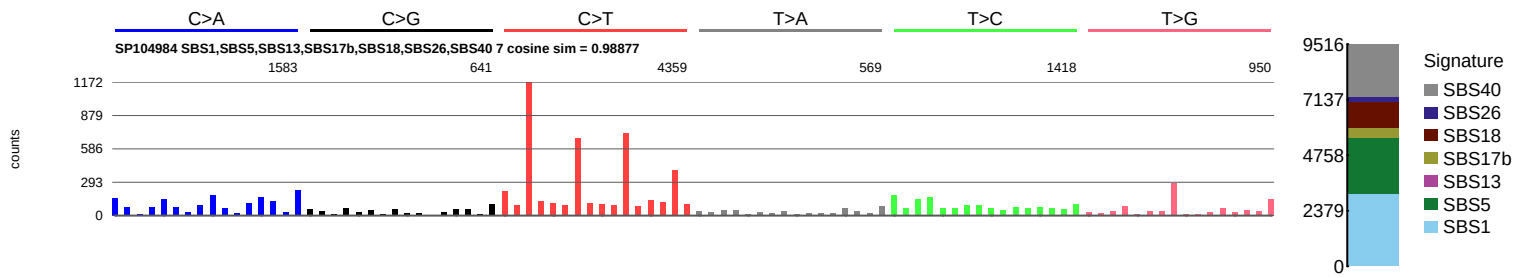
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98879
1582
633
4364
572
1416
952
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS41 6 cosine sim = 0.98878
1525
539
4413
633
1481
925
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS1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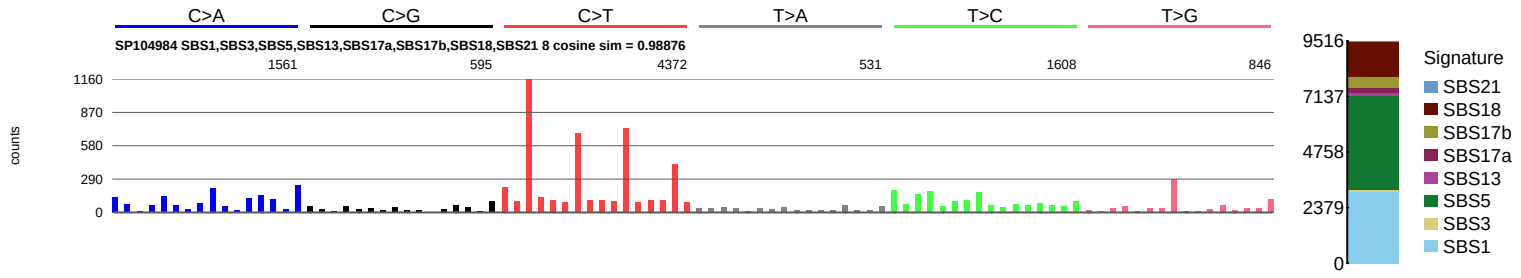
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98877
1583
641
4359
569
1418
950
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS1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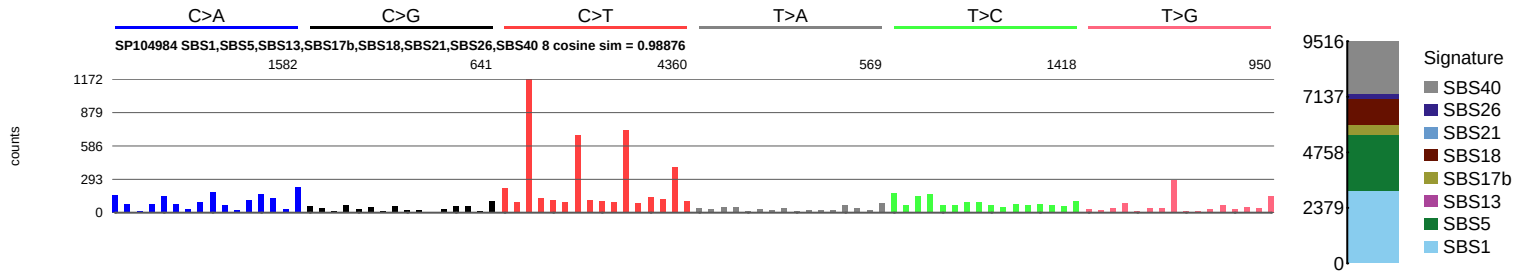
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98876
1561
595
4372
531
1608
846
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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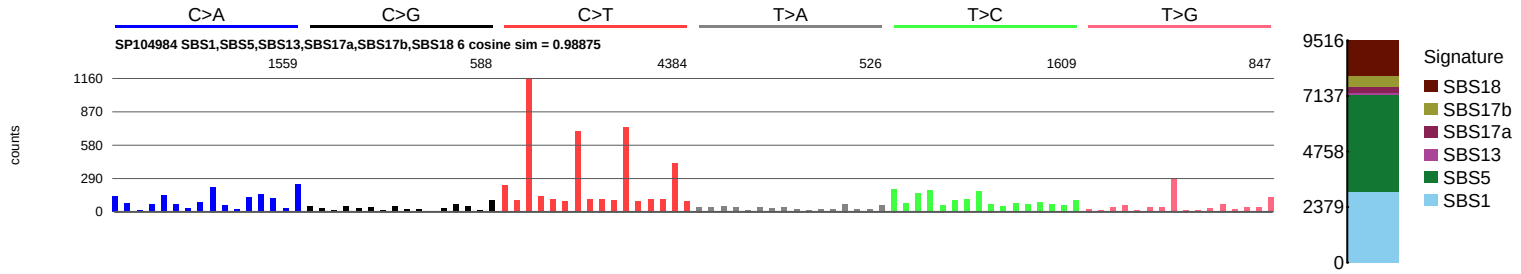
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98876
1582
641
4360
569
1418
950
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1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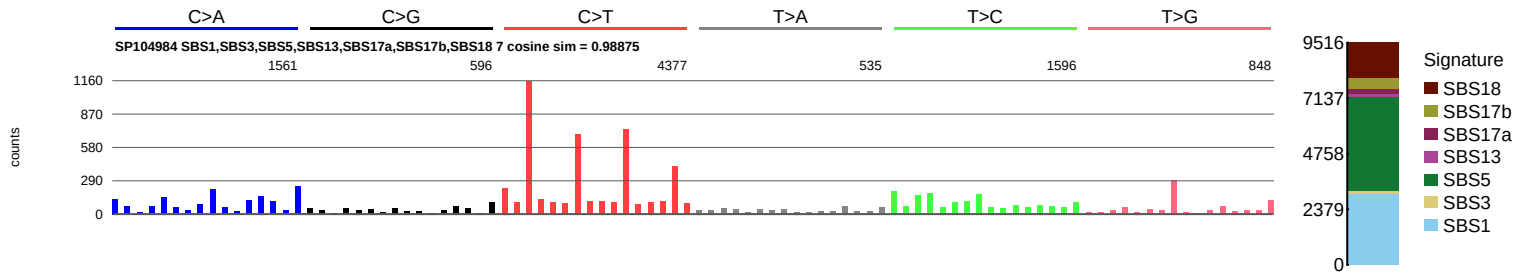
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98875
1559
588
4384
526
1609
847
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98875
1561
596
4377
535
1596
848
counts
1160
870
580
290
0
9516
7137
4758
2379
0
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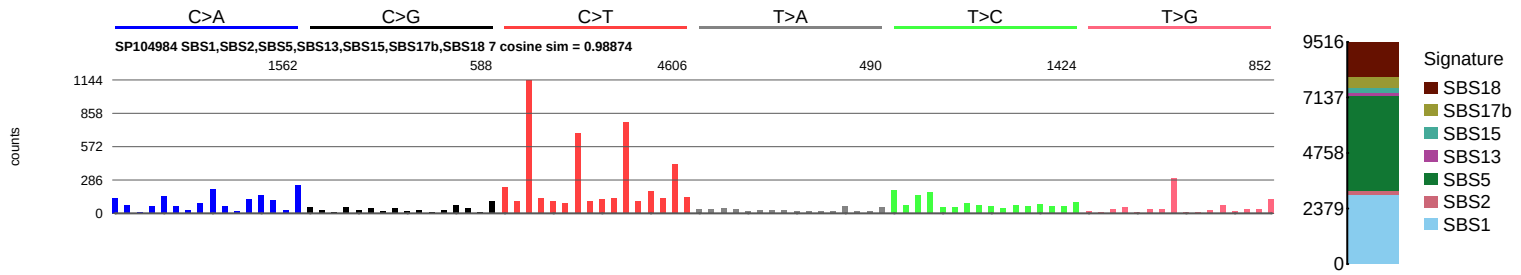
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18 7 cosine sim = 0.98874
1562
588
4606
490
1424
852
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
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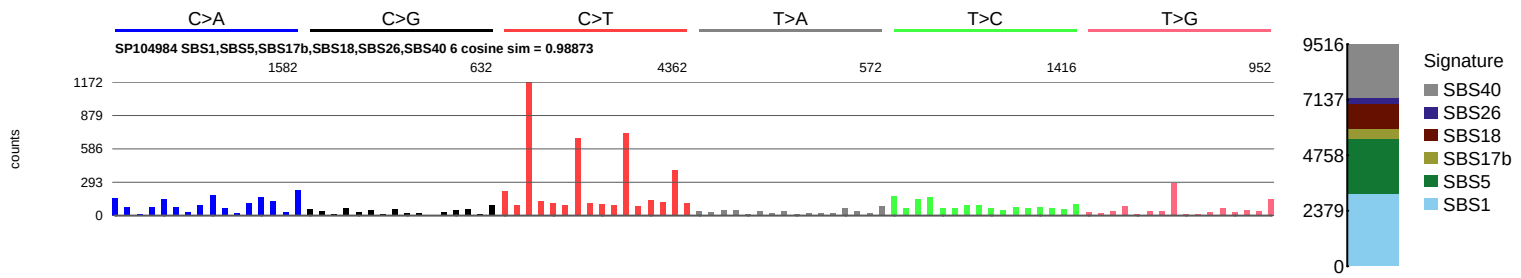
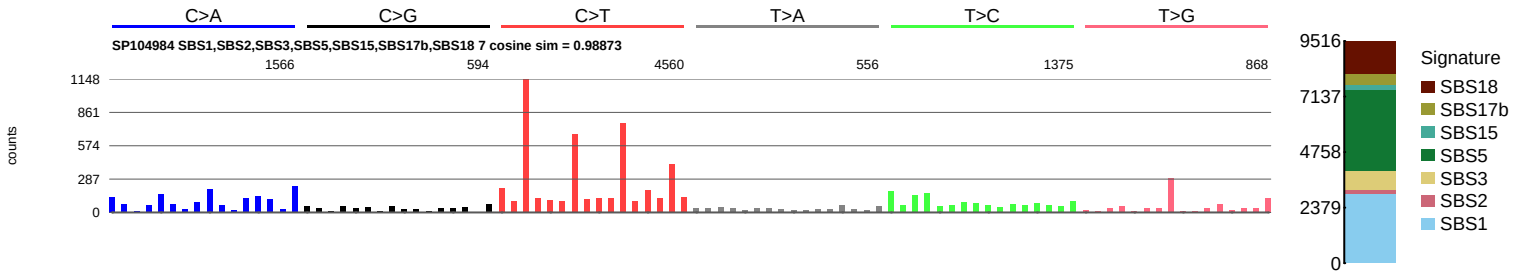
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98873
1582
632
4362
572
1416
952
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18 7 cosine sim = 0.98873
1566
594
4560
556
1375
868
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
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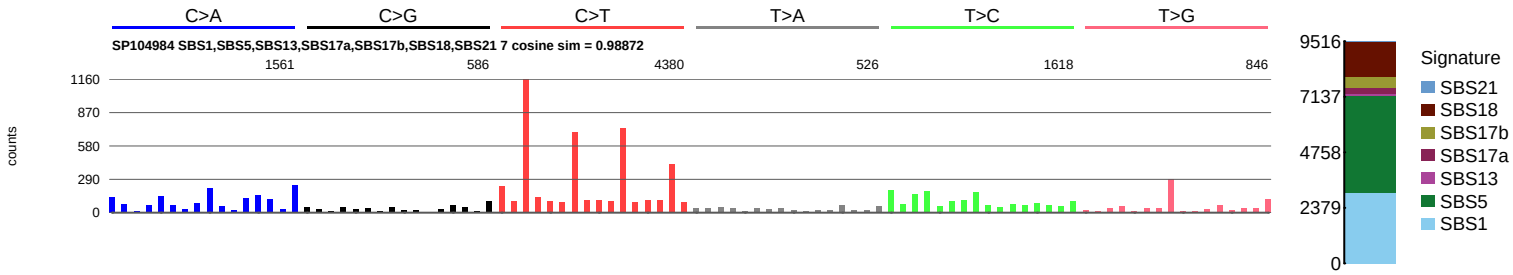
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98872
1561
586
4380
526
1618
846
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
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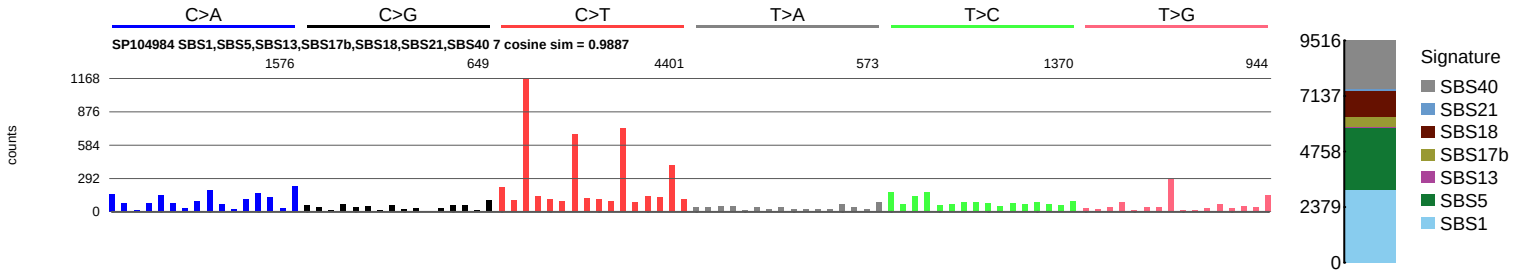
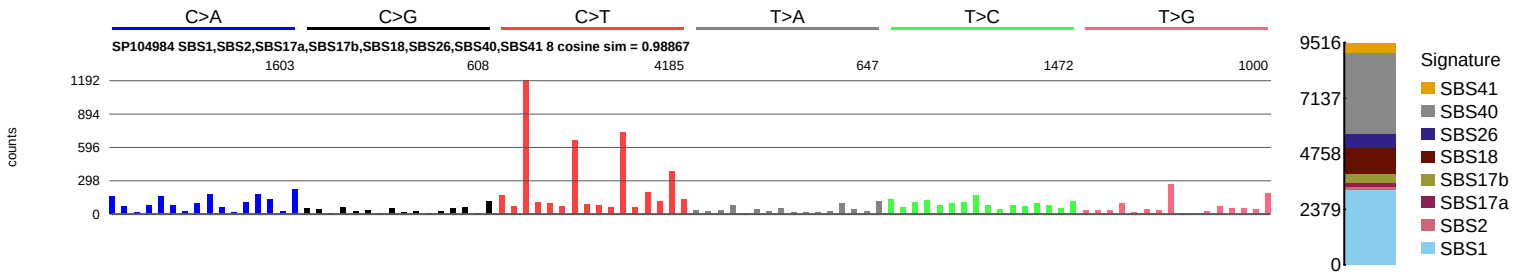
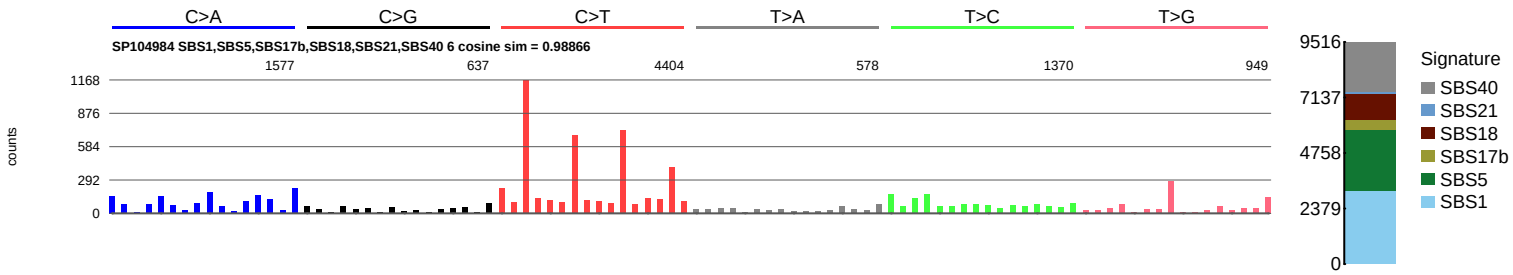
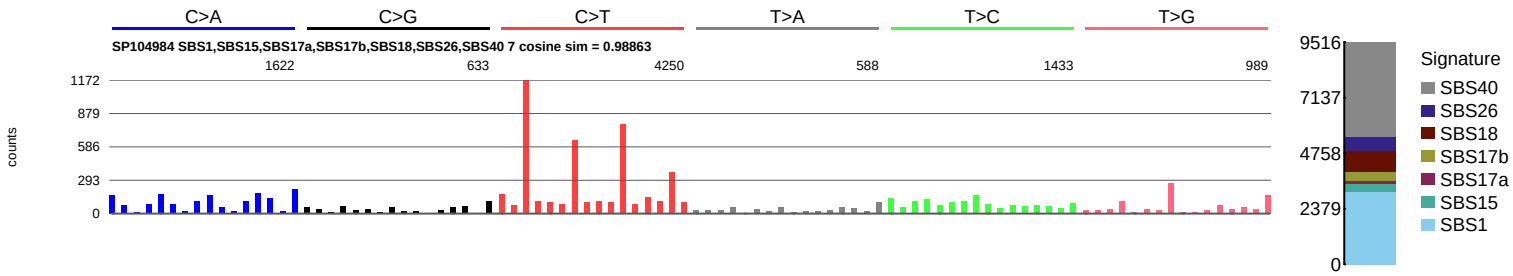
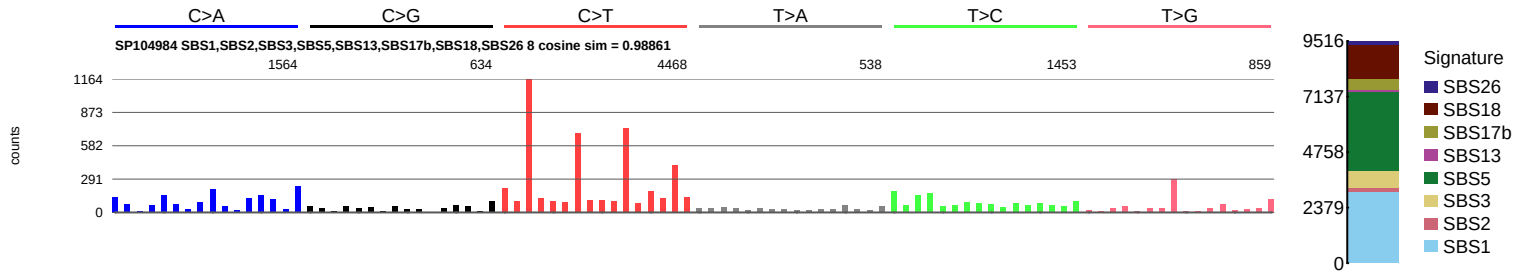
SBS5
SBS1

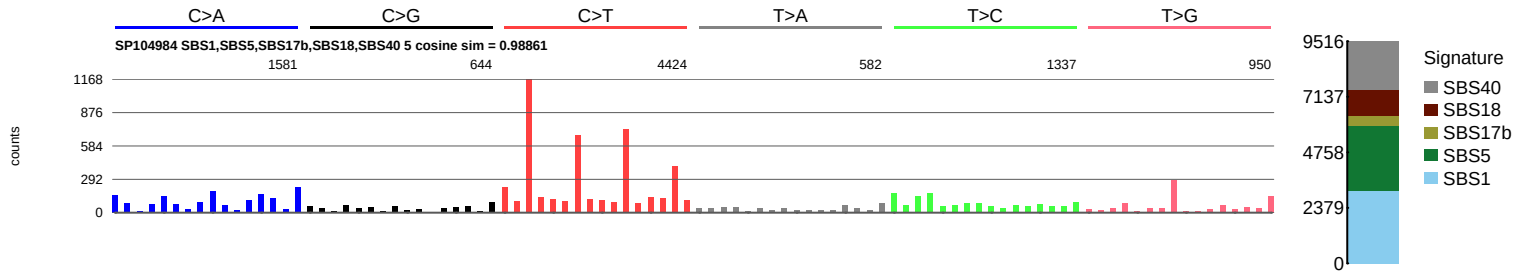

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.9887
1576
649
4401
573
1370
944
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1

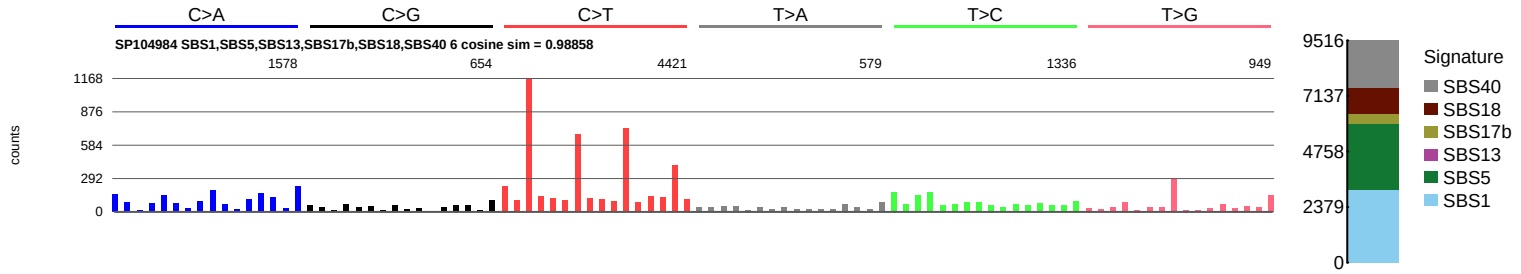

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98867
1603
608
4185
647
1472
1000
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

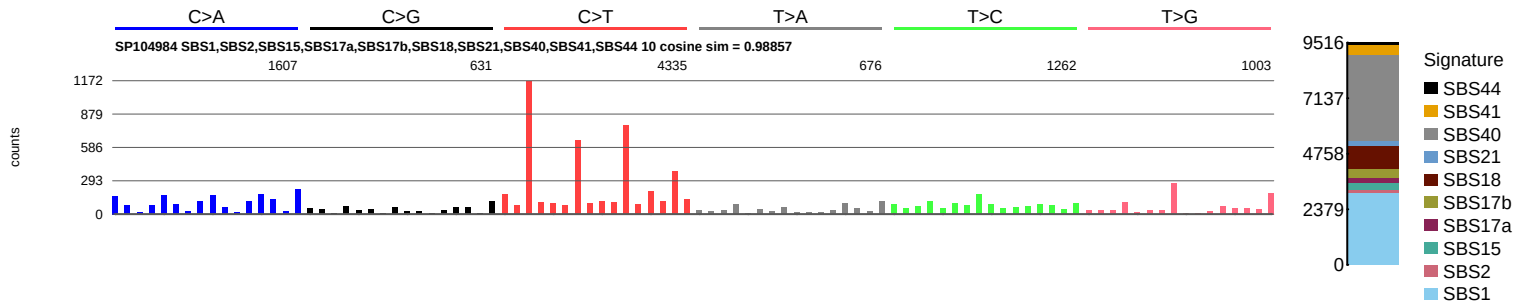

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98866
1577
637
4404
578
1370
949
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1

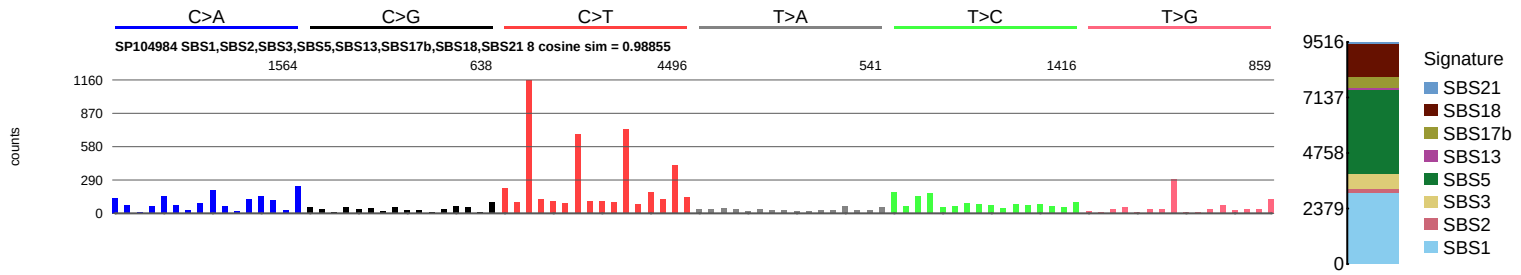

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98863
1622
633
4250
588
1433
989
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26 8 cosine sim = 0.98861
1564
634
4468
538
1453
859
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS18,SBS40 5 cosine sim = 0.98861
1581
644
4424
582
1337
950
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS40 6 cosine sim = 0.98858
1578
654
4421
579
1336
949
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98857
1607
631
4335
676
1262
1003
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.98855
1564
638
4496
541
1416
859
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
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SBS17b
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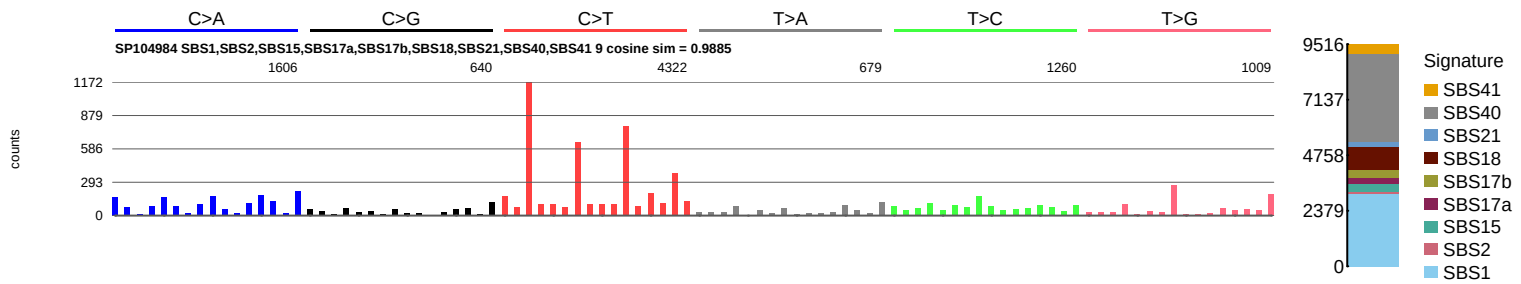
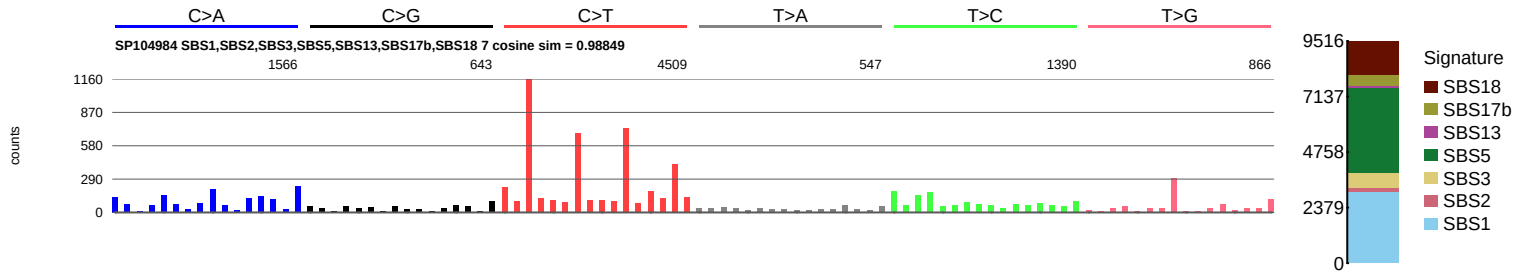
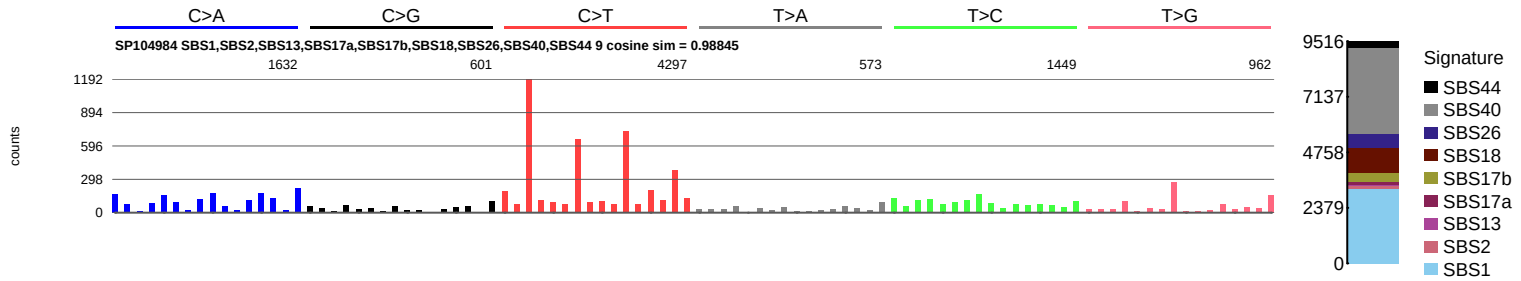
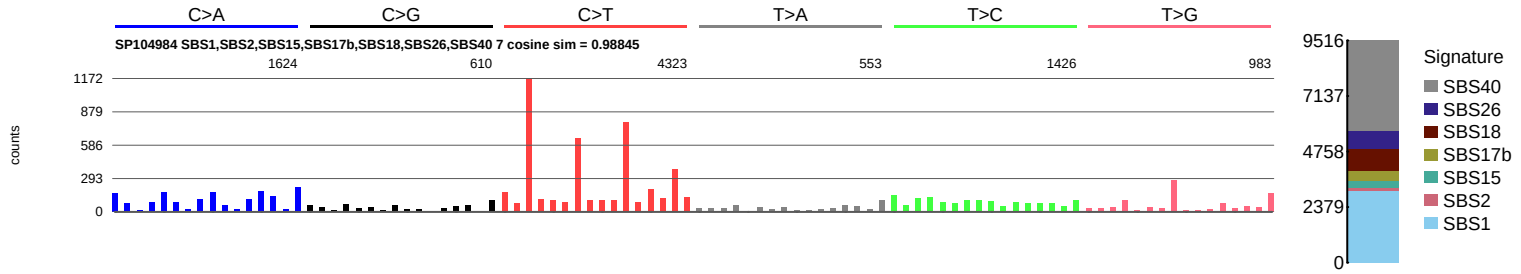
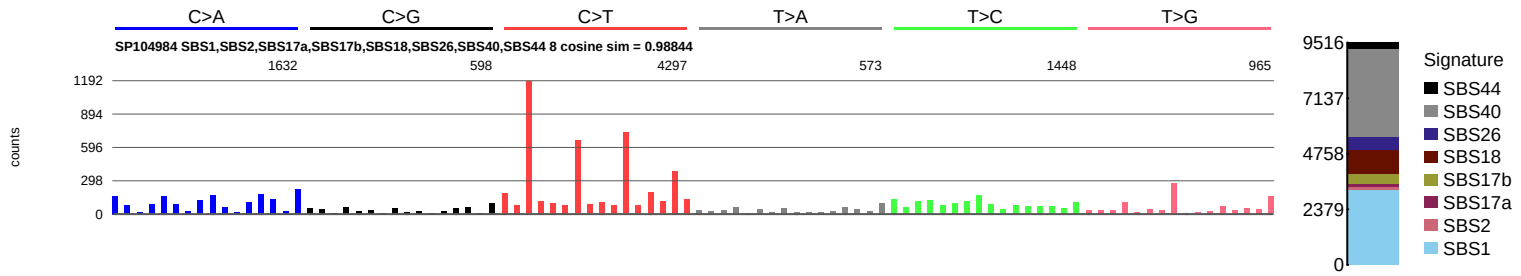
SBS3
SBS2
SBS1

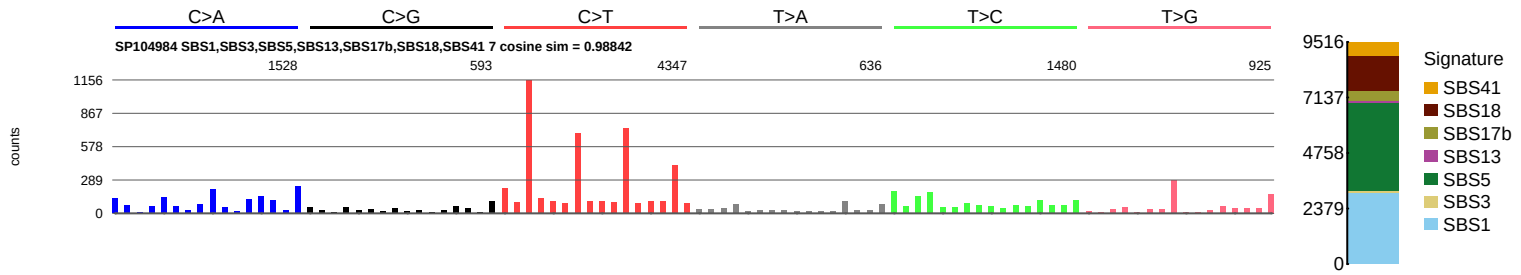

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.9885
1606
640
4322
679
1260
1009
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

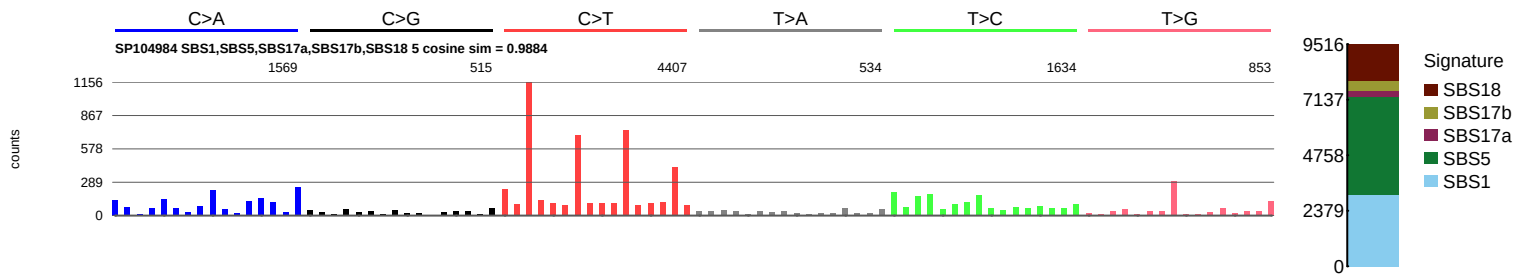

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.98849
1566
643
4509
547
1390
866
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

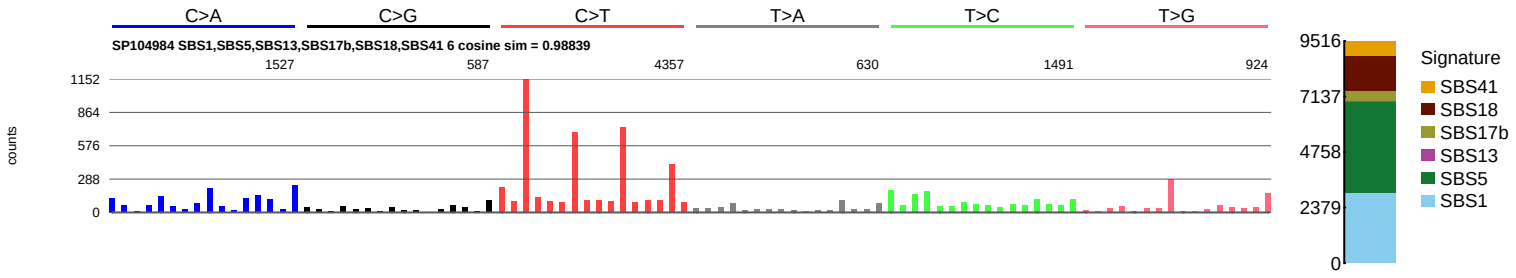

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98845
1632
601
4297
573
1449
962
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

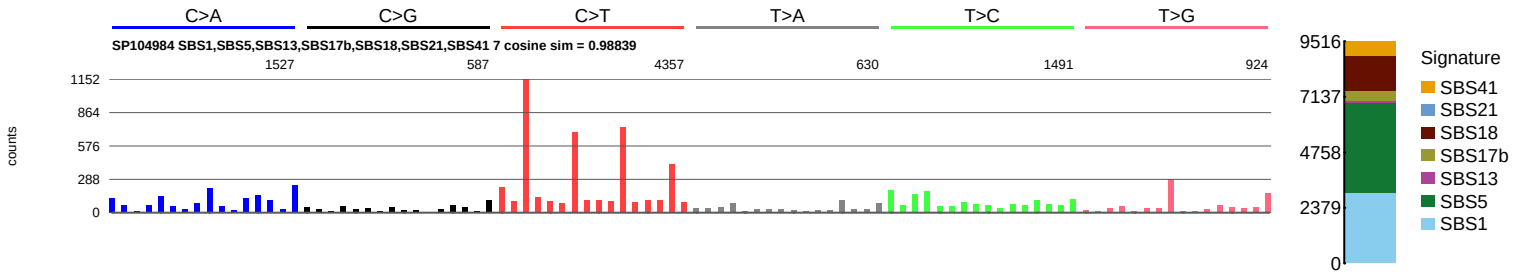

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98845
1624
610
4323
553
1426
983
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1

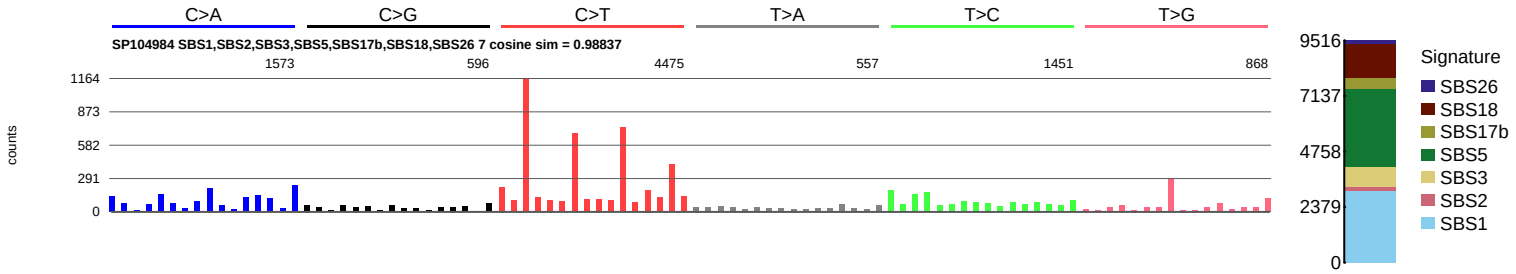

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98844
1632
598
4297
573
1448
965
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.98842
1528
593
4347
636
1480
925
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.9884
1569
515
4407
534
1634
853
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS41 6 cosine sim = 0.98839
1527
587
4357
630
1491
924
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98839
1527
587
4357
630
1491
924
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1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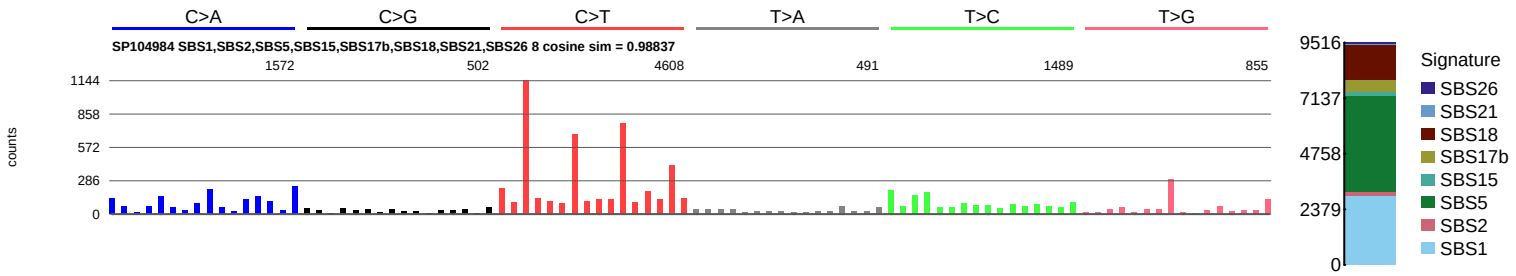
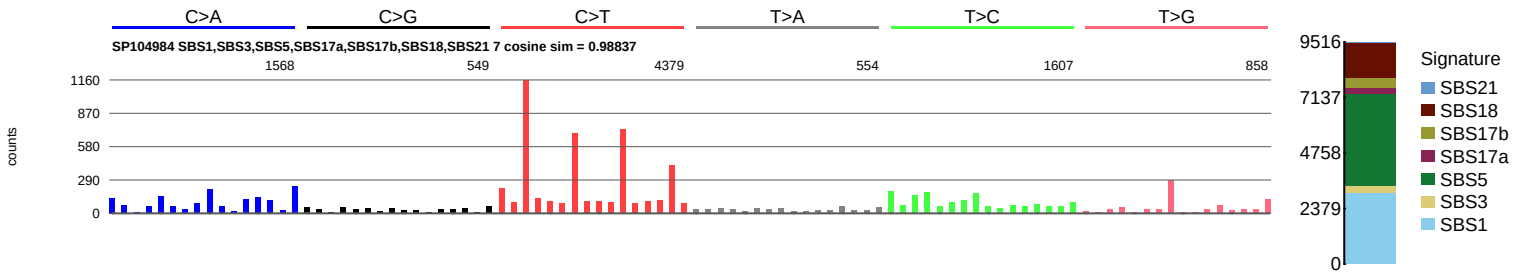
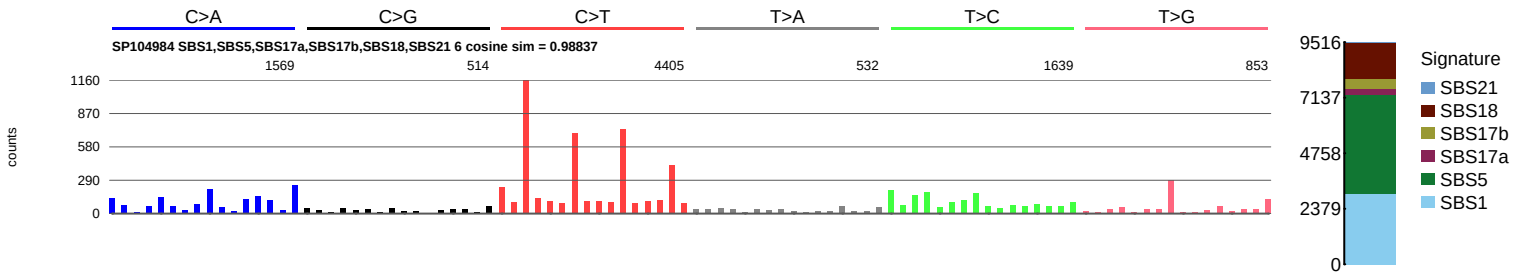
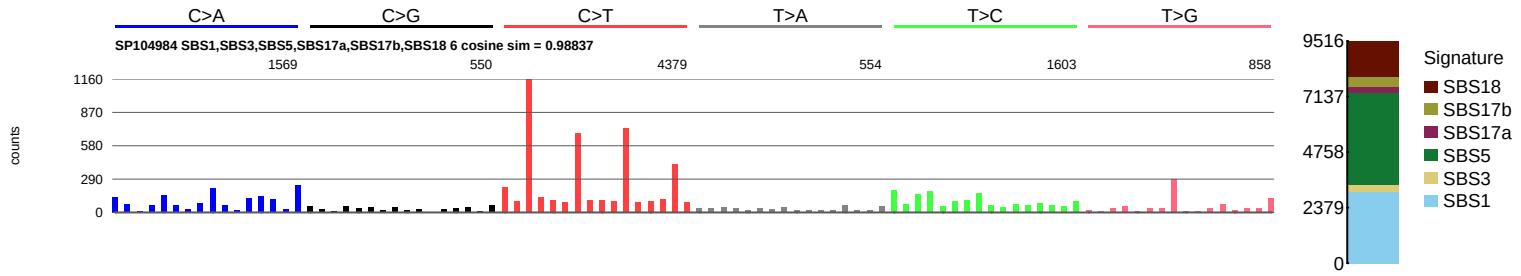
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26 7 cosine sim = 0.98837
1573
596
4475
557
1451
868
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98837
1572
502
4608
491
1489
855
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98837
1568
549
4379
554
1607
858
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98837
1569
514
4405
532
1639
853
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98837
1569
550
4379
554
1603
858
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
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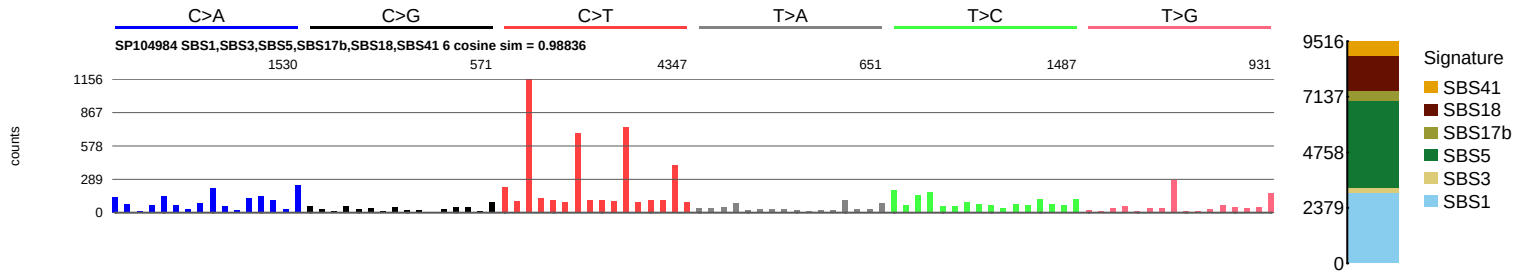
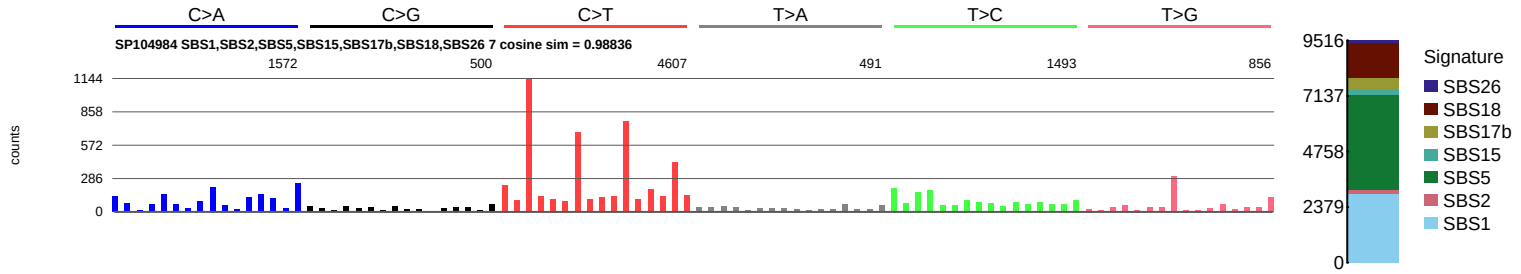
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS17b,SBS18,SBS41 6 cosine sim = 0.98836
1530
571
4347
651
1487
931
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26 7 cosine sim = 0.98836
1572
500
4607
491
1493
856
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
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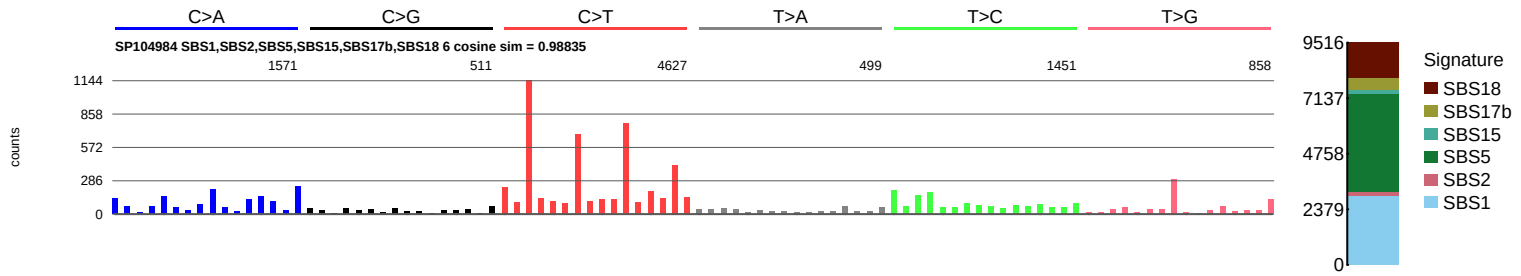
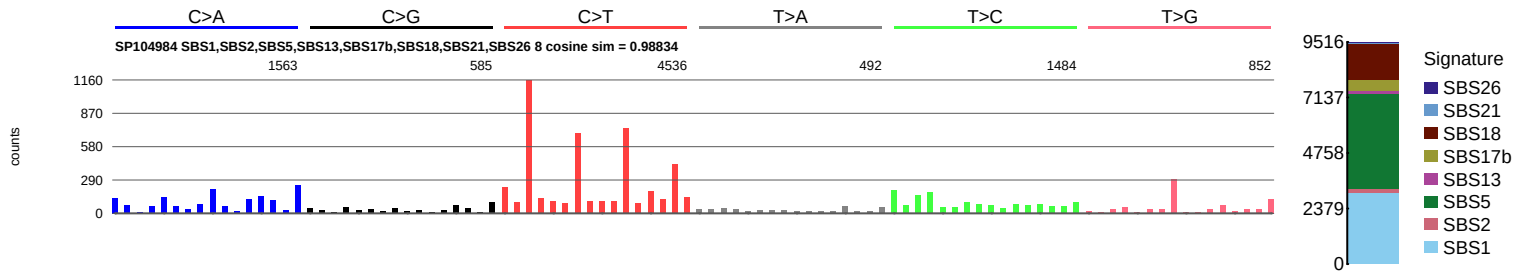
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18 6 cosine sim = 0.98835
1571
511
4627
499
1451
858
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98834
1563
585
4536
492
1484
852
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS26
SBS21
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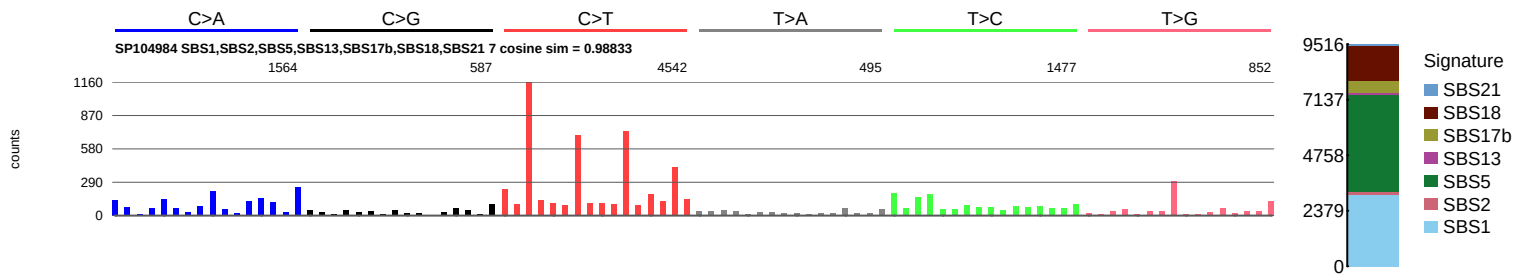
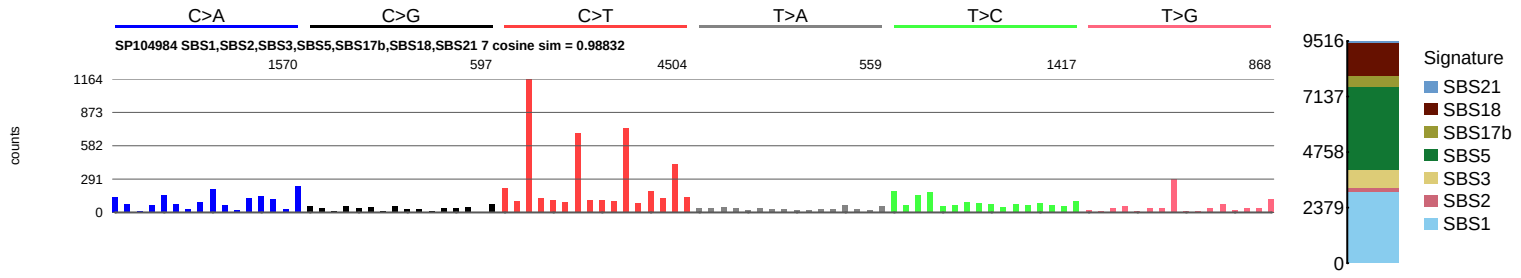
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21 7 cosine sim = 0.98833
1564
587
4542
495
1477
852
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
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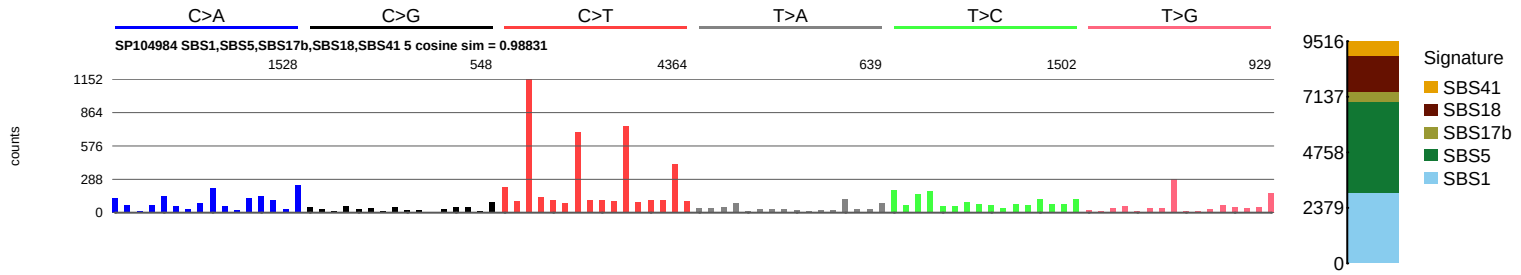
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21 7 cosine sim = 0.98832
1570
597
4504
559
1417
868
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>G
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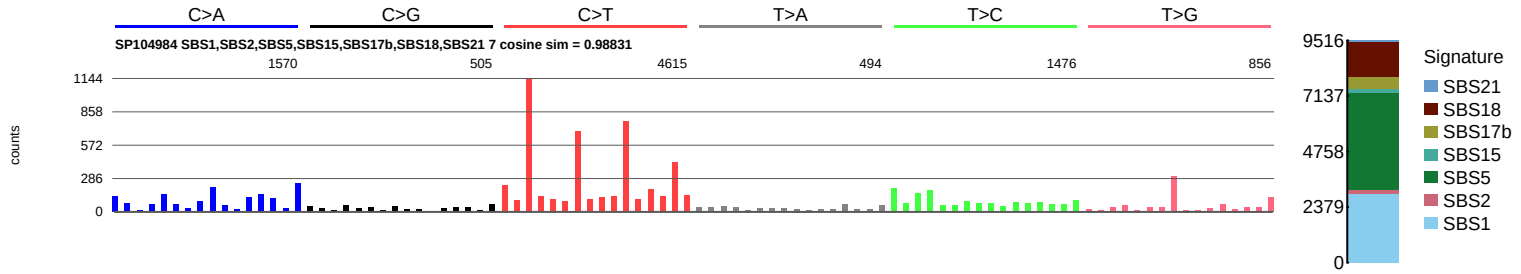
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS18,SBS41 5 cosine sim = 0.98831
1528
548
4364
639
1502
929
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS1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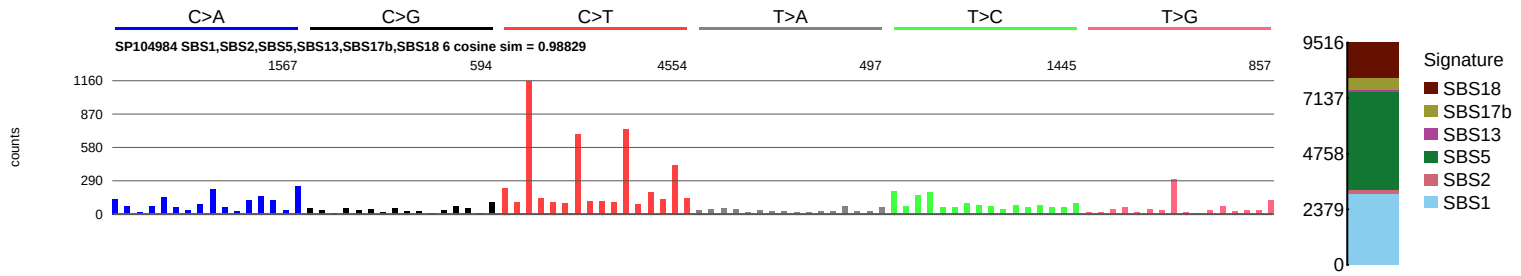
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21 7 cosine sim = 0.98831
1570
505
4615
494
1476
856
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
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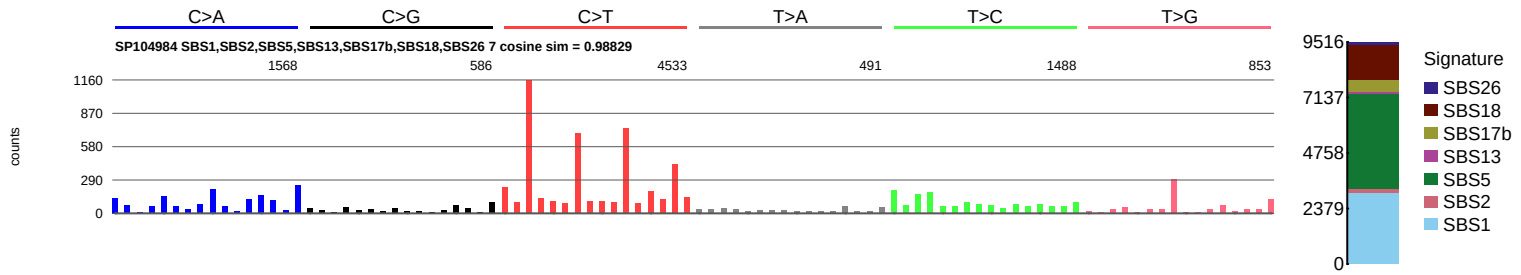
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18 6 cosine sim = 0.98829
1567
594
4554
497
1445
857
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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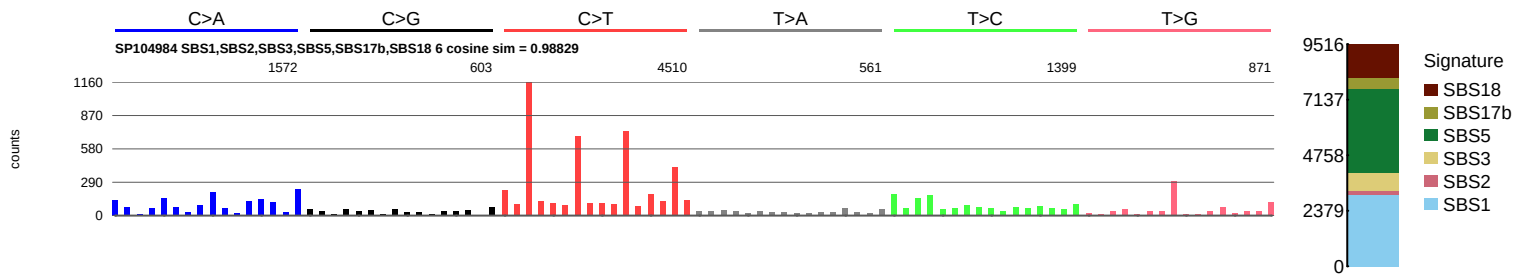
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26 7 cosine sim = 0.98829
1568
586
4533
491
1488
853
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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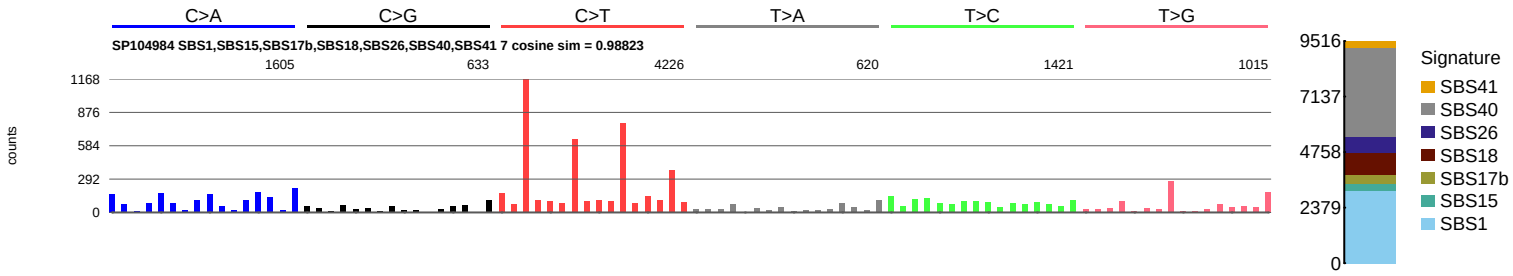
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18 6 cosine sim = 0.98829
1572
603
4510
561
1399
871
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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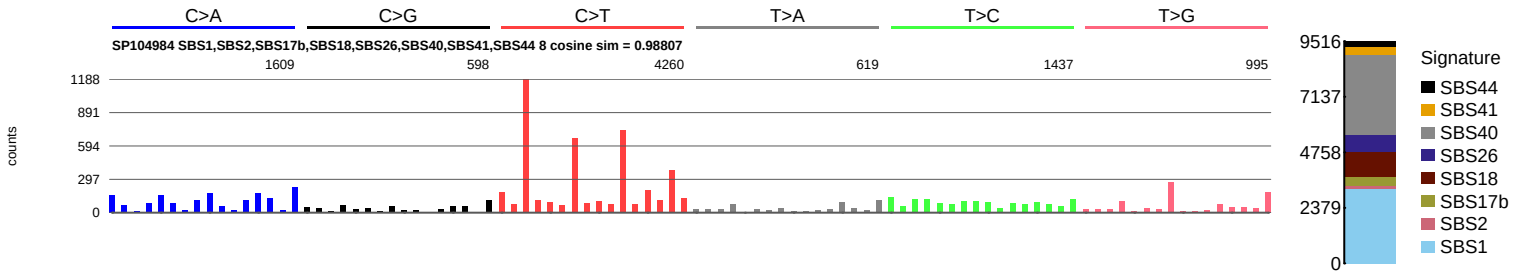
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98823
1605
633
4226
620
1421
1015
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS1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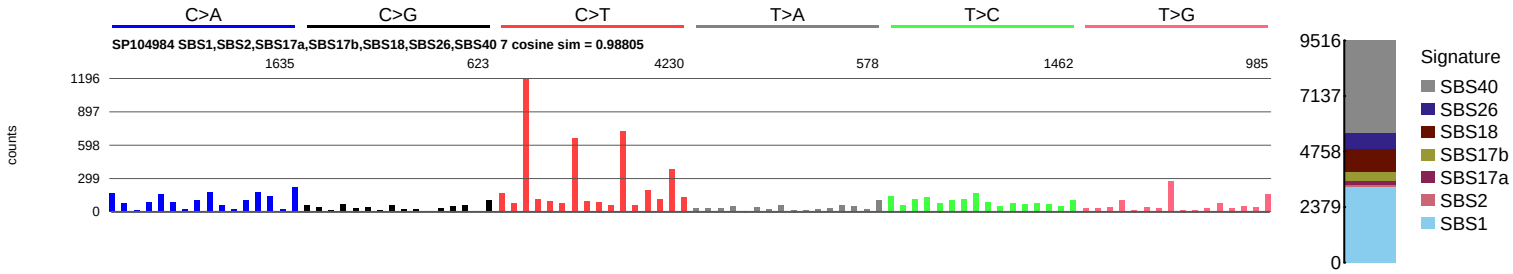
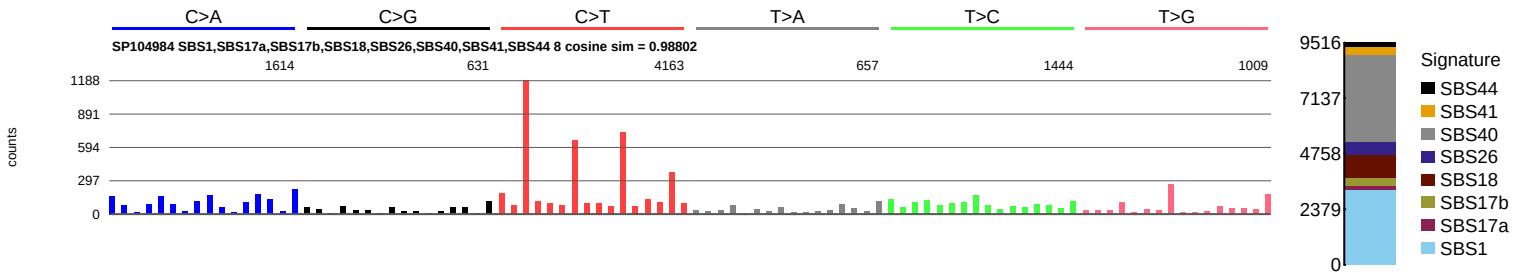
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98807
1609
598
4260
619
1437
995
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98805
1635
623
4230
578
1462
985
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98802
1614
631
4163
657
1444
1009
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
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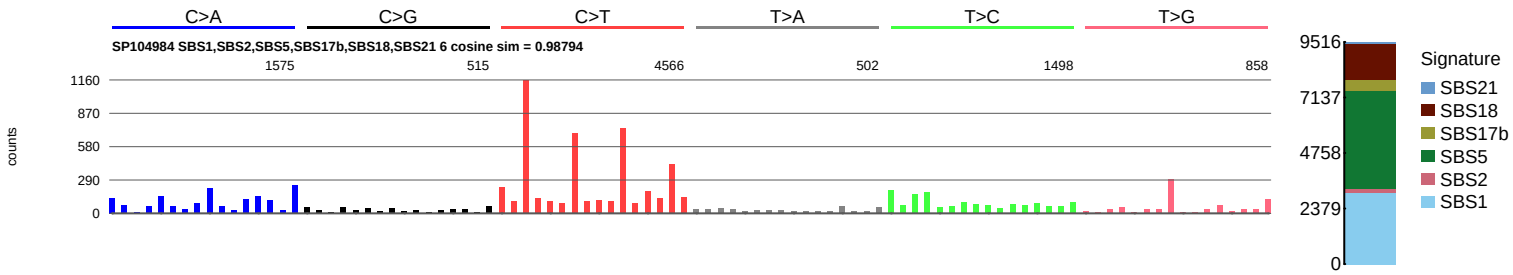
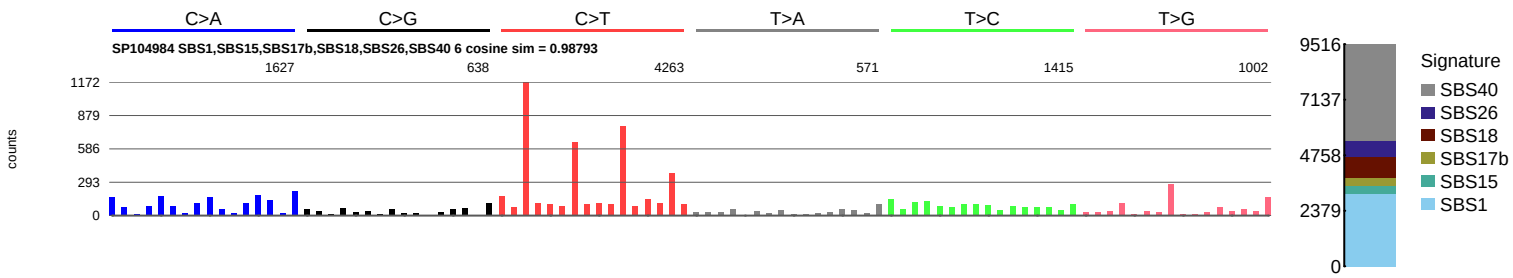
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21 6 cosine sim = 0.98794
1575
515
4566
502
1498
858
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98793
1627
638
4263
571
1415
1002
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
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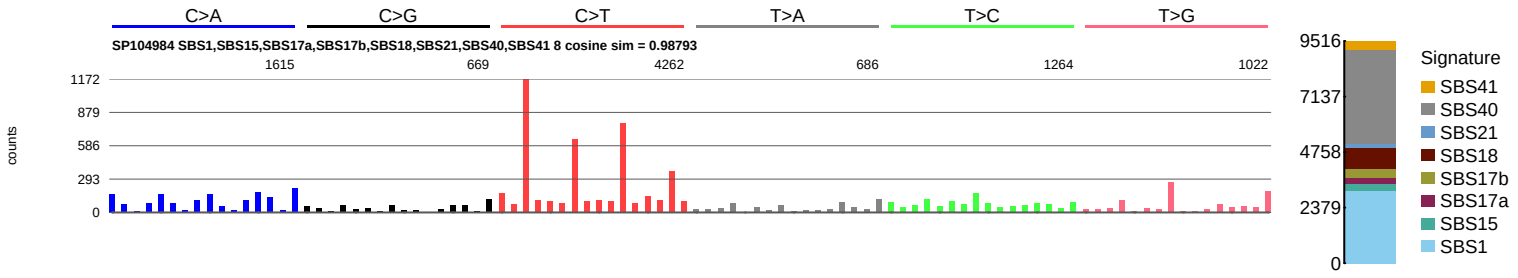
SBS26
SBS18
SBS17b
SBS15
SBS1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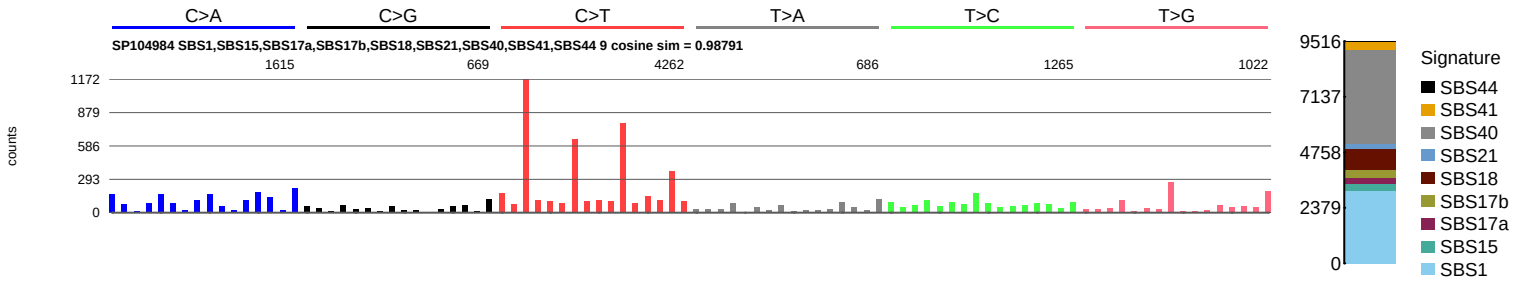
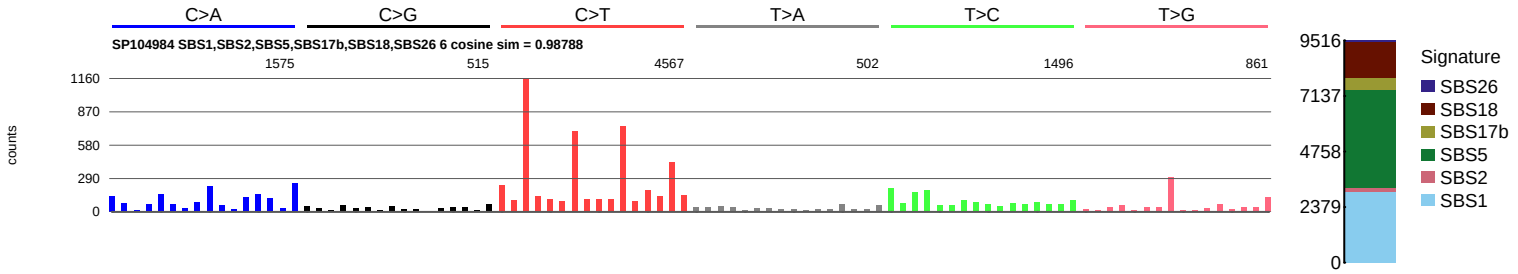
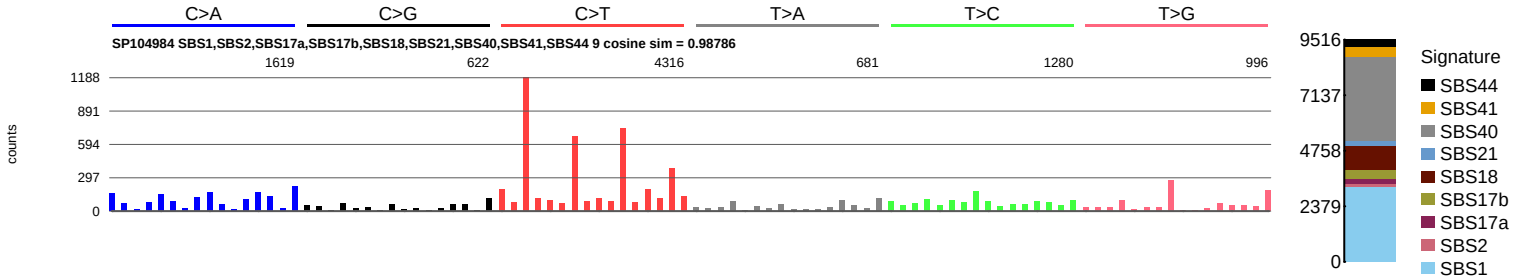
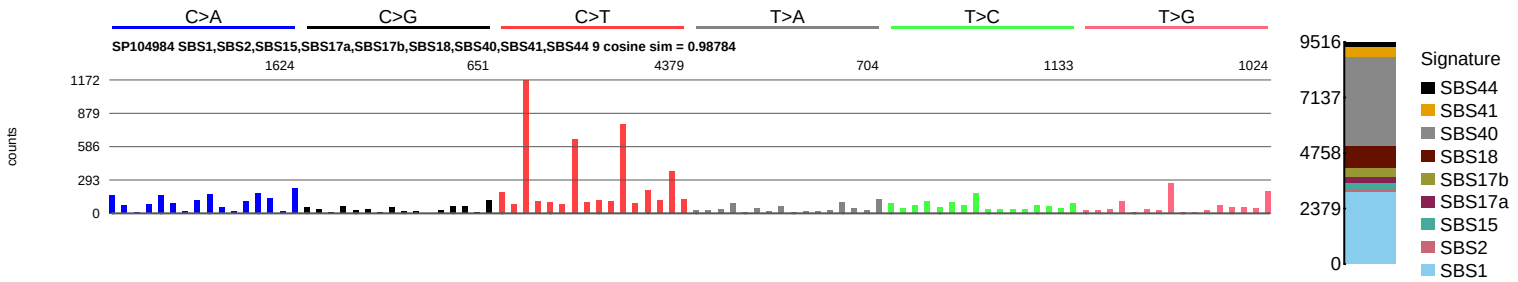
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98793
1615
669
4262
686
1264
1022
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98791
1615
669
4262
686
1265
1022
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26 6 cosine sim = 0.98788
1575
515
4567
502
1496
861
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98786
1619
622
4316
681
1280
996
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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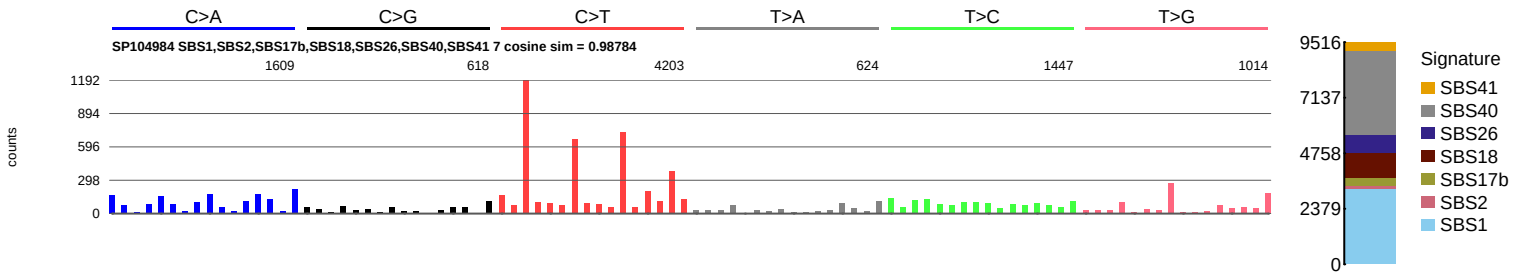
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98784
1624
651
4379
704
1133
1024
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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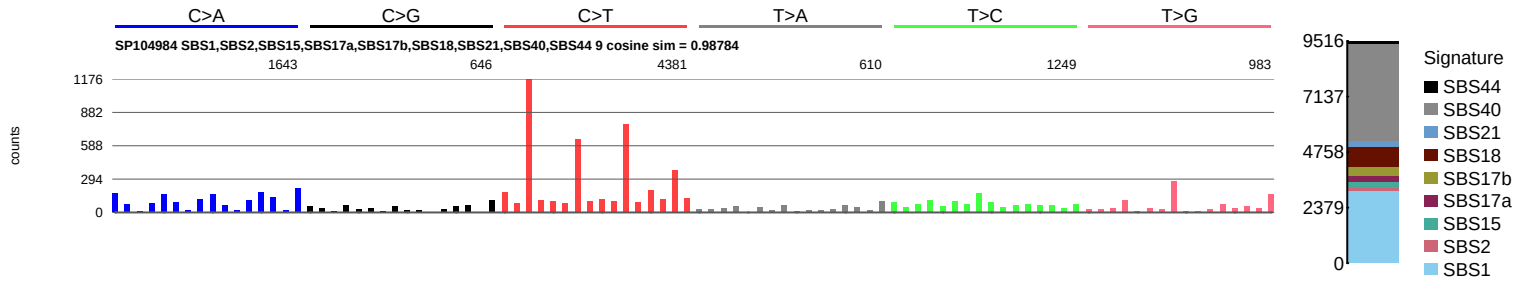
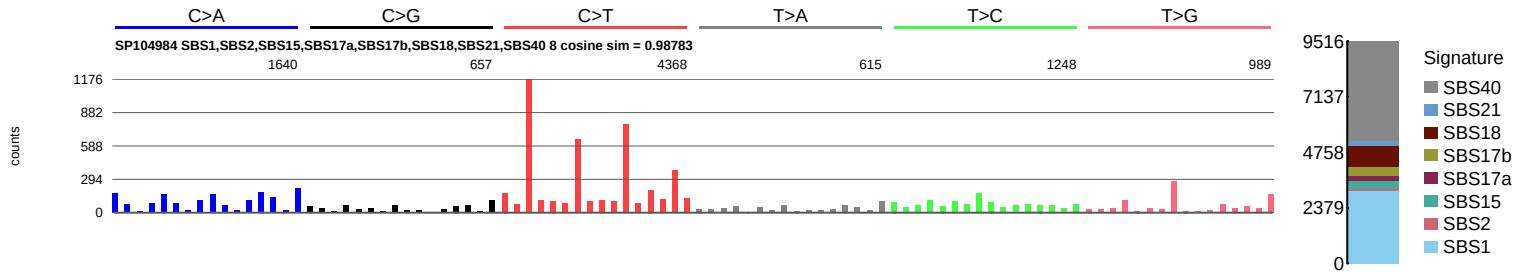
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98784
1609
618
4203
624
1447
1014
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98784
1643
646
4381
610
1249
983
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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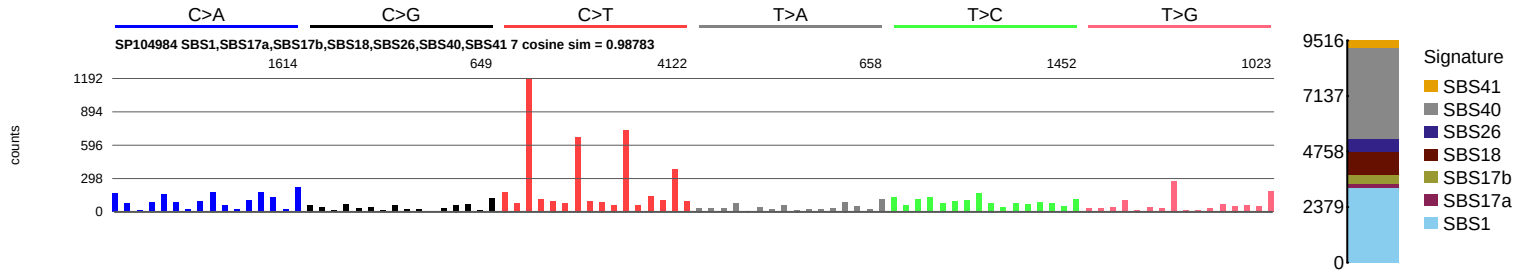
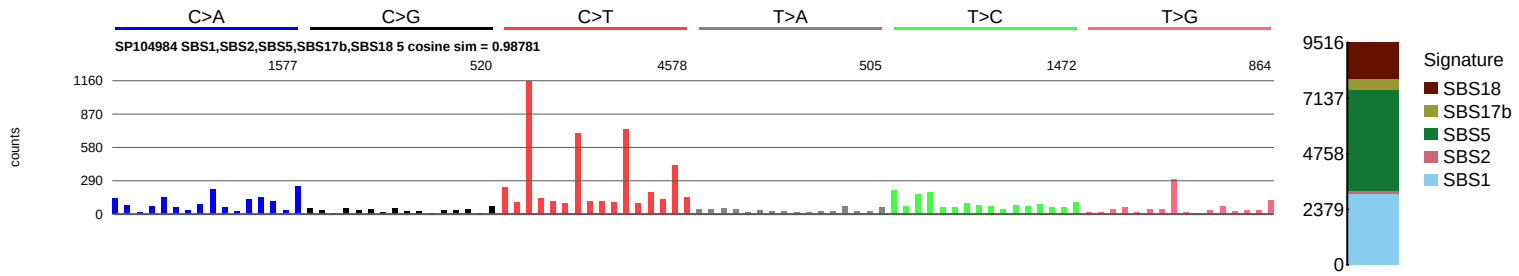
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98783
1640
657
4368
615
1248
989
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98783
1614
649
4122
658
1452
1023
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS18 5 cosine sim = 0.98781
1577
520
4578
505
1472
864
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
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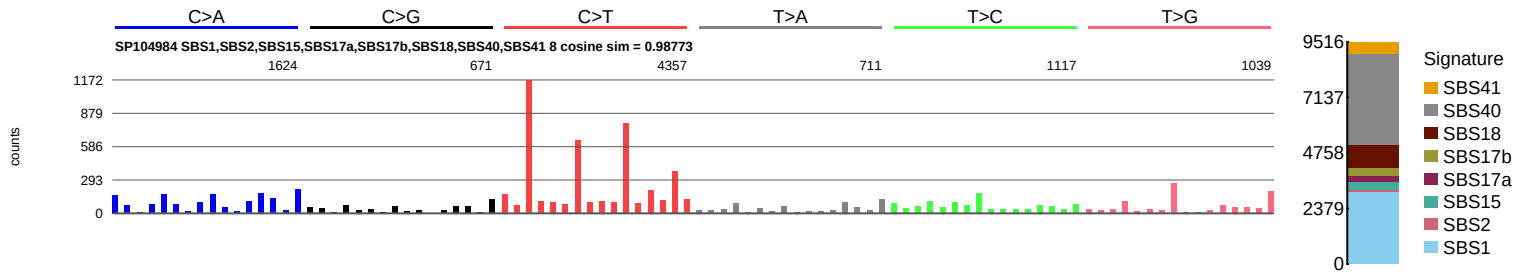
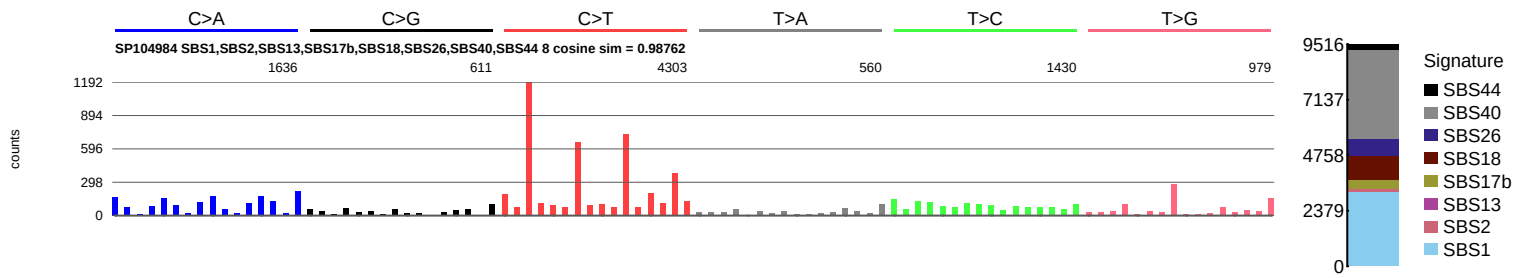
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98773
1624
671
4357
711
1117
1039
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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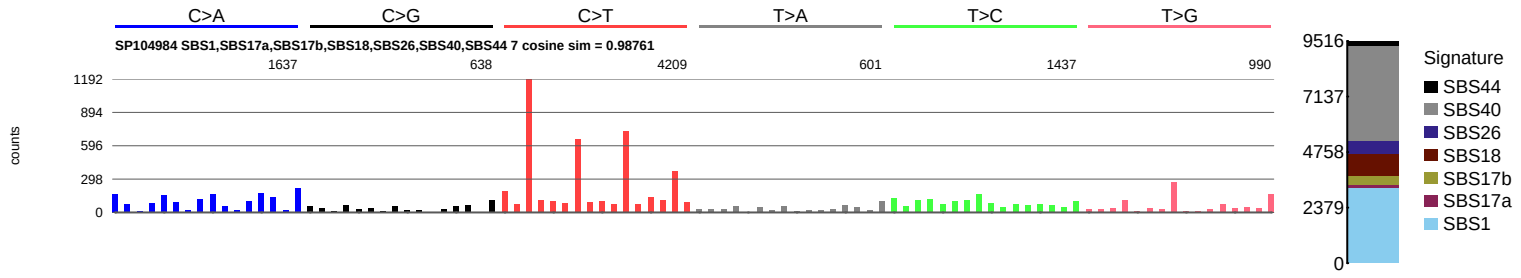
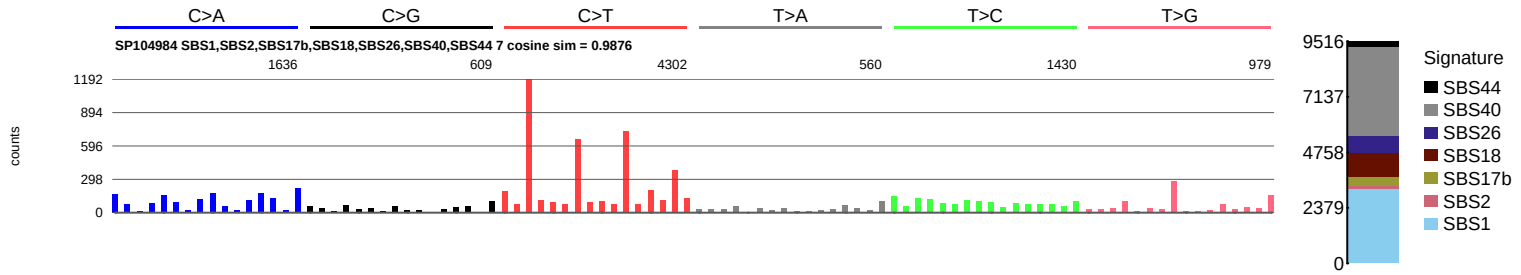
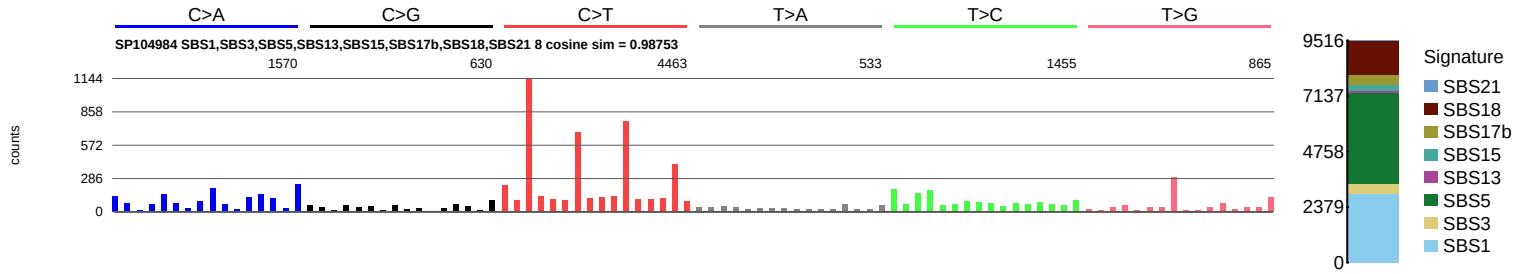
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98762
1636
611
4303
560
1430
979
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98761
1637
638
4209
601
1437
990
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.9876
1636
609
4302
560
1430
979
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1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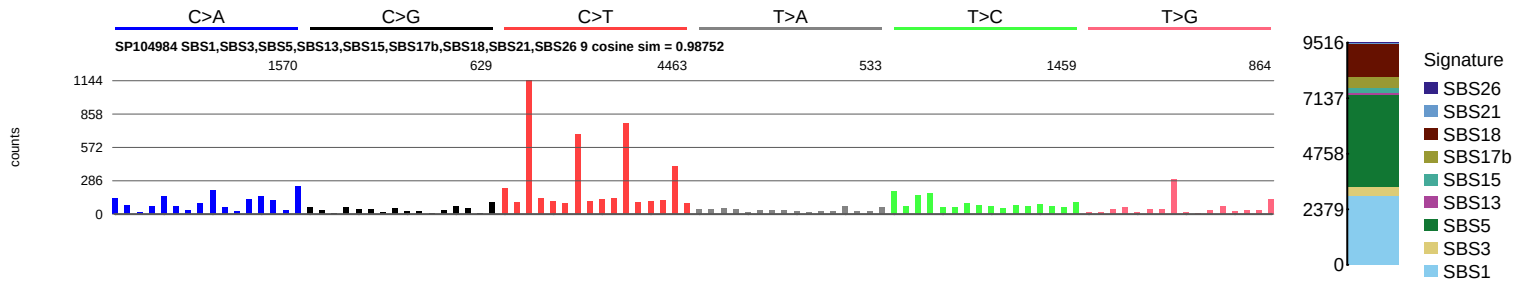
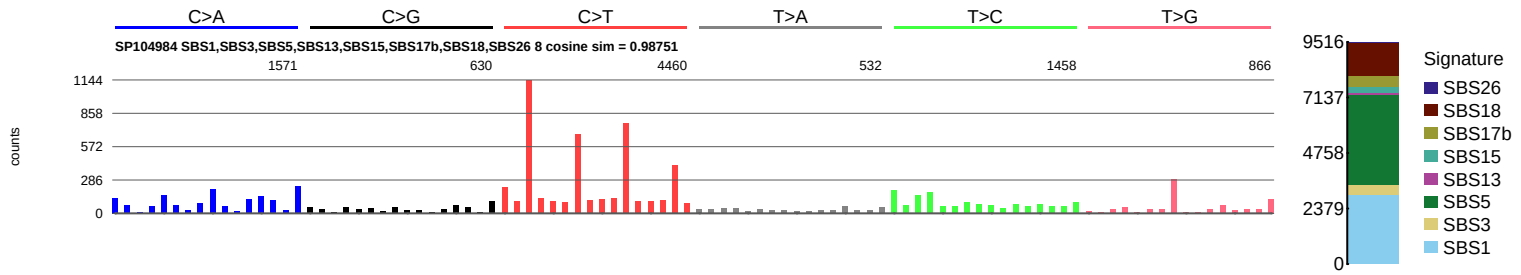
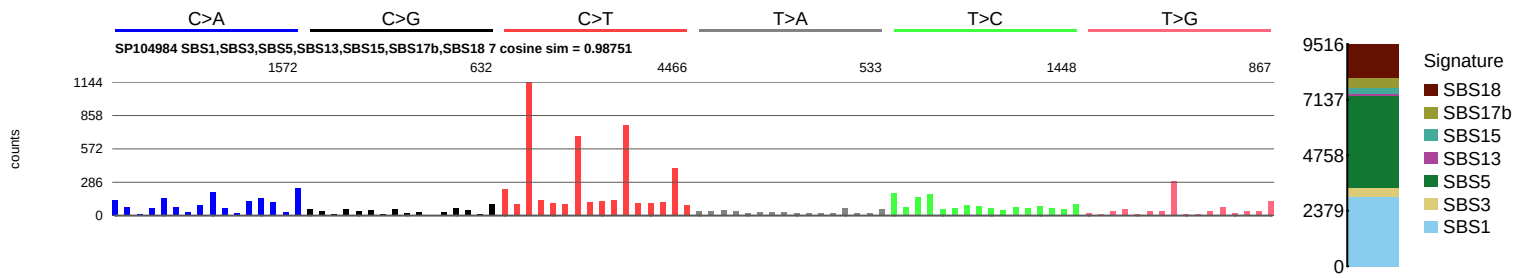
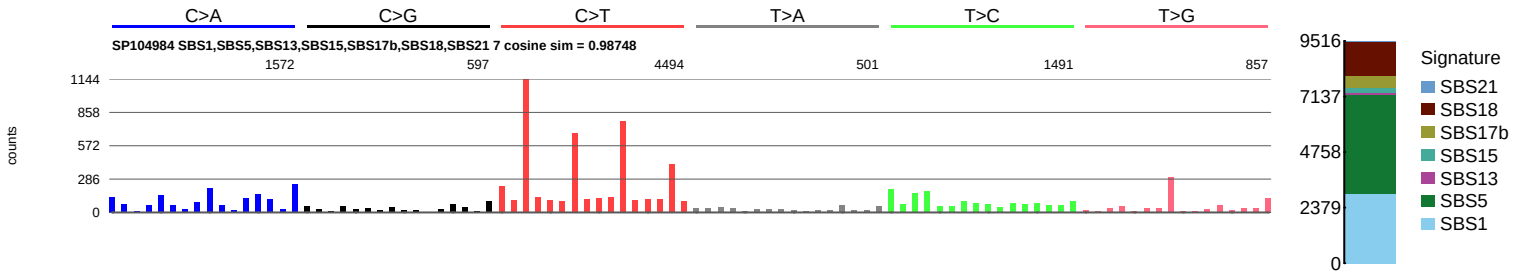
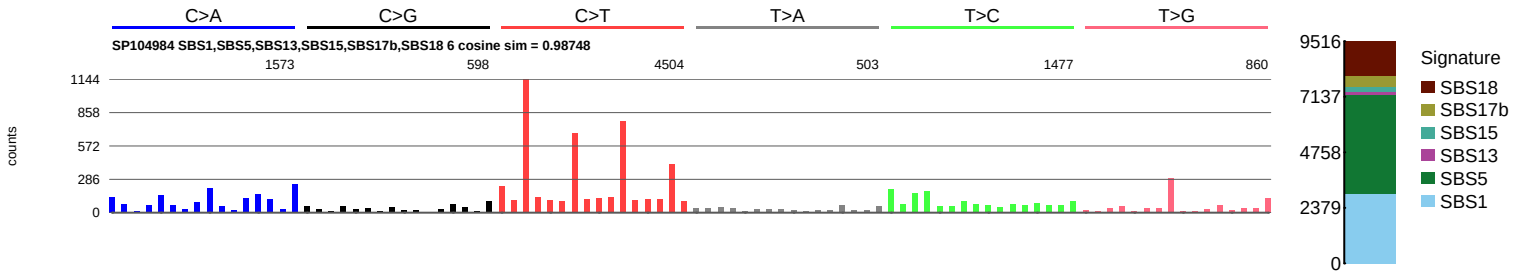
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98753
1570
630
4463
533
1455
865
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1

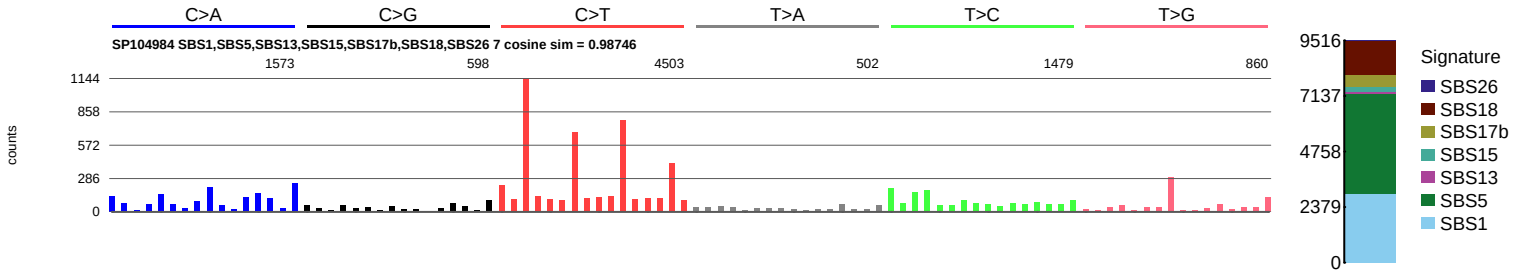

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98752
1570
629
4463
533
1459
864
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1

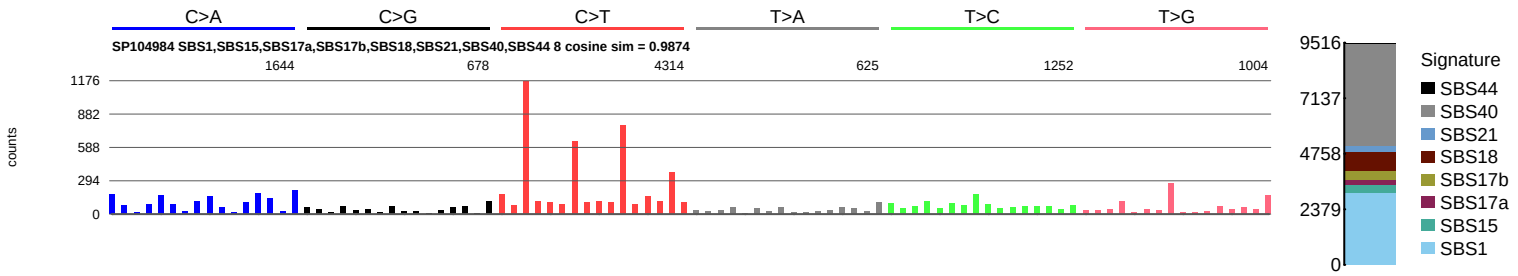

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26 8 cosine sim = 0.98751
1571
630
4460
532
1458
866
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1

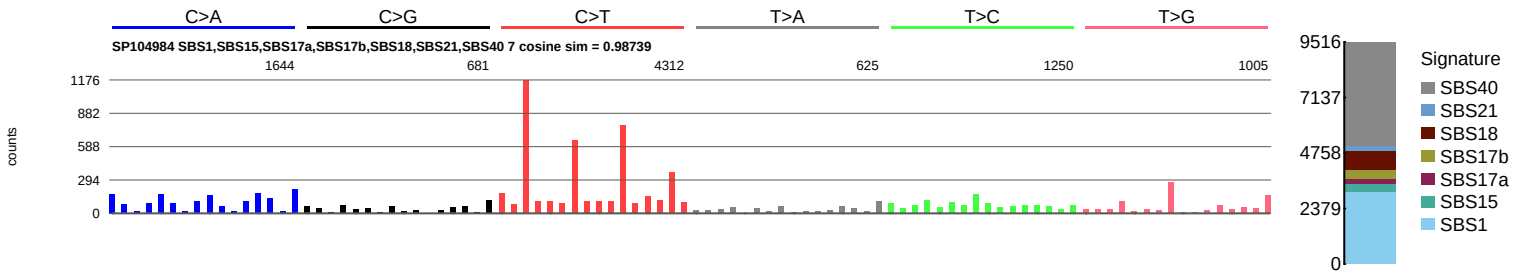

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 7 cosine sim = 0.98751
1572
632
4466
533
1448
867
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS1

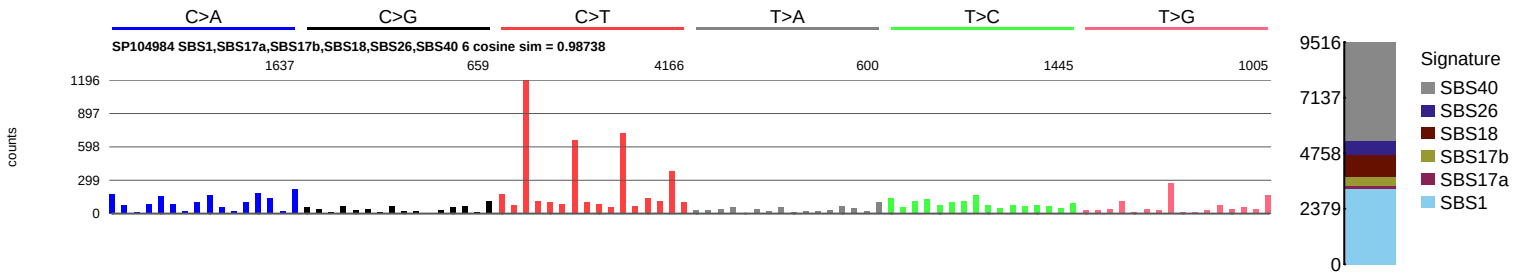

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 7 cosine sim = 0.98748
1572
597
4494
501
1491
857
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1

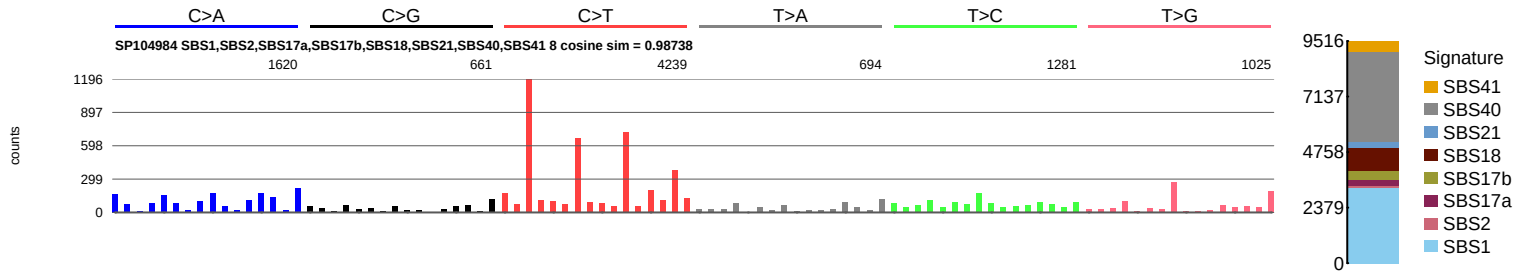

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18 6 cosine sim = 0.98748
1573
598
4504
503
1477
860
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1

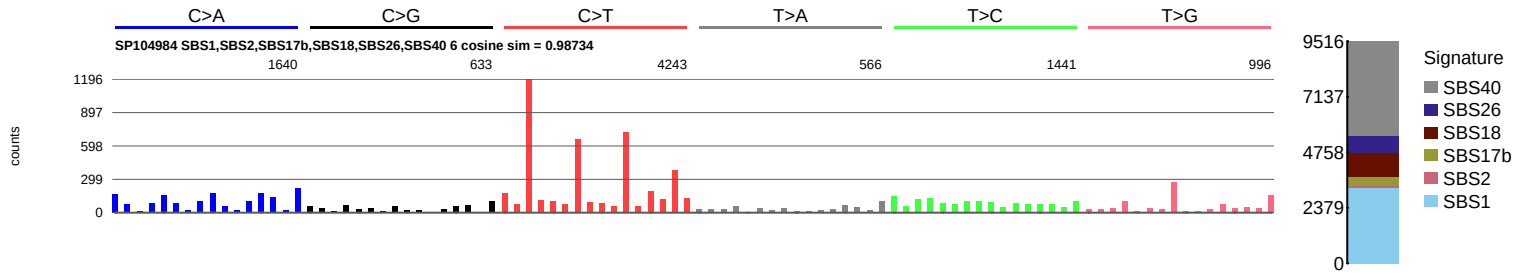

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26 7 cosine sim = 0.98746
1573
598
4503
502
1479
860
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1

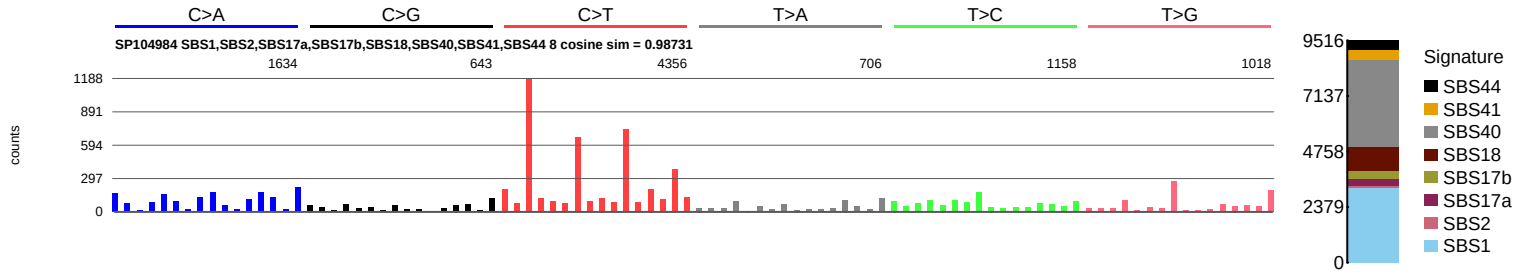

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9874
1644
678
4314
625
1252
1004
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

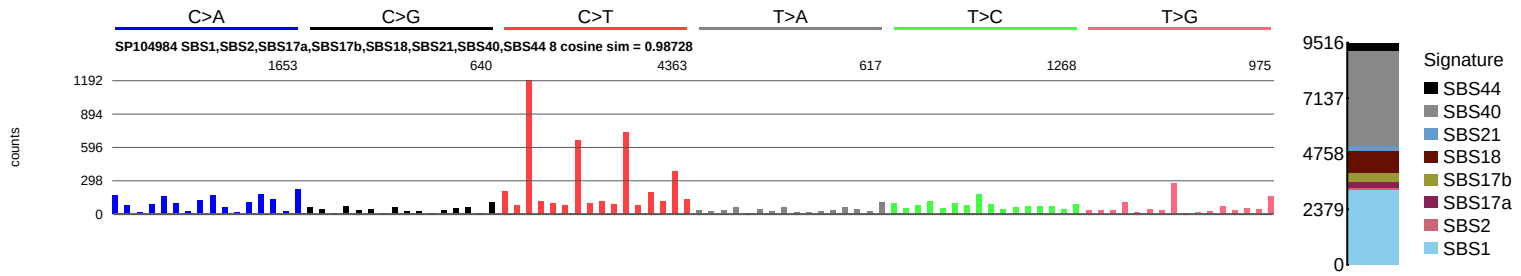

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98739
1644
681
4312
625
1250
1005
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

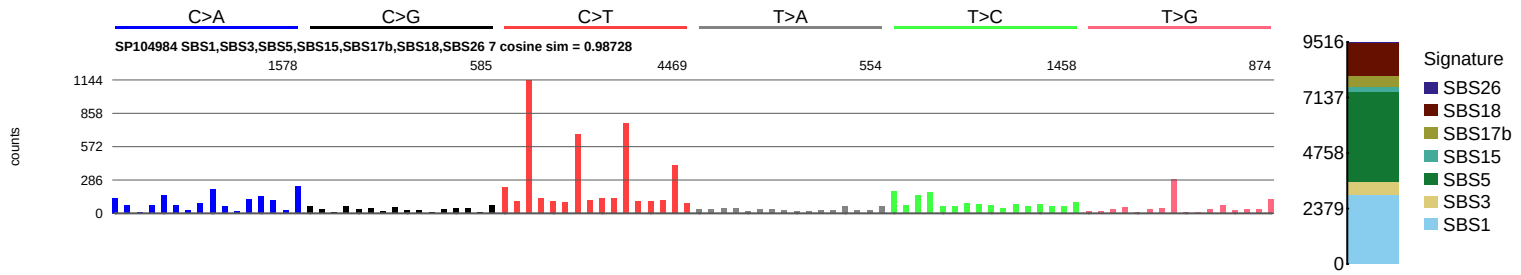

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98738
1637
659
4166
600
1445
1005
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

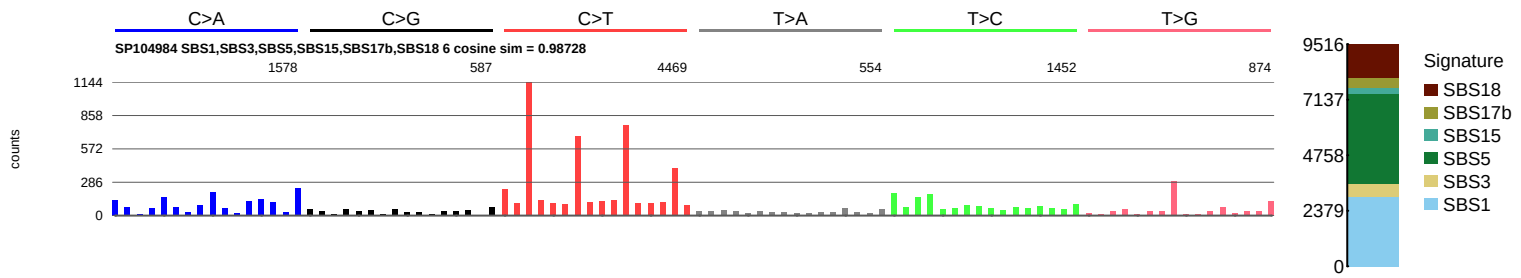

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98738
1620
661
4239
694
1281
1025
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

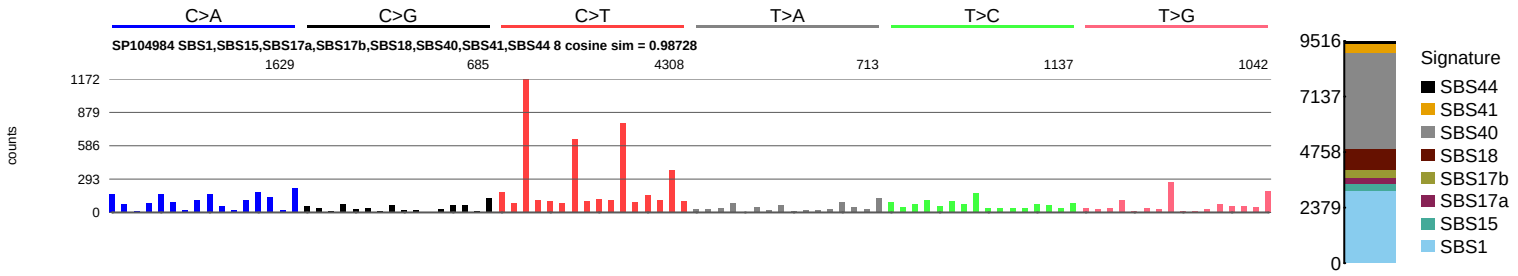

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98734
1640
633
4243
566
1441
996
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1

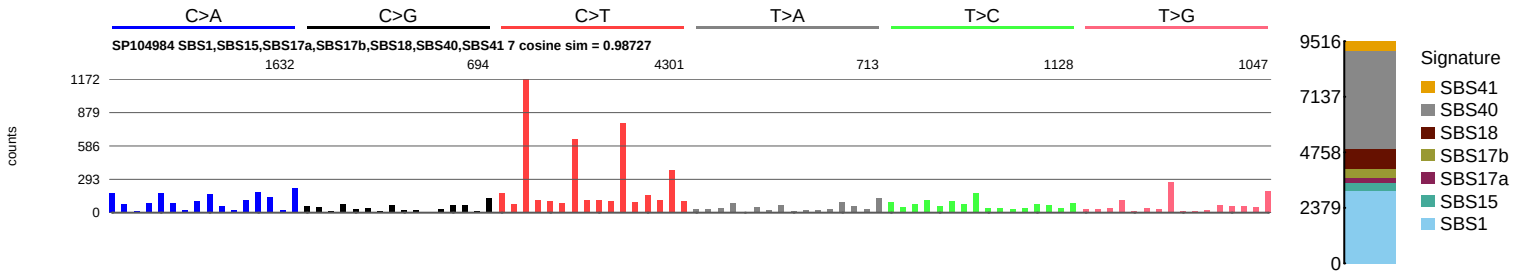

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98731
1634
643
4356
706
1158
1018
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

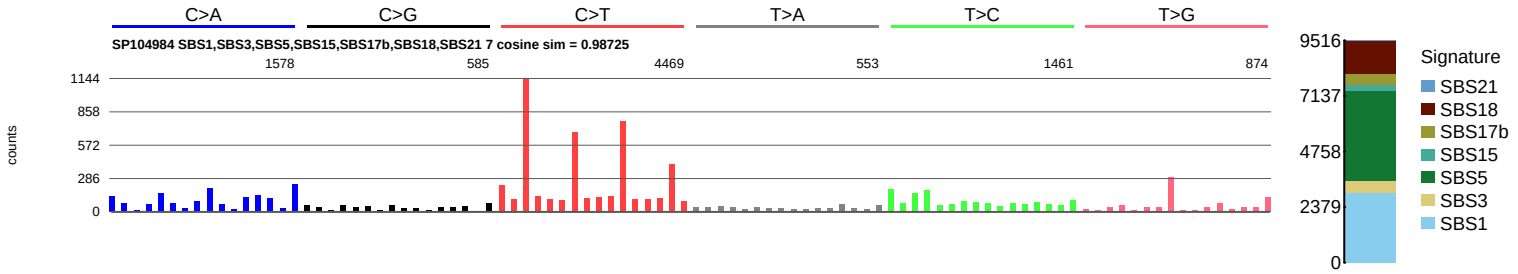

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98728
1653
640
4363
617
1268
975
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

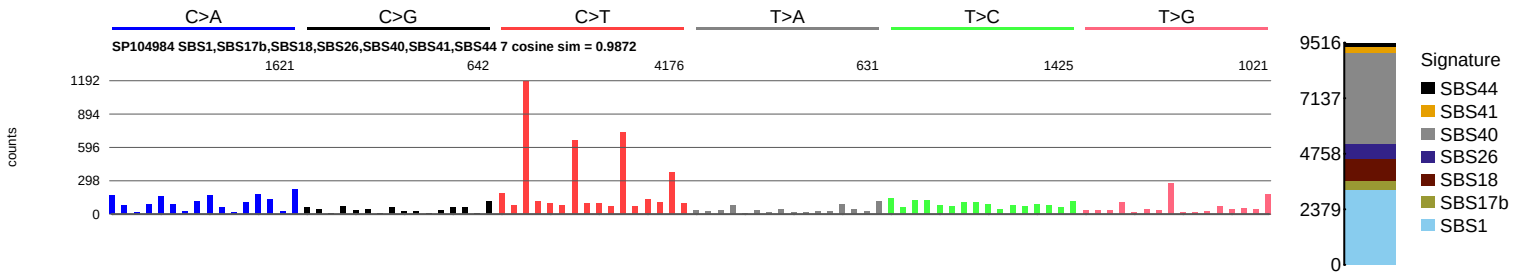

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26 7 cosine sim = 0.98728
1578
585
4469
554
1458
874
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1

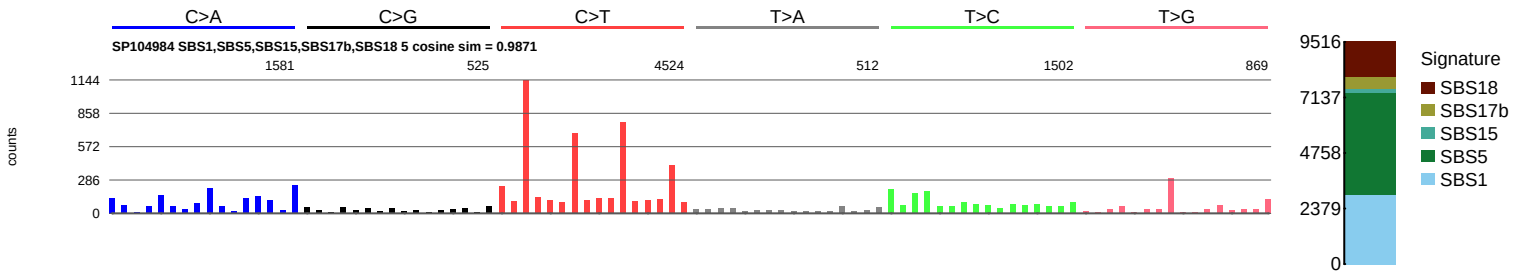

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18 6 cosine sim = 0.98728
1578
587
4469
554
1452
874
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1

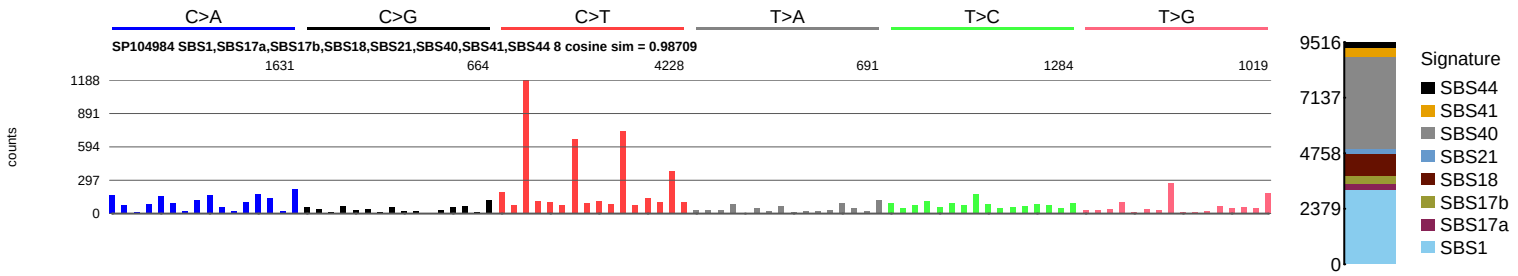

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98728
1629
685
4308
713
1137
1042
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

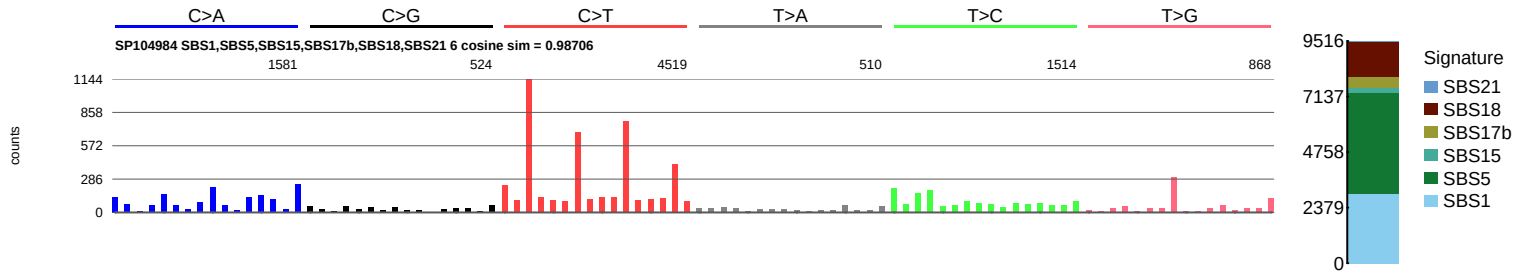

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98727
1632
694
4301
713
1128
1047
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

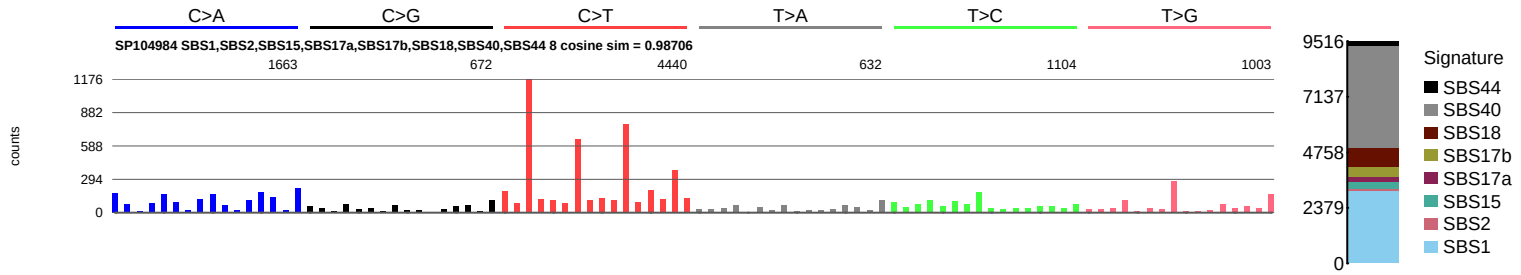

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21 7 cosine sim = 0.98725
1578
585
4469
553
1461
874
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1

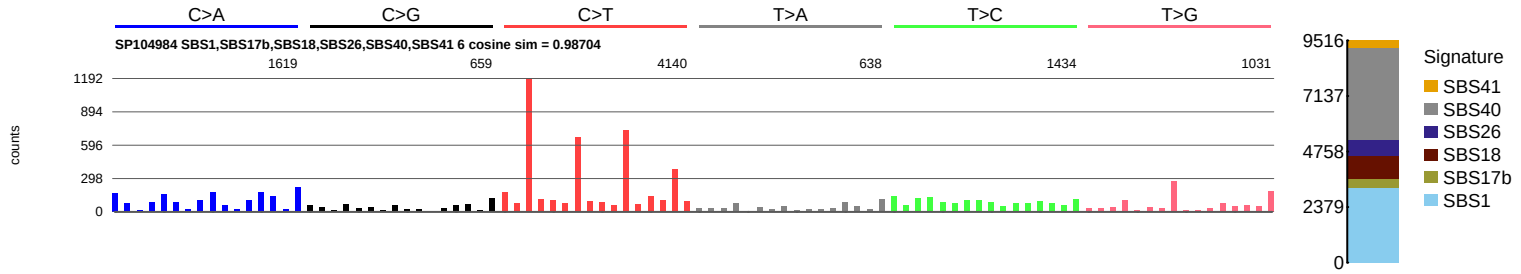

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.9872
1621
642
4176
631
1425
1021
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS1

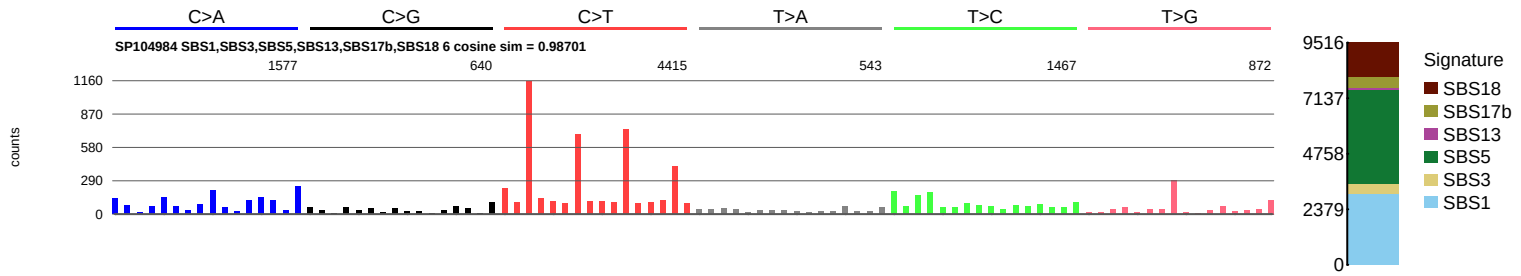

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS18 5 cosine sim = 0.9871
1581
525
4524
512
1502
869
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98709
1631
664
4228
691
1284
1019
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21 6 cosine sim = 0.98706
1581
524
4519
510
1514
868
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98706
1663
672
4440
632
1104
1003
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98704
1619
659
4140
638
1434
1031
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS1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C>G
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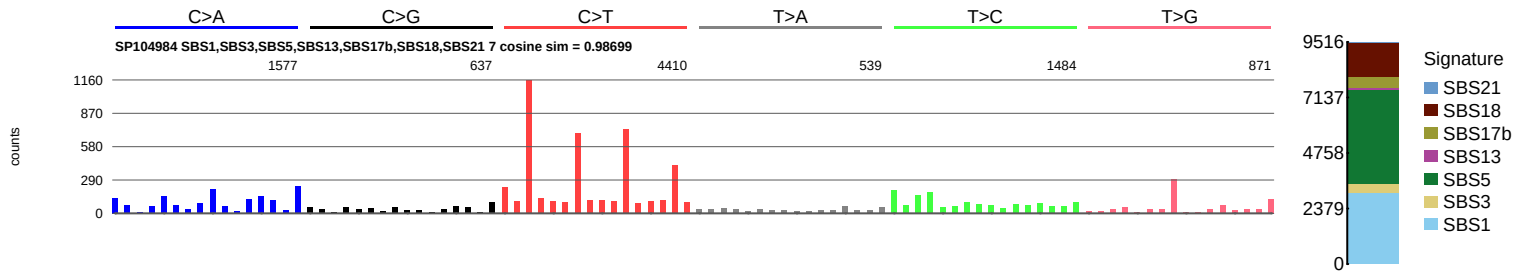
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18 6 cosine sim = 0.98701
1577
640
4415
543
1467
872
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1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C>G
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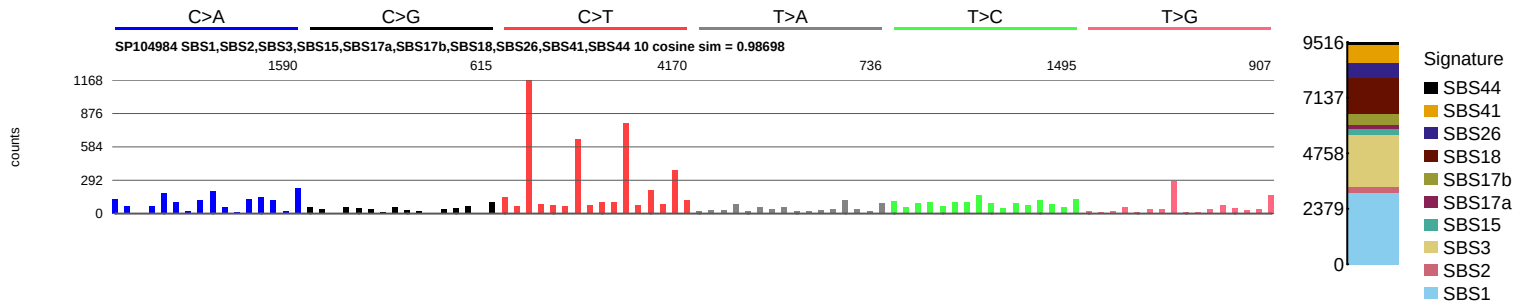
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 7 cosine sim = 0.98699
1577
637
4410
539
1484
871
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS1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C>G
C>T
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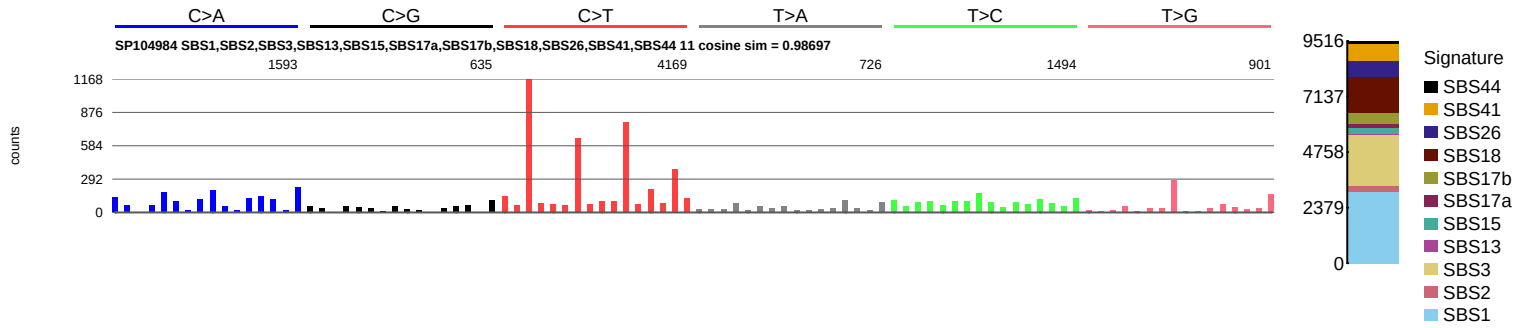
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98698
1590
615
4170
736
1495
907
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>T
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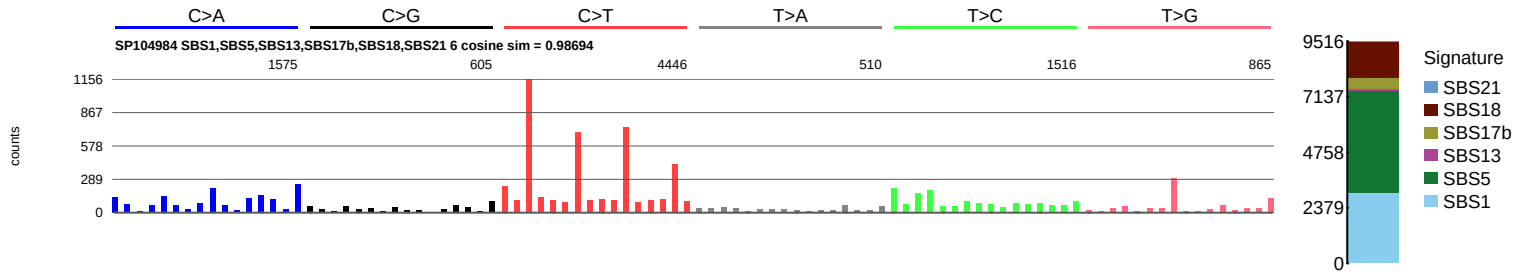
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98697
1593
635
4169
726
1494
901
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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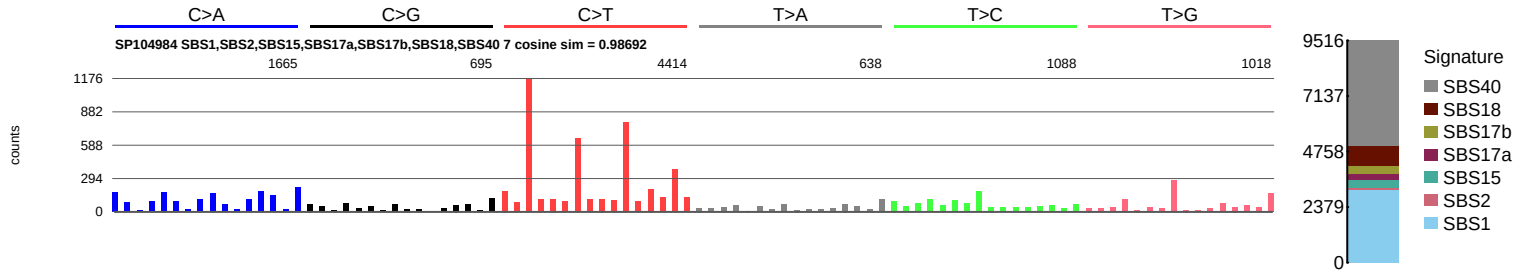
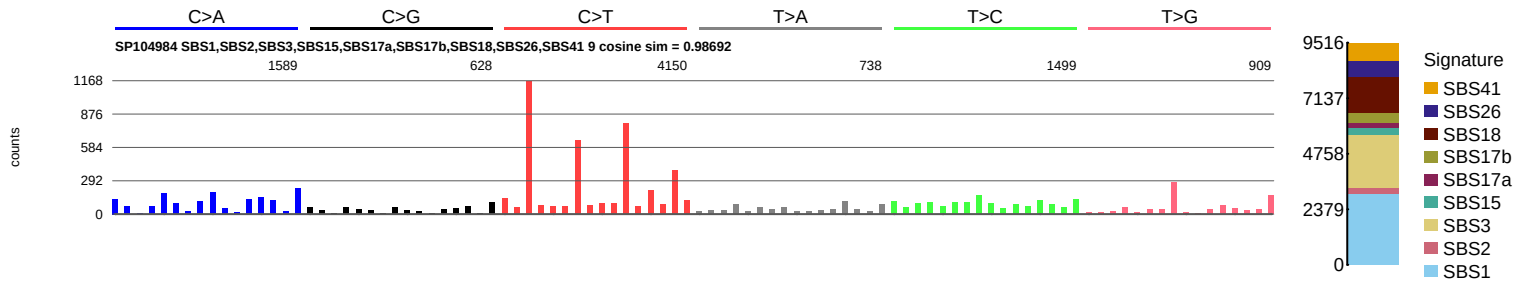
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21 6 cosine sim = 0.98694
1575
605
4446
510
1516
865
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98692
1665
695
4414
638
1088
1018
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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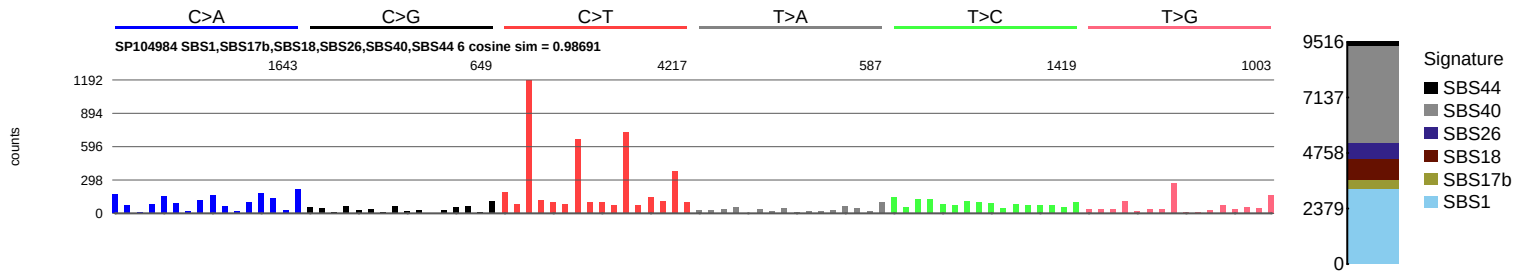
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98692
1589
628
4150
738
1499
909
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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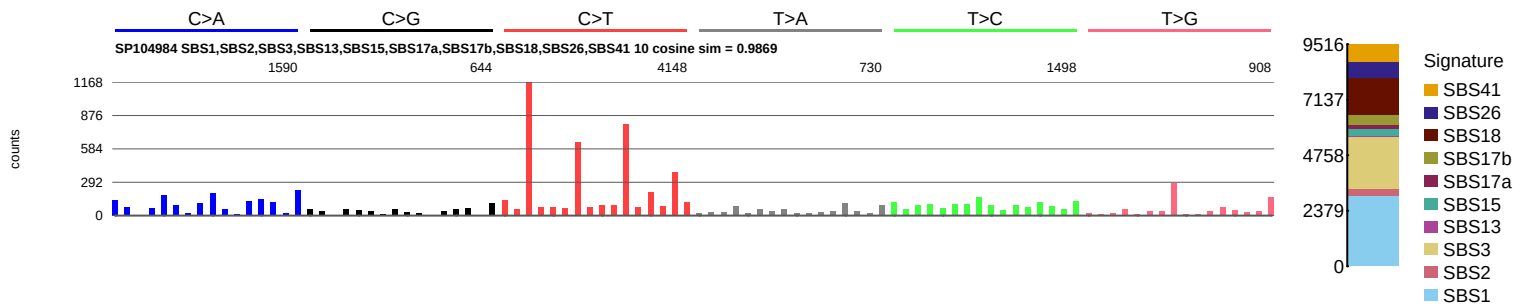
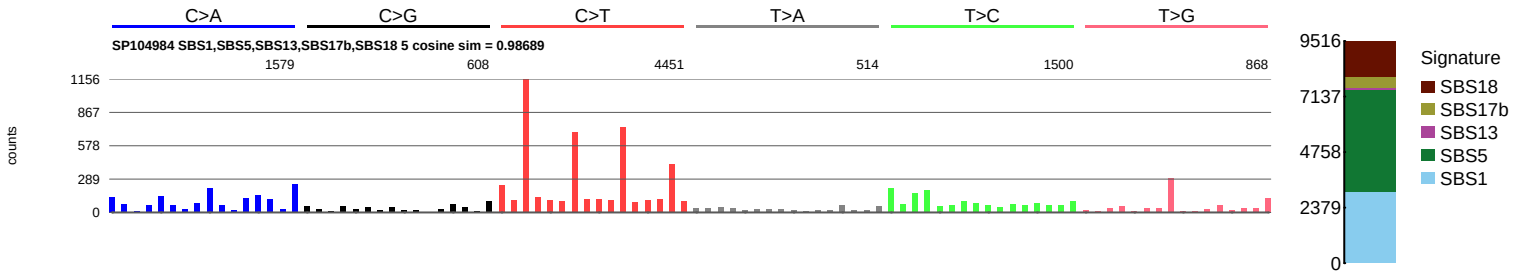
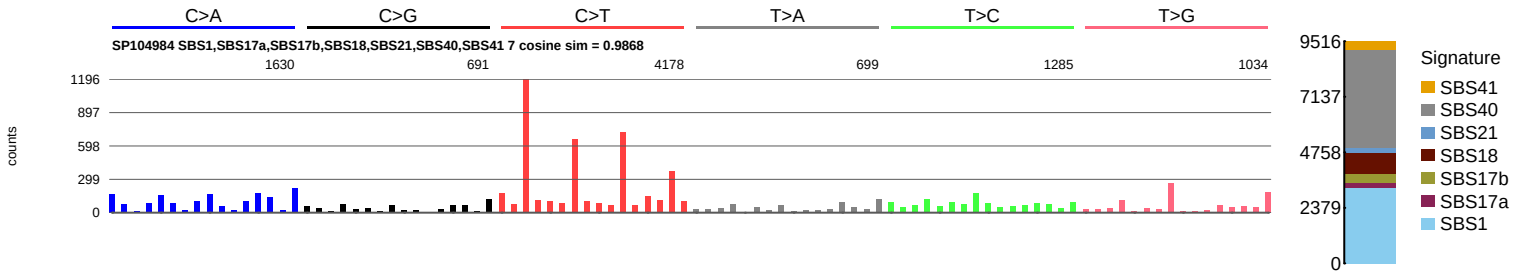
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98691
1643
649
4217
587
1419
1003
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.9869
1590
644
4148
730
1498
908
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17b,SBS18 5 cosine sim = 0.98689
1579
608
4451
514
1500
868
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.9868
1630
691
4178
699
1285
1034
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
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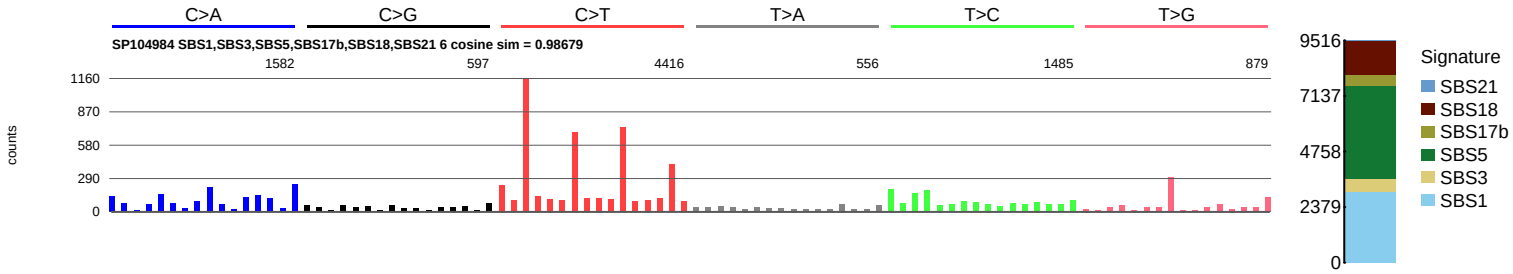
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS17b,SBS18,SBS21 6 cosine sim = 0.98679
1582
597
4416
556
1485
879
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS21
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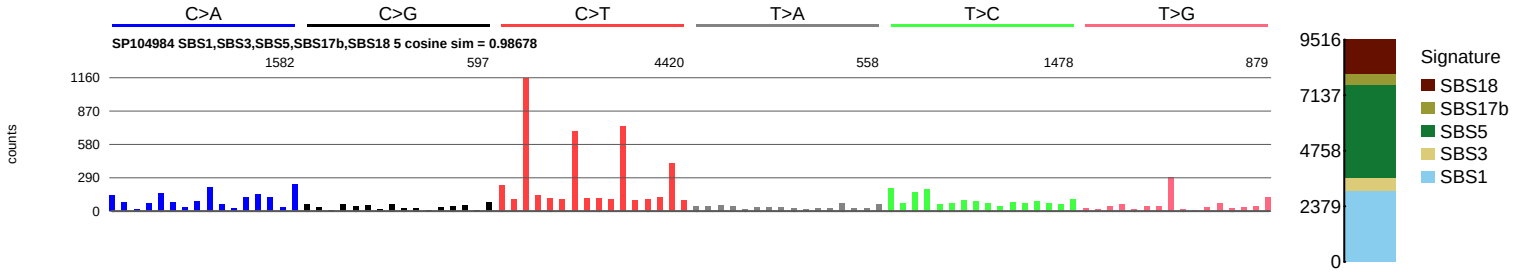
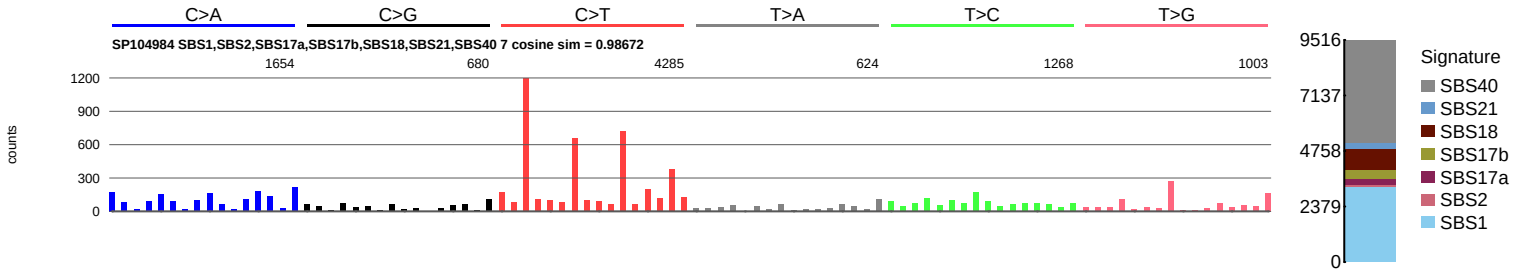
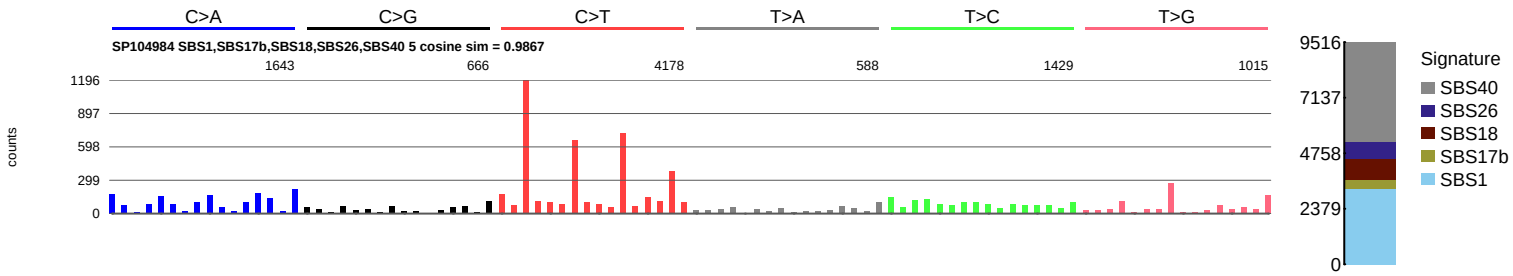
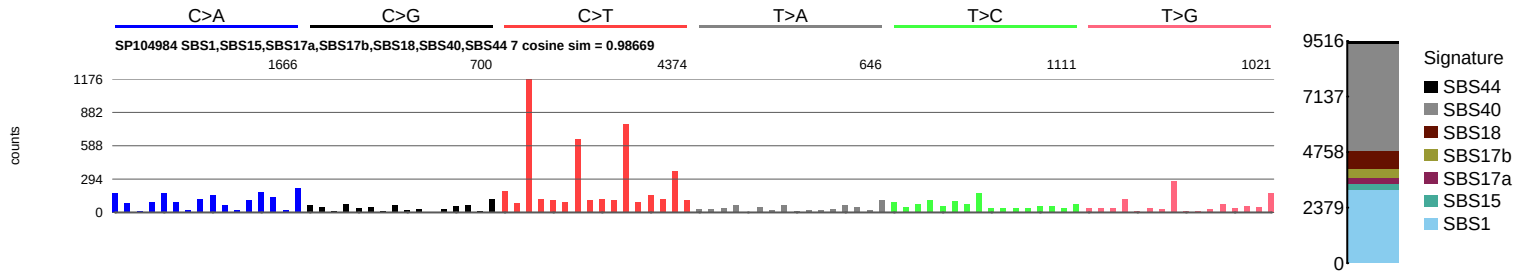
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS17b,SBS18 5 cosine sim = 0.98678
1582
597
4420
558
1478
879
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98672
1654
680
4285
624
1268
1003
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS26,SBS40 5 cosine sim = 0.9867
1643
666
4178
588
1429
1015
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS1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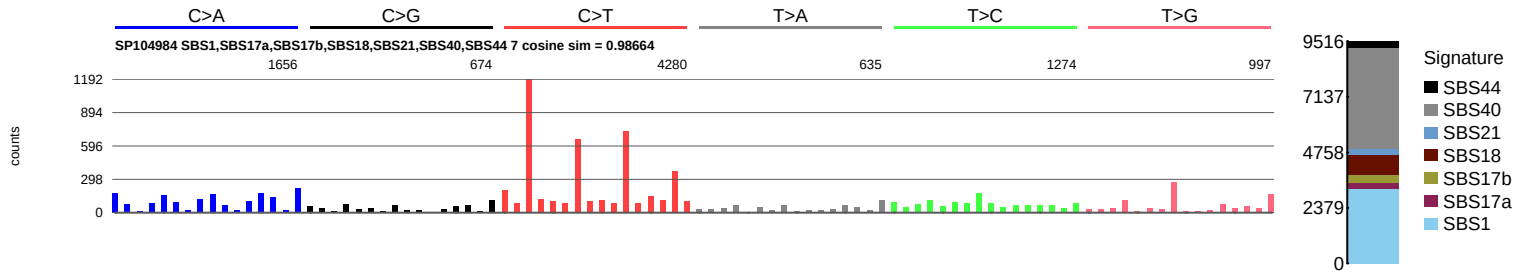
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98669
1666
700
4374
646
1111
1021
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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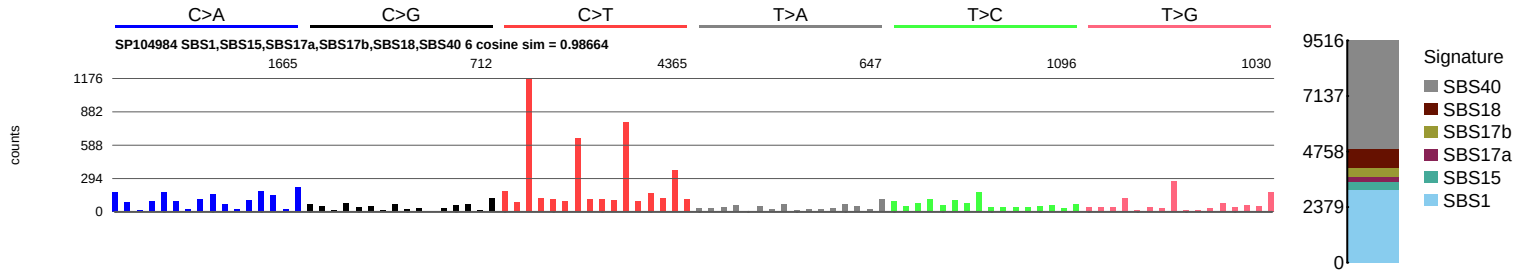
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98664
1656
674
4280
635
1274
997
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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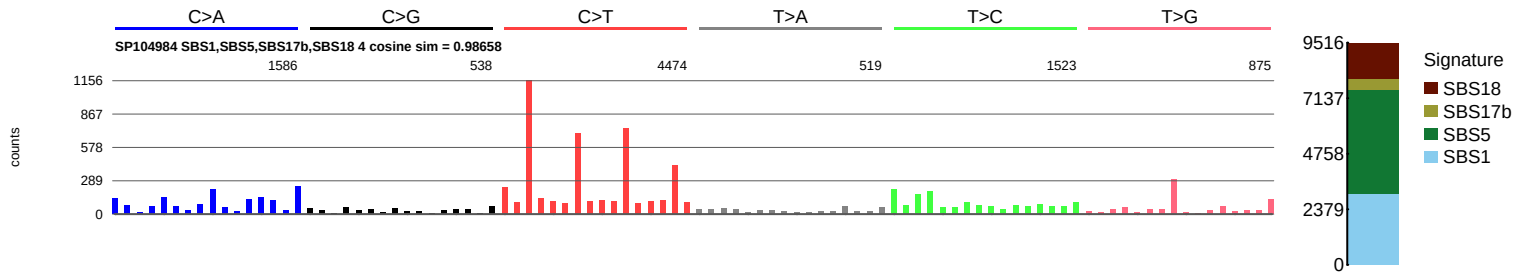
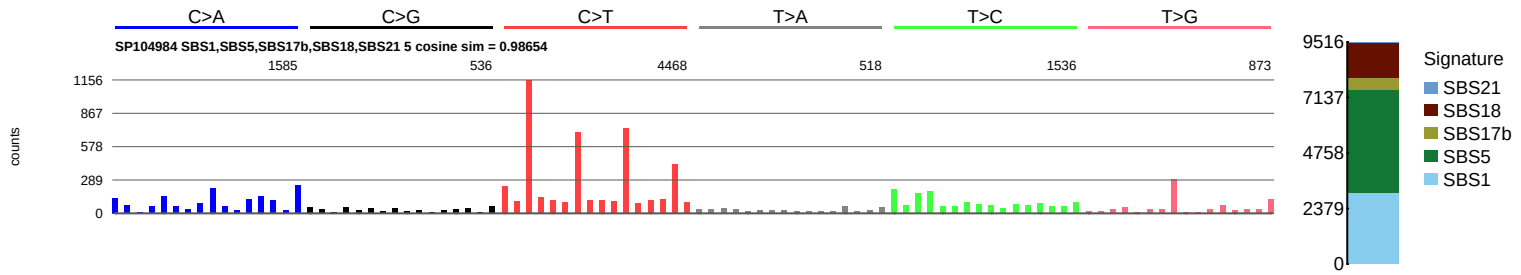
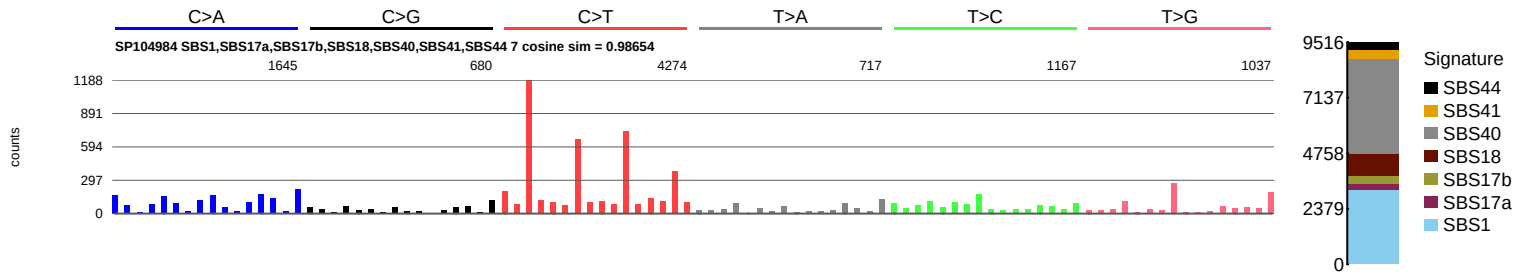
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98664
1665
712
4365
647
1096
1030
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS18 4 cosine sim = 0.98658
1586
538
4474
519
1523
875
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS18,SBS21 5 cosine sim = 0.98654
1585
536
4468
518
1536
873
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98654
1645
680
4274
717
1167
1037
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
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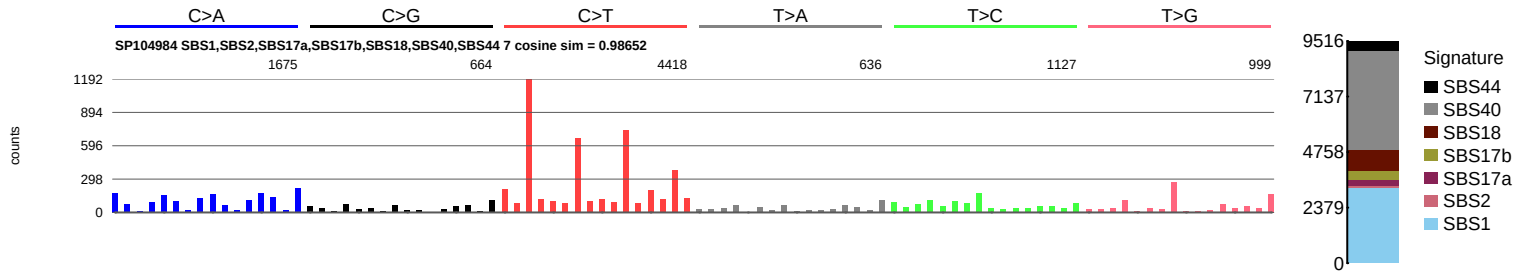
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98652
1675
664
4418
636
1127
999
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
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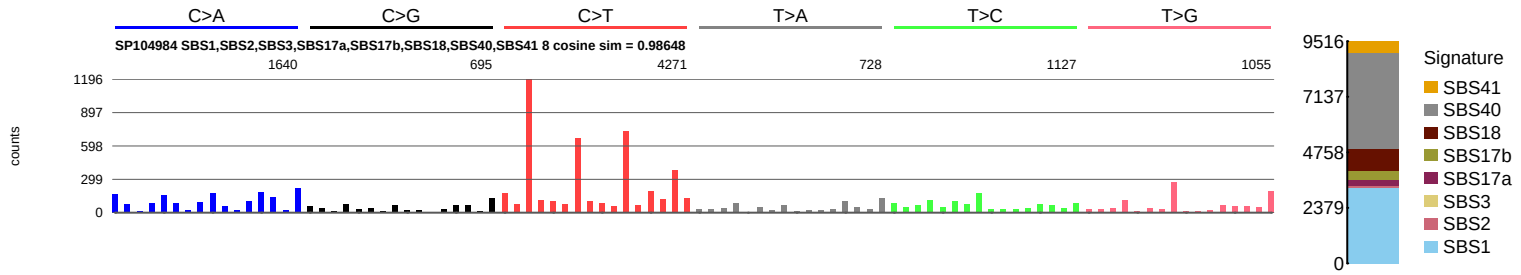
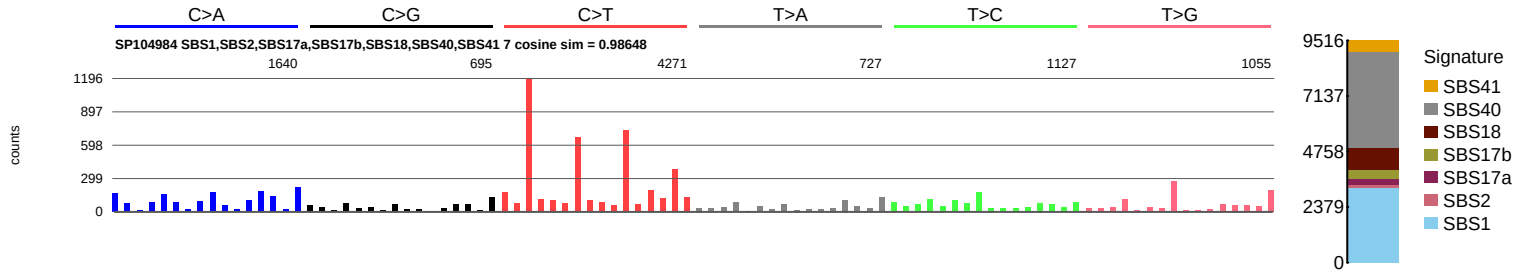
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98648
1640
695
4271
728
1127
1055
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98648
1640
695
4271
727
1127
1055
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
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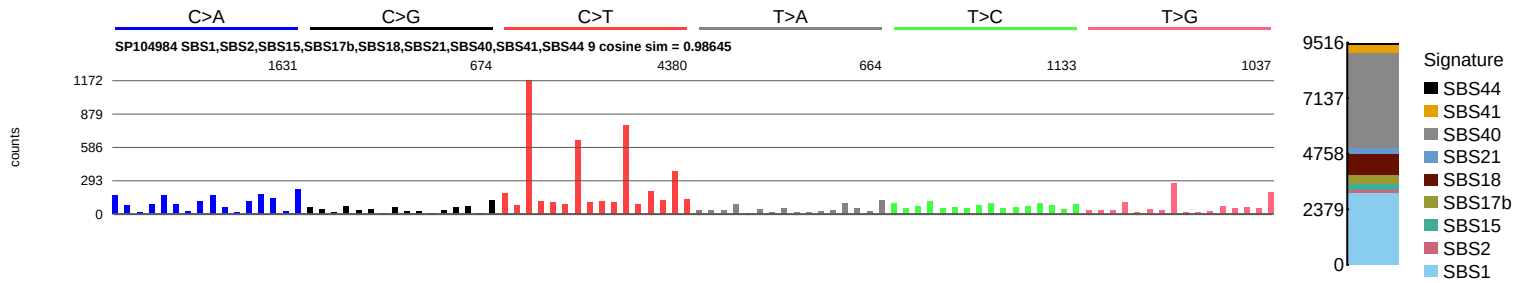
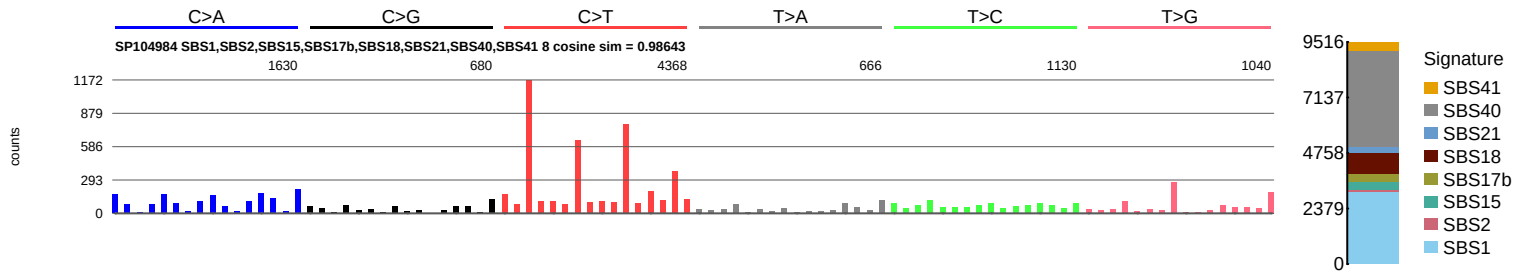
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98645
1631
674
4380
664
1133
1037
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98643
1630
680
4368
666
1130
1040
counts
1172
879
586
293
0
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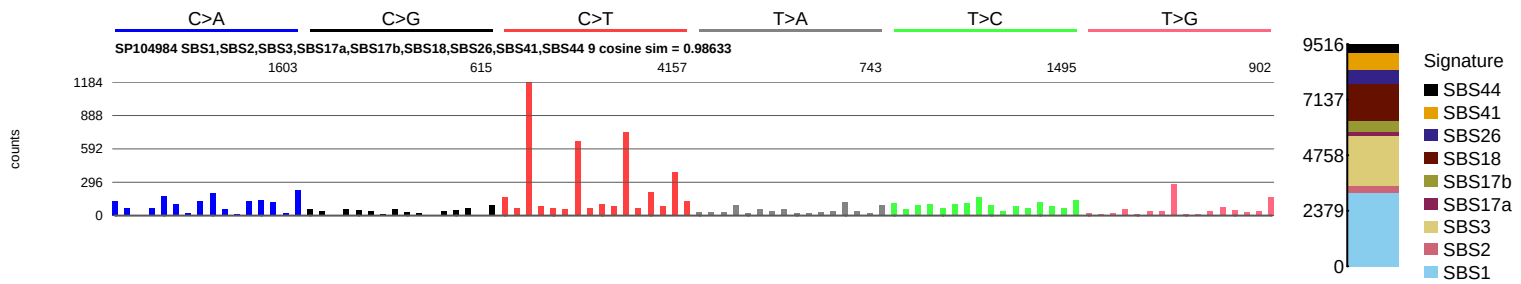
SBS1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SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98633
1603
615
4157
743
1495
902
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
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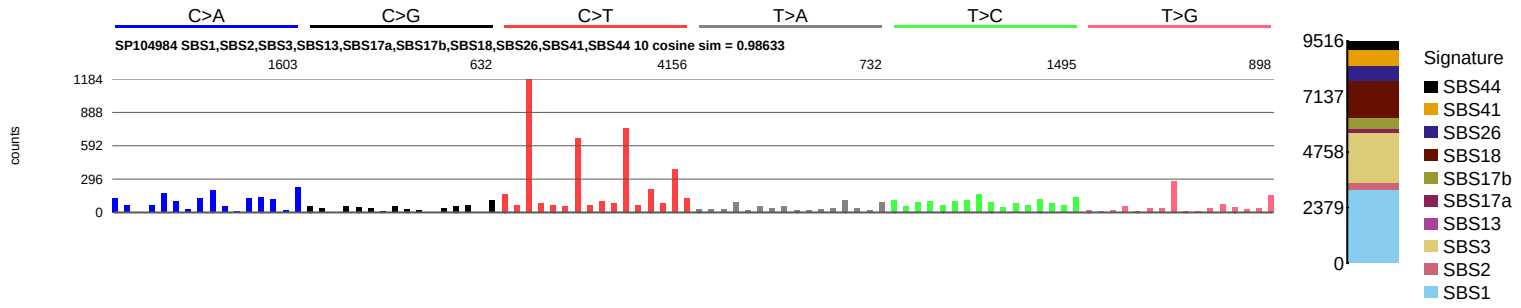
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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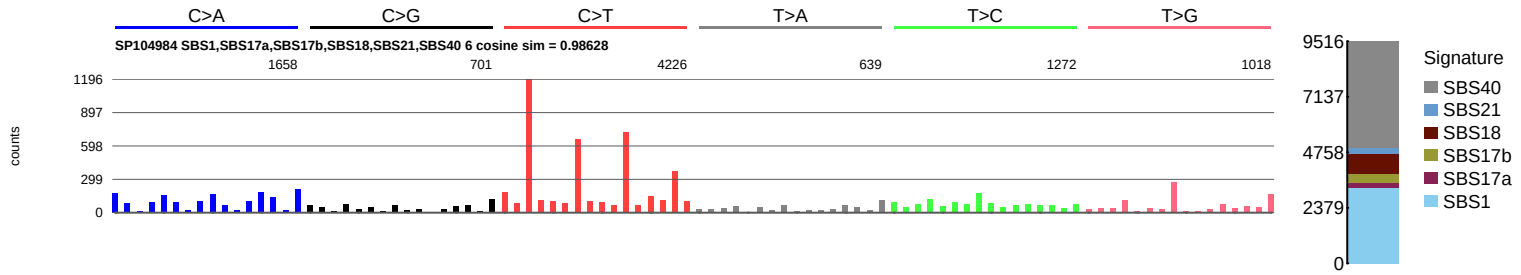
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98633
1603
632
4156
732
1495
898
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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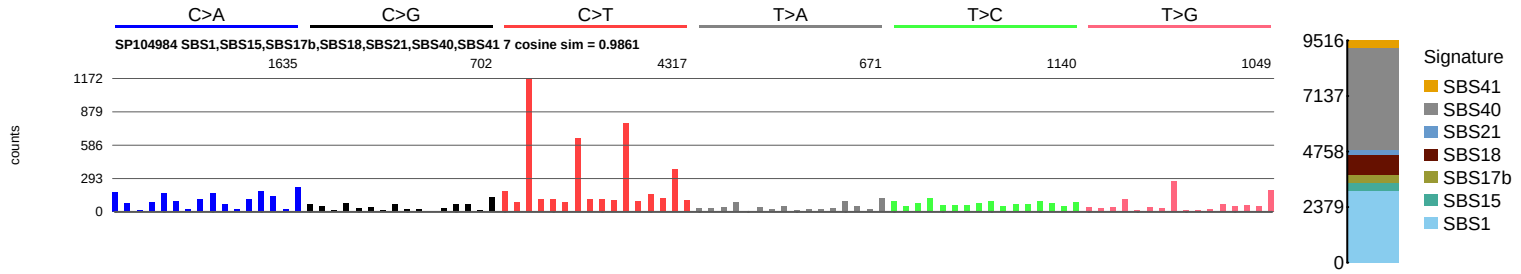
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98628
1658
701
4226
639
1272
1018
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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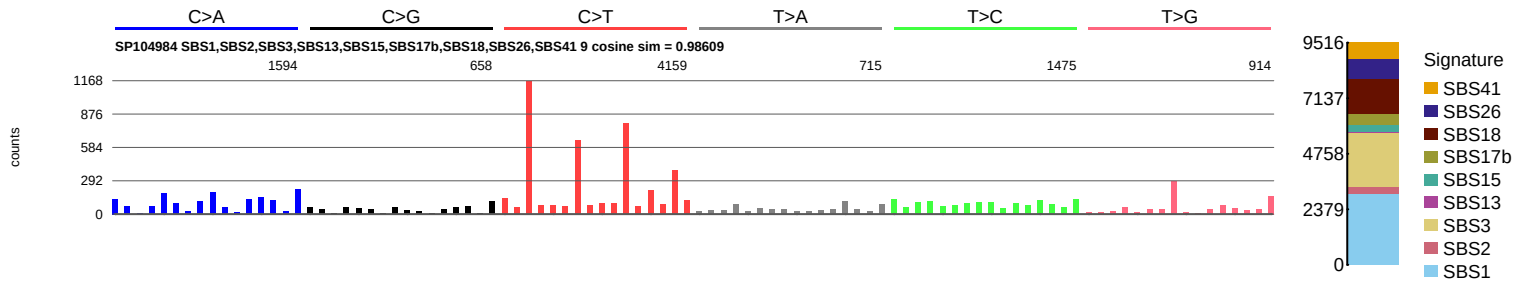
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.9861
1635
702
4317
671
1140
1049
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS1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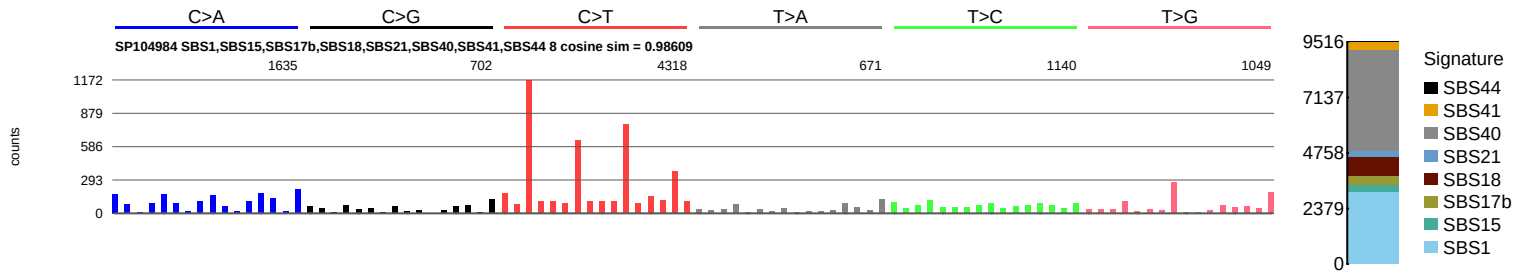
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98609
1594
658
4159
715
1475
914
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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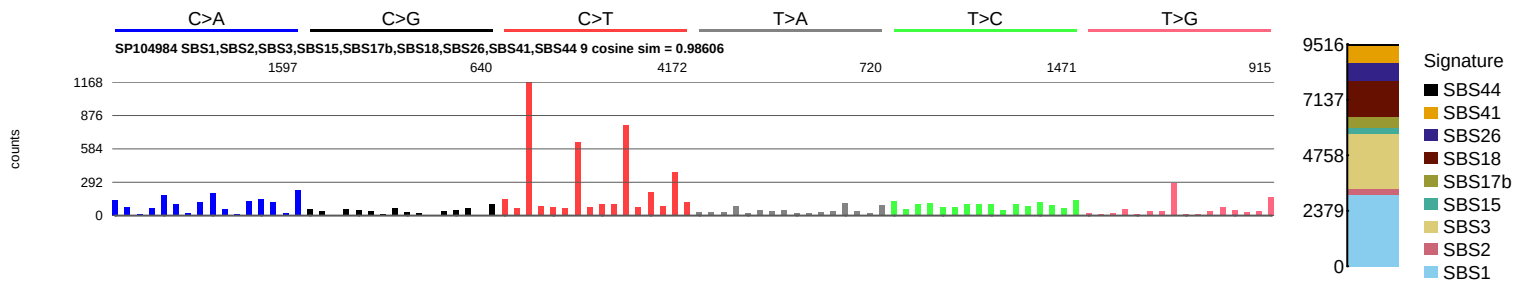
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98609
1635
702
4318
671
1140
1049
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS1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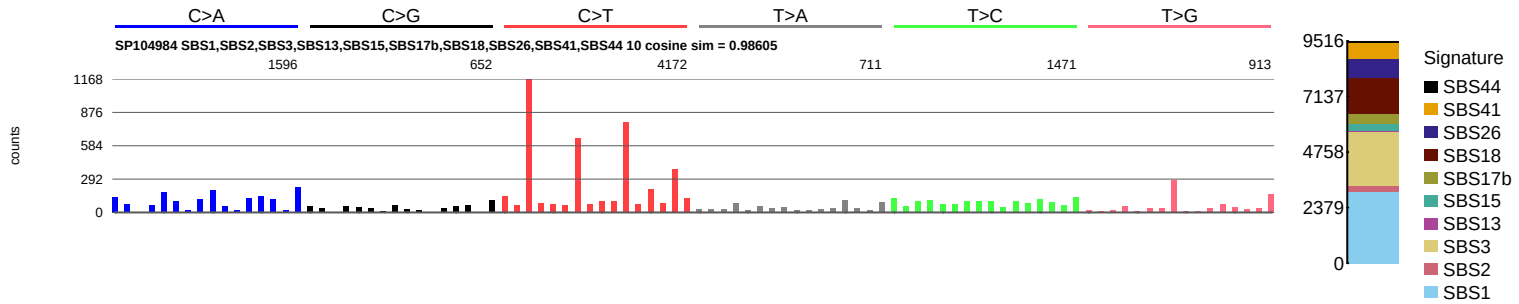
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98606
1597
640
4172
720
1471
915
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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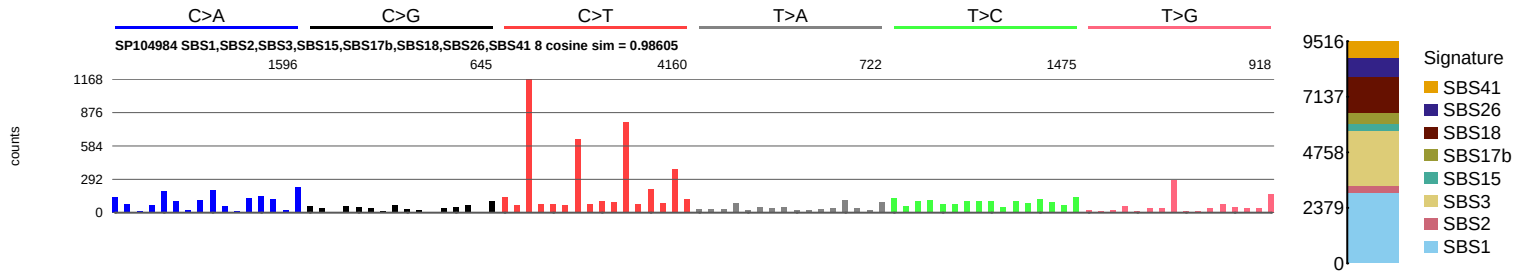
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98605
1596
652
4172
711
1471
913
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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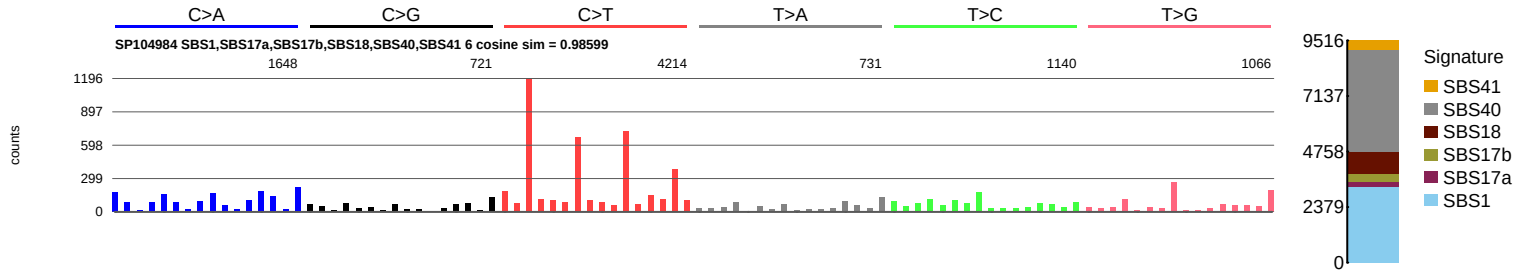
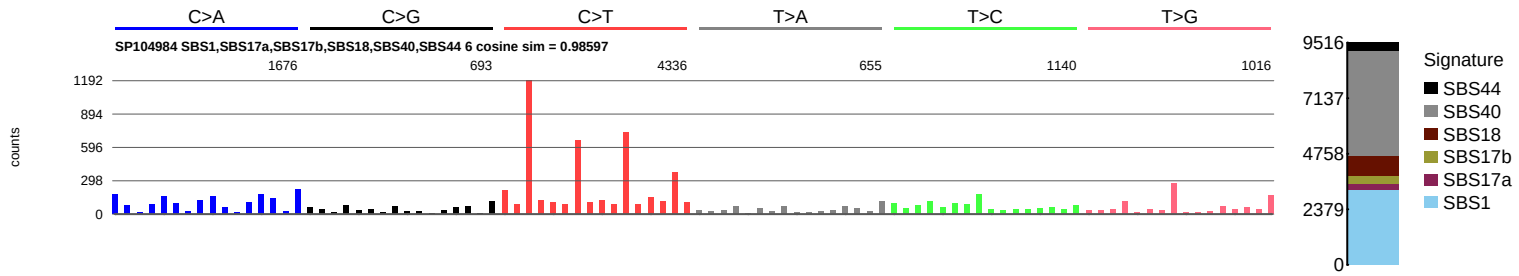
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98605
1596
645
4160
722
1475
918
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98599
1648
721
4214
731
1140
1066
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98597
1676
693
4336
655
1140
1016
counts
1192
894
596
298
0
9516
7137
4758
2379
0
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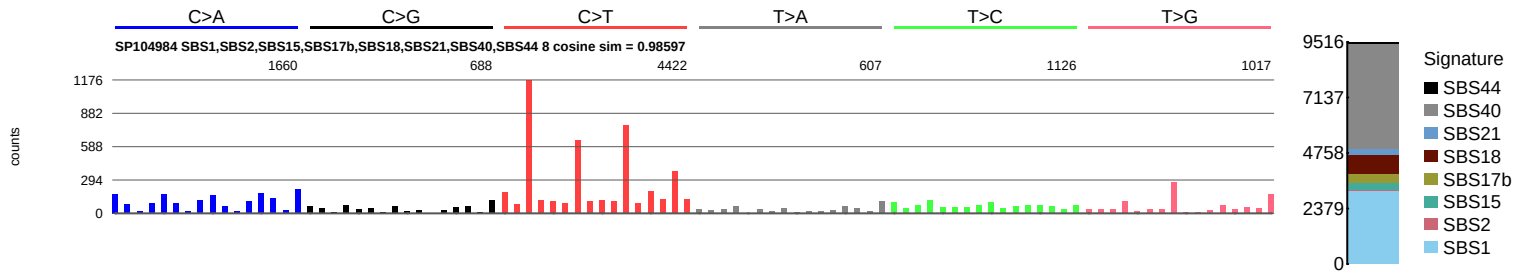
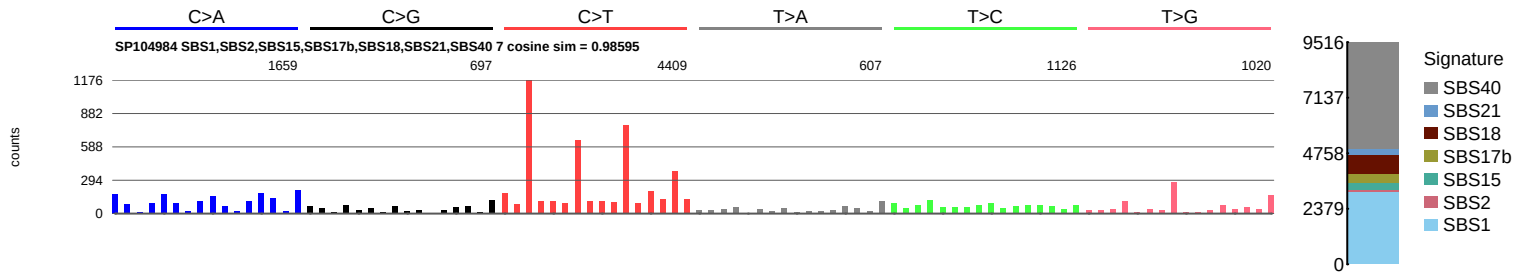
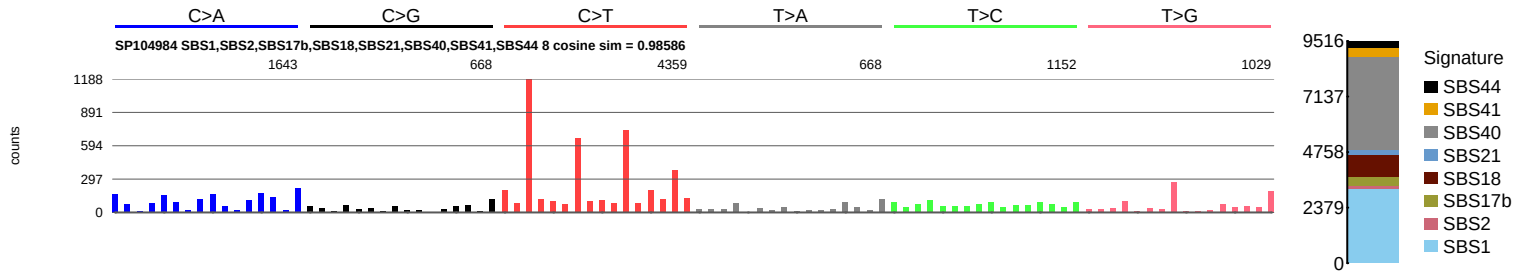
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98597
1660
688
4422
607
1126
1017
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98595
1659
697
4409
607
1126
1020
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1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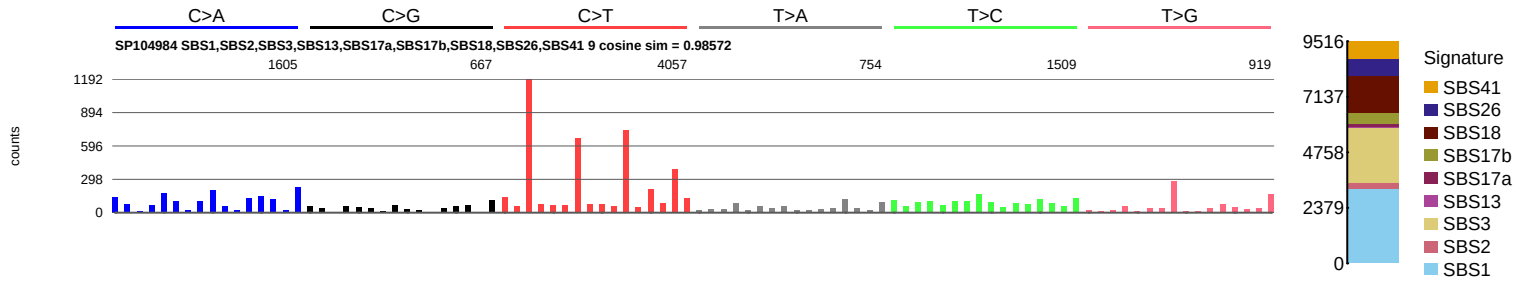
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98586
1643
668
4359
668
1152
1029
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1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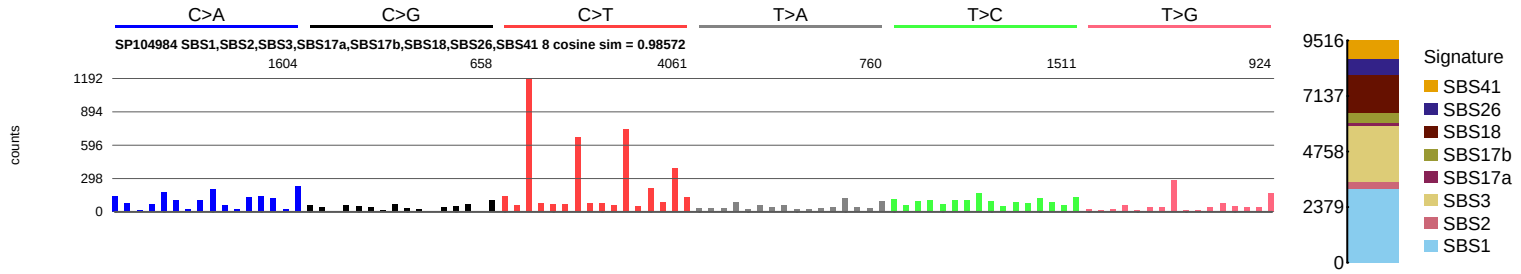
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98572
1605
667
4057
754
1509
919
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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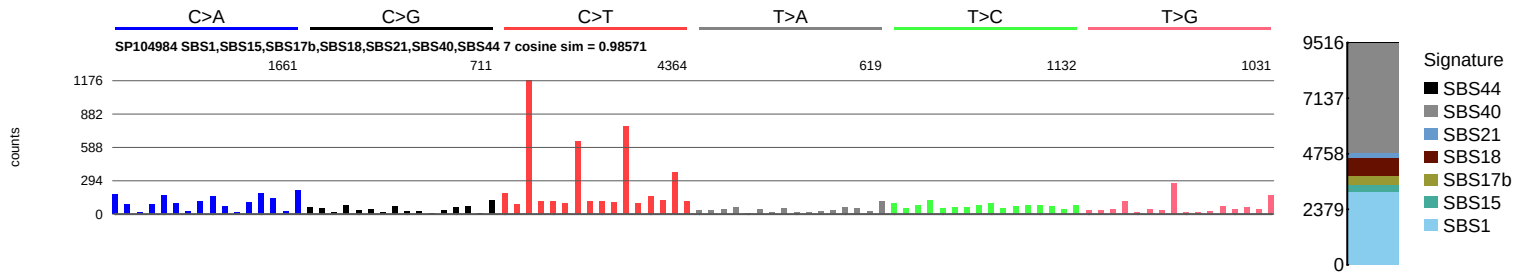
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98572
1604
658
4061
760
1511
924
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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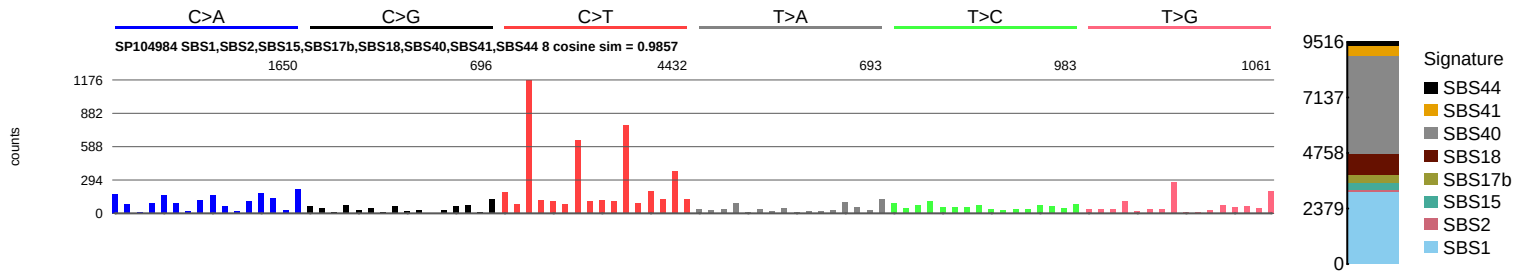
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98571
1661
711
4364
619
1132
1031
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS1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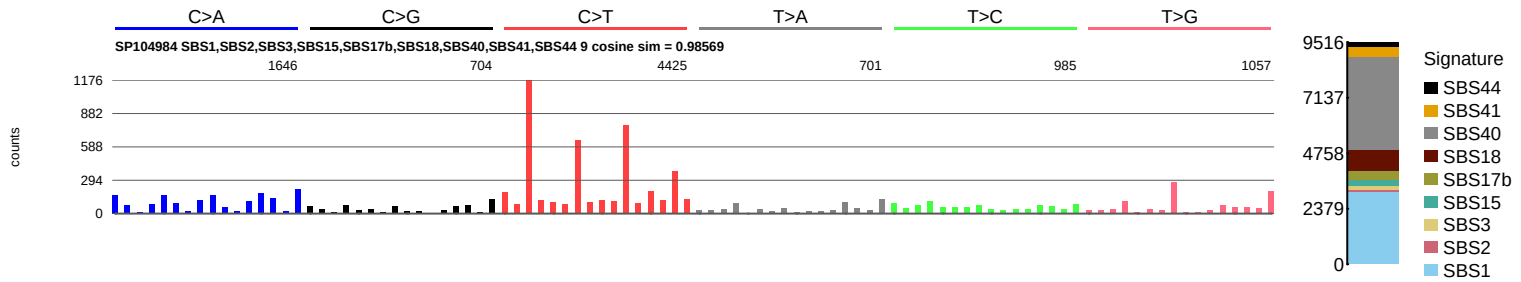
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9857
1650
696
4432
693
983
1061
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98569
1646
704
4425
701
985
1057
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
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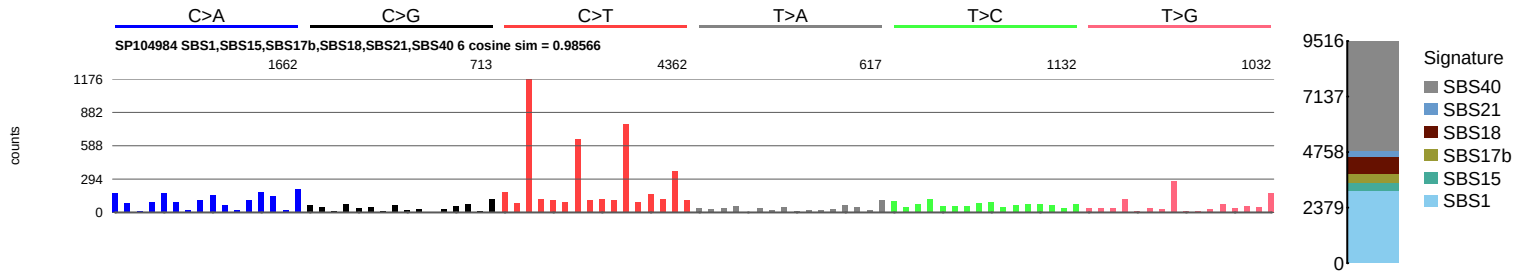
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98566
1662
713
4362
617
1132
1032
counts
1176
882
588
294
0
9516
7137
4758
2379
0
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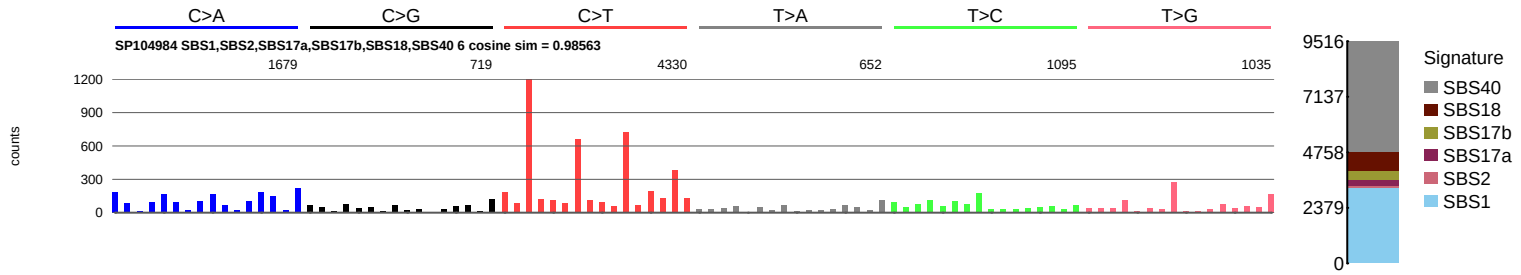
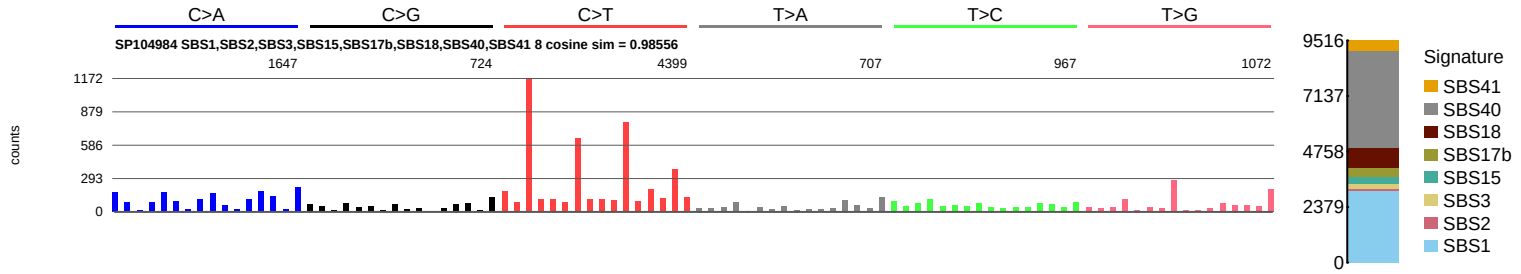
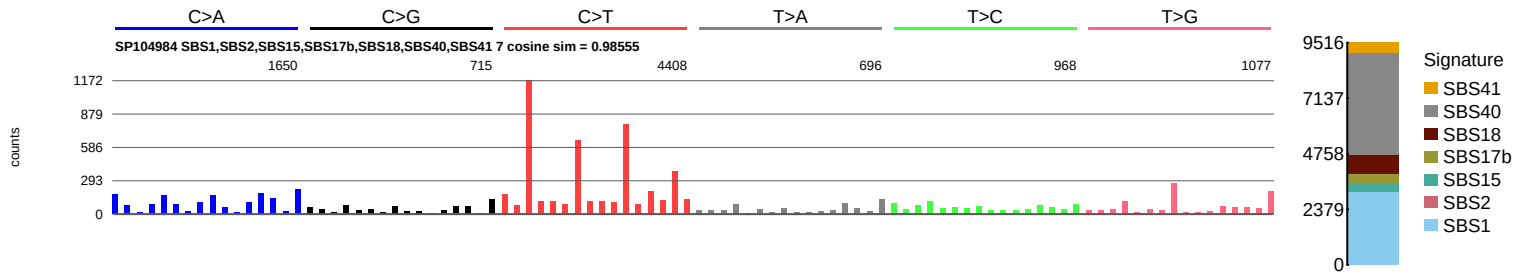
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98563
1679
719
4330
652
1095
1035
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98556
1647
724
4399
707
967
1072
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98555
1650
715
4408
696
968
1077
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
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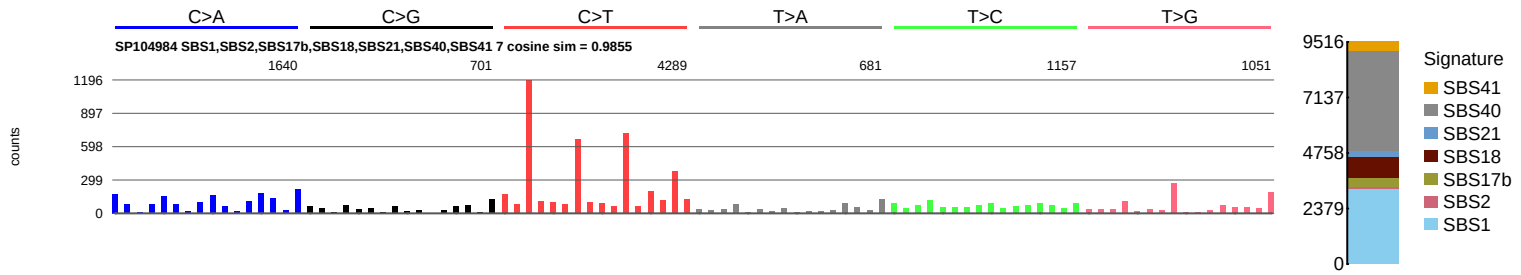
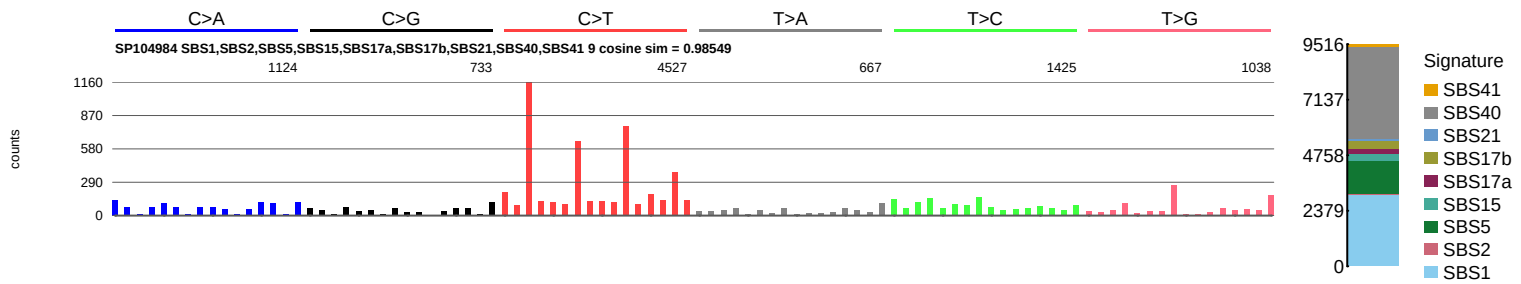
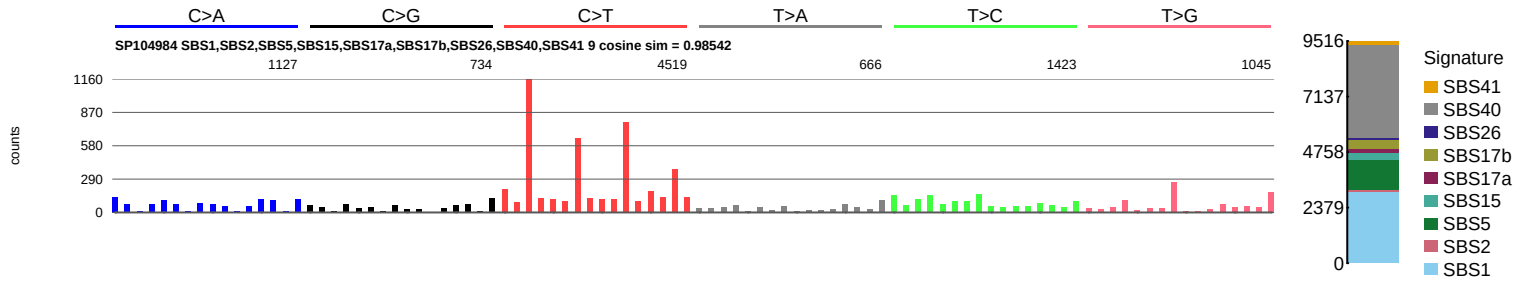
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.9855
1640
701
4289
681
1157
1051
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98549
1124
733
4527
667
1425
1038
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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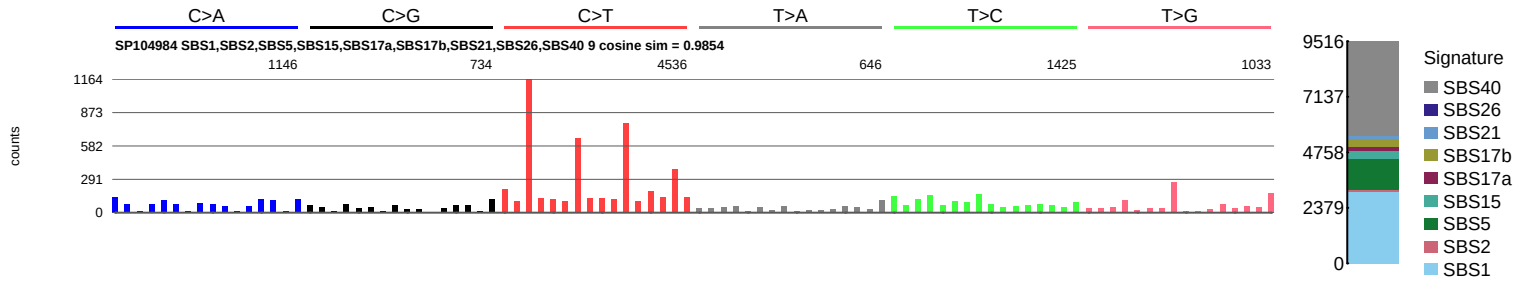
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98542
1127
734
4519
666
1423
1045
1160
870
580
290
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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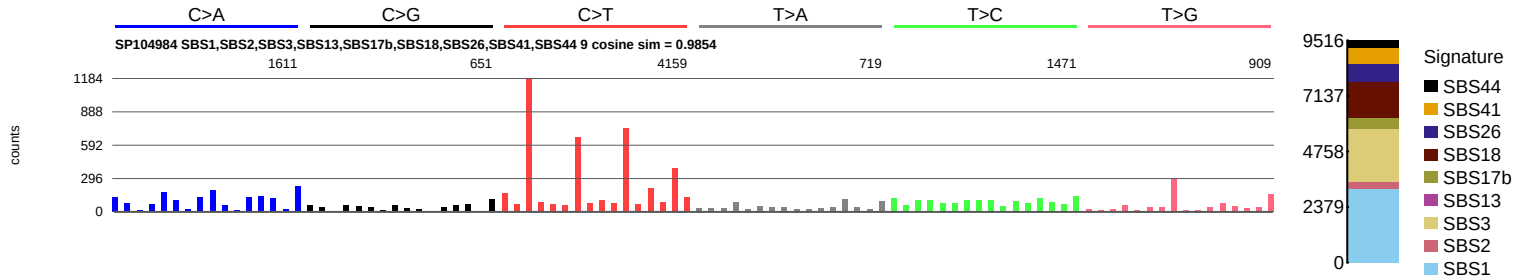
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.9854
1146
734
4536
646
1425
1033
1164
873
582
291
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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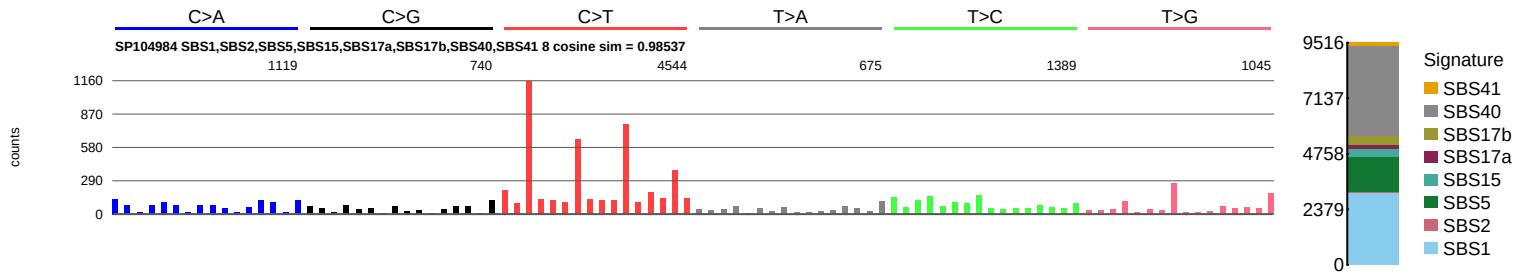
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.9854
1611
651
4159
719
1471
909
1184
888
592
296
0
counts
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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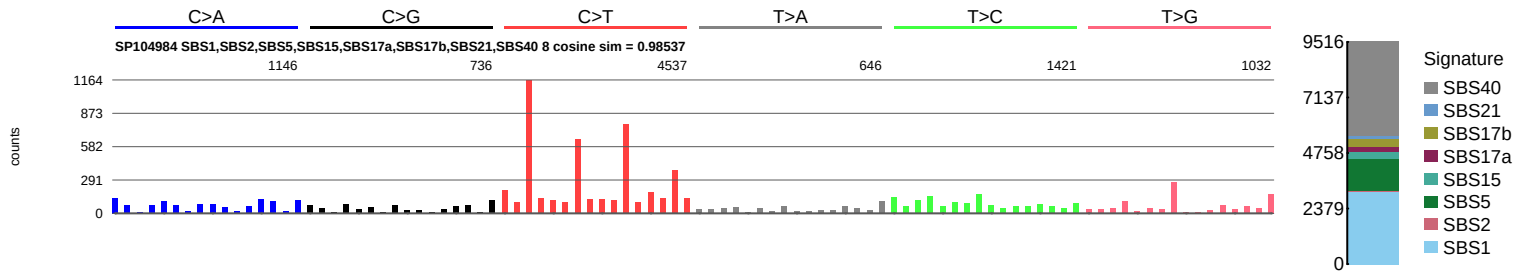
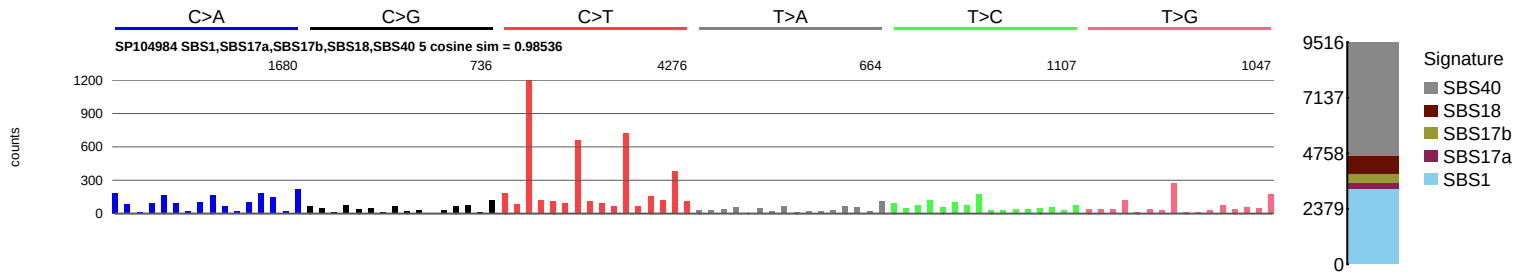
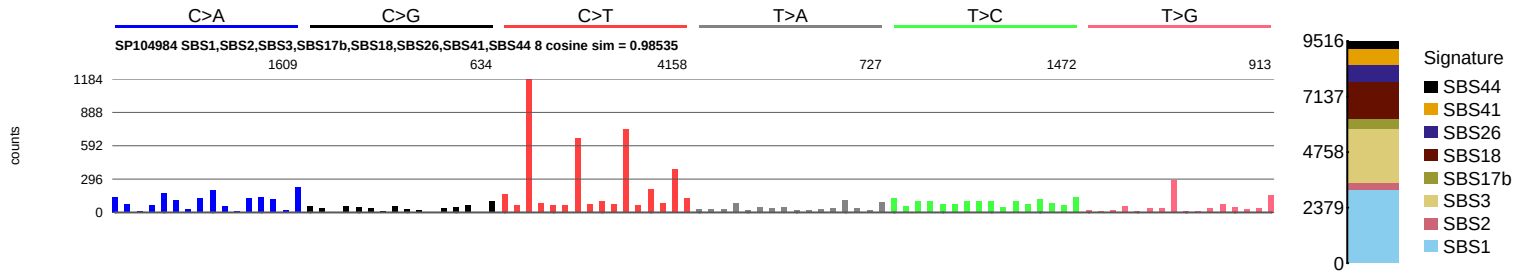
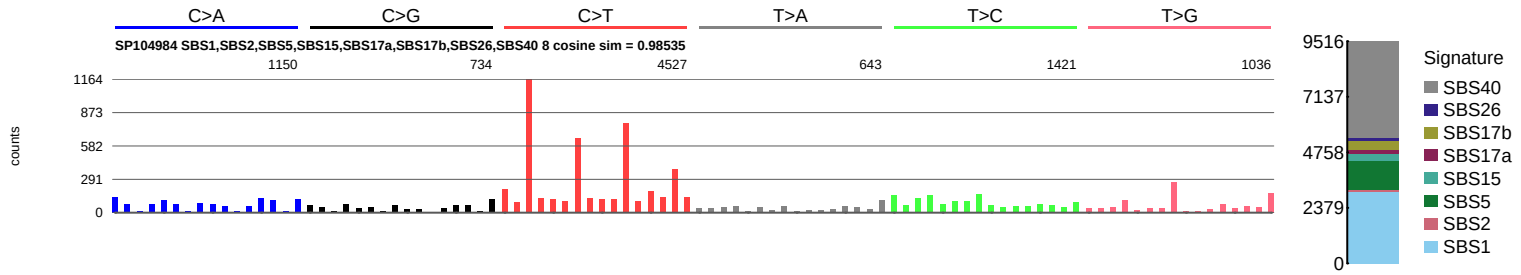
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98537
1119
740
4544
675
1389
1045
1160
870
580
290
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98537
1146
736
4537
646
1421
1032
1164
873
582
291
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98536
1680
736
4276
664
1107
1047
1200
900
600
300
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98535
1609
634
4158
727
1472
913
1184
888
592
296
0
counts
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98535
1150
734
4527
643
1421
1036
1164
873
582
291
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS17a
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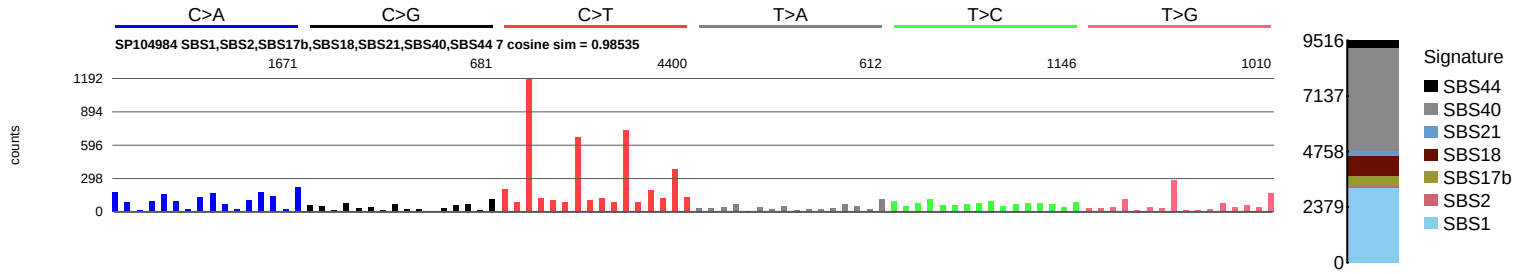
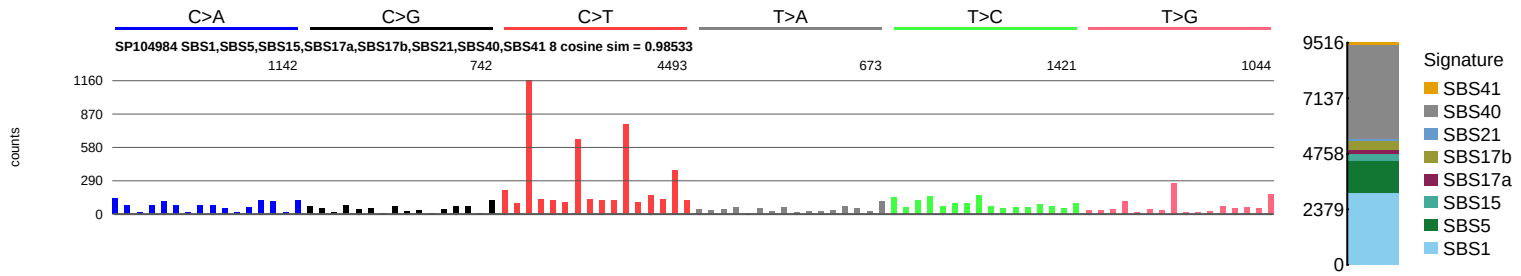
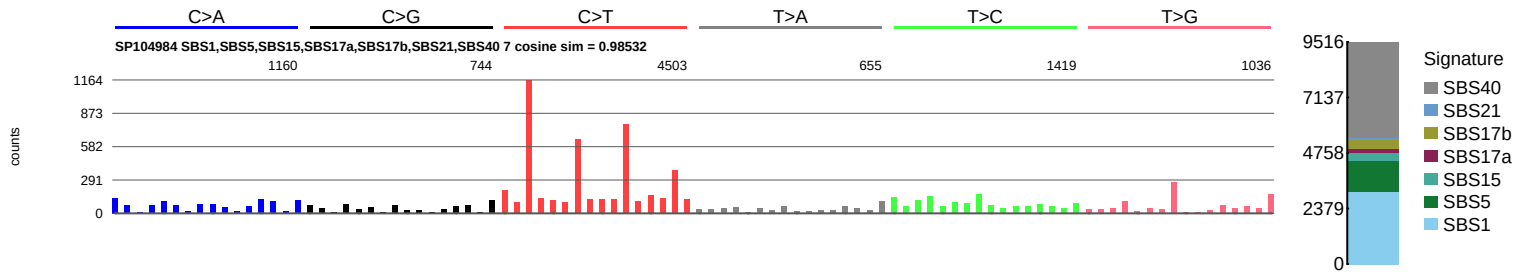
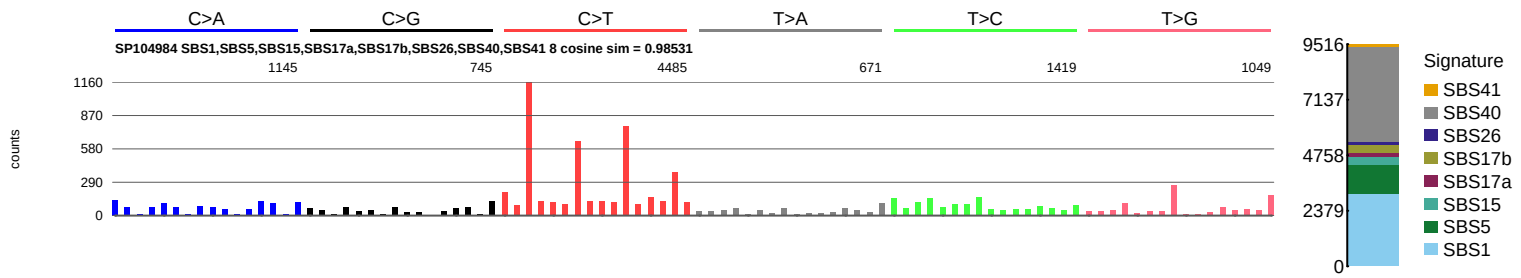
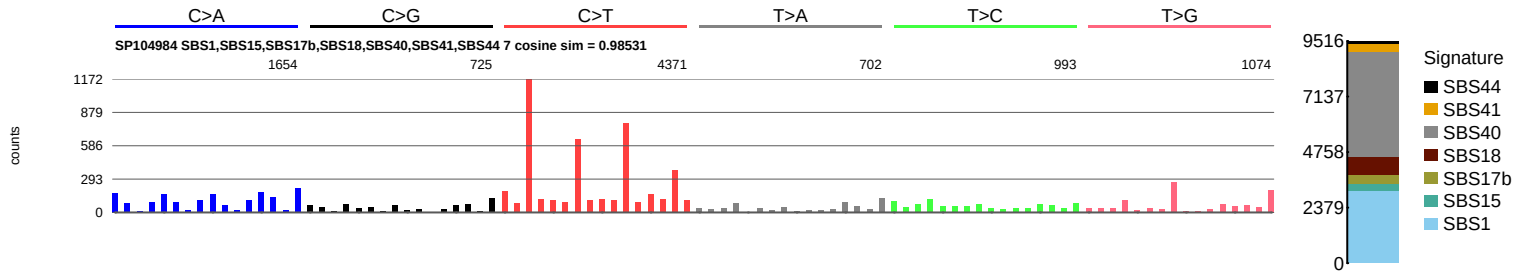
SBS5
SBS2
SBS1

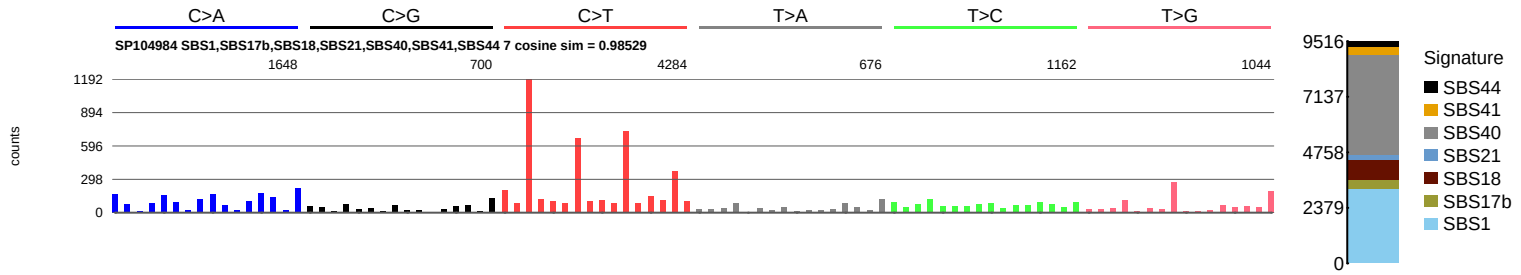

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98535
1671
681
4400
612
1146
1010
1192
894
596
298
0
counts
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1

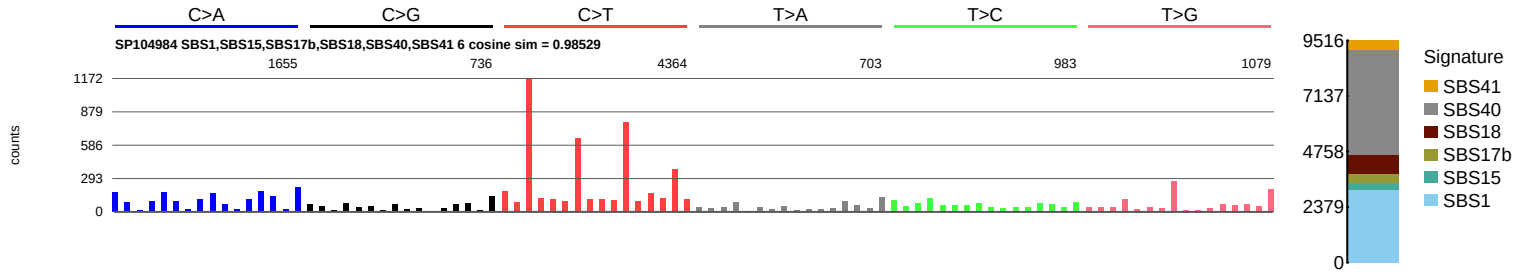

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98533
1142
742
4493
673
1421
1044
1160
870
580
290
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

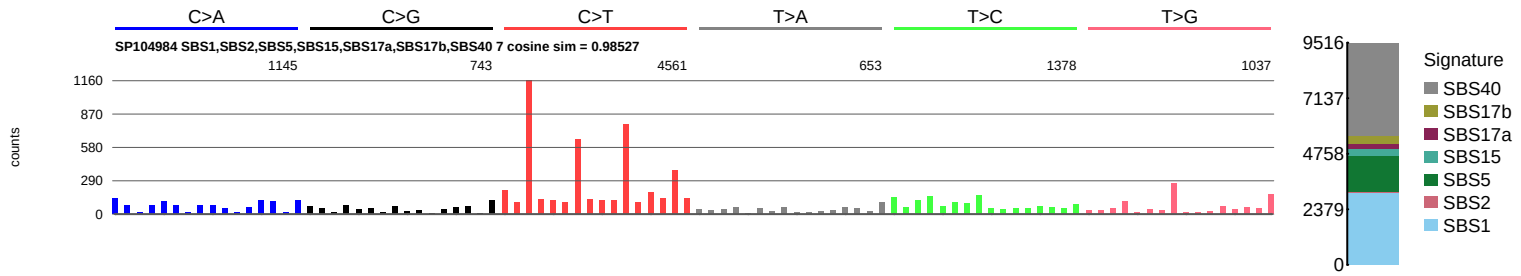

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98532
1160
744
4503
655
1419
1036
1164
873
582
291
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98531
1145
745
4485
671
1419
1049
1160
870
580
290
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98531
1654
725
4371
702
993
1074
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98529
1648
700
4284
676
1162
1044
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98529
1655
736
4364
703
983
1079
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS1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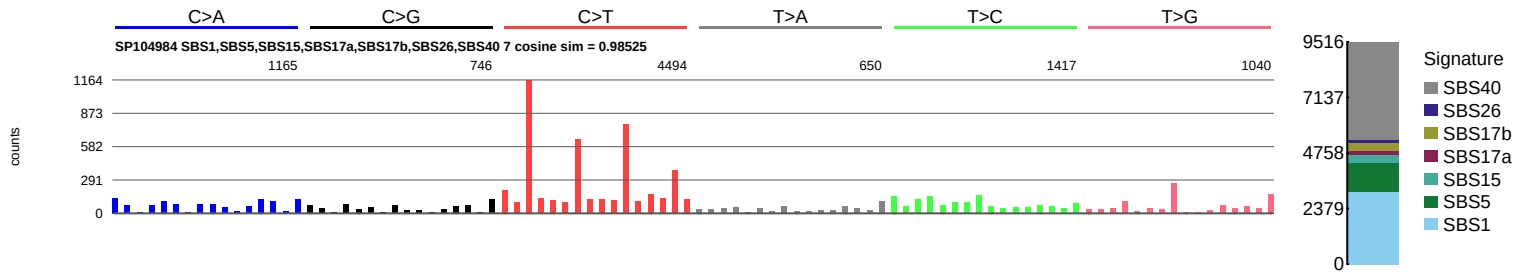
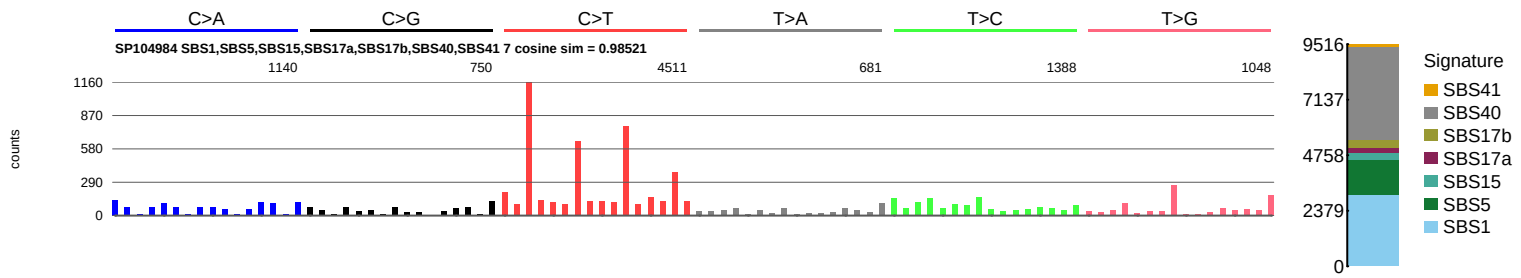
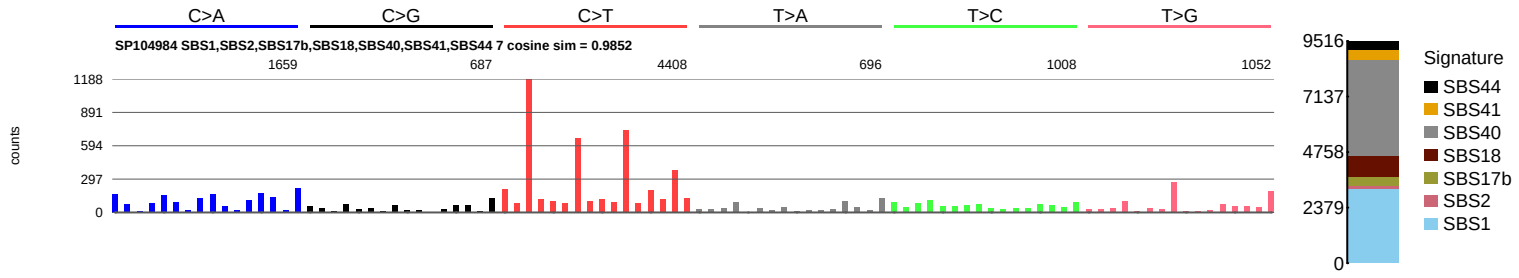
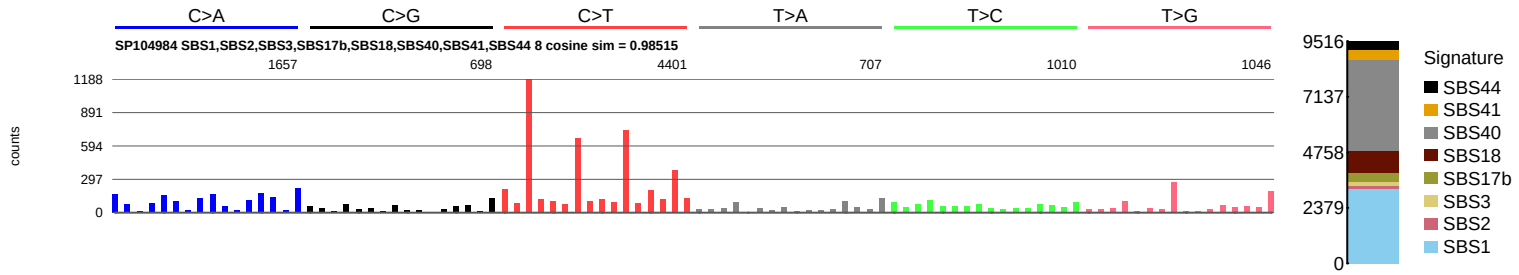
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98527
1145
743
4561
653
1378
1037
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98525
1165
746
4494
650
1417
1040
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98521
1140
750
4511
681
1388
1048
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.9852
1659
687
4408
696
1008
1052
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98515
1657
698
4401
707
1010
1046
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
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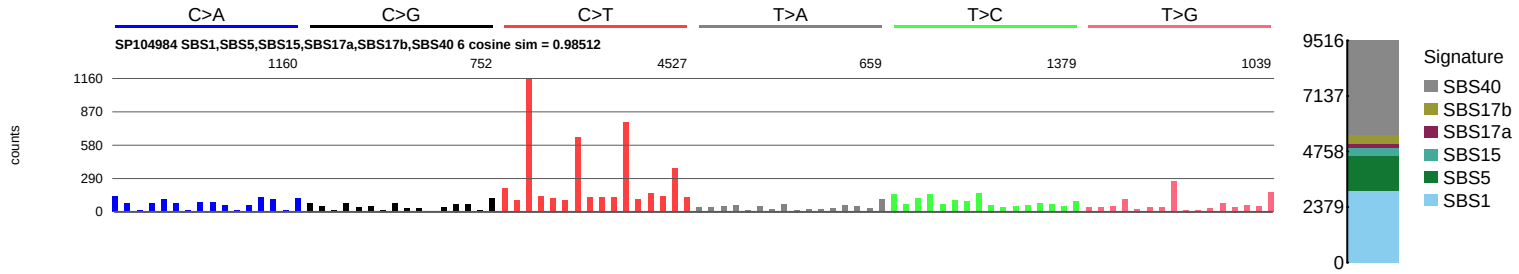
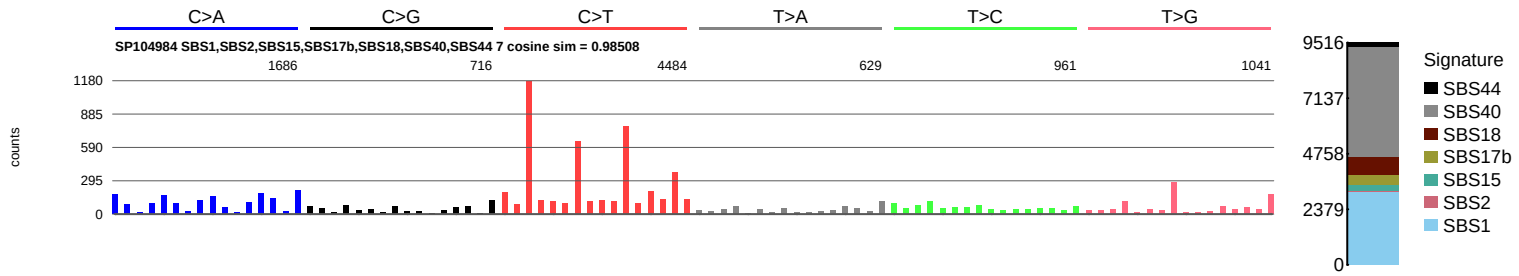
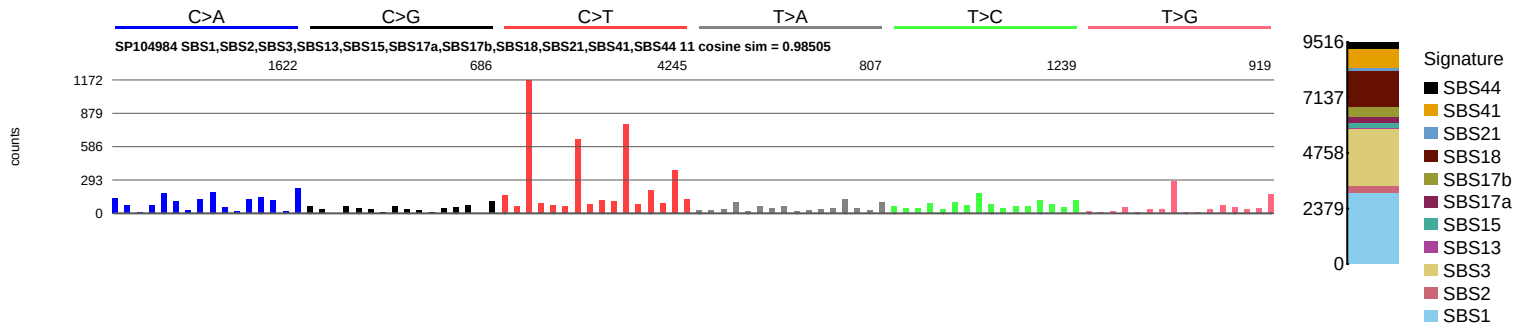
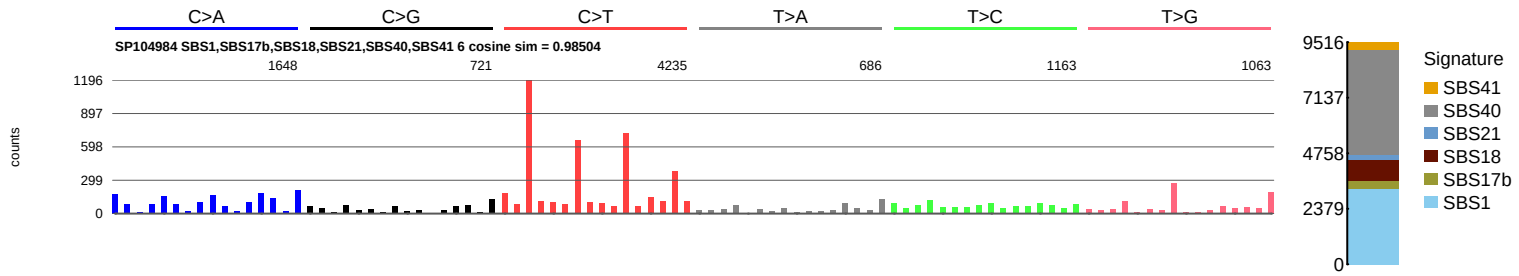
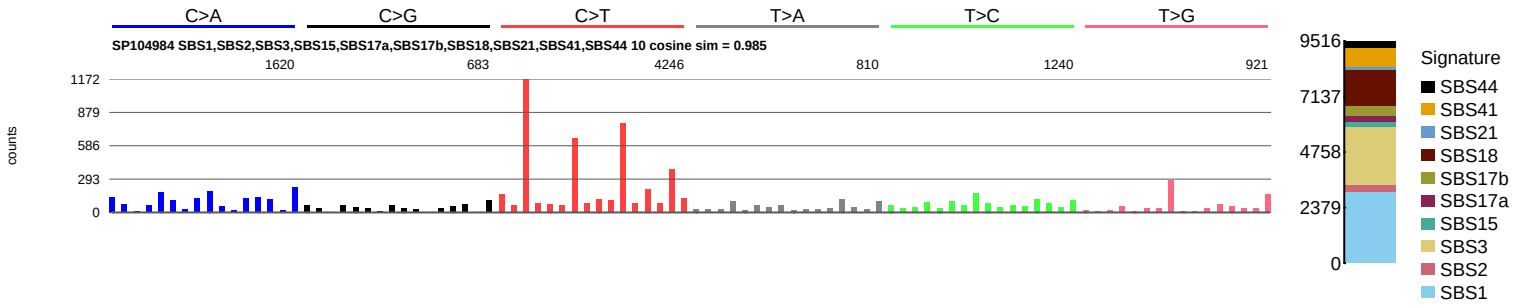
SBS3
SBS2
SBS1

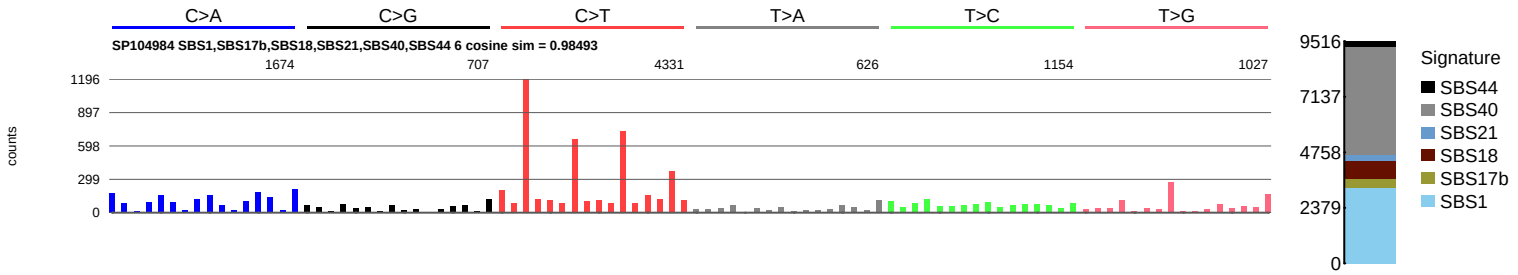

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98512
1160
752
4527
659
1379
1039
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98508
1686
716
4484
629
961
1041
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98505
1622
686
4245
807
1239
919
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98504
1648
721
4235
686
1163
1063
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.985
1620
683
4246
810
1240
921
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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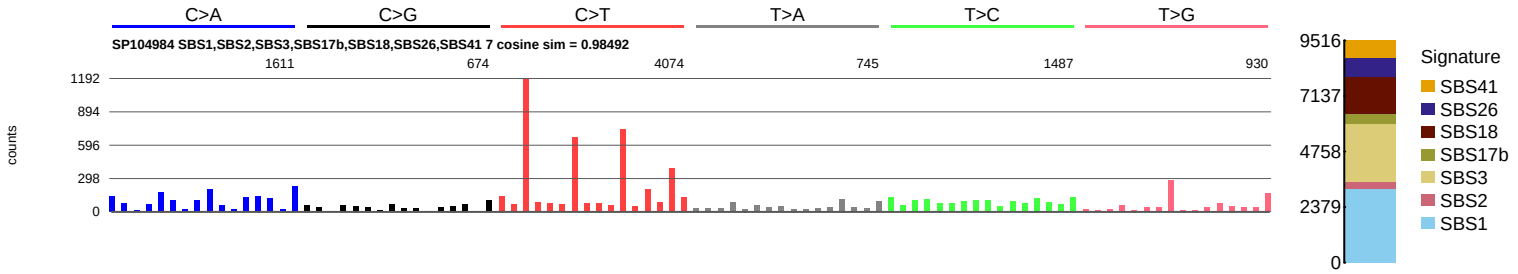
T>G
SP104984 SBS1,SBS17b,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.98493
1674
707
4331
626
1154
1027
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS1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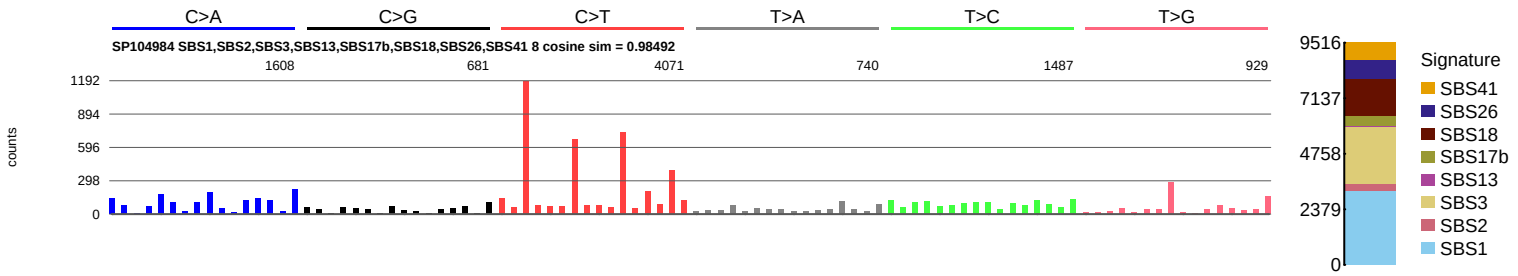
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98492
1611
674
4074
745
1487
930
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1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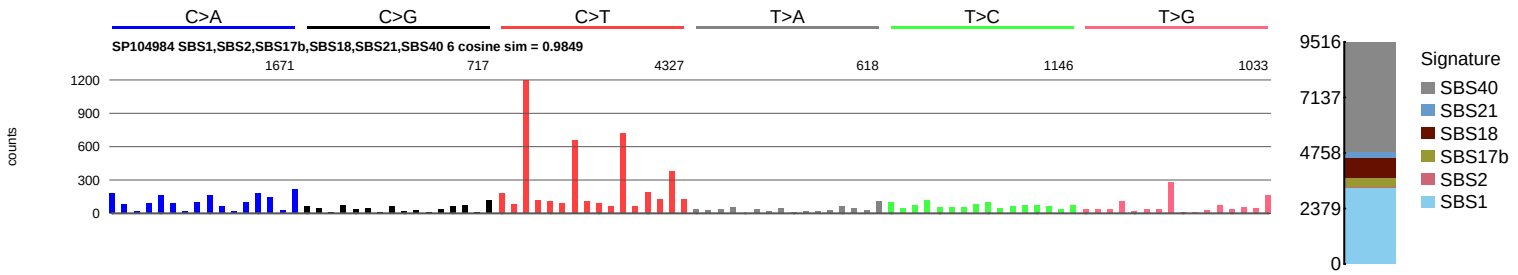
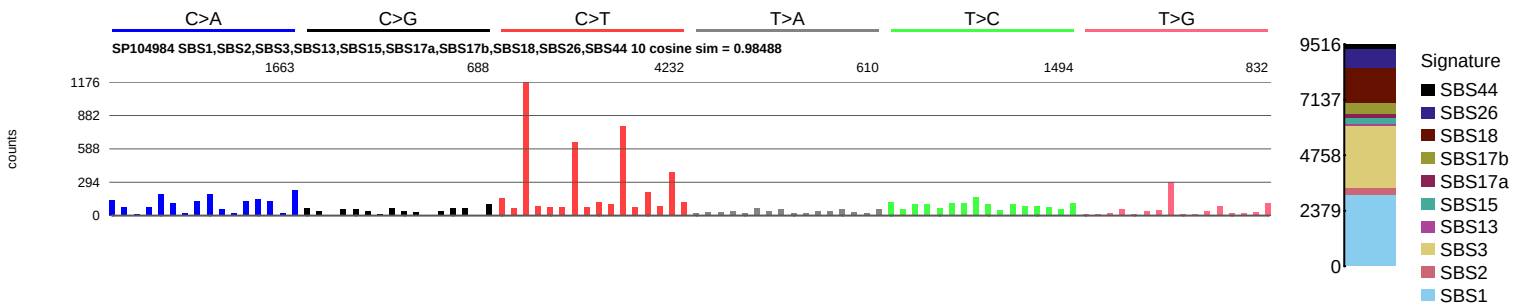
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98492
1608
681
4071
740
1487
929
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.9849
1671
717
4327
618
1146
1033
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1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C>G
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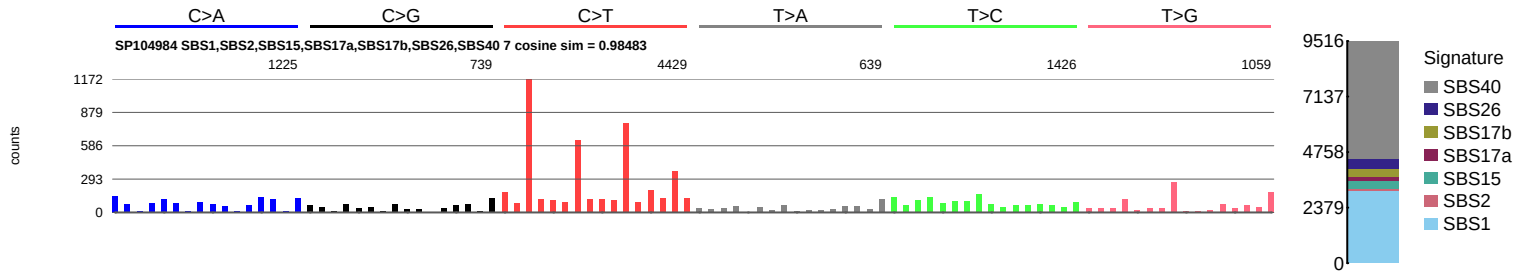
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98488
1663
688
4232
610
1494
832
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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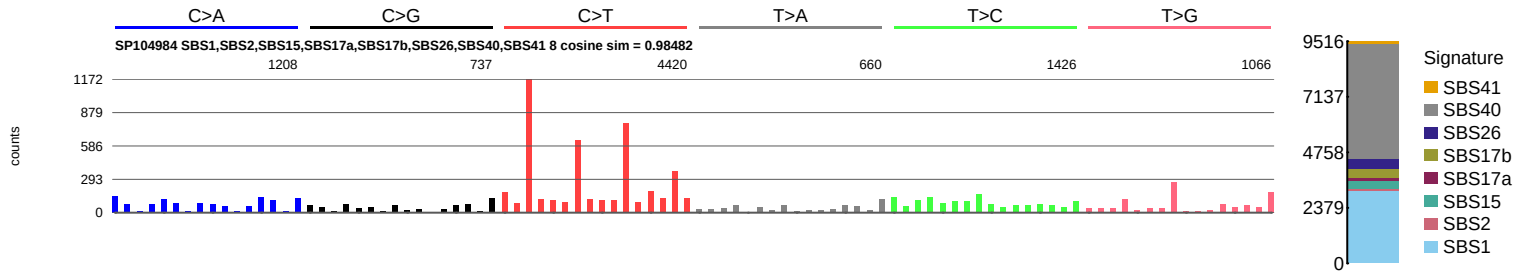
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98483
1225
739
4429
639
1426
1059
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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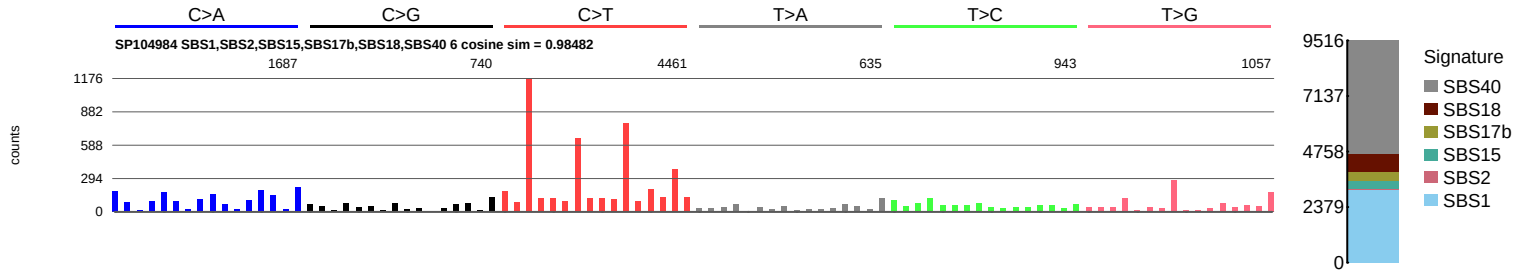
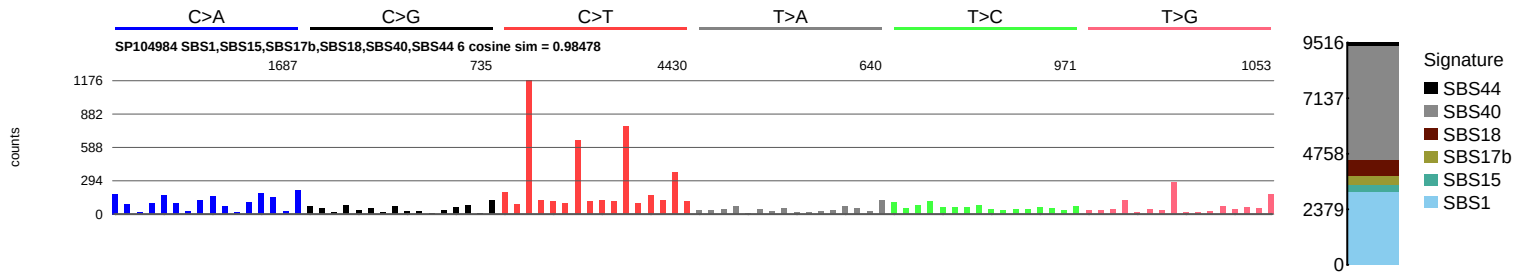
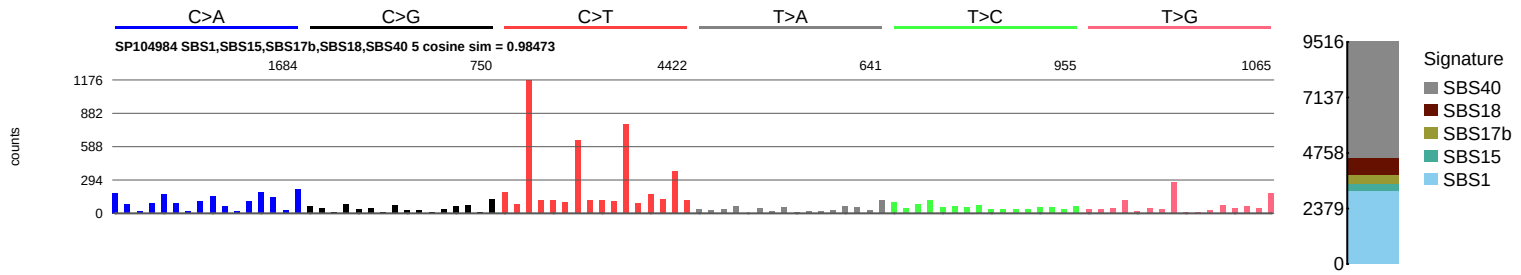
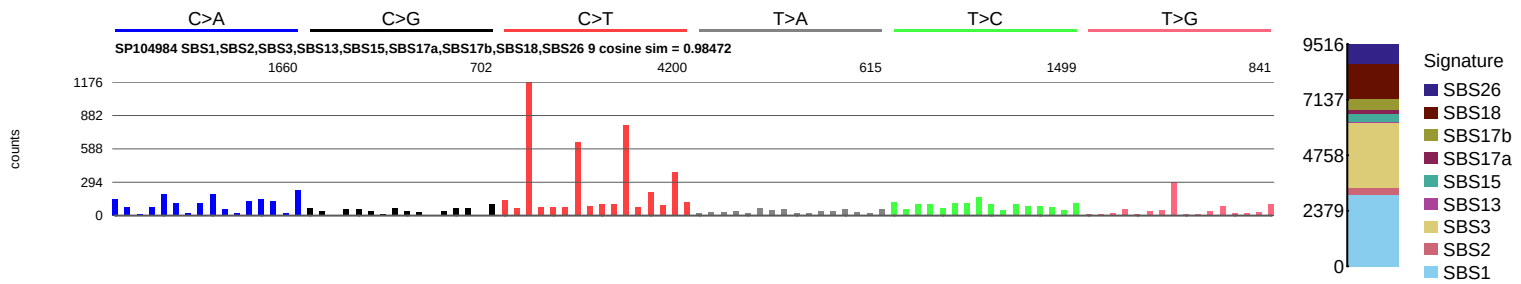
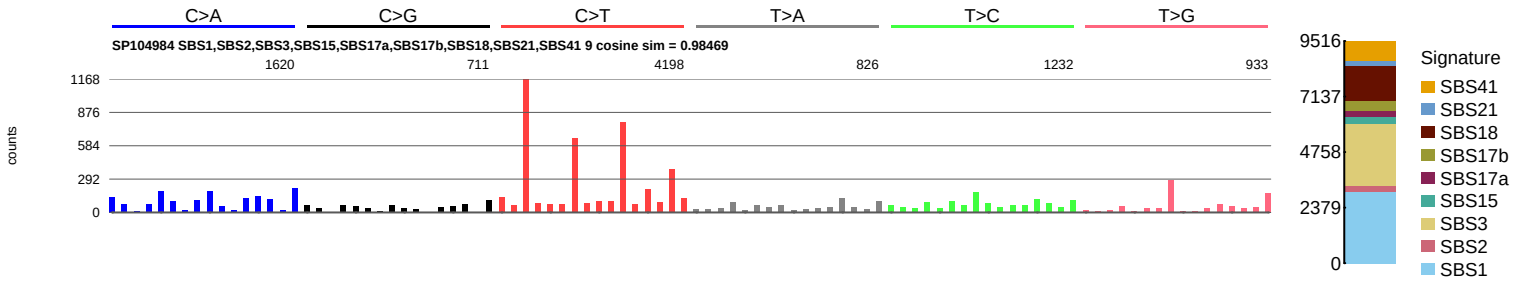
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98482
1208
737
4420
660
1426
1066
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40 6 cosine sim = 0.98482
1687
740
4461
635
943
1057
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98478
1687
735
4430
640
971
1053
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS18,SBS40 5 cosine sim = 0.98473
1684
750
4422
641
955
1065
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98472
1660
702
4200
615
1499
841
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98469
1620
711
4198
826
1232
933
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
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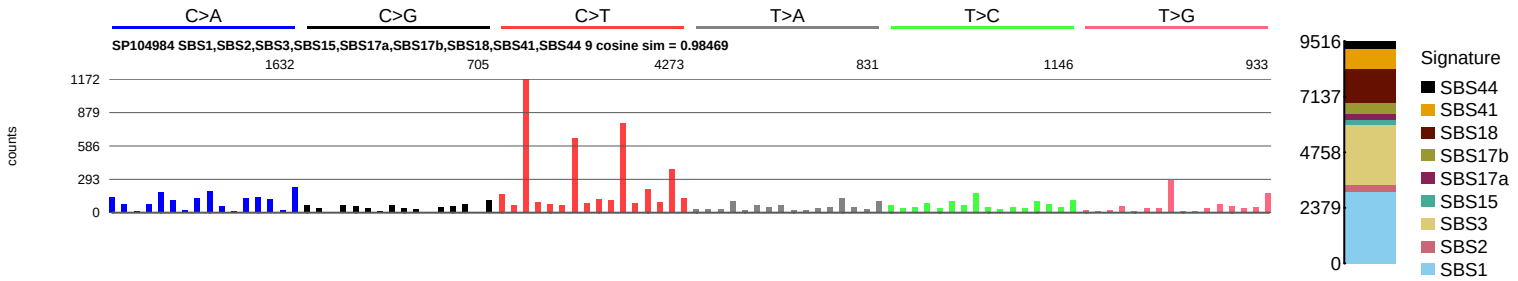
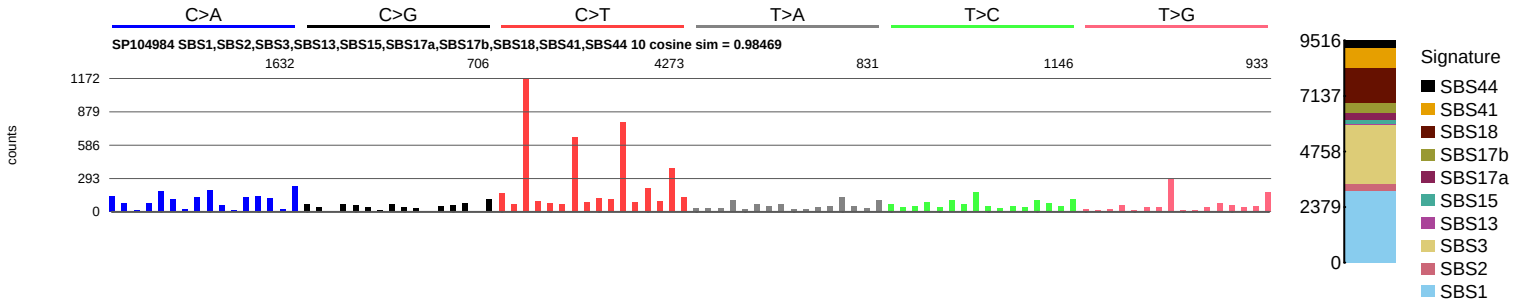
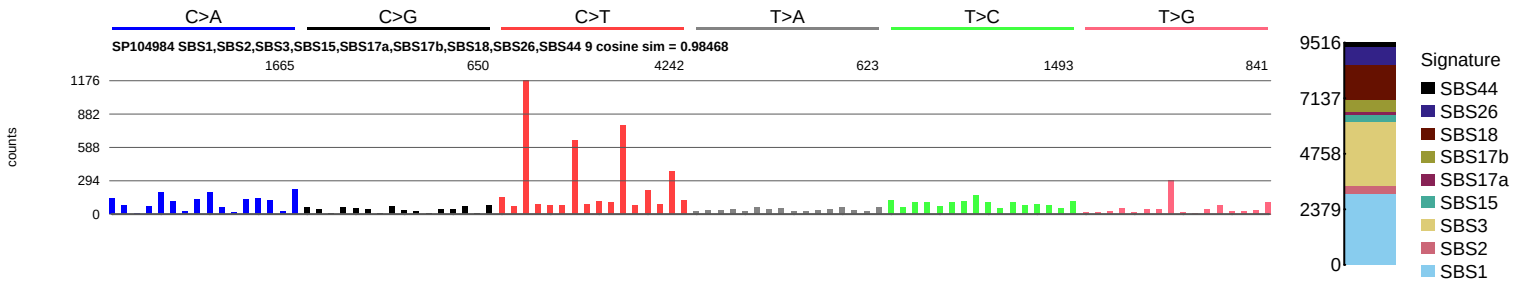
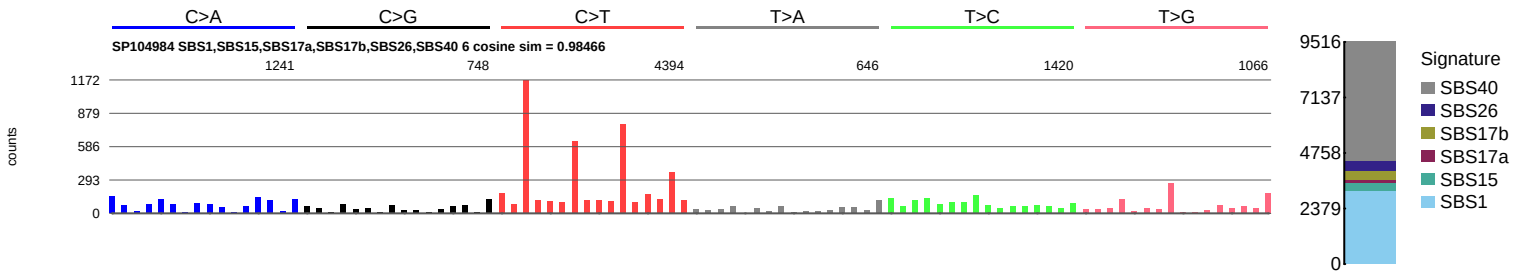
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98469
1632
705
4273
831
1146
933
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98469
1632
706
4273
831
1146
933
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98468
1665
650
4242
623
1493
841
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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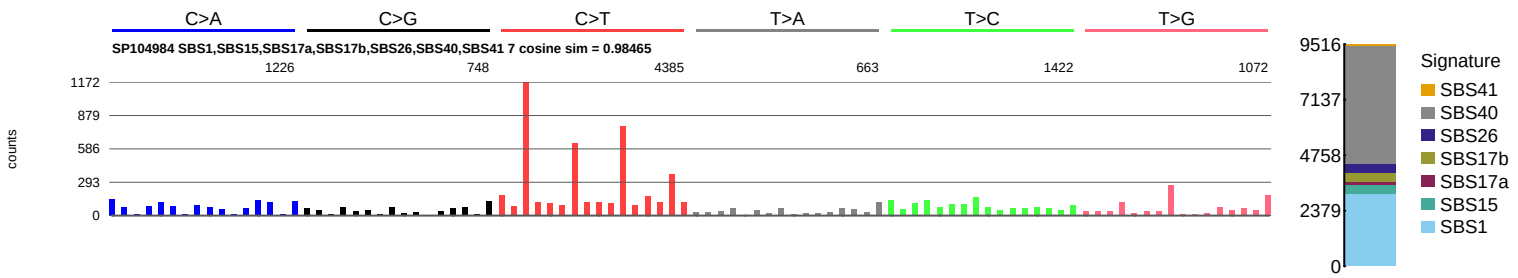
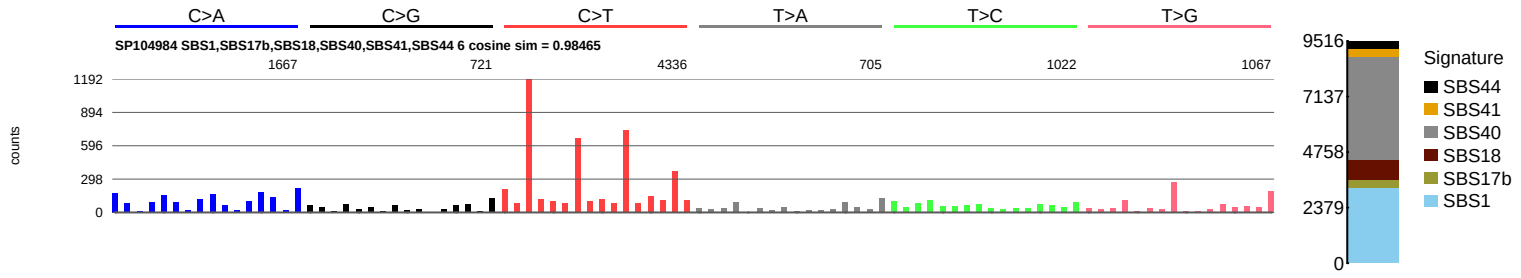
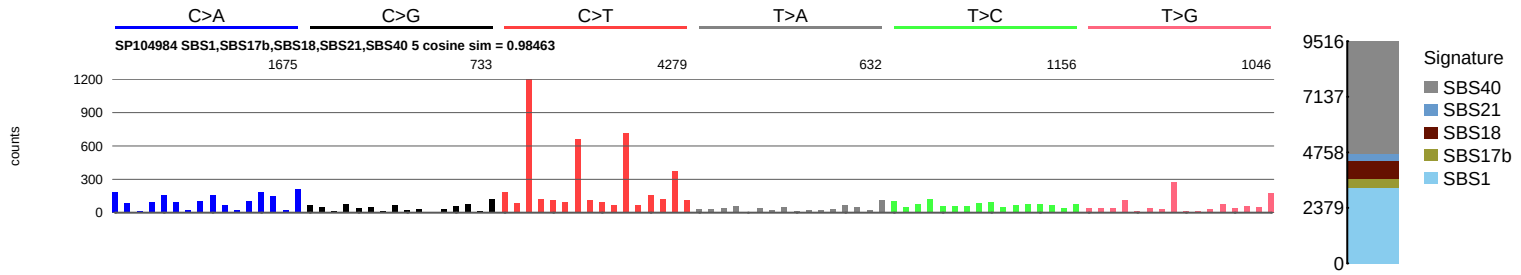
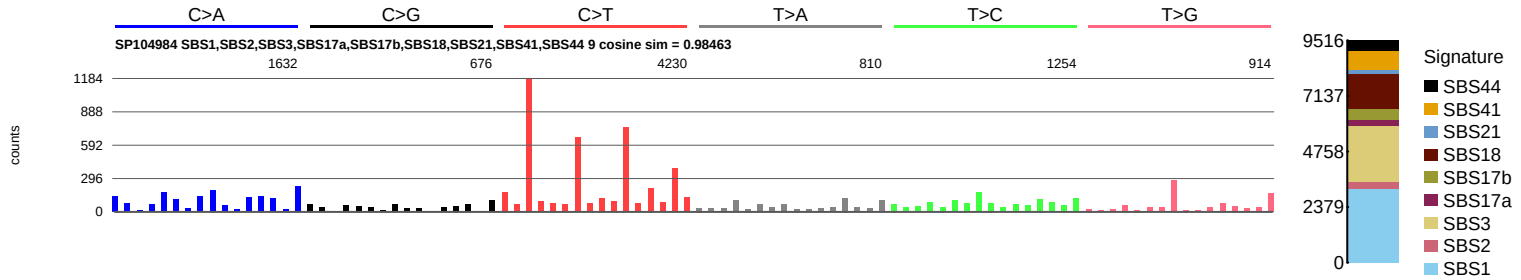
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98466
1241
748
4394
646
1420
1066
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98465
1226
748
4385
663
1422
1072
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS40,SBS41,SBS44 6 cosine sim = 0.98465
1667
721
4336
705
1022
1067
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS21,SBS40 5 cosine sim = 0.98463
1675
733
4279
632
1156
1046
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS1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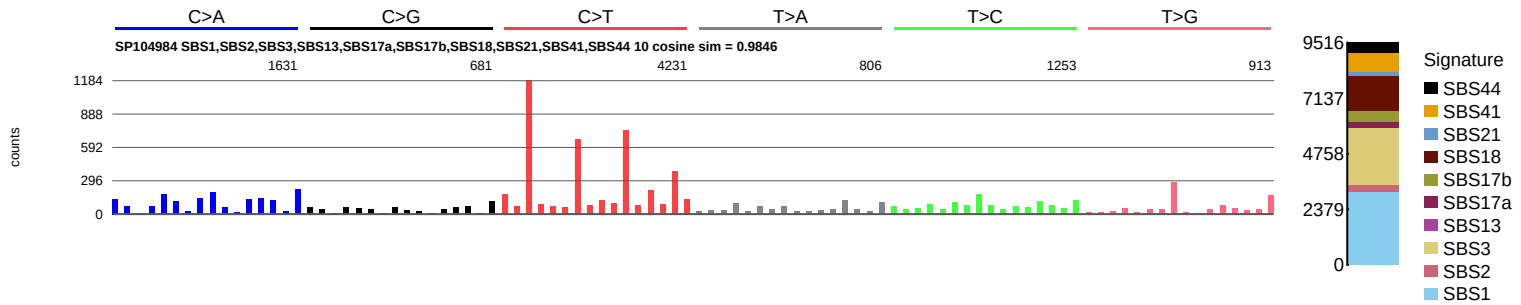
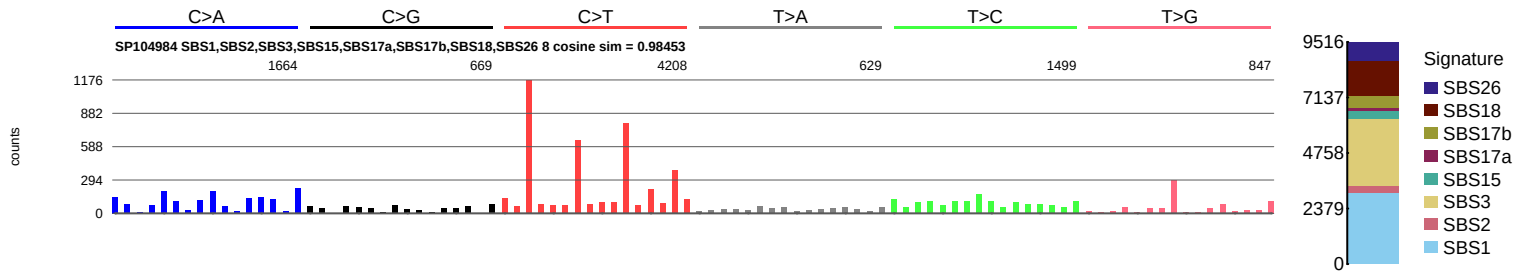
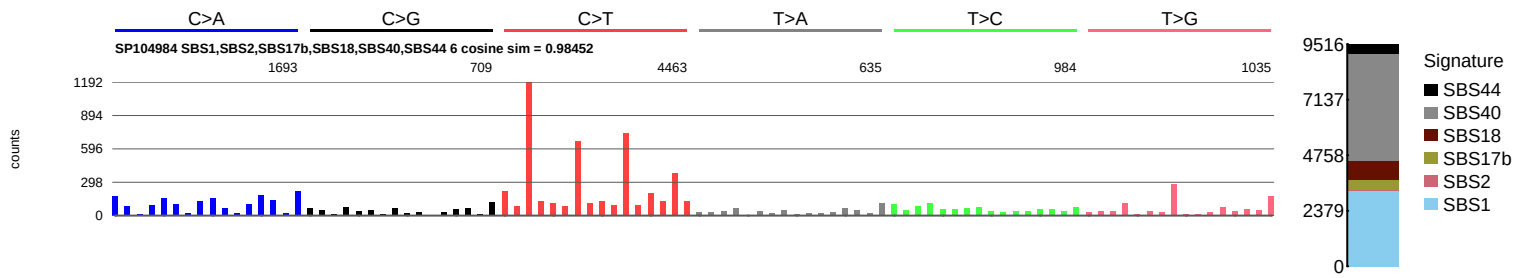
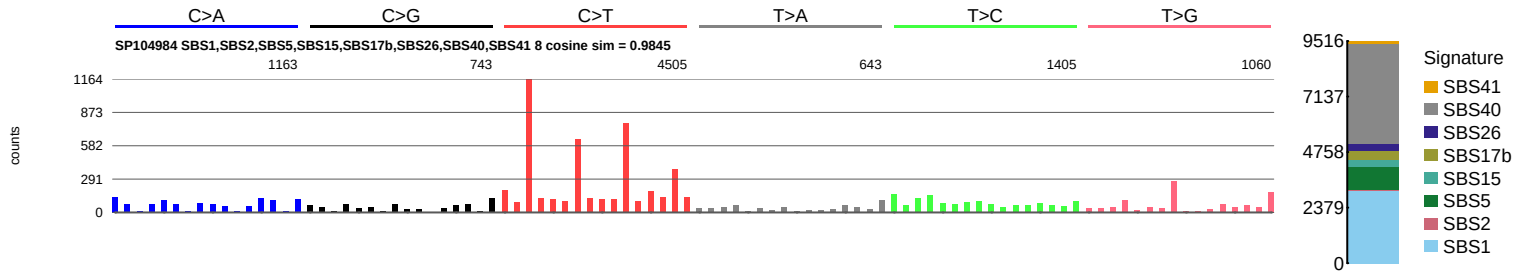
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98463
1632
676
4230
810
1254
914
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9846
1631
681
4231
806
1253
913
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98453
1664
669
4208
629
1499
847
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98452
1693
709
4463
635
984
1035
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.9845
1163
743
4505
643
1405
1060
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
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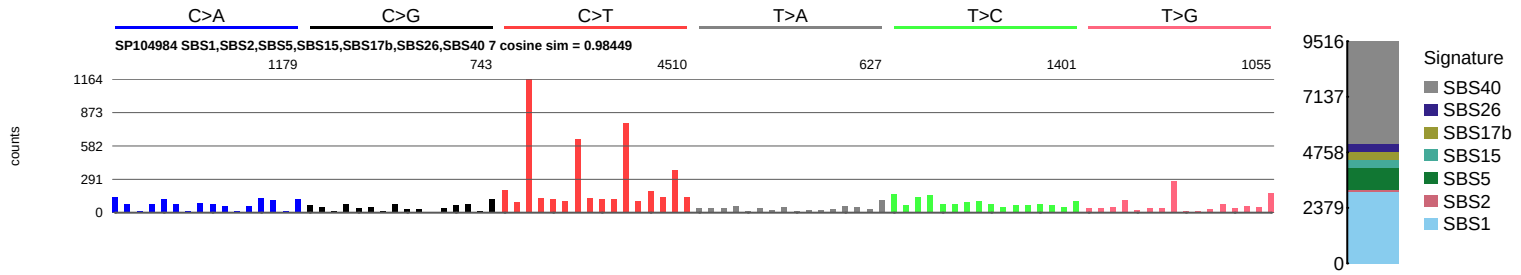
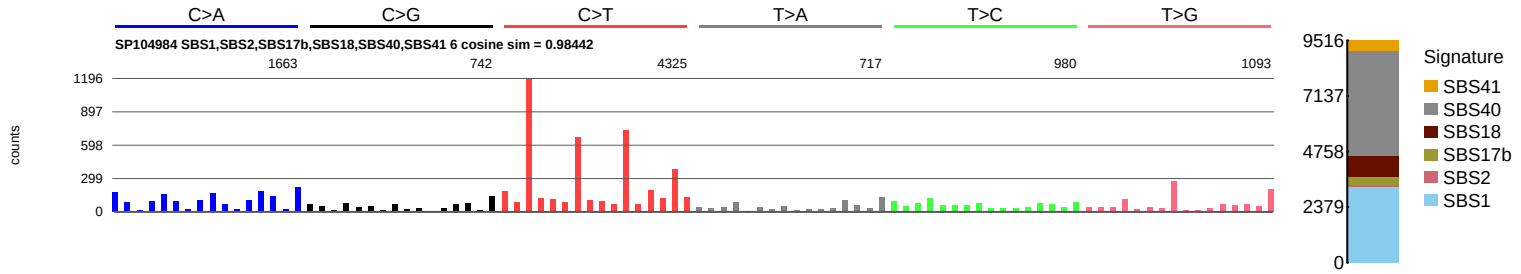
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS26,SBS40 7 cosine sim = 0.98449
1179
743
4510
627
1401
1055
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS2
SBS1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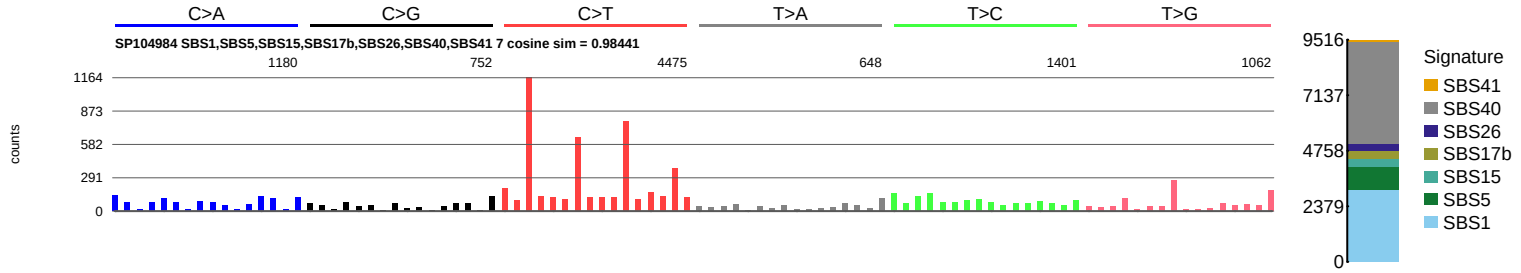
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98442
1663
742
4325
717
980
1093
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS2
SBS1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C>G
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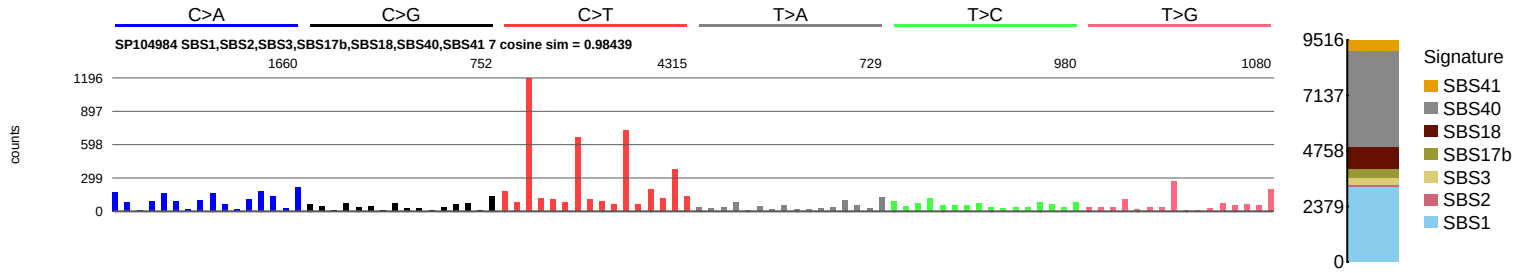
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98441
1180
752
4475
648
1401
1062
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS1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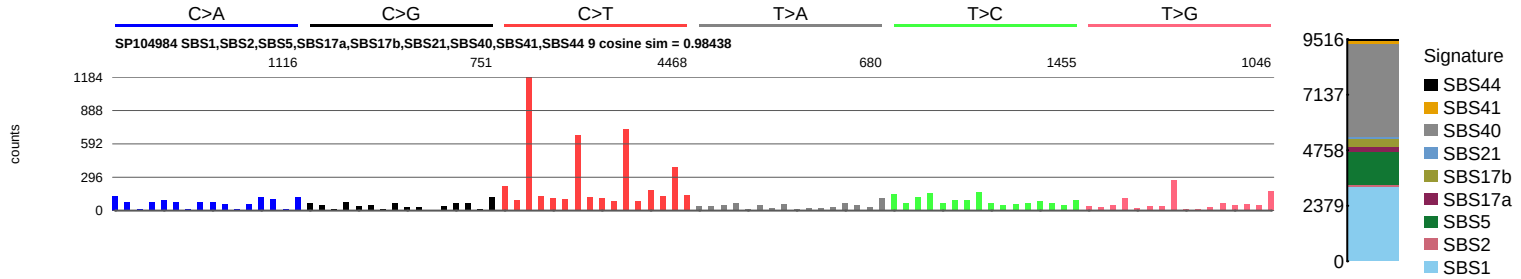
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98439
1660
752
4315
729
980
1080
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1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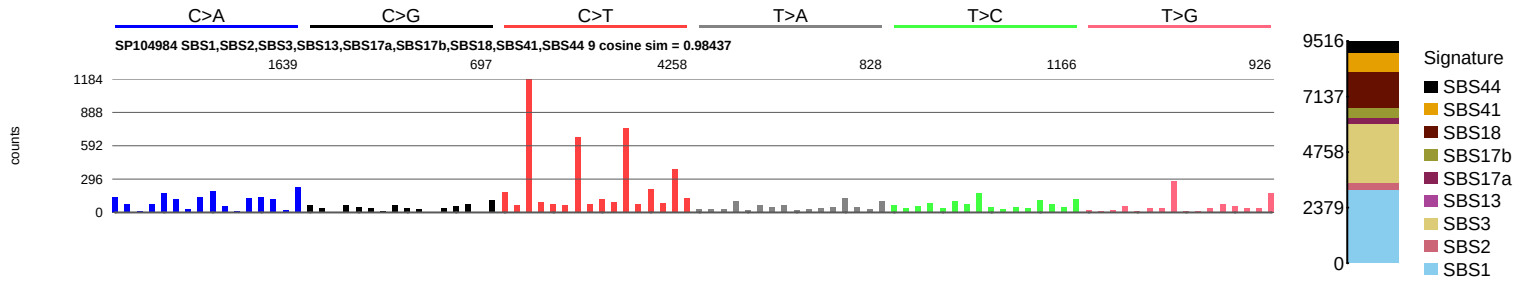
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98438
1116
751
4468
680
1455
1046
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1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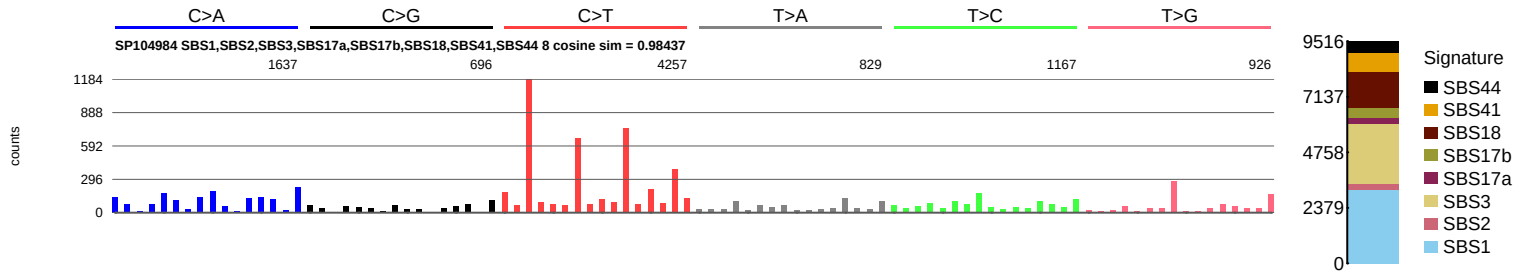
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98437
1639
697
4258
828
1166
926
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98437
1637
696
4257
829
1167
926
counts
1184
888
592
296
0
9516
7137
4758
2379
0
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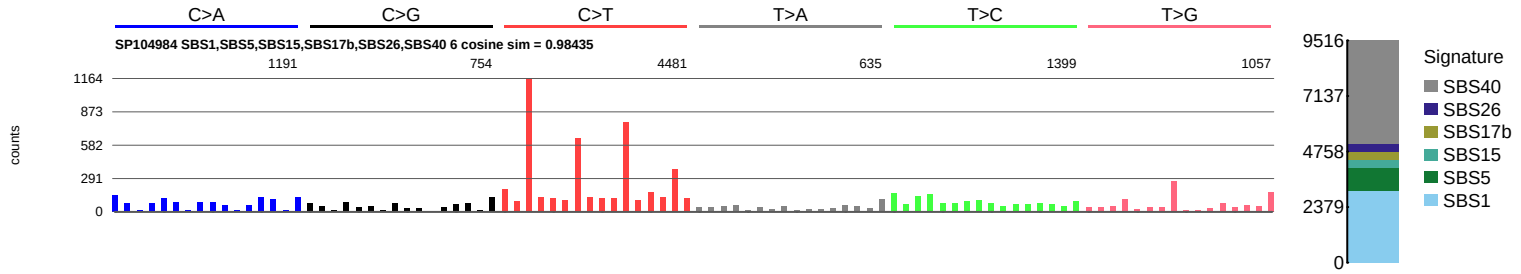
SBS2
SBS1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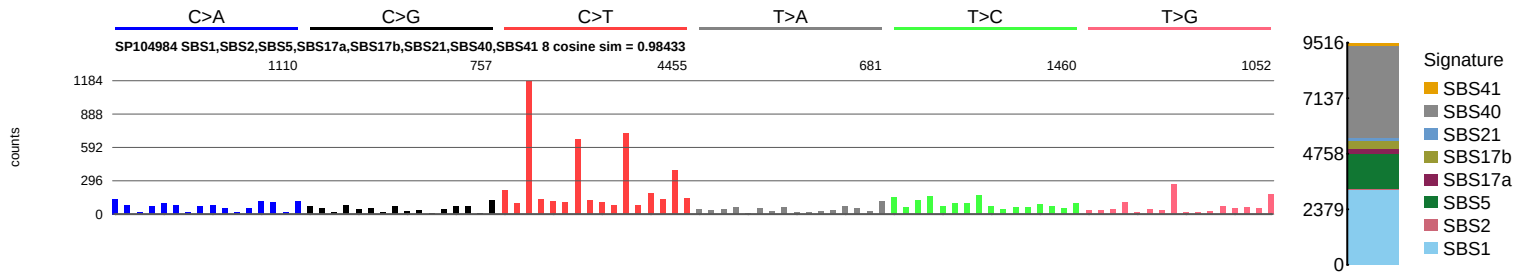
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS26,SBS40 6 cosine sim = 0.98435
1191
754
4481
635
1399
1057
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS1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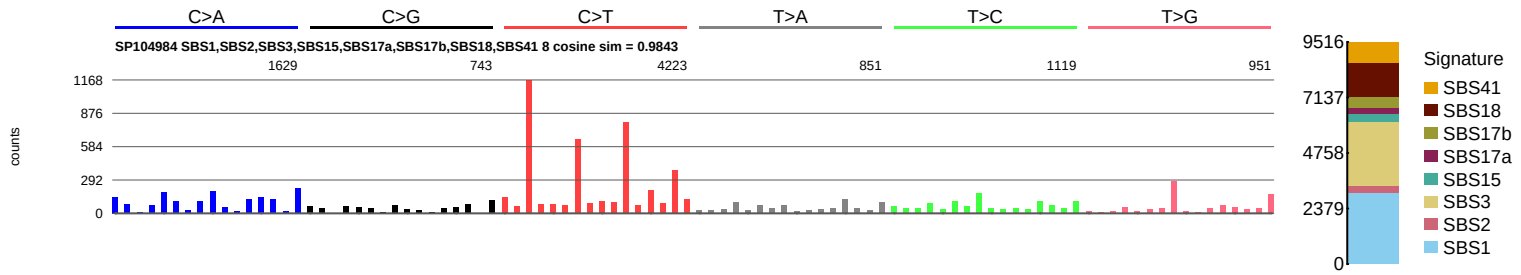
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98433
1110
757
4455
681
1460
1052
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1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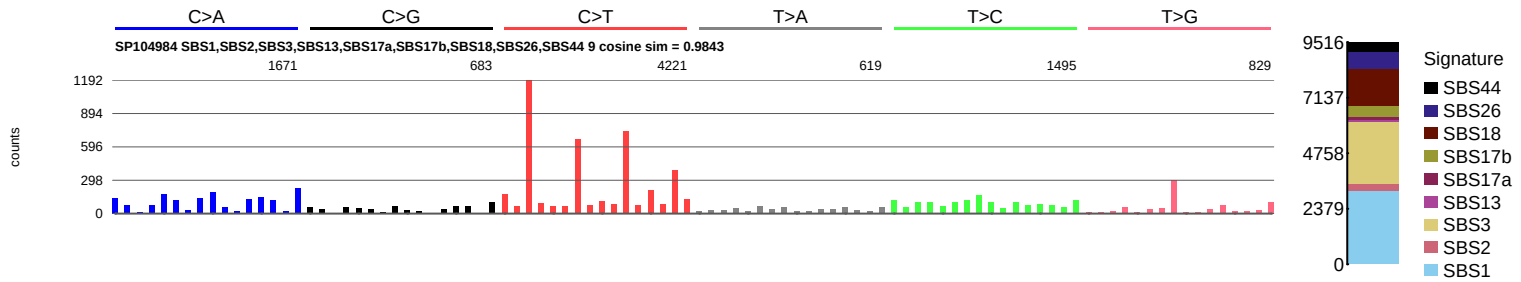
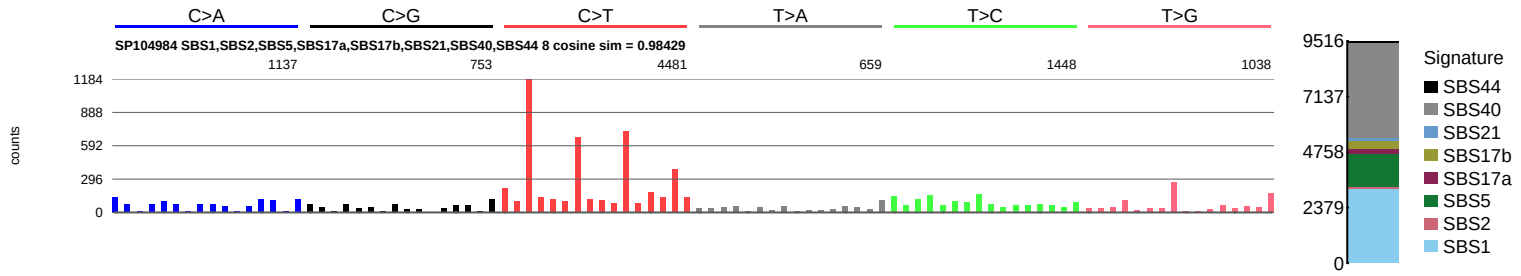
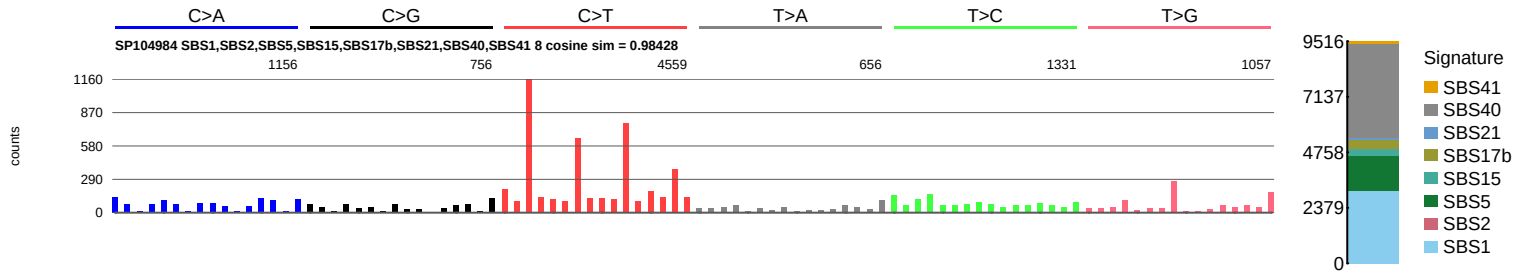
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9843
1629
743
4223
851
1119
951
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.9843
1671
683
4221
619
1495
829
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98429
1137
753
4481
659
1448
1038
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1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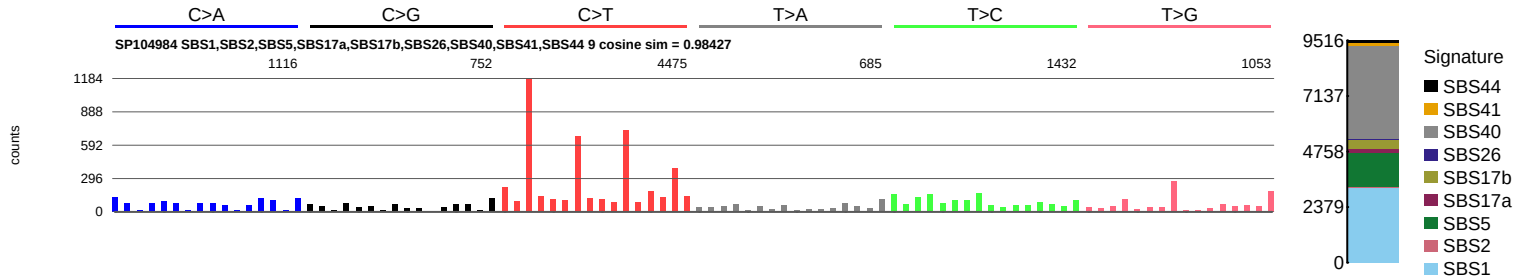
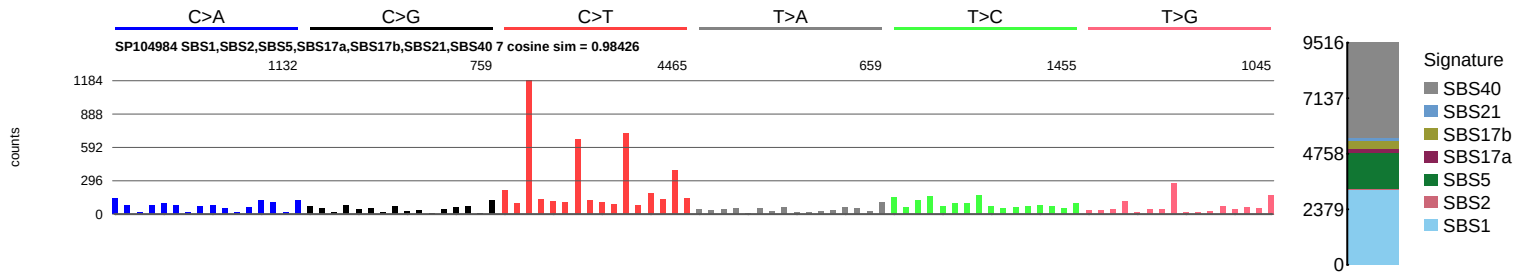
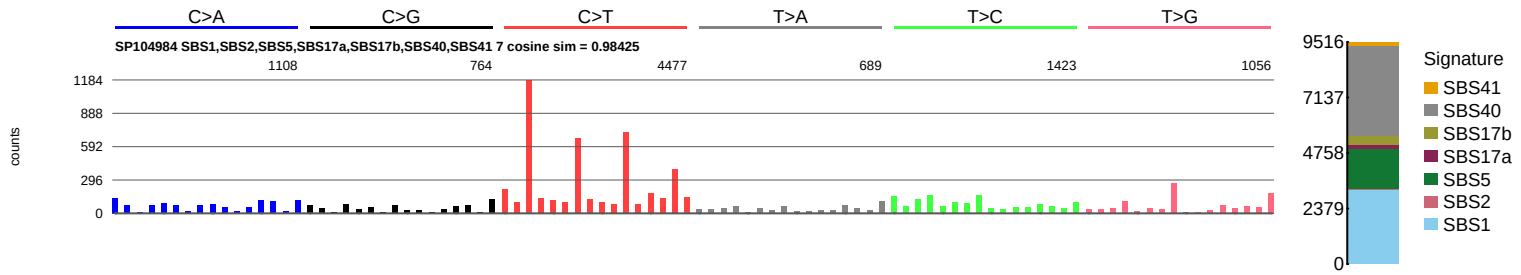
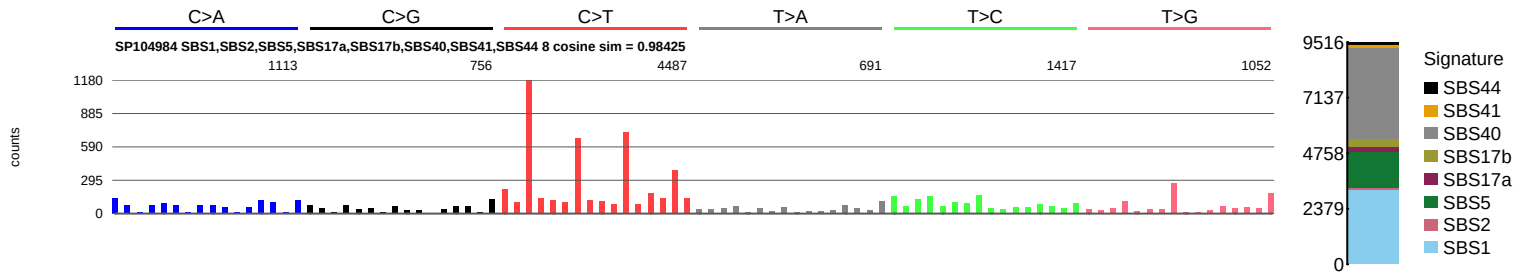
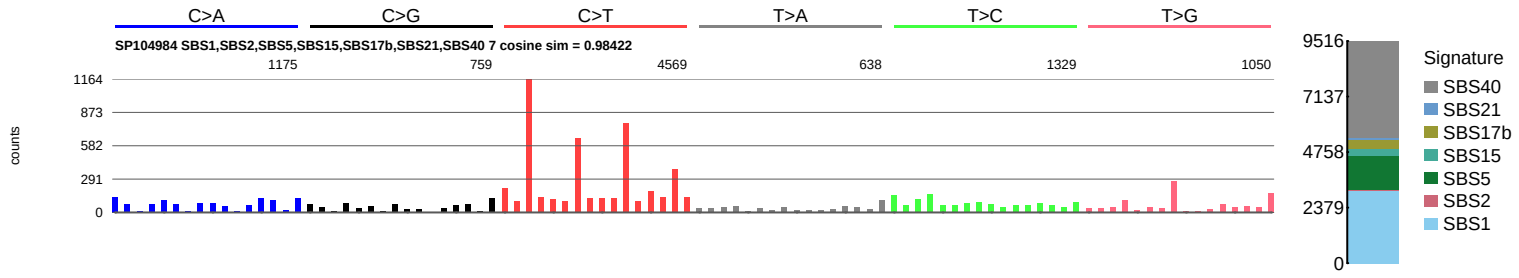
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98428
1156
756
4559
656
1331
1057
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98427
1116
752
4475
685
1432
1053
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98426
1132
759
4465
659
1455
1045
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98425
1108
764
4477
689
1423
1056
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98425
1113
756
4487
691
1417
1052
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1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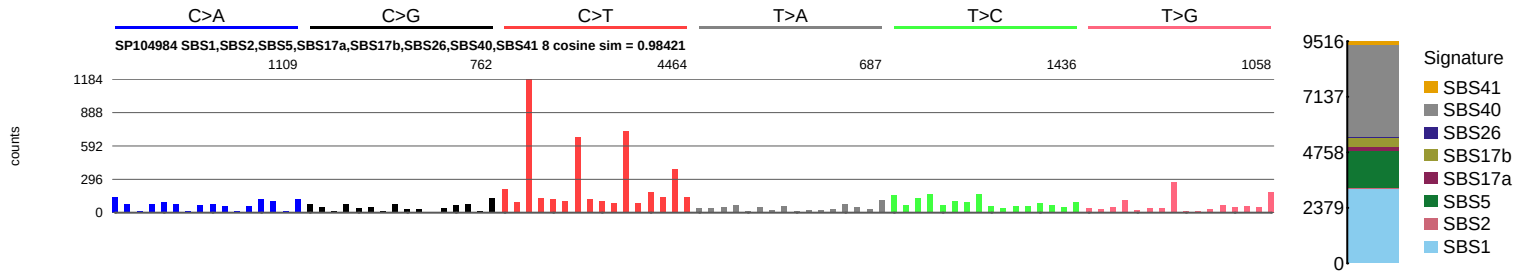
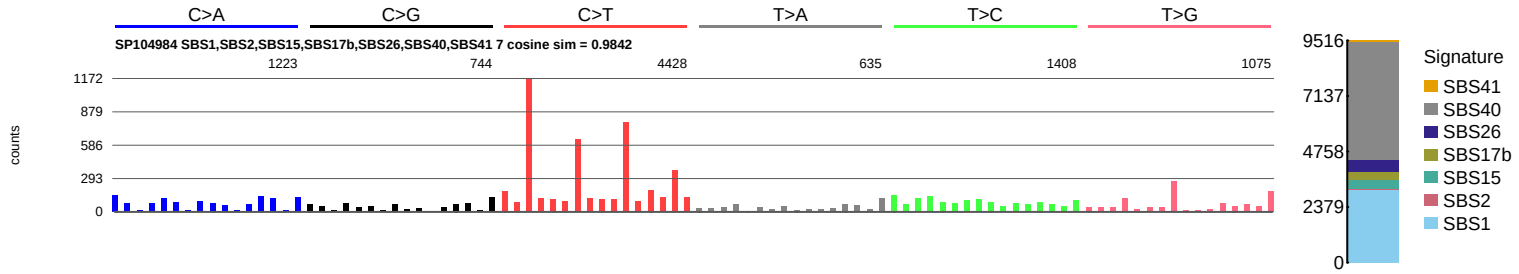
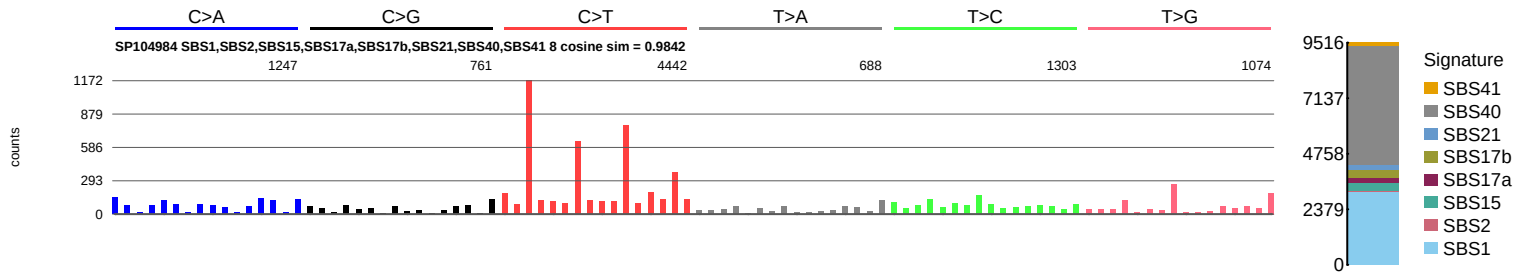
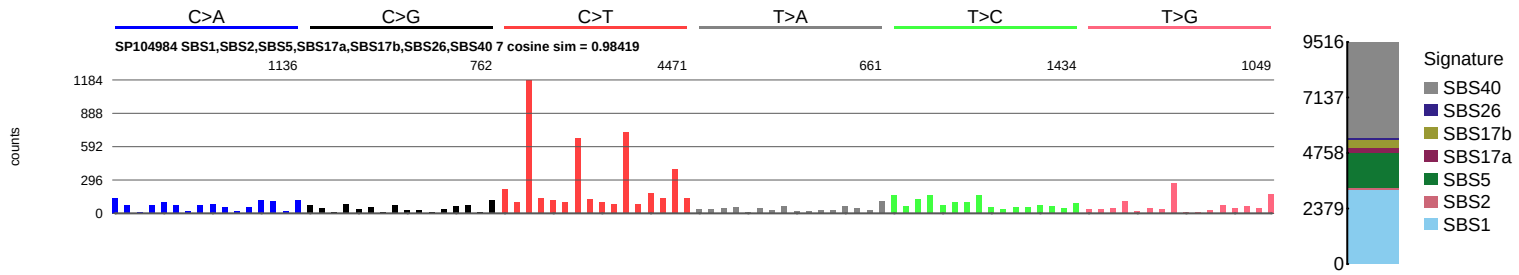
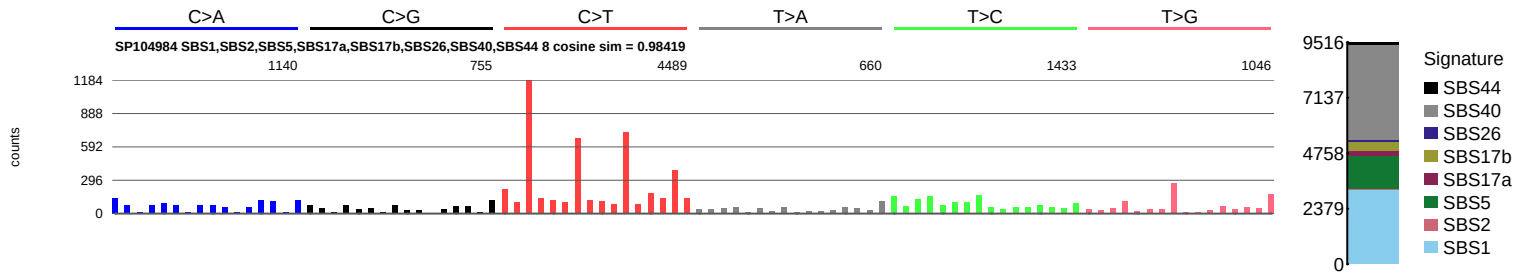
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS21,SBS40 7 cosine sim = 0.98422
1175
759
4569
638
1329
1050
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS5
SBS2
SBS1

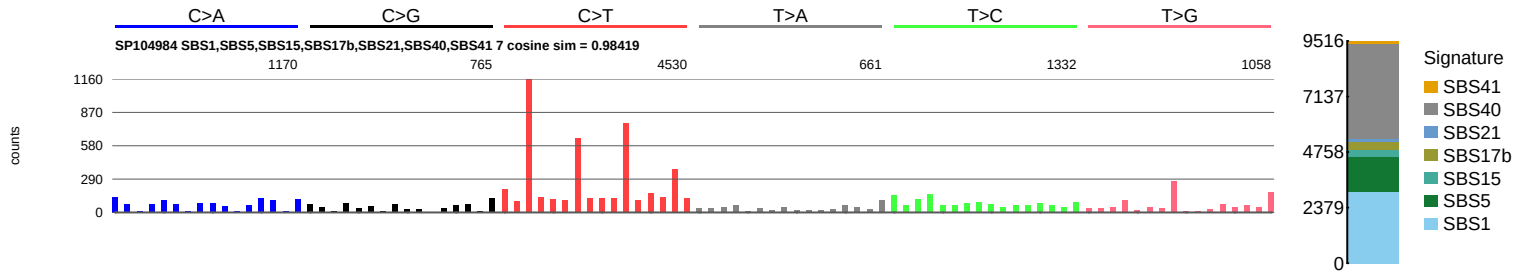

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98421
1109
762
4464
687
1436
1058
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1

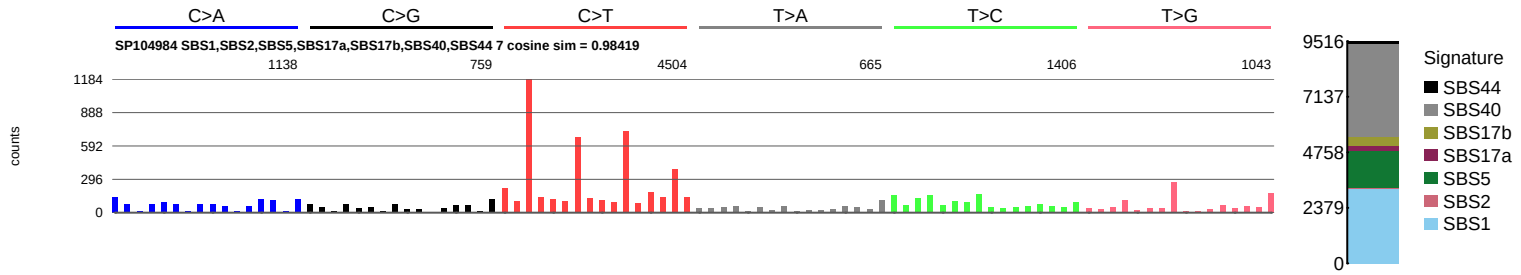

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.9842
1223
744
4428
635
1408
1075
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS2
SBS1

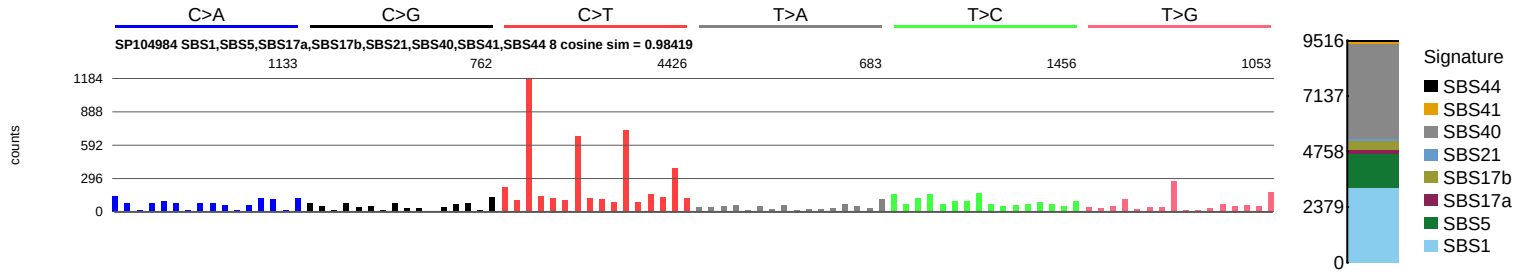

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.9842
1247
761
4442
688
1303
1074
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

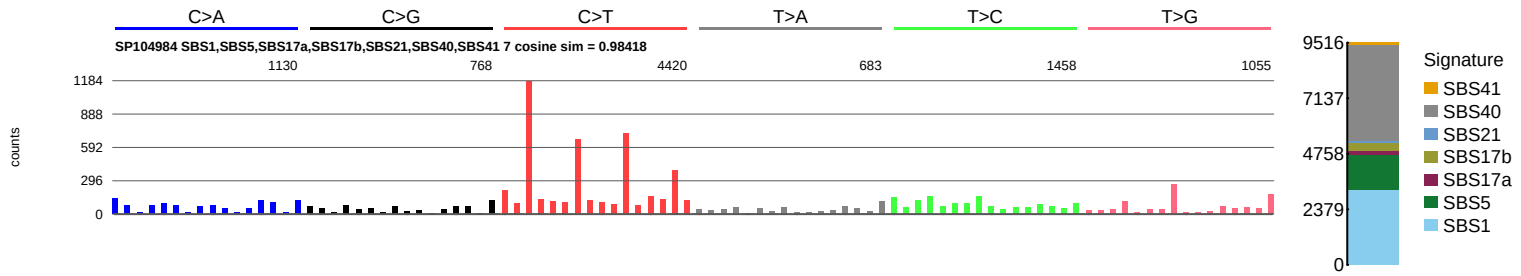

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98419
1136
762
4471
661
1434
1049
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98419
1140
755
4489
660
1433
1046
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98419
1170
765
4530
661
1332
1058
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98419
1138
759
4504
665
1406
1043
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98419
1133
762
4426
683
1456
1053
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
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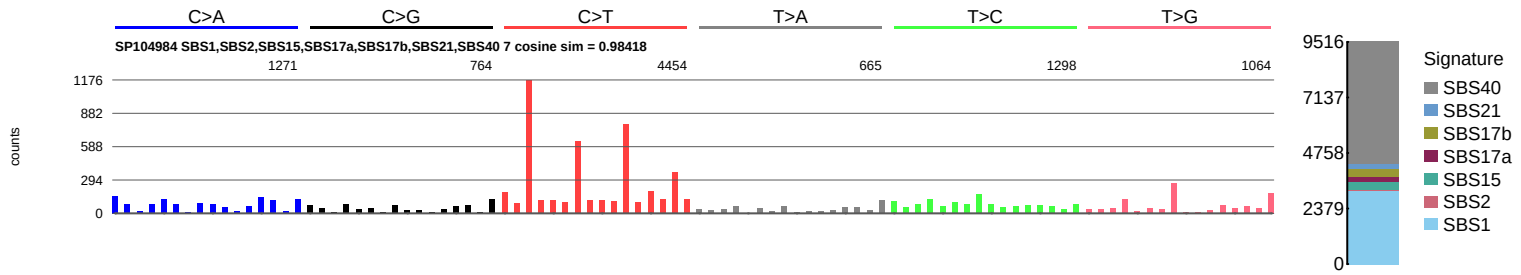
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98418
1130
768
4420
683
1458
1055
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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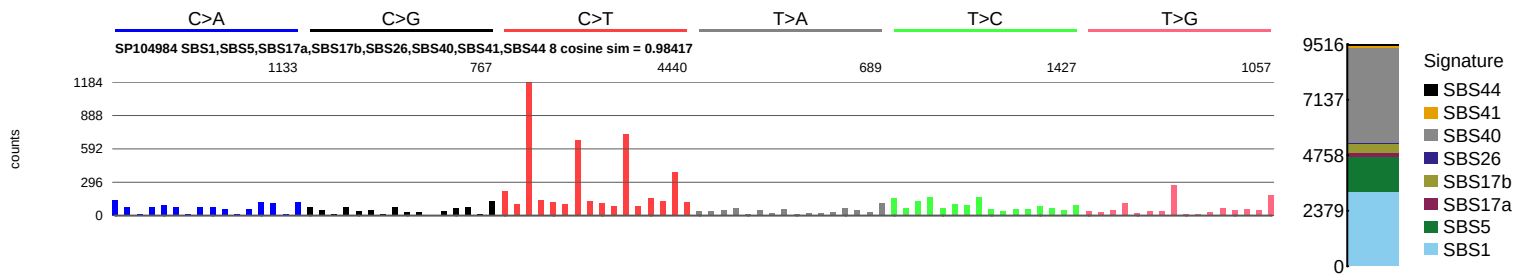
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98418
1271
764
4454
665
1298
1064
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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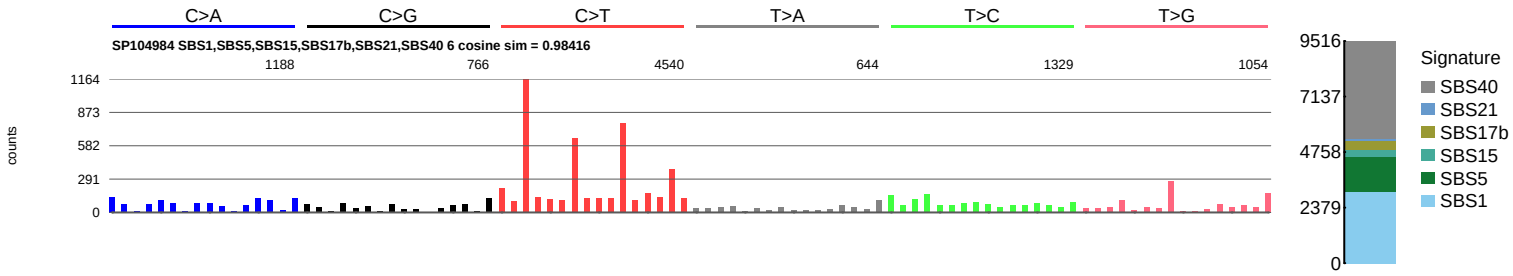
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98417
1133
767
4440
689
1427
1057
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1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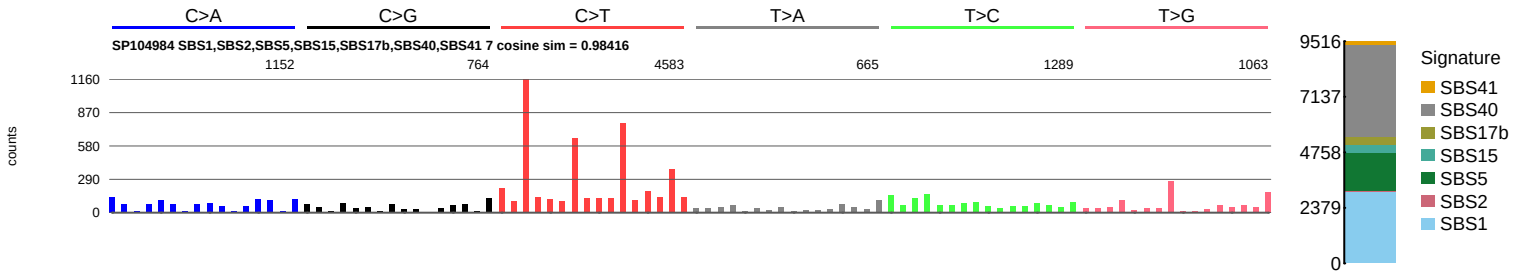
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS21,SBS40 6 cosine sim = 0.98416
1188
766
4540
644
1329
1054
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS5
SBS1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C>G
C>T
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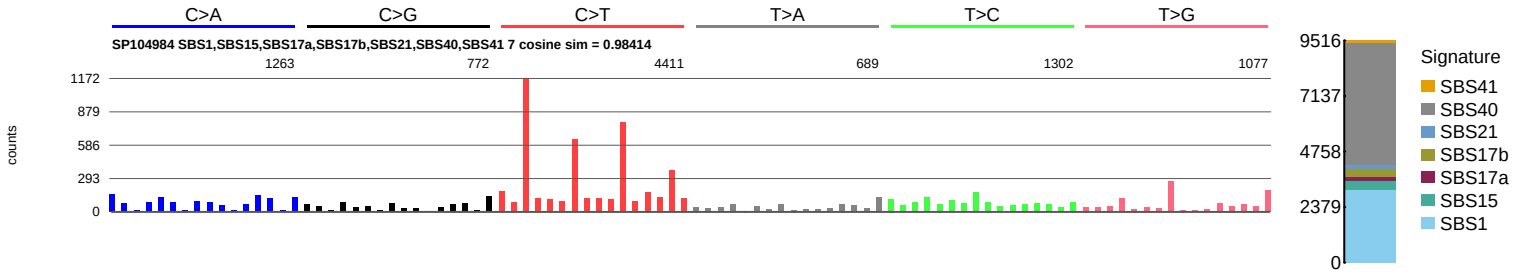
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS40,SBS41 7 cosine sim = 0.98416
1152
764
4583
665
1289
1063
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS5
SBS2
SBS1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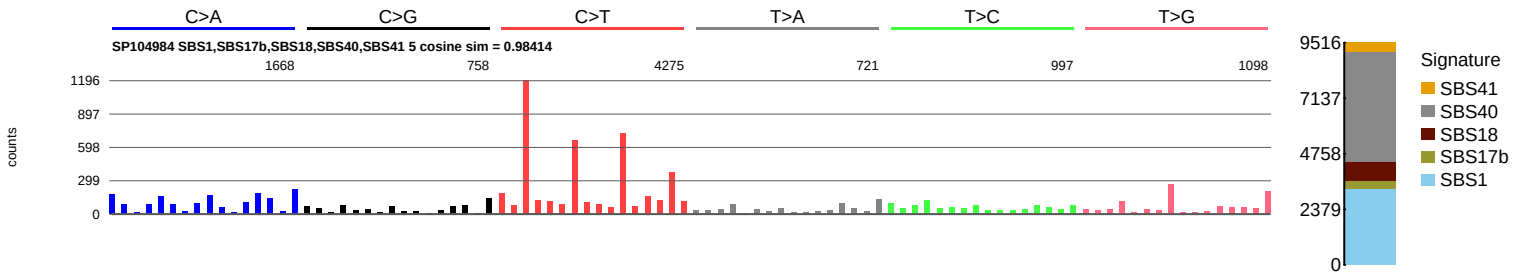
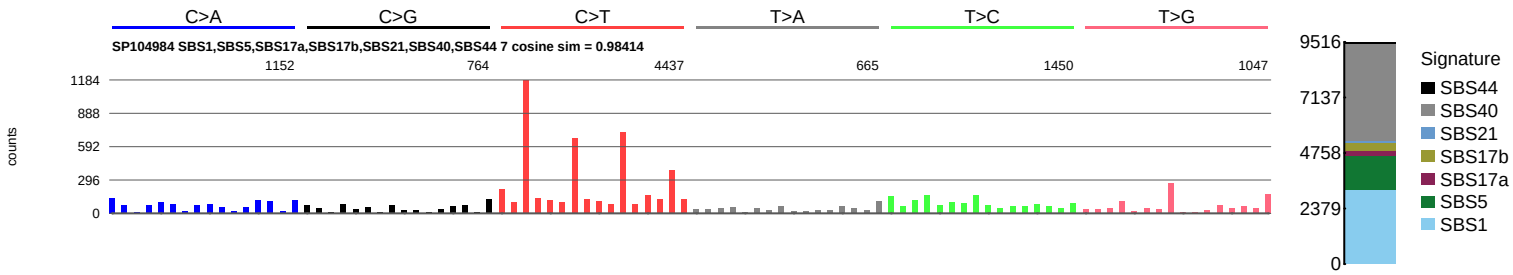
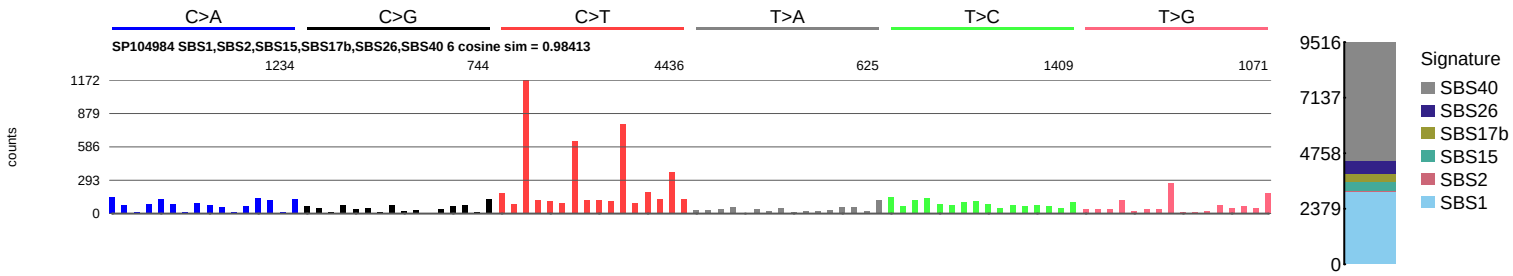
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98414
1263
772
4411
689
1302
1077
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS40,SBS41 5 cosine sim = 0.98414
1668
758
4275
721
997
1098
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98414
1152
764
4437
665
1450
1047
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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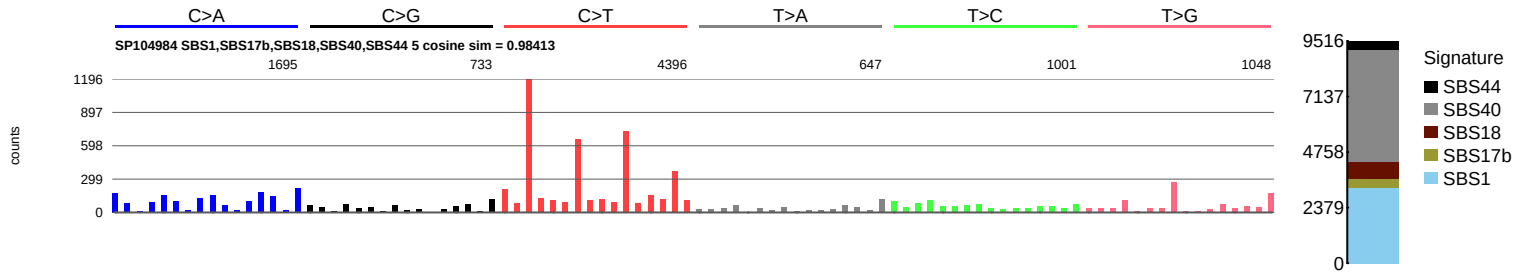
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS26,SBS40 6 cosine sim = 0.98413
1234
744
4436
625
1409
1071
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS15
SBS2
SBS1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C>G
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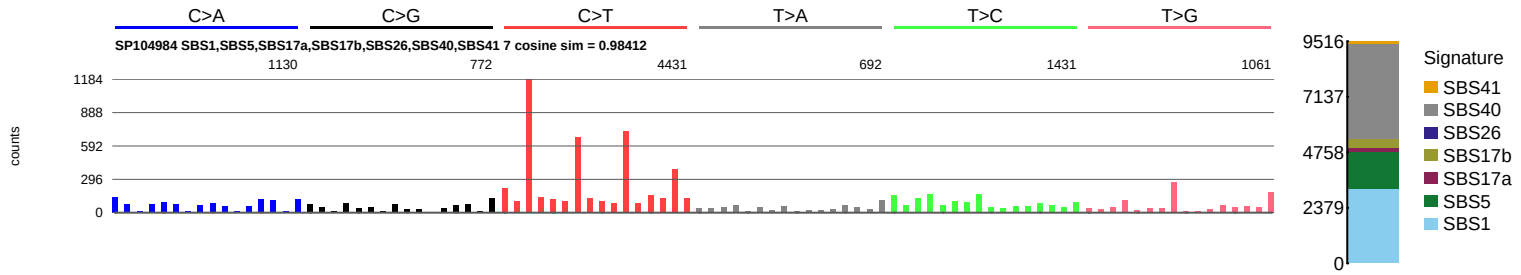
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS40,SBS44 5 cosine sim = 0.98413
1695
733
4396
647
1001
1048
1196
897
598
299
0
counts
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS1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C>G
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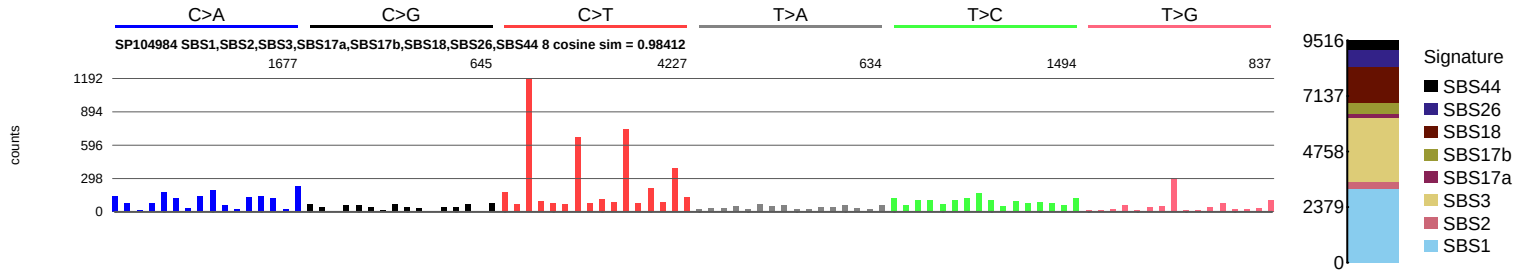
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98412
1130
772
4431
692
1431
1061
1184
888
592
296
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1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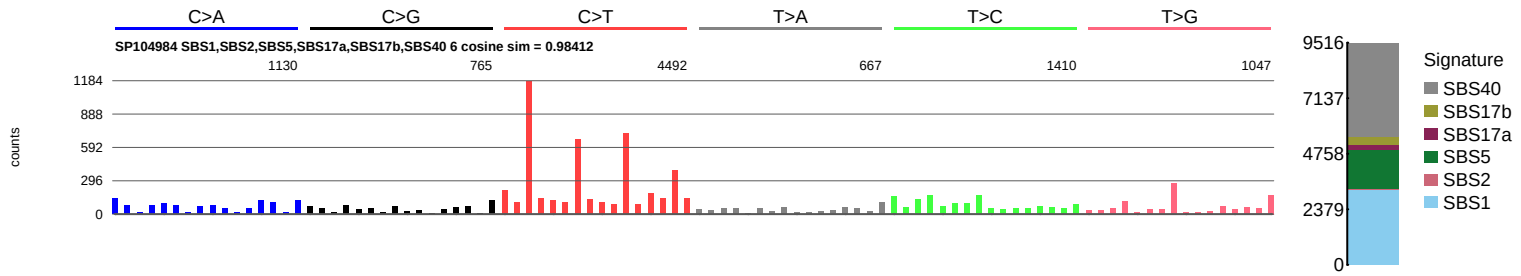
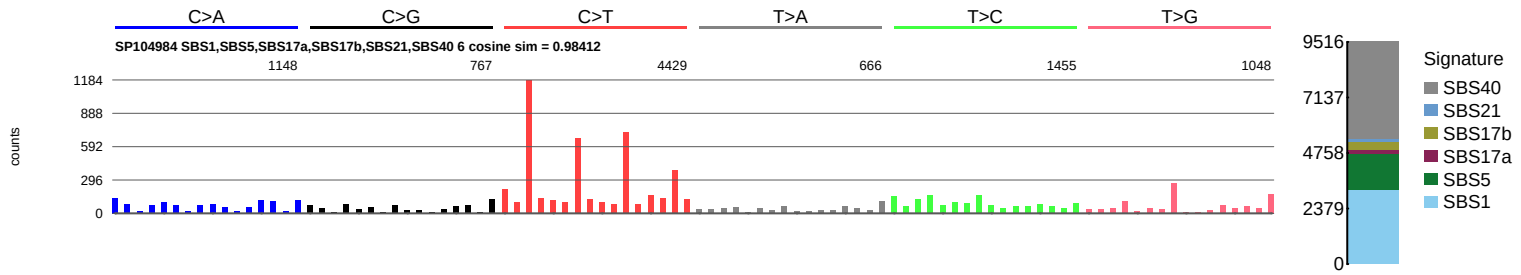
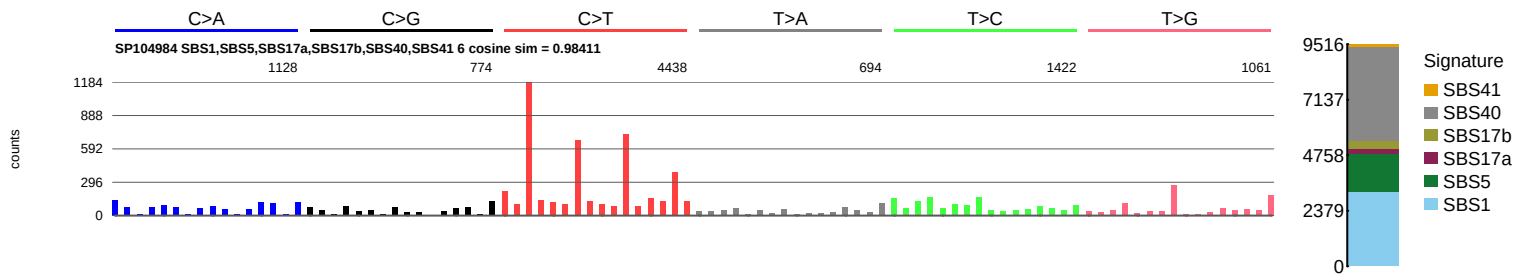
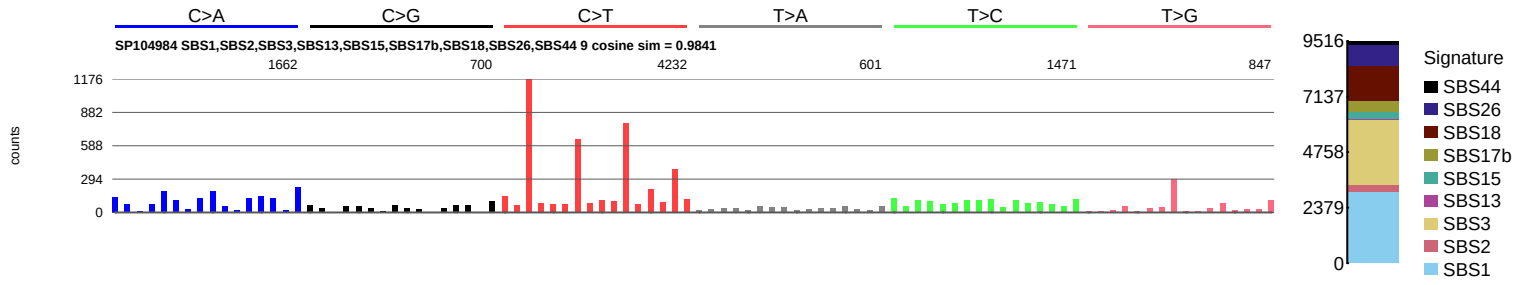
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98412
1677
645
4227
634
1494
837
1192
894
596
298
0
counts
9516
7137
4758
2379
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98412
1130
765
4492
667
1410
1047
1184
888
592
296
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98412
1148
767
4429
666
1455
1048
1184
888
592
296
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98411
1128
774
4438
694
1422
1061
1184
888
592
296
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1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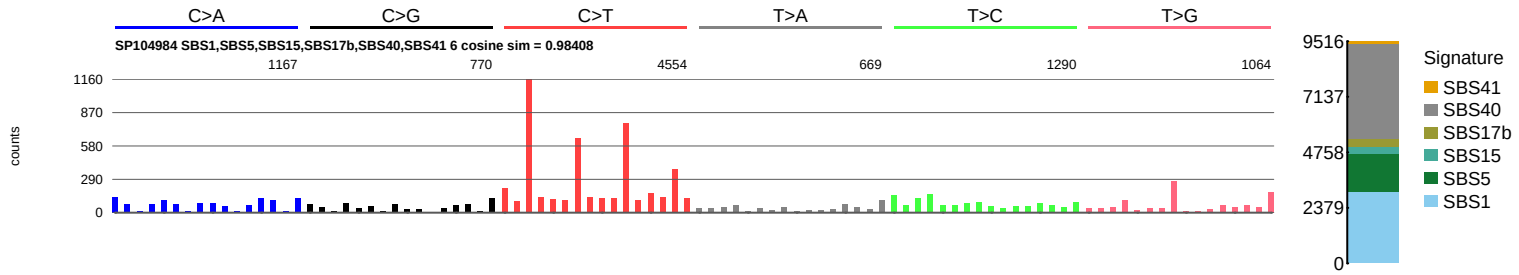
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.9841
1662
700
4232
601
1471
847
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1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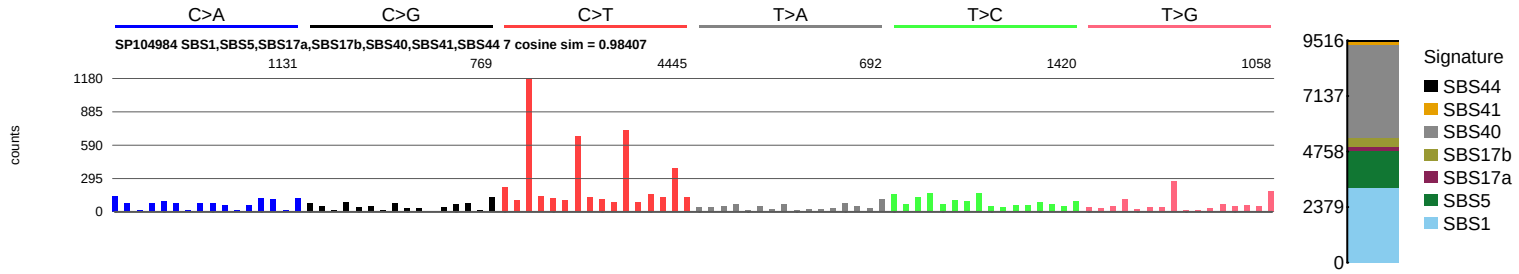
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS40,SBS41 6 cosine sim = 0.98408
1167
770
4554
669
1290
1064
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS5
SBS1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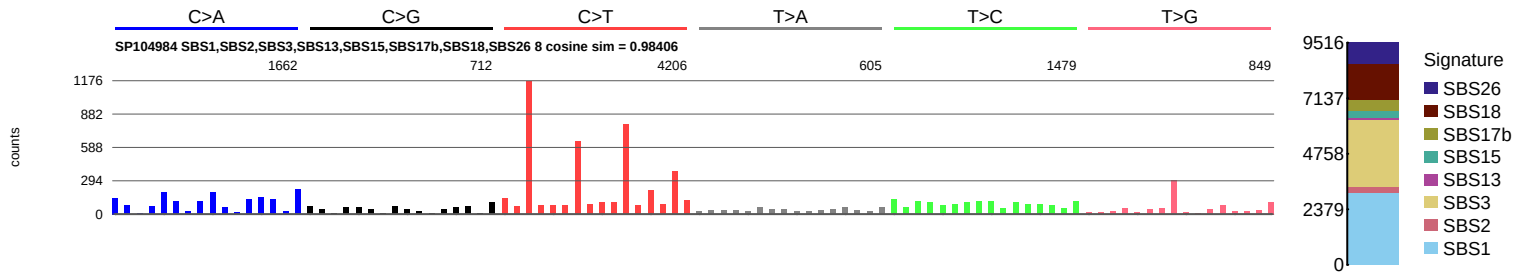
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98407
1131
769
4445
692
1420
1058
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26 8 cosine sim = 0.98406
1662
712
4206
605
1479
849
counts
1176
882
588
294
0
9516
7137
4758
2379
0
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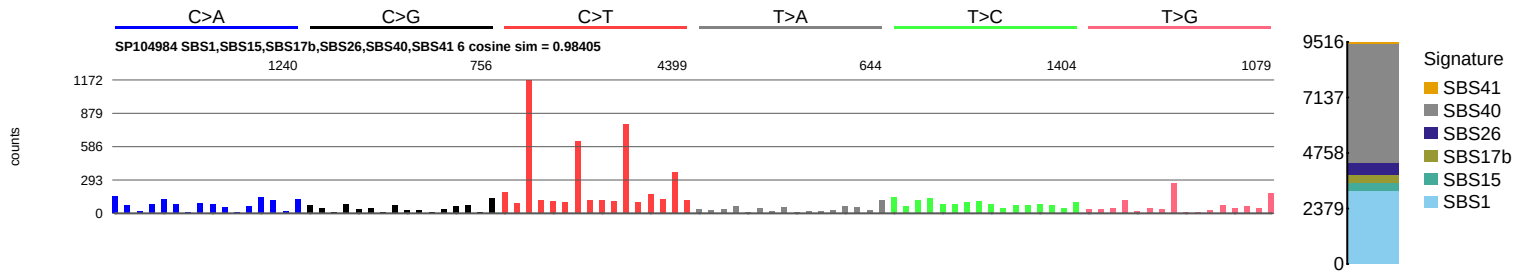
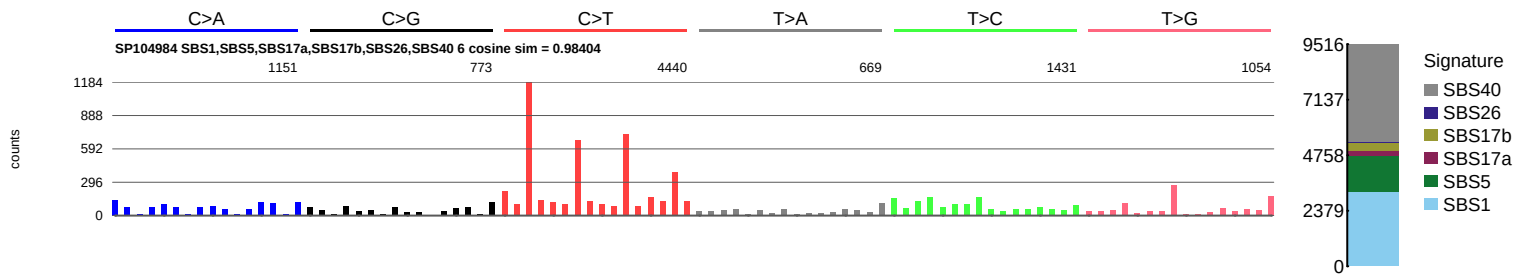
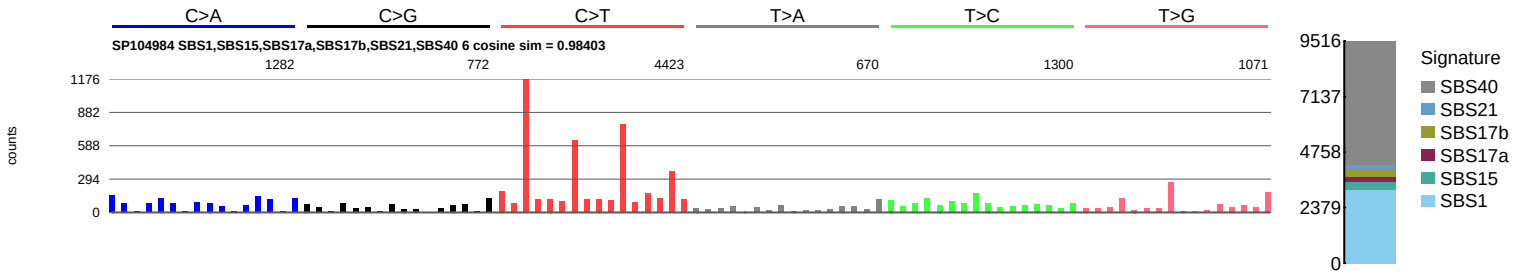
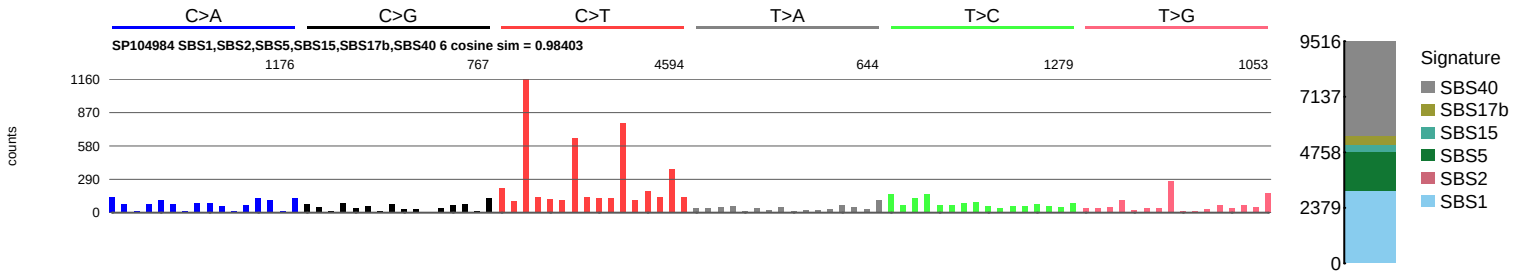
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98405
1240
756
4399
644
1404
1079
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98404
1151
773
4440
669
1431
1054
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98403
1282
772
4423
670
1300
1071
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS40 6 cosine sim = 0.98403
1176
767
4594
644
1279
1053
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
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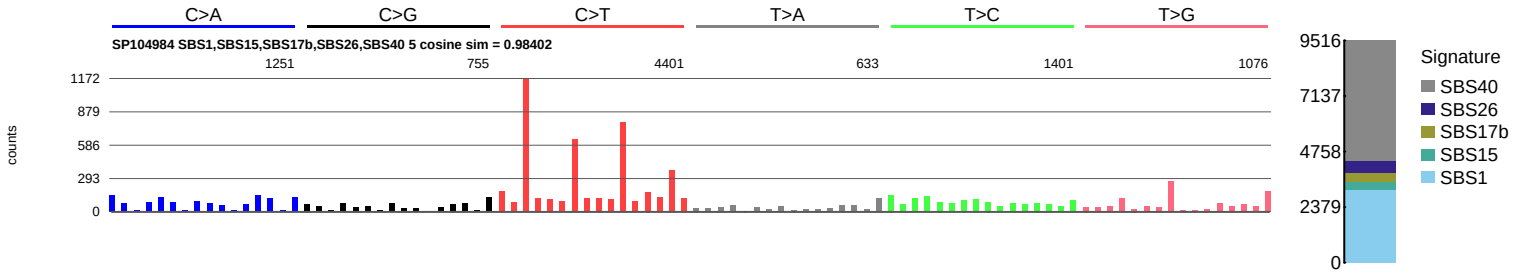
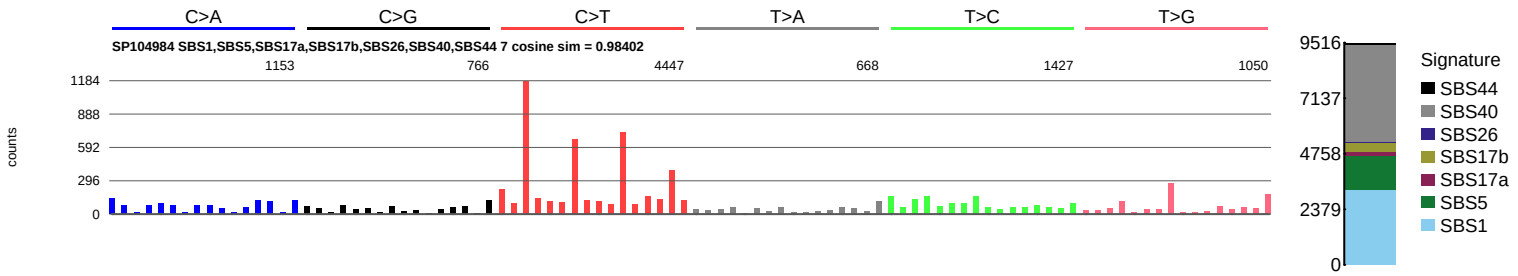
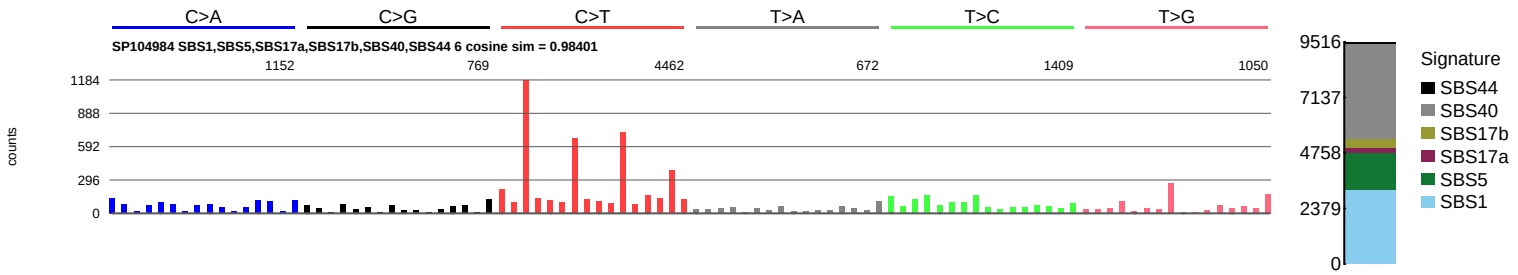
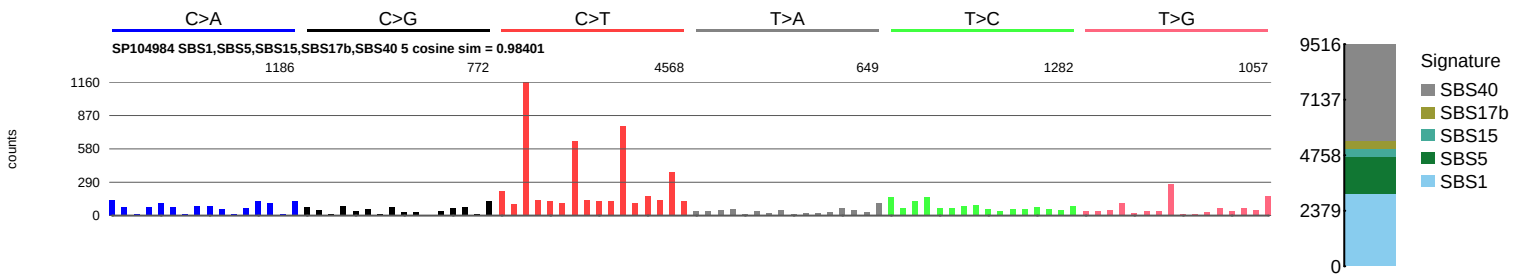
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS26,SBS40 5 cosine sim = 0.98402
1251
755
4401
633
1401
1076
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98402
1153
766
4447
668
1427
1050
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98401
1152
769
4462
672
1409
1050
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
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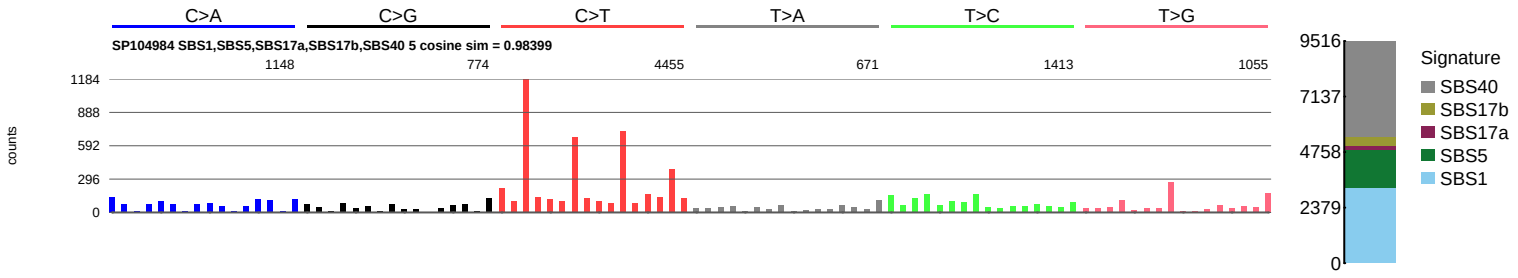
T>G
SP104984 SBS1,SBS5,SBS15,SBS17b,SBS40 5 cosine sim = 0.98401
1186
772
4568
649
1282
1057
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS15
SBS5
SBS1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C>G
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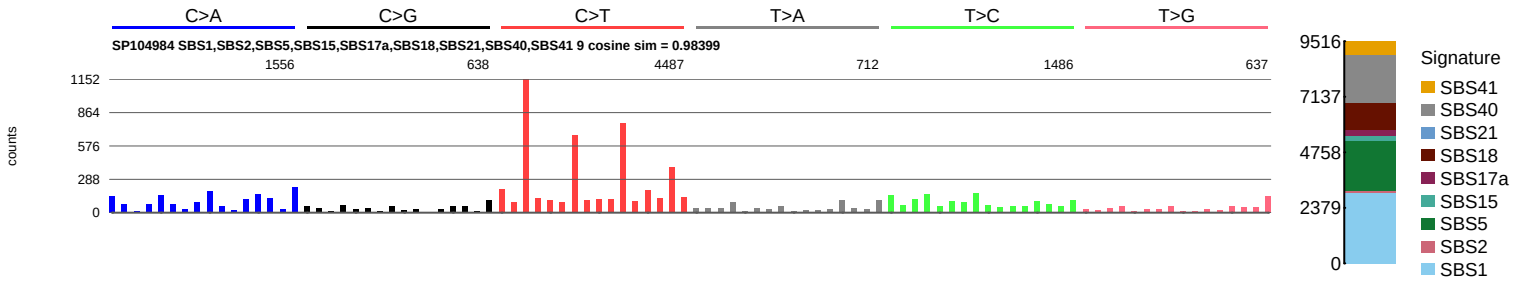
T>G
SP104984 SBS1,SBS5,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98399
1148
774
4455
671
1413
1055
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS1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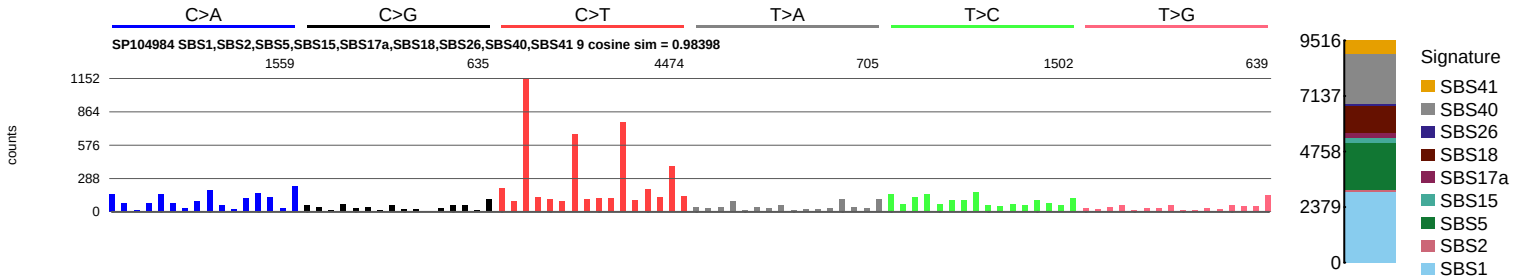
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98399
1556
638
4487
712
1486
637
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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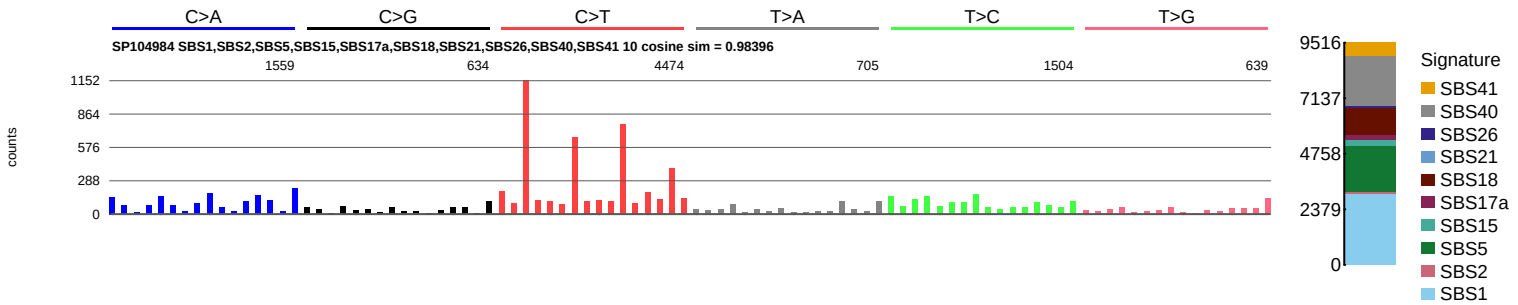
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98398
1559
635
4474
705
1502
639
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98396
1559
634
4474
705
1504
639
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
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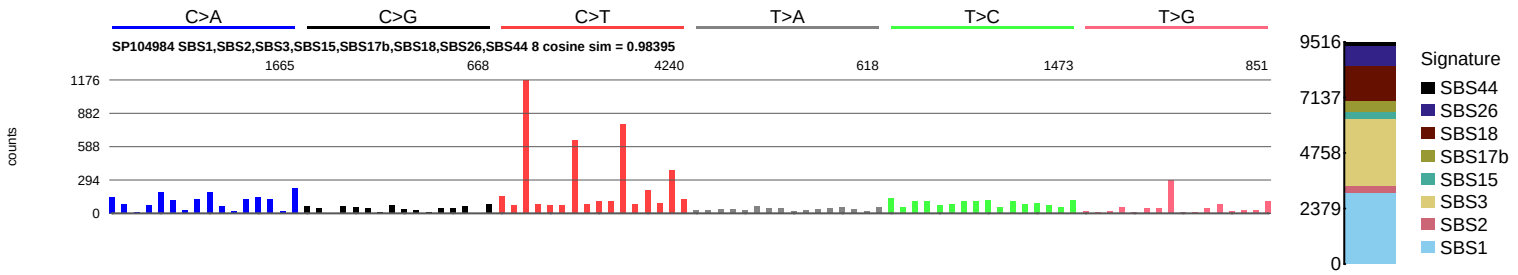
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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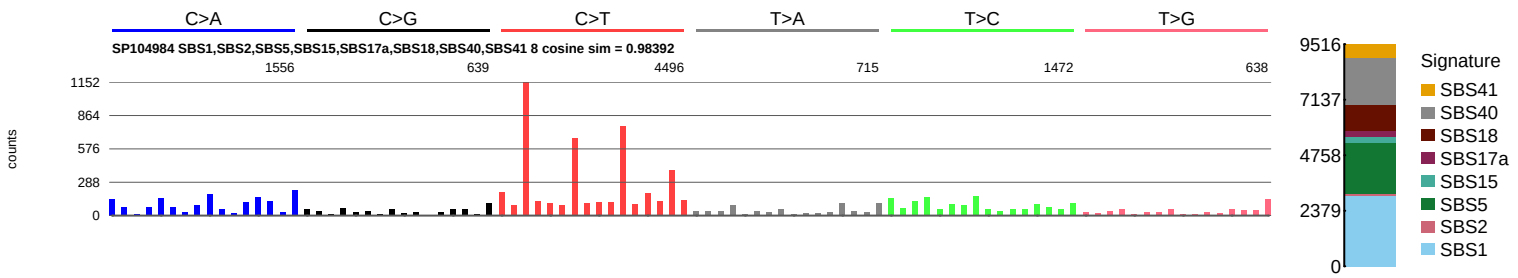
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98395
1665
668
4240
618
1473
851
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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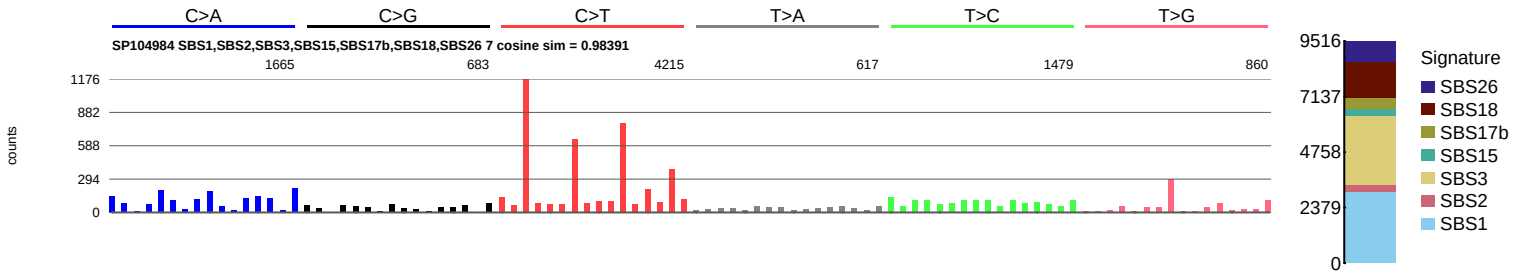
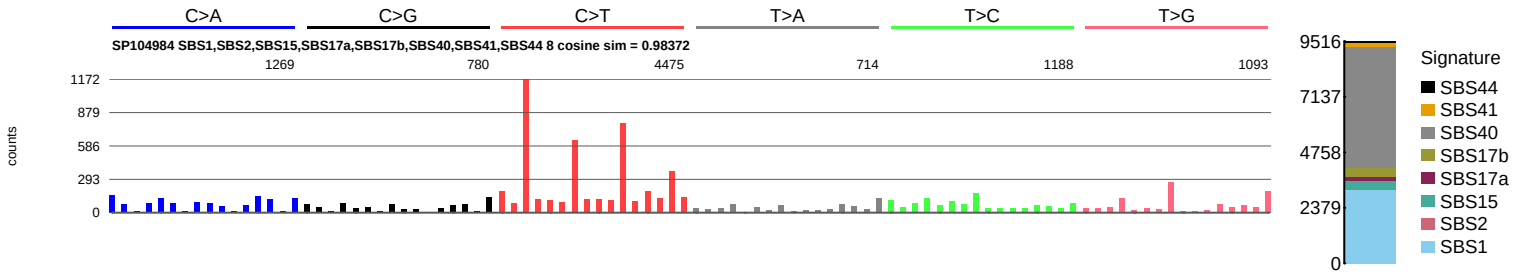
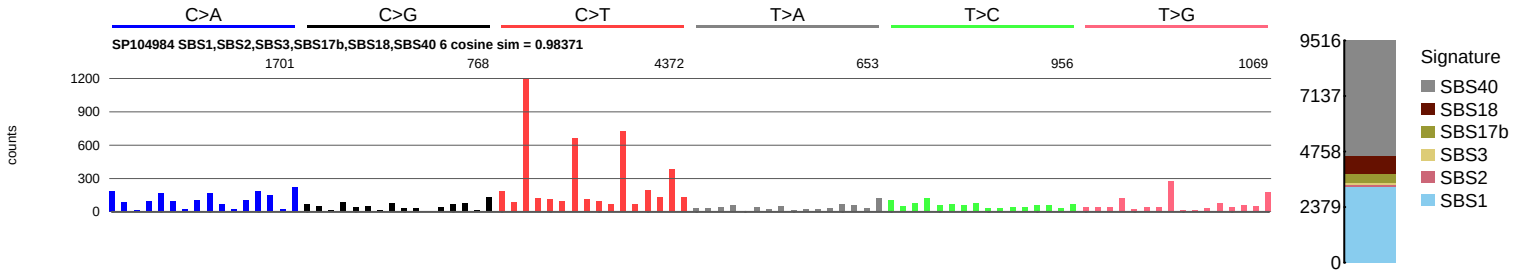
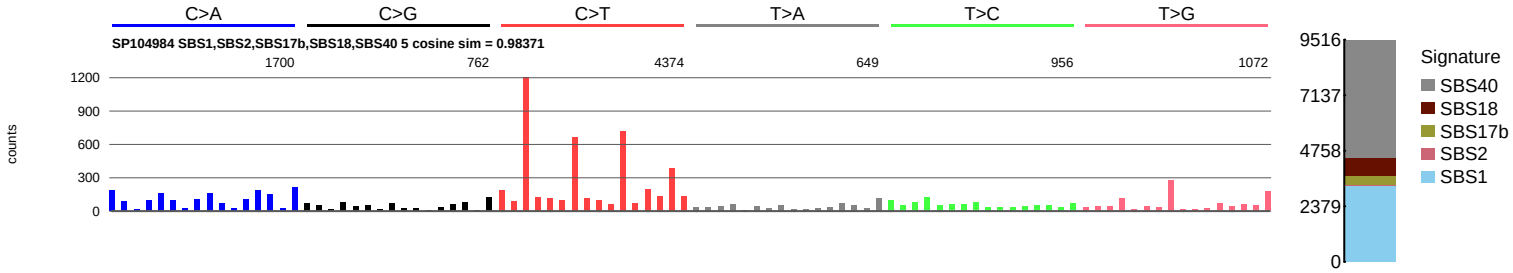
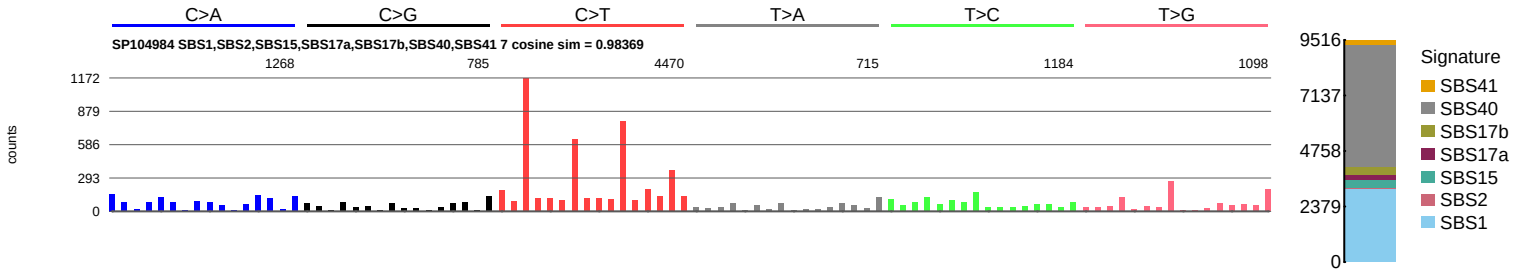
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98392
1556
639
4496
715
1472
638
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

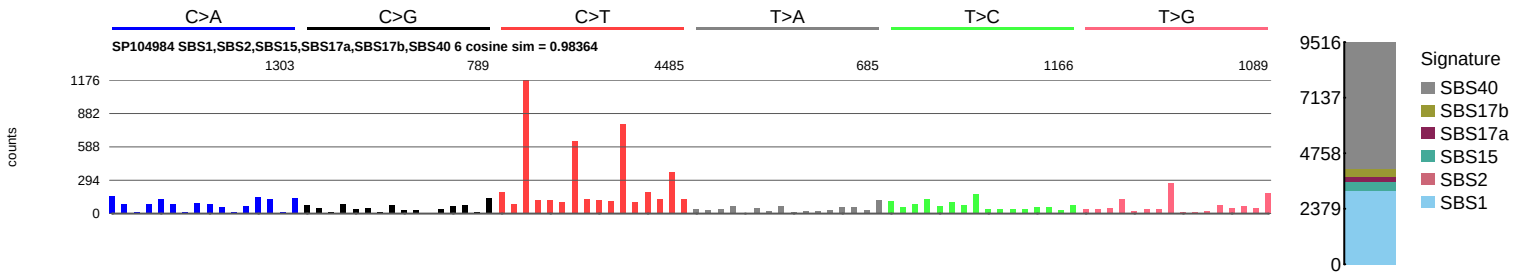

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26 7 cosine sim = 0.98391
1665
683
4215
617
1479
860
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

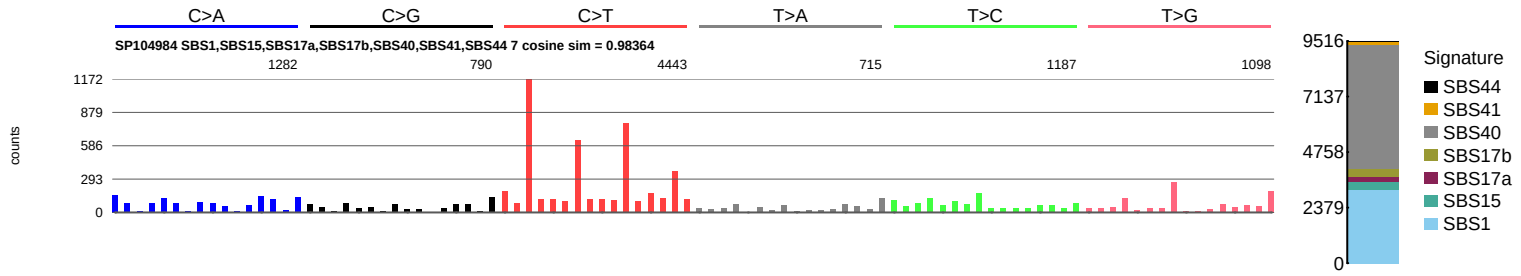

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98372
1269
780
4475
714
1188
1093
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1

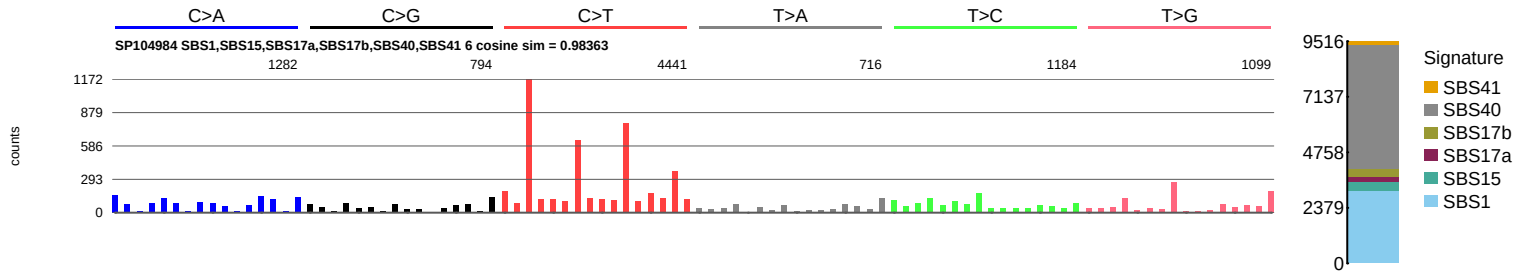

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS40 6 cosine sim = 0.98371
1701
768
4372
653
956
1069
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS3
SBS2
SBS1

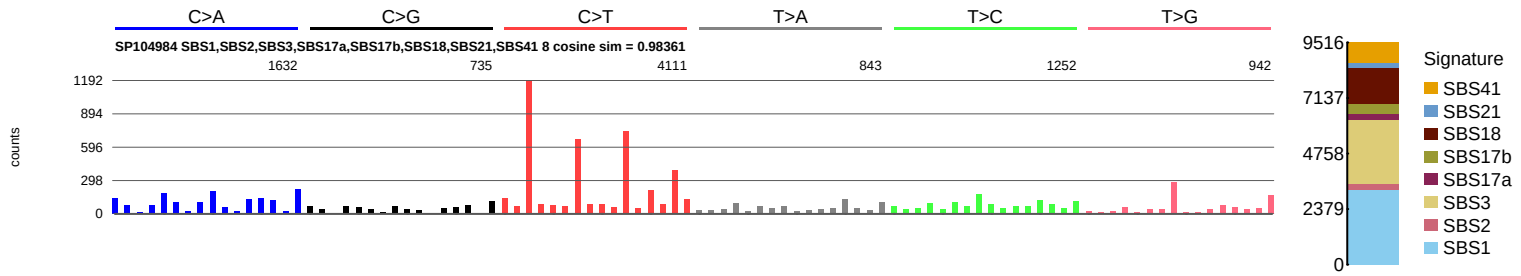

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS18,SBS40 5 cosine sim = 0.98371
1700
762
4374
649
956
1072
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS2
SBS1

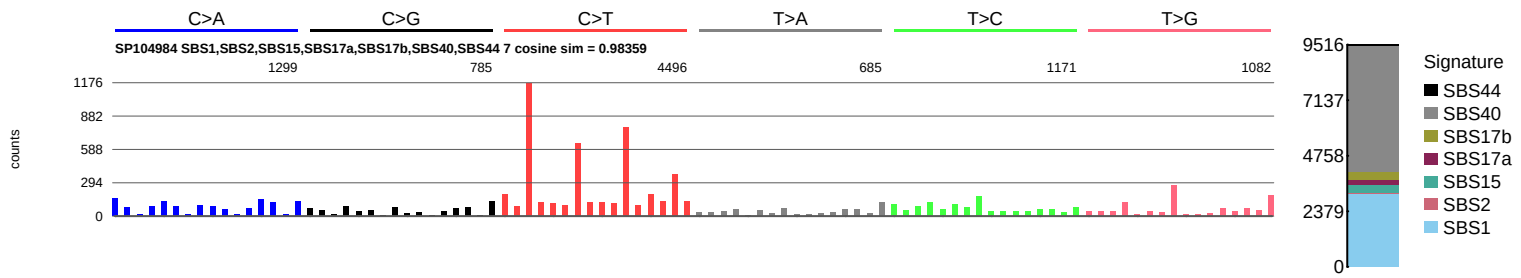

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98369
1268
785
4470
715
1184
1098
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1

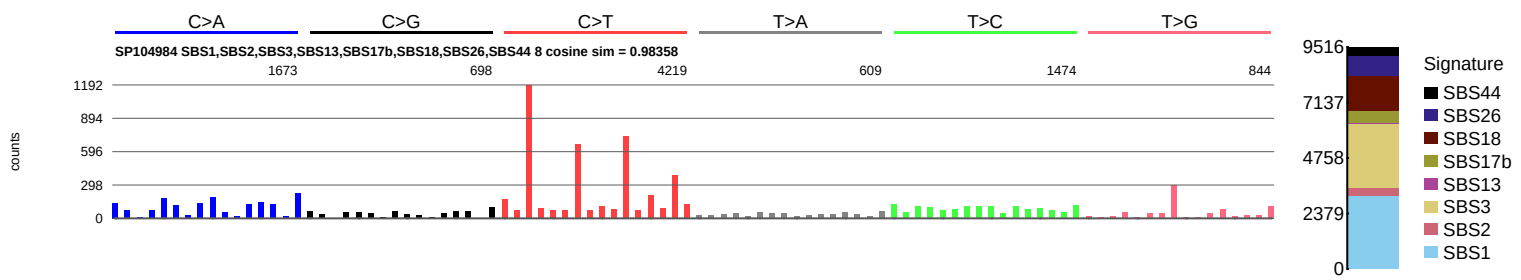

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98364
1303
789
4485
685
1166
1089
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1

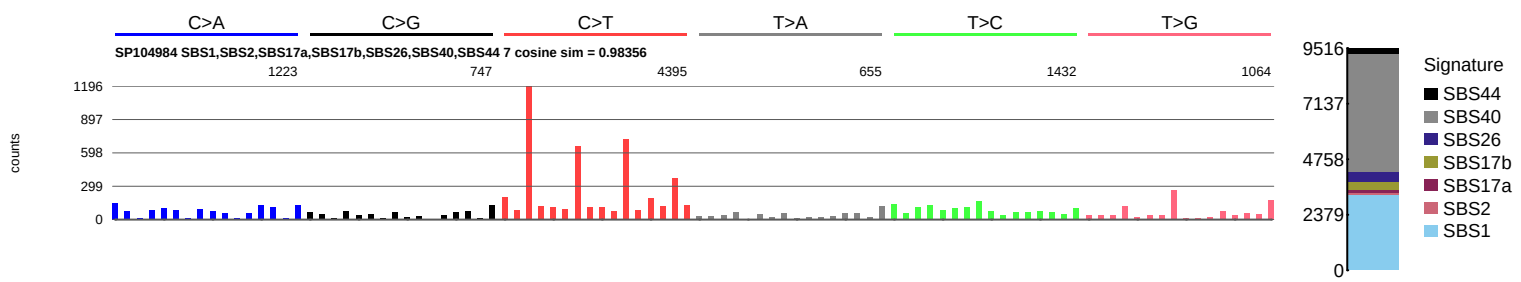

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98364
1282
790
4443
715
1187
1098
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1

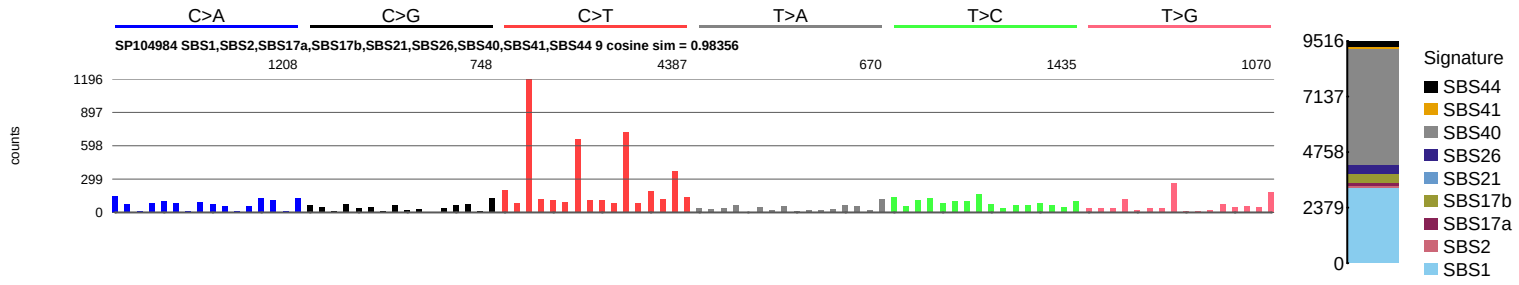

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98363
1282
794
4441
716
1184
1099
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1

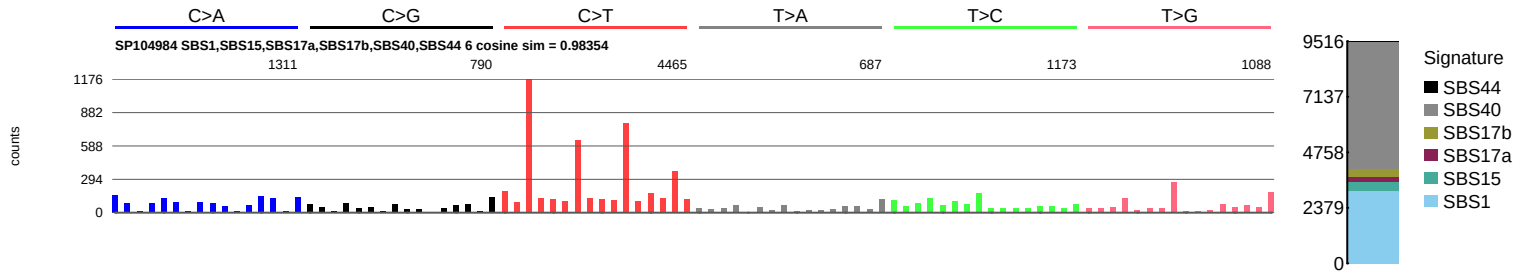

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98361
1632
735
4111
843
1252
942
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

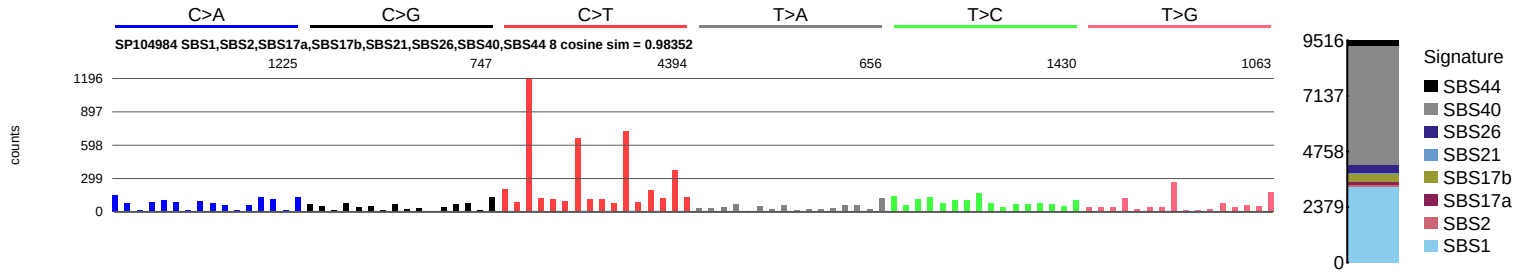

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98359
1299
785
4496
685
1171
1082
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98358
1673
698
4219
609
1474
844
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98356
1223
747
4395
655
1432
1064
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98356
1208
748
4387
670
1435
1070
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98354
1311
790
4465
687
1173
1088
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98352
1225
747
4394
656
1430
1063
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
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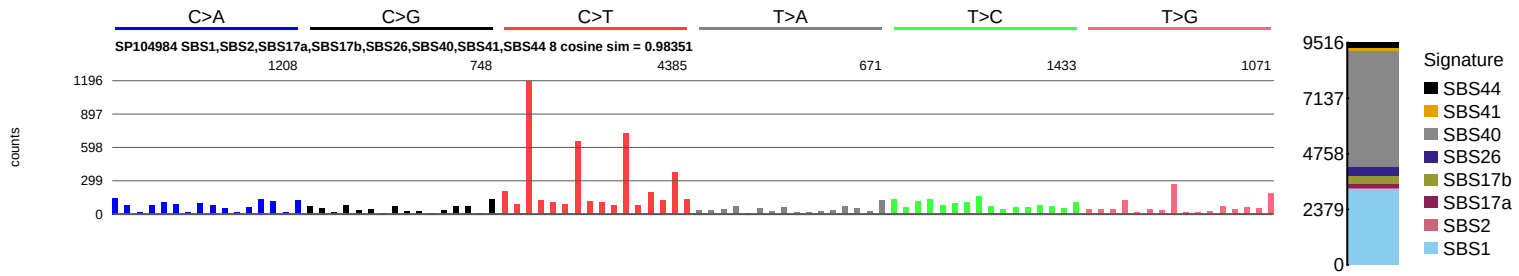
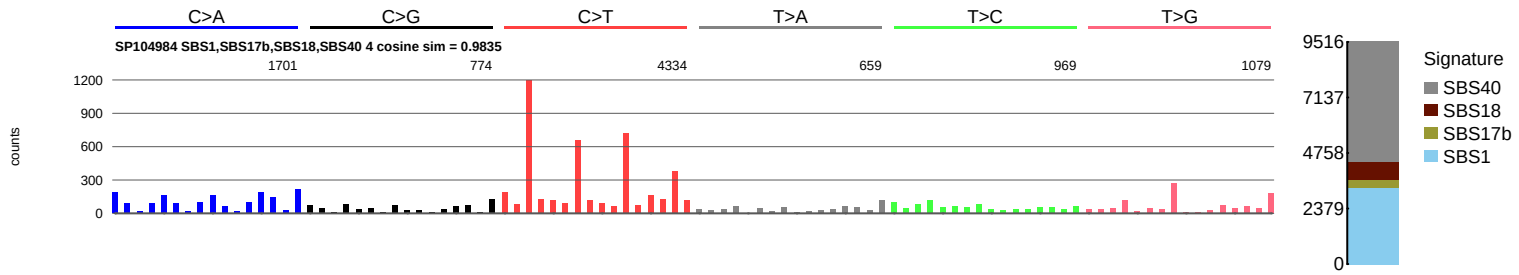
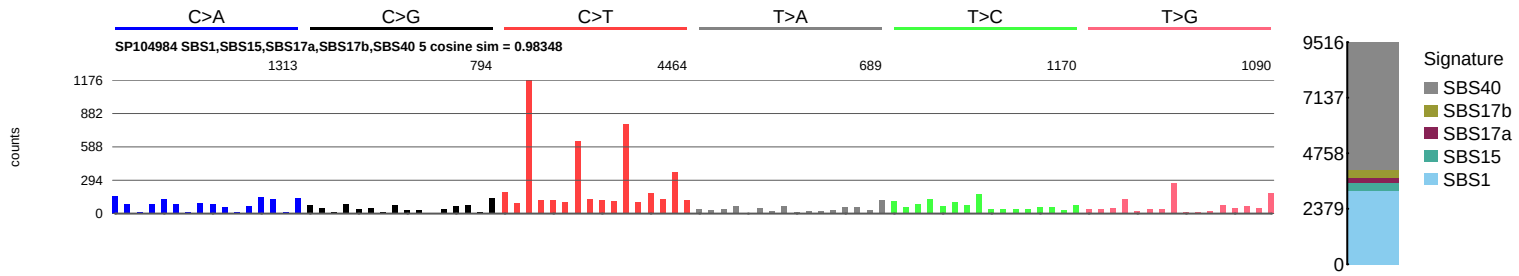
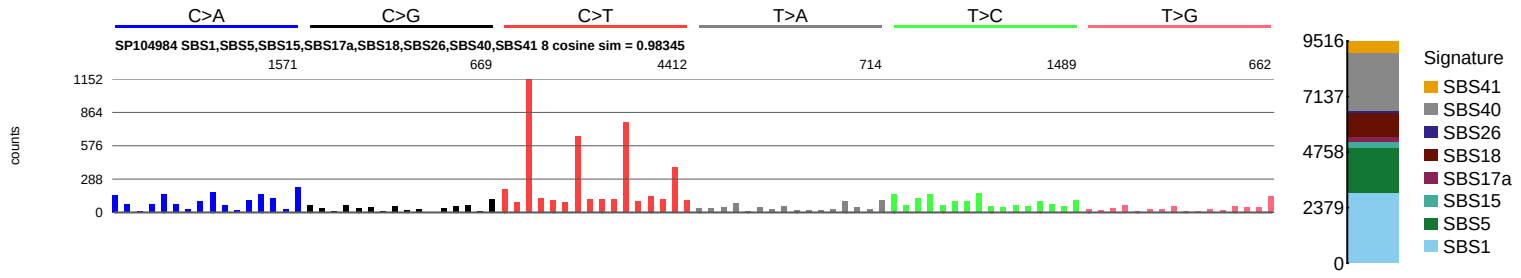
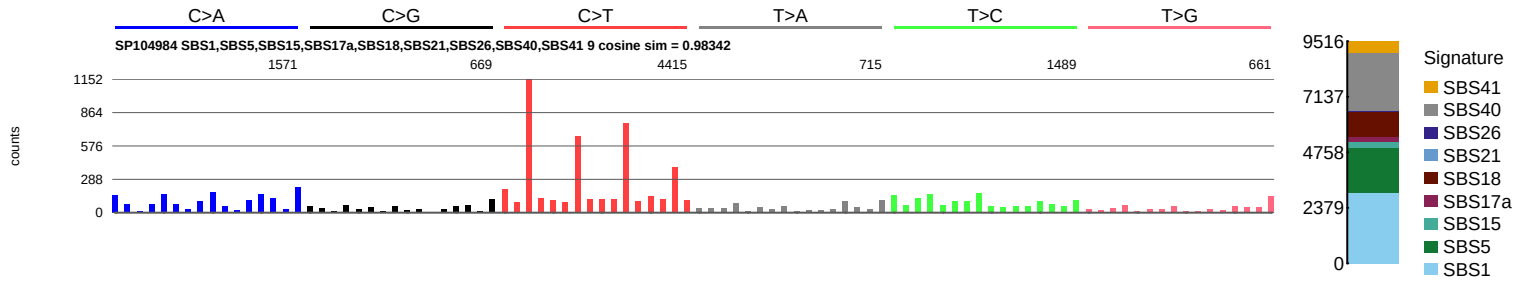
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98351
1208
748
4385
671
1433
1071
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS18,SBS40 4 cosine sim = 0.9835
1701
774
4334
659
969
1079
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98348
1313
794
4464
689
1170
1090
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98345
1571
669
4412
714
1489
662
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS1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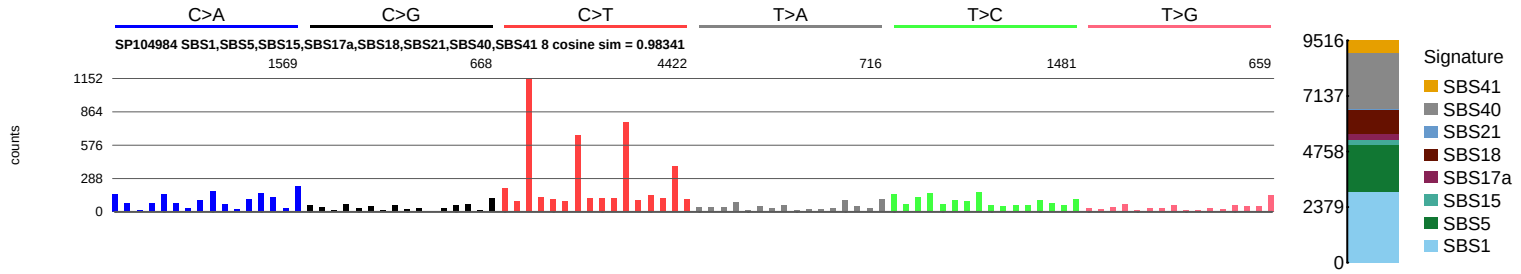
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98342
1571
669
4415
715
1489
661
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1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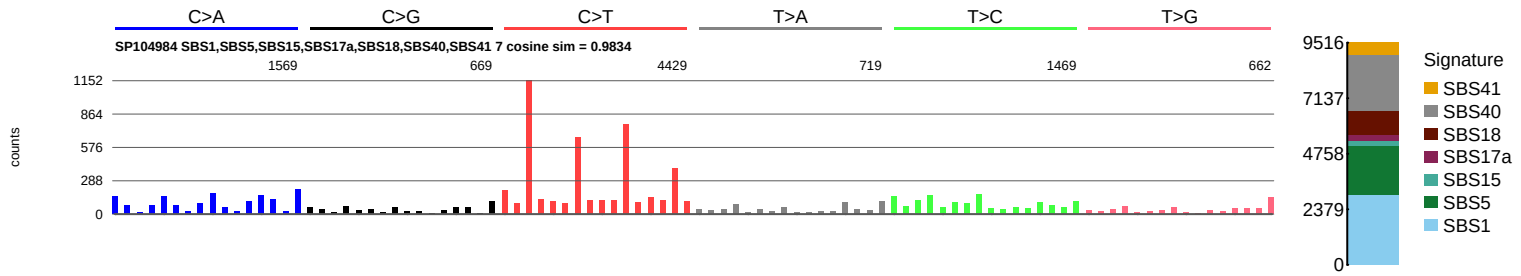
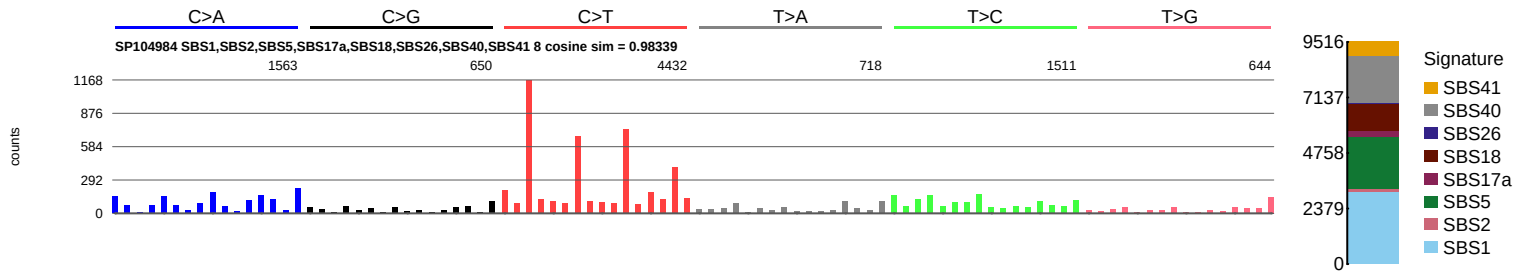
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98341
1569
668
4422
716
1481
659
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.9834
1569
669
4429
719
1469
662
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS1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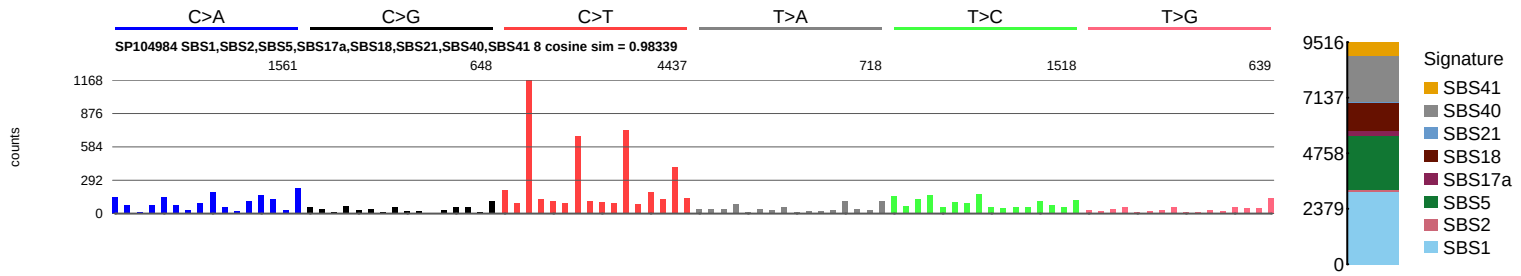
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98339
1563
650
4432
718
1511
644
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98339
1561
648
4437
718
1518
639
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
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SBS21
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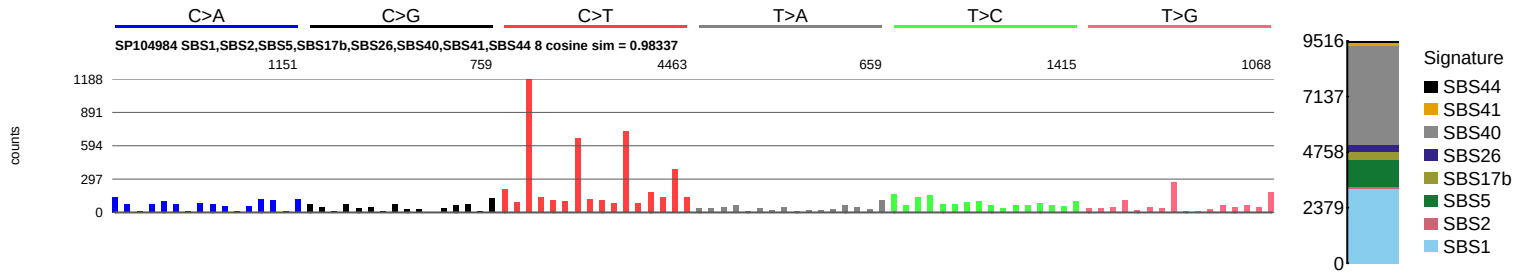
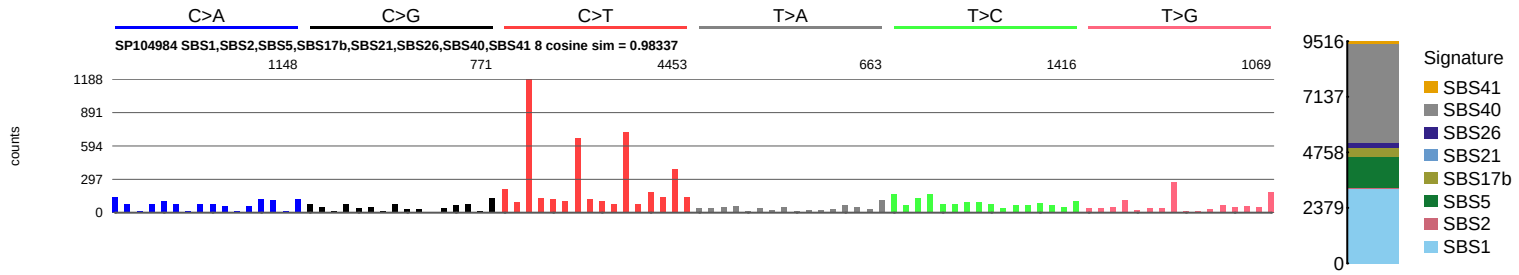
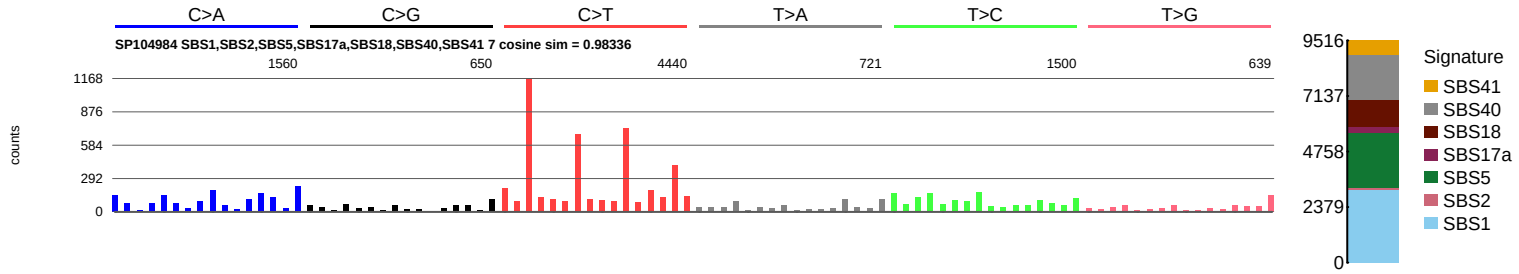
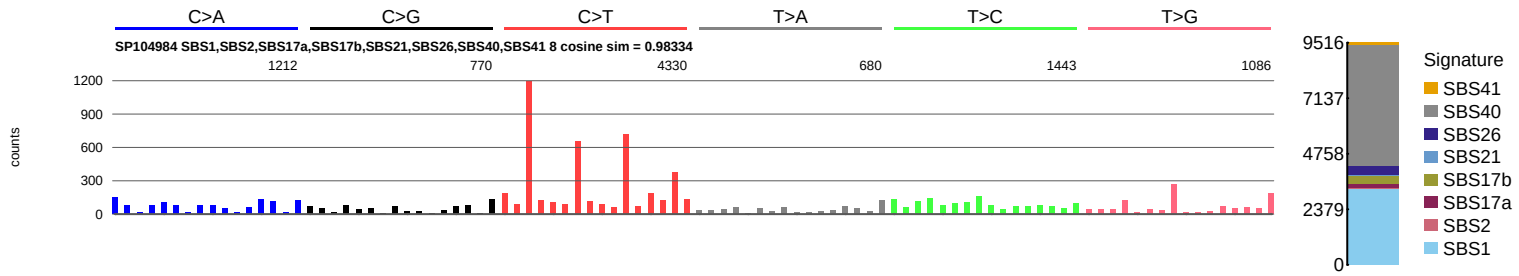
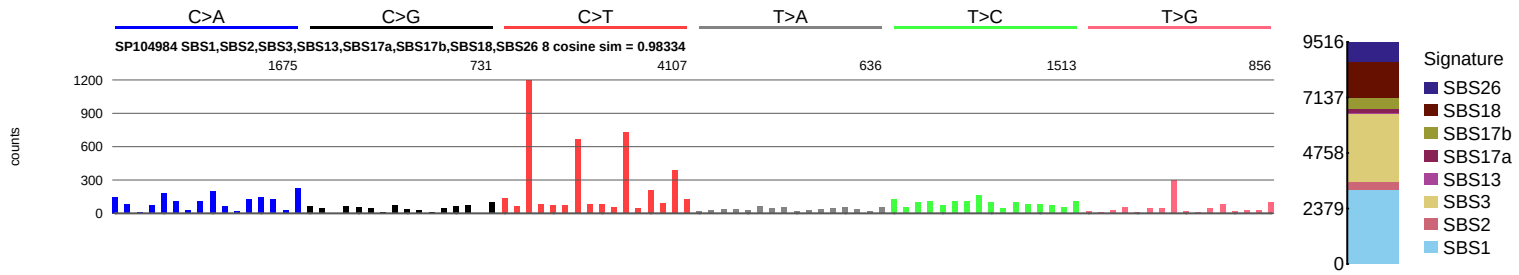
SBS5
SBS2
SBS1

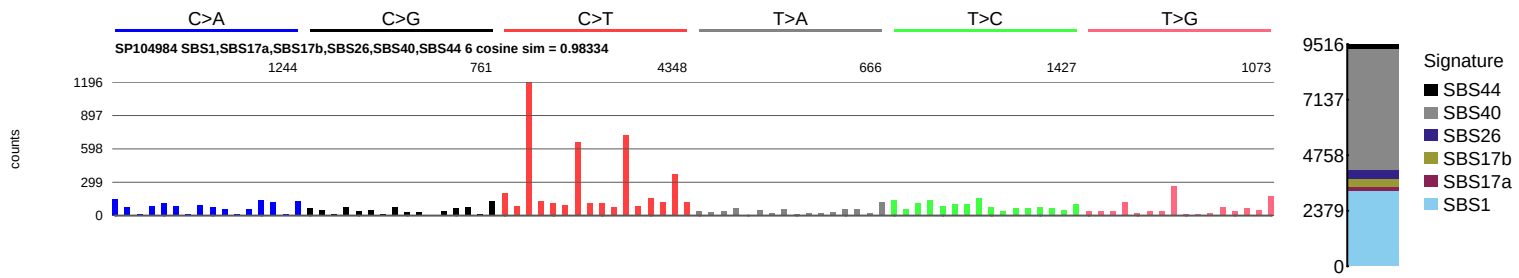

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98337
1151
759
4463
659
1415
1068
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS5
SBS2
SBS1

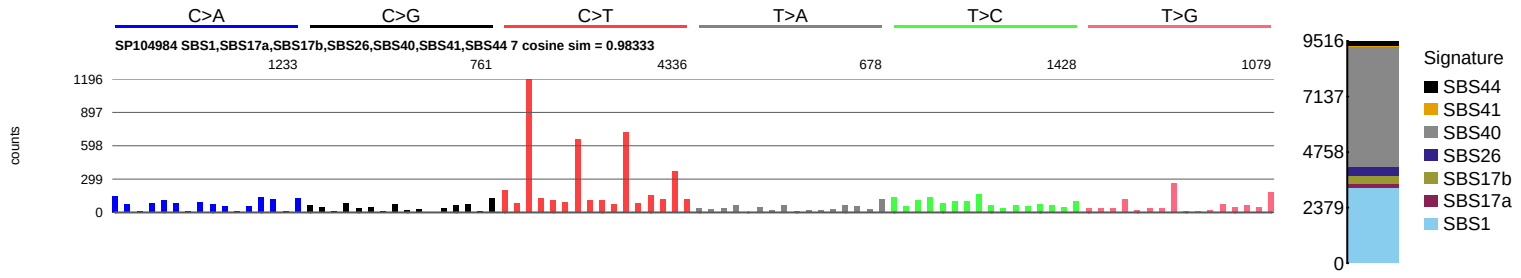

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98337
1148
771
4453
663
1416
1069
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS5
SBS2
SBS1

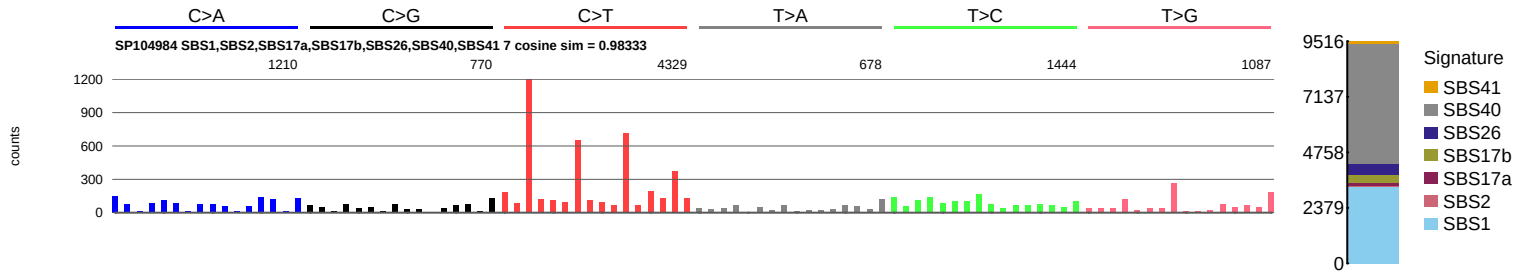

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98336
1560
650
4440
721
1500
639
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1

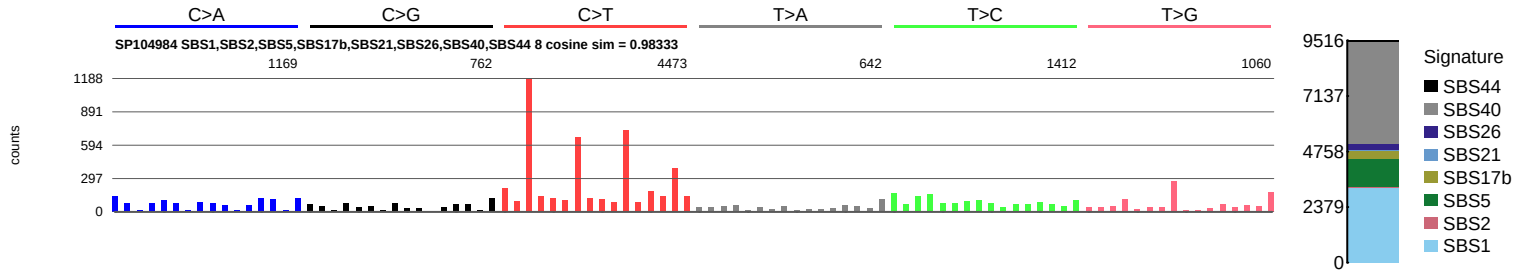

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98334
1212
770
4330
680
1443
1086
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1

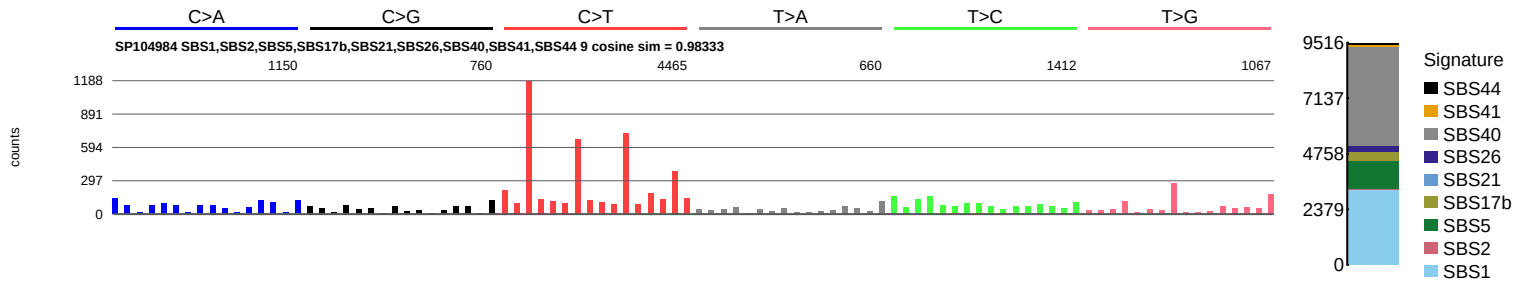

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98334
1675
731
4107
636
1513
856
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

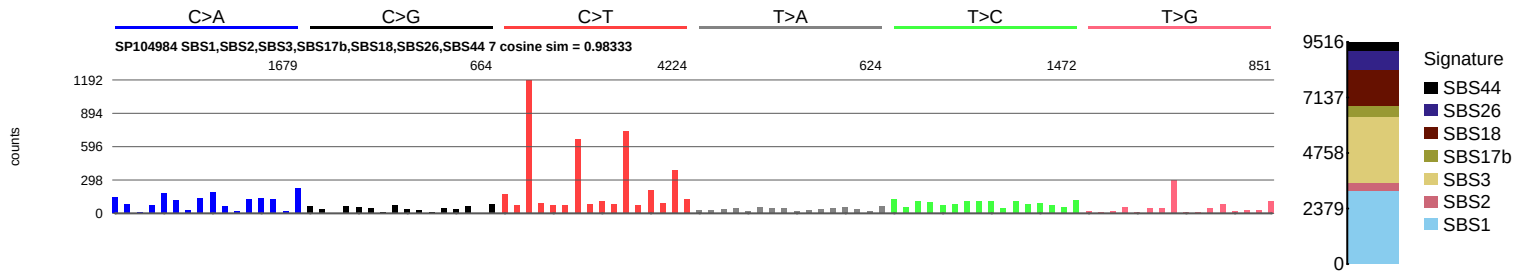

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98334
1244
761
4348
666
1427
1073
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS1

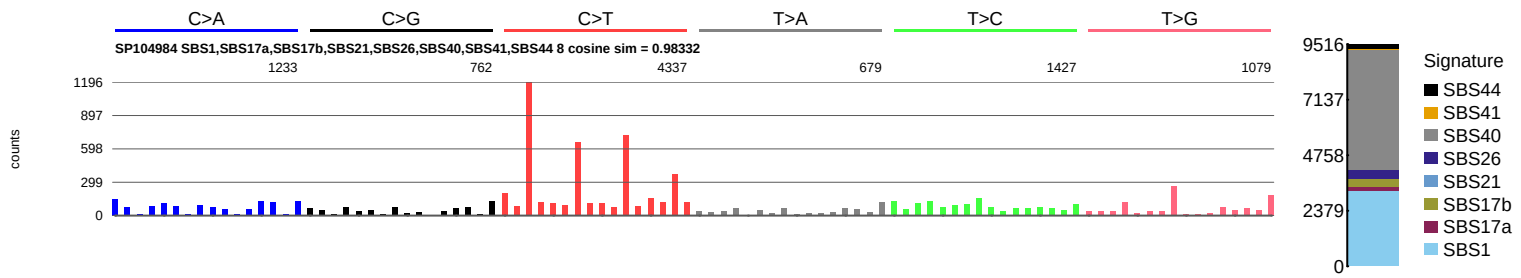

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98333
1233
761
4336
678
1428
1079
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1

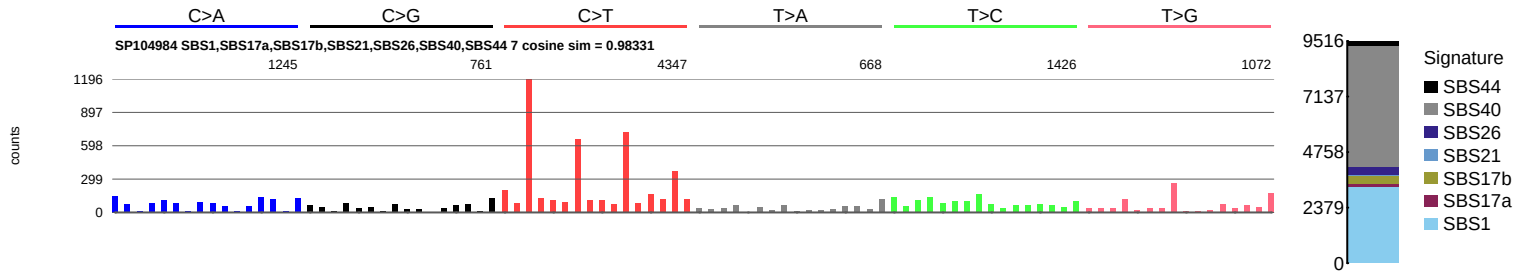

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98333
1210
770
4329
678
1444
1087
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1

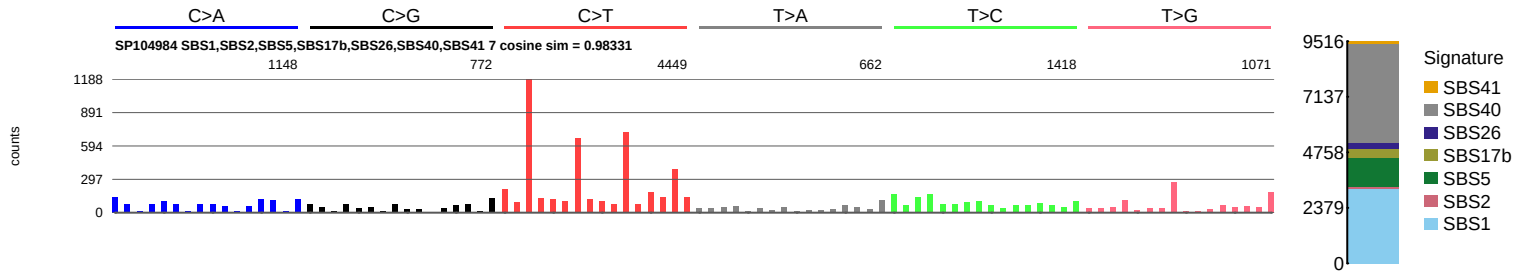

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98333
1169
762
4473
642
1412
1060
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS5
SBS2
SBS1

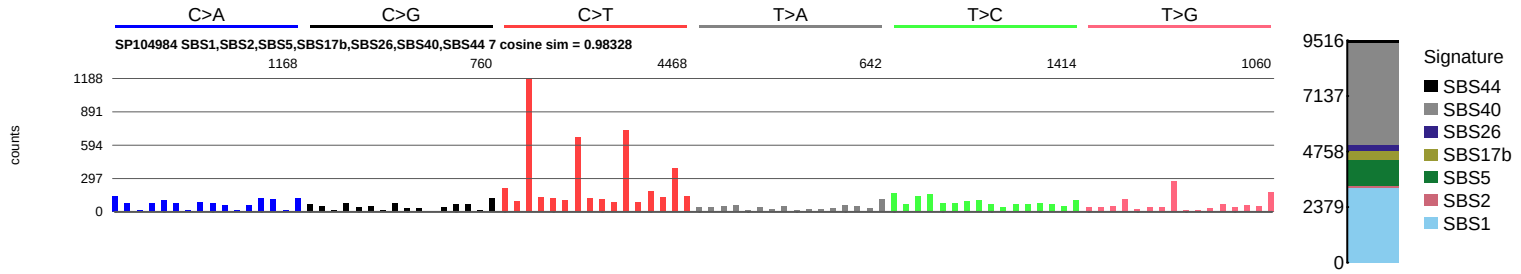

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98333
1150
760
4465
660
1412
1067
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS5
SBS2
SBS1

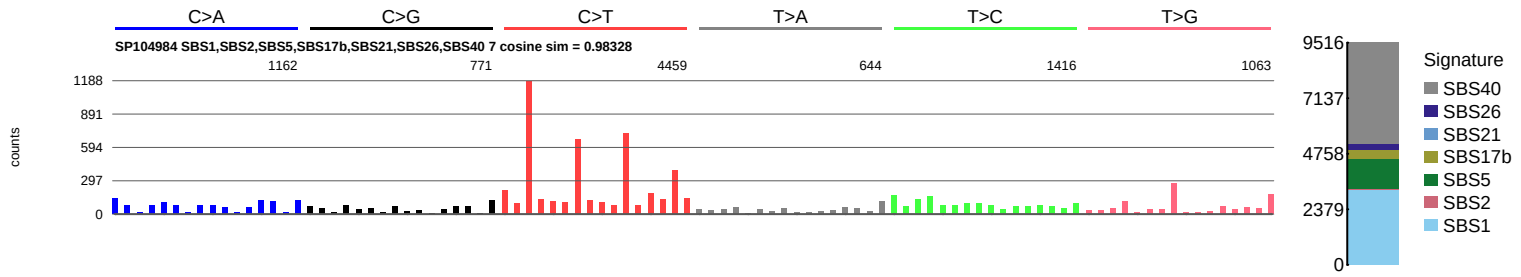

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98333
1679
664
4224
624
1472
851
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1

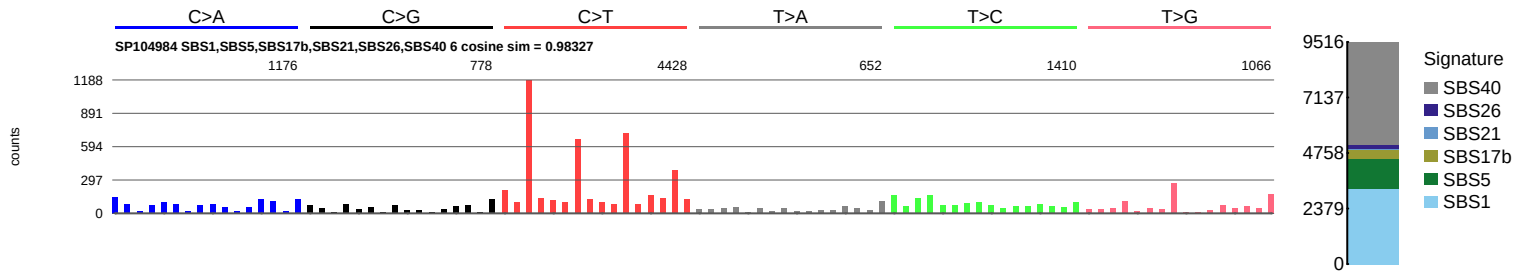

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98332
1233
762
4337
679
1427
1079
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1

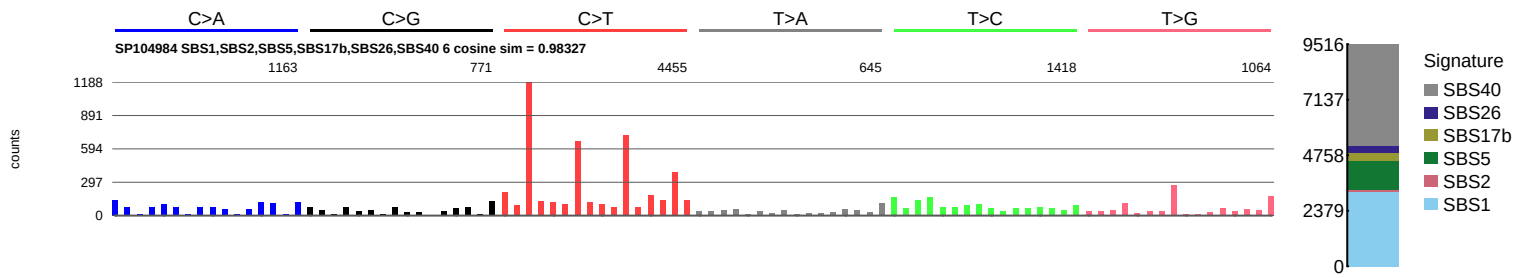

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98331
1245
761
4347
668
1426
1072
1196
897
598
299
0
counts
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1

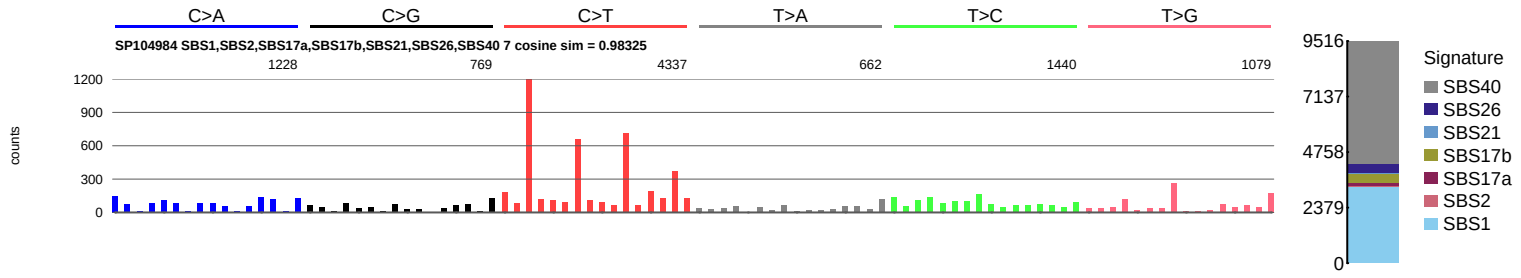

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98331
1148
772
4449
662
1418
1071
1188
891
594
297
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS5
SBS2
SBS1

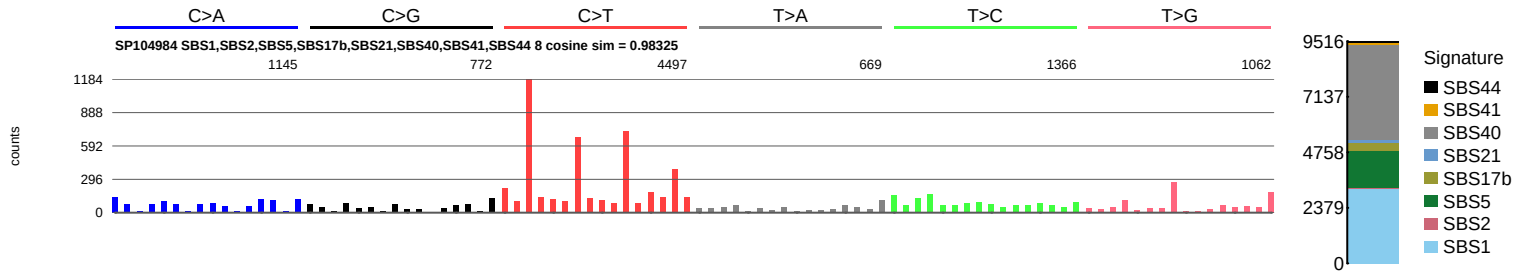

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98328
1168
760
4468
642
1414
1060
1188
891
594
297
0
counts
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS5
SBS2
SBS1

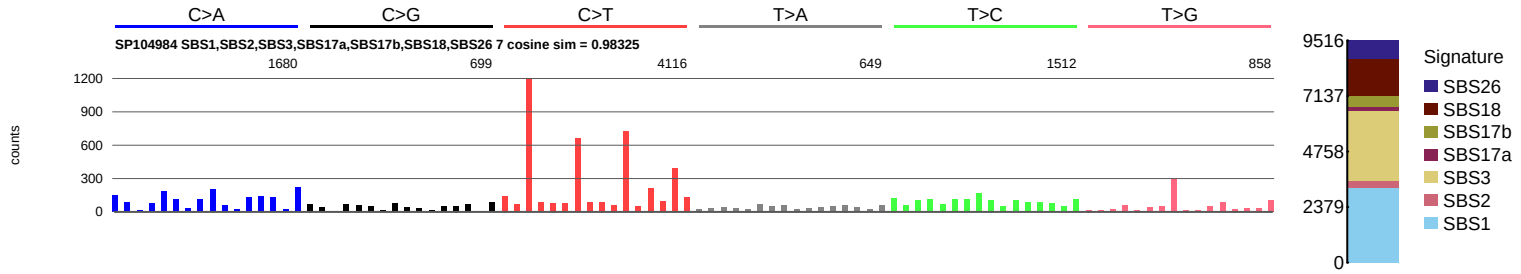

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98328
1162
771
4459
644
1416
1063
1188
891
594
297
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98327
1176
778
4428
652
1410
1066
1188
891
594
297
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS26,SBS40 6 cosine sim = 0.98327
1163
771
4455
645
1418
1064
1188
891
594
297
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98325
1228
769
4337
662
1440
1079
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98325
1145
772
4497
669
1366
1062
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS5
SBS2
SBS1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C>T
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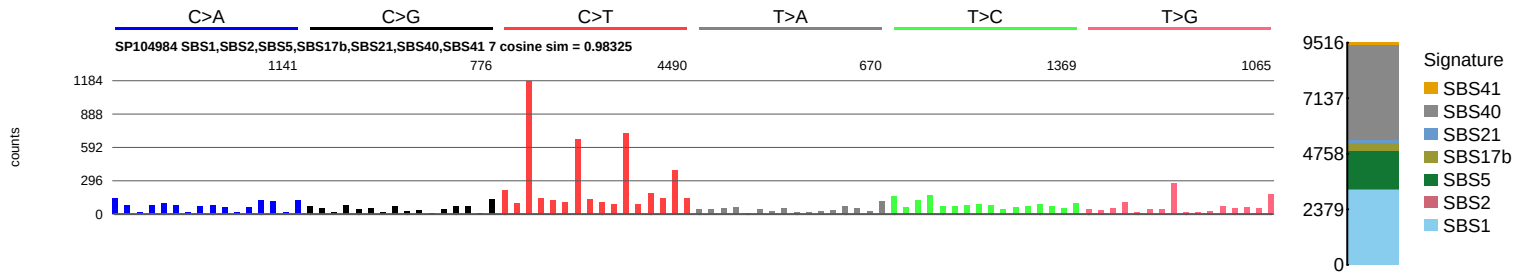
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98325
1680
699
4116
649
1512
858
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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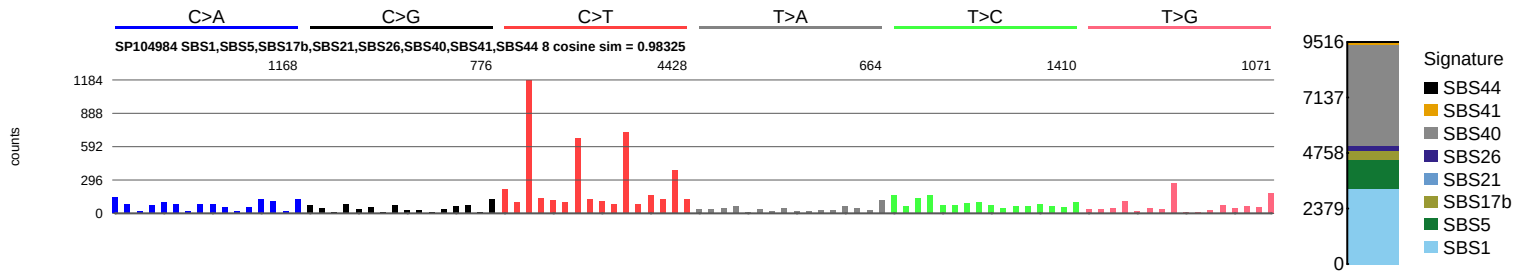
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98325
1141
776
4490
670
1369
1065
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS5
SBS2
SBS1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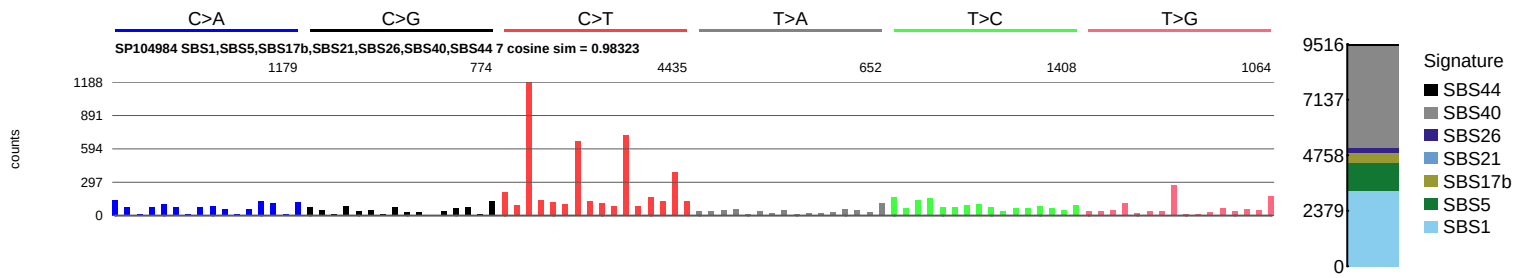
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98325
1168
776
4428
664
1410
1071
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS5
SBS1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C>G
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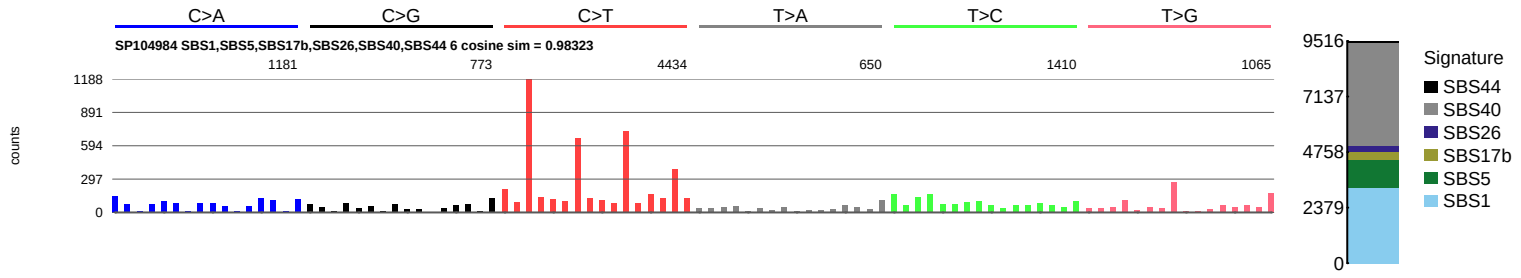
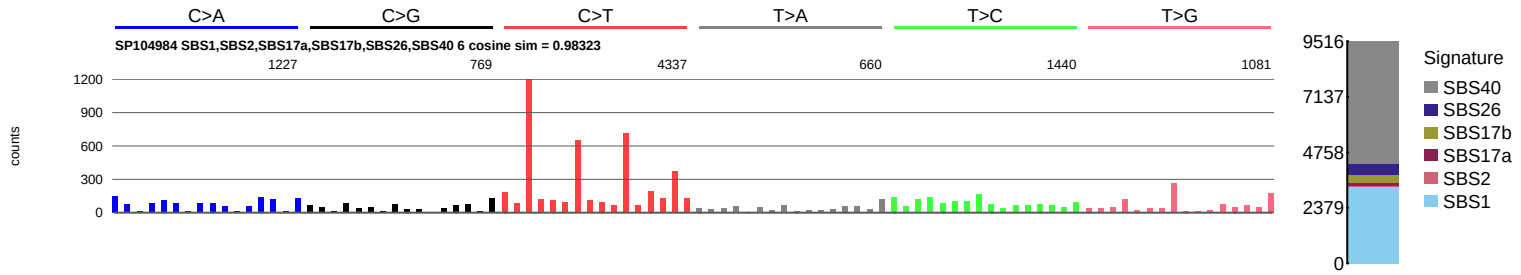
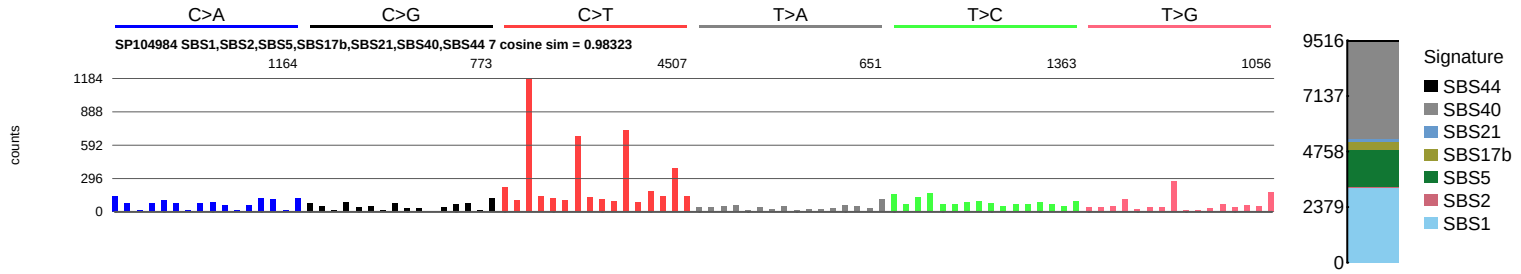
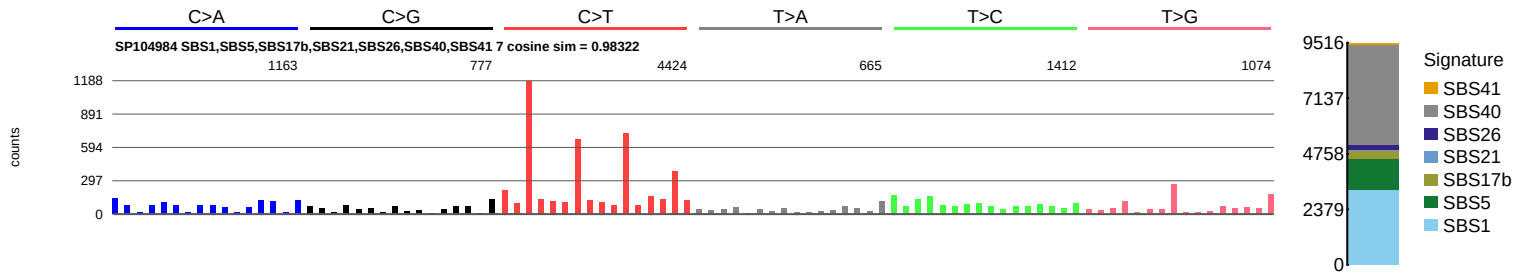
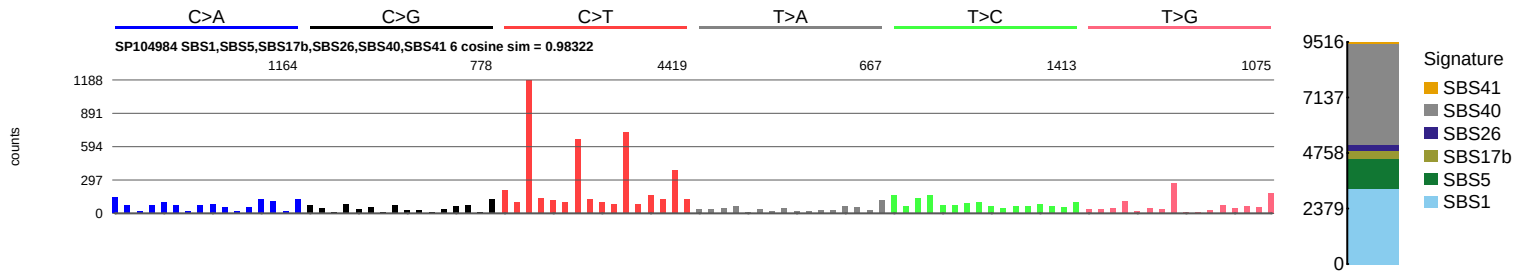
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98323
1179
774
4435
652
1408
1064
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98323
1181
773
4434
650
1410
1065
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98323
1227
769
4337
660
1440
1081
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98323
1164
773
4507
651
1363
1056
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98322
1163
777
4424
665
1412
1074
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS5
SBS1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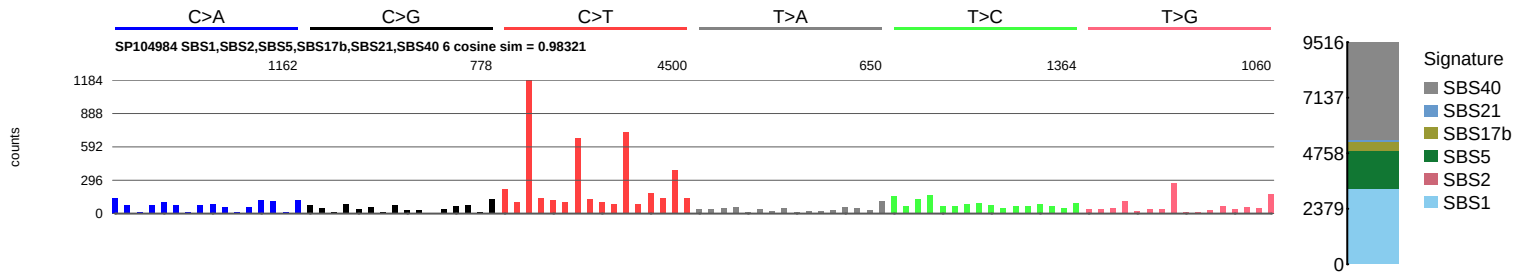
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98322
1164
778
4419
667
1413
1075
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS5
SBS1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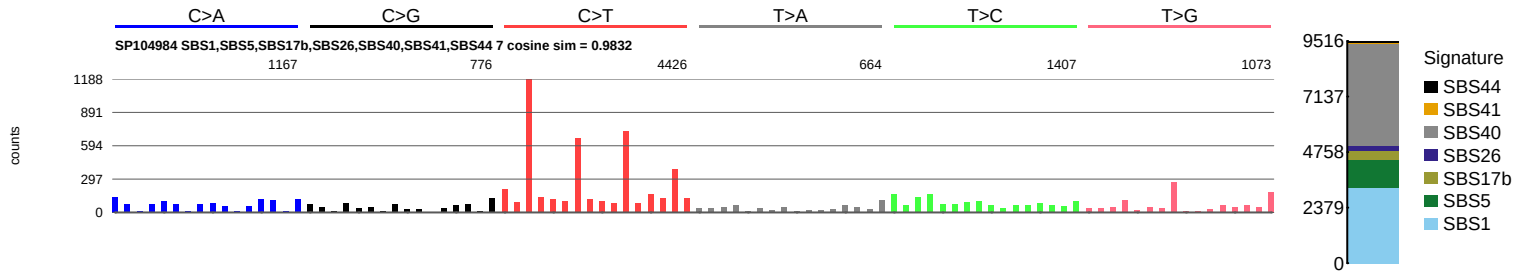
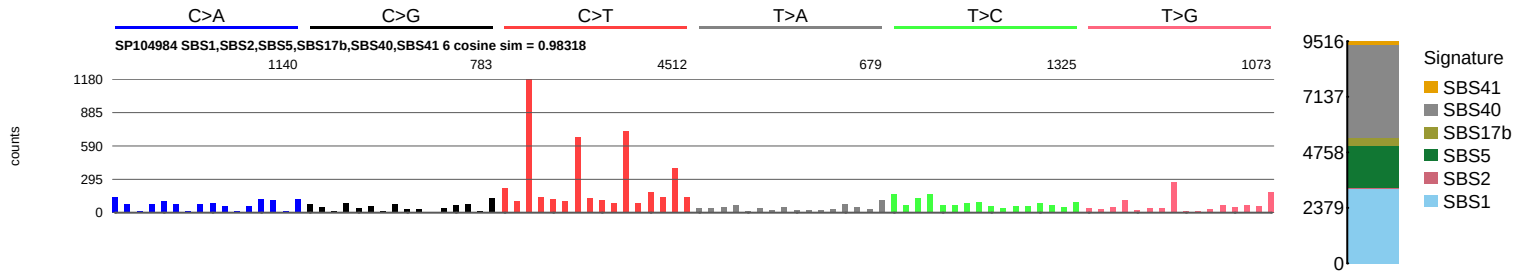
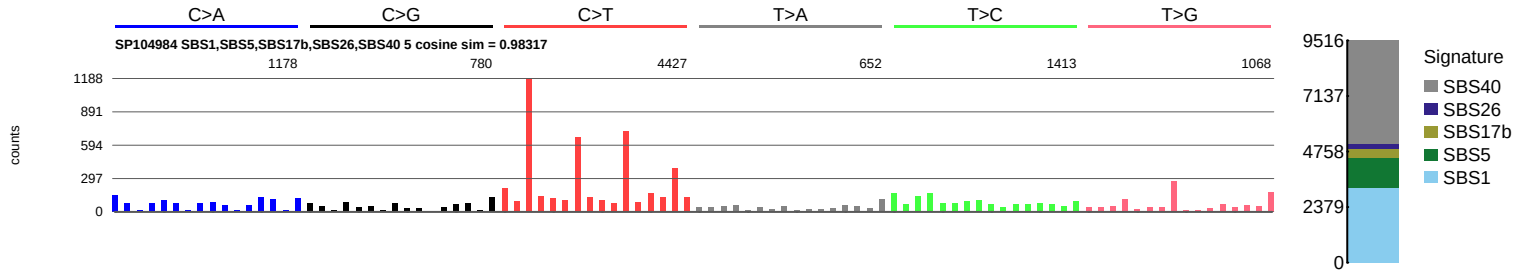
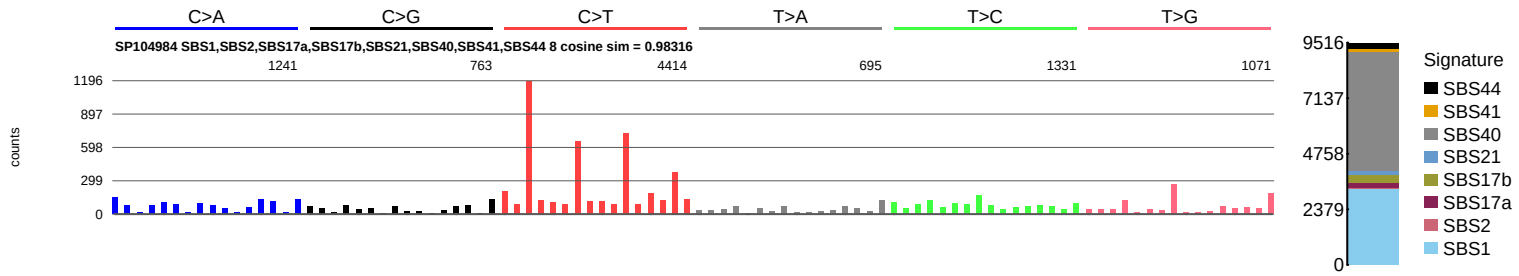
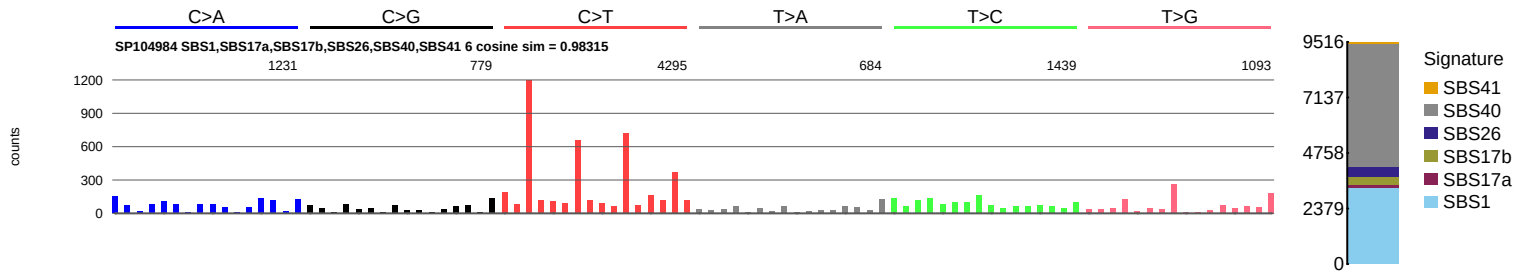
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS21,SBS40 6 cosine sim = 0.98321
1162
778
4500
650
1364
1060
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS5
SBS2
SBS1

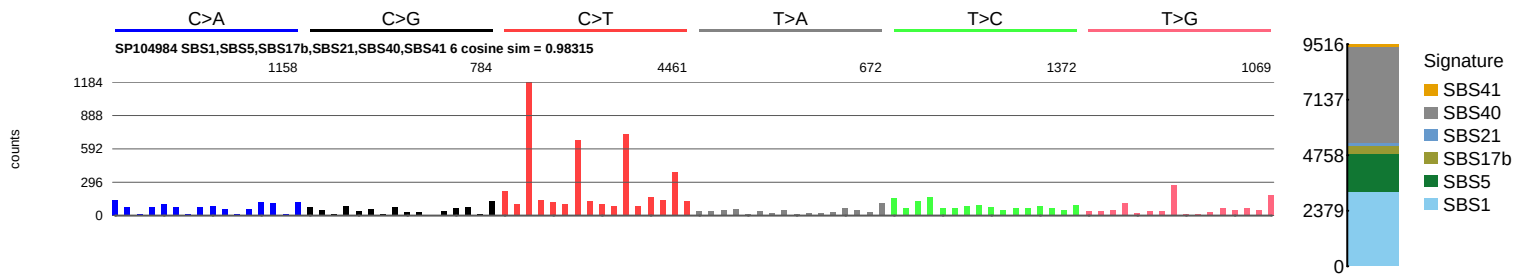

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.9832
1167
776
4426
664
1407
1073
1188
891
594
297
0
counts
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS40,SBS41 6 cosine sim = 0.98318
1140
783
4512
679
1325
1073
1180
885
590
295
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS26,SBS40 5 cosine sim = 0.98317
1178
780
4427
652
1413
1068
1188
891
594
297
0
counts
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98316
1241
763
4414
695
1331
1071
1196
897
598
299
0
counts
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98315
1231
779
4295
684
1439
1093
1200
900
600
300
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1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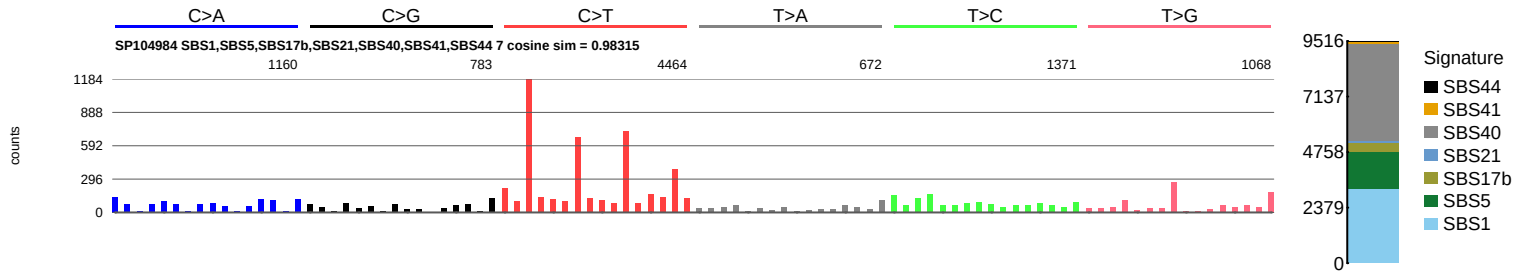
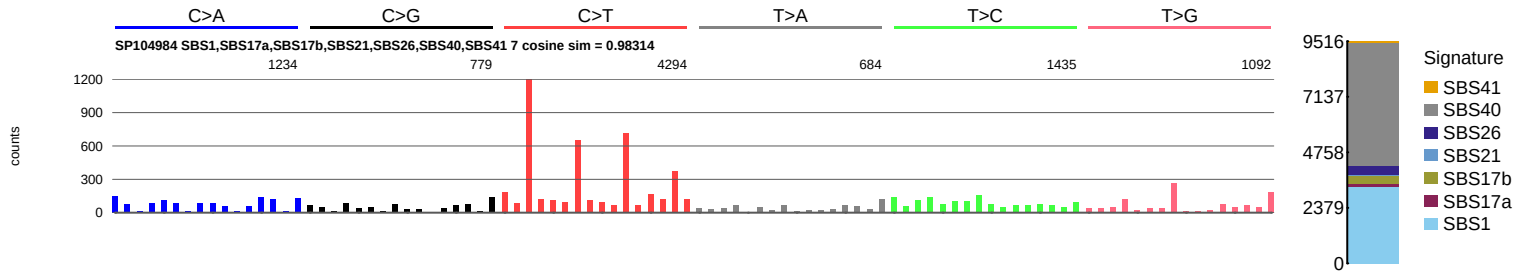
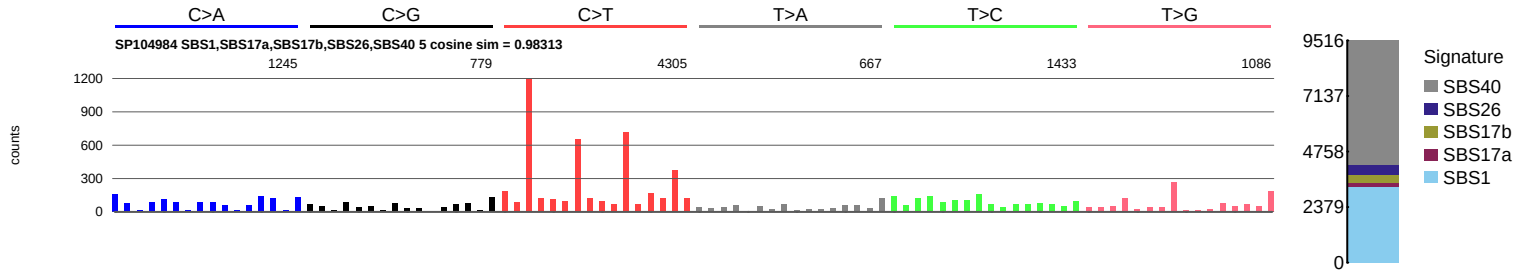
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98315
1158
784
4461
672
1372
1069
1184
888
592
296
0
counts
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98315
1160
783
4464
672
1371
1068
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98314
1234
779
4294
684
1435
1092
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98313
1245
779
4305
667
1433
1086
counts
1200
900
600
300
0
9516
7137
4758
2379
0
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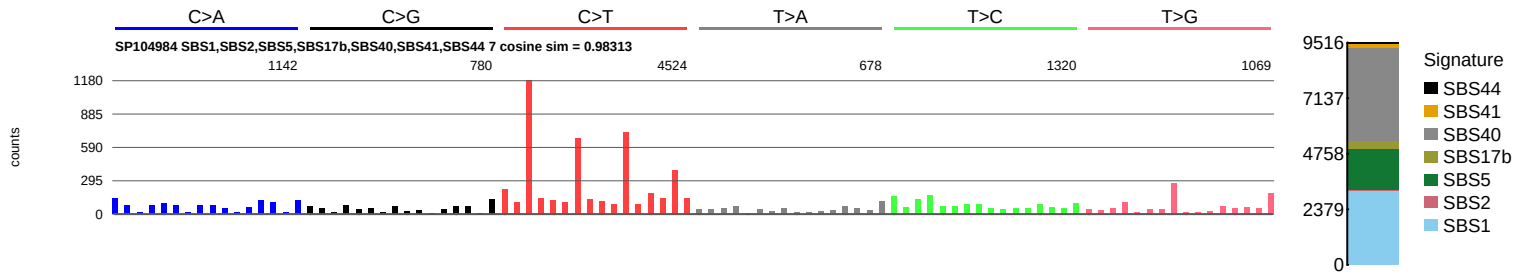
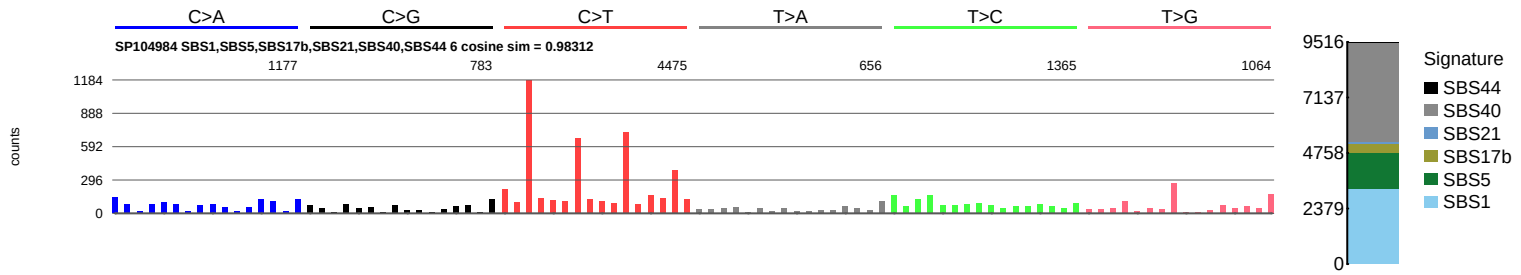
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98313
1142
780
4524
678
1320
1069
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98312
1177
783
4475
656
1365
1064
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
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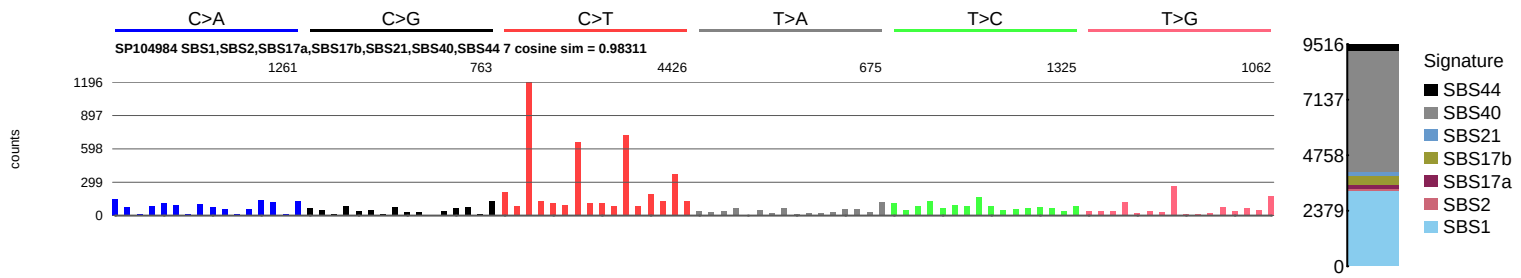
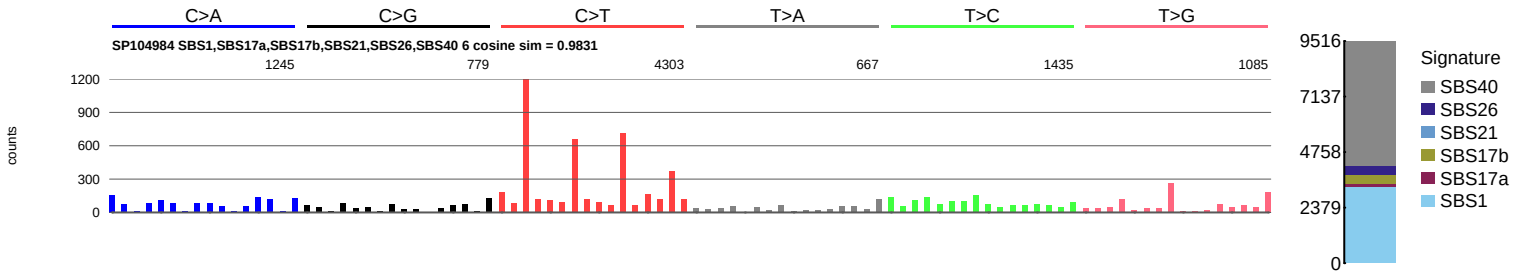
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98311
1261
763
4426
675
1325
1062
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.9831
1245
779
4303
667
1435
1085
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
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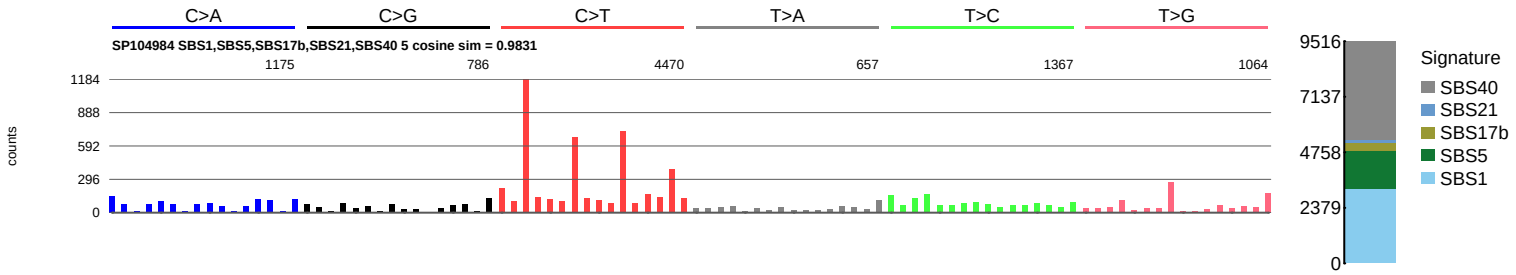
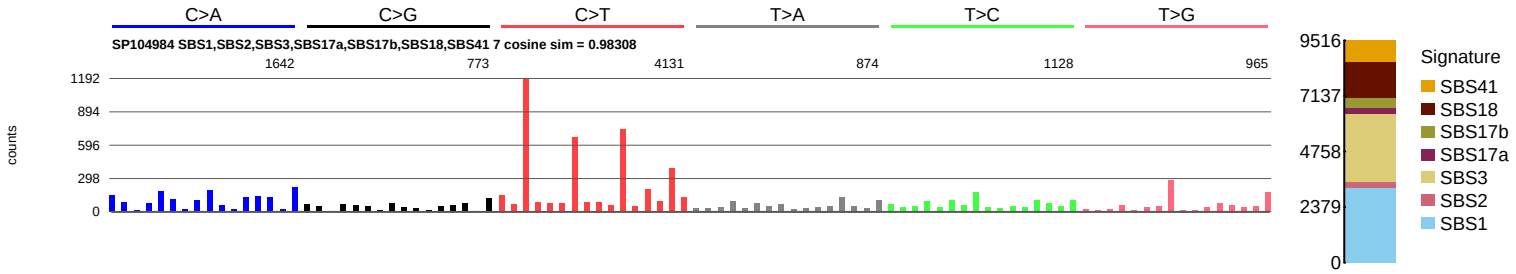
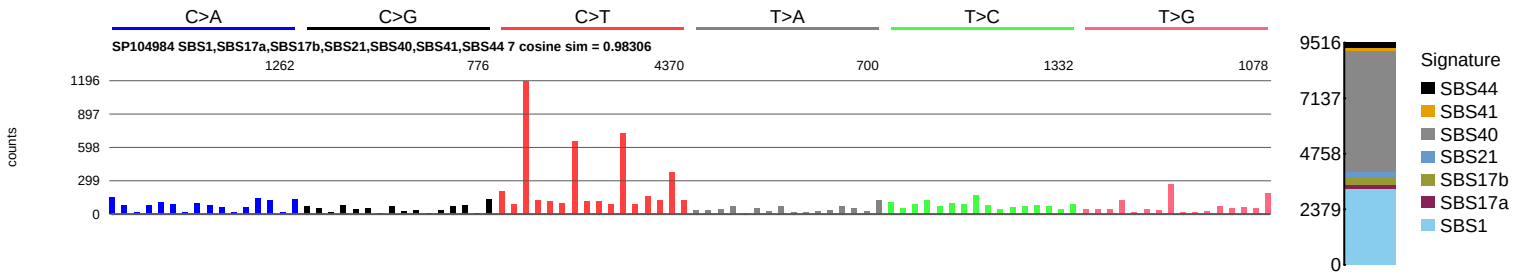
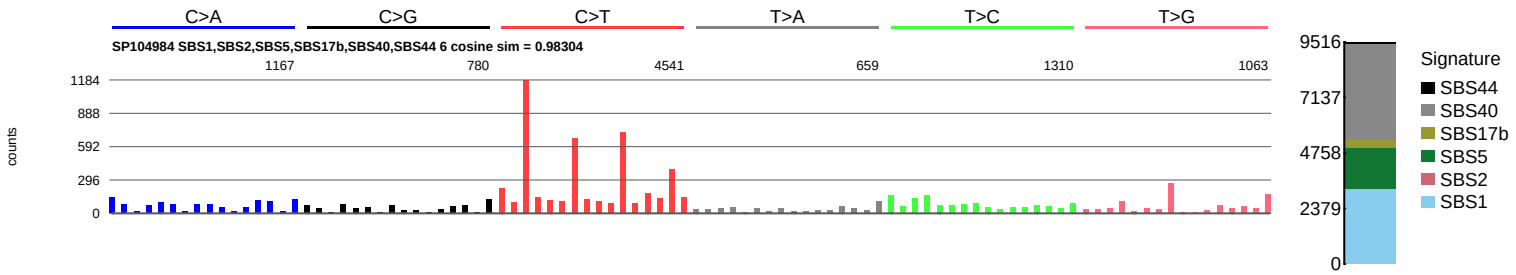
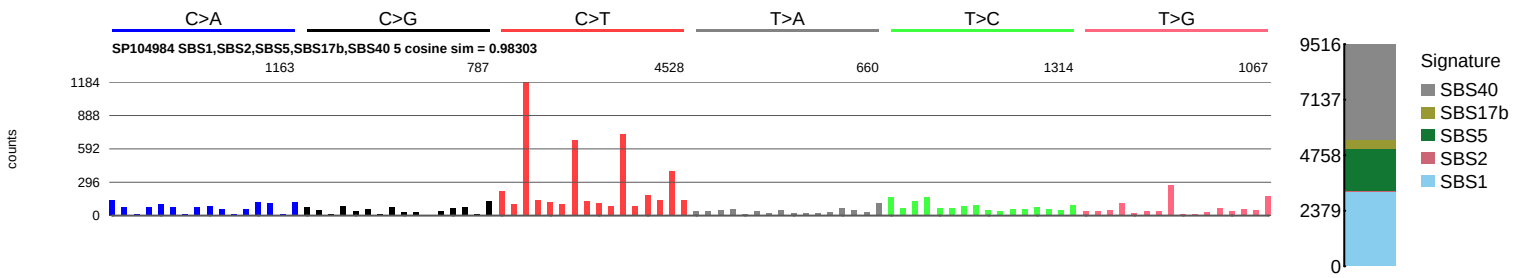
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS21,SBS40 5 cosine sim = 0.9831
1175
786
4470
657
1367
1064
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98308
1642
773
4131
874
1128
965
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98306
1262
776
4370
700
1332
1078
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS40,SBS44 6 cosine sim = 0.98304
1167
780
4541
659
1310
1063
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS5
SBS2
SBS1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C>G
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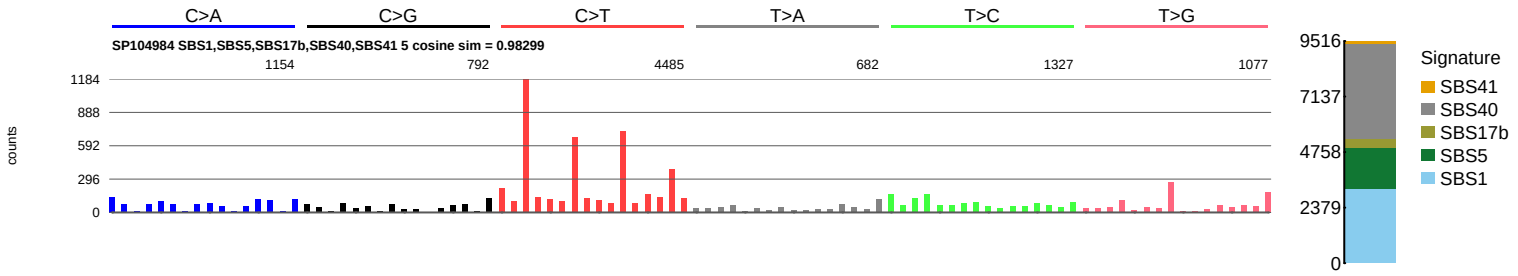
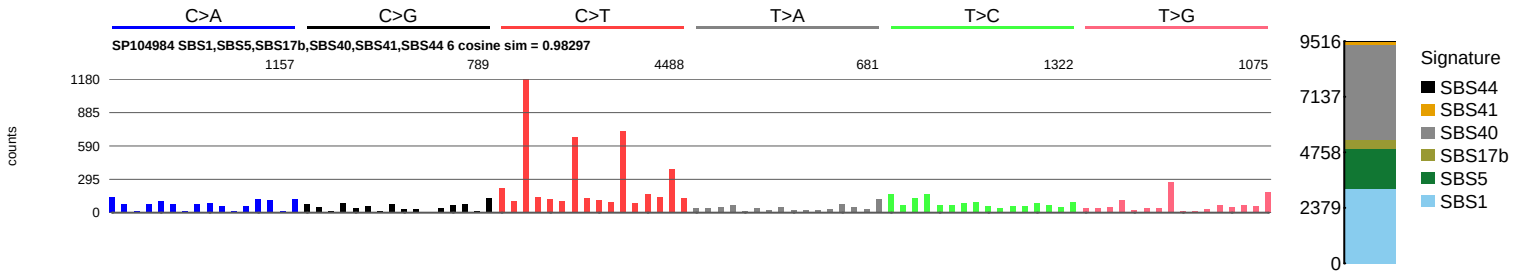
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17b,SBS40 5 cosine sim = 0.98303
1163
787
4528
660
1314
1067
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS40,SBS41 5 cosine sim = 0.98299
1154
792
4485
682
1327
1077
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS5
SBS1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C>G
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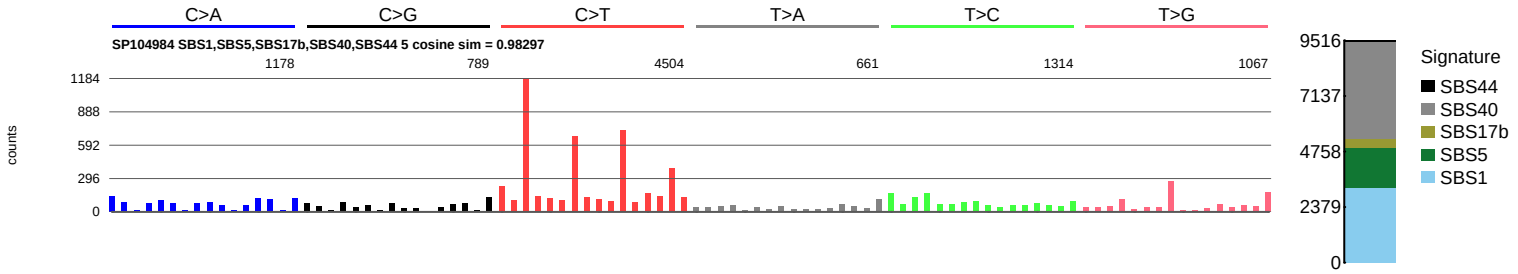
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98297
1157
789
4488
681
1322
1075
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS5
SBS1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C>G
C>T
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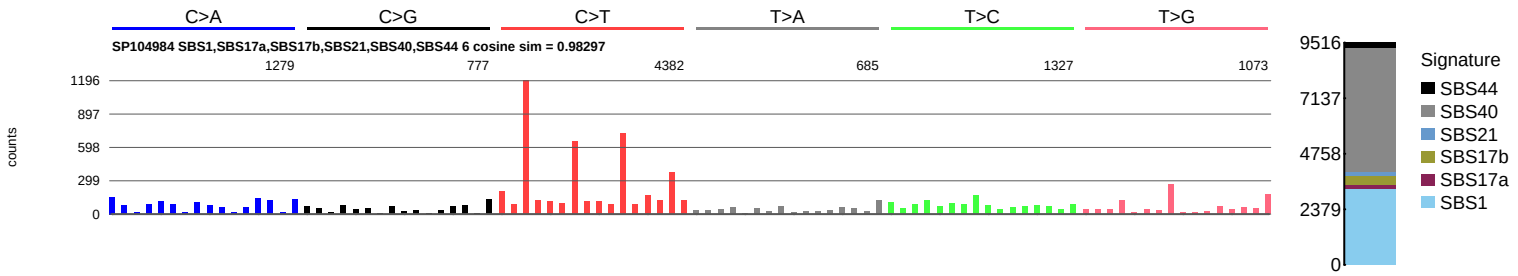
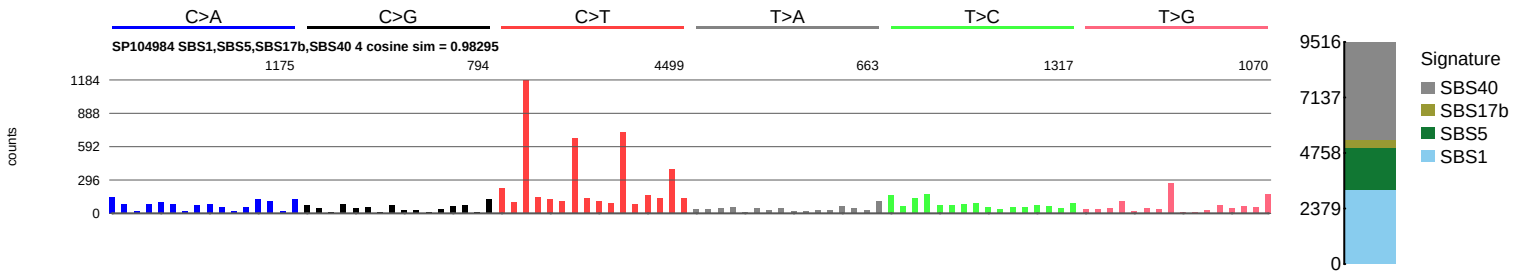
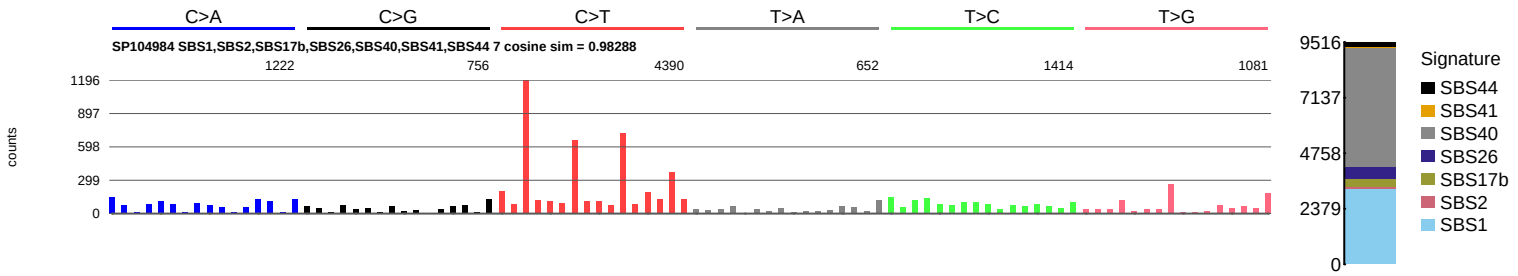
T>G
SP104984 SBS1,SBS5,SBS17b,SBS40,SBS44 5 cosine sim = 0.98297
1178
789
4504
661
1314
1067
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98297
1279
777
4382
685
1327
1073
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17b,SBS40 4 cosine sim = 0.98295
1175
794
4499
663
1317
1070
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS5
SBS1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C>G
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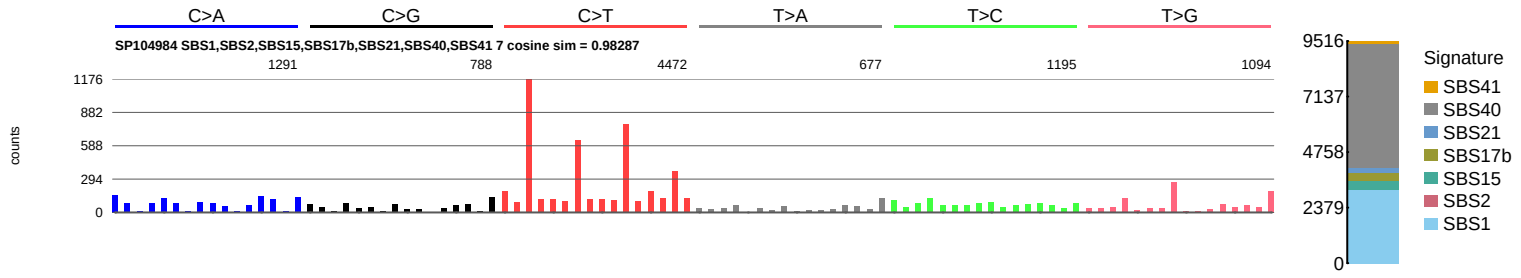
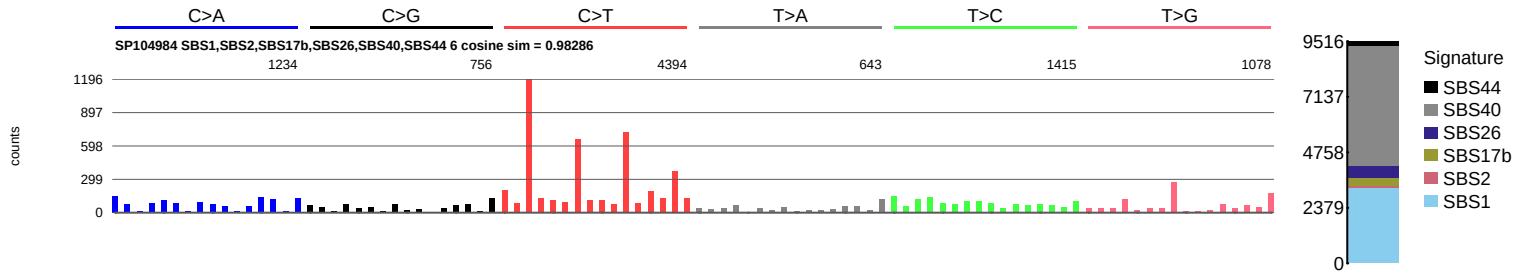
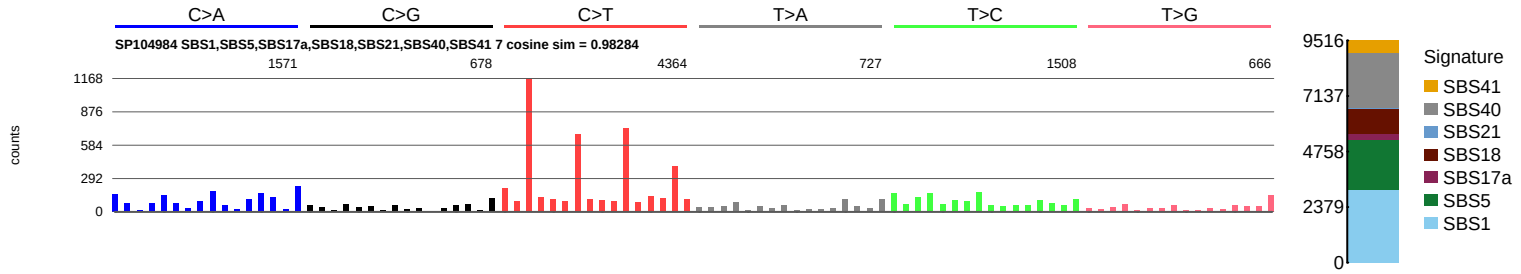
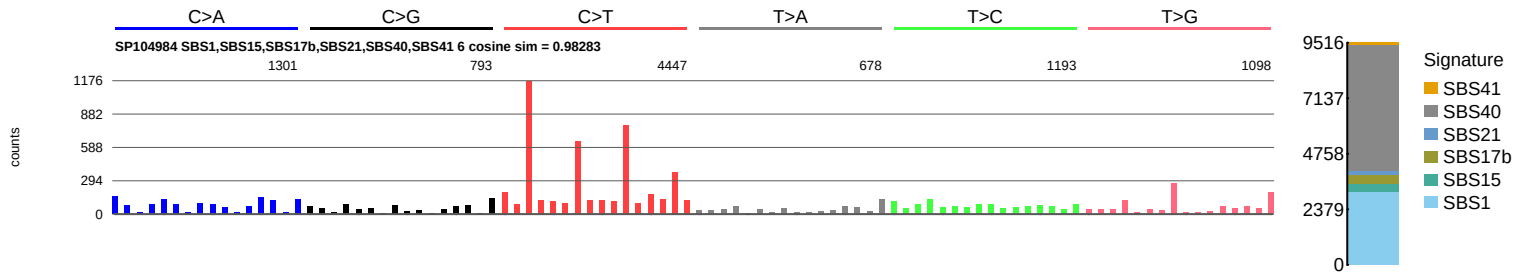
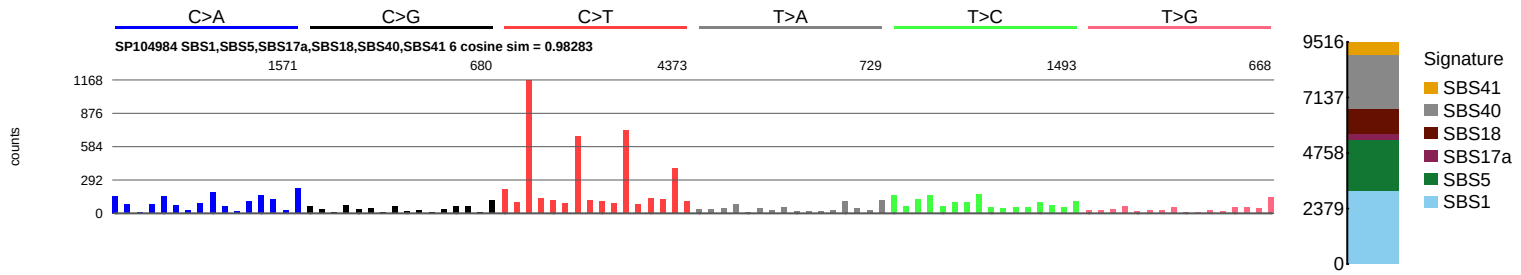
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98288
1222
756
4390
652
1414
1081
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS2
SBS1

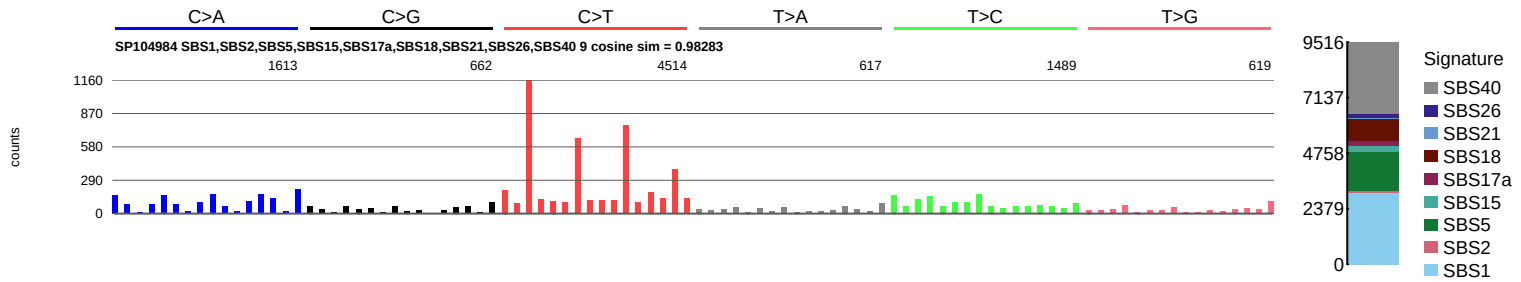

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98287
1291
788
4472
677
1195
1094
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98286
1234
756
4394
643
1415
1078
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98284
1571
678
4364
727
1508
666
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98283
1301
793
4447
678
1193
1098
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS18,SBS40,SBS41 6 cosine sim = 0.98283
1571
680
4373
729
1493
668
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS1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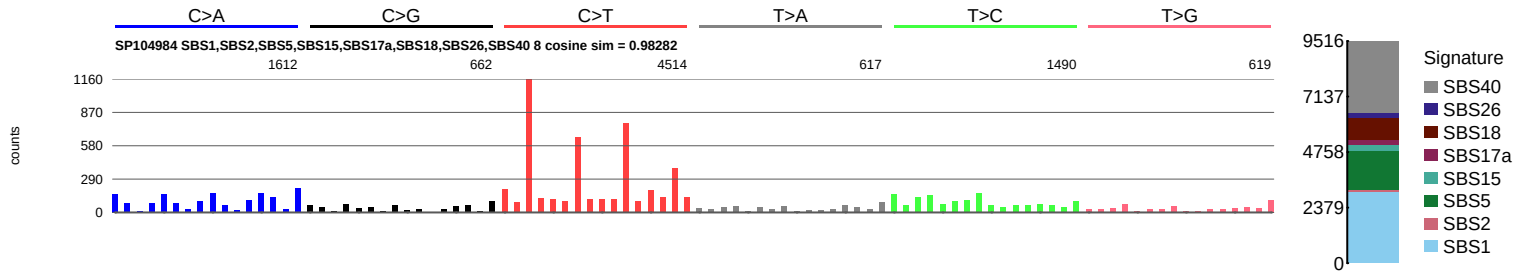
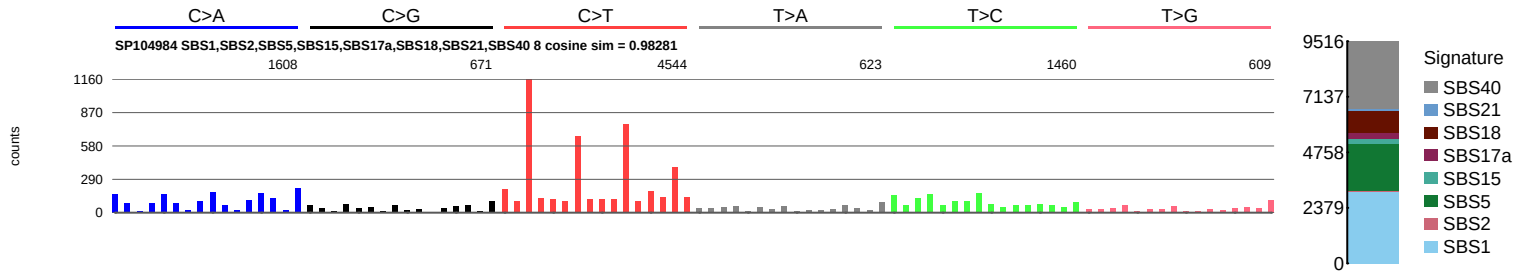
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98283
1613
662
4514
617
1489
619
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98282
1612
662
4514
617
1490
619
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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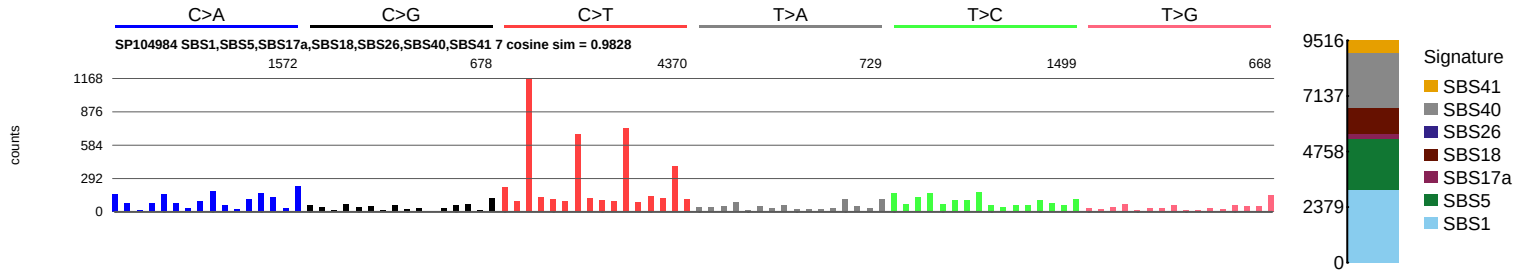
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98281
1608
671
4544
623
1460
609
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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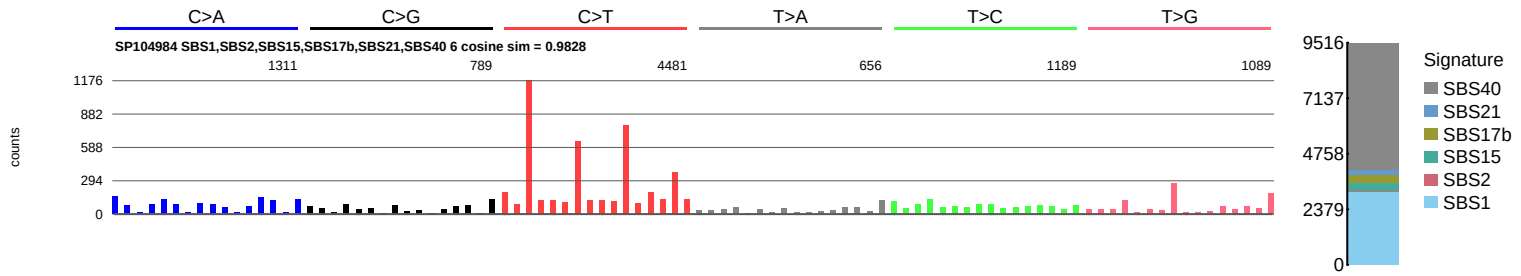
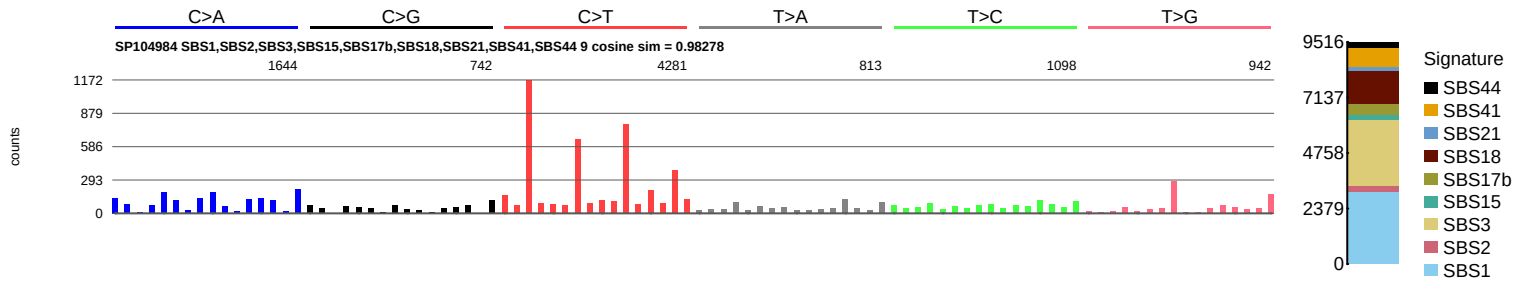
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.9828
1572
678
4370
729
1499
668
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS21,SBS40 6 cosine sim = 0.9828
1311
789
4481
656
1189
1089
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS15
SBS2
SBS1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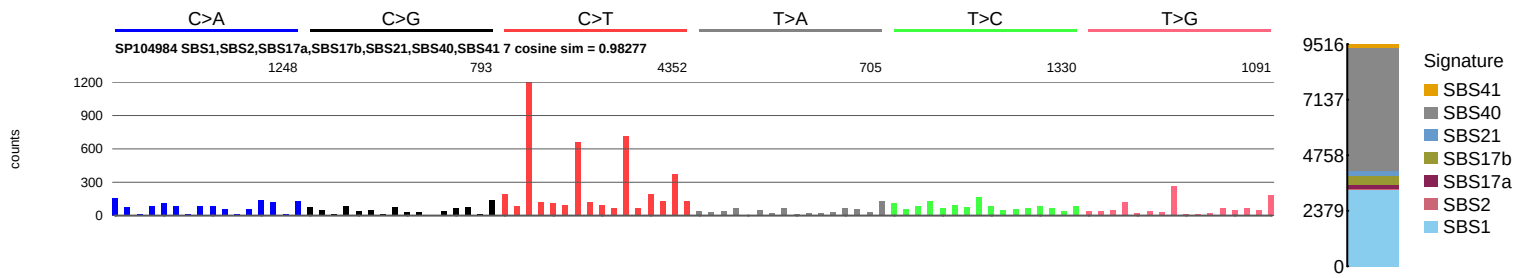
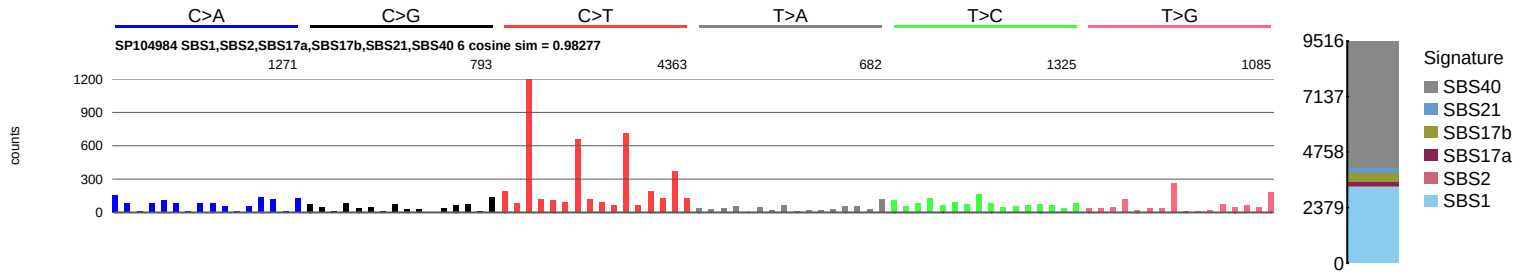
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98278
1644
742
4281
813
1098
942
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98277
1248
793
4352
705
1330
1091
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98277
1271
793
4363
682
1325
1085
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
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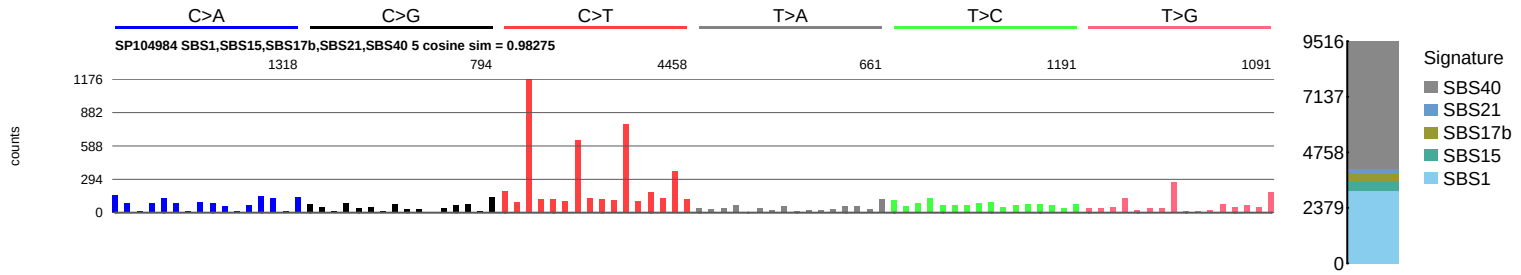
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS21,SBS40 5 cosine sim = 0.98275
1318
794
4458
661
1191
1091
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
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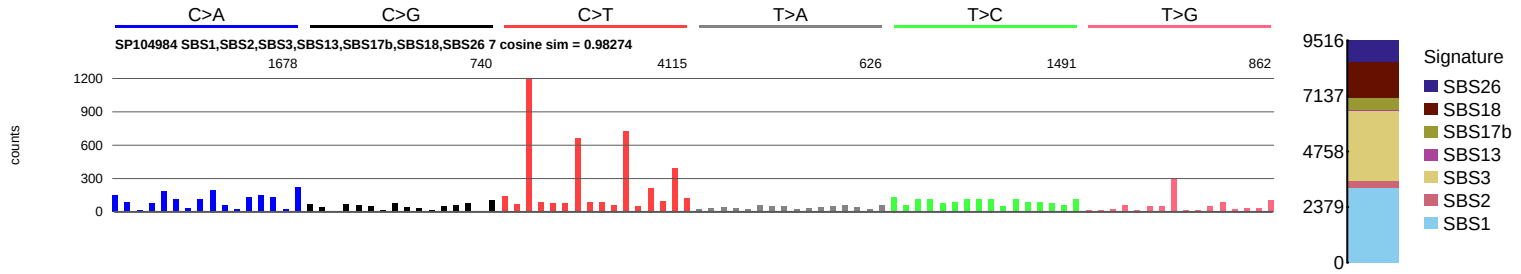
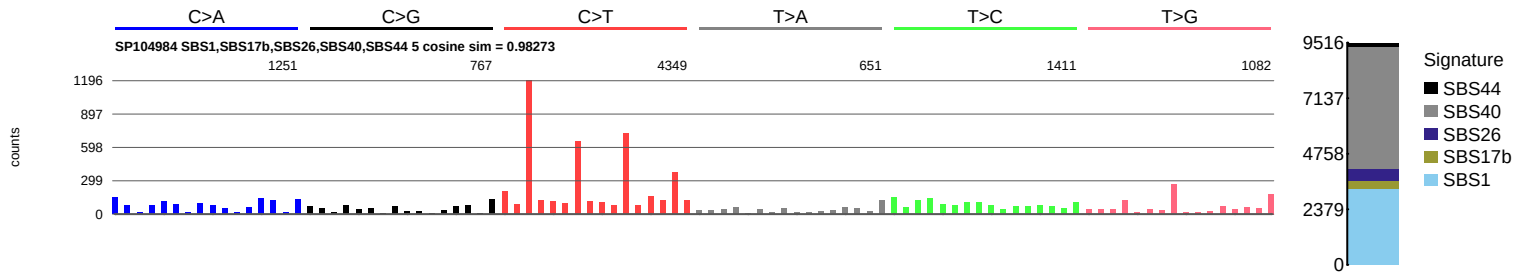
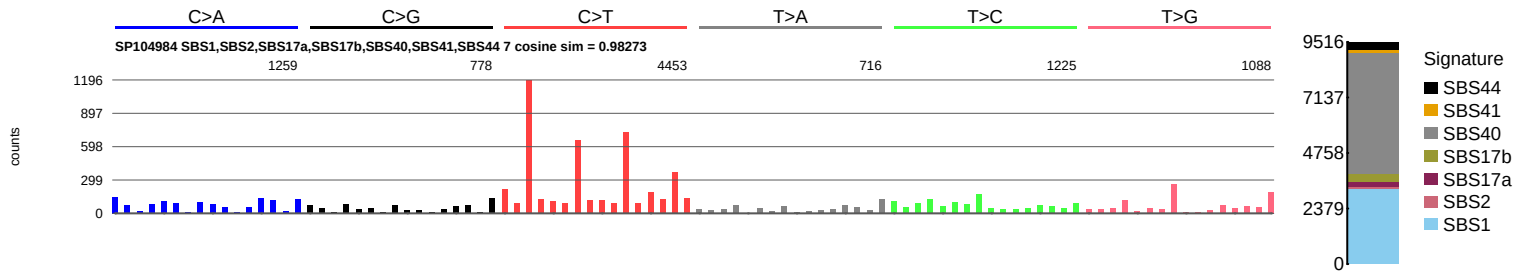
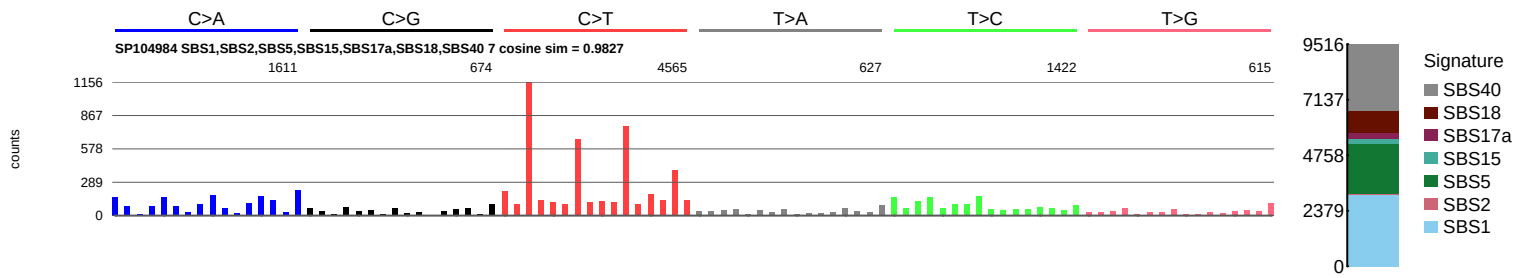
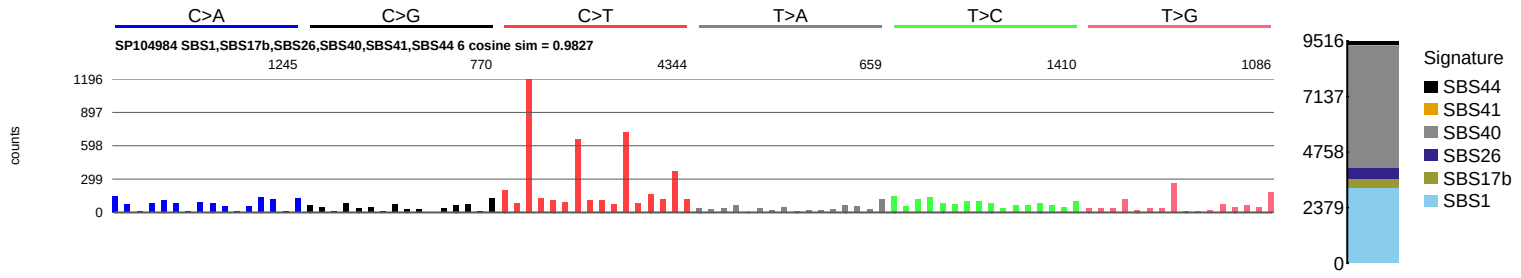
SBS17b
SBS15
SBS1

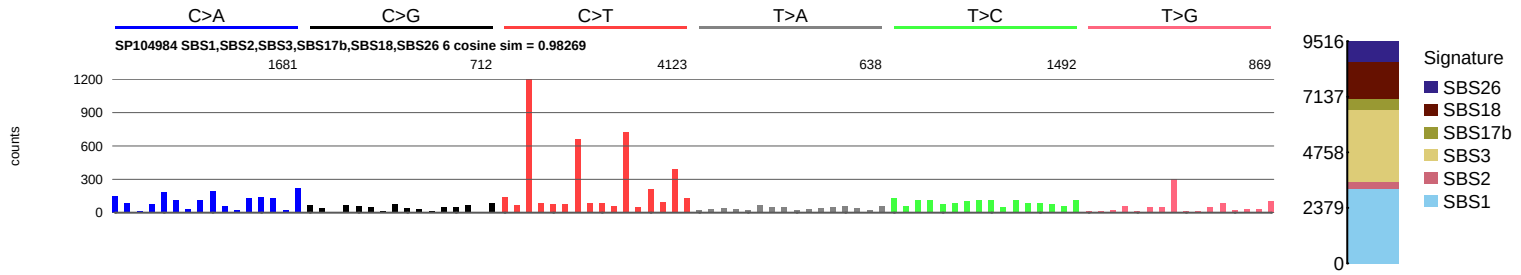

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26 7 cosine sim = 0.98274
1678
740
4115
626
1491
862
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS26,SBS40,SBS44 5 cosine sim = 0.98273
1251
767
4349
651
1411
1082
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98273
1259
778
4453
716
1225
1088
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40 7 cosine sim = 0.9827
1611
674
4565
627
1422
615
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS26,SBS40,SBS41,SBS44 6 cosine sim = 0.9827
1245
770
4344
659
1410
1086
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS1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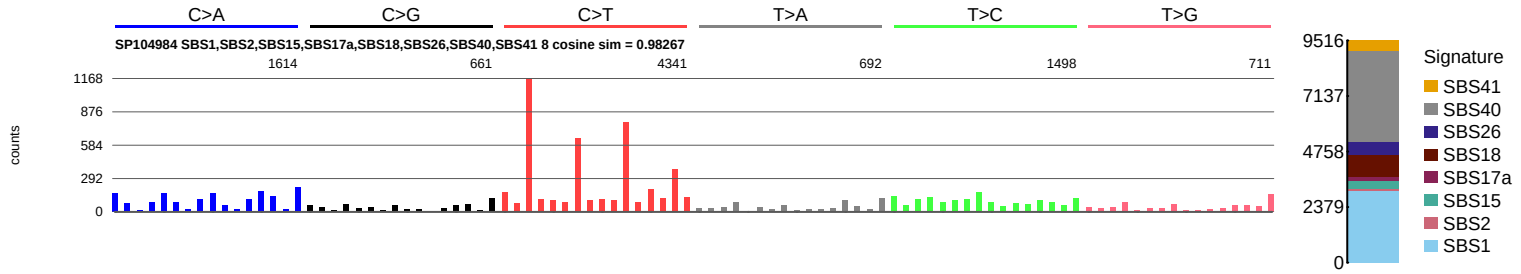
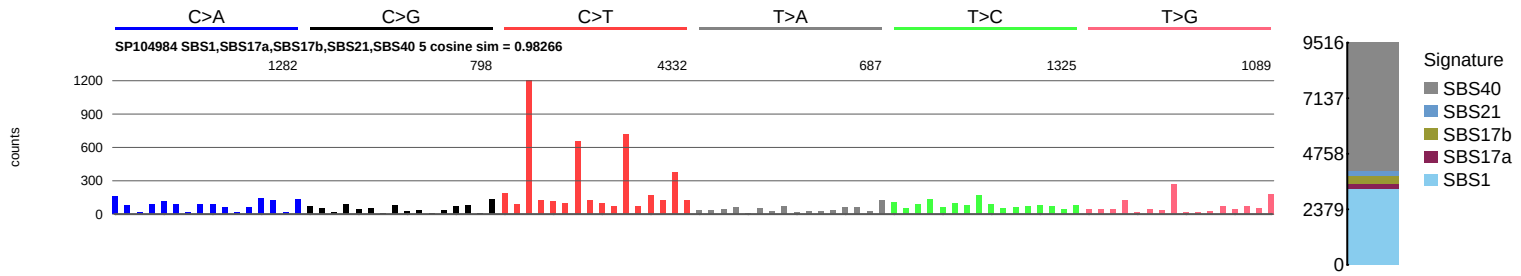
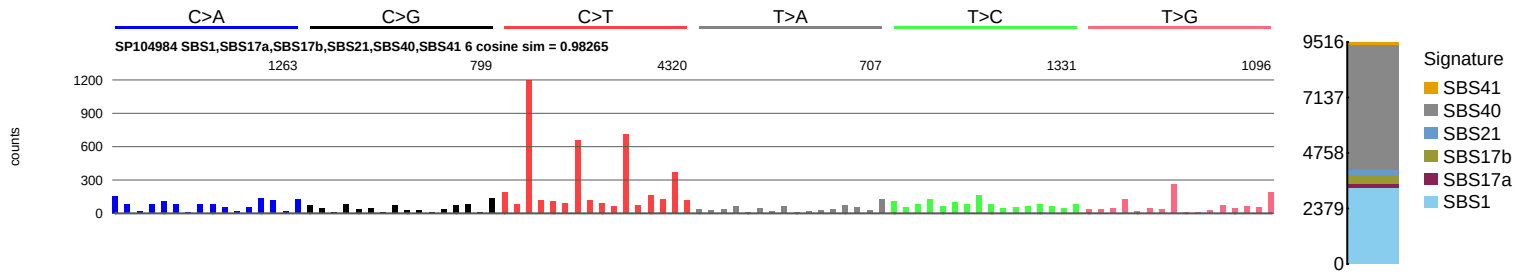
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26 6 cosine sim = 0.98269
1681
712
4123
638
1492
869
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98267
1614
661
4341
692
1498
711
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98266
1282
798
4332
687
1325
1089
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS1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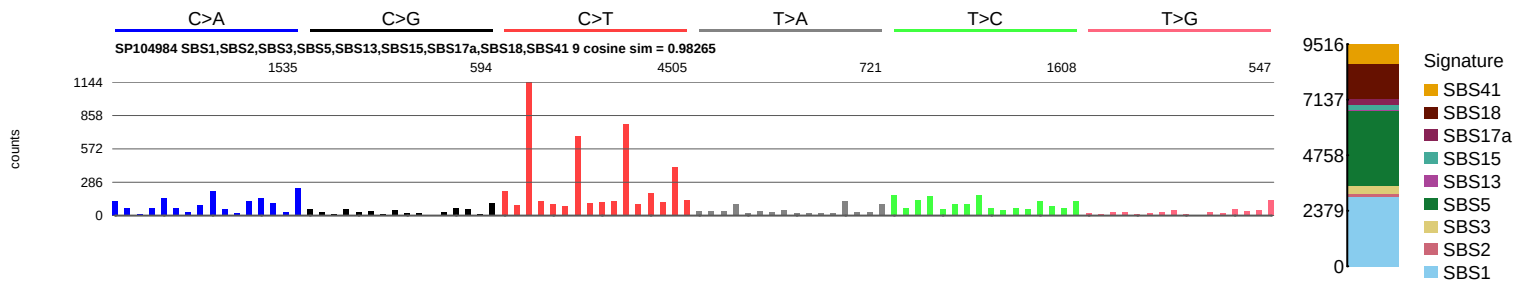
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98265
1263
799
4320
707
1331
1096
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1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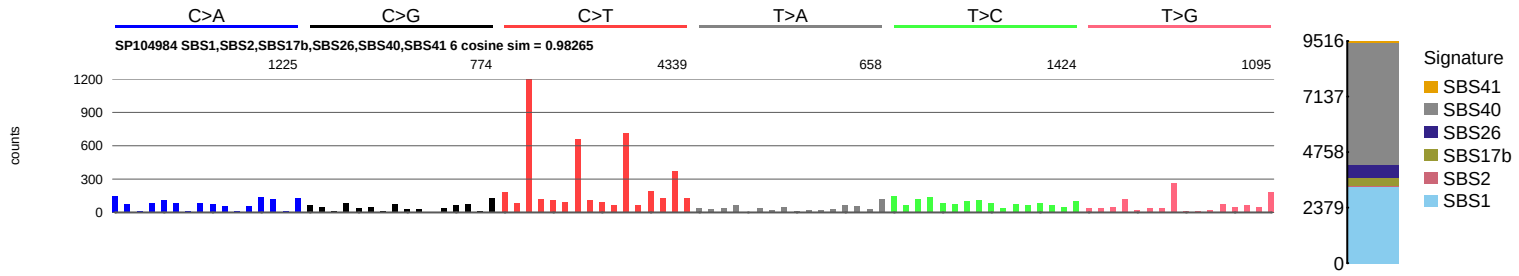
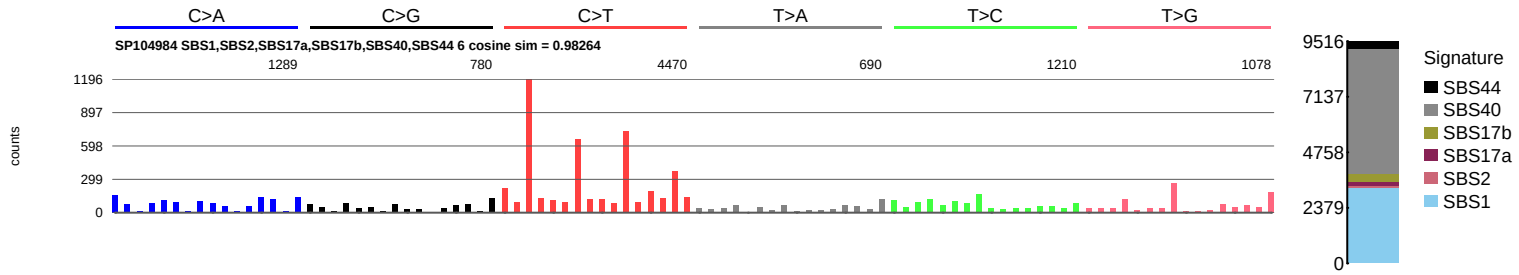
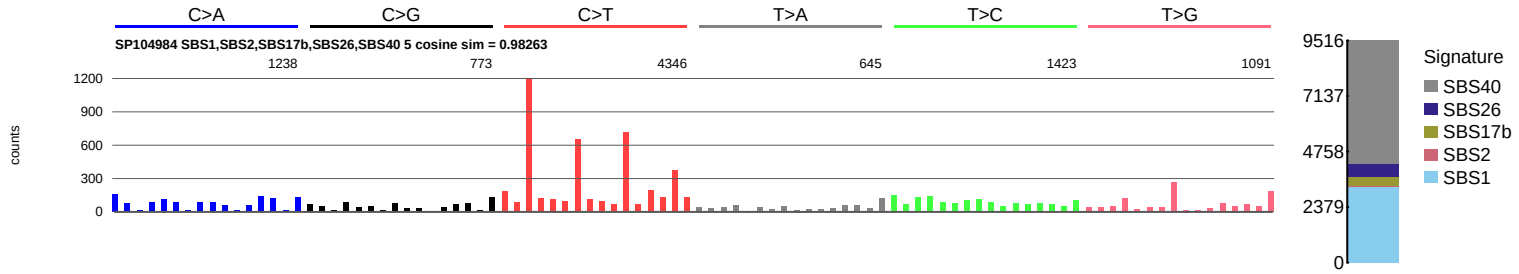
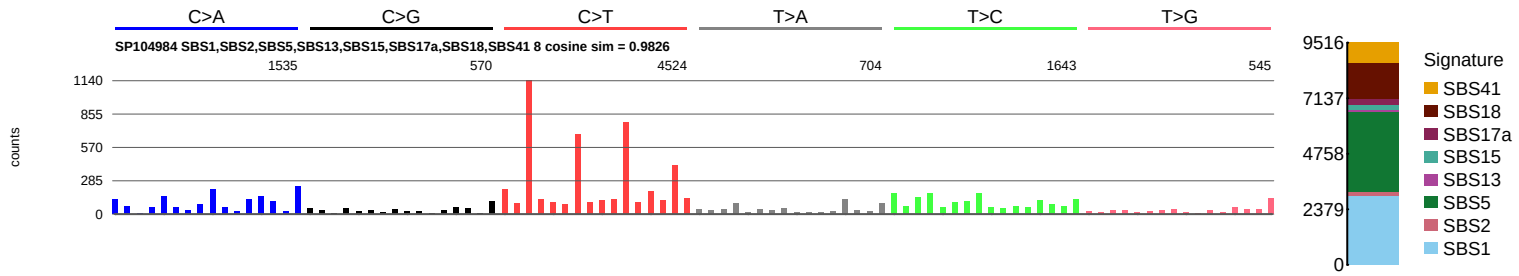
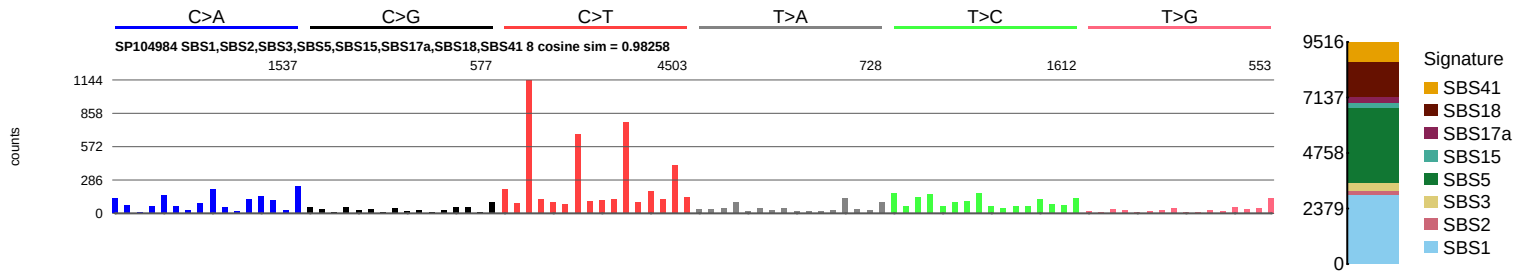
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 9 cosine sim = 0.98265
1535
594
4505
721
1608
547
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

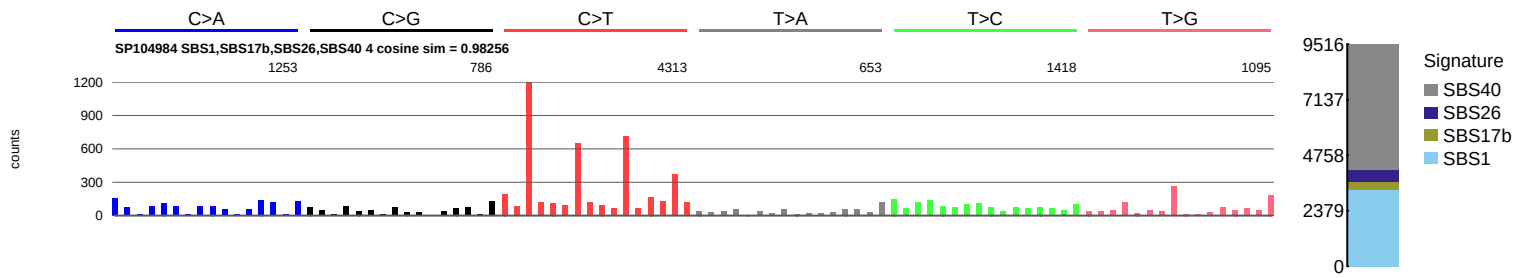

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98265
1225
774
4339
658
1424
1095
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS2
SBS1

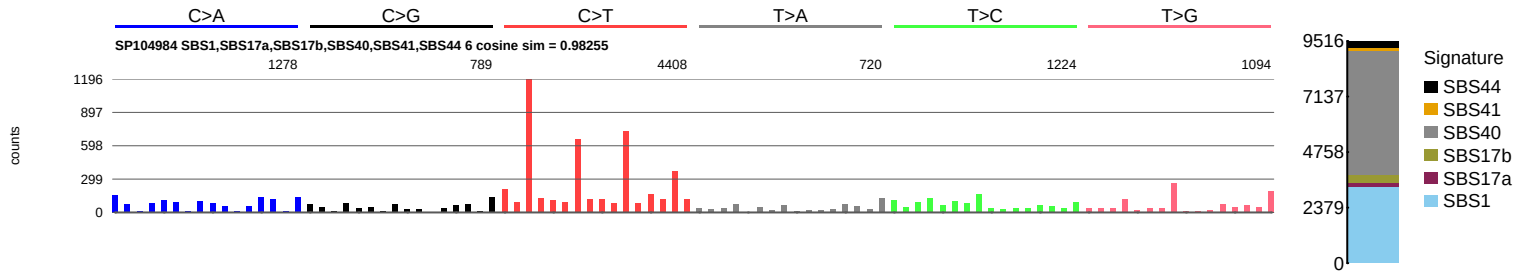

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98264
1289
780
4470
690
1210
1078
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS2
SBS1

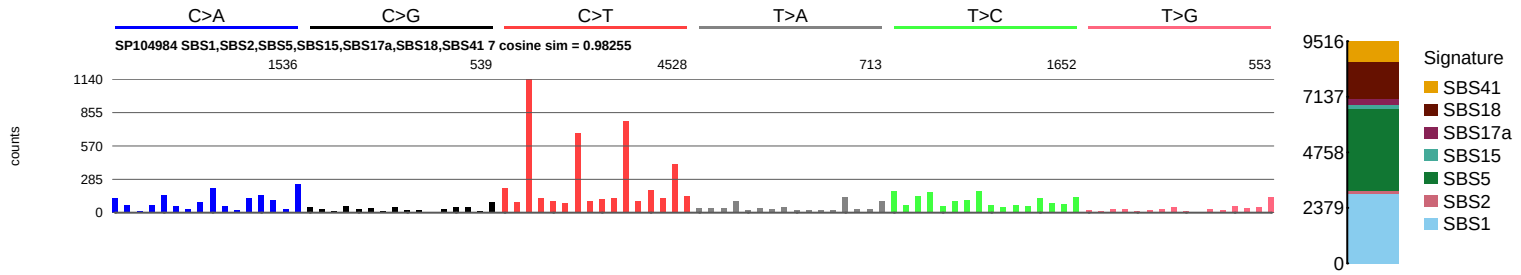

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS26,SBS40 5 cosine sim = 0.98263
1238
773
4346
645
1423
1091
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS2
SBS1

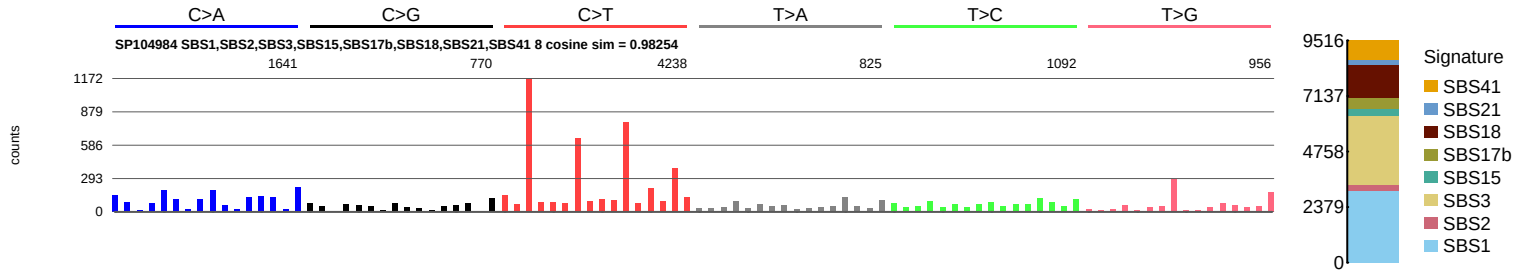

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.9826
1535
570
4524
704
1643
545
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

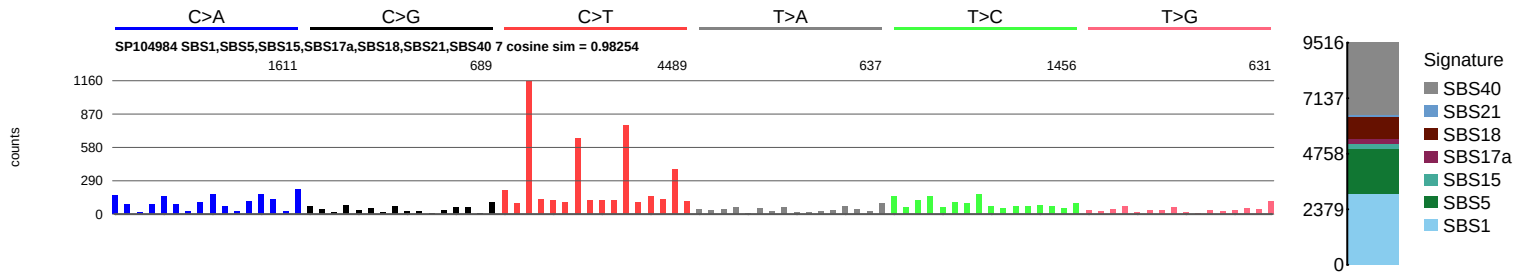

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98258
1537
577
4503
728
1612
553
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

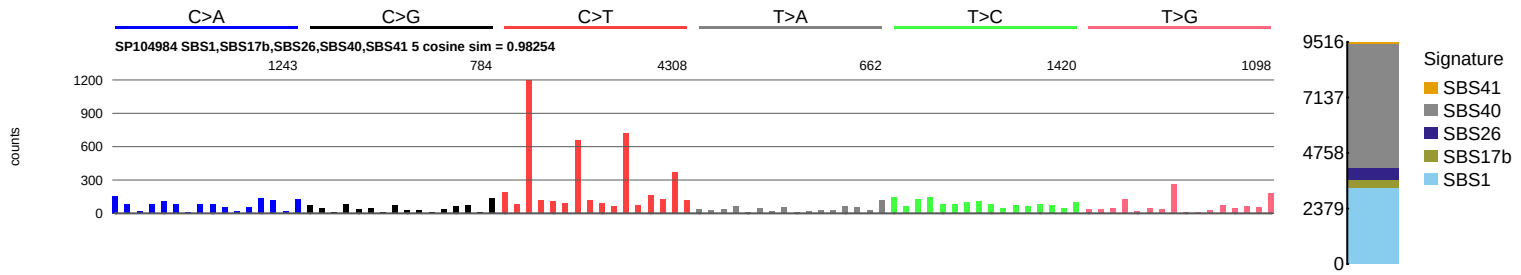

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS26,SBS40 4 cosine sim = 0.98256
1253
786
4313
653
1418
1095
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS17b
SBS1

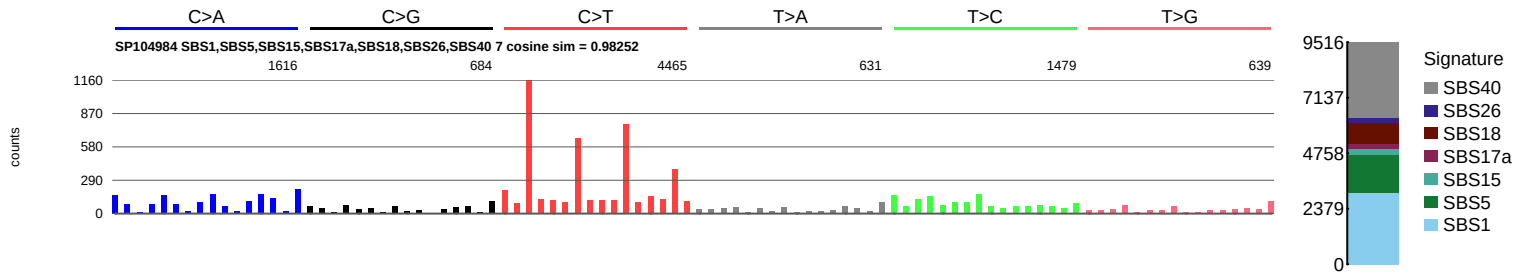

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98255
1278
789
4408
720
1224
1094
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS1

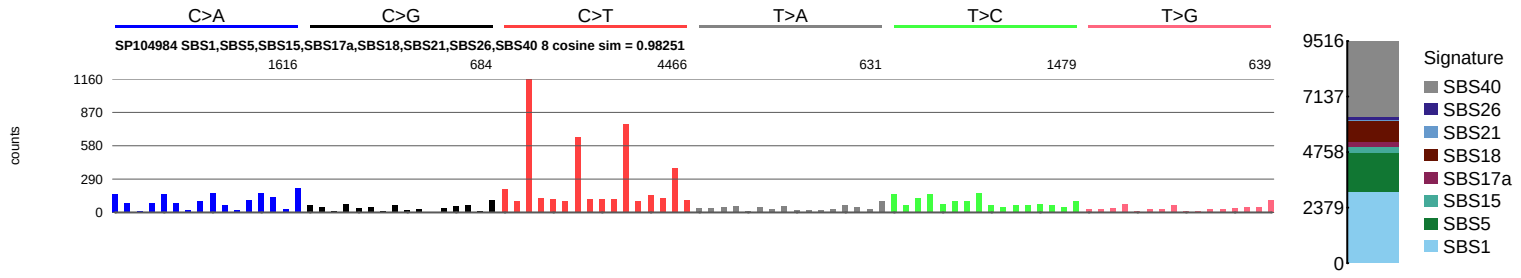

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS41 7 cosine sim = 0.98255
1536
539
4528
713
1652
553
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

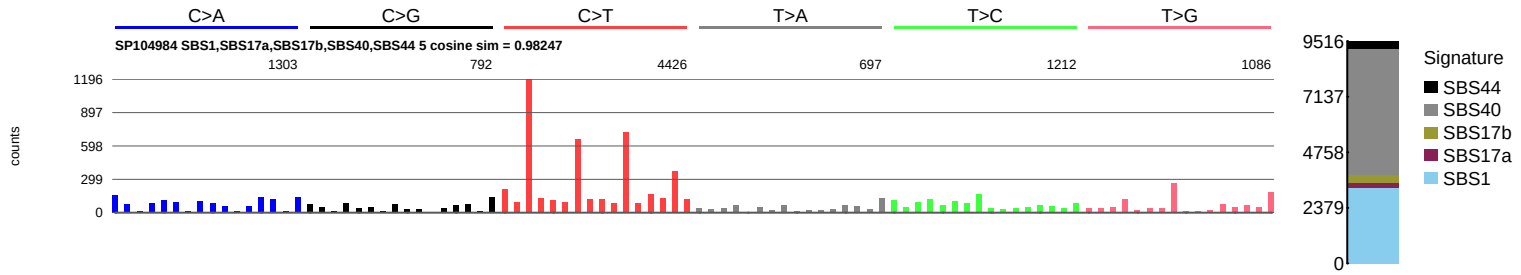

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98254
1641
770
4238
825
1092
956
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

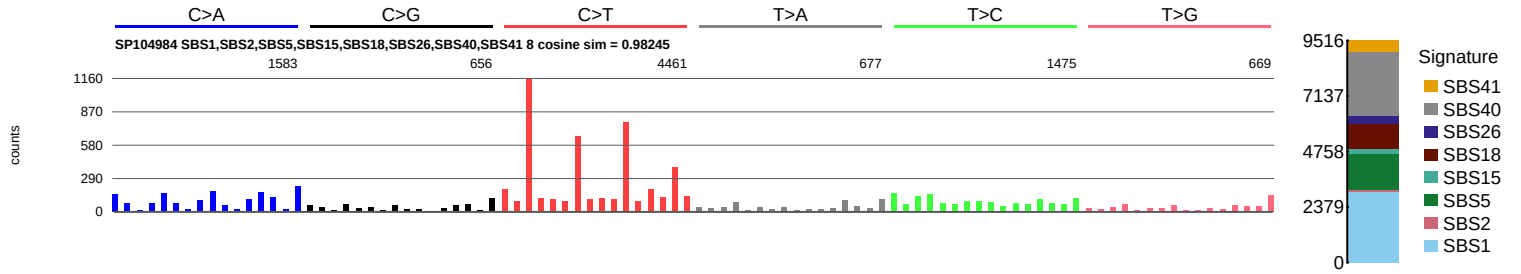

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98254
1611
689
4489
637
1456
631
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS26,SBS40,SBS41 5 cosine sim = 0.98254
1243
784
4308
662
1420
1098
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98252
1616
684
4465
631
1479
639
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98251
1616
684
4466
631
1479
639
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98247
1303
792
4426
697
1212
1086
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98245
1583
656
4461
677
1475
669
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
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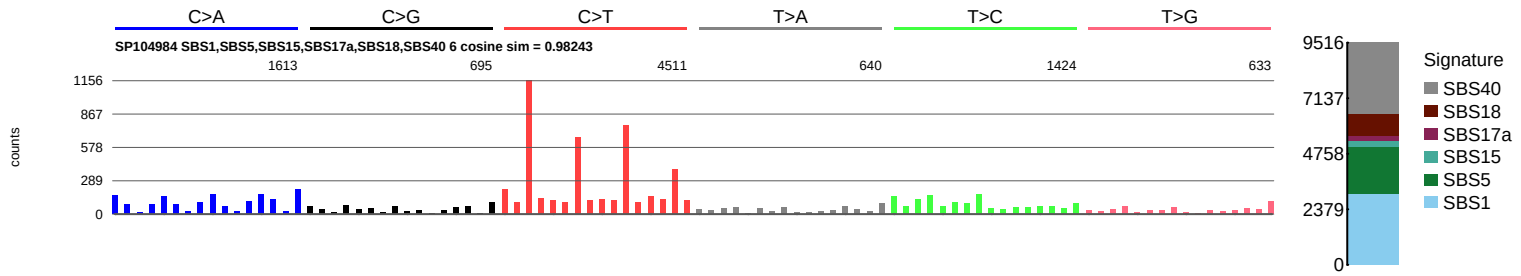
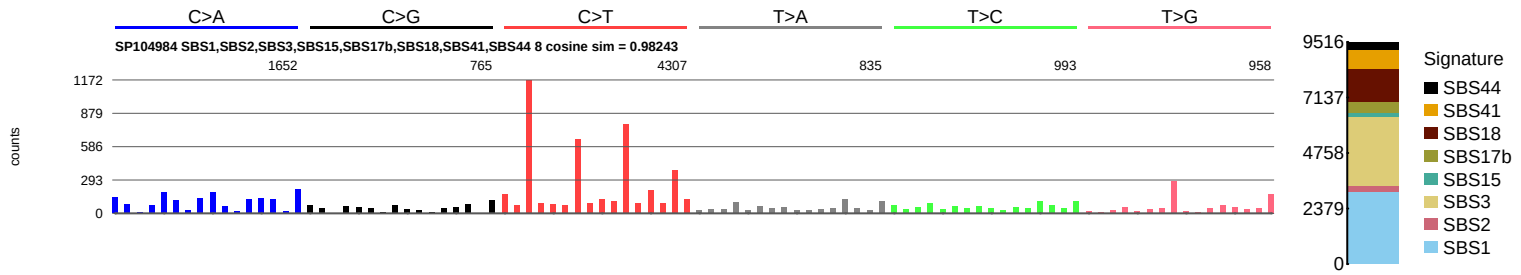
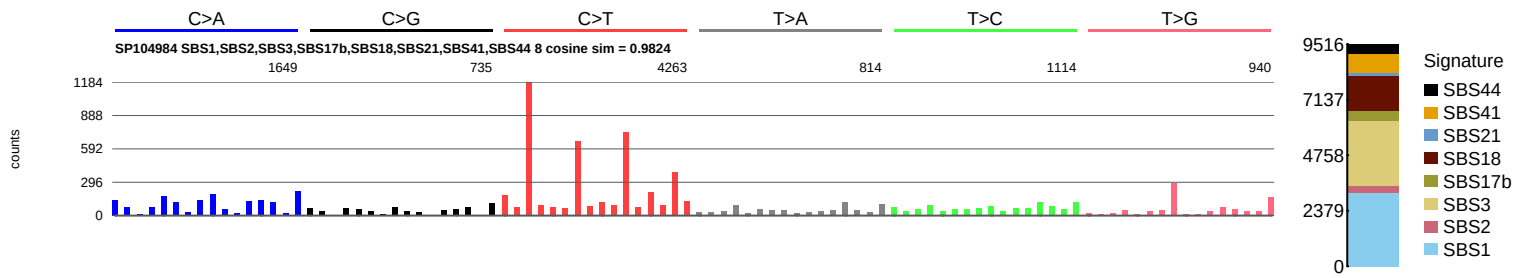
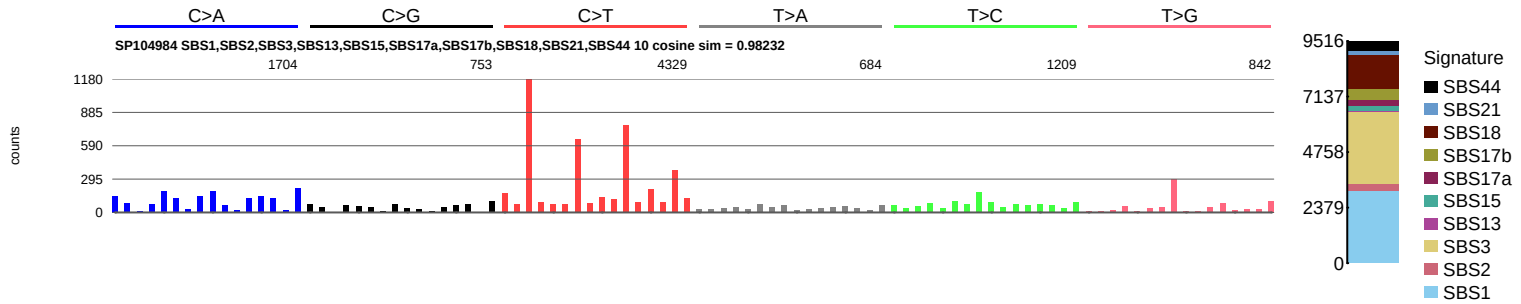
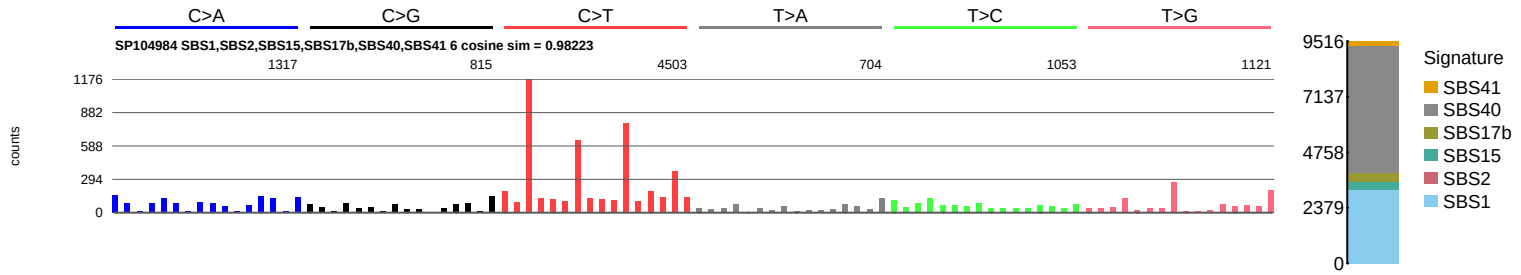
SBS5
SBS2
SBS1

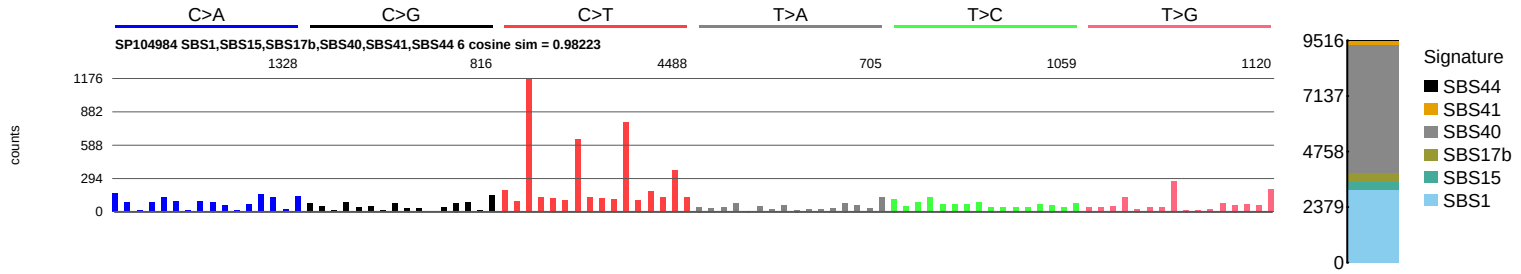

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS40 6 cosine sim = 0.98243
1613
695
4511
640
1424
633
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98243
1652
765
4307
835
993
958
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.9824
1649
735
4263
814
1114
940
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98232
1704
753
4329
684
1209
842
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS40,SBS41 6 cosine sim = 0.98223
1317
815
4503
704
1053
1121
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS2
SBS1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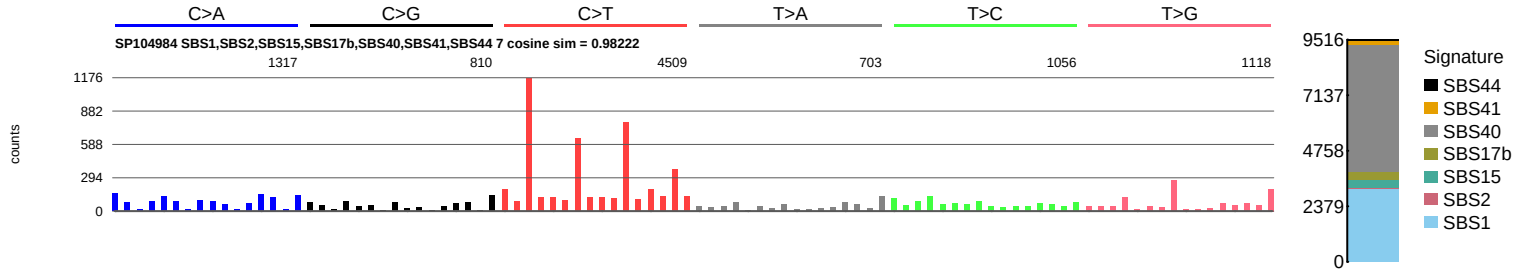
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98223
1328
816
4488
705
1059
1120
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS15
SBS1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C>G
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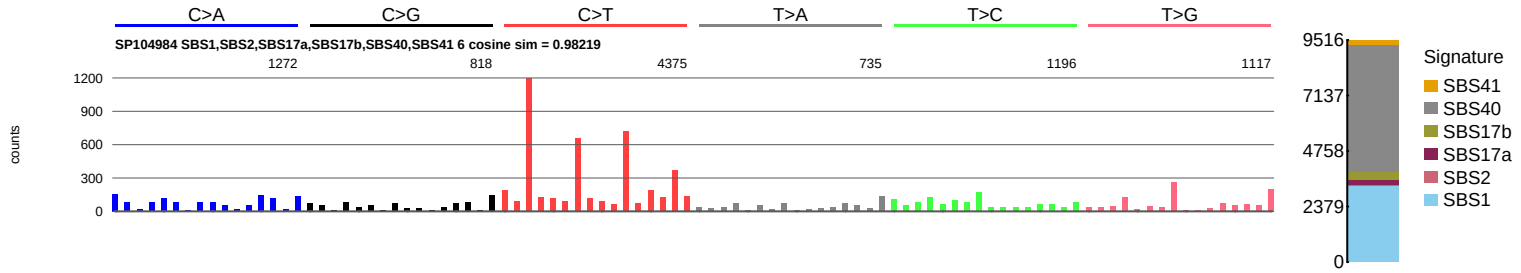
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98222
1317
810
4509
703
1056
1118
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS15
SBS2
SBS1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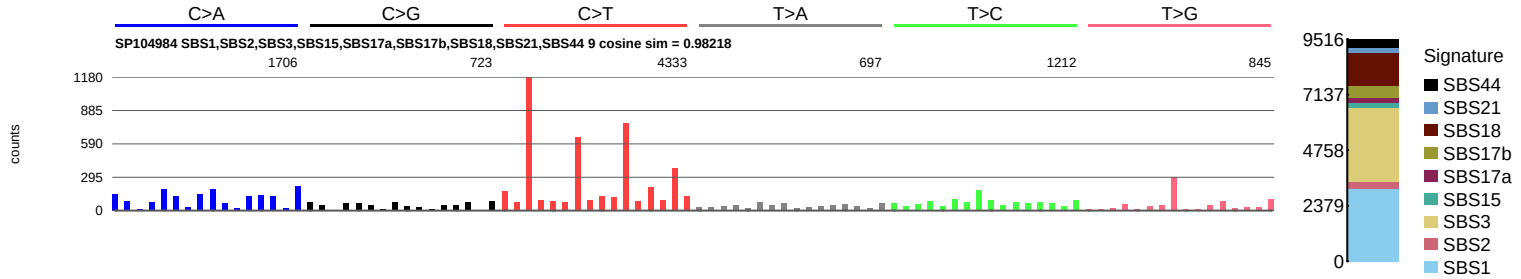
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98219
1272
818
4375
735
1196
1117
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1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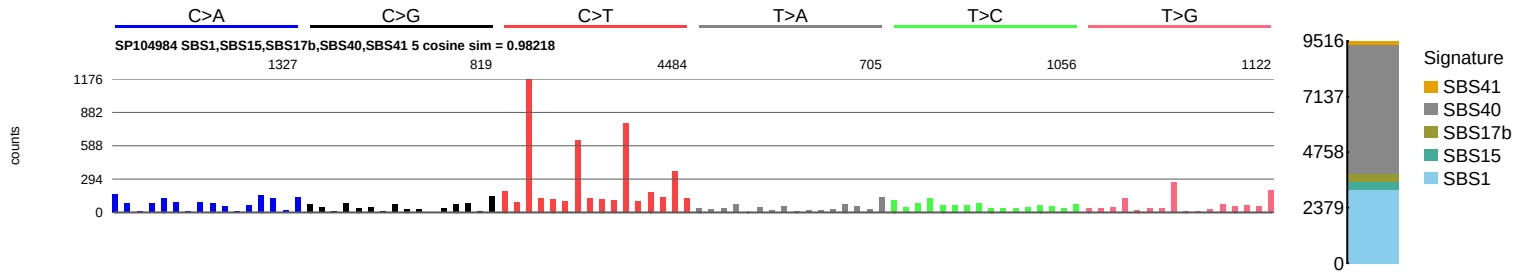
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98218
1706
723
4333
697
1212
845
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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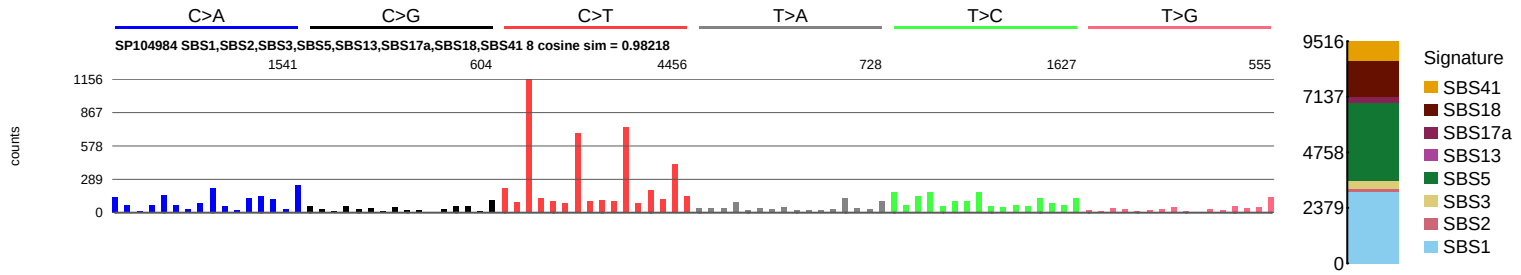
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS40,SBS41 5 cosine sim = 0.98218
1327
819
4484
705
1056
1122
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS15
SBS1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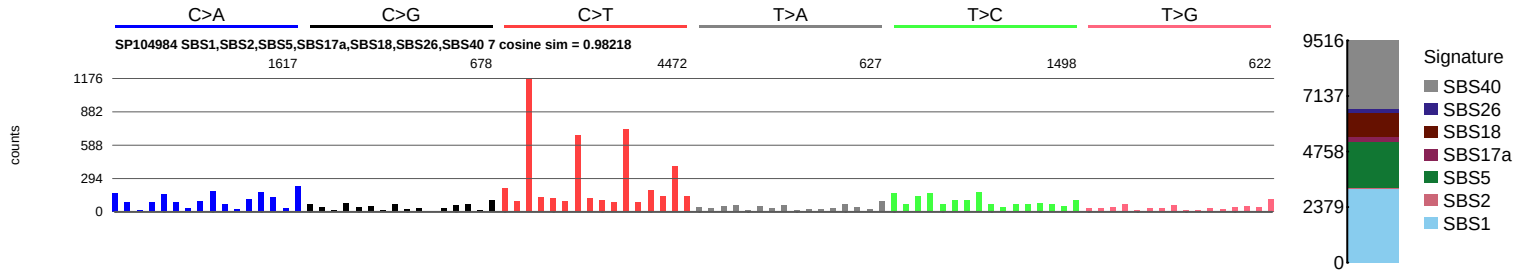
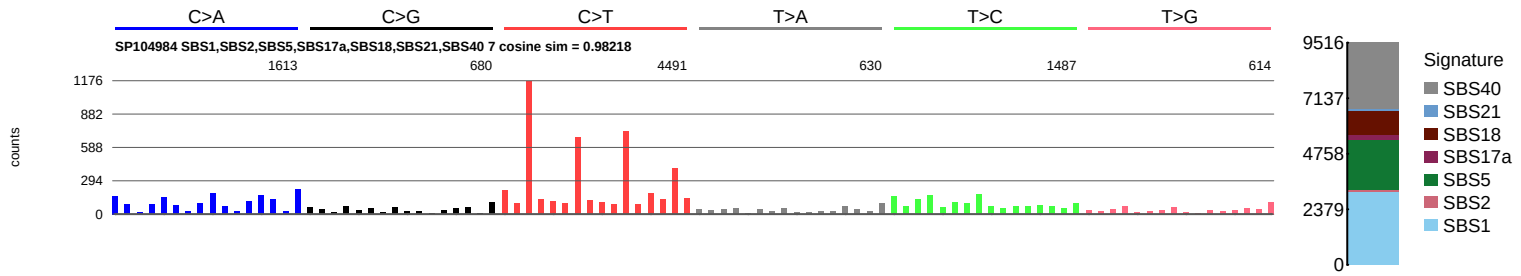
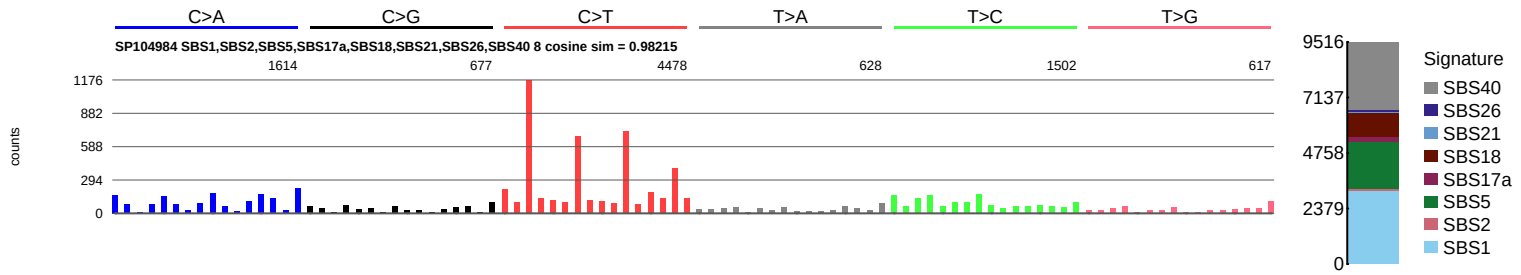
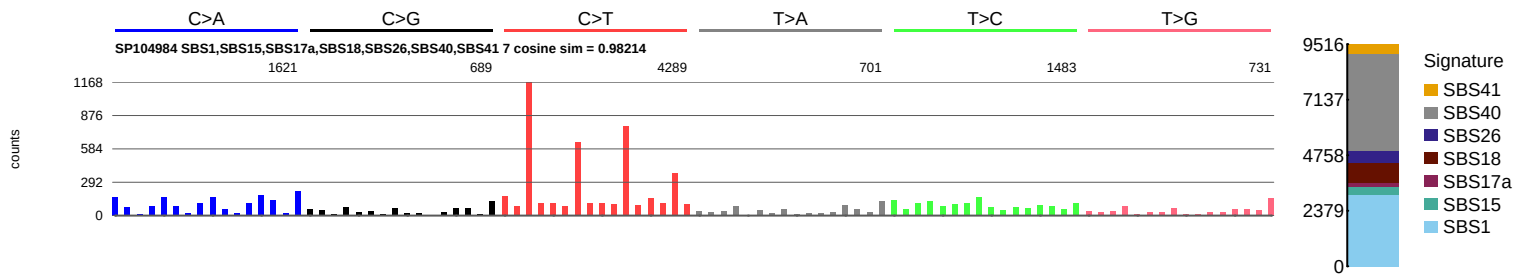
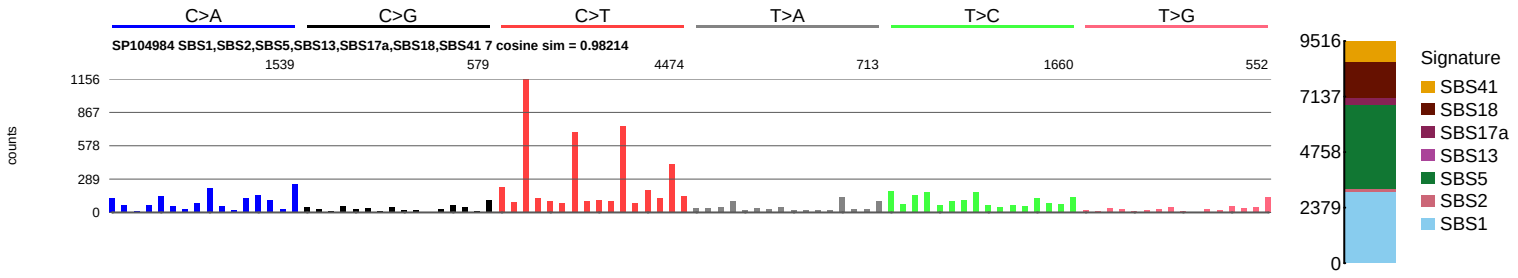
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41 8 cosine sim = 0.98218
1541
604
4456
728
1627
555
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

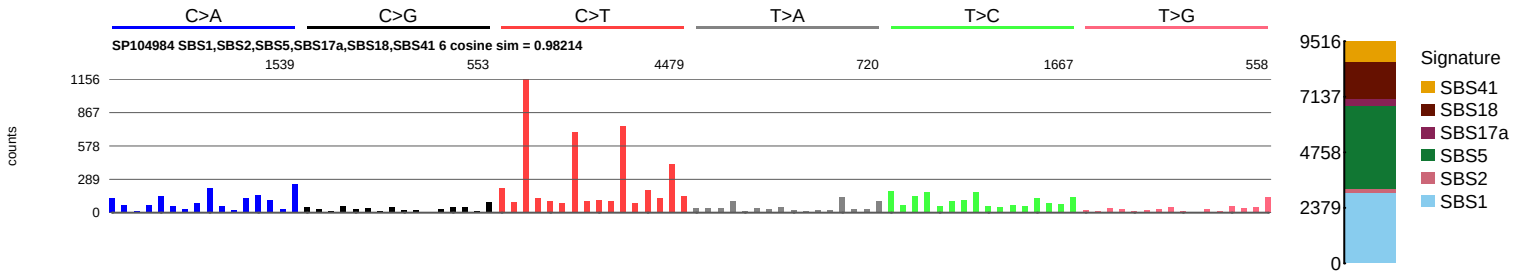

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98218
1617
678
4472
627
1498
622
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS2
SBS1

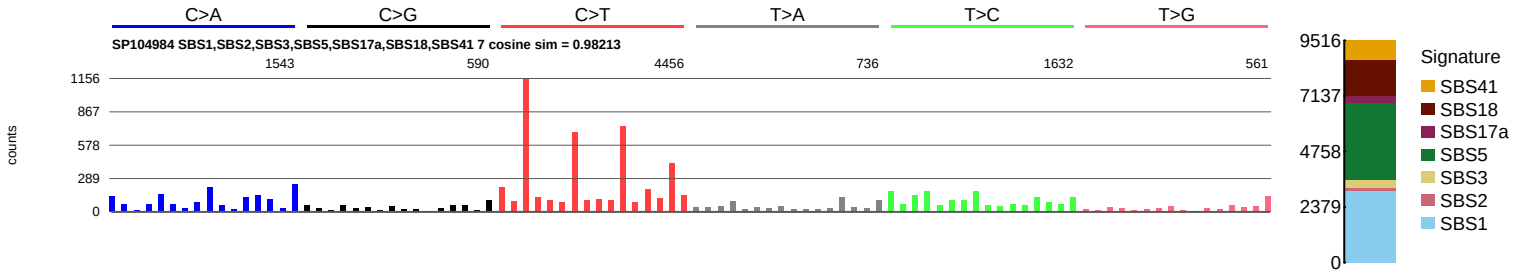

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98218
1613
680
4491
630
1487
614
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1

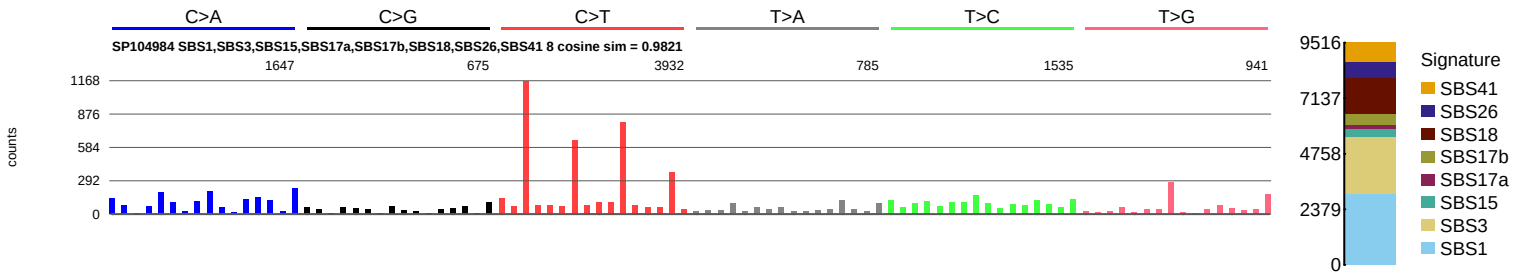

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98215
1614
677
4478
628
1502
617
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1

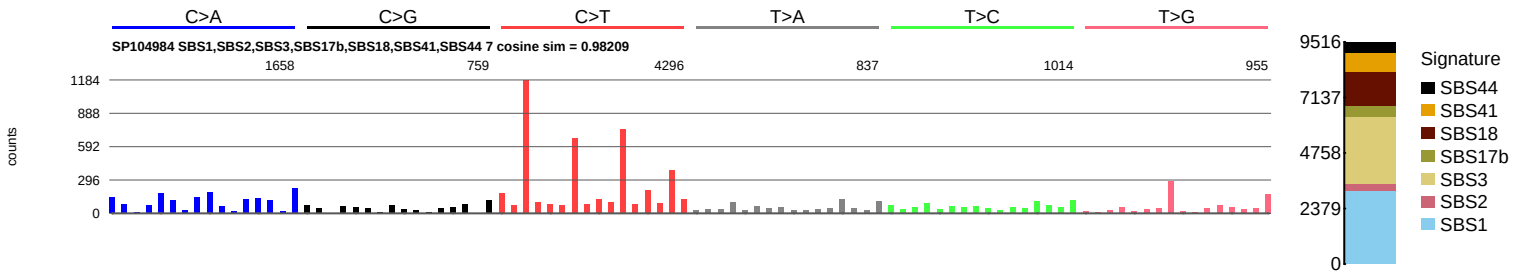

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98214
1621
689
4289
701
1483
731
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS1

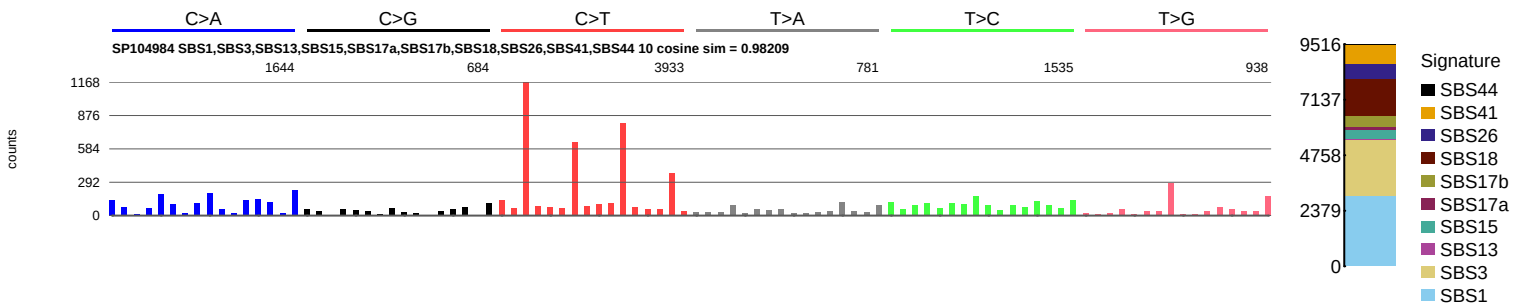

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS41 7 cosine sim = 0.98214
1539
579
4474
713
1660
552
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS41 6 cosine sim = 0.98214
1539
553
4479
720
1667
558
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS41 7 cosine sim = 0.98213
1543
590
4456
736
1632
561
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.9821
1647
675
3932
785
1535
941
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98209
1658
759
4296
837
1014
955
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS3
SBS2
SBS1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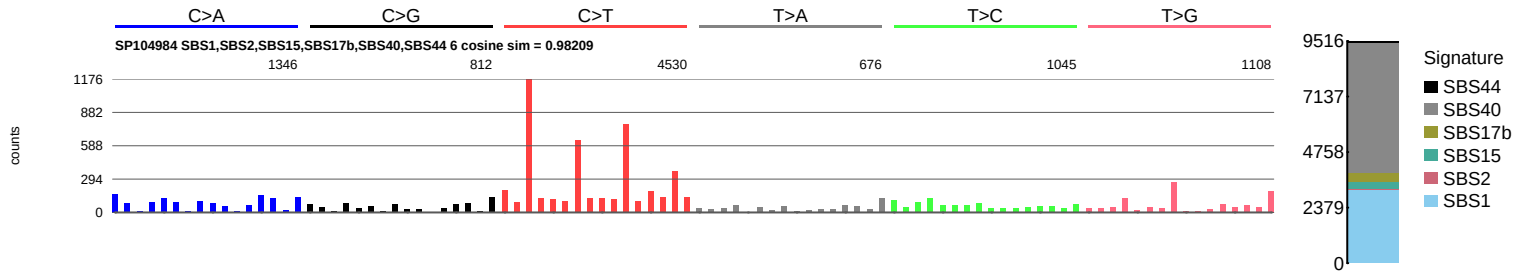
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98209
1644
684
3933
781
1535
938
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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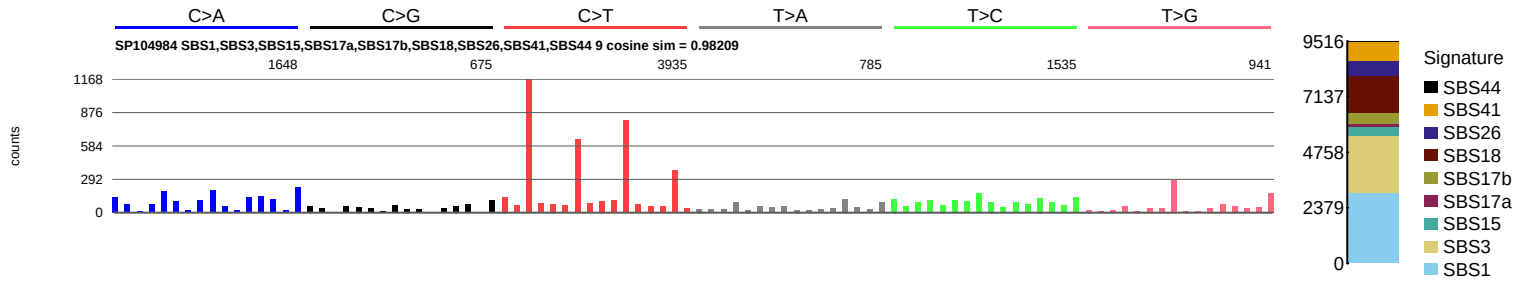
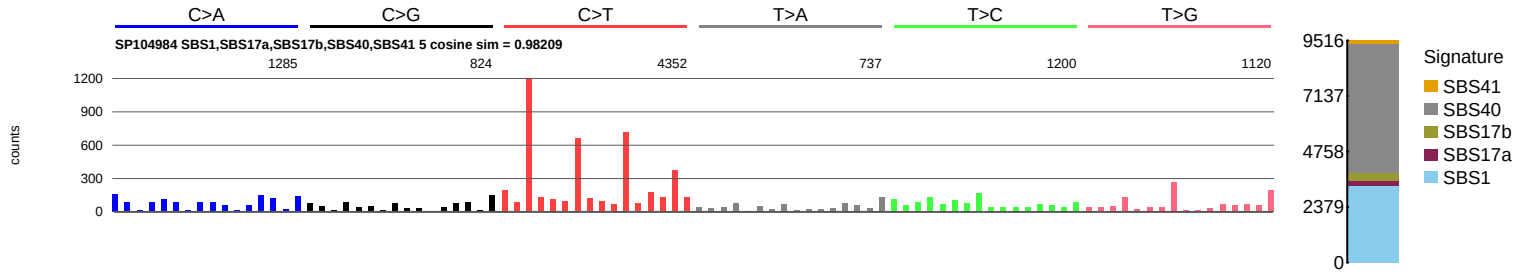
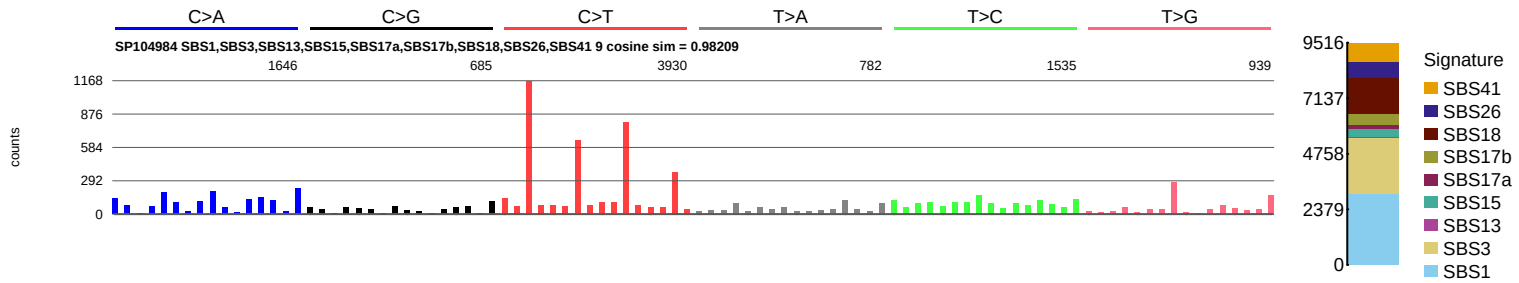
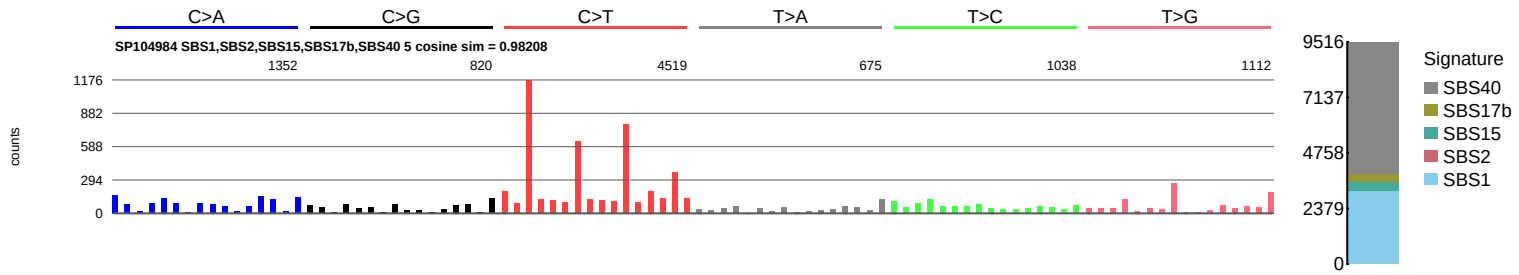
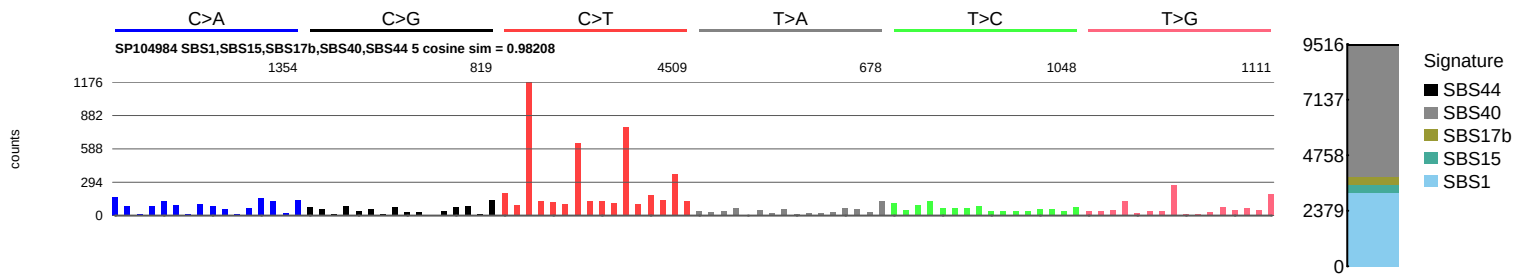
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS40,SBS44 6 cosine sim = 0.98209
1346
812
4530
676
1045
1108
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS2
SBS1

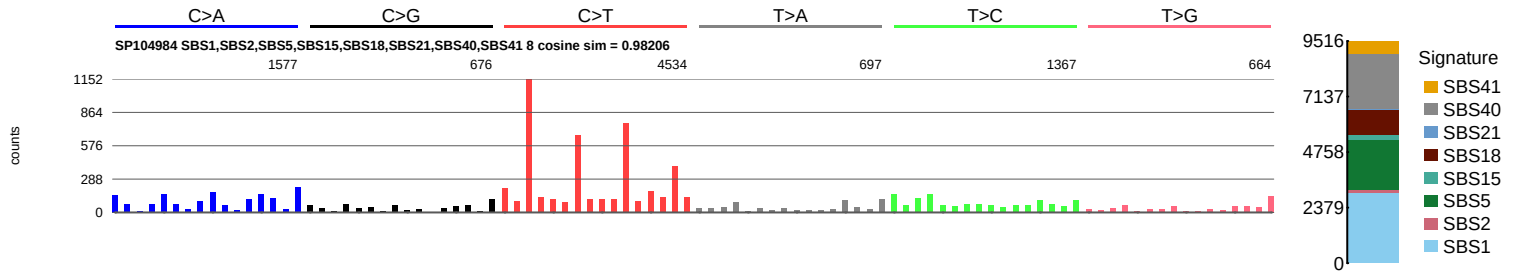

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98209
1648
675
3935
785
1535
941
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98209
1285
824
4352
737
1200
1120
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98209
1646
685
3930
782
1535
939
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17b,SBS40 5 cosine sim = 0.98208
1352
820
4519
675
1038
1112
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS40,SBS44 5 cosine sim = 0.98208
1354
819
4509
678
1048
1111
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98206
1577
676
4534
697
1367
664
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
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SBS40
SBS21
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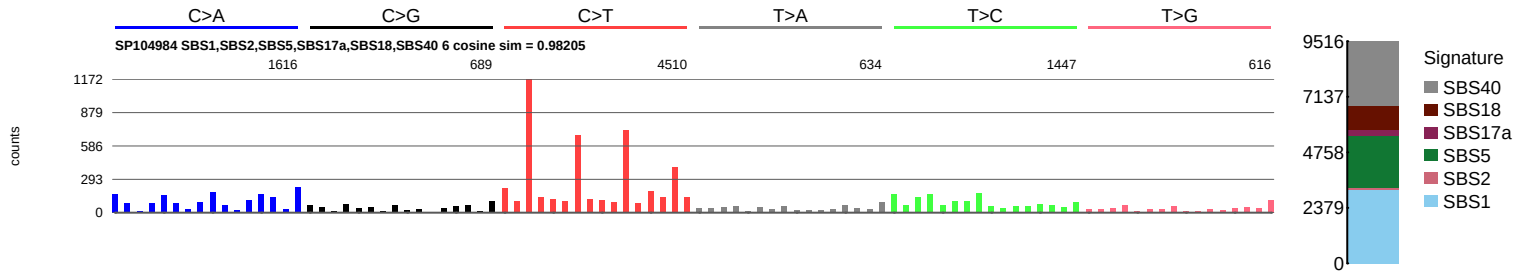
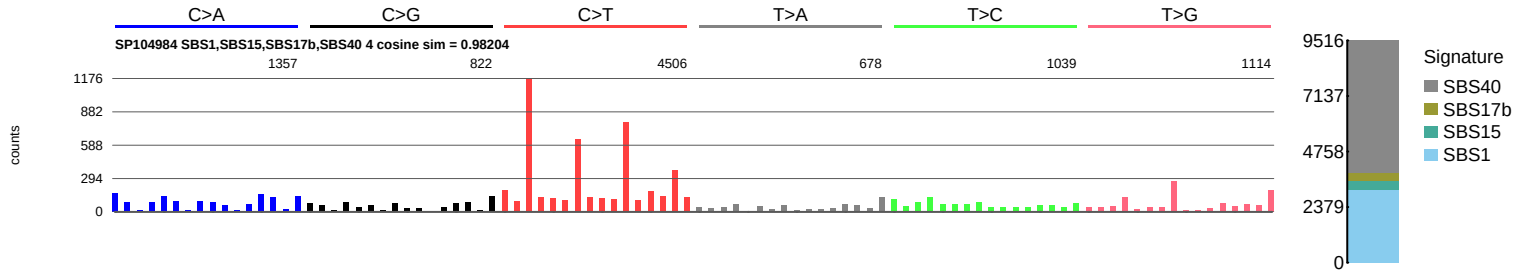
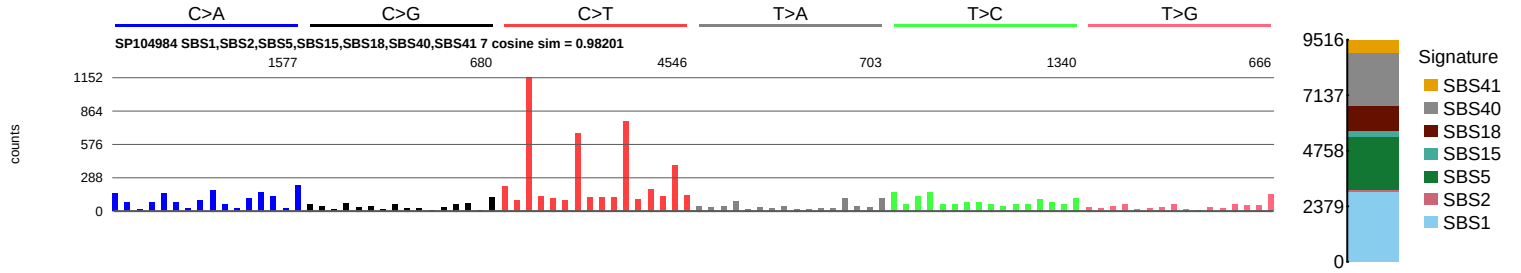
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40 6 cosine sim = 0.98205
1616
689
4510
634
1447
616
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17b,SBS40 4 cosine sim = 0.98204
1357
822
4506
678
1039
1114
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98201
1577
680
4546
703
1340
666
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
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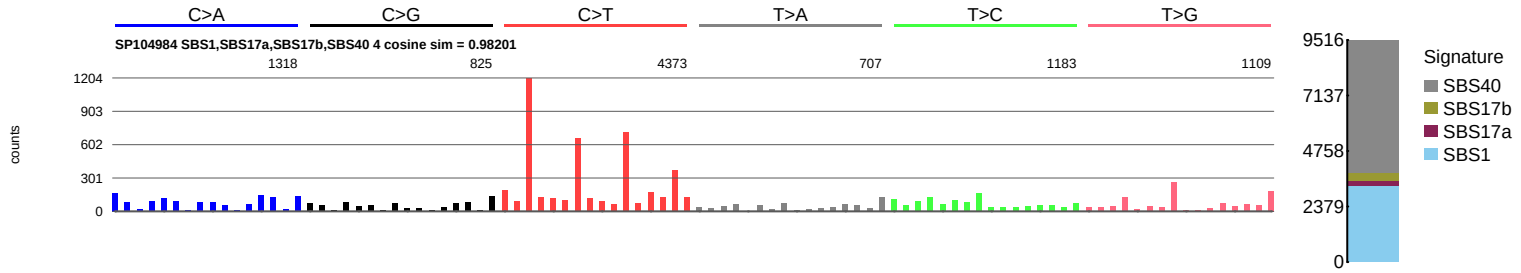
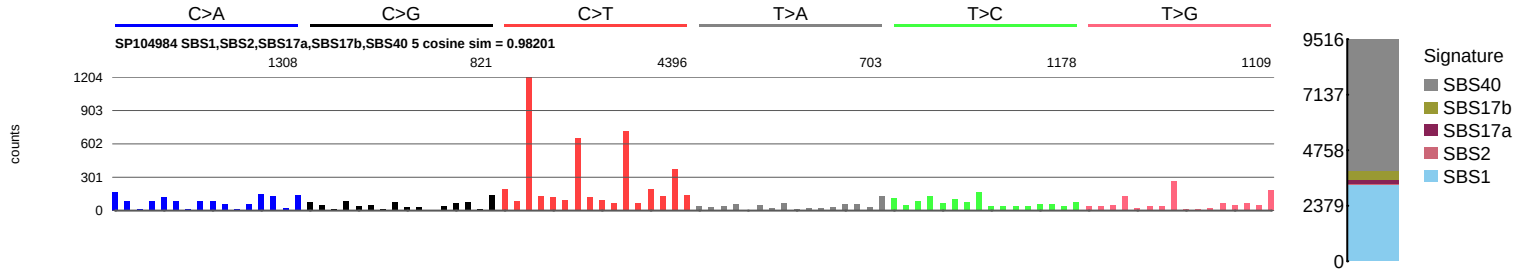
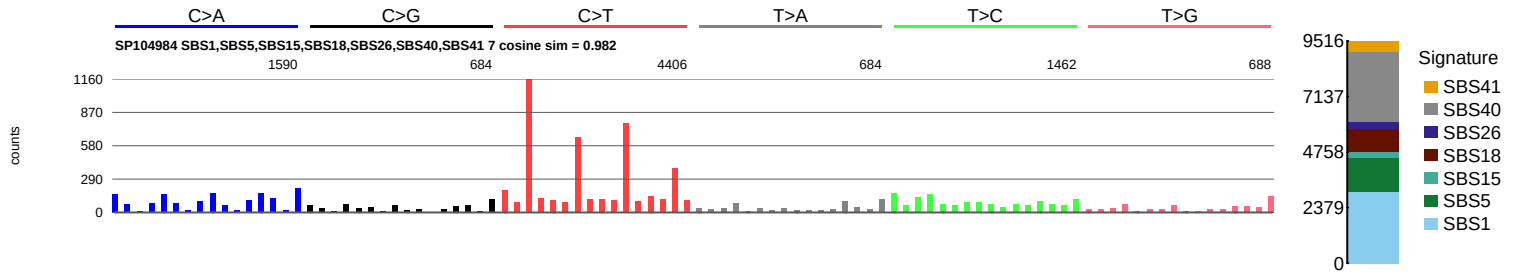
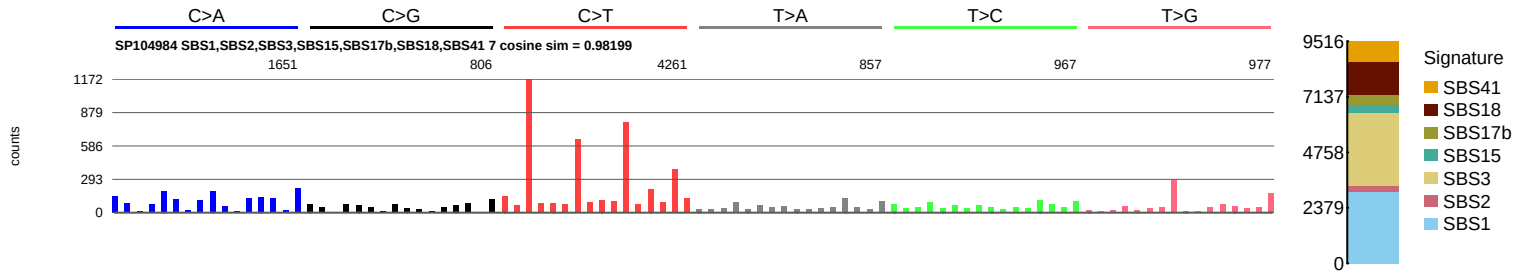
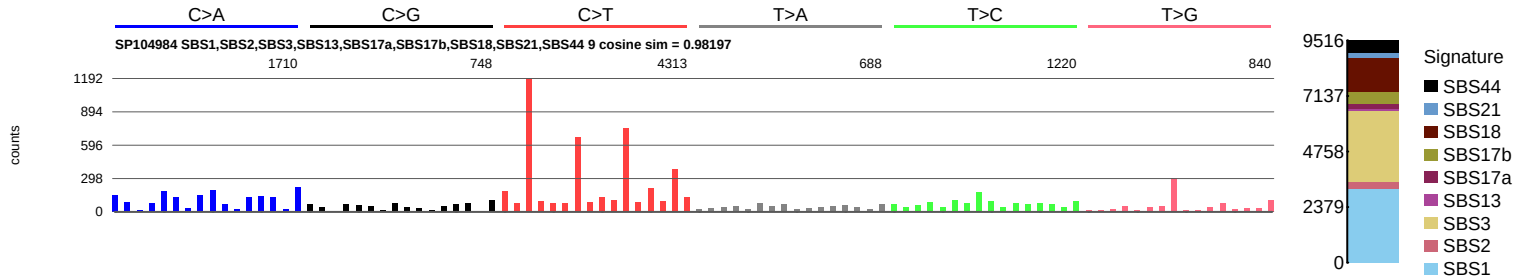
SBS5
SBS2
SBS1

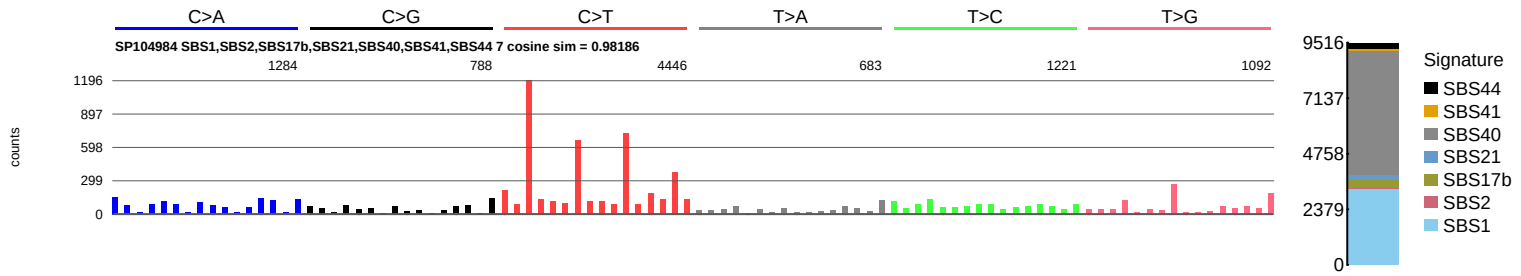

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98201
1318
825
4373
707
1183
1109
counts
1204
903
602
301
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS17a
SBS1

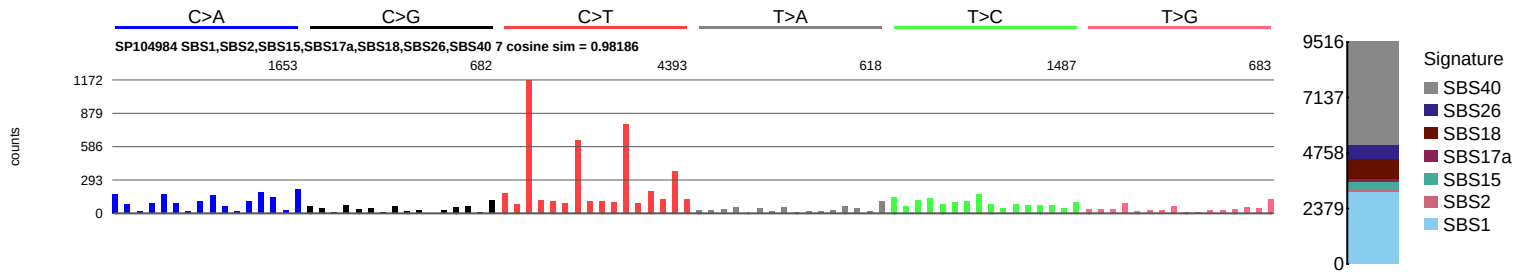

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98201
1308
821
4396
703
1178
1109
counts
1204
903
602
301
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS17a
SBS2
SBS1

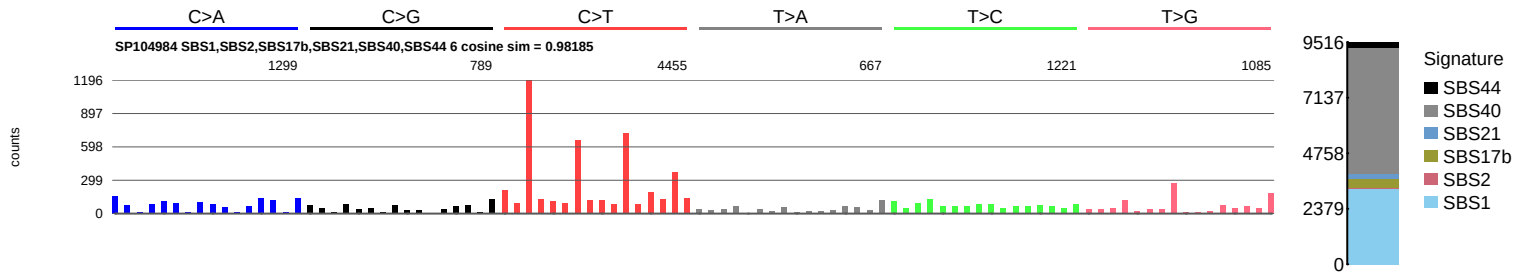

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.982
1590
684
4406
684
1462
688
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS1

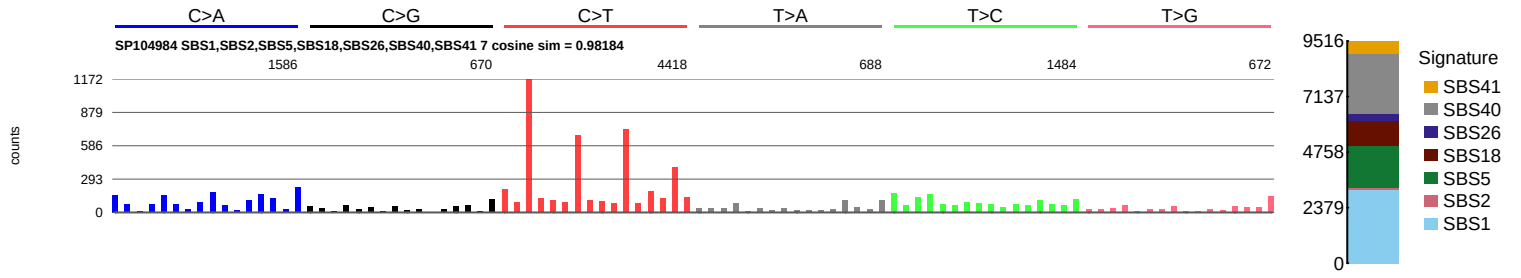

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS41 7 cosine sim = 0.98199
1651
806
4261
857
967
977
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

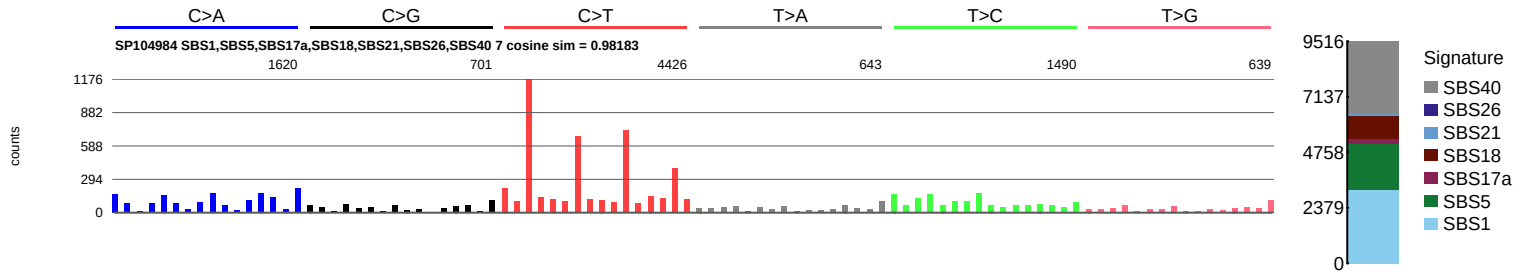

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98197
1710
748
4313
688
1220
840
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

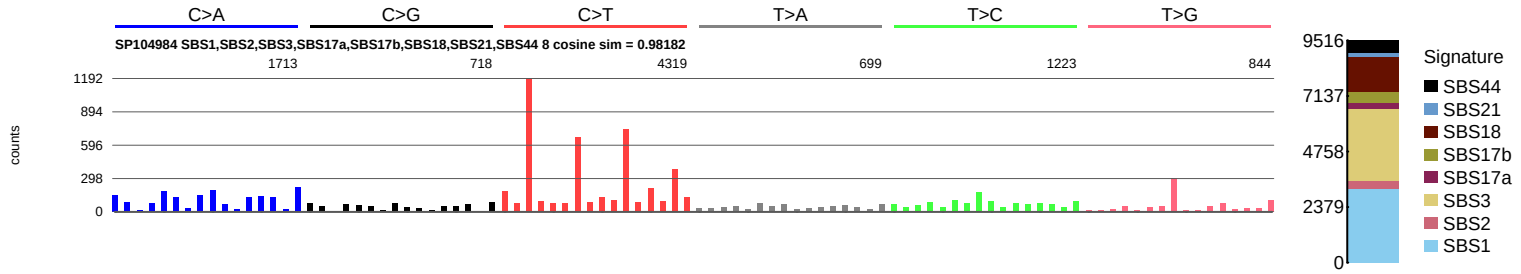

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98186
1284
788
4446
683
1221
1092
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS2
SBS1

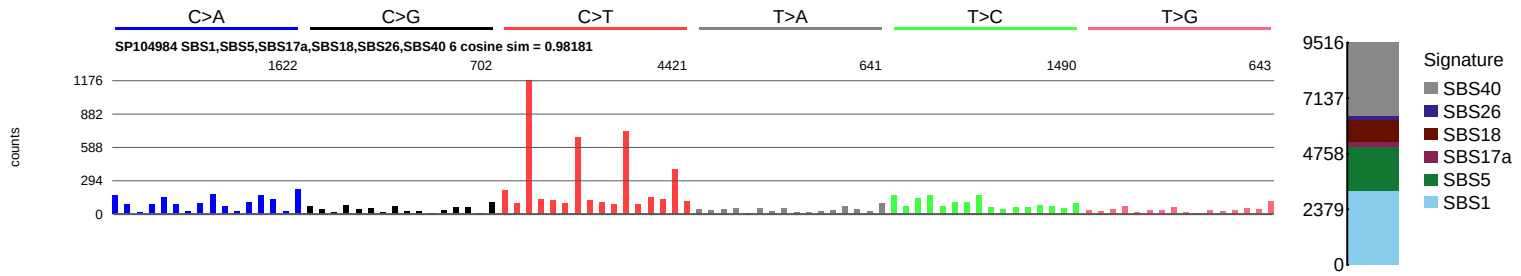

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98186
1653
682
4393
618
1487
683
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1

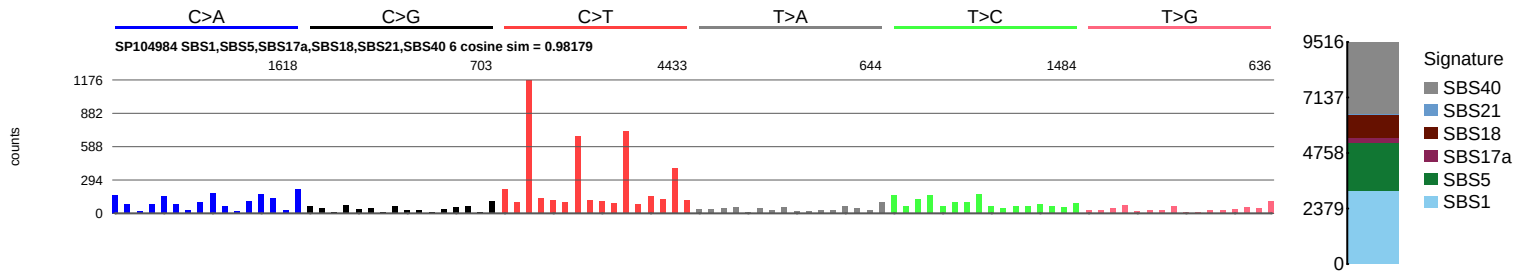

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98185
1299
789
4455
667
1221
1085
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98184
1586
670
4418
688
1484
672
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98183
1620
701
4426
643
1490
639
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98182
1713
718
4319
699
1223
844
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS18,SBS26,SBS40 6 cosine sim = 0.98181
1622
702
4421
641
1490
643
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS18,SBS21,SBS40 6 cosine sim = 0.98179
1618
703
4433
644
1484
636
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
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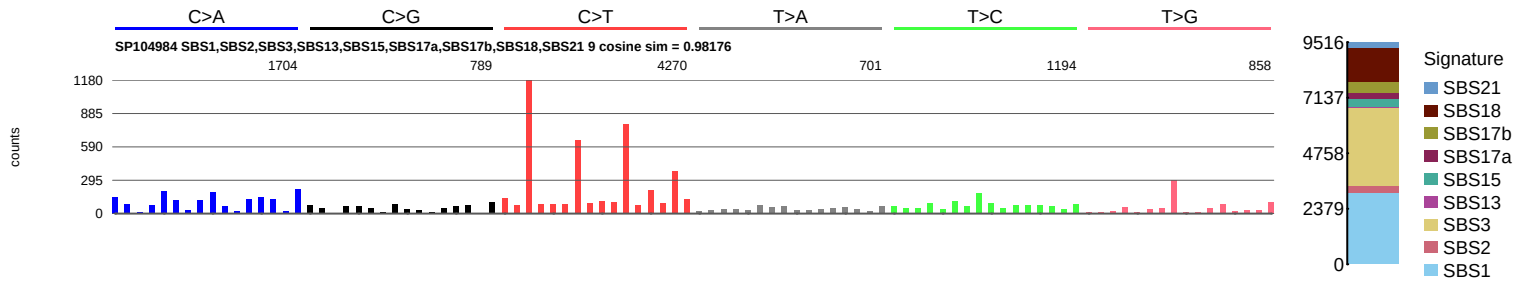
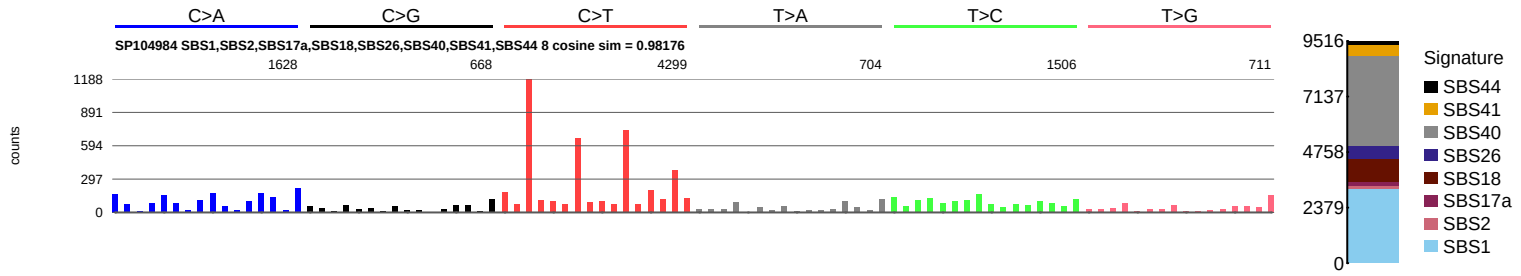
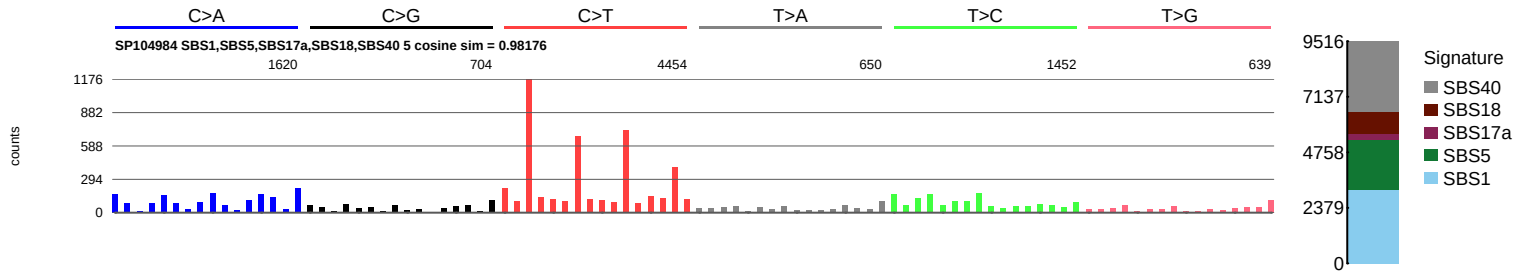
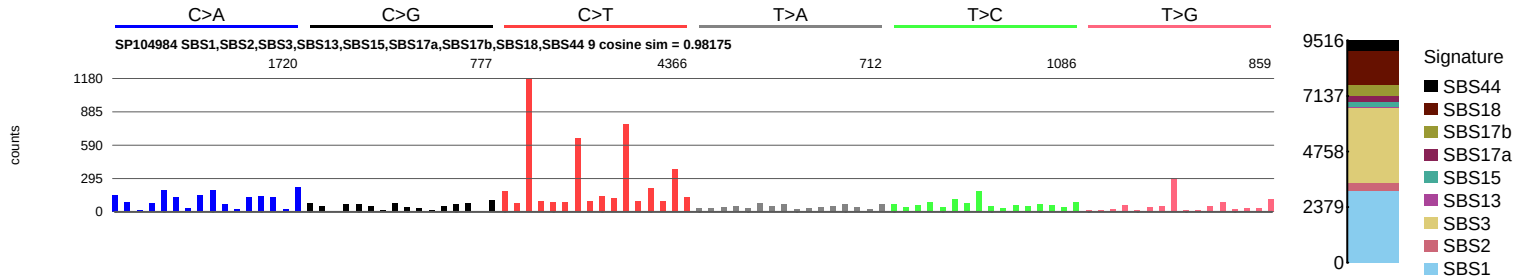
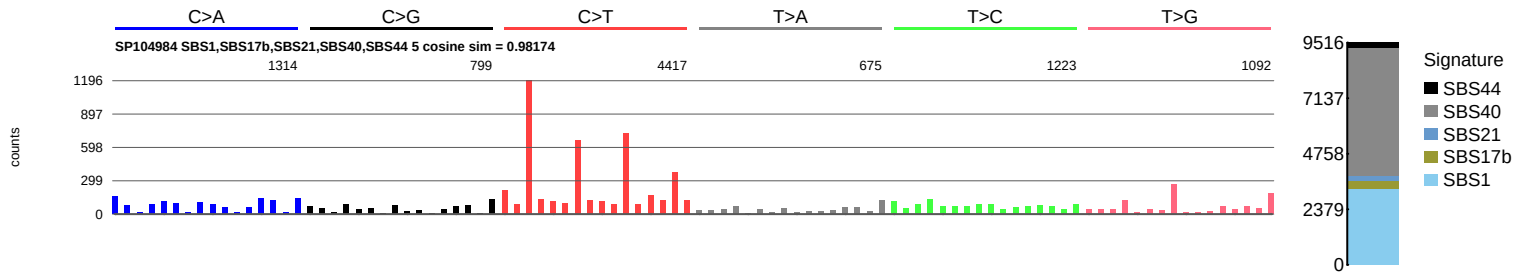
SBS17a
SBS5
SBS1

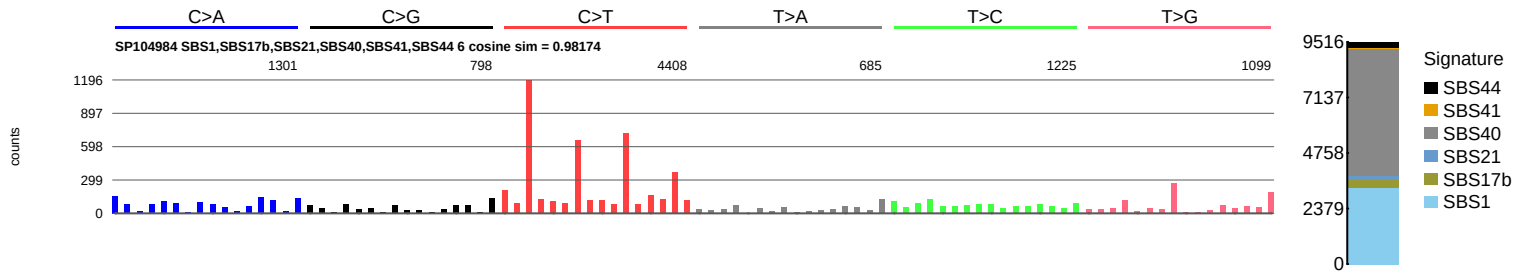

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98176
1704
789
4270
701
1194
858
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98176
1628
668
4299
704
1506
711
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS18,SBS40 5 cosine sim = 0.98176
1620
704
4454
650
1452
639
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98175
1720
777
4366
712
1086
859
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS21,SBS40,SBS44 5 cosine sim = 0.98174
1314
799
4417
675
1223
1092
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS21,SBS40,SBS41,SBS44 6 cosine sim = 0.98174
1301
798
4408
685
1225
1099
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
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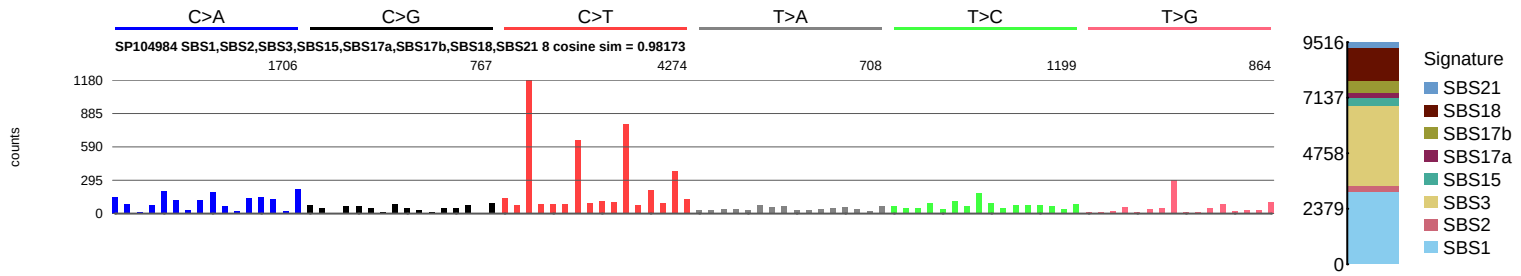
SBS21
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98173
1706
767
4274
708
1199
864
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS21
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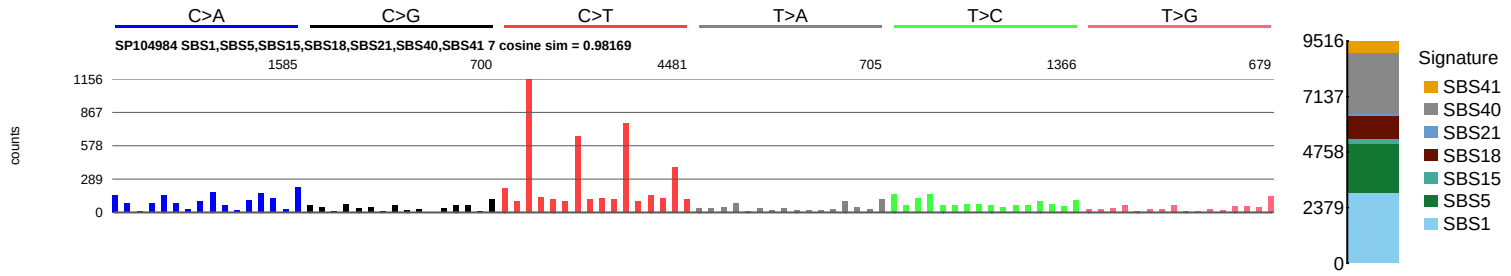
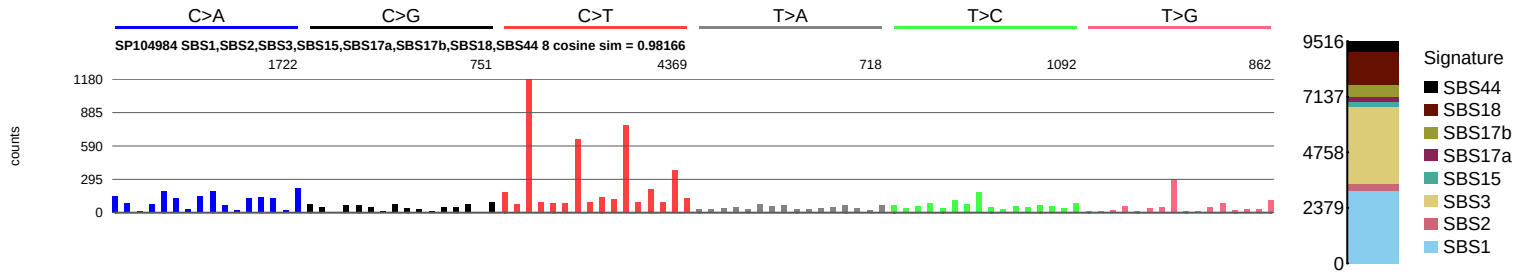
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98169
1585
700
4481
705
1366
679
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98166
1722
751
4369
718
1092
862
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS18
SBS17b
SBS17a
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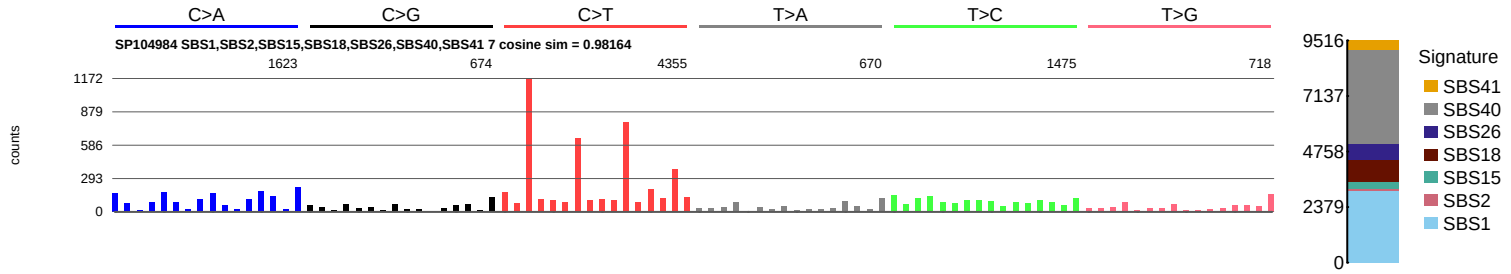
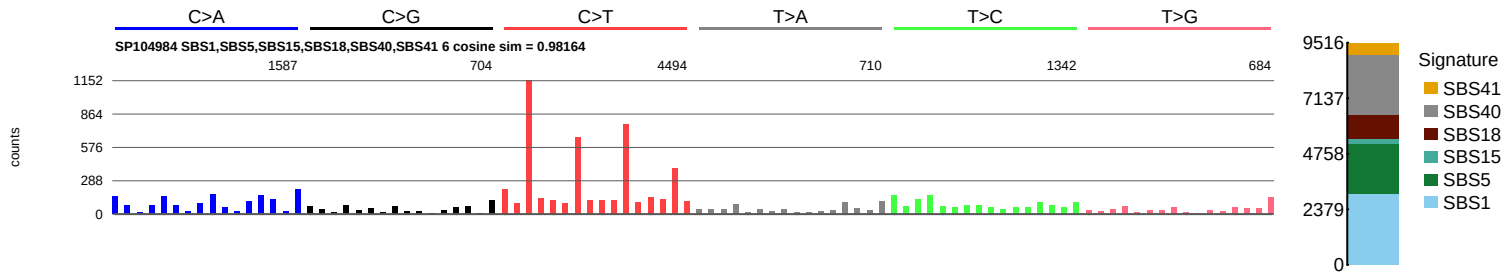
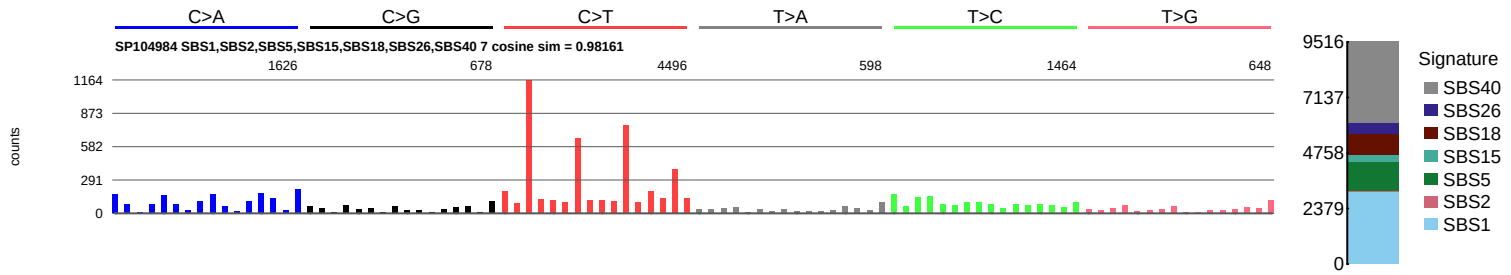
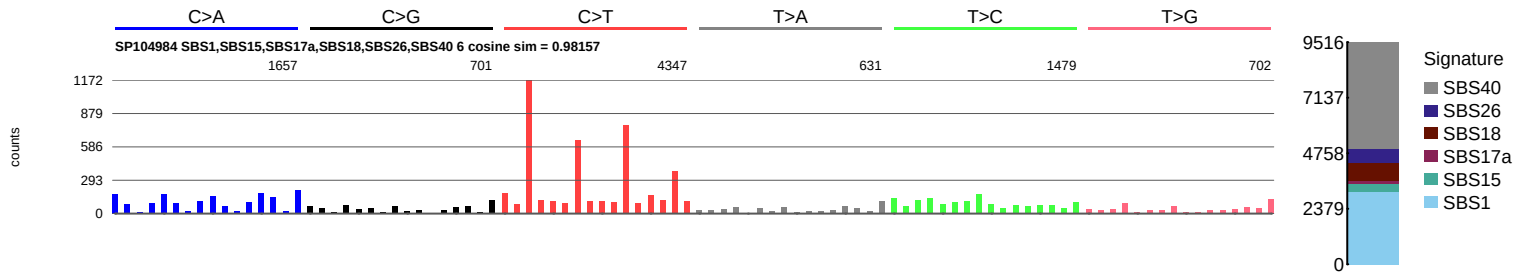
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98164
1623
674
4355
670
1475
718
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS18,SBS40,SBS41 6 cosine sim = 0.98164
1587
704
4494
710
1342
684
counts
1152
864
576
288
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.98161
1626
678
4496
598
1464
648
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS40 6 cosine sim = 0.98157
1657
701
4347
631
1479
702
counts
1172
879
586
293
0
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7137
4758
2379
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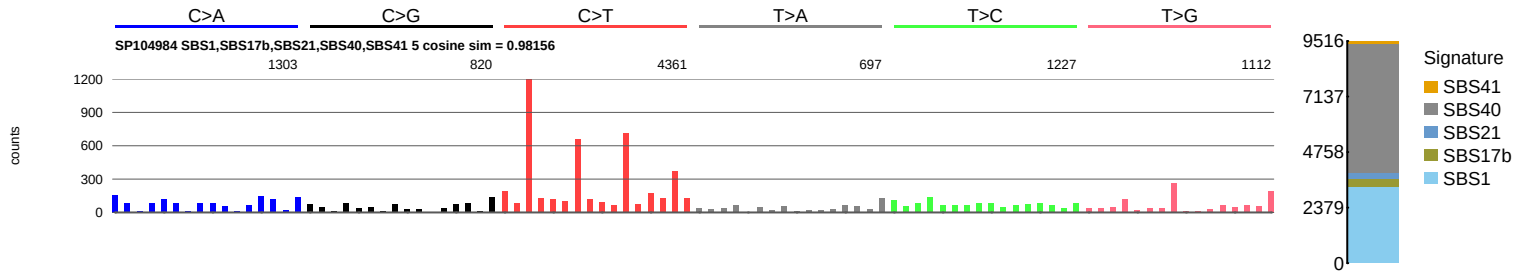
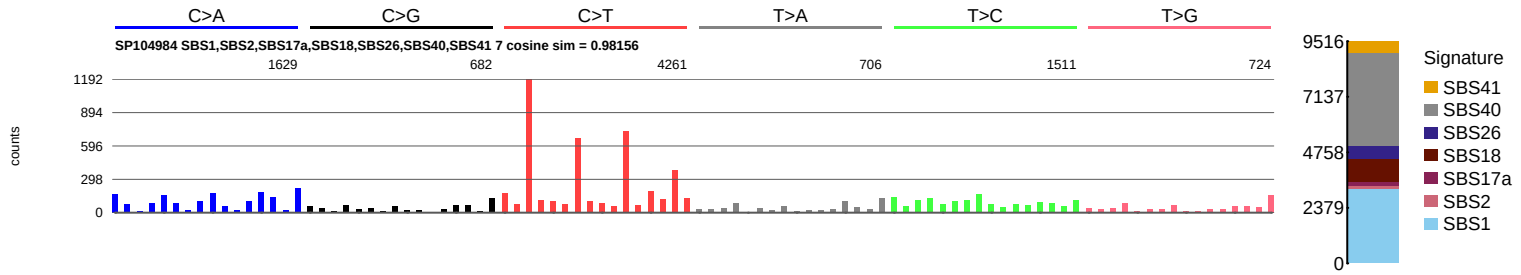
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS21,SBS40,SBS41 5 cosine sim = 0.98156
1303
820
4361
697
1227
1112
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS1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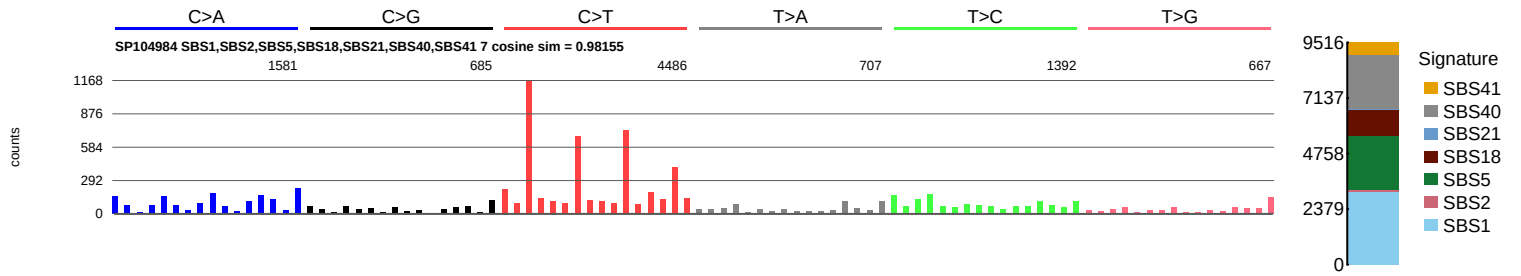
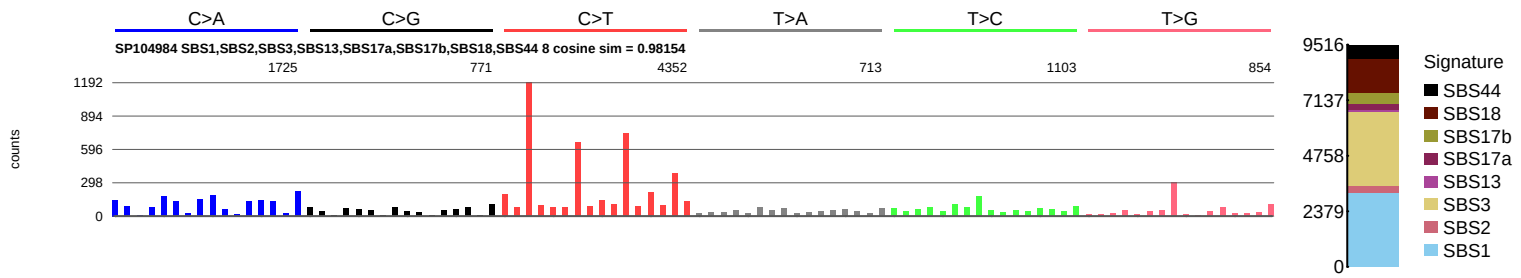
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98156
1629
682
4261
706
1511
724
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98155
1581
685
4486
707
1392
667
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1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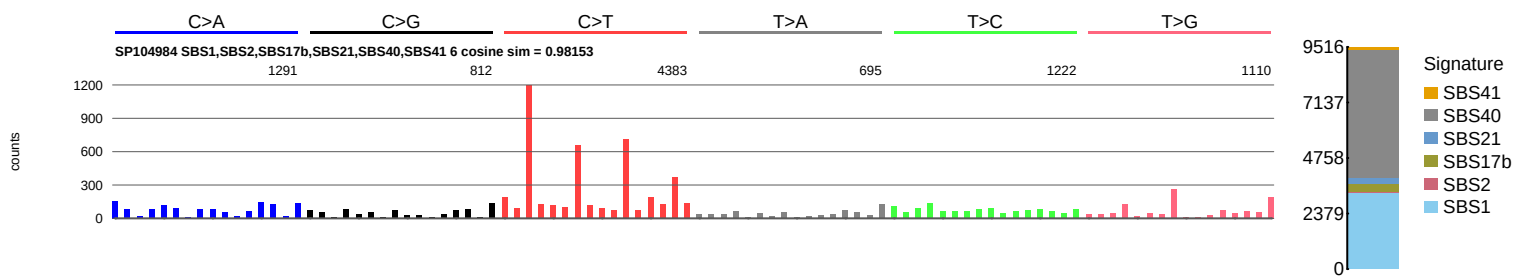
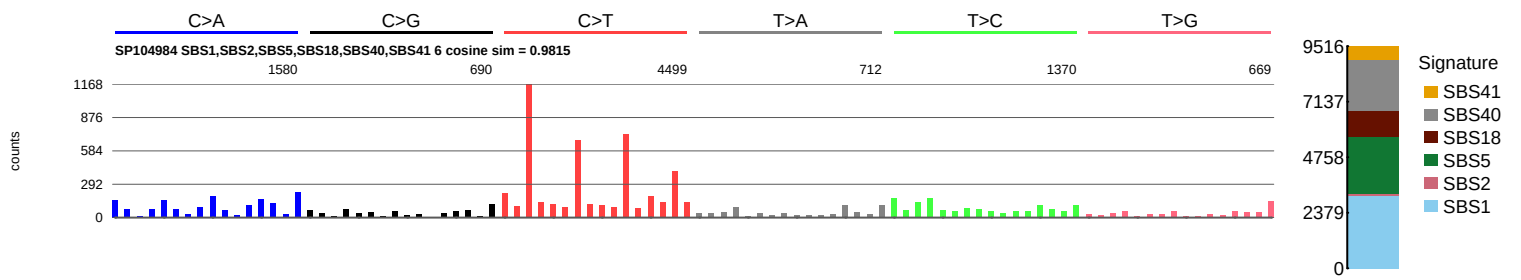
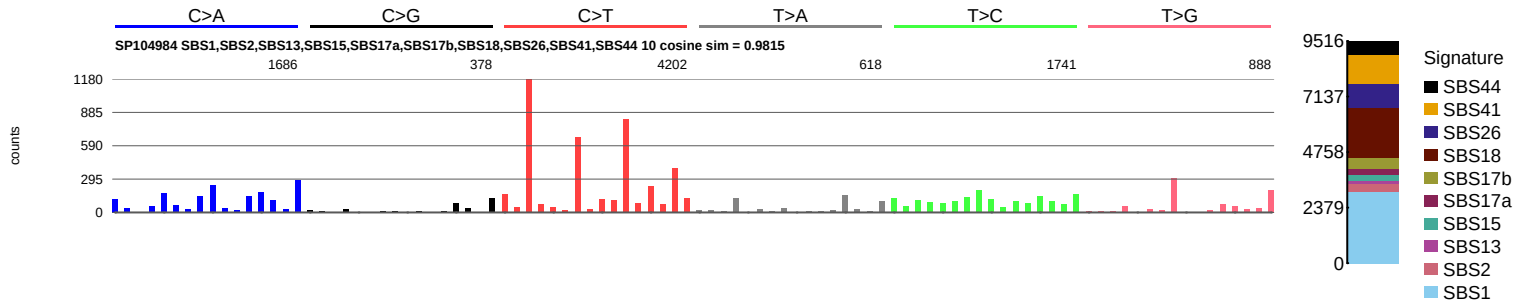
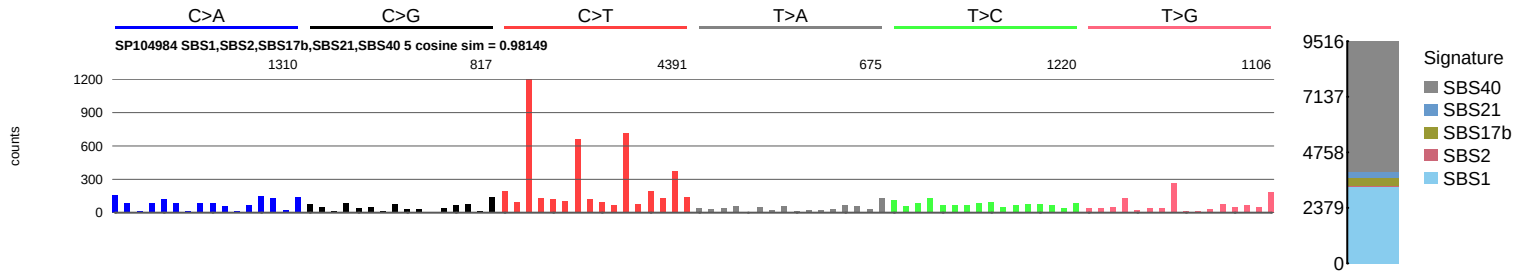
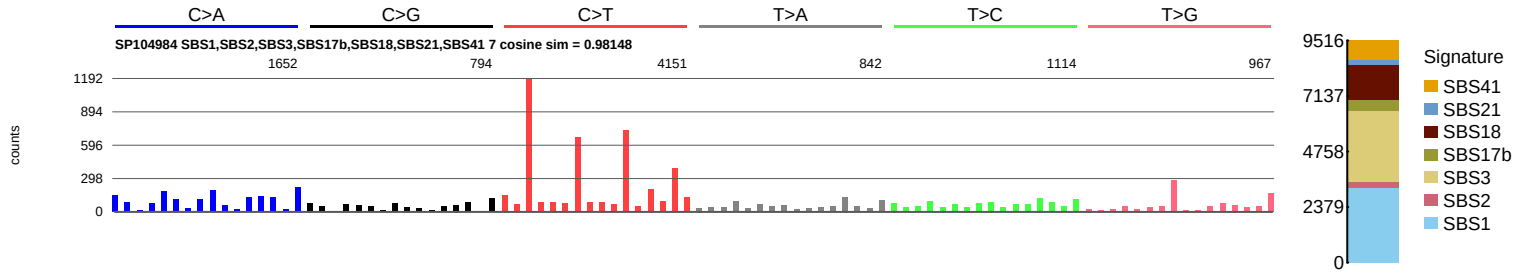
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98154
1725
771
4352
713
1103
854
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

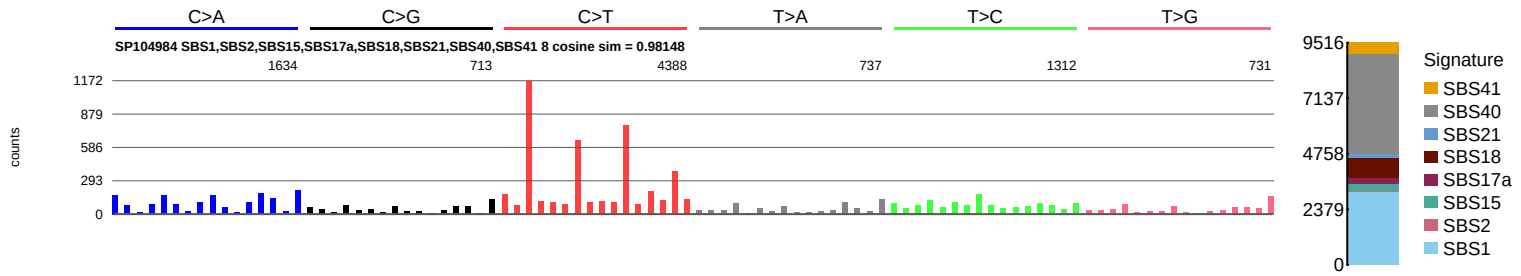

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98153
1291
812
4383
695
1222
1110
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS2
SBS1

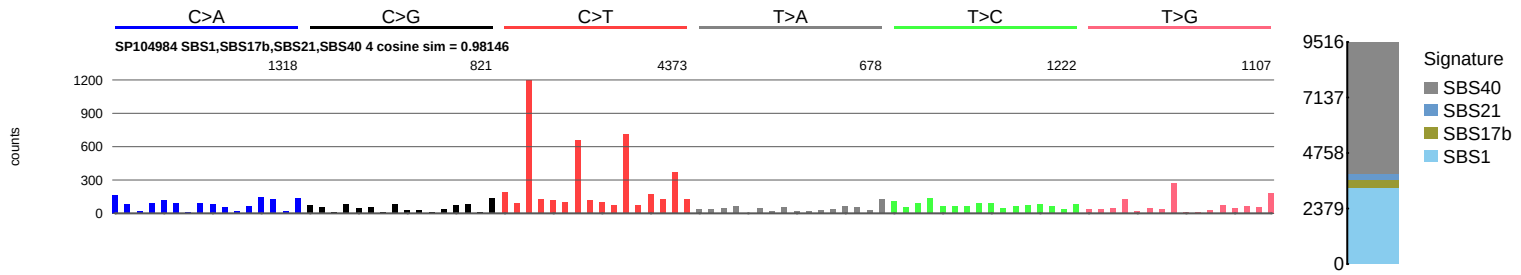

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS18,SBS40,SBS41 6 cosine sim = 0.9815
1580
690
4499
712
1370
669
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS5
SBS2
SBS1

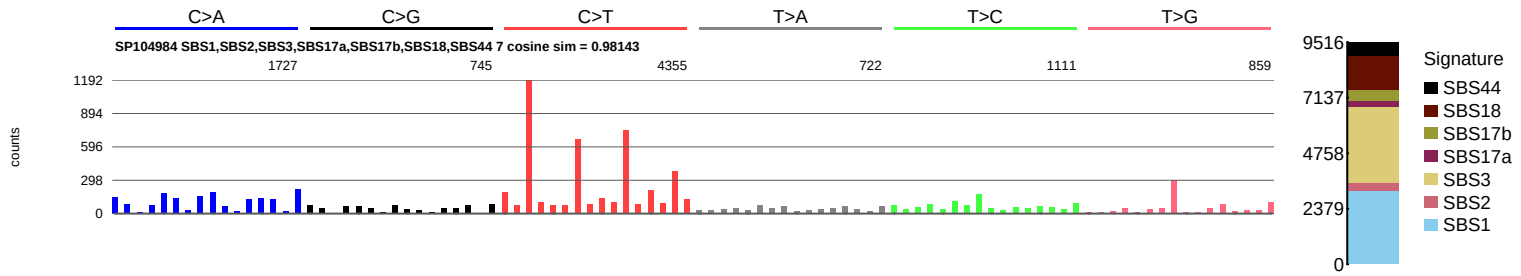

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.9815
1686
378
4202
618
1741
888
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

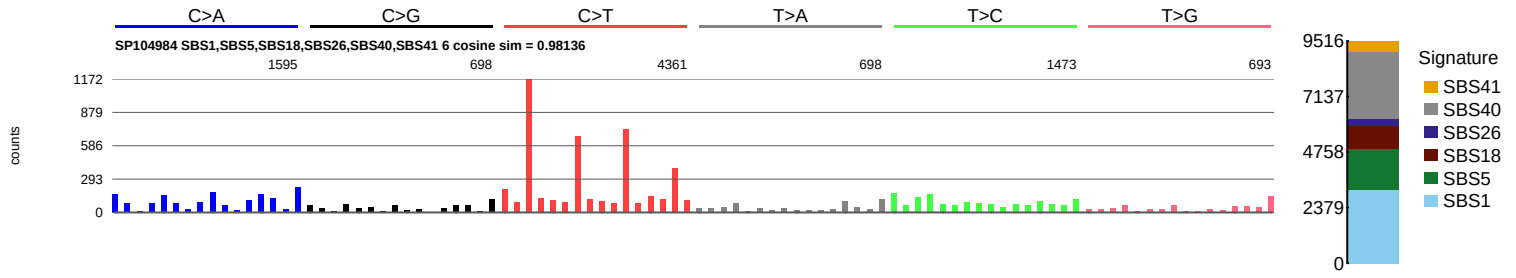

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS21,SBS40 5 cosine sim = 0.98149
1310
817
4391
675
1220
1106
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS2
SBS1

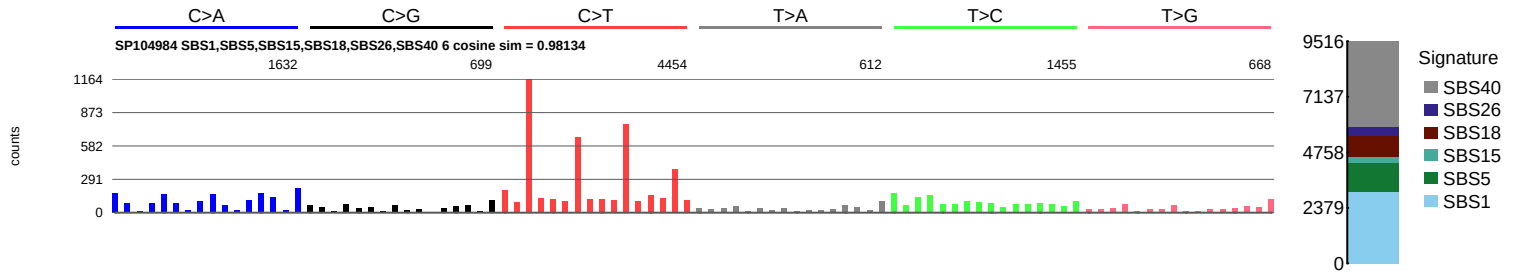

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98148
1652
794
4151
842
1114
967
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS3
SBS2
SBS1

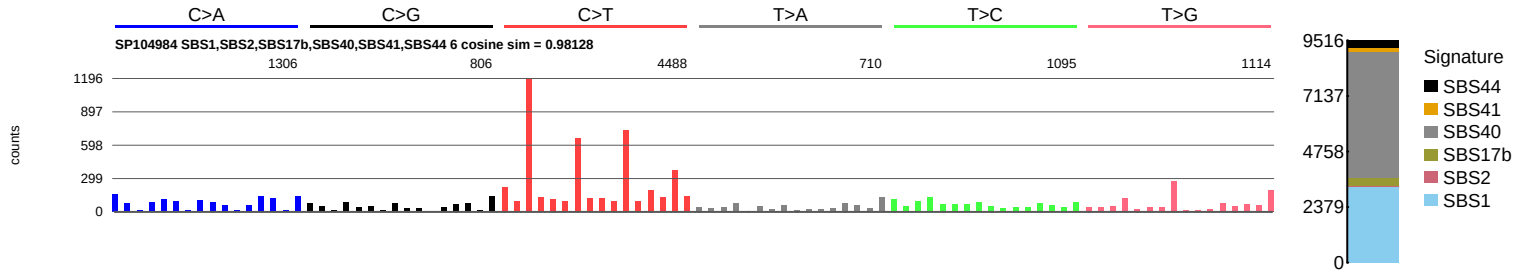

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98148
1634
713
4388
737
1312
731
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1

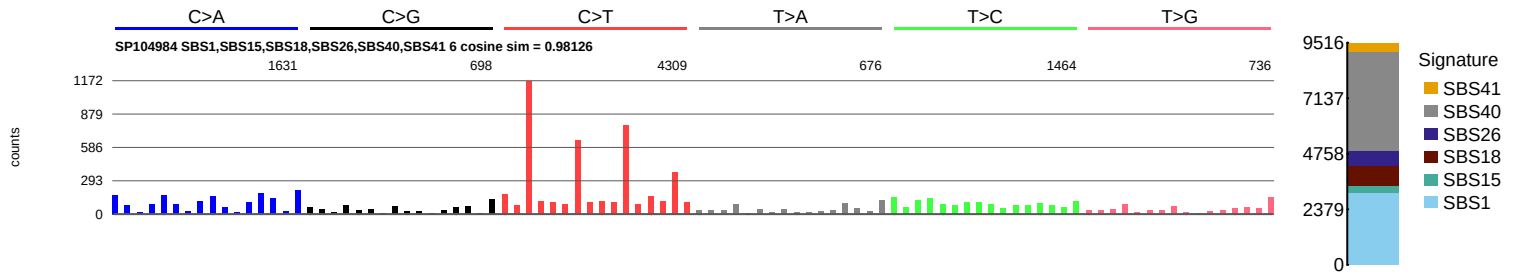

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS21,SBS40 4 cosine sim = 0.98146
1318
821
4373
678
1222
1107
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS17b
SBS1

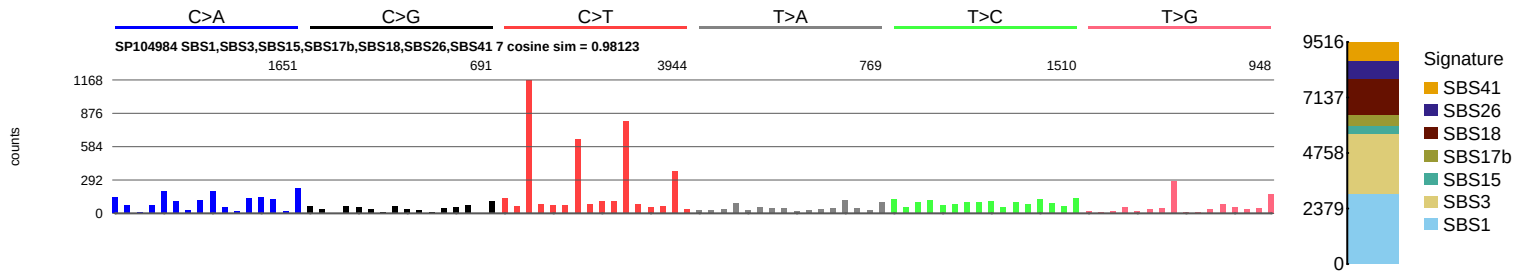

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98143
1727
745
4355
722
1111
859
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

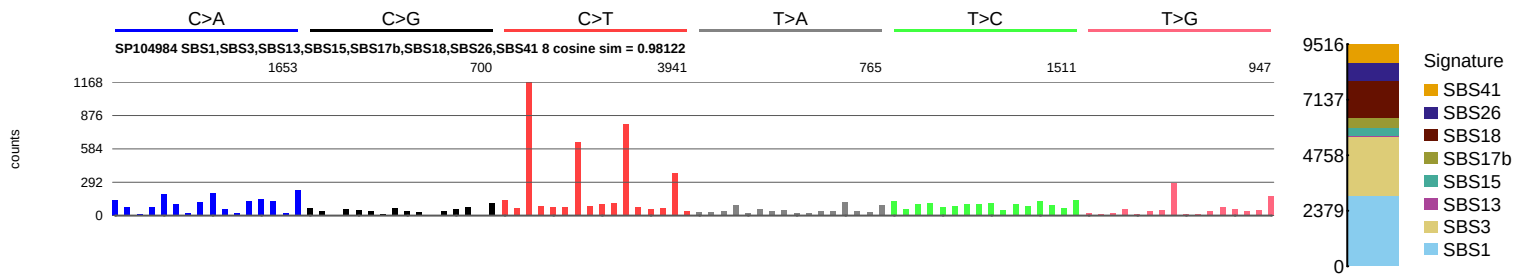

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98136
1595
698
4361
698
1473
693
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS5
SBS1

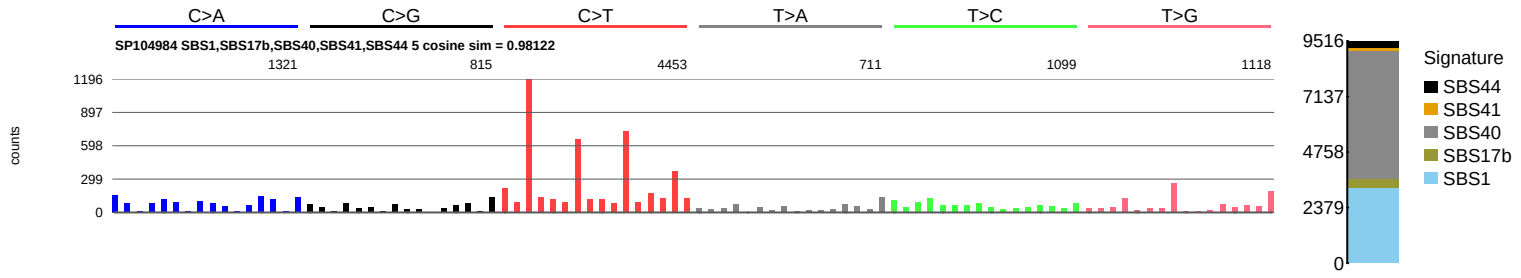

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS18,SBS26,SBS40 6 cosine sim = 0.98134
1632
699
4454
612
1455
668
counts
1164
873
582
291
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS5
SBS1

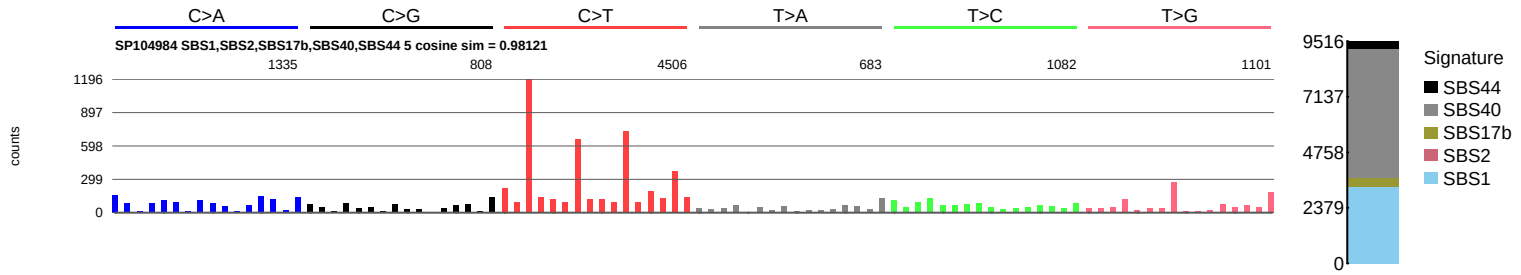

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98128
1306
806
4488
710
1095
1114
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS2
SBS1

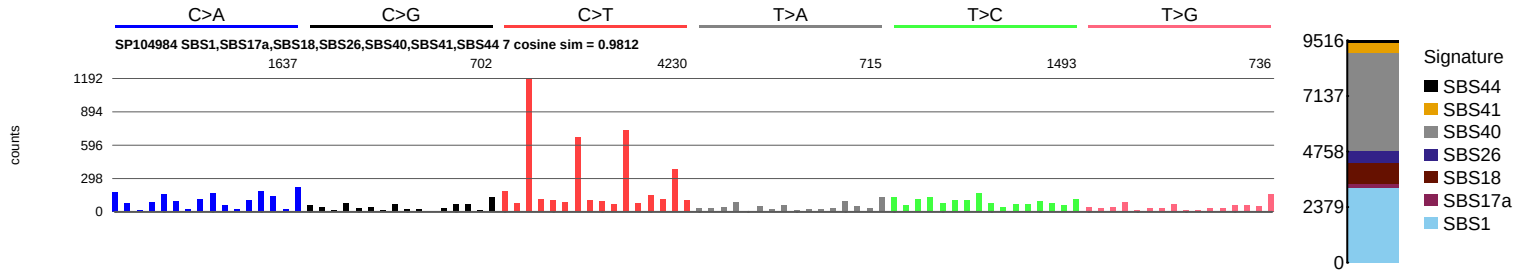

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98126
1631
698
4309
676
1464
736
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS1

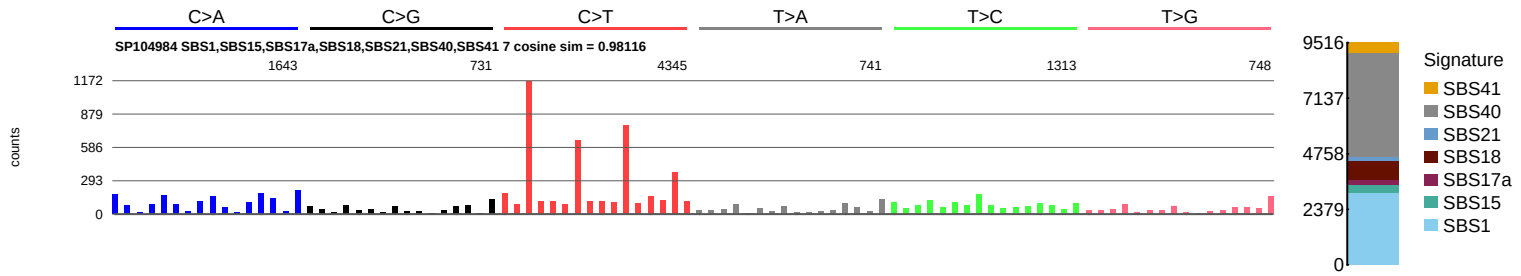

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98123
1651
691
3944
769
1510
948
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS1

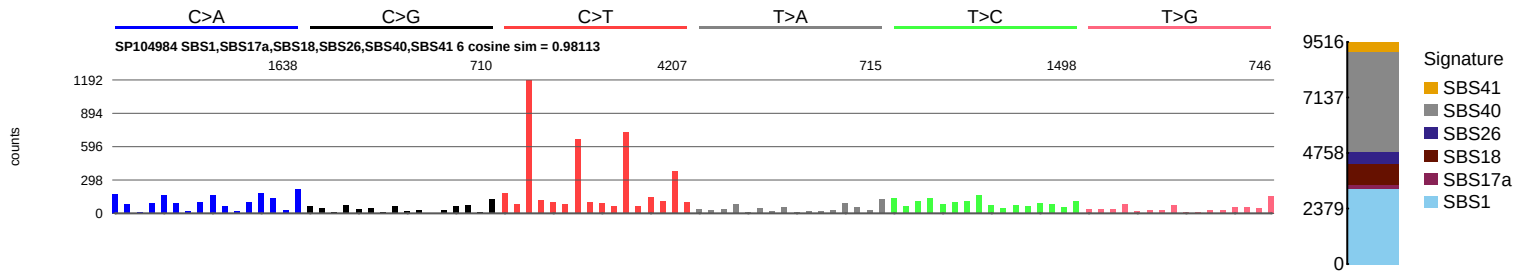

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98122
1653
700
3941
765
1511
947
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS1

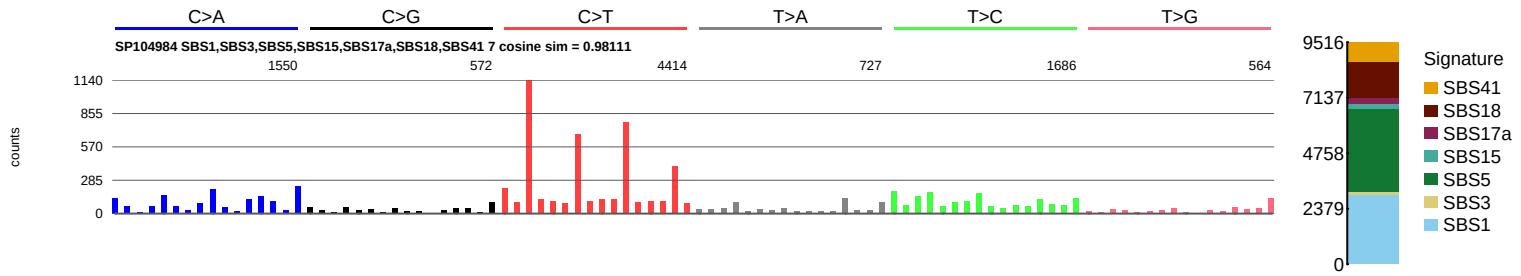

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS40,SBS41,SBS44 5 cosine sim = 0.98122
1321
815
4453
711
1099
1118
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS1

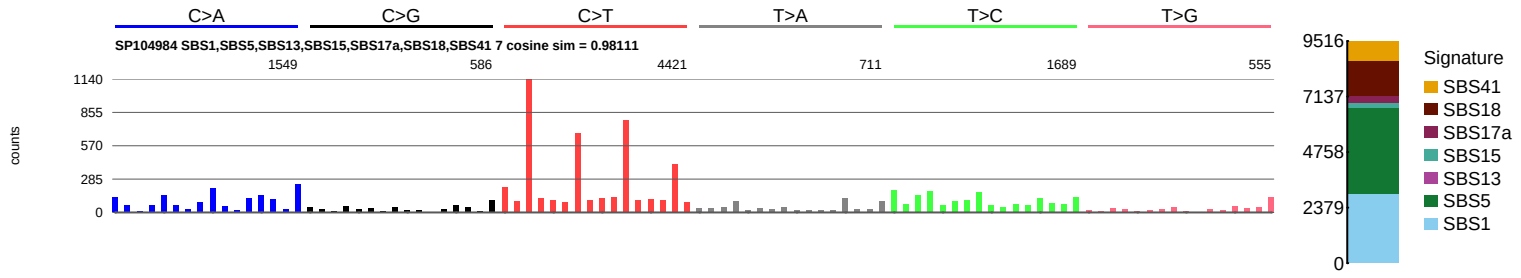

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS40,SBS44 5 cosine sim = 0.98121
1335
808
4506
683
1082
1101
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS2
SBS1

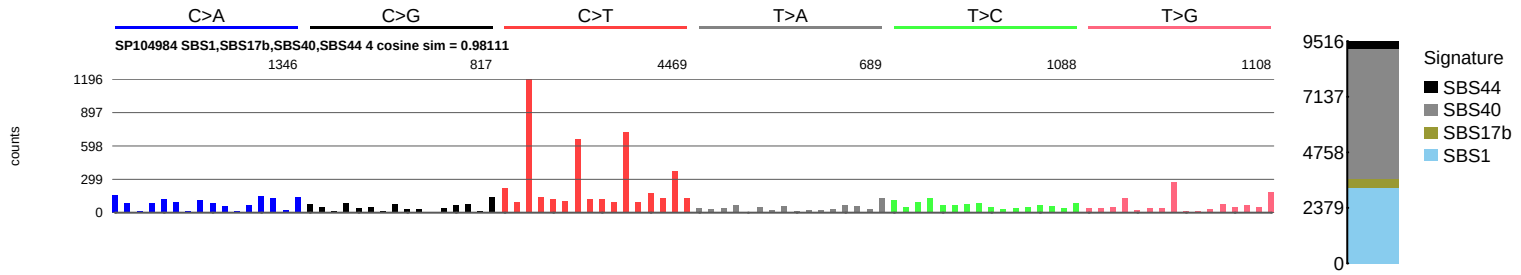

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.9812
1637
702
4230
715
1493
736
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS1

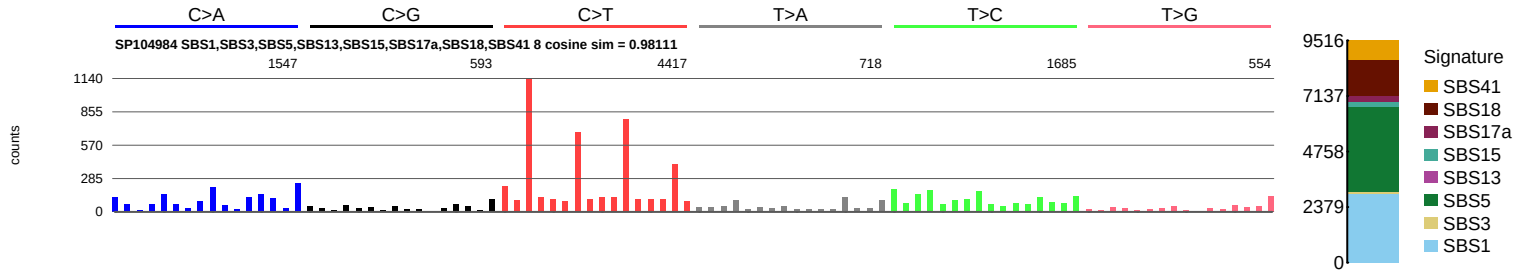

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98116
1643
731
4345
741
1313
748
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98113
1638
710
4207
715
1498
746
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS41 7 cosine sim = 0.98111
1550
572
4414
727
1686
564
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 7 cosine sim = 0.98111
1549
586
4421
711
1689
555
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS40,SBS44 4 cosine sim = 0.98111
1346
817
4469
689
1088
1108
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS17b
SBS1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C>G
C>T
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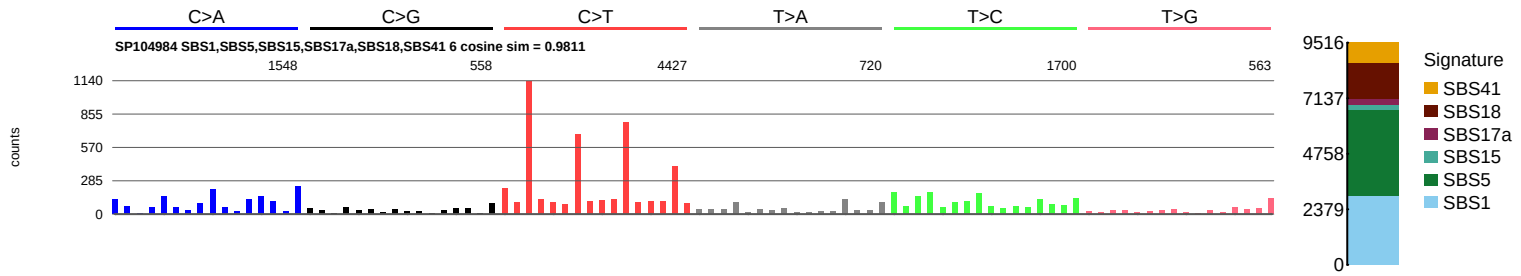
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98111
1547
593
4417
718
1685
554
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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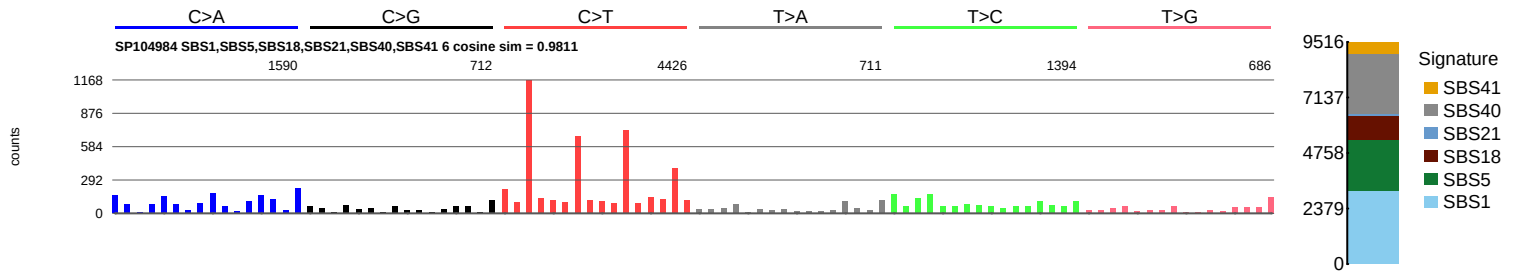
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS41 6 cosine sim = 0.9811
1548
558
4427
720
1700
563
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
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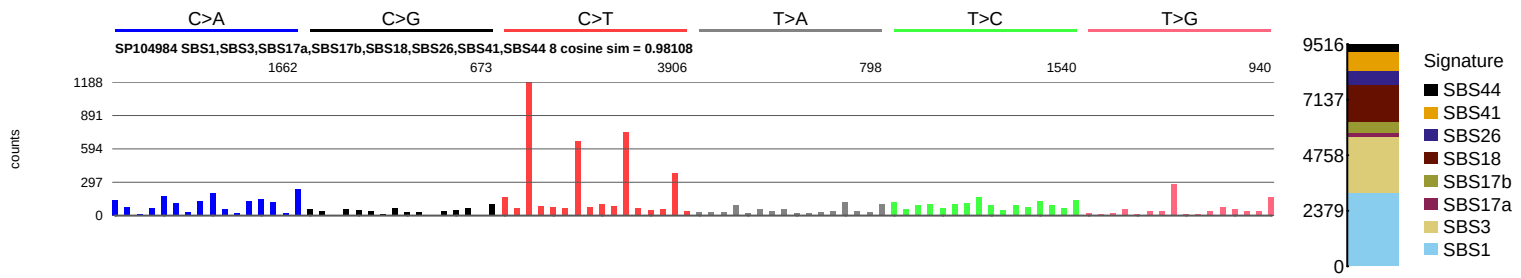
T>C
T>G
SP104984 SBS1,SBS5,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.9811
1590
712
4426
711
1394
686
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS5
SBS1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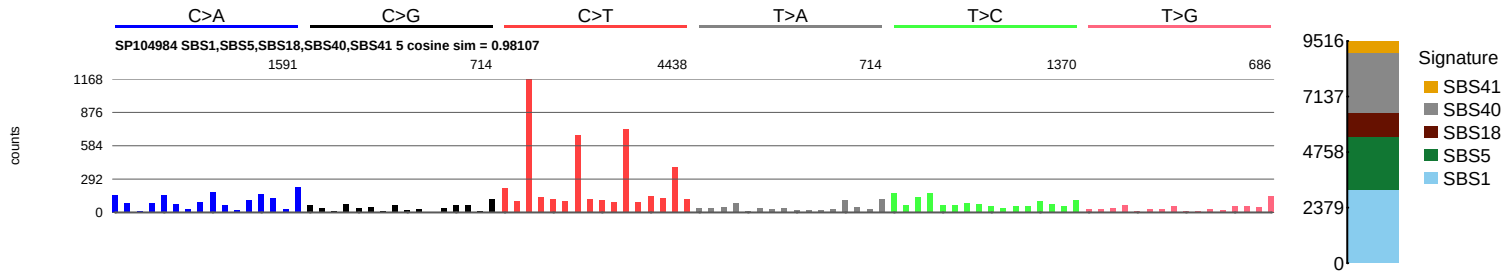
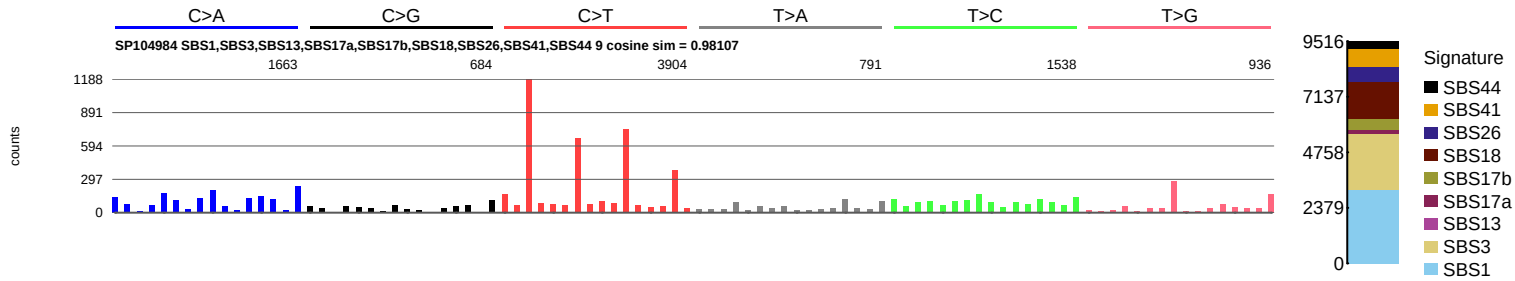
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98108
1662
673
3906
798
1540
940
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS18,SBS40,SBS41 5 cosine sim = 0.98107
1591
714
4438
714
1370
686
counts
1168
876
584
292
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS5
SBS1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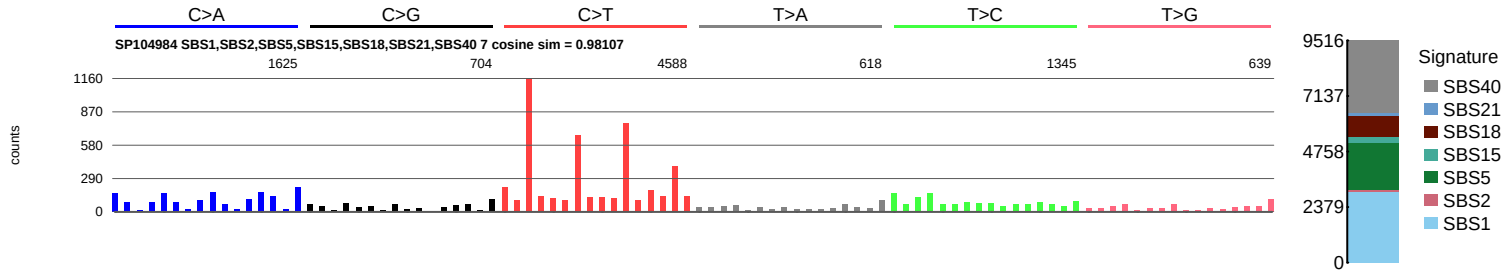
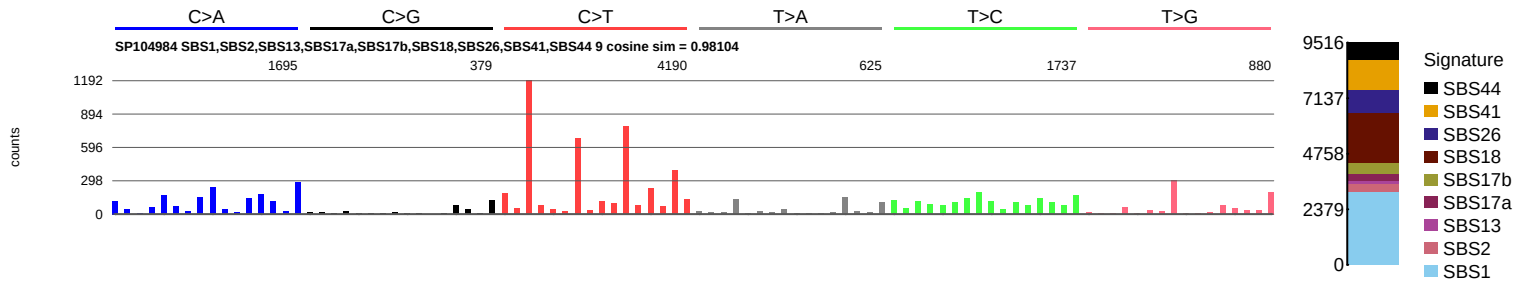
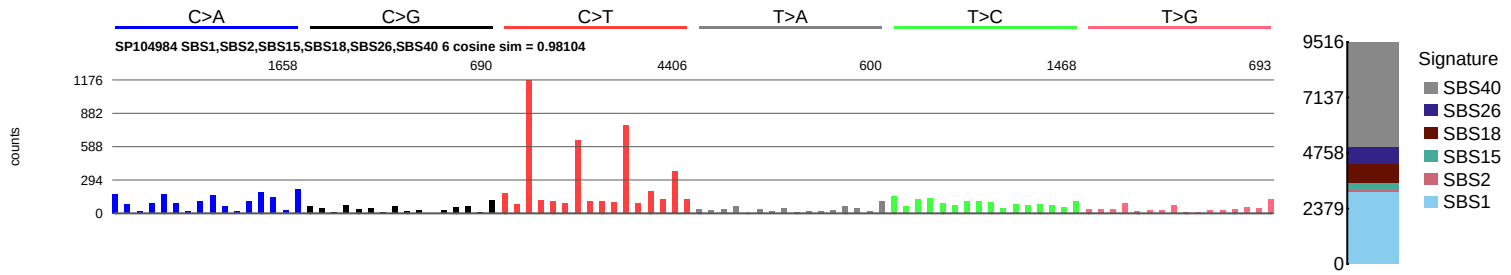
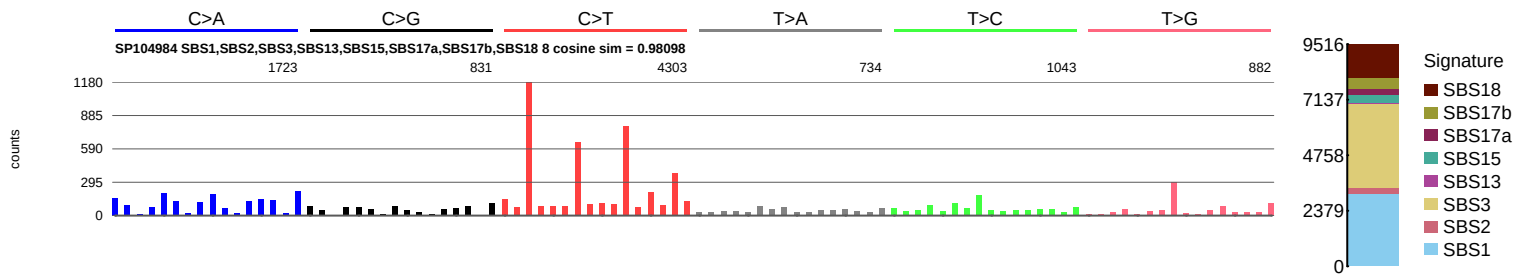
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98107
1663
684
3904
791
1538
936
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.98107
1625
704
4588
618
1345
639
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98104
1695
379
4190
625
1737
880
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40 6 cosine sim = 0.98104
1658
690
4406
600
1468
693
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1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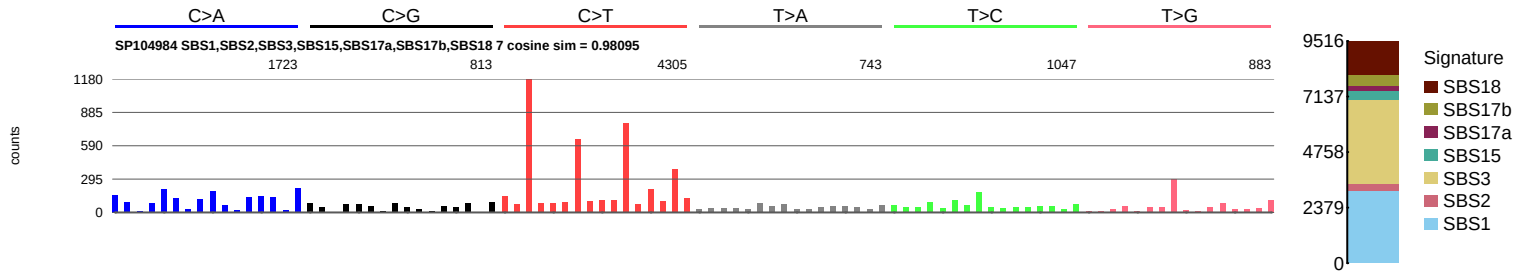
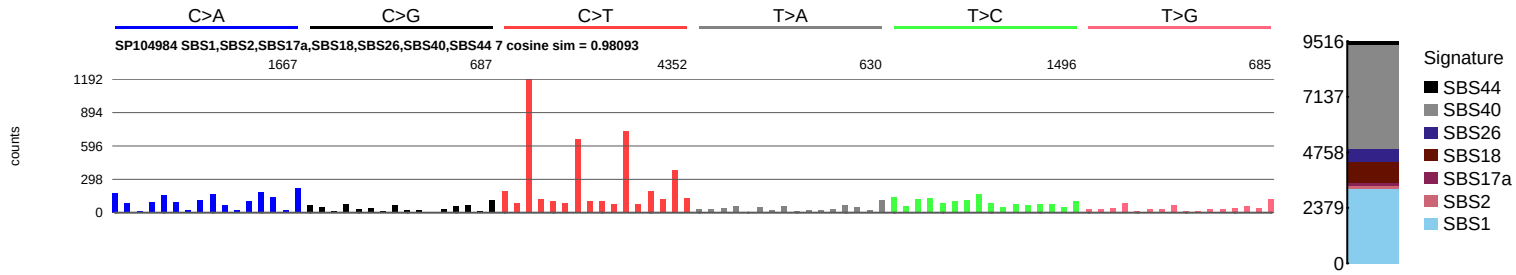
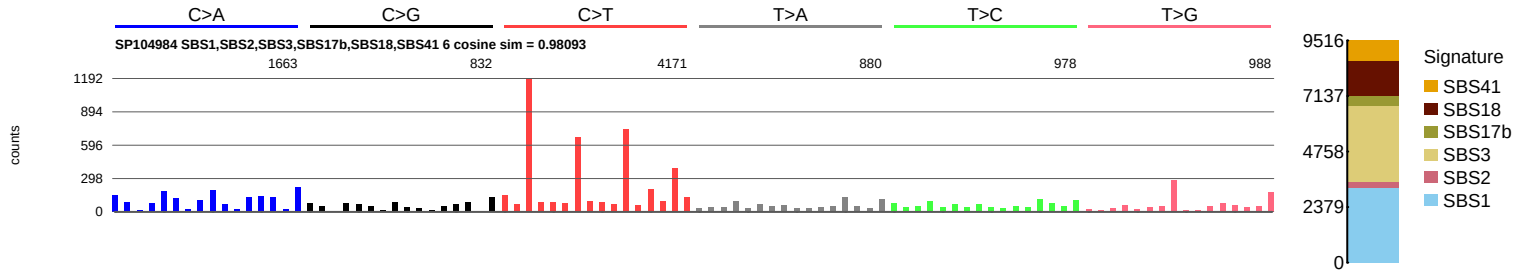
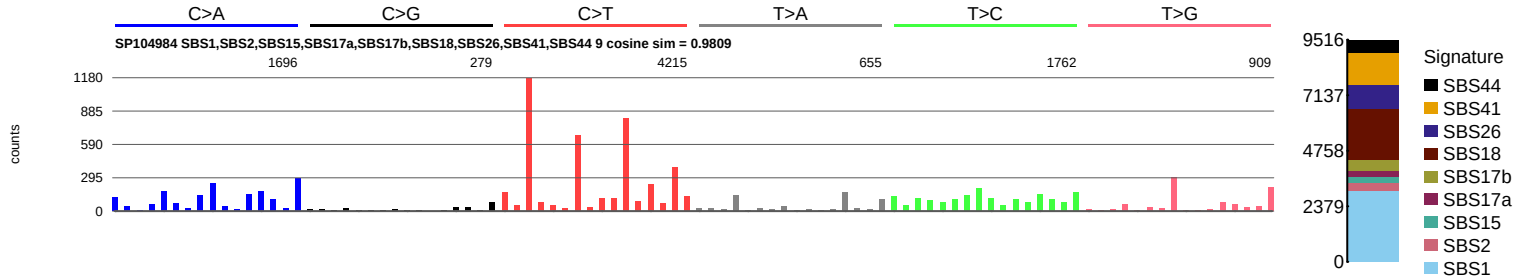
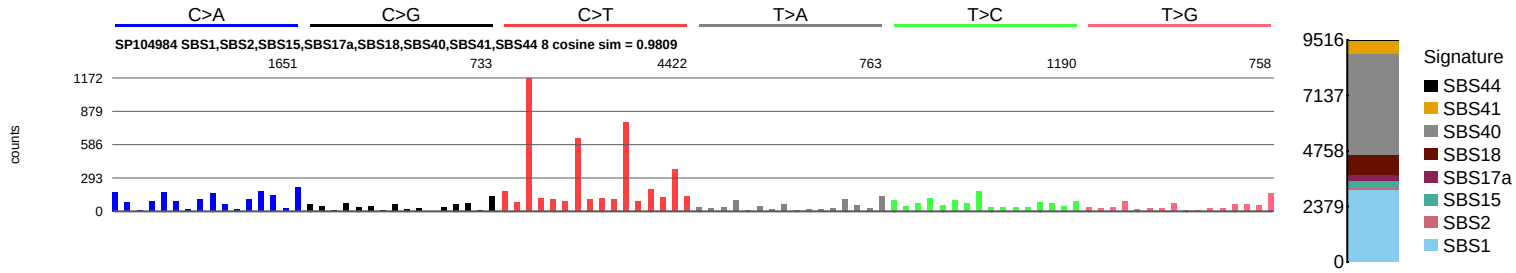
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98098
1723
831
4303
734
1043
882
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98095
1723
813
4305
743
1047
883
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98093
1667
687
4352
630
1496
685
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS41 6 cosine sim = 0.98093
1663
832
4171
880
978
988
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17b
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.9809
1696
279
4215
655
1762
909
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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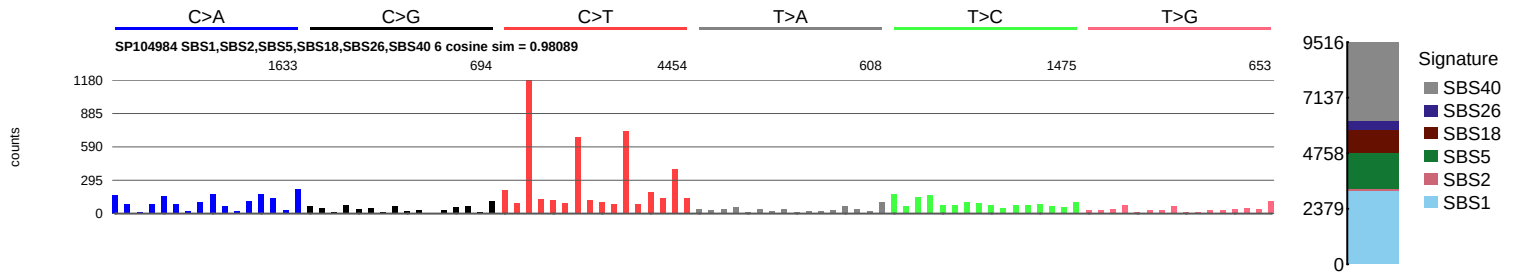
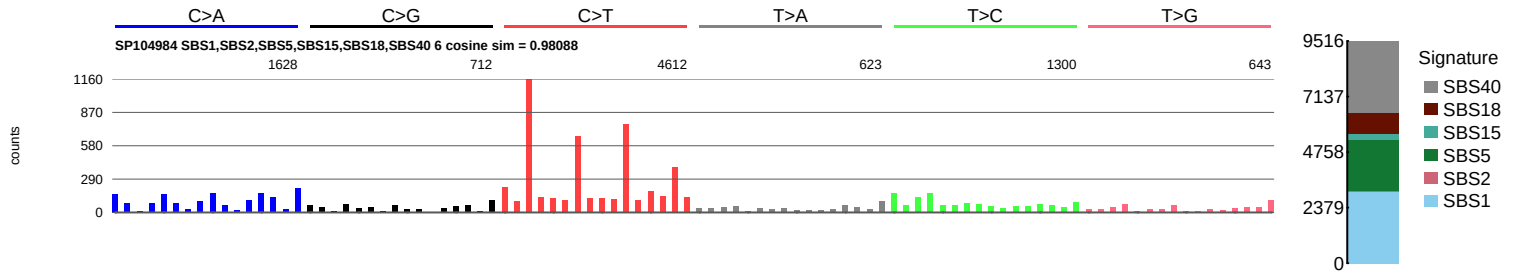
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.9809
1651
733
4422
763
1190
758
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40 6 cosine sim = 0.98089
1633
694
4454
608
1475
653
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40 6 cosine sim = 0.98088
1628
712
4612
623
1300
643
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
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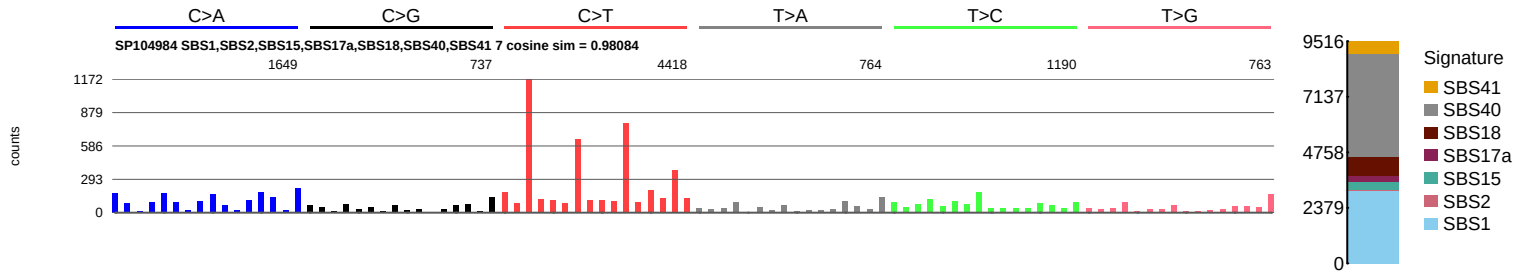
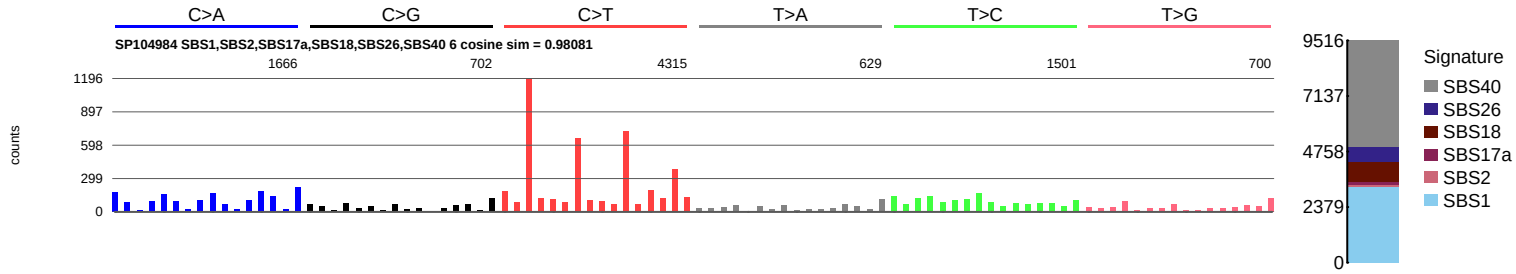
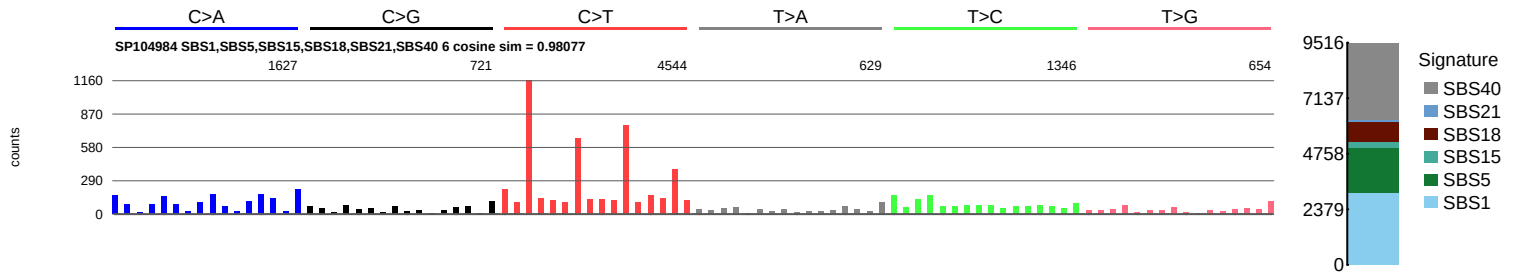
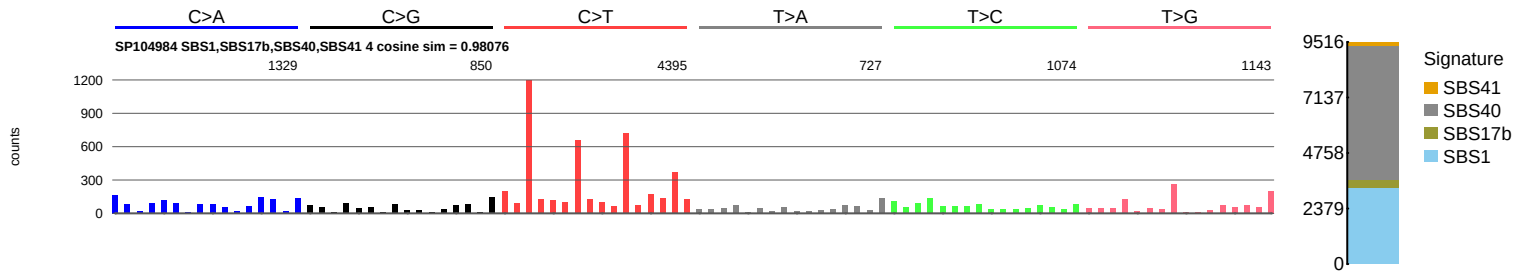
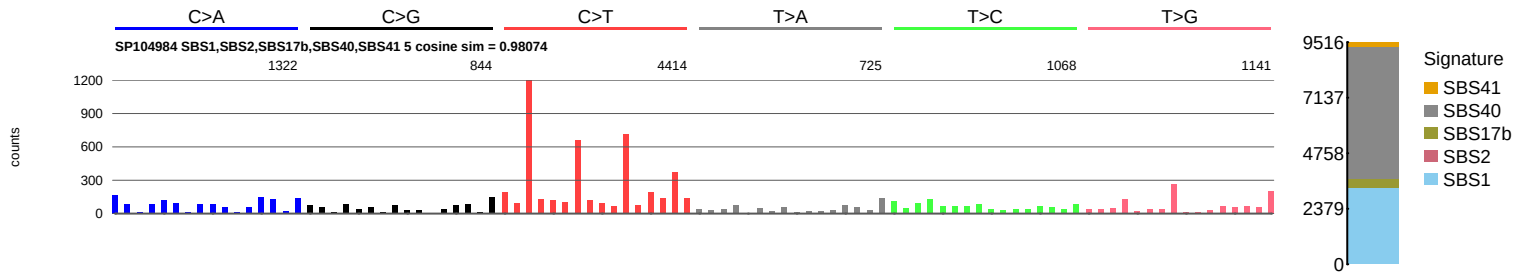
SBS2
SBS1

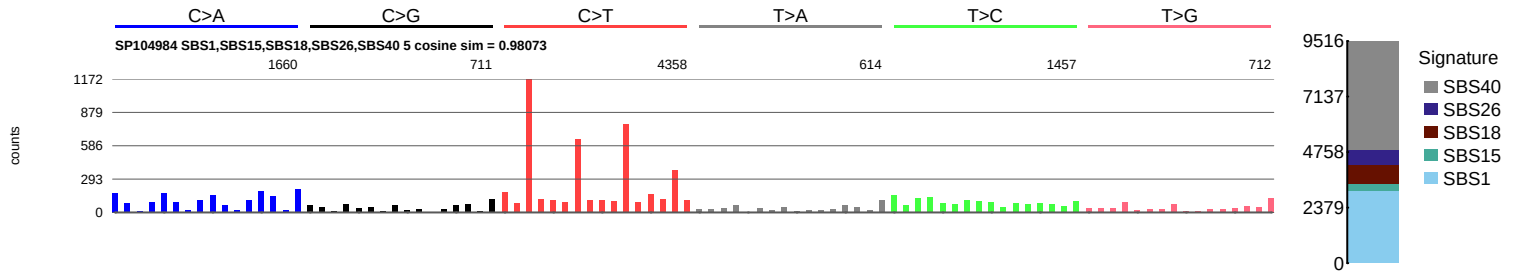

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98084
1649
737
4418
764
1190
763
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1

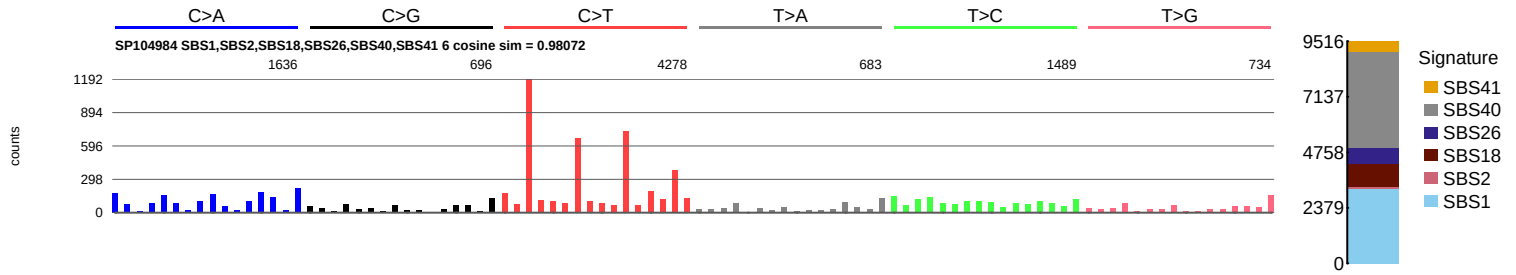

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40 6 cosine sim = 0.98081
1666
702
4315
629
1501
700
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1

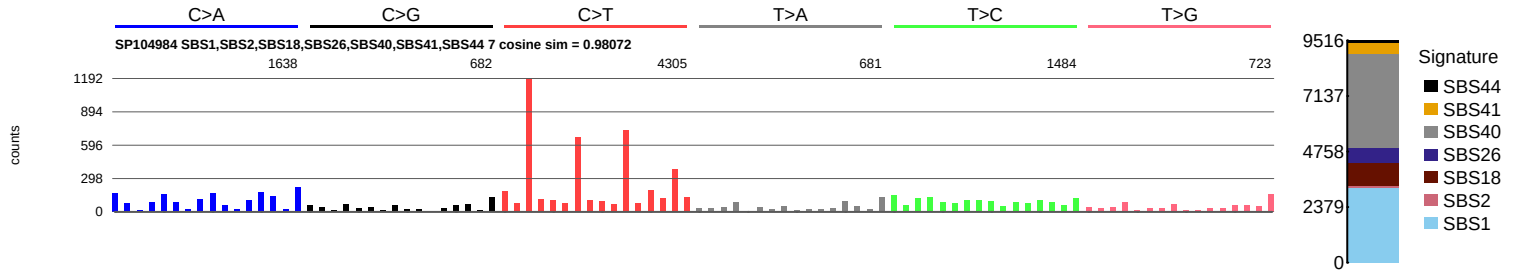

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS18,SBS21,SBS40 6 cosine sim = 0.98077
1627
721
4544
629
1346
654
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS5
SBS1

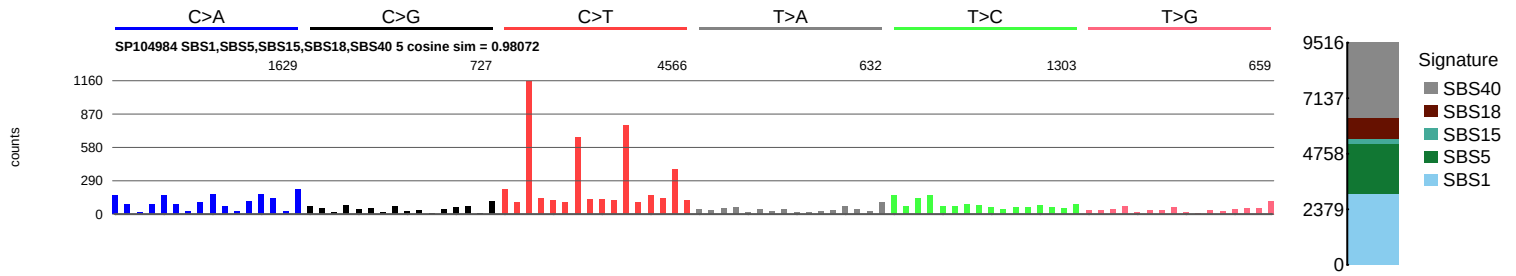

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS40,SBS41 4 cosine sim = 0.98076
1329
850
4395
727
1074
1143
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS1

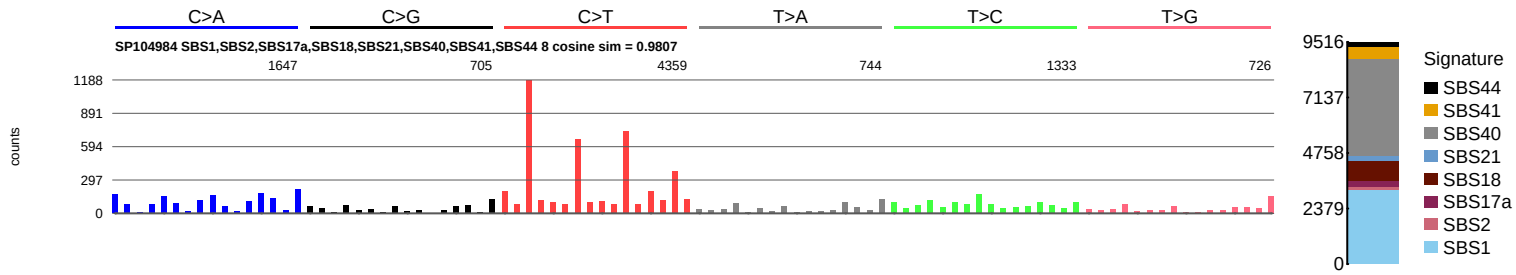

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS40,SBS41 5 cosine sim = 0.98074
1322
844
4414
725
1068
1141
counts
1200
900
600
300
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS17b
SBS2
SBS1

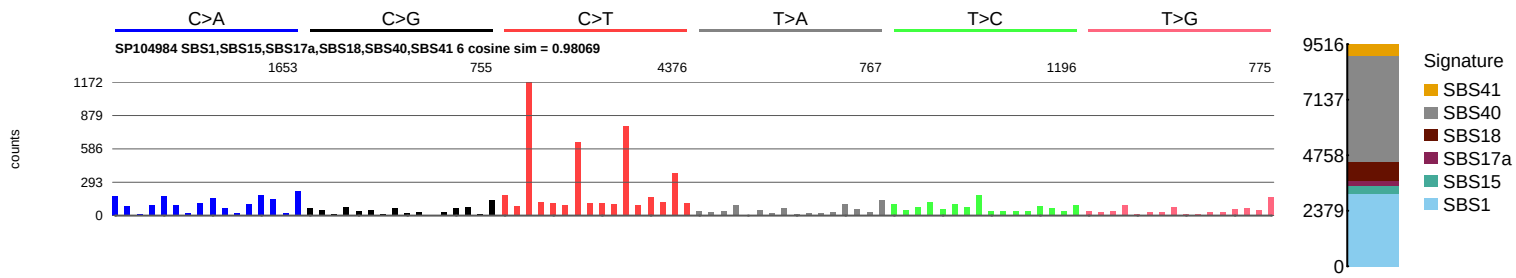

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS18,SBS26,SBS40 5 cosine sim = 0.98073
1660
711
4358
614
1457
712
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS1

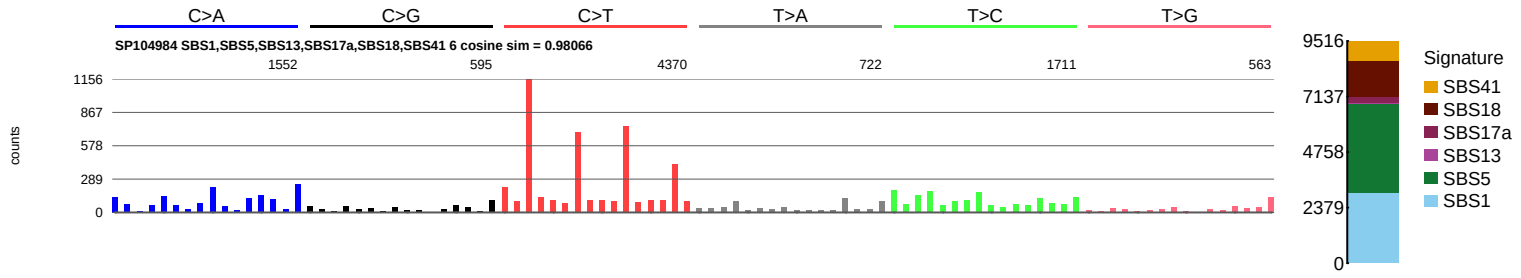

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98072
1636
696
4278
683
1489
734
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1

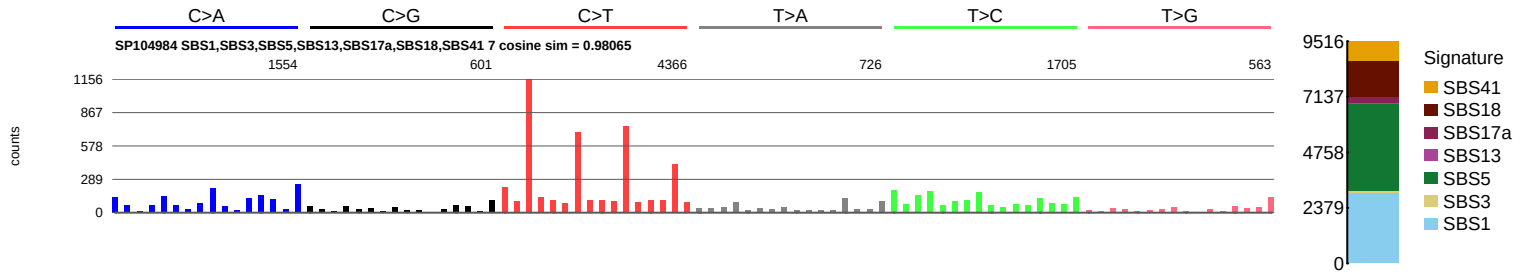

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98072
1638
682
4305
681
1484
723
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1

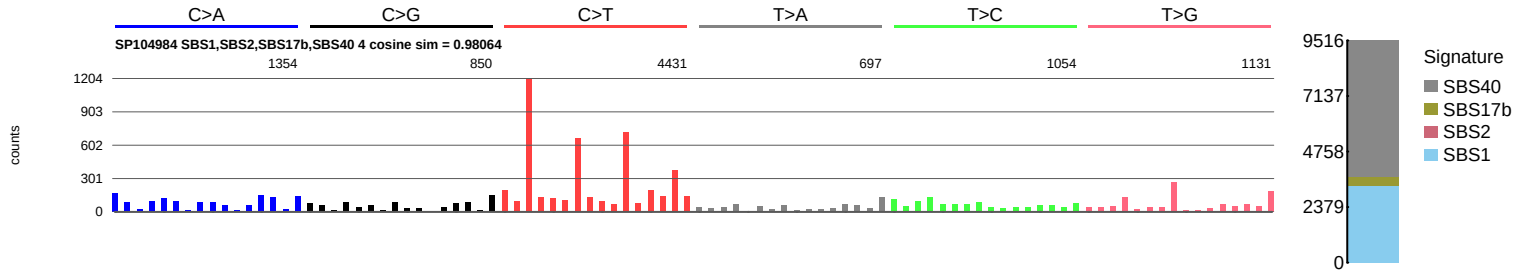

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS15,SBS18,SBS40 5 cosine sim = 0.98072
1629
727
4566
632
1303
659
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS15
SBS5
SBS1

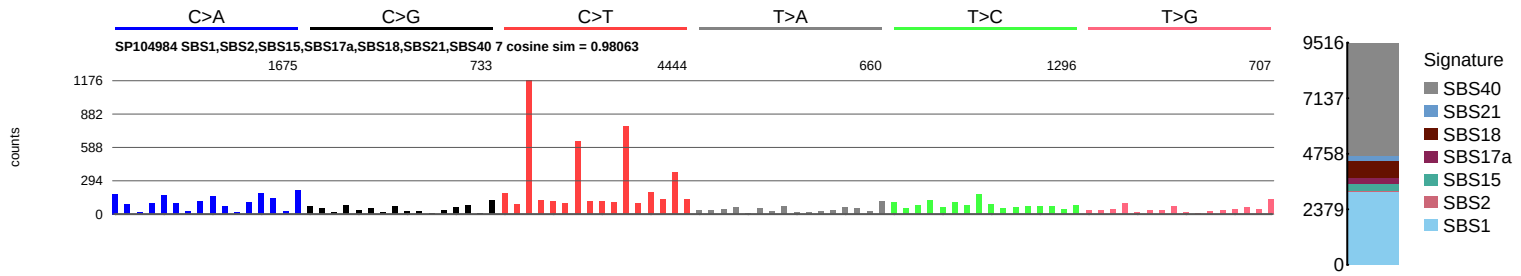

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9807
1647
705
4359
744
1333
726
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1

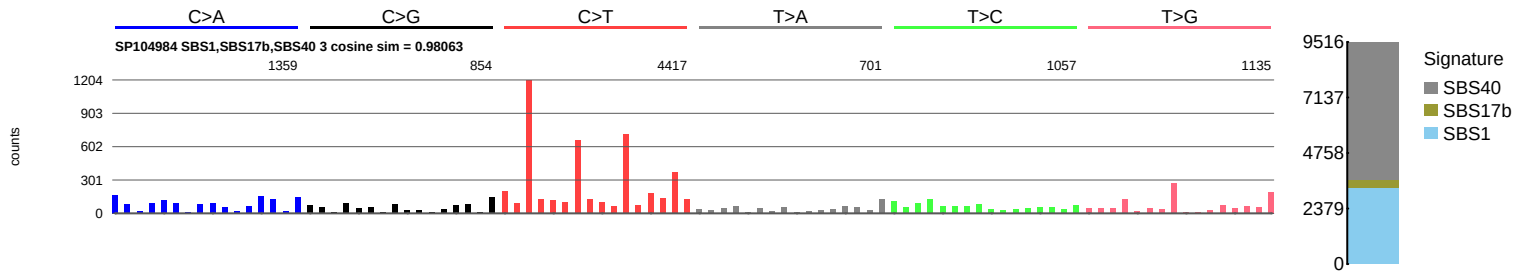

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS18,SBS40,SBS41 6 cosine sim = 0.98069
1653
755
4376
767
1196
775
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS1

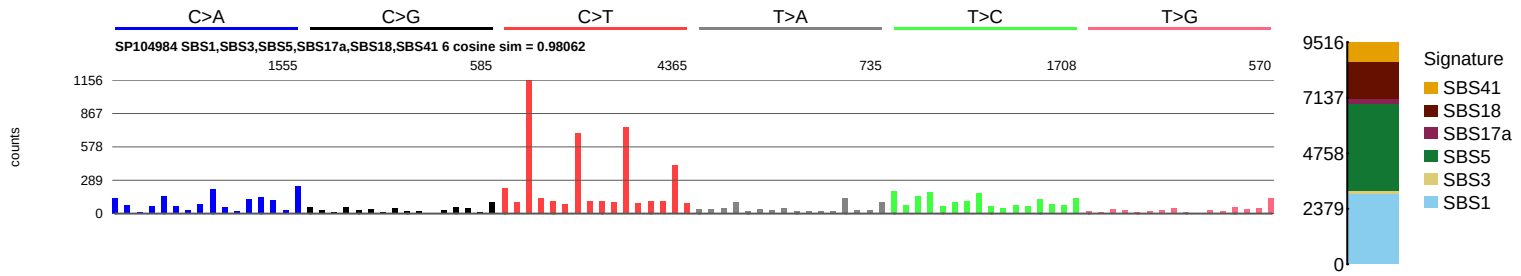

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS13,SBS17a,SBS18,SBS41 6 cosine sim = 0.98066
1552
595
4370
722
1711
563
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS1

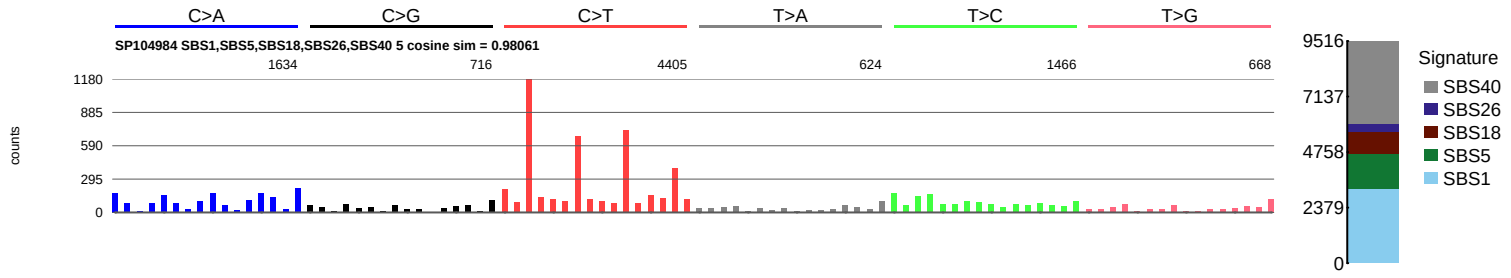

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41 7 cosine sim = 0.98065
1554
601
4366
726
1705
563
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS1

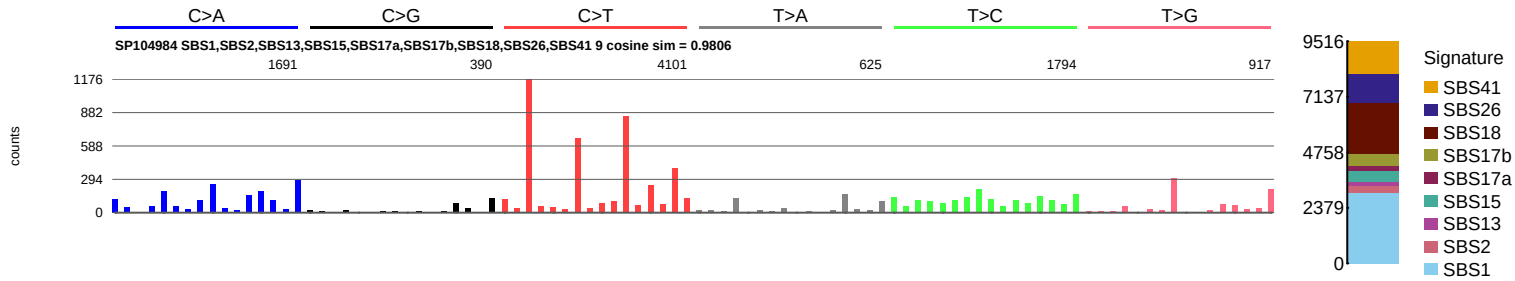

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17b,SBS40 4 cosine sim = 0.98064
1354
850
4431
697
1054
1131
counts
1204
903
602
301
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS2
SBS1

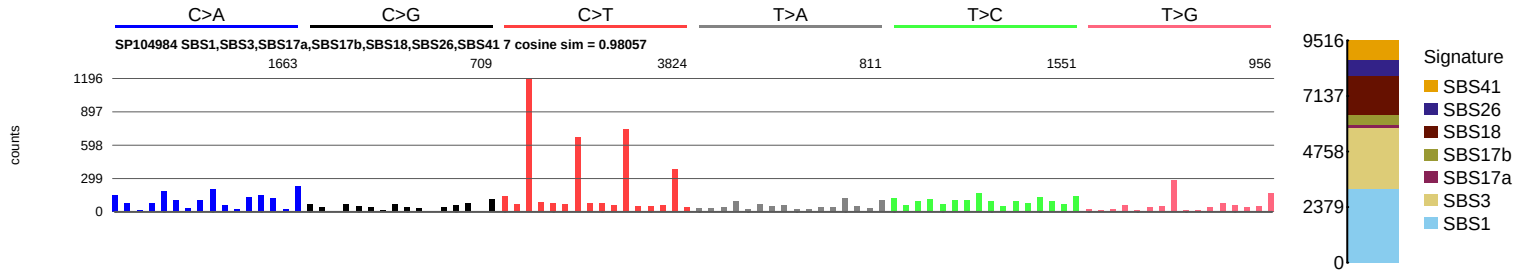

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98063
1675
733
4444
660
1296
707
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1

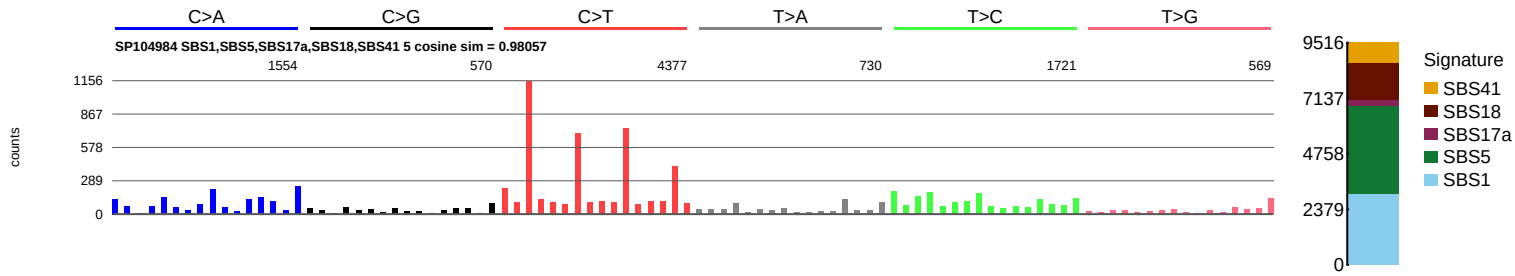

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17b,SBS40 3 cosine sim = 0.98063
1359
854
4417
701
1057
1135
counts
1204
903
602
301
0
9516
7137
4758
2379
0
Signature
SBS40
SBS17b
SBS1

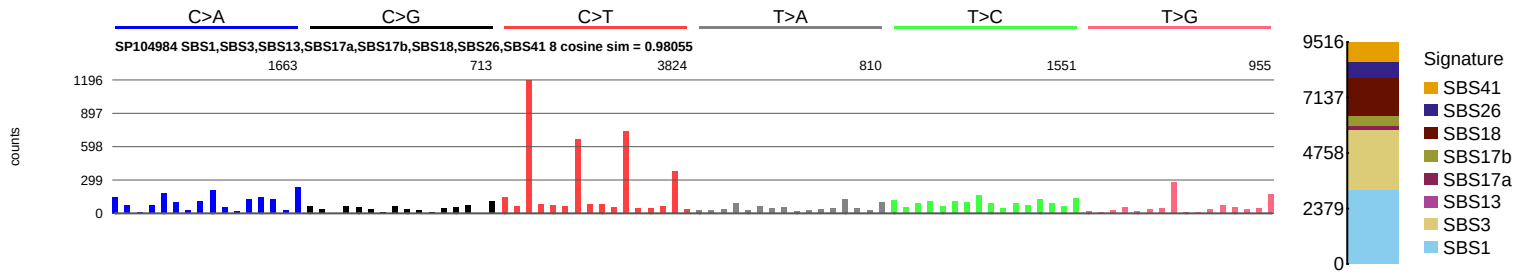

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS5,SBS17a,SBS18,SBS41 6 cosine sim = 0.98062
1555
585
4365
735
1708
570
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS5
SBS3
SBS1

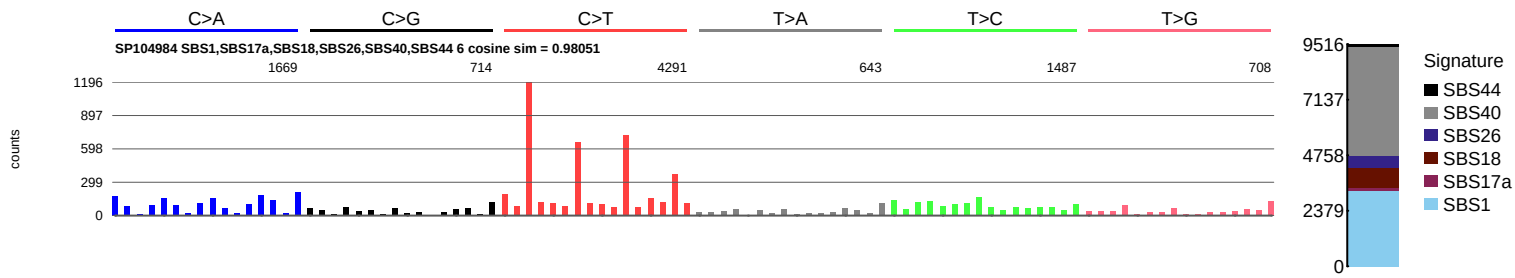

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS18,SBS26,SBS40 5 cosine sim = 0.98061
1634
716
4405
624
1466
668
counts
1180
885
590
295
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS5
SBS1

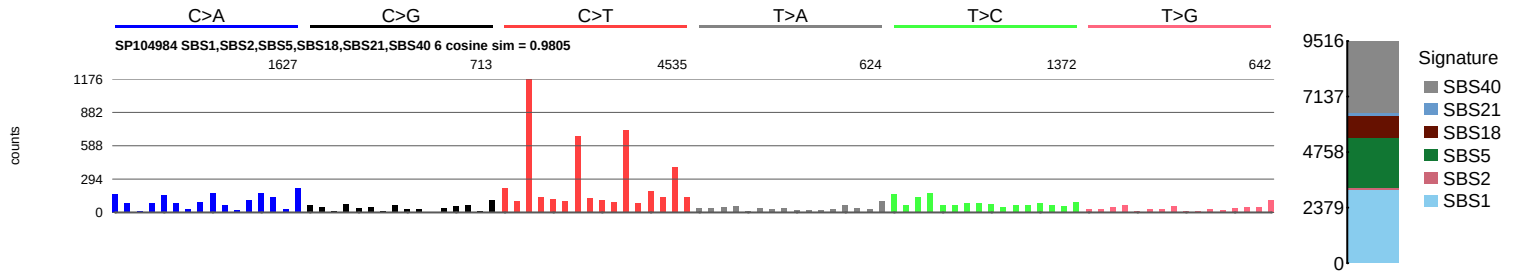

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.9806
1691
390
4101
625
1794
917
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

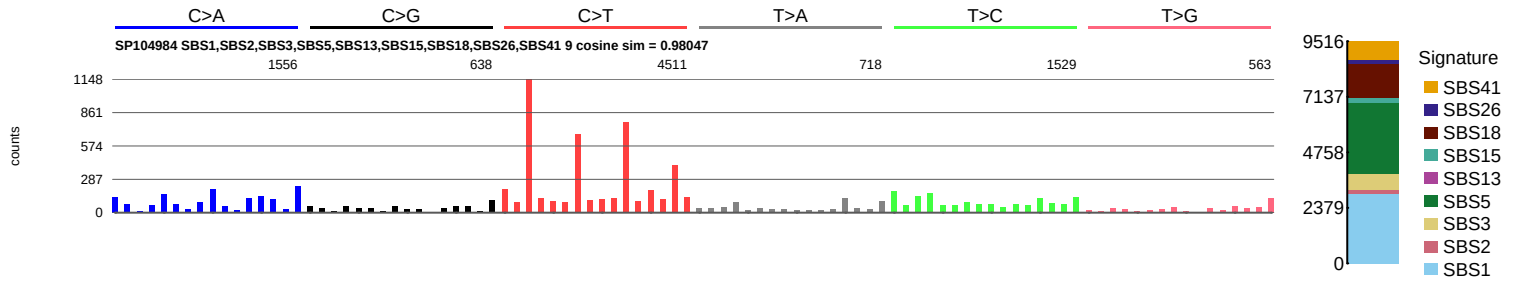

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98057
1663
709
3824
811
1551
956
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

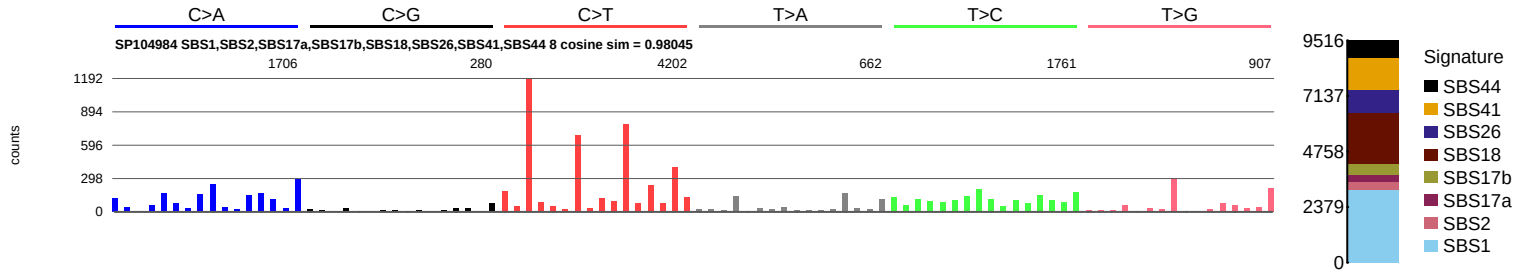

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS17a,SBS18,SBS41 5 cosine sim = 0.98057
1554
570
4377
730
1721
569
counts
1156
867
578
289
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS17a
SBS5
SBS1

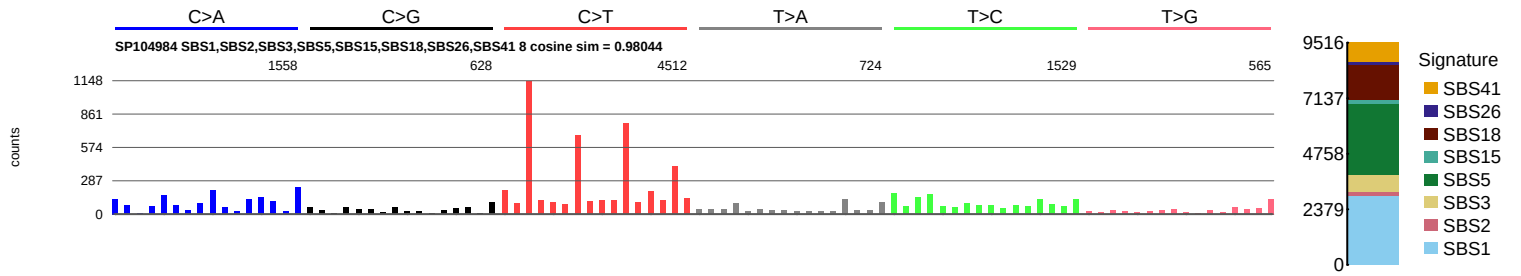

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98055
1663
713
3824
810
1551
955
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

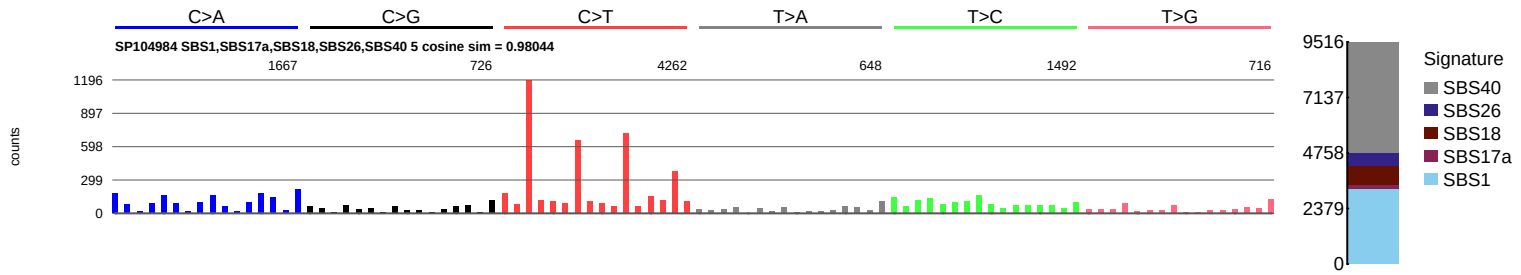

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98051
1669
714
4291
643
1487
708
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS1

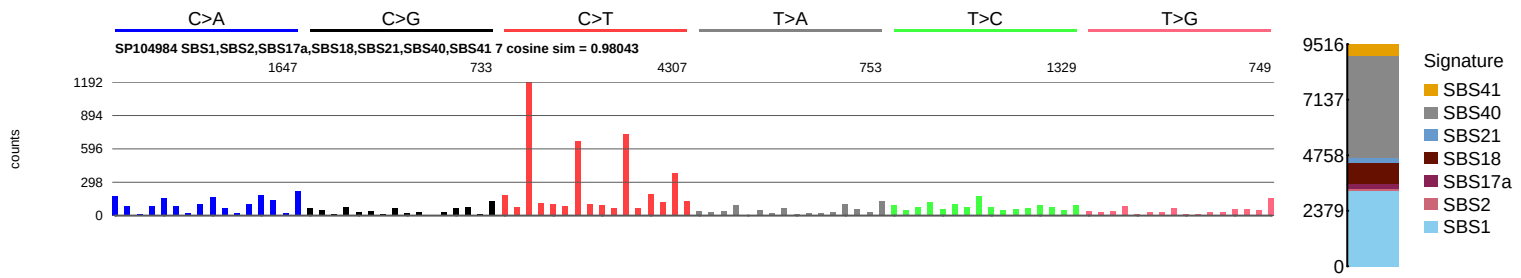

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40 6 cosine sim = 0.9805
1627
713
4535
624
1372
642
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1

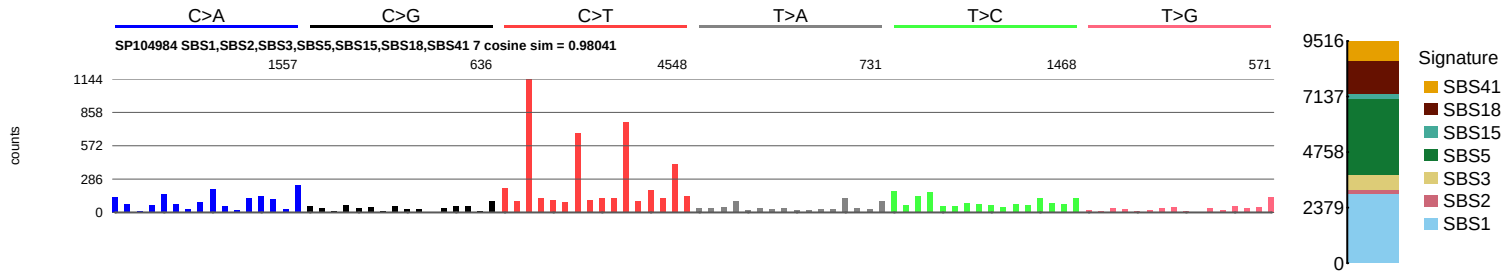

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS26,SBS41 9 cosine sim = 0.98047
1556
638
4511
718
1529
563
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

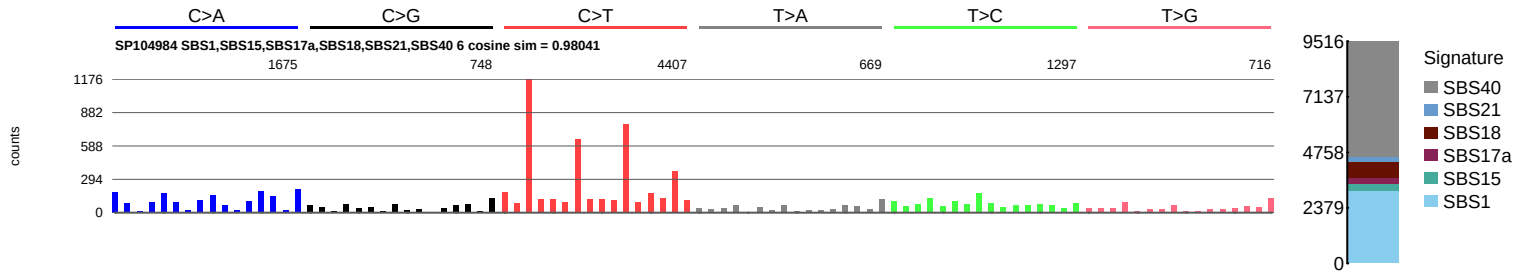

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98045
1706
280
4202
662
1761
907
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

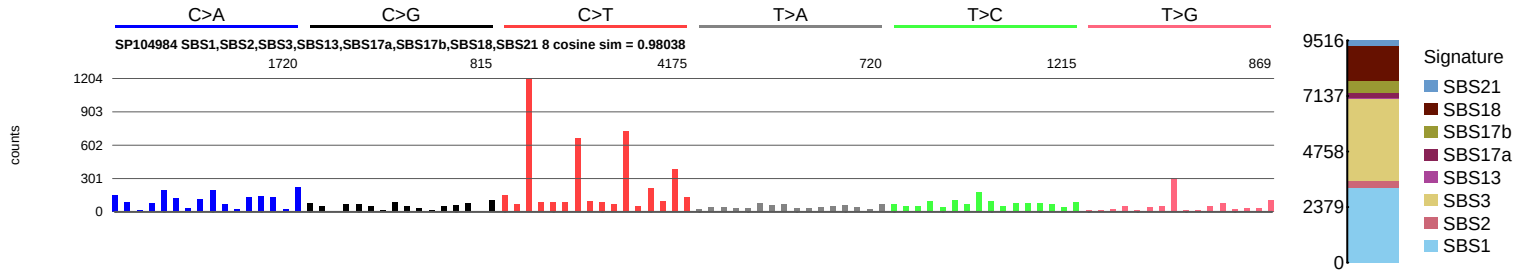

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS26,SBS41 8 cosine sim = 0.98044
1558
628
4512
724
1529
565
counts
1148
861
574
287
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1

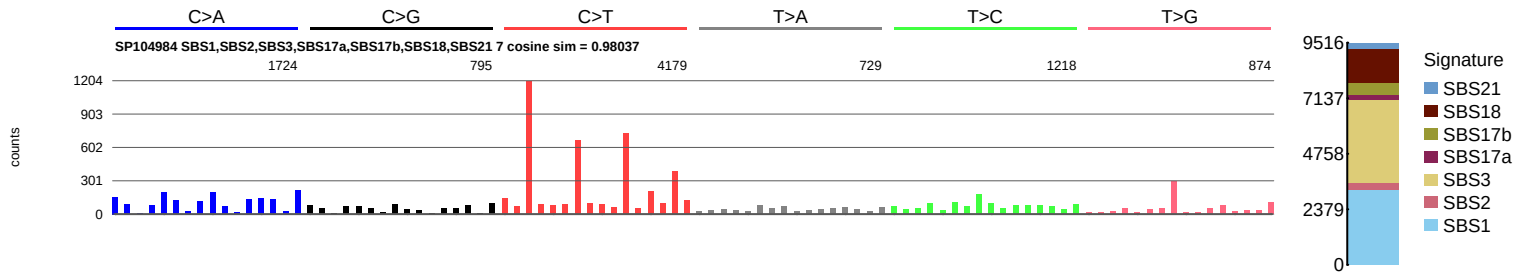

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS18,SBS26,SBS40 5 cosine sim = 0.98044
1667
726
4262
648
1492
716
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS1

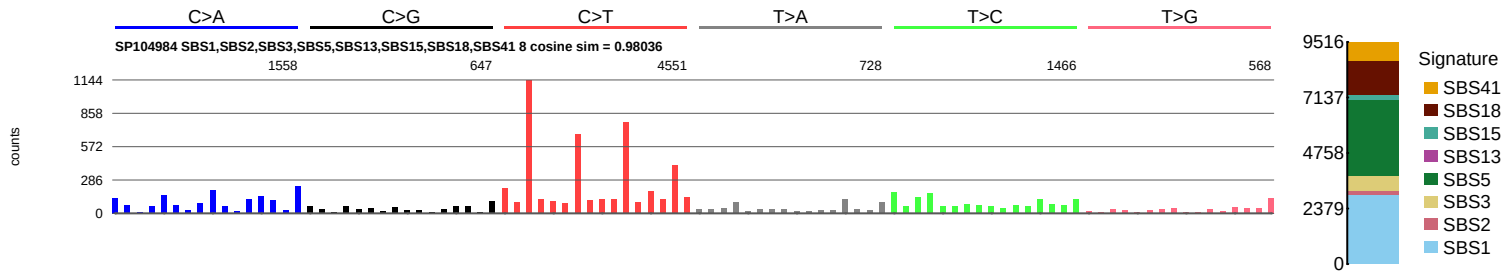

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98043
1647
733
4307
753
1329
749
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1

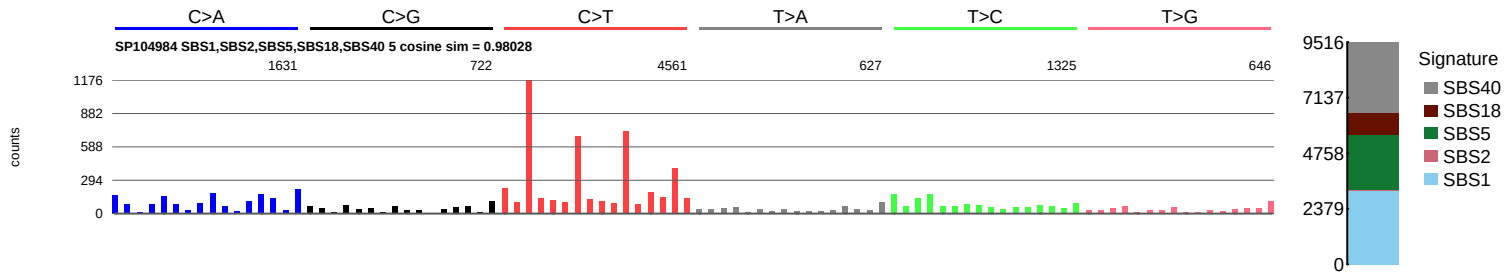

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS15,SBS18,SBS41 7 cosine sim = 0.98041
1557
636
4548
731
1468
571
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS15
SBS5
SBS3
SBS2
SBS1

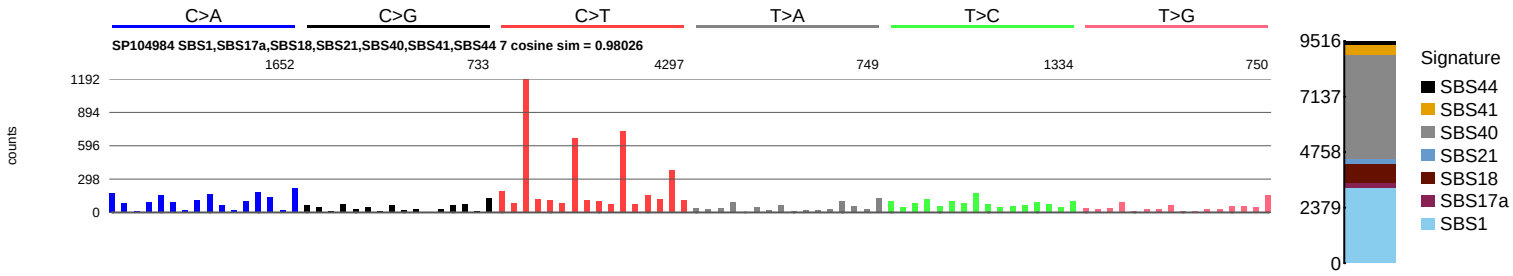

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS40 6 cosine sim = 0.98041
1675
748
4407
669
1297
716
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS1

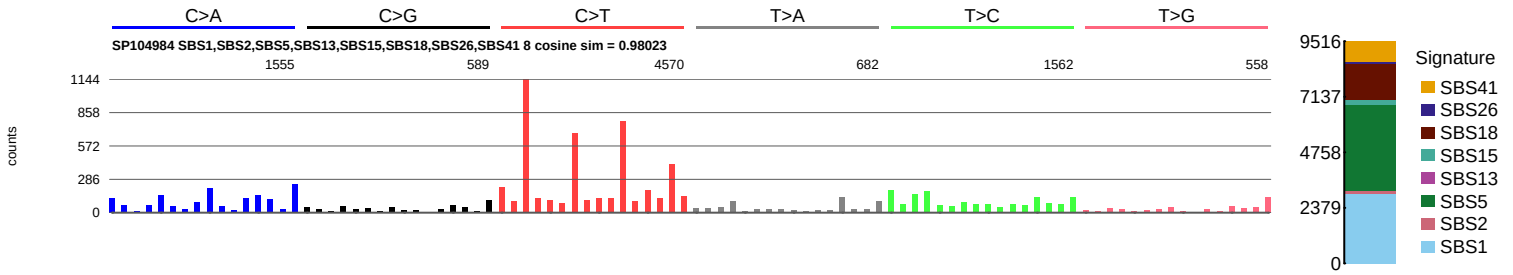

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98038
1720
815
4175
720
1215
869
counts
1204
903
602
301
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98037
1724
795
4179
729
1218
874
counts
1204
903
602
301
0
9516
7137
4758
2379
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS18,SBS41 8 cosine sim = 0.98036
1558
647
4551
728
1466
568
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS18,SBS40 5 cosine sim = 0.98028
1631
722
4561
627
1325
646
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98026
1652
733
4297
749
1334
750
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS1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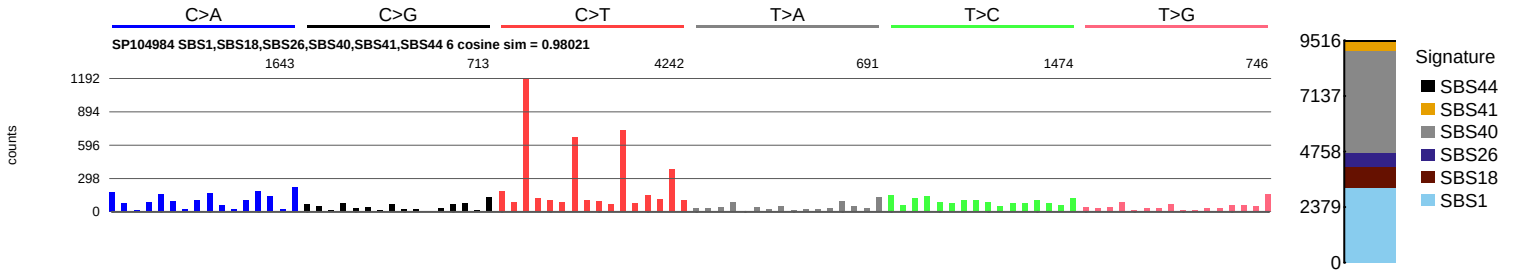
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS41 8 cosine sim = 0.98023
1555
589
4570
682
1562
558
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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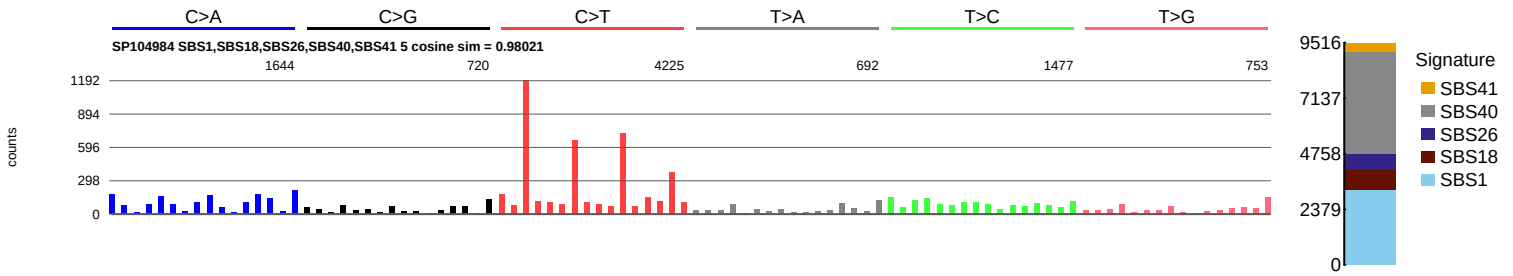
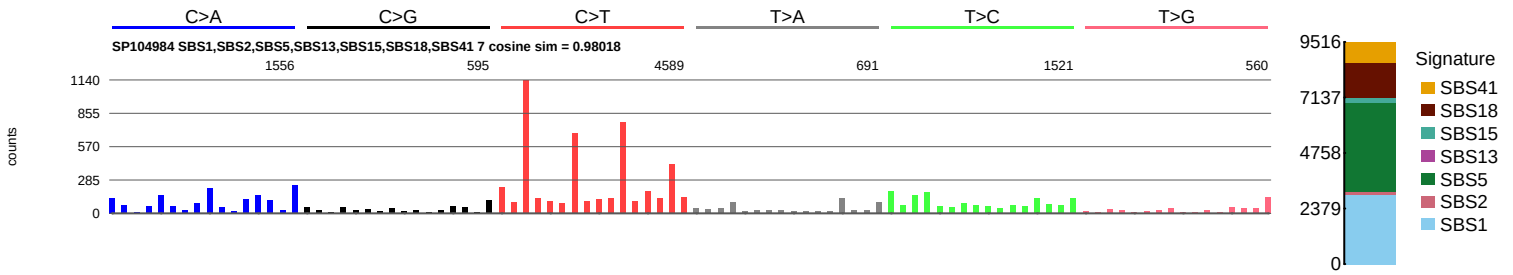
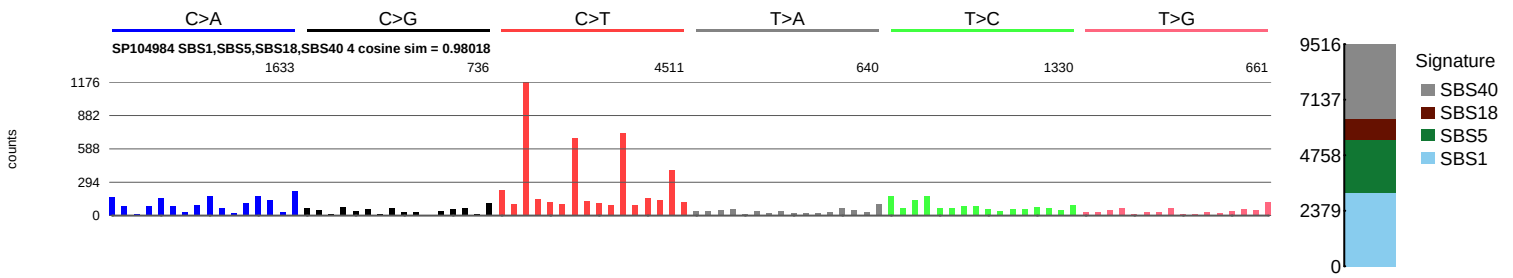
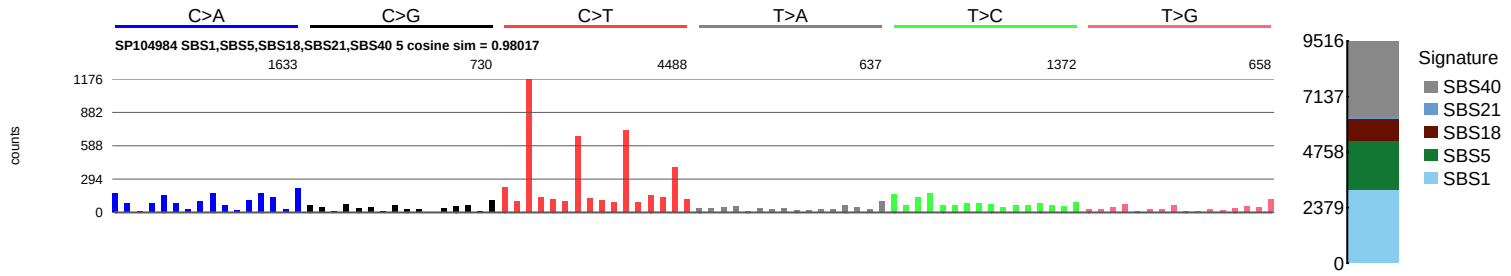
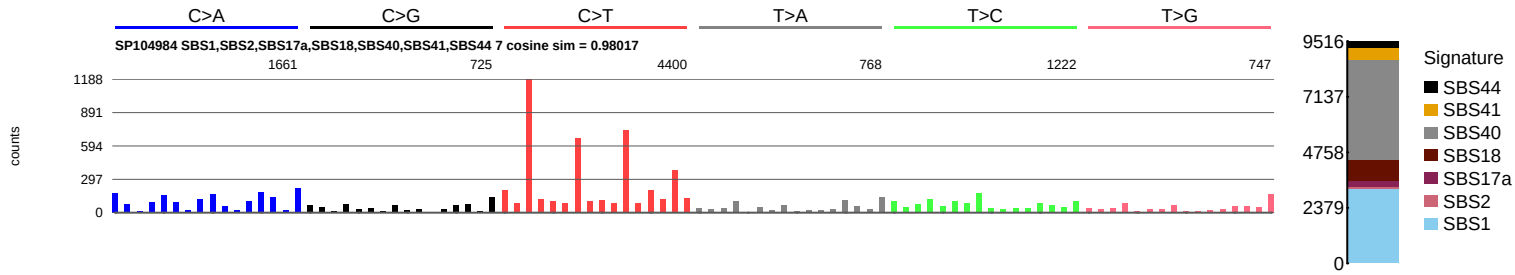
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS18,SBS26,SBS40,SBS41,SBS44 6 cosine sim = 0.98021
1643
713
4242
691
1474
746
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS1

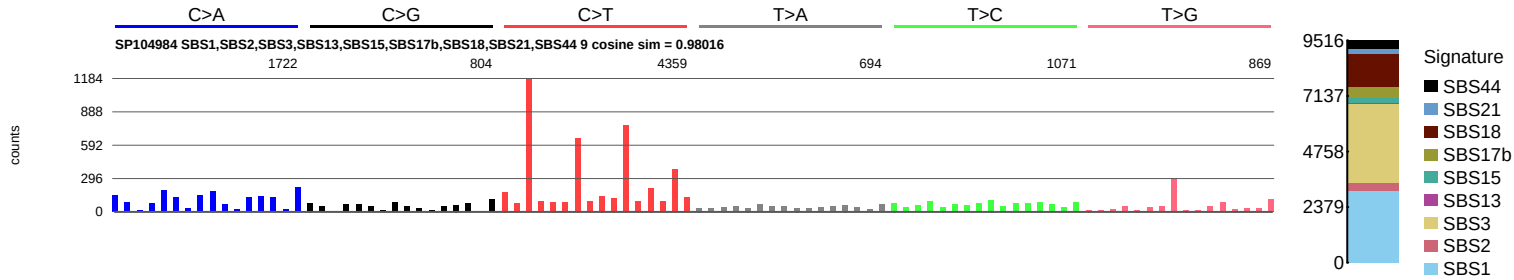

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS18,SBS26,SBS40,SBS41 5 cosine sim = 0.98021
1644
720
4225
692
1477
753
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS1

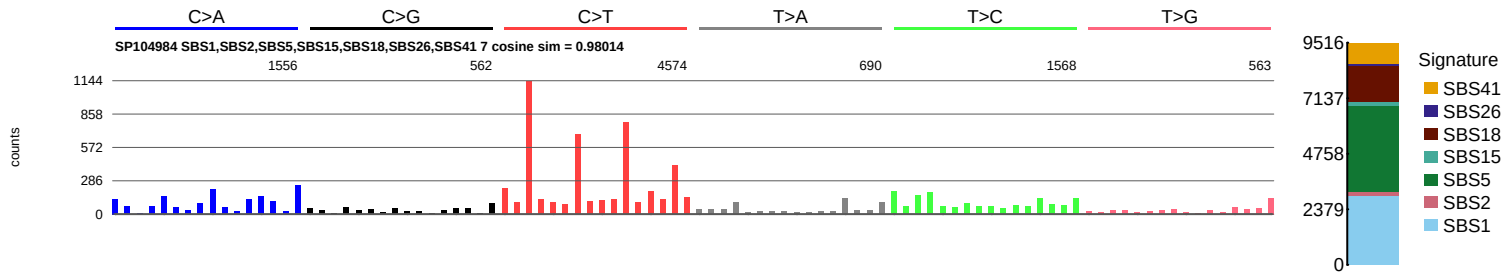

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS41 7 cosine sim = 0.98018
1556
595
4589
691
1521
560
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

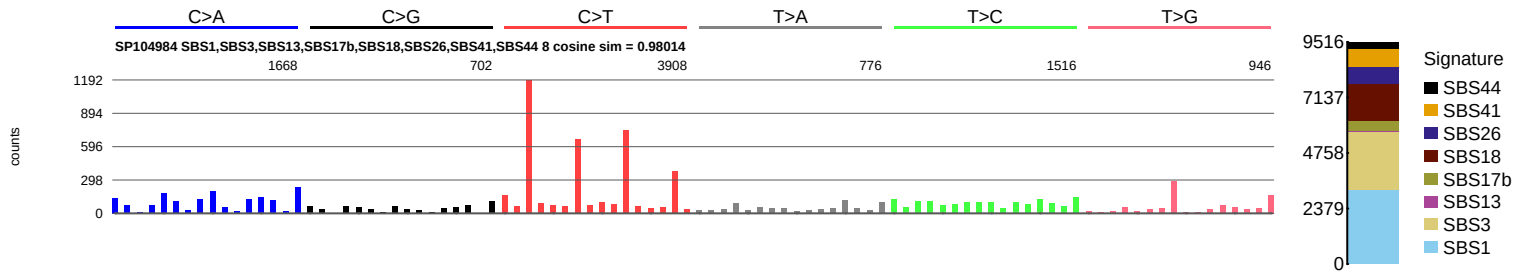

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS18,SBS40 4 cosine sim = 0.98018
1633
736
4511
640
1330
661
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS18
SBS5
SBS1

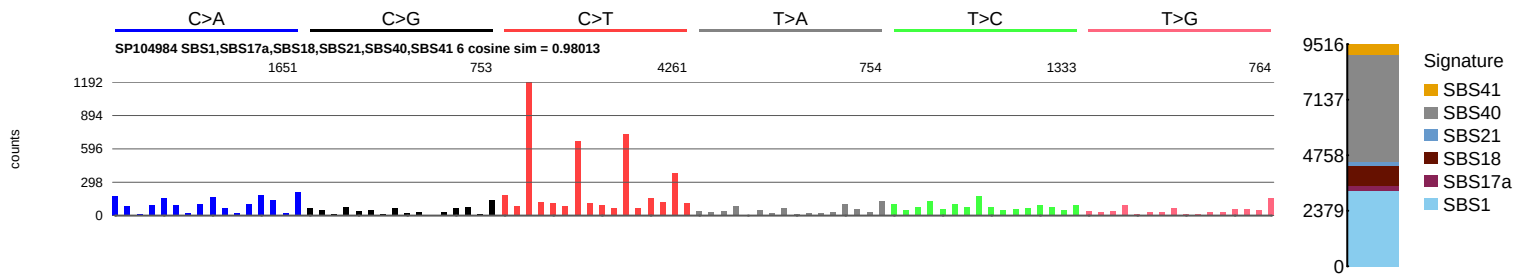

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS5,SBS18,SBS21,SBS40 5 cosine sim = 0.98017
1633
730
4488
637
1372
658
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS40
SBS21
SBS18
SBS5
SBS1

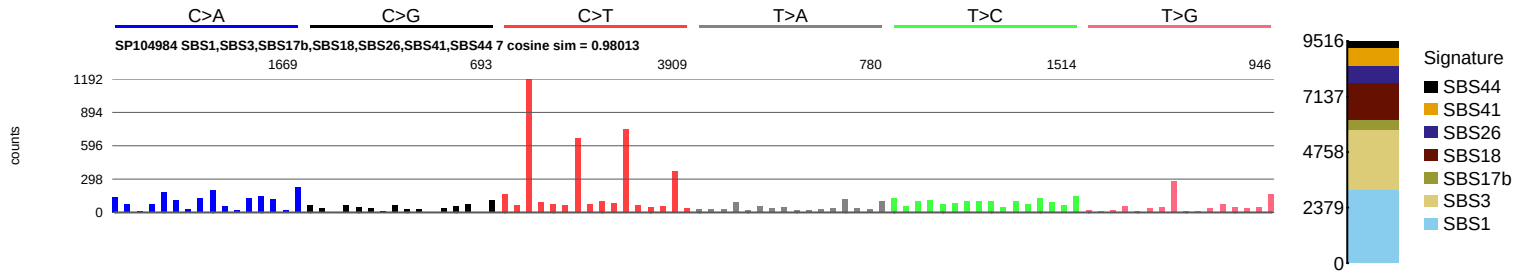

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS17a,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98017
1661
725
4400
768
1222
747
counts
1188
891
594
297
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS2
SBS1

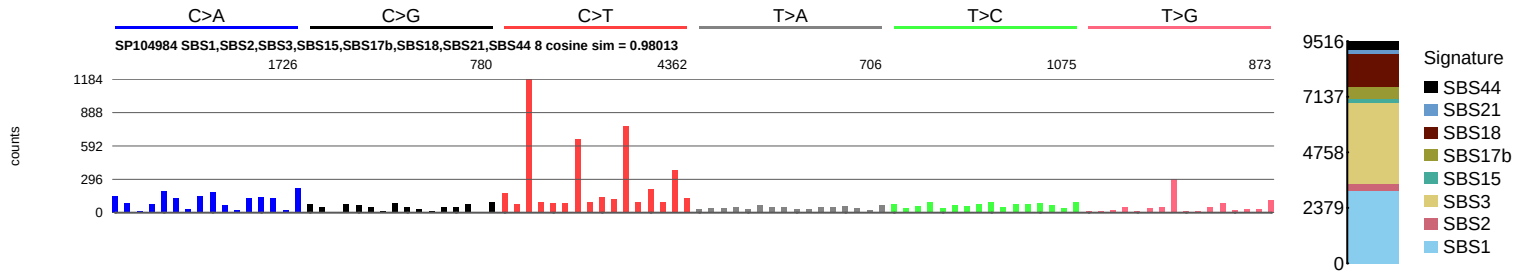

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98016
1722
804
4359
694
1071
869
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

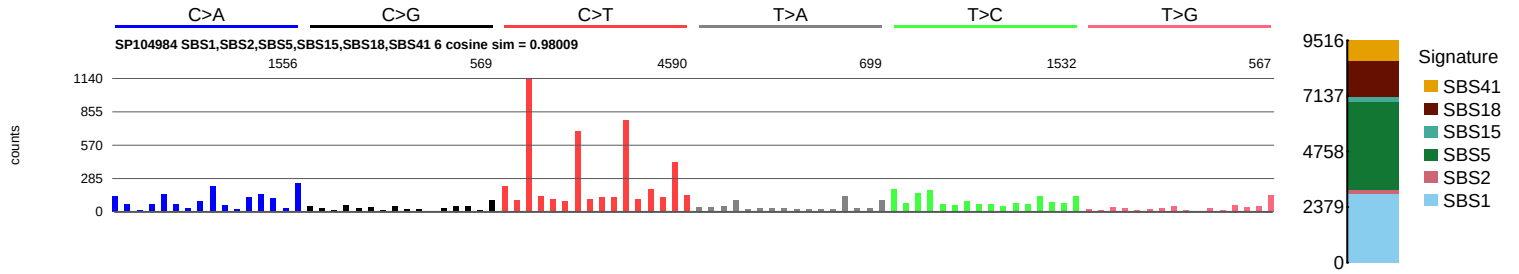

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS41 7 cosine sim = 0.98014
1556
562
4574
690
1568
563
counts
1144
858
572
286
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1

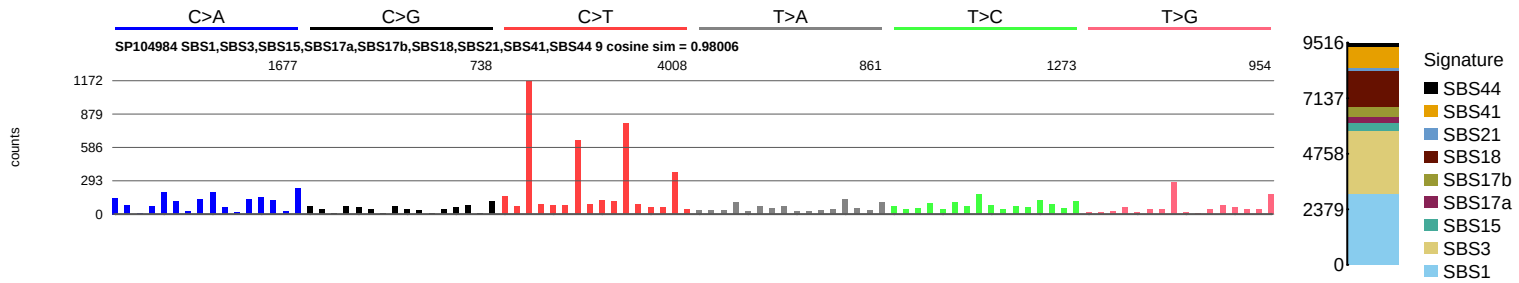

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98014
1668
702
3908
776
1516
946
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS1

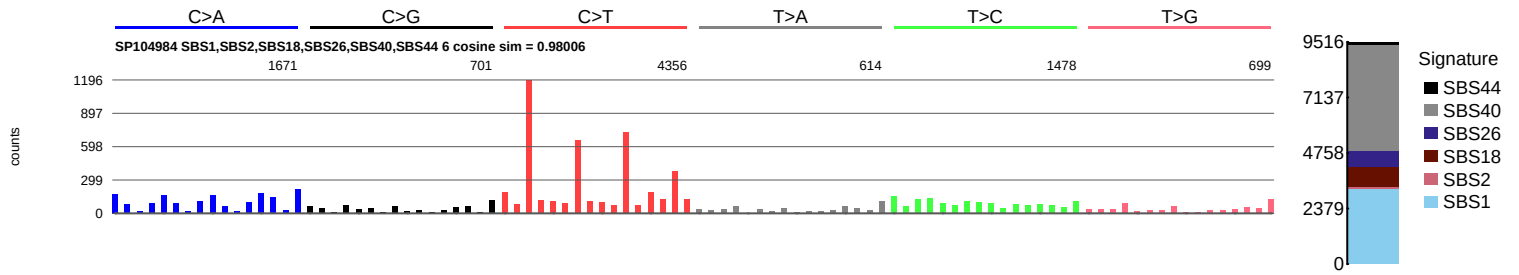

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS17a,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98013
1651
753
4261
754
1333
764
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS1

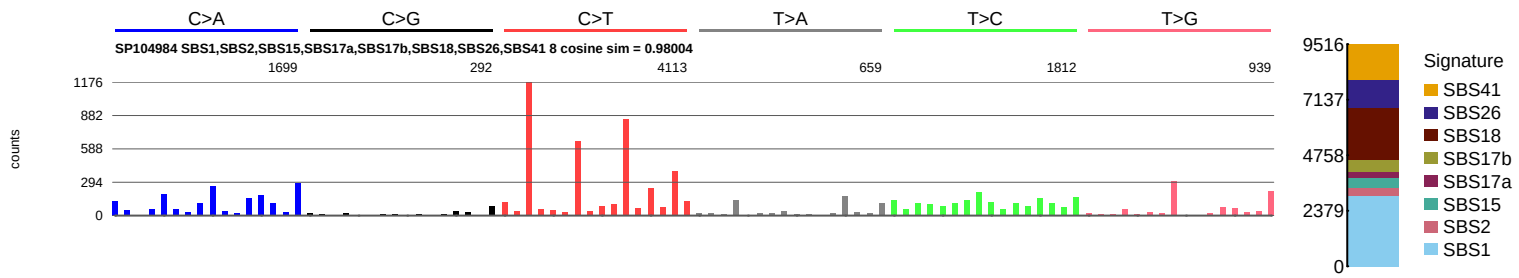

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98013
1669
693
3909
780
1514
946
counts
1192
894
596
298
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS3
SBS1

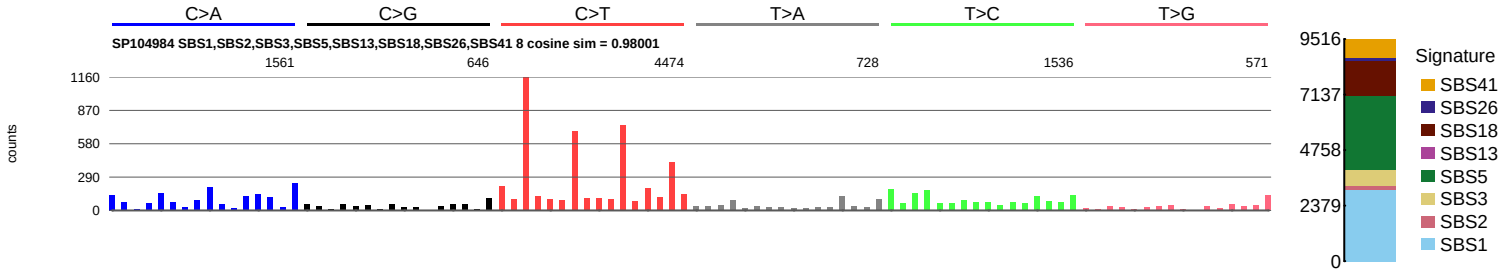

C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98013
1726
780
4362
706
1075
873
counts
1184
888
592
296
0
9516
7137
4758
2379
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS5,SBS15,SBS18,SBS41 6 cosine sim = 0.98009
1556
569
4590
699
1532
567
counts
1140
855
570
285
0
9516
7137
4758
2379
0
Signature
SBS41
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98006
1677
738
4008
861
1273
954
counts
1172
879
586
293
0
9516
7137
4758
2379
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98006
1671
701
4356
614
1478
699
counts
1196
897
598
299
0
9516
7137
4758
2379
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98004
1699
292
4113
659
1812
939
counts
1176
882
588
294
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP104984 SBS1,SBS2,SBS3,SBS5,SBS13,SBS18,SBS26,SBS41 8 cosine sim = 0.98001
1561
646
4474
728
1536
571
counts
1160
870
580
290
0
9516
7137
4758
2379
0
Signature
SBS41
SBS26
SBS18
SBS13
SBS5
SBS3
SBS2
SBS1